Capability Matters

A CASEBOOK

Learning Engineering for Next-Generation Systems

Capability

Matters

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LEARNING DESIGN AND TECHNOLOGY

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An ongoing artifact of the LENS Practice Flywheel — Identify, Activate, Prototype, Analyze, Transition.

DEDICATION

To everyone who believes in a dream and refuses to give up.

COLOPHON

Learning Engineering for Next-Generation Systems: Capability Matters
A Casebook.
First edition, 2026.
Edited by William Gray-Roncal, PhD and James Diamond, PhD.
Compiled for the LENS specialization within the Learning Design and Technology program at the Johns Hopkins University School of Education.
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HOW THIS VOLUME WAS MADE

This casebook was assembled using AI tools as part of an iterative learning-engineering process — the same Practice Flywheel (Identify → Activate → Prototype → Analyze → Transition) that the volume's case studies argue for. The methodology treats AI as the dual entity the curriculum names it: a creative partner that accelerated drafting, cross-referencing, layout, and citation lookup, and an epistemic risk that had to be hand-checked against the record. Every case in the book was reviewed by the editors and hand-checked by students for accuracy: confirming that the sources cited exist and are findable, that the quoted attributions are fairly represented, and that the case studies are accurate accounts of the incidents and the investigations they draw on. Items where the source could not be confirmed are marked "Paraphrasing…" so the attribution is honest about what is the author's reconstruction and what is verbatim from the record.

The volume is not finished. It is the first iteration of a reference dataset that LENS will continue to develop. We will revise the book based on reader feedback, reviewer findings, and our own ongoing practice of learning engineering — adding new cases as the operational record grows, correcting our record where it can be improved, and updating the curriculum crosswalk as the program itself iterates. Corrections, new cases, and reader feedback are welcomed.

SOURCES, ATTRIBUTION & LEGAL NOTICES

All cases reference publicly reported incidents and published investigations. Quoted material is attributed to source investigations, court documents, and academic publications. Reflection questions are original to this volume. The audit record of citations checked and changes made is maintained alongside the source repository.

References to convictions, settlements, and findings reflect the public record at the time of writing. Where individuals are named, they are named only as participants in investigations, court proceedings, or public inquiries already in the public domain. The casebook makes no legal characterization beyond what those records establish.

Corrections, requests for clarification, and notice of any inaccuracy can be sent to LDT / LENS at the Johns Hopkins University School of Education. Substantive corrections will be incorporated in subsequent printings and noted in the casebook's public errata.

WELCOME

Learning engineering is a toolbox, not a single tool.

At its center it is a dialogue between two traditions — the engineering disciplines that design, build, and sustain complex systems, and the learning sciences and education that study how people come to know and do. Around that dialogue gather many other fields: cognitive and human-factors engineering, measurement, implementation science, design, data and computation, and the operational domains themselves. The work is plural by design — many disciplines at once — and it draws on domain knowledge, coding, theory, systems thinking, design, teaching, and analysis.

It welcomes everyone who brings a real tool to it. It is not, for that, a free-for-all: learning engineering stands in an intellectual tradition with its own methods, evidence standards, and lineage, and it holds the work accountable to them. No one carries every tool. This is a team sport — capability gets built at the seams between people who each see part of the system clearly and trust the others to see the rest.

So you don't need to know everything before you start. You need to know what you bring, stay honest about what you don't, and find the people who carry the tools you're missing. What you bring is yours to decide. The cases that follow are an invitation to find your part in the work.

— THE EDITORS

INTRODUCTION

Capability is a system parameter.

Every complex system depends on what people can do inside it. That dependency is measurable, designable, and too important to leave to chance.

> **01.** Systems don't fail because the technology breaks. They fail because someone couldn't do what the system required.
>
> **02.** We engineer every parameter of a system except the people.
>
> **03.** Capability is not a soft problem. It is a systems engineering problem.
>
> **04.** Decision-grade evidence. Not opinions about training.
>
> **05.** Capability without agency is automation. The goal is operators who can perform, adapt, and lead — not comply.
>
> **06.** Sometimes the constraints make the goal unreachable. Saying so early is a result, not a failure.

The cases in this book are not, individually, hard to understand. A pilot recovered from a stall the wrong way. A nurse could not stop a doctor who skipped a step. A safety operator was watching television on her phone when a pedestrian crossed the road. A captain shot down a civilian airliner because the radar data and the operational framing pointed in opposite directions, and the framing won. The proximate cause is usually obvious. The first investigator on the scene almost always finds it within hours.

What is hard to understand — and what this book exists to make legible — is the system that produced the gap into which each incident fell. In every case the proximate actor was operating inside an architecture of training, procedure, authority, measurement, and incentive that had, for years, been quietly degrading. The pilot had never trained to recover a stall at cruise altitude. The nurse had no procedural path to intervene. The safety operator had been hired into a role nobody had figured out how to make performable. The radar operator had been trained to a tempo and a presumption that the system above him no longer shared with the system around him.

Each of these architectures was designed. Each was funded. Each was reviewed. Each carried, written somewhere in its specifications, the assumption that the people inside it would be able to do what the system required of them when the system required it. That assumption is what this book calls the **capability parameter**. It is a property of the entire sociotechnical system, not of any individual operator. Capability lives at the **interface** between what a system requires of its operators and the impact the system has to deliver — and when that interface is wrong, the gap may originate in human development, in system design, in organizational performance, or in the interaction among them. Distinguishing which is itself a measurement and engineering problem; the casebook calls it **gap attribution**, and treats it as the diagnostic move that the rest of the discipline depends on. When the interface fails — wherever the source — the consequences are paid in lives, in dollars, in trust, and in time the institution will never recover.

The premise of the Johns Hopkins School of Education's Learning Design & Technology program — and of the Learning Engineering for Next-Generation Systems specialization that has grown out of it — is that the capability parameter is engineerable. Not by accident. Not by declaration. By the same kind of evidence-grounded, methods-grounded, cross-domain discipline that built modern reliability engineering, modern epidemiology, and modern systems safety. The discipline does not yet exist at the scale the evidence demands. LDT and LENS are organized to help build it.

I · THE COST OF THE GAP

The Institute of Medicine's **To Err Is Human** (1999) estimated that preventable medical error caused between forty-four thousand and ninety-eight thousand deaths a year in the United States — already, at the lower bound, a top-ten cause of death. The report catalyzed a generation of patient-safety work: surgical checklists, central-line bundles, team training, computerized order entry. Some of those interventions are documented in Part II of this book as successes. And yet in 2016 Makary and Daniel returned to the question from the same institution that produces this casebook and concluded that medical error remains the third leading cause of death in the United States, killing more than two hundred fifty thousand people a year [1].

The interval between the IOM report and the Makary update is approximately the canonical gap implementation science has identified between research evidence and clinical practice: seventeen years [2]. The average. The median. Only about fourteen percent of research findings ever reach practice at all [3]. The system to surface them, vet them, deploy them, sustain them, and measure their effect at the scale at which they would matter has never been built as an institution. It has been built as a series of grants. The difference is the point.

This is not only a healthcare problem. The U.S. Navy lost seventeen sailors in two avoidable destroyer collisions in 2017 because in 2003 it had cut a sixteen-week classroom-and-simulator course down to a set of CD-ROMs [4]. The world's most expensive aircraft program is operating at half of its design readiness in 2026 because the platform was fielded faster than the maintainer pipeline could be built [5]. One hundred million dollars of educational infrastructure dissolved in fourteen months because the institution building it did not engineer the governance and trust that the technology presupposed [6]. A predictive grading algorithm downgraded a third of an entire nation's high-school students in a single afternoon, and was withdrawn within a week, because nobody inside the agency could be specifically held to account for what the measurement instrument was measuring [7].

None of these failures is a failure of effort. None is a failure of intelligence. Each is a failure of the capability infrastructure to match the system it was operating. In each case, the gap was diagnosable in advance. In most cases, it was actually diagnosed in advance, by someone — and the diagnosis did not produce the remediation. The cost of the unrepaired diagnosis is why **capability** matters.

II · WHAT AN ENGINEERABLE DISCIPLINE LOOKS LIKE

Crew Resource Management did not exist as a discipline in 1976. In March 1977 two 747s collided on a foggy runway at Tenerife and 583 people died. Within five years United Airlines had operationalized the first CRM curriculum [8] within twenty years the Commercial Aviation Safety Team had built the closed-loop hazard-identification system that lets the FAA know whether CRM is still working [9]. The result over the next two decades was an eighty-three percent reduction in U.S. commercial aviation fatality risk and a ninety-five percent reduction in fatalities per hundred million passengers [10]. CRM works not because it taught individual airmanship — pilots were already excellent — but because it redesigned the **system of interaction** in the cockpit: the authority gradient, the communication protocol, the standard brief and debrief. The discipline treated team coordination as an engineerable property of the system.

The same pattern shows up in every domain where the cost of failure has been high enough to fund the discipline. After Three Mile Island the U.S. nuclear industry stood up the Institute of Nuclear Power Operations within nine months — before the Kemeny Commission's report was even finalized — because every utility had recognized that an accident at any single plant would affect every operator's license to operate [11]. INPO and the National Academy for Nuclear Training, founded in 1985, have presided over more than four decades of zero significant releases at U.S. commercial reactors. The comparison with the surface Navy, which cut its training during the same era, is the cleanest available test of what happens when capability is engineered versus when it is deferred.

Healthcare has its own version of this arc, but only partially. Peter Pronovost's central-line checklist, paired with the nurses' authority to enforce it, drove bloodstream infections to near zero across 103 Michigan ICUs and saved an estimated fifteen hundred lives in eighteen months [12]. Atul Gawande's nineteen-item surgical safety checklist, paired with three required pauses in the operative timeline, halved the surgical mortality rate across eight hospitals in eight countries [13]. TeamSTEPPS, jointly developed by AHRQ and DoD, translated five decades of high-reliability research from aviation and the military into clinical practice in years rather than decades — because the implementation infrastructure was funded as part of the intervention rather than as an afterthought [14].

These are not stories about individual hospitals or individual ships. They are stories about the difference between a domain that has organized itself to engineer capability and one that has not. The pattern is consistent across all of them: a catastrophe makes the capability gap visible; an institution is built that treats the gap as the institution's responsibility; the institution funds the measurement architecture that lets it know whether the gap is closing; and twenty years later the death rate is half what it was. The intervention is always paired. A technical artifact — a checklist, a brief, a cord — combined with a cultural artifact — protected authority to use it, a no-blame protocol, a credible governance body. Neither alone moves the curve. Both together move it dramatically.

III · THE DISCIPLINE WE HAVE, THE DISCIPLINE WE NEED

The intellectual material to build this discipline already exists. The learning sciences have produced four decades of converging evidence on how skill, knowledge, and judgment develop and decay [15]. Cognitive engineering has produced rigorous methods for analyzing human-machine systems [16]. Human factors and ergonomics have produced an evidence base for interface and workflow design [17]. Implementation science has produced frameworks for taking effective interventions to scale [18].

Systems safety engineering has produced both diagnostic tools — Reason's swiss-cheese model, Rasmussen's migration-to-the-boundary, Leveson's STAMP — and control-theoretic methods for design [19]. High-reliability- organization theory has documented what the institutions that have solved the capability problem actually do [20]. The discipline is not missing its evidence base. It is missing its integration.

Learning engineering is the integration. The term, in its modern usage, traces to Bror Saxberg and was elaborated by Goodell, Kolodner, and the IEEE Industry Connections Industry Consortium on Learning Engineering (ICICLE) [21]. The IEEE Learning Engineering Body of Knowledge (LEBoK) and its associated process model are the most developed available specification of what the discipline does [22]. The premise is that the work of building, deploying, and sustaining capability at scale is an engineering activity: it requires explicit requirements, evidence-based design, instrumented measurement, and feedback-driven iteration. It is not exclusively a research activity. It is not exclusively a managerial activity. It is engineering, and it can be taught and practiced as engineering.

LENS — Learning Engineering for Next-Generation Systems — is a specialization within the Johns Hopkins School of Education's Learning Design & Technology program that operationalizes this premise for high-consequence operational domains. Its core competencies — capability analysis and requirements, evidence and analytics, human-AI teaming, knowledge transfer, bias and governance, computational and AI methods — are organized to produce graduates who can do the work the cases in this book required, and who can build the institutions that the success cases in Part II had to invent. The curriculum threading commitments — implementation science as a through-line, equity as a design commitment rather than an audit, decision-grade evidence as a deliverable rather than a report, communication and system-of-systems integration as engineerable properties of the system, the disciplined use of learning- technology aids (from XR to adaptive platforms) as instruments whose evaluation is part of the work, and **AI as both a creative partner and an epistemic risk** — leveraging AI's generative reach while guarding against offloading the thinking — are drawn directly from the patterns the cases reveal.

Three of those commitments deserve a moment of their own — they are the substrate the rest of the curriculum runs on. The first is **communication, translation, and integration within and across disparate systems**. A striking number of cases in this book are, at their proximate cause, translation or integration failures across boundaries — failures of language, convention, unit, role, agency, or discipline to carry meaning reliably from one part of a system to another, or failures of subsystems built to different assumptions to compose into a working whole. Mars Climate Orbiter was a metric-versus- imperial translation failure (Case 9). Tenerife was a translation failure at the takeoff-clearance boundary between two cockpits and a tower under fog and time pressure (Case 70). The Patriot at Dhahran failed because the manufacturer's assumption about run time and the operator's actual run time were on opposite sides of a documentation boundary that had not been engineered to be crossed (Case 8). Eagle Claw failed because the rotary-wing, fixed-wing, ground, and command components were assembled across service boundaries that had no shared operating practice (Case 5). 9/11 was a cross-agency translation failure measured in thousands of dead (Case 167). Boeing 737 MAX was, at one level, an integration failure: a single-string sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together (Case 7). The successful cases — AlphaFold (Case 132), MICrONS, CRM (Case 70 paired with Case 72), Keystone (Case 71), the paired-intervention examples in Chapter 8 — are, equivalently, cases of disciplined translation and integration: biology into computation, science into operational practice, technical reform into cultural reform, multiple subsystems into a working whole. Engineering capability across these boundaries — language, discipline, agency, subsystem-to-subsystem, system-of-systems — is itself a designable target. The LENS curriculum treats it as such.

The second is **technology aids and learning platforms**. The discipline has at its disposal a fast-evolving toolkit: extended- reality (XR) simulation environments — VR, AR, and mixed-reality systems used for procedural training, spatial-reasoning practice, and high-fidelity rehearsal under conditions that would be unsafe or impossible in the field — together with learning- management systems, adaptive learning platforms (the lineage from Cognitive Tutor and ASSISTments to current ITS work; Cases 38, 33), intelligent tutoring frameworks such as GIFT, learning- analytics infrastructures such as xAPI and the Total Learning Architecture (Case 171), game-based learning environments, LLM-augmented tutors and authoring tools, and high-fidelity simulators in aviation, surgery, defense, and process industries. These tools are not the discipline. They are the discipline's instruments — and the cases in this book show, repeatedly, that powerful instruments do not by themselves engineer capability (Cases 146, 149, 61, 33, 79, 80). The LENS curriculum teaches the practitioner to choose, configure, evaluate, and govern these tools with the same evidentiary discipline applied to any other intervention, and to recognize when the binding constraint is the surrounding architecture rather than the tool itself.

The third — and the design constraint on all the others — is **human agency**. The capability discipline can be misread as optimization: produce operators whose behavior matches a specification. That reading collapses the very property the cases reveal as load-bearing. Every successful institution in Part II — INPO operators trained to challenge their reactor team, Keystone ICU nurses authorized to halt a procedure (Case 71), the Nuclear Navy's questioning attitude (Case 18), the first-officer authority CRM built into the cockpit (Case 70) — engineered for capability **and** for the operator's capacity to exercise independent judgment, to adapt to conditions the system was not designed for, and to shape the surrounding architecture rather than only execute inside it. Capability without that capacity is automation. LENS treats the preservation and development of agency — the room the system leaves its operators to think, decide, and lead — as a design constraint on every intervention, including those in which AI is the partner. The paired risk runs the same direction: an AI collaborator can accelerate the flywheel or quietly offload the thinking, and which one it does is a design and governance decision, not a property of the tool.

IV · THE METHOD · UNPACKING THE CASE

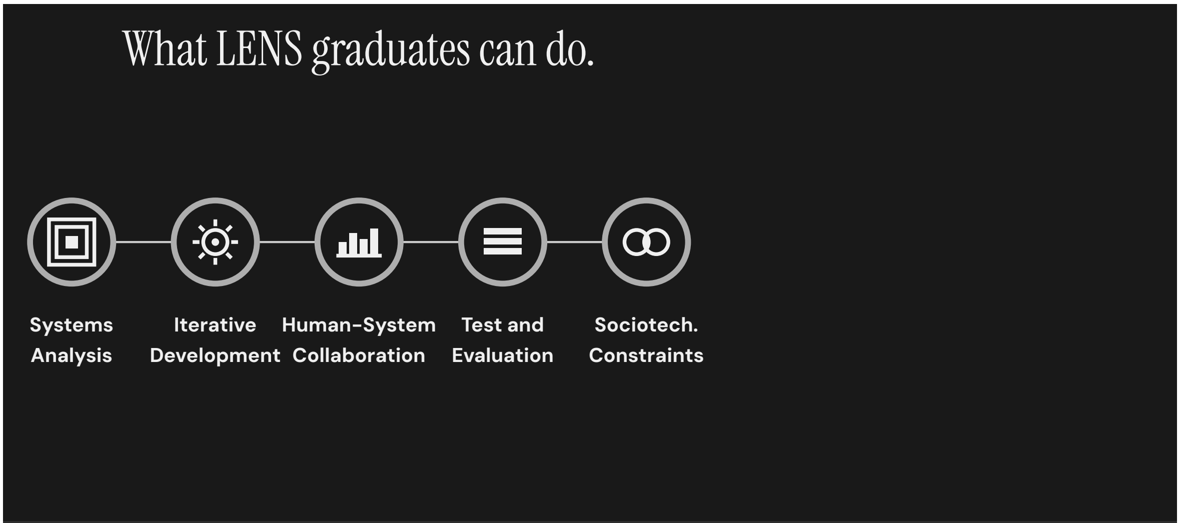
Learning engineering is not a one-time act of design. Its defining feature, as articulated in the IEEE Industry Connections Industry Consortium on Learning Engineering's process model, is iteration: understand the operational context, specify capability requirements, prototype an intervention, instrument it, measure, learn, redesign. The work cycles. The cases in this book that produced sustained outcomes — CRM and CAST, INPO, Keystone ICU, the WHO Surgical Safety Checklist, TeamSTEPPS — all share that loop structure. They were not built once and left alone. They were built, measured, rebuilt, remeasured, and rebuilt again. The institutional capability to sustain the loop **is** what distinguishes them from interventions whose effects decayed.

The loop also has a legitimate negative result, and the discipline has to be honest enough to reach it. Capability engineering always operates inside constraints — budget, schedule, regulation, institutional will, politics, ethics, public trust — and many of the binding ones are not technical. Part of the work is **managing those constraints and the risk they carry**: reading the constraint space accurately enough to recognize when external, non-technical factors have narrowed the solution space to the point where a particular capability goal is no longer realistic to pursue. Reaching that conclusion — and documenting **why** — is itself a valid outcome of a project, not a failure of one. It redirects effort toward goals that are actually reachable, surfaces the non-technical barriers for the stakeholders who can change them, and prevents the far more expensive failure of driving an infeasible target all the way to the operational record before it collapses there. A number of cases in this book are, in part, stories of institutions that could not say “not like this, not yet” in time.

The casebook turns this loop into a reading method. Each case is read, diagnosed, paired with a learning-engineering response, and tested in studio against students’ own operational domains. The cycle — framing the system whose capability is at stake, eliciting requirements from operational analysis, building the evidence architecture that lets an institution know whether its interventions worked, examining the human and AI roles inside the designed system, and running the loop end-to-end on a real problem — is the same discipline the casebook applies to every case it reads. The cases that return as success stories in Part II are the cases whose institutions completed the loop. The cases in Part I are the cases whose institutions did not — or did not loop fast enough. Unpacking the case **is** the method: identify, instrument, and close the next such loop before the next case is written.

V · THE ANALYTIC LENS · THE FIVE CAPABILITIES THE CASES TURN ON

The cases in this book turn on five capabilities. They are the analytic lens the casebook reads each case through, and the cases that follow are tagged to them: where capability was absent, one or more of these five was the capability that was missing; where it was engineered, one or more of these five was the capability that was built. Named as concentration learning outcomes, they are also what the discipline sets out to teach.



WHAT LENS GRADUATES CAN DO — FIVE COMPETENCIES

1 · SYSTEMS ANALYSIS — SEE THE WHOLE SYSTEM

SYSTEMS THINKING AND ANALYSIS

Decompose system performance requirements into measurable human capability requirements; apply systems-engineering lifecycle models to scope, sequence, and evaluate capability interventions **within and across disparate systems**; analyze and communicate the interdependencies between human performance and system performance to predict operational impact at scale.

2 · ITERATIVE DEVELOPMENT — BUILD, TEST, REFINE

LEARNING ENGINEERING DESIGN AND IMPLEMENTATION

Design capability-development interventions that integrate learning-sciences principles with measurable outcomes and system-design constraints, and that function at **speed and scale** in operational environments; construct and communicate implementation plans that address adoption, sustainment, and lifecycle integration across organizational and system boundaries.

3 · HUMAN-SYSTEM COLLABORATION — WORK TOGETHER

HUMAN-SYSTEM COLLABORATION AND ADAPTIVE SYSTEMS

Design role architectures, interface and alert systems, mode/state transparency, authority gradients, and recoverability mechanisms that engineer collaborative capability at the human-system boundary — including human-AI partnership as one sub-pattern; evaluate the measured impact of system-mediated work on human performance, naming automation bias, cognitive offloading, and skill atrophy; specify **delegation with revocation** — the disconfirming evidence that would revoke a delegation — and measure collaboration as the unit of analysis.

4 · TEST AND EVALUATION — SHOW WHAT WORKS

DATA, MEASUREMENT, AND EVALUATION

Design ethical instrumentation strategies that produce measurable evidence in regulated or high-consequence environments; construct **decision-grade evidence** artifacts — dashboards, reports, governance documentation — under irreducible uncertainty, naming what is known, what is assumed, and what would change the decision; differentiate capability gaps from system-design and organizational-performance problems using **gap attribution**; demonstrate that omitting a protected attribute does not establish fairness.

5 · NAVIGATING SOCIOTECHNICAL CONSTRAINTS — MAKE IT WORK IN THE REAL WORLD

CONTEXT AND DOMAIN FLUENCY

Analyze the regulatory, organizational, and cultural constraints that shape capability development in healthcare, defense, or education contexts; apply human-systems-integration frameworks to evaluate the measurable impact of capability approaches on operational environments; synthesize and communicate stakeholder requirements across disciplinary and institutional boundaries into coherent design specifications, including cross-regime / platform-dependency

> governance where capability is deployed on a platform governed by a different regime than the one operating it.

See the whole system. Build, test, refine. Work together. Show what works. Make it work in the real world.

Three phrases recur across those five capabilities and across the cases that follow: **within and across disparate systems** (the scope at which capability is actually engineered), **speed and scale** (the operational tempo the discipline must match), and **gap attribution between learning, system design, and organizational performance problems** (the diagnostic that decides what kind of intervention will actually move the outcome). They are the working language the cases are read in.

VI · HOW TO READ THIS BOOK

Part I, "The Failure Modes," organizes the failure cases under six recurring patterns: training gap, capability designed out, normalization of deviance, interface failure, governance failure, and knowledge loss. A seventh chapter — the Evidence Gap — anchors the measurement question. The taxonomy is not a theory. It is a finding: six categories that account for almost every well-documented case in the literature of preventable failure in complex sociotechnical systems, and that recur across healthcare, defense, education, energy, transportation, and government.

Part II, "What Works," collects the cases in which the capability parameter was engineered rather than left to drift. Every success case in the dataset shares a structural feature: a **paired intervention.** A technical artifact paired with a cultural authority. A measurement instrument paired with a governance body that listens to it. A curriculum paired with the institutional architecture to sustain it. Neither half alone moves the curve. The pair is irreducible. This is the strongest empirical pattern in the book and it directly informs the LENS pedagogical commitment to co-optimization across technical and human design.

Part III, "The Frontier," concerns the cases where the discipline is being asked to do work it has not done before — particularly the human-AI teaming cases that anticipate the next decade of capability engineering. The cases are deliberately less settled. The discipline is being asked to specify what good looks like before the catastrophe arrives. That is the work the program exists to teach.

Each case occupies a two-page spread. The left page tells what happened — narrative, evidence, attribution. The right page is the **Learning Engineering Lens:** a synthesis of what the case teaches, how the LENS curriculum addresses the pattern, and reflection questions designed for studio discussion. The casebook is built to be taught from, read straight through, or consulted as a reference. The curriculum crosswalk in the closing matter maps each case to the specific courses in the program that use it as a worked example.

> **How these cases were chosen.** The collection is purposive, not systematic. The cases are exemplars — drawn from the documented public record and from the editors' own practice — selected because each makes a capability-engineering pattern legible, not because they constitute a representative or exhaustive sample. The corpus over-represents catastrophic, well-investigated failures, which generate

> the richest public record, and under-represents the everyday, program-scale work that fills most of a practitioner's career; a deliberate expansion has begun to close that gap. Read the book as a teaching instrument rather than a systematic review: the claim is that these patterns recur and are engineerable — not that this is every case, or the average one.

HOW TO USE THIS BOOK

A field manual for capability engineering.

This book collects a growing record of real incidents in which capability — what people could or could not do inside a complex system — determined whether the system worked. Some are failures: lives lost, systems wrecked, billions written off. Some are successes: deliberate interventions that engineered capability into the architecture of the work. Together they form an evidence base for the argument that capability is a designable, measurable property of every complex system.

Each case occupies a two-page spread. The left-hand page tells the story: what happened, what investigators found, and the documentary record. The right-hand page is the **Learning Engineering Lens** — what the case teaches about capability as a system parameter, how the LENS curriculum addresses the pattern, and a short set of reflection questions designed for studio discussion.

The Lens page also carries a **Who Builds This** note: the mix of expertise and tools a team would bring to engineer the fix, read off the case's failure modes. A Training-Gap case pulls in learning scientists and instructional designers; an Interface case, human- factors and interaction designers; a Governance case, policy, ethics, and measurement. Most cases need several at once, held together by domain experts and a learning engineer. No single discipline carries a case alone — reading the modes as a staffing list is itself a LENS skill, and what **you** bring is yours to decide.

READING THE TAXONOMY

Each case is tagged with one or more failure modes. Most cases involve several. The primary mode is listed first; contributing modes follow.

- **T** Training Gap
  Personnel were never taught what they needed to know.
- **D** Designed Out
  Human capability was deliberately removed from the system.
- **N** Normalization of Deviance
  Known risks were accepted as routine.
- **H** Human-System Interface
  Correct performance was made unreasonably difficult.
- **G** Governance & Trust
  Ethics, accountability, and stakeholder trust were afterthoughts.
- **K** Knowledge & Institutional Memory
  Organizational knowledge degraded or was never captured.

LENS COURSE CODES

Each case maps to one or more courses in the LENS specialization:

SHARED LDT FOUNDATIONS · ADDITIONAL CONTEXT

F1	**Learning Sciences Studio.** How learning, motivation, and skill develop — applied to technology-mediated design.
F2	**Critical Perspectives on Educational Technology.** How power, equity, and society shape — and are shaped by — learning tech.

LENS-DESIGNED COURSES

LEN 1	Principles of LE for Complex Systems
LEN 2	Human-AI Teaming and Adaptive Systems
LEN 3	Learning Engineering Systems
LEN 4	Evidence, Analytics, and Measurement
LEN 5	Human Capability Analysis and Requirements
LEN 6	Applied Problem Solving in LE
LEN 7	Bias, Risk, and Governance

LEN 8	Knowledge Transfer and Organizational Learning
LEN 9	Computational and AI Methods
LEN 10	Learning Engineering Project (capstone)

> *The strongest cases pair a catastrophic failure with a systematic capability intervention. Together they show both the cost of the gap and the tractability of the solution. The discipline that closes that gap is learning engineering.*

THE CASE MATRIX

The cases at a glance

Gold numbers indicate failures and systemic conditions; teal numbers indicate paired-intervention successes and the open closing case.

1	USS Fitzgerald & USS John S. McCain 2017	T·K·N
2	Marine Corps Training in the INDOPACOM AOR ongoing	T·K
3	F-35 Sustainment & Maintainer Shortage ongoing	T·K·D
4	Military Fratricide — Desert Storm to Afghanistan 1991 – 2014	T·H·K
5	Operation Eagle Claw 1980	T·K
6	AeroPerú Flight 603 1996	H·T
7	Boeing 737 MAX / MCAS 2018 – 2019	D·T·H
8	Patriot Missile / Dhahran 1991	D·H·K
9	Mars Climate Orbiter — Unit Mismatch 1999	D·K
10	Knight Capital Trading Loss 2012	D·K
11	Sago Mine Disaster 2006	N·T·K
12	Boeing 737 Rudder Hardovers 1991, 1994	H·D
13	Glass-Cockpit Transition in Light Aircraft — Technology Outran Training 2010	H·N·K
14	Healthcare.gov Launch 2013	K·T·G
15	Ariane 5 Flight 501 1996	D·K·H
16	EU Human Brain Project — Top-Down Vision That Unraveled 2013 – 2023	G·N
17	GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data 2022 (GAO-22-104533)	G·K
18	U.S. Nuclear Navy / Rickover Training Model 1954 – present	T·K·N
19	Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable 1963 – present	G·K·H
20	California Nurse Staffing Ratios — Legislating a Capability Requirement 2004 – 2010	G·K
21	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard 2020 revision (series since 1968)	G·K
22	FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule 1988 – 2010s	G·D·K
23	FAA NextGen and the ADS-B Out Transition 2003 – 2020	G·D·K

50	The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension 2016 – 2025	T·K·D
51	Zhang/Scardamalia — Knowledge Building Across Three Cohorts 2009	T·K·D
52	Chen et al. — Rural China AI Devices and the Equity-Direction Finding 2025	T·K·D
53	Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account 2023	T·K·N
54	USS Vincennes & Iran Air Flight 655 1988	H·T
55	Kegworth / British Midland 92 1989	T·H·K
56	Asiana Airlines Flight 214 2013	T·H
57	Helios Airways Flight 522 2005	T·H
58	TransAsia Airways Flight 235 2015	T·H
59	Therac-25 1985 – 1987	H·D·G
60	Northeast Blackout 2003	H·K
61	EHR / CPOE Implementation 2005 – present	H·D·G
62	Uber ATG / Tempe Fatality 2018	T·N·G·H
63	Eastern Air Lines Flight 401 1972	H·T
64	Stanislav Petrov / 1983 False Alert 1983	H·T
65	NASA EVA-23 — Water Intrusion Inside a Spacesuit 2013	H·N·D
66	UK Post Office Horizon Scandal 1999 – 2015	G·H·K
67	Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018	D·K·N
68	Air Canada Chatbot Liability — Delegation Without Revocation 2022 – 2024	D·K·N
69	Three Mile Island 1979	T·H
70	Crew Resource Management & CAST 1981 – present	T·H·N
71	Keystone ICU / Pronovost Checklist 2004 – present	T·N
72	Korean Air Safety Transformation 2000 – present	T·N
73	Toyota Production System / Andon Cord 1950s – present	N·G
74	TREWS / Sepsis Watch 2018 – 2022	H·K·G
75	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff 2024 – 2025	H·K·N
76	INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet 2010 – present	D·H·K
77	Hybrid Human-AI Tutoring — Augmentation, Not Delegation 2024	T·K·D
78	Tesla Autopilot — Recurring Fatalities 2016 – present	T·N·G·H
79	ChatGPT in Healthcare — Hallucination Cases 2023 – present	H·D
80	AI-Augmented Coding Tools 2021 – present	T·H

111	Aviation Safety Reporting System (ASRS)	1976 – present	T·K·N
112	Bristol Heart Babies Reform	1984 – present	G·N
113	Removing the Race Coefficient from eGFR	2021	D·G·N
114	Pulse Oximetry Across Skin Tones	1990 – 2020 – 2025	D·G·N
115	Open University 'Ethical Use of Student Data' and OU Analyse	2014 – 2025	G·K·N
116	Anesthesia Monitoring Standards and the APSF — The Mortality Decline	1986 – present	H·K·G
117	Interprofessional Education and the Evidence Gap	2013 – 2015	K·N
118	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation	1996 – 2002	H·K·G
119	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson	1981 – 2008 (TCAS II Version 7.1)	H·K·G
120	Bar-Code Medication Administration — Cue at the Point of Care	2010	H·K·D
121	Surgical Skill Video Peer-Rating Predicts Complications	2013	H·D·K
122	Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units	2009 – 2016	T·K·H
123	MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil)	2024	K·N
124	Brinkerhoff Success Case Method — Tails as the Evaluation Instrument	2005 – present	K·N
125	Annual-Screening UI Redesign + CDS at University of Missouri Health Care	2020	T·D·N
126	Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal	2019 – 2024	T·D·N
127	Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive	2007 – 2014	T·K·D
128	OU Analyse — Predictive Learning Analytics and Teacher Use at Scale	2019 – 2023	T·K·D
129	COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case	2022 – 2024	T·K·D
130	Radiology AI Miscalibration	2018 – present	H·K·D
131	Predictive Policing — PredPol	2011 – present	G·H·D
132	AlphaFold — Protein Structure Prediction	2020 – present	T
133	Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction	2024	D·K·N
134	Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models	2021 – 2024	D·K·N
135	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models	2025	T·K·N

136	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback 2020	T·K·D
137	Texas City BP Refinery Explosion 2005	N·T·K·G
138	Mark 14 Torpedo Failures 1941 – 1943	T·K·G
139	GM Ignition Switch 2002 – 2014	D·G
140	Challenger & Columbia 1986 / 2003	N·K·G
141	V-22 Osprey 1991 – present	T·H·N
142	Davis-Besse Reactor Head Corrosion 2002	N·K·G
143	Fukushima Daiichi 2011	N·G·K
144	CrowdStrike Falcon Outage 2024	D·K·G
145	TSB Bank IT Migration 2018	H·G
146	inBloom 2014	G
147	Bhopal 1984	T·K·N·G
148	Grenfell Tower 2017	G·T·K·N
149	Summit Learning / Personalized Learning Rollout 2014–2019	G·T·K
150	Theranos 2003 – 2018	G·D
151	Cambridge Analytica / Facebook 2014 – 2018	G
152	Equifax Data Breach 2017	G·K
153	Hyatt Regency Walkway Collapse 1981	D·G
154	Camp Fire / PG&E 2018	G·N·K
155	Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds 2014 – 2020	D·G·N
156	Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment 2023	G·K·N
157	Enrollment-Algorithm Yield Optimization Across U.S. Higher Education 2010s – present (Brookings synthesis 2021)	T·K·N
158	Crisis Point — Merit-Aid Leveraging at Public Flagships 2001 – 2017 (Burd IPEDS analysis); 2024 (Lifting the Veil)	T·K·N
159	GAO Online Program Manager Oversight Gap (GAO-22-104463) 2022	D·K·N
160	USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC) 2010s – 2024	D·K·N
161	In re 2U, Inc. Securities Class Action — When Yield Management Crashes Into Disclosure 2019 – 2022	D·K·N
162	Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict 2016 – 2023	D·G·N
163	Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model 2019 – 2021	D·K·N
164	NASA Saturn V Documentation 1972 – present	K
165	Boeing Starliner 2019 – 2024	K·D
166	Bernard Madoff / SEC Failures 1992 – 2008	G·K·N

167	9/11 Intelligence Sharing Failures 1996 – 2001	G·K
168	iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms 2006 – 2011	G·D·N
169	Data Privacy and Learning Analytics on the African Continent 2022	K·G·N
170	Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions 2025	T·K·N
171	xAPI / Total Learning Architecture — Interoperability Gap ongoing	K·G
172	INPO and the Nuclear Academy 1979 – present	T·K·G
173	TeamSTEPPS 2006 – present	T·N
174	Tylenol Recall 1982	G·N
175	Singapore Airlines Safety Transformation 1980s – present	T·N
176	Launching the BRAIN Initiative — Governance of a Grand Challenge 2011 – 2015 – present	G·K·N
177	CIRAS — Confidential Incident Reporting for UK Rail 1996 – present	G·K·N
178	Team Science Training for Clinical and Translational Scientists 2020 – 2022	K·N
179	Implementation-Science Training — Stated Goals Outrunning Operational Practice 2020s	K·N
180	FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike 1992 – 2000s	G·K·N
181	LALA — Building Learning-Analytics Governance Capacity Across Latin America 2017 – 2020	K·N
182	Norway's National Expert Commission on Learning Analytics 2022 – 2023	G·K·N
183	Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact 2023 / 2025	G·K·N
184	CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver 2020 – 2024	G·K·D
185	Singapore SkillsFuture — National Workforce Capability at Scale 2015 – present	G·K·D
186	Australian Hospital-Pharmacy Technician Role Redesign 2016	D·N·H
187	Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India 2018 – 2025	G·N·H
188	Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface 2013 – 2018	H·N
189	Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD 2014 – present	G·N
190	CARE Principles — Indigenous Data Governance as a Replaced Regime 2019 – 2020 (principles); ongoing	G·N
191	NYC Local Law 144 — Bias Audits as Governance Artifact 2023 – present	D·G·K
192	Cruise Robotaxi — Pedestrian Drag 2023	G·D·H

Modes. T Training Gap · D Designed Out · N Normalization of Deviance · H Human-System Interface · G Governance & Trust · K Knowledge & Institutional Memory

Courses. The LDT / LENS courses each case serves as a worked example for are indexed separately in **The cases by LENS course** in the appendix.

Parts. Part II — **What Works** — opens at Chapter 8 with the paired-intervention successes. Part III — **The Frontier** — and the contemporary expansion carry the human-AI teaming cases; the open question closes the volume at Case 194.

PART I — THE FAILURE MODES

Chapter 1

Systems Analysis — What Fails

When the work the system requires goes unspecified.

Capability is the interface between what a system requires of its operators and the impact of that system.

THE CAPABILITY MATTERS VOLUME THESIS.

CASE 1 DEFENSE 2017

USS Fitzgerald & USS John S. McCain

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT 17 sailors killed in two avoidable destroyer collisions in the Western Pacific within nine weeks

IN BRIEF

In the summer of 2017 two U.S. Navy destroyers of the forward-deployed Seventh Fleet collided with merchant ships nine weeks apart, killing seventeen sailors — seven on the Fitzgerald, ten on the John S. McCain. Their crews' seamanship, navigation, and console skills had eroded under a decade of relentless operational tempo and the self-study "training" that replaced the in-person school the Navy cut in 2003. Investigations by the Navy and the NTSB judged both collisions avoidable and traced them to insufficient training and weak oversight; on the McCain, a confusing touch-screen helm let a watch team believe it had lost steering it never actually lost. The case is this book's canonical training gap: stated readiness and real readiness diverged for years, and the system could no longer see the difference.

BACKGROUND

The destroyers belonged to the forward-deployed Seventh Fleet in Japan, kept at the fleet's highest tempo — where training was the first thing spent. Unlike home-ported ships, which rotate through a dedicated work-up cycle before deploying, the forward-deployed force was expected to be continuously available, so there was rarely a quiet stretch in which to recover a lapsed qualification. In 2003 the Navy replaced its in-person Surface Warfare Officers School course with self-study CD-ROMs — "SWOS in a Box" — sending newly commissioned officers directly to their ships to learn the trade as time and the ship's qualified watchstanders allowed.[0] The change was framed as a cost-saving modernization; in practice it transferred junior-officer education from the schoolhouse to the bridge at the moment the operational tempo on those bridges left the least slack to absorb it. By 2017 the GAO found 37% of the Japan-based ships' warfare certifications — including basic seamanship — expired, a fivefold rise since 2015, with the lapses routinely waived rather than fixed.[1]

WHAT HAPPENED

On 17 June 2017 the Fitzgerald, the give-way vessel in a busy shipping channel, took no early action and was struck by the container ship ACX Crystal off the coast of Japan; the sea poured into a berthing compartment where the crew slept, and seven sailors drowned before they could escape.[2] The NTSB later described a bridge team that had lost track of converging traffic on a clear night, and an officer of the deck whose qualifications and recency on the very procedures the situation demanded had lapsed under the waiver regime. Nine weeks later the John S. McCain collided with the tanker Alnic MC near Singapore, killing ten: while shifting throttle control between consoles, a watchstander unknowingly handed off steering to another station, the ship turned across the strait's traffic, and no one on the bridge understood the touch-screen helm well enough to recognize what had happened.[3] For more than a minute the bridge team believed the ship had lost steering it had never lost — a misdiagnosis the interface invited and the training had not equipped anyone to overturn.

THE INVESTIGATION

The Navy's Comprehensive Review (2017) judged both collisions avoidable, citing failures in basic seamanship, navigation, and operating the ships' own equipment.[4] The NTSB found the McCain's probable cause to be a lack of Navy oversight that produced insufficient training and inadequate bridge procedures,[5] and faulted the design of the touch-screen helm, installed to modernize the bridge, whose blending of steering and throttle made an unintended transfer of control easy to trigger and hard to notice — a trap waiting for an under-trained crew. The companion Strategic Readiness Review, commissioned by the Secretary of the Navy and led by retired Admiral Phil Balisle, reached further into the institution: a decade of "can-do" responses to mounting demand had eroded the manning, certification, and maintenance margins the surface force was built on, and the readiness reports senior leaders relied on had stopped reflecting the conditions on the ships.[6] Watchbills and crew-day logs gathered after the collisions showed watchstanders routinely averaging fewer than five hours of sleep on patrol — a finding the NTSB folded into its causal chain.

THE CAPABILITY GAP

The gap was invisible from inside. The Strategic Readiness Review described risks that "accumulated over time and did so insidiously," the system no longer able to see that the processes meant to surface shortfalls had themselves failed.[7] Each individual waiver was locally defensible — a deadline met, a deployment kept — but in aggregate they hollowed out the force while every readiness dashboard still reported green. Stated and actual readiness had diverged for a decade; the cost arrived as seventeen lives, not a failed inspection. Two collisions, nine weeks apart, ruled out the comforting reading that one was an outlier: Fitzgerald taught that an under-trained bridge team could fail at the most basic give-way rules, and McCain taught that an unfamiliar interface could be the precipitating mechanism through which the same gap reached the hull. The pair, not either case alone, made the diagnosis structural rather than individual.

AFTERMATH & REFORM

The reforms were the deepest in a generation: the in-person officer pipeline was rebuilt as a multi-phase Basic Division Officer Course, reinstating classroom and simulator instruction the 2003 CD-ROM model had displaced; a Ready-for-Sea Assessment and a Naval Surface Group Western Pacific stood up to give forward-deployed units the independent certification cycle home-ported ships already had; circadian watchbills were adopted fleet-wide to fight fatigue; and the touch-screen helm was slated for replacement by a conventional wheel and throttle across the destroyer fleet.[8] Each change conceded that the training and the interface had been real requirements all along — and that trading them for tempo had only moved the cost onto two ships full of people.[9]

SURFACE WARFARE OFFICER TRAINING, 2003 → 2017

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THE LEARNING ENGINEERING LENS

> *The risks that were taken in the Western Pacific accumulated over time and did so insidiously.*
>
> U.S. NAVY STRATEGIC READINESS REVIEW, 2017

LE INSIGHT

Fitzgerald/McCain is the canonical Training Gap case because the gap was invisible from inside the system. Operational tempo and self-study "training" combined to produce a fleet whose stated readiness and actual readiness diverged for more than a decade. The two collisions nine weeks apart converted what could have been read as an outlier into a structural diagnosis: an under-trained bridge team failing the most basic give-way rules, and an unfamiliar interface that made the same gap reach the hull. Seven sailors died on Fitzgerald; ten on McCain. The cost of the divergence was paid in lives long after it could have been measured in dollars or inspections.

LENS APPROACH

Fitzgerald/McCain is the worked example of induced sub-competency 1.1 (engineered vs. stated requirements) and the LENS D1/PT3 pairing — Systems Analysis applied to the capability-engineering problem of underway watchstanding. Students reconstruct the capability requirements from operational analysis, then design the evidence architecture (LENS D3) that would have surfaced the gap before the collisions and the sociotechnical reforms (LENS D4) that would keep waivers from quietly hollowing the requirement out. The case pairs with INDOPACOM Marine Corps training (Case 2) as the live counterpart where the same divergence between declared and engineered priority is visible on a theater scale, and with Three Mile Island (Case 69) as the failure that engineered durable industry reform through INPO. CLO mapping: CLO-1 (Systems Analysis) primary, CLO-5 (Sociotechnical Constraints) for the waiver-and-reporting institutional dynamics.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify watchstanding proficiency as a measured deliverable — define the competency, instrument it, and gate qualification on demonstrated skill rather than sea time.
- Keep human-factors review in the procurement loop: validate that any interface change (e.g., the bridge console) is tested against operator performance before it is fielded.
- Design the readiness signal to report demonstrated capability, not self-attested course completion.

IN OPERATION / AFTER

- Audit the gap between reported and actual readiness with independent assessment (the Ready-for-Sea model) — with authority to pull a unit offline.
- Protect training time against operational tempo so the capability cannot erode silently when schedules tighten.
- Track leading indicators — qualification currency, near-miss reports — so divergence is visible before it is paid for in lives.

REFLECTION QUESTIONS

1. The Navy replaced classroom and simulator instruction with CD-ROM self-study in 2003. What measurement would have detected the capability shortfall before 2017?
2. The Strategic Readiness Review found the readiness-reporting system had itself stopped working. Design a capability-evidence pipeline that would not normalize its own gaps.
3. Identify a capability in your organization that is certified by completion (a checked box) rather than demonstrated performance. What measure would convert it to evidence — and who would hold the authority to act on a red signal?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 2 DEFENSE ONGOING

Marine Corps Training in the INDOPACOM AOR

T Training Gap K Knowledge & Institutional Memory

IMPACT Structural readiness gap in DoD's stated top-priority theater

IN BRIEF

The 2022 National Defense Strategy names the Indo-Pacific the Pentagon's priority theater and China its "pacing challenge" — yet the theater so designated has among the least mature live-training infrastructure in the force. For nearly a decade the U.S. Marine Corps has been unable to meet its training requirements at Indo-Pacific ranges, papering over the gap with rotations back to U.S. ranges, virtual substitutes, and multinational exercises pressed into proxy duty. In May 2024 the GAO documented the decade-long shortfall and urged the Corps to analyze its unmet requirements and build a remediation plan. The case is this book's live, ongoing counterpart to its historical failures: a fully recognized, repeatedly documented capability gap that declared priority has not closed, because engineered priority — ranges, basing, certified units — is slow and expensive.

BACKGROUND

The 2022 National Defense Strategy names the Indo-Pacific the priority theater and China the "pacing challenge."[0] Set against that is an awkward fact: the theater so designated has among the least mature live-training infrastructure in the force — the place called most important is, where forces can actually rehearse, one of the least built out, so the strategy's top priority and the physical means to prepare for it point in opposite directions, a contradiction the briefings do not resolve.[1]

WHAT HAPPENED

The failure is a condition, not an event. For nearly a decade the Marine Corps has been unable to meet its training requirements at Indo-Pacific ranges, papering over the shortfall with rotations back to U.S. ranges, virtual substitutes, and multinational exercises pressed into proxy duty. The workarounds keep units partially trained; the structural gap does not close, because each substitute buys a single cycle of readiness without building the ranges, basing, or instrumented airspace that would let the theater train its own force — treating a permanent shortfall as a series of temporary ones.[2]

THE INVESTIGATION

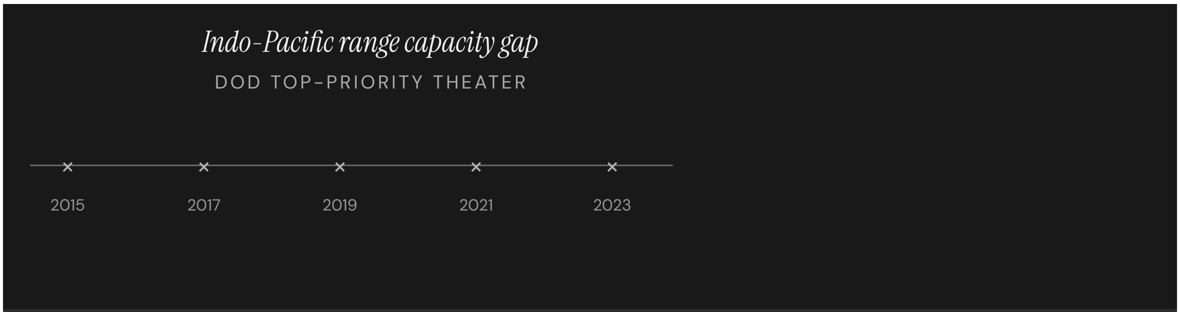
In May 2024 the Government Accountability Office documented the unmet requirements, the decade over which they had gone unmet, and the workarounds standing in for the real thing, and recommended the Corps formally analyze its unmet requirements and build a remediation plan for Indo-Pacific ranges; the Department concurred.[3] It was not the first warning — GAO has pressed the same readiness recommendations for years, and flagged the gap between overseas training and readiness reporting two decades ago, so the 2024 finding documents not a fresh discovery but the durability of a shortfall that survived repeated diagnosis, concurrence, and the passage of time without being engineered away.[4]

THE CAPABILITY GAP

What makes INDOPACOM diagnostic is the inverse correlation between stated and engineered priority. A theater can be named the pacing challenge in every briefing while its training infrastructure stays short for ten years, because capability follows where construction and dollars flow, not where strategy points. Declared priority is cheap; engineered priority — ranges, instrumented airspace, basing, certified units — is slow and expensive, and the gap between them is the real measure of intent. A strategy document can be rewritten in a season; a range complex takes years of construction and budget, so the decade-long persistence of the shortfall says more about resourcing than any reassertion of priority could.

AFTERMATH & REFORM

The reform remains mostly prospective: the GAO recommendations were open as of 2024, an analysis and a funded remediation plan still to come.[5] Whether the gap closes will be decided not by another strategy document but by whether the recommendation becomes programmed ranges, dollars, and schedule — converting a concurrence on paper into construction and certified units that a future review could actually measure. It sits at the front of this book as the live counterpart to its historical failures — a gap fully recognized, repeatedly documented, and still not engineered away.



DECADE OF UNMET TRAINING IN DOD'S PRIORITY THEATER

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THE LEARNING ENGINEERING LENS

> *Without meeting these requirements, the Marine Corps cannot ensure that its forces will be ready for combat.*
>
> PARAPHRASING GAO-24-107463, MILITARY READINESS, 2024

LE INSIGHT

INDOPACOM illustrates the difference between declared priority and engineered priority. A theater designated as DoD's pacing challenge while the training infrastructure to operate in it remains structurally short is not a resourcing oversight — it is a capability architecture failure. Capability follows where resources actually flow, not where briefings name as critical.

LENS APPROACH

LENS uses INDOPACOM in LEN 5 to teach the gap between stated requirements and engineered requirements, and in LEN 8 to examine the organizational dynamics that allow declared priorities to coexist with unfunded capability gaps for a decade. The case is also a live test for any student's claim about how to engineer cross-service capability at scale.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Tie any declared-priority designation to a funded engineering plan — ranges, basing, instrumented airspace, certified units — so strategy and resourcing cannot diverge unnoticed.
- Treat live-training infrastructure for a priority theater as a programmed deliverable with schedule and budget, not a workaround filled by rotations and exercises.
- Define the capability requirement for the theater explicitly, so the gap between what is needed and what is built is visible at the point of decision.

IN OPERATION / AFTER

- Audit declared versus engineered priority annually — measure where construction and dollars actually flowed against where strategy named as critical.
- Track recommendation status to closure with an authority that can escalate when a concurrence does not become programmed work.
- Monitor whether substitute training (rotations, virtual, multinational) is masking a structural shortfall rather than retiring it.

REFLECTION QUESTIONS

1. In your domain, what is the gap between **declared** priority and **engineered** priority? How would you measure it?
2. Construct the capability requirements artifact for a theater you do not currently operate in. What would it cost, and who would sign for it?
3. GAO documented this shortfall for years, the Department concurred, and the gap stayed open. Design the accountability mechanism that would convert a concurrence into programmed dollars and schedule, with a signal that fires when the plan slips.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines

CASE 3 DEFENSE ONGOING

F-35 Sustainment & Maintainer Shortage

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT **Fleet mission-capable rate about 55% (March 2023), far short of program goals; lifecycle cost exceeds $1.7T, with ~$1.3T in operating and support; maintainer, technical-data, and depot shortfalls are the binding readiness constraint (GAO-23-105341)**

IN BRIEF

The F-35 is the most expensive weapons program in history — roughly 2,500 jets planned and a lifecycle cost above $1.7 trillion, about $1.3 trillion of it in operating and support. The hard part was never the airplane but keeping a global fleet ready, and that half has lagged from the start. As of March 2023 the fleet's mission-capable rate was about 55%, far short of goal, with more than 10,000 components in the repair queue and depots behind schedule. GAO traced the shortfall to maintainer shortages, the military's lack of access to technical data, and contractor dependency, and urged a full reassessment of the sustainment strategy. It is the book's cleanest case of a platform fielded faster than the capability infrastructure to sustain it.

BACKGROUND

The F-35 is the most expensive weapons program in history: the Pentagon plans to buy nearly 2,500, at a lifecycle cost exceeding $1.7 trillion — roughly $1.3 trillion of it not the aircraft but the decades of operating and sustaining them.[0] The flyaway jet is the finite part; keeping a global fleet ready — maintainers, technical data, depots — is the open-ended part, and the part that lagged, because the cost that dominates the program is not buying the aircraft but the decades of sustaining them, the very work that received the least attention as the jets rolled off the line.

WHAT HAPPENED

The failure is a standing condition. As of March 2023 the fleet's mission-capable rate was about 55%, far short of goal; more than 10,000 components waited in the repair queue, and the depots averaged about 72 days per repair while still behind schedule in standing up the capacity to do the work at all — a backlog and a turnaround time that compound, since parts stuck in the queue keep jets grounded and the under-built depots cannot clear the queue fast enough to recover the mission-capable rate.[1]

THE INVESTIGATION

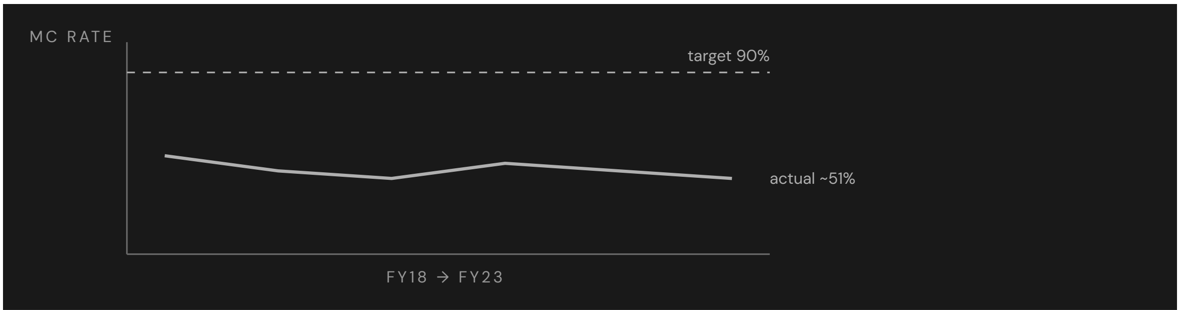
GAO's September 2023 review was bluntly titled: the Department and services **need to reassess the future sustainment strategy**. It traced the shortfall to maintainer shortages, the military's lack of access to the technical data needed to do its own repairs, and the resulting dependence on the prime contractor.[2] None of it was new — GAO has repeated the same diagnosis year after year, through a troubled logistics-software backbone and slow progress, even as procurement continued and readiness stayed flat, so the program kept buying more jets it could not fully sustain while the same three shortfalls were named in review after review without being closed.[3]

THE CAPABILITY GAP

The F-35 is the cleanest modern case of a platform fielded faster than the capability infrastructure to sustain it. Aircraft arrived on schedule; the maintainers, technical data, and depots were treated as follow-on costs rather than deliverables that had to field with the jets. The hardest part is the data: much of what is needed to repair the aircraft stayed controlled by the contractor, so the services cannot freely write procedures, qualify depots, or compete the work, which locks the fleet into a single source for the knowledge needed to keep it flying — merely expensive in peacetime, dangerous in war, when a contractor-dependent sustainment chain is exactly the kind of bottleneck an adversary would seek to exploit.[4]

AFTERMATH & REFORM

GAO urged a full reassessment of the sustainment strategy — government access to technical data, depot capacity, and a maintainer pipeline — rather than more patching.[5] Those recommendations remain a work in progress, with later reviews showing costs still rising and readiness still below goal. The F-35 sits in this book as the live argument for treating capability infrastructure — people, data, the means to sustain them — as a fielding gate, not an afterthought. The bill for skipping that gate does not disappear; it compounds, arriving as grounded jets, a swelling repair queue, and a sustainment chain the services cannot control rather than as a line item caught early enough to fix cheaply.



MISSION-CAPABLE RATE VS. FLEET TARGET

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THE LEARNING ENGINEERING LENS

> *Organizational-level maintenance has been affected by a number of issues, including a lack of technical data and training.*
>
> PARAPHRASING GAO-23-105341, F-35 AIRCRAFT, 2023

LE INSIGHT

F-35 is the live, current example of fielding a platform faster than its capability infrastructure can be built. The aircraft is the easy part; the maintainers are the hard part. A decade of program schedules treated maintainer training as a follow-on cost, not a fielding gate. The fleet is now operating at half of its design readiness, and the program of record is more than a trillion dollars over its 2018 estimate.

LENS APPROACH

LENS treats the F-35 in LEN 5 as the canonical case of **Capability- System Misalignment at Transition**, the program's first canonical problem type. Students design the capability infrastructure that should have accompanied each lot of aircraft delivered, including the training pipeline, technical-data deliverables, and depot establishment.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make sustainment infrastructure — maintainers, technical data, depot capacity — a contractual deliverable that fields with each lot of aircraft, not a follow-on cost.
- Secure government rights to the technical data needed to write procedures, qualify depots, and compete the work, so the fleet is not locked to a single source.
- Gate fielding on demonstrated sustainment capacity, not just aircraft delivery, so readiness is engineered alongside the platform.

IN OPERATION / AFTER

- Audit the mission-capable rate against repair-queue depth and depot turnaround, and act when the backlog signals a structural, not transient, shortfall.
- Track recurring GAO-style diagnoses to closure, refusing to let procurement outpace the sustainment fixes named year after year.
- Monitor contractor dependency as a wartime risk, and sustain a government maintainer pipeline and depot capacity that can hold under pressure.

REFLECTION QUESTIONS

1. Pick a current technology platform in your domain. Estimate the capability infrastructure that must field with it. What happens if half of that infrastructure is years behind the hardware?
2. The F-35 program treated maintainer training as follow-on cost. Design a fielding gate that would prevent that decision being available to a future program manager.
3. Much of the data needed to repair the F-35 stayed controlled by the contractor, foreclosing the government's ability to compete or qualify the work. For a platform in your domain, what data rights would you make a contractual deliverable up front, and how would you verify they were actually delivered?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · task analysis · design prototyping

CASE 4 DEFENSE 1991 – 2014

Military Fratricide — Desert Storm to Afghanistan

T Training Gap H Human-System Interface K Knowledge & Institutional Memory

IMPACT 24% of U.S. KIA in Desert Storm from friendly fire (35 of 146) — an order of magnitude above the historical baseline

IN BRIEF

Friendly fire killed an unusual share of coalition forces in the 1991 Gulf War: 35 of 146 U.S. combat deaths (24%) and 72 of 467 wounded (15%). (The often-cited "2% historical baseline" from Shrader's 1982 study is contested — later estimates run nearer 10–15%, and Shrader stepped back from it.) Post-war reviews blamed the chaos of combat, weak situational awareness and fire-control discipline, and combat-identification failures — and noted the military lacked a shared record to even study its own pattern. Fratricide is the failure of several systems to integrate; despite a generation of combat-ID investment, recurrences in Afghanistan and after show the rate never fell to a confidently low level, because integration is the hardest property to engineer by program.

BACKGROUND

Friendly fire is as old as war, but its true rate is hard to pin down. The most-cited estimate, from Charles Shrader's 1982 study Amicicide, put it under 2% of battle casualties — a figure later analysts challenged as far too low (nearer 10–15%), and one Shrader himself stepped back from. The disputed baseline mattered because it became the yardstick against which a modern war would be measured — and a yardstick set too low makes any later rate look like a catastrophe.[0]

WHAT HAPPENED

The 1991 Gulf War made the question grim and concrete: of 146 U.S. service members killed in action, 35 — about 24% — died by friendly fire, and 72 of 467 wounded (15%) were hit by their own side.[1] An A-10 strike on U.S. LAV-25s near Khafji killed seven Marines; a single A-10 attack on British Warrior vehicles killed nine — each an aircraft firing on friendly vehicles it had failed to identify, the recurring shape of the problem. Whatever the true baseline, a quarter of American combat deaths from one's own forces could not be waved off as the ordinary friction of battle.[2]

THE INVESTIGATION

Post-war reviews converged on familiar causes: the chaos of combat, inadequate situational awareness, weak adherence to fire-control measures, and combat-identification failures.[3] One finding cut deeper — the military lacked a comprehensive, shared record of fratricide incidents, so it could not study its own pattern, separate training failures from doctrine or equipment, or tell whether a fix was working. Without a common database, every incident was investigated in isolation and the aggregate signal that might have driven reform never formed. The capability to **learn** was itself missing, a second-order gap beneath the first.[4]

THE CAPABILITY GAP

Fratricide is not one problem but the failure of several systems to integrate — situational awareness, fire-control discipline, combat identification, joint coordination — each the subject of dedicated programs, yet the rate never fell to anything negligible. In Afghanistan a satellite-guided strike went wrong after a controller's device reset its coordinates, and in 2014 a B-1's targeting pod could not detect the infrared strobes marking U.S. troops, killing five. Each contributor was worked on; the integration **across** them — which is what actually keeps friendly forces from killing each other — is the hardest thing to engineer by program, because no single office owns it and no single procurement can deliver it — and it is where the capability kept failing.[5]

AFTERMATH & REFORM

The response was a generation of combat-identification investment: better IFF systems, blue-force-tracking networks, and changes to fire-control doctrine and training. The improvements were real, yet fratricide never dropped to a confidently low, stable rate, and even measuring it remained contested.[6] That is the lesson: where capability is an emergent property of many systems working together, no single program closes the gap — improving each contributor in isolation still leaves the integration between them unaddressed — and progress has to be measured against an honest baseline rather than a convenient one.

FRIENDLY-FIRE RATE, U.S. CASUALTIES, DESERT STORM VS. HISTORICAL

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THE LEARNING ENGINEERING LENS

> *The lack of a comprehensive and accessible automated database prevented thorough examination of the problem.*
>
> PARAPHRASING POST–DESERT STORM FRATRICIDE REVIEWS, C. 1993

LE INSIGHT

Fratricide is a multi-decade capability problem that resists single- intervention solutions. Each of the contributing causes — situational awareness, fire-control discipline, combat identification, joint coordination — has been the subject of dedicated programs. The rate persists because the integration across the contributors is itself the capability, and integration is the hardest property to engineer by program.

LENS APPROACH

LENS uses fratricide in LEN 5 to teach systems-of-systems capability analysis and in LEN 2 to introduce combat identification as a Human-AI Teaming problem (operators rely on automated IFF systems whose limitations are not in their training). LEN 8 examines why decades of awareness have not produced a sustained reduction.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat combat identification as a systems-of-systems requirement with a single owner for the integration, not as separate IFF, situational-awareness, and fire-control programs.
- Build a comprehensive, shared incident database from the outset so the pattern across events can be studied and fixes tested against it.
- Design fire-control discipline and identification into joint coordination, so the integration that prevents fratricide is engineered rather than left emergent.

IN OPERATION / AFTER

- Audit the friendly-fire rate against an honest baseline, distinguishing a temporary dip from a structural reduction before crediting any program.
- Monitor each contributor and the integration across them, since improving components in isolation can leave the joint failure mode untouched.
- Sustain the learning channel so later-conflict recurrences feed back into doctrine and training rather than being investigated in isolation.

REFLECTION QUESTIONS

1. Identify a capability problem in your domain that has been the subject of repeated programs without sustained improvement. What is the integration gap that the programs do not address?
2. Design a measurement system that would distinguish a temporary improvement in fratricide rate from a structural one.
3. The military lacked a shared incident database, so it could not study its own pattern. Identify a recurring harm in your domain that is investigated case-by-case but never aggregated, and design the shared record that would let the pattern become visible.

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CASE 5 DEFENSE 1980

Operation Eagle Claw

T Training Gap K Knowledge & Institutional Memory

IMPACT 8 servicemembers killed in Iran; mission to rescue 52 American hostages aborted; catalyst for the founding of U.S. Special Operations Command

IN BRIEF

The April 1980 mission to rescue fifty-two American hostages held at the U.S. embassy in Tehran was aborted at a desert staging point when three of eight helicopters were lost to mechanical and weather problems. During the withdrawal a helicopter collided with a refueling C-130, killing eight servicemembers; the hostages remained captive for another nine months. The Holloway Commission found the operation had been assembled ad hoc: each service contributed its own units, equipment, and command relationships, the aircrews had not trained together, and there was no standing joint special-operations command to own the mission. The capability had to be improvised, and the improvisation failed. Eagle Claw catalyzed the creation of JSOC, the Goldwater-Nichols reforms, and ultimately U.S. Special Operations Command.

BACKGROUND

In November 1979, Iranian revolutionaries seized the U.S. embassy in Tehran and took fifty-two Americans hostage. After months of failed diplomacy, the Carter administration authorized a military rescue. No standing joint special-operations command existed to plan or execute it; the force had to be assembled from units drawn separately from the Army, Navy, Marines, and Air Force — each bringing its own equipment, procedures, and chain of command to a mission that demanded they act as one, with no institution whose job it was to make them cohere before the night of the raid.[0]

WHAT HAPPENED

On 24 April 1980, the mission staged at a remote site code-named Desert One. Three of the eight RH-53 helicopters were disabled by a dust storm and mechanical failure, dropping the force below the minimum needed; the commander aborted. During the withdrawal a helicopter collided with a C-130 tanker, and the resulting fire killed eight servicemembers. The mission failed before reaching Tehran — undone not by enemy action but by attrition and a chaotic withdrawal among aircraft and crews that had never operated together, the predictable cost of integration improvised under pressure rather than built and rehearsed in advance.[1]

THE INVESTIGATION

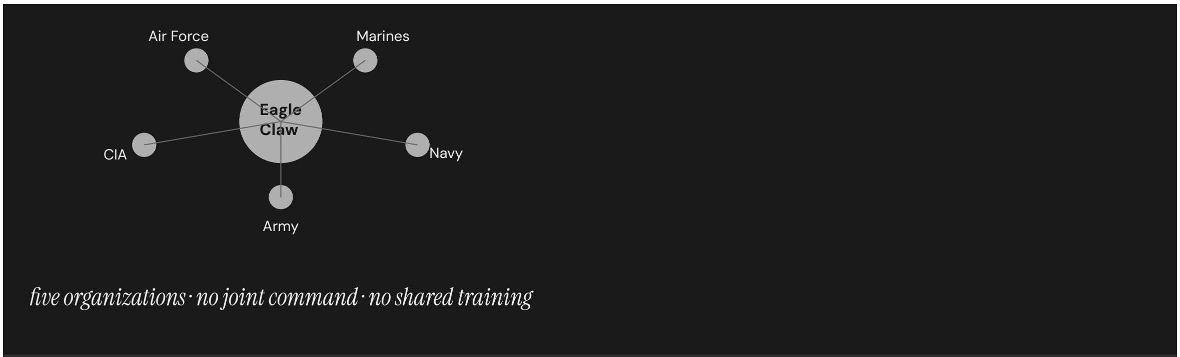
The Holloway Special Operations Review Group identified the operation's "ad hoc-ery" as central: each service contributed its own units, equipment, command relationships, and communications; the aircrews had not trained together as a unit; the RH-53D had been selected partly for a minesweeping cover story rather than for fitness for a desert rescue. There was no standing organization to own the mission end to end — no single authority responsible for the force's training, equipment fit, and command architecture as a whole, so each gap was someone's problem in part and no one's in full.[2]

THE CAPABILITY GAP

Each service was competent inside its own boundary. The capability that did not exist was the integration across them — a joint command, a common communications architecture, and a force that had trained together. That integration had to be improvised for a single high-stakes mission, and the improvisation could not hold — because cross-service cohesion is not summoned on demand but accrued through standing structure and repeated joint training, neither of which existed when the order came.[3]

AFTERMATH & REFORM

Eagle Claw produced the Holloway Commission, the creation of the Joint Special Operations Command in 1980, and — alongside the 1983 Beirut bombing and the Grenada invasion — the impetus for the Goldwater-Nichols Act of 1986 and the Nunn-Cohen Amendment establishing U.S. Special Operations Command in 1987. The reform built the institution the mission had needed and not had — converting a capability that had been improvised once and failed into a standing command with its own forces, training, and authority, so the next mission would inherit cohesion rather than assemble it from scratch.[4]



THE ABSENCE OF AN INSTITUTION THAT COULD OWN THE MISSION

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THE LEARNING ENGINEERING LENS

> *The mission was ad hoc — assembled from units, equipment, and command relationships that had never operated together.*
>
> PARAPHRASING THE HOLLOWAY SPECIAL OPERATIONS REVIEW GROUP, 1980

LE INSIGHT

Eagle Claw is the canonical case in U.S. defense for the absence of institutionalized cross-service capability. Each service was competent inside its own boundary; the integration across them did not exist as a deliverable — until the reform that followed built the institution the mission had needed and not had.

LENS APPROACH

LENS uses Eagle Claw in LEN 5 as a worked case for cross- organizational capability requirements and in LEN 8 for the institutional reform that followed; it pairs with INPO (Case 172) as the defense analog of failure-driven institution building. The seven-year arc to Goldwater-Nichols (1986) and USSOCOM (1987) is itself a measurement: the institutional response time when the fix requires statutory action.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Stand up a single command that owns the cross-organizational mission end to end — its force, equipment fit, communications, and training — rather than assembling it per mission.
- Require the contributing units to train together as one force on a common communications architecture before they are committed.
- Select equipment for fitness to the actual mission, not for a convenient cover story or parent-service availability.

IN OPERATION / AFTER

- Audit whether a cross-organizational capability that exists on paper has ever actually operated as one force, and treat the absence of joint reps as an unfilled gap.
- Sustain the standing institution and its training pipeline so the next mission inherits cohesion rather than improvising it.
- Monitor for the recurrence of ad-hoc assembly, since the conditions that produced Desert One reappear whenever no single authority owns the joint mission.

REFLECTION QUESTIONS

1. Where in your domain does a cross-organizational capability exist on paper but not in practice? What would force its institutionalization?
2. Eagle Claw produced USSOCOM and Goldwater-Nichols six years later. Sketch the institutional design that an equivalent failure in your domain would force into existence.
3. The Holloway Commission named the mission's ad-hoc-ery as the diagnosis. What standing capability — institution, command, training pipeline — does your domain currently lack that an Eagle-Claw-class failure would force into existence?

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CASE 6 AVIATION 1996

AeroPerú Flight 603

H Human-System Interface T Training Gap

IMPACT 70 killed; Boeing 757 crashed into the Pacific after maintenance tape was left over static ports

IN BRIEF

While polishing the fuselage of a Boeing 757 in Lima in October 1996, maintenance staff taped over the static ports and failed to remove the tape before flight. With the static system blocked, the aircraft's altimeters, airspeed indicators, and altitude-alert and overspeed warnings all read inconsistently. The crew received a cascade of contradictory warnings — overspeed, stick shaker, ground proximity — and could not determine their true altitude or speed. Believing they were higher than they were, they flew into the Pacific, killing all 70 aboard. The investigation named the maintenance error as the primary cause but stressed that every cockpit instrument depended on the same blocked sensor: the apparent redundancy was an illusion, and the crew's training had no procedure for "everything you see is wrong."

BACKGROUND

AeroPerú 603 was a Boeing 757 night departure from Lima to Santiago. Before the flight, ground crew had covered the aircraft's static ports with adhesive tape during cleaning and polishing of the fuselage, and the tape was not removed. The static ports feed the instruments that tell the crew their altitude and airspeed — a single physical source upstream of nearly every primary cockpit display, so blocking it corrupted not one instrument but the whole air-data picture the crew would rely on in the dark.[0]

WHAT HAPPENED

On 2 October 1996, the aircraft took off into darkness over the ocean with its static system blocked. Altimeters, airspeed indicators, and the altitude-alert and overspeed systems all produced false and contradictory readings. The crew received simultaneous overspeed and stall warnings and ground-proximity alerts they could not reconcile. Believing they were higher than they were, they descended and struck the Pacific, killing all 70 aboard — the contradictory warnings offering no way to tell which, if any, instrument to trust, because every one of them drew on the same blocked source and so failed in concert rather than disagreeing usefully.[1]

THE INVESTIGATION

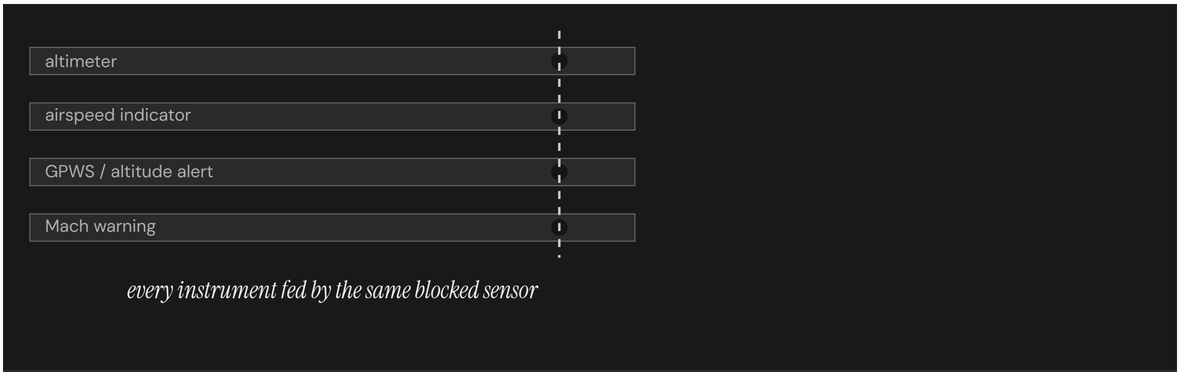
The Peruvian accident investigation identified the failure to remove the static-port tape as the primary cause, but emphasized that the cockpit had no independent reference by which to detect the inconsistency. Every instrument that should have flagged the failure drew on the same blocked source. The crew's training had no procedure for the case in which all primary instruments are simultaneously wrong — that failure had been assumed away rather than planned for, leaving the crew to improvise against a problem the system design had quietly made possible and the curriculum never named.[2]

THE CAPABILITY GAP

AeroPerú 603 is the case for redundancy that is not redundancy. The cockpit's apparent instrument redundancy was an illusion at the source — a common-cause failure that defeated every downstream display at once. The missing capability was both a maintenance control that made the blocked port impossible to miss and a procedure for flying when the instruments themselves cannot be trusted — the first preventing the common-cause failure at its source, the second giving the crew somewhere to stand when the apparent redundancy in front of them collapsed all at once.[3]

AFTERMATH & REFORM

The accident reinforced industry maintenance practices for conspicuous covering and mandatory removal-verification of static ports and pitot tubes, and entered the human-factors literature as a worked example of common-cause failure and the limits of crew training under total instrument corruption. It is paired with other pitot-static events in discussions of air-data integrity — its lesson being that redundancy counted at the display tells you nothing if the redundant paths share a single upstream point that can fail them all together.[4]



AEROPERÚ 603 — REDUNDANCY THAT WAS NOT REDUNDANCY

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> *The crew received contradictory indications they could neither reconcile nor override.*
>
> PARAPHRASING THE PERUVIAN DGAC / COMISIÓN INVESTIGADORA DE ACCIDENTES DE AVIACIÓN REPORT ON AEROPERÚ 603, 1996

LE INSIGHT

AeroPerú 603 is the case for redundancy that is not redundancy. Every cockpit indicator depended on the same physical sensor. The apparent redundancy in the cockpit was an illusion at the source. The training did not include the failure case because the failure case had been assumed not to occur.

LENS APPROACH

LENS uses AeroPerú in LEN 5 to teach the difference between apparent and actual redundancy and in LEN 2 for the human role when all interface data is unreliable. Studio projects examine the “trust nothing” procedure design.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Map redundant displays back to their upstream sources, and eliminate or instrument any common point whose failure would corrupt them all at once.
- Make critical sensor covers conspicuous and require verified removal, so a blocked static port cannot reach takeoff unnoticed.
- Provide an independent reference (e.g., a source not fed by the common component) so the crew can detect inconsistency at the source.

IN OPERATION / AFTER

- Audit maintenance controls for tasks that cover or block critical sensors, confirming each has a conspicuous marker and a removal-verification step.
- Train and rehearse a “trust nothing” procedure for total instrument corruption, the case the curriculum had assumed away.
- Monitor for common-cause failure modes whose probability was assumed negligible, and revisit that assumption when the consequence is catastrophic.

REFLECTION QUESTIONS

1. Identify a redundant interface in your domain whose redundancy depends on a common upstream component. What is the operator procedure when the upstream component fails?
2. Design the “trust nothing” procedure for the AeroPerú crew. What information remains reliable when all instruments are corrupted?
3. A taped-over static port survived to takeoff because nothing made it conspicuous or forced its removal to be verified. Design the maintenance control in your domain that would make a blocked or covered critical component impossible to overlook before operation.

WHO BUILDS THIS human factors & interaction design · learning science & instructional design — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · scenario simulation · competency models

CASE 7 AVIATION 2018 – 2019

Boeing 737 MAX / MCAS

D Designed Out T Training Gap H Human-System Interface

IMPACT 346 killed across two crashes — Lion Air 610 and Ethiopian Airlines 302; 20-month worldwide grounding; ~\$20B direct cost; FAA delegated-authority reform under the Aircraft Certification, Safety, and Accountability Act (2020)

IN BRIEF

The Boeing 737 MAX was a re-engined 737 meant to fly without new pilot training — the commercial promise that sold it against the Airbus A320neo. Its bigger, forward-mounted engines changed the jet's pitch behavior, so Boeing added software (MCAS) to mask the difference, then kept it out of the manuals and training and let it fire repeatedly on a single angle-of-attack sensor. When that sensor failed on Lion Air 610 (October 2018) and Ethiopian 302 (March 2019), MCAS forced the nose down and crews who had never heard of it could not recover; 346 people died and the fleet was grounded for twenty months. Five major investigations — the NTSB-supported KNKT (Lion Air) and EAIB (Ethiopian) reports, the US House Transportation Committee final report, the DOT Inspector General review, and the multinational Joint Authorities Technical Review (JATR) — converged on the same diagnosis: the training omission was a deliberate cost-avoidance choice, executed through an FAA delegated-authority regime in which Boeing reviewed much of its own safety judgment. The MAX is the book's inverse case: human capability engineered out of a system to save the price of sustaining it, with the elision underwritten by a certification process that did not have the independence to catch it.

BACKGROUND

The 737 MAX was Boeing's answer to the Airbus A320neo: a re-engined version of its best-seller that airlines could fly without retraining pilots on a new type. But the MAX's larger, forward-mounted engines changed how it pitched at high angles of attack, so Boeing added software — the Maneuvering Characteristics Augmentation System, MCAS — to make it handle like the older 737 NG, papering over an aerodynamic change with a control law so the airframe would feel identical from the cockpit. The whole commercial logic depended on that software staying invisible: the less of a "new function" MCAS seemed, the less the MAX would trigger costly new training, and the lower the airline's switching cost away from the A320neo. Boeing reportedly promised Southwest a rebate of about a million dollars per jet if simulator training proved necessary — a clause that turned the training requirement into a direct line on the program's ledger, and one the House investigation later cited as a structural reason the "no new training" promise behaved as a binding constraint on the engineering rather than as an aspiration.[0]

WHAT HAPPENED

To keep MCAS in the background, it was left out of the manual and the type-rating training, and allowed to fire on a single angle-of-attack sensor with no second source to cross-check it. The system's authority was also expanded late in development — the trim it could command grew from 0.6° to 2.5° per cycle, and its firing was made repeatable rather than one-shot — without the corresponding reassessment of the failure modes that change implied. When the sensor failed on Lion Air Flight 610 in October 2018, MCAS repeatedly forced the nose down; the crew, never told the system existed, fought it cycle after cycle until the jet dove into the Java Sea, killing all 189. Five months later Ethiopian Airlines Flight 302 repeated almost exactly the same sequence — sensor failure, repeated trim commands, unrecoverable nose-down attitude — killing 157, for 346 dead across the two crashes, and the entire MAX fleet grounded worldwide for what would become twenty months.[1]

THE INVESTIGATION

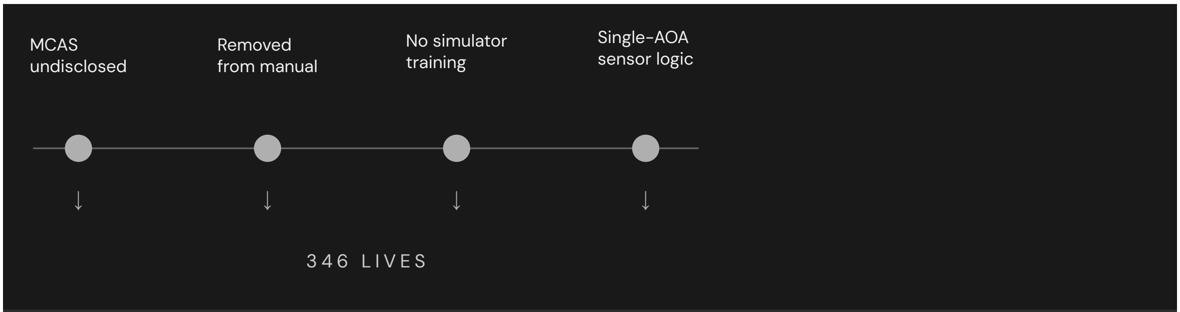
The investigations made the decisions legible. The Indonesian KNKT and Ethiopian EAIB accident reports, with NTSB participation, established the accident sequences and the single-sensor design as the proximate cause. The multinational Joint Authorities Technical Review (JATR), convened by the FAA in 2019 with international regulators, concluded that MCAS had not been evaluated as the novel flight-control function it actually was. Internal 2013 Boeing meeting minutes showed employees noting that calling MCAS a new function would bring "greater certification and training impact" — the cost the program was built to avoid, written down years before either crash; a 2016 internal survey found 39% of certification staff felt undue management pressure. The House Transportation Committee's 2020 final report concluded Boeing's assumption that simulator training was unnecessary "diminished safety, minimized the value of pilot training, and inhibited technical design improvements," naming the omission as a choice rather than an oversight.[2] The DOT Inspector General traced how the single-sensor design and the training omission passed through a certification process in which the FAA had delegated much of the safety judgment back to Boeing itself under the Organization Designation Authorization (ODA) program — so the company effectively reviewed its own most consequential trade-offs, and the regulator lacked both the technical depth and the institutional independence to challenge the assessment.[3]

THE CAPABILITY GAP

The MAX inverts this book's usual case. Human capability was not overlooked by accident; it was deliberately engineered out to avoid the cost of the training that would have created it. The pilots were not undertrained by oversight but by design — the absence of training was, in effect, a contract deliverable promised to customers and protected by a rebate. The single-sensor architecture was a second engineered-out capability: a redundant cross-check on the angle-of-attack signal would have triggered the kind of design and certification work the program was built to avoid, and was left out for the same reason. Seen together, the crashes are not a training failure that befell a good airplane but exactly what the commercial and engineering decisions specified: a flight-control system that depended on pilots reacting correctly, in seconds, to a failure they had been guaranteed never to learn about, with no sensor cross-check that could keep the failure from arriving in the first place.[4]

AFTERMATH & REFORM

The MAX was grounded for some twenty months — the longest grounding of a US-certified airliner in the jet era. MCAS was redesigned to use both AOA sensors, to fire once rather than repeatedly, and with limited authority bounded by airspeed — restoring the margins the original had stripped away. The FAA's 2020 return-to-service directive required the simulator training the airplane had been built to avoid. The Aircraft Certification, Safety, and Accountability Act of 2020 then tightened FAA oversight of the ODA delegation regime, required disclosure to pilots of flight-control systems that act on a single sensor input, and funded FAA engineering capacity that the delegation system had let atrophy. Boeing entered a Deferred Prosecution Agreement with the DOJ on a fraud charge tied to its disclosures to the FAA's Aircraft Evaluation Group.[5] The reform conceded the point the program had spent years resisting: the training was a real requirement all along, the sensor cross-check was a real requirement all along, and removing them from the paperwork only deferred the cost — and then moved it onto two airplanes full of people.



MCAS — CAPABILITY ENGINEERED OUT AT FOUR DECISION POINTS

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THE LEARNING ENGINEERING LENS

> *Boeing's assumption that simulator training was unnecessary diminished safety, minimized the value of pilot training, and inhibited technical design improvements.*
>
> HOUSE TRANSPORTATION COMMITTEE REPORT, 2020

LE INSIGHT

The 737 MAX is the documentary record of a design choice to remove human capability to avoid the regulatory and commercial cost of sustaining it. The pilots were not undertrained by accident — they were undertrained as a **requirement**. The omission of training was a contract deliverable. Once that is understood, the accident becomes legible as an engineering decision rather than a training failure.

LENS APPROACH

The 737 MAX is the canonical engineered-out-capability failure (induced 1.1; LENS D1+D3/PT4). LENS uses it in Domain 1 (Systems Analysis) for the CLO-1 work of decomposing system performance requirements into measurable human-capability requirements: the elided pilot-training requirement is the traceable artifact a capability-requirements analysis would have surfaced before the omission could become a contract deliverable. LENS uses it in Domain 5 (Navigating Sociotechnical Constraints; CLO-5) for the delegated-authority regulatory regime in which Boeing reviewed its own most consequential safety judgments — the case is the governance counterpart to Therac-25 (Case 59) at the safeguard-removed-with-nothing-in-its-place layer, and pairs with Patriot/Dhahran (Case 8) at the layer of design assumptions that do not travel with a change of context.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat any control law that masks a handling change as a new function in its own right — document it, surface it to pilots, and size its training as part of the design, not after it.
- Forbid safety-critical authority from resting on a single sensor; require an independent source or a cross-check before software can command the flight surfaces.
- Quarantine commercial commitments — rebates, "no new training" promises — from the engineering judgment about what capability the airplane actually requires.

IN OPERATION / AFTER

- Audit delegated certification so the party making a trade-off is never the same party that signs off on its safety, restoring independent review of the riskiest decisions.
- Monitor in-service AOA-disagree and uncommanded-trim events as leading indicators, with a route that reaches engineering before a second airframe is lost.
- Re-validate the "no new training needed" claim against real fleet incidents, and reopen the training requirement the moment operational data contradicts the assumption.

REFLECTION QUESTIONS

1. If a customer contract required removing a training requirement you judged necessary, what artifact would you produce, who would you route it to, and what would you do if the routing failed?

2. Reconstruct the MAX accident as a **capability-requirements** failure: what was the elided requirement, at what life-cycle stage was it elided, and who had the authority to insert it?

3. MCAS was permitted to act on a single sensor with no cross-check. Identify a system you work on that takes a safety-critical action from one unverified input, and specify the redundancy or human confirmation it lacks.

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CASE 8 DEFENSE 1991

Patriot Missile / Dhahran

D Designed Out H Human-System Interface K Knowledge & Institutional Memory

IMPACT 28 U.S. soldiers killed in their barracks; roughly 100 wounded

IN BRIEF

On 25 February 1991 a Scud missile struck a U.S. barracks in Dhahran, Saudi Arabia, killing 28 soldiers and wounding about a hundred; the Patriot battery defending the area never engaged it. Designed for short engagements against Soviet aircraft in Europe, the Patriot tracked time in a register whose tiny rounding error grew with every hour of uptime. After about a hundred hours of continuous operation the Gulf War demanded, the radar's range gate was off by a third of a second — enough to look in the wrong place. Israeli operators had flagged the drift two weeks earlier, and a patch reached Dhahran the day after the strike. The capability to hold accuracy under sustained use was assumed away at design time, and no one carried that assumption forward when the mission changed.

BACKGROUND

The Patriot was built to defend Western Europe against Soviet aircraft in engagements of minutes — switched on, fired, switched off, then moved before it could be targeted. In the 1991 Gulf War it was doing something else: defending fixed sites in Saudi Arabia against ballistic missiles, in batteries left running continuously for more than a hundred hours because the threat came without warning and could not be scheduled. The mismatch between the original concept of operations and the new use was real, and invisible to the operators, because no one had told them the machine had been built around an assumption they were now violating.[0]

WHAT HAPPENED

The system tracked time in a 24-bit fixed-point register, and a tiny rounding error — invisible in any short engagement — accumulated with every hour of uptime. After about a hundred hours the timing was off by roughly a third of a second, which seems negligible until it is multiplied by a ballistic missile's speed: enough for the radar's range gate to look in the wrong place and reject the real track as noise. On 25 February 1991 an incoming Scud arrived where the Patriot was no longer searching, passed unengaged, and struck a barracks in Dhahran, killing twenty-eight soldiers of the 14th Quartermaster Detachment and wounding about a hundred.[1]

THE INVESTIGATION

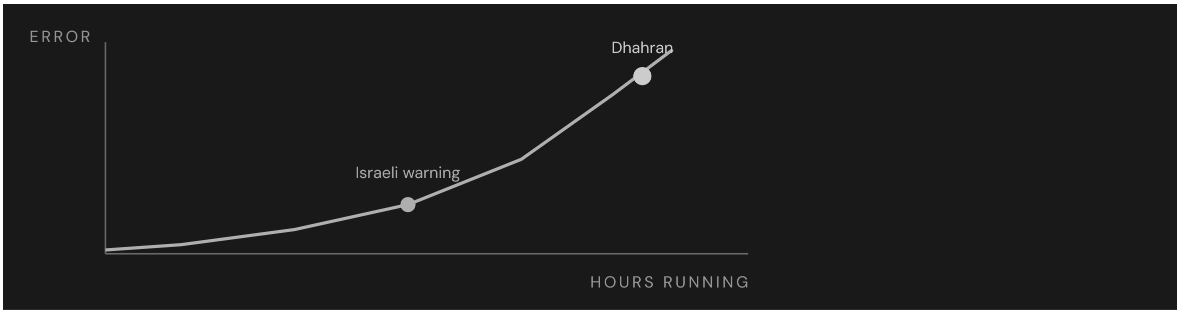
The drift was not unknown: Israeli operators had flagged it two weeks earlier from their own sustained use, and engineers had a patch already in hand — which reached Dhahran the day after the strike, too late to matter.[2] The only field mitigation was an advisory to reboot after "very long" run times, never defining "very long," so a crew could obey the instruction to the letter and still drift into the danger band. The General Accounting Office found the Army had simply presumed no one would run a battery continuously for so long, and so never treated the accumulating error as a hazard worth specifying or warning against.[3]

THE CAPABILITY GAP

The capability to hold accuracy under sustained operation was never built in, because the original concept of operations did not require it — a defensible omission for the system as first imagined, fighting in short bursts and moving on. The failure came when the concept changed and the assumption did not travel with it. Nothing carried the design's hidden premise forward to the soldiers in Dhahran; no artifact, briefing, or warning told them "this system was built assuming you would turn it off." The system did not malfunction — it did exactly what it was built to do; the transition between operating contexts did, with nothing in place to catch the broken premise.[4]

AFTERMATH & REFORM

The patch was distributed and concrete reboot intervals defined in place of the vague advisory, and the defect — a fixed-point truncation error growing without bound as uptime increased — became a staple of numerical-analysis and software-engineering teaching.[5] The harder lesson is the one this chapter turns on: a capability quietly assumed away at design time does not announce itself when the context shifts. It waits, intact and invisible, until the day the assumption is wrong — and by then the people relying on the system have no way to know the ground has moved beneath them, because the premise was never written where they could read it.



TRACKING ERROR GREW WITH EACH HOUR OF CONTINUOUS OPERATION

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THE LEARNING ENGINEERING LENS

> *Army officials presumed that the users would not continuously run the batteries for such extended periods.*
>
> GAO/IMTEC-92-26, 1992

LE INSIGHT

Patriot is the canonical Designed-Out case from defense. The capability to maintain accuracy under sustained operation was omitted because the original concept of operations did not anticipate it. When the concept changed, the assumption did not travel. The system did not fail; the transition from one operational context to another did, and there was no capability infrastructure to catch it.

LENS APPROACH

LENS treats Patriot as the textbook example of **Capability Degradation Under System Change**. LEN 5 methods require operators of any transitioning system to have current capability-relevant documentation specifying what assumptions of the original design have changed. LEN 8 examines how organizational knowledge about design constraints travels from engineering to operations — and why, here, it arrived a day late.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make every load-bearing design assumption — here, that the system would be cycled off — an explicit, recorded requirement so a later change of use cannot silently violate it.
- Bound numerical error over the full intended operating envelope, including run times far beyond the original concept, rather than only the durations first imagined.
- Design the system to detect and bound its own accumulating drift, degrading or warning before the error reaches a magnitude that defeats the mission.

IN OPERATION / AFTER

- Issue concrete, testable operating limits — exact reboot intervals, not “very long” — derived from the design data and propagated to every crew.
- Establish a path for field reports of anomalous behavior, like the Israeli warning, to reach engineering and trigger fleet-wide action faster than a patch can lose a race with the threat.
- Require a capability-transition review whenever a system is redeployed to a new concept of operations, surfacing which original assumptions the new mission breaks.

REFLECTION QUESTIONS

1. The Patriot’s design assumed short engagements. What assumption in a current system you work with would become lethal if the operational context shifted, and how would operators learn of the shift?

2. Construct the capability-transition artifact that should have accompanied the redeployment of Patriot batteries from Europe to Saudi Arabia. What would it have said, and who would have signed it?

3. The field advisory said to reboot after “very long” run times without defining the term. Rewrite a vague operational instruction in your domain into a specific, testable limit, and name the design data the limit must be derived from.

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CASE 9 SPACE / AEROSPACE 1999

Mars Climate Orbiter — Unit Mismatch

D Designed Out K Knowledge & Institutional Memory

IMPACT ~$125M spacecraft lost on approach to Mars; ground software produced thrust output in pound-force while navigation expected newtons

IN BRIEF

The Mars Climate Orbiter, a NASA spacecraft built by Lockheed Martin and navigated by JPL, was lost on arrival at Mars in September 1999. Lockheed's ground software reported thruster impulse in pound-force seconds; JPL's navigation expected newton-seconds. The conversion — a factor of about 4.45 — was never applied at the boundary between the two teams, so every trajectory correction was mis-modeled and the error accumulated over the cruise. The orbiter arrived too low, into the atmosphere, and burned up; the orbiter and its $125 million were lost to a unit mismatch. The investigation found no individual blunder — both contractors did their own work correctly. What failed was the interface between them, which had no owner, specification, or verification step. It is the canonical case of interface-as-requirement.

BACKGROUND

The Mars Climate Orbiter, launched in December 1998 to study the Martian atmosphere, was one of NASA's "faster, better, cheaper" missions, built on a compressed budget that trimmed margins across the program. Lockheed Martin built the spacecraft and its ground software; JPL navigated it from Earth. Across the months of interplanetary cruise the two teams exchanged data so JPL could command the small trajectory corrections that keep a spacecraft on course — an exchange that crossed a software interface between the two organizations.[0]

WHAT HAPPENED

Lockheed's ground software reported thruster impulse in pound-force seconds; JPL's navigation expected newton-seconds. The conversion — a factor of about 4.45 — was never applied at the boundary where the two systems met. Each firing was mis-modeled by that factor, and the error accumulated steadily over the long cruise until the predicted and actual trajectories had quietly diverged. When the orbiter reached Mars on 23 September 1999 it arrived far too low, deep into the atmosphere it was built to study from orbit, and was destroyed. The orbiter, and its roughly $125 million, were lost to a unit conversion no one made.[1]

THE INVESTIGATION

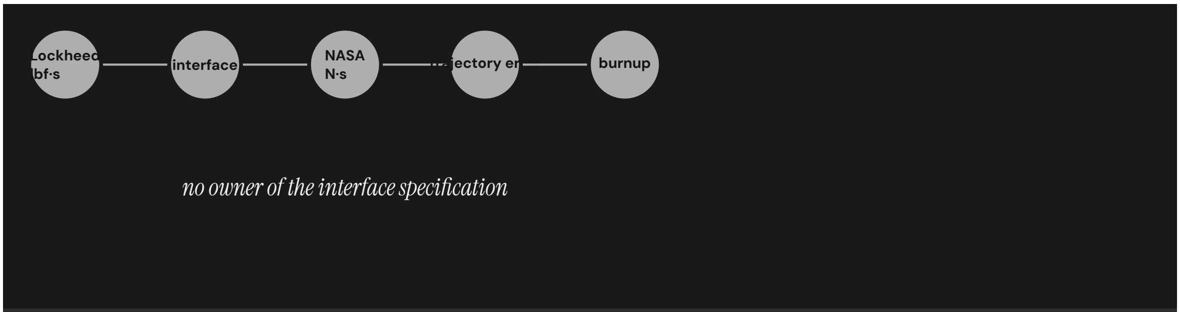
The Mishap Investigation Board put the proximate cause exactly there — the failed English-to-metric translation in ground software — but was careful to name the deeper one rather than stop at the bug. No individual blundered; both contractors did their own work correctly within their own assumptions. What failed was the boundary between them: there was no specified, verified interface fixing the units and checking that both sides agreed, and no end-to-end validation of the data flowing across the seam. Navigators had even noticed odd trajectory behavior in cruise, but the concern was never run fully to ground before arrival, and the chance to catch it passed.[2]

THE CAPABILITY GAP

The missing capability was ownership of the interface itself. Where a system is split across two organizations, the place they meet is not a documentation footnote but an engineering deliverable in its own right — with an owner, a specification, and a verification step. Here it had none of the three. Each team treated its own side as complete and the boundary as someone else's concern, so the one assumption that had to be shared and checked — what units are we speaking in? — was precisely the one no one verified. The spacecraft did not fail; it performed as built; the seam between the two halves of the organization did.[3]

AFTERMATH & REFORM

NASA tightened interface management and end-to-end verification and treated the loss as a cautionary tale about how far "faster, better, cheaper" could be pushed before the corners being cut turned out to be load-bearing. The orbiter became the canonical systems-engineering example of interface-as-requirement — the civilian, software parallel to the Patriot's untraveled assumption (Case 8): two competent halves of a system, a boundary nobody owned or verified, and a small unspecified thing at that boundary that quietly destroyed the whole.[4]



MARS CLIMATE ORBITER — THE UNOWNED INTERFACE

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> *The root cause was the failed translation of English units to metric units in a segment of ground-based, navigation-related mission software.*
>
> PARAPHRASING THE NASA MARS CLIMATE ORBITER MISHAP INVESTIGATION BOARD, 1999

LE INSIGHT

Mars Climate Orbiter is the textbook case for interface boundaries as engineering deliverables. The contractors did their work. The boundary between them did not have an owner. The capability that was missing was the interface-specification verification step.

LENS APPROACH

LENS uses Mars Climate Orbiter in LEN 5 to teach interface-as- requirement and in LEN 8 to discuss cross-contractor capability boundaries. The case is the foundational software-engineering parallel to Patriot (Case 8).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make the interface between two organizations an owned deliverable with a written specification of units and formats, not a footnote each side assumes the other handles.
- Fix and verify the unit convention at the boundary explicitly, requiring both sides to confirm agreement before data crosses the seam.
- Run end-to-end validation across the combined system, since each half can be correct within its own assumptions while their composition is wrong.

IN OPERATION / AFTER

- Treat in-cruise anomalies, like the noticed trajectory behavior, as signals to run to ground before a point of no return rather than concerns to revisit after arrival.
- Monitor predicted-versus-actual trajectory continuously so an accumulating boundary error surfaces while there is still time to correct it.
- Audit cross-contractor seams for unowned assumptions whenever a program trims margins under “faster, better, cheaper,” confirming the corners cut are not load-bearing.

REFLECTION QUESTIONS

1. Identify a contractor-to-contractor interface in your domain whose specification ownership is unclear. What would the equivalent unit-mismatch look like there?
2. Design the interface-verification deliverable that would have caught the pound-force / newton boundary before launch.
3. Navigators noticed odd trajectory behavior in cruise but the concern was never run to ground before arrival. Specify the threshold and process by which an in-flight anomaly in your domain must be resolved before a point of no return, rather than carried past it.

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CASE 10 TECHNOLOGY 2012

Knight Capital Trading Loss

D Designed Out K Knowledge & Institutional Memory

IMPACT $440M loss in 45 minutes; firm forced to seek emergency capital and was effectively acquired within months

IN BRIEF

On 1 August 2012 Knight Capital, a major U.S. market maker, deployed new order-routing software to seven of its eight servers and missed the eighth. The new code reused a flag that, on the eighth server's old software, reactivated long-dead "Power Peg" test code never removed from the repository. At the opening bell it fired millions of unintended orders; in about 45 minutes Knight lost roughly $440 million — more than the firm was worth — and was effectively acquired within months. The SEC found Knight had no procedure to verify the deployment across all servers and no controls to halt the runaway orders. The capability designed out was deployment verification; the dead code was technical debt that exercised its option at the worst possible moment.

BACKGROUND

Knight Capital was one of the largest market makers in U.S. equities, routing an enormous share of retail order flow through automated systems wired directly into the market. On 1 August 2012 the New York Stock Exchange was launching a new Retail Liquidity Program, and Knight had updated its order-routing software to participate in it from the opening bell. The update went out to production the way countless updates had gone before — a routine deployment, on an ordinary morning, handled as unremarkable.[0]

WHAT HAPPENED

A technician deployed the new code to seven of the eight routing servers and missed the eighth, leaving one node running stale software. The new code reused a configuration flag that, on the old software still on that eighth server, had once activated long-dormant "Power Peg" test code — retired years earlier but never removed from the repository. At the open the dead code woke and began firing millions of unintended orders into the market. In about forty-five minutes Knight amassed a vast unwanted position; losses passed $170 million almost at once and reached about $440 million — more than the firm itself was worth. It survived only on emergency capital and was effectively acquired within months.[1]

THE INVESTIGATION

The Securities and Exchange Commission's September 2013 enforcement order — In re Knight Capital Americas LLC — found Knight had no written procedure requiring a second technician to verify the deployment across all hosts, and no automated check that all eight servers were running the same code — nor controls able to recognize and halt the flood of erroneous orders once it began. The order catalogued specific violations of the SEC's Market Access Rule (Rule 15c3-5), and Knight settled for a $12 million penalty.[2] The dead Power Peg code was the proximate trigger and the reused flag the match that lit it; but the underlying cause was the absence of the verification and risk controls that should surround any change to a system wired directly into the live market. The order is unusually explicit that the institutional gap was financial-engineering practice catching up to software-engineering practice — release management, configuration control, and pre-trade risk limits were not yet treated as first-class deliverables on the trading desk.[3]

THE CAPABILITY GAP

The capability designed out was deployment verification — confirming, every time, that what runs in production is exactly what was intended, on every node, before it touches live money. The dead code was technical debt in its most literal form: a retired function left in the repository is a standing option on a future failure, and reusing its flag exercised that option at the worst possible moment. As at the Mars Climate Orbiter's interface, the boundary that mattered — between "deployed" and "verified as deployed everywhere" — had no owner and no automated check, and a large institution kept walking across it routinely until the day the floor gave way.[4]

AFTERMATH & REFORM

Knight became the canonical case in modern software-operations practice for why deployment is itself an engineering deliverable, not a clerical step: automated verification that every host runs the intended build, disciplined removal of dead code from the repository, pre-trade risk limits, and kill switches that can stop a runaway process in seconds. Regulators sharpened their attention to automated market-access controls in response. The SEC's 2014 Regulation Systems Compliance and Integrity (Regulation SCI) extended formal system-development, testing, change-management, and incident-reporting obligations to registered exchanges, clearing agencies, and large alternative trading systems — the institutional response that pulled the deployment discipline Knight had lacked into the regulatory perimeter. The lesson rhymes with the orbiter's across a forty-year, civilian-to-financial gap: a small, unowned boundary inside a large automated system is precisely where the institution is most exposed, and least watching — and as algorithmic trading continued to grow after 2012, the same class of fragility kept producing new incidents that the Knight template explained.[5]

$440M

LOSS IN 45 MINUTES · 7 OF 8 SERVERS UPDATED

dead code on the eighth server, repurposed flag, no deployment verification

KNIGHT CAPITAL — THE COST OF AN UNVERIFIED DEPLOYMENT

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THE LEARNING ENGINEERING LENS

> *Knight did not have written procedures requiring a second technician to verify the deployment.*
>
> PARAPHRASING THE SEC ORDER AGAINST KNIGHT CAPITAL, 2013

LE INSIGHT

Knight Capital is the financial-industry version of Mars Climate Orbiter (Case 9): a small, unspecified boundary inside a large system that took the institution down. The capability that was missing was deployment verification. The dead code was the proximate trigger; the absent procedure was the cause.

LENS APPROACH

Knight Capital is the canonical change-control-and-disclosure governance case (induced 5.4; LENS D1/PT3). LENS uses it in LEN 5 to teach deployment-as-capability — students design the deployment deliverable that would have caught the eighth server — and in LEN 9 for the technical-debt argument: every line of dead code carries an option on a future failure. Adjacent to Mars Climate Orbiter (Case 9) at the small-boundary-no-owner layer and to Regulation SCI as the institutional response that codified the missing controls.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Remove retired code and dormant flags from the repository as a matter of discipline, since every dead function left behind is a standing option on a future failure.
- Never reuse a configuration flag whose old meaning still exists somewhere in the fleet; treat flag semantics as a versioned, owned interface.
- Design pre-trade risk limits and a kill switch into any system wired to the live market, so a runaway process can be stopped in seconds rather than minutes.

IN OPERATION / AFTER

- Require automated verification that every host runs the intended build before a change touches live money, with a second technician confirming the deployment across all nodes.
- Monitor for the runaway condition — an abnormal order rate — and halt automatically, rather than relying on humans to notice and intervene mid-flood.
- Audit deployments for the orphaned node: confirm no host is left on stale code, closing the gap between "deployed" and "verified as deployed everywhere."

REFLECTION QUESTIONS

1. Identify a deployment procedure in your domain whose verification step depends on convention rather than on a designed check. What is the eighth server?

2. Design the deployment deliverable that would prevent a Knight Capital-equivalent loss in your organization.

3. A retired “Power Peg” function left in the repository became the trigger years later when its flag was reused. Identify dead code or a dormant feature flag in a system you run that is still an option on a future failure, and specify the policy that should have removed it.

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CASE 11 ENERGY INDUSTRIAL 2006

Sago Mine Disaster

N Normalization of Deviance T Training Gap K Knowledge & Institutional Memory

IMPACT 12 killed in a West Virginia coal-mine explosion; emergency-response failures compounded the initial event; MINER Act of 2006

IN BRIEF

On 2 January 2006 lightning ignited methane in a sealed area of the Sago Mine in West Virginia; the seals failed, the explosion propagated, and thirteen miners were trapped behind it. Twelve died of carbon-monoxide poisoning over the hours that followed; only one, Randal McCloy Jr., survived. A communications failure briefly told the nation — and the families — that twelve had been found alive, when the opposite was true. The MSHA investigation found seals built to an inadequate design, an inadequate emergency plan, and lapsed self-rescue training — each marginal for years, all inadequate together in the one window that mattered. Sago drove the MINER Act of 2006, strengthening mine-rescue requirements and mandating underground refuge chambers.

BACKGROUND

The Sago Mine in West Virginia had sealed off a mined-out area behind barriers built to a design that had not kept pace with current standards. Its emergency plan and the miners' self-rescue training were, like the seals, marginally adequate — good enough on an ordinary day, untested against the worst one.[0] On any ordinary day each of these margins was invisible precisely because it was never tested at its edge, so the mine could run for years with every defense quietly thin and nothing to signal that the thinness was accumulating.

WHAT HAPPENED

On 2 January 2006 lightning ignited methane in the sealed area. The seals failed and the explosion propagated, trapping thirteen miners behind it. Over the hours that followed twelve died of carbon-monoxide poisoning; only one, Randal McCloy Jr., survived. Compounding the tragedy, garbled communications briefly told the nation and the families that twelve had been found alive — the opposite of the truth.[1] The hours of carbon-monoxide exposure were exactly the window the emergency plan and self-rescue provisions existed to bridge, so the marginal defenses failed in the one stretch of time their adequacy was supposed to guarantee.

THE INVESTIGATION

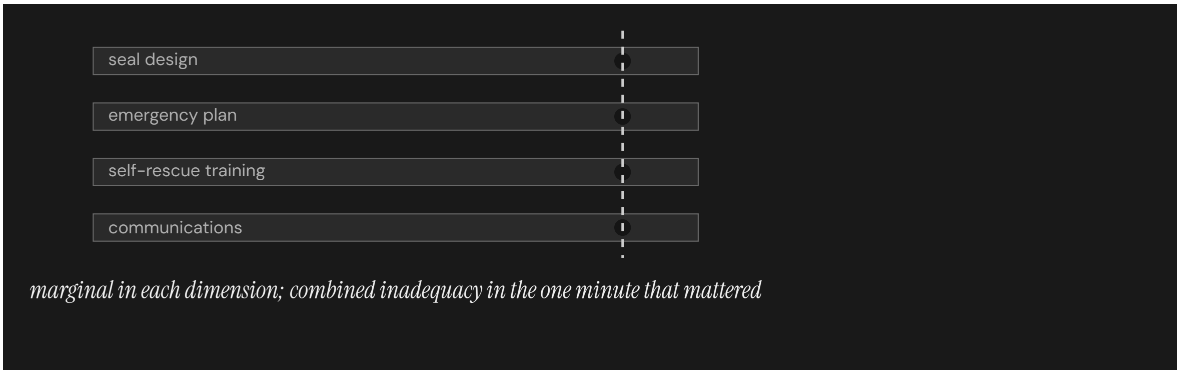
The Mine Safety and Health Administration found the seals had been built to a design that did not meet then-current standards, the mine's emergency plan was inadequate, and self-rescue training had not been kept current.[2] The miners "faced multiple equipment, training, and emergency-response shortcomings that compounded their initial trapping" — no single failure decisive, the combination lethal.[3] That each shortcoming was real but none was solely decisive is the finding's whole weight: an investigation looking for one nameable cause would have found several survivable ones and missed the lethal way they combined.

THE CAPABILITY GAP

Sago is the cumulative-inadequacy pattern. Each defense — seal design, emergency plan, self-rescue training, communications — was marginally adequate for years and recoverable on its own; none was the dramatic, nameable cause. They failed together in the only minute that mattered, which is exactly how normalization works: a system drifts within tolerance on several fronts until the tolerances align.[4] The hazard lived not in any one margin but in their simultaneity, which no inspection of a single defense could surface, because each looked acceptable on its own and the danger was a property only of the set.

AFTERMATH & REFORM

Sago drove the federal MINER Act of 2006, which strengthened mine-rescue requirements, tightened seal standards, improved communications and tracking, and mandated breathable-air refuge chambers underground.[5] The reform addressed the combination rather than a single cause — the right response to a failure whose lesson is that marginal-everywhere is itself a system-level hazard, even when no single margin looks alarming. The refuge-chamber mandate in particular conceded that survivable air over those carbon-monoxide hours had to be engineered in advance, not left to the chain of marginal defenses that failed together at Sago.



SAGO — CUMULATIVE INADEQUACY ACROSS MULTIPLE DEFENSES

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THE LEARNING ENGINEERING LENS

> *The miners faced multiple equipment, training, and emergency-response shortcomings that compounded their initial trapping.*
>
> MINE SAFETY AND HEALTH ADMINISTRATION, SAGO INVESTIGATION REPORT, 2007

LE INSIGHT

Sago is the case for the cumulative inadequacy pattern in industrial accidents. No single failure caused the disaster. Each failure on its own would have been recoverable. The combination was not, and the combination had been the operating condition of the mine for years.

LENS APPROACH

LENS uses Sago in LEN 5 for cumulative-inadequacy analysis and in LEN 8 for the legislative-reform arc that followed. Studio projects compare Sago and Upper Big Branch (Case 87) as paired cases.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assess defenses jointly against the worst-case window, since each margin looks acceptable alone and the hazard lives in their simultaneity.
- Engineer a guaranteed survivable resource — like breathable air over the rescue window — rather than relying on a chain of marginal provisions.
- Keep seal design, emergency plans, and self-rescue training current to standards as a coupled set, not as independently deferred items.

IN OPERATION / AFTER

- Track how many defenses sit at the margin simultaneously, treating marginal-everywhere as a measurable system-level hazard.
- Stress-test the emergency response against the exact window it exists to bridge, so untested margins are exposed before an event.
- Investigate near-misses for combined inadequacy, not a single nameable cause, so survivable contributors are not dismissed individually.

REFLECTION QUESTIONS

1. Identify a process in your domain that is marginally adequate across multiple parameters. What is the cumulative failure mode?
2. Sago produced the MINER Act. What legislative change would your domain require if a Sago-equivalent occurred?
3. The danger was a property of the set of defenses, not any single one. What assessment in your domain would evaluate defenses jointly rather than one at a time?

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CASE 12 AVIATION 1991, 1994

Boeing 737 Rudder Hardovers

H Human-System Interface D Designed Out

IMPACT 157 killed across United 585 (Colorado Springs, 1991) and USAir 427 (Pittsburgh, 1994); rudder Power Control Unit malfunctions traced to a thermal-shock condition

IN BRIEF

Two Boeing 737s fell out of controlled flight when their rudders reversed — moving opposite to the pilots' input: United 585 near Colorado Springs in 1991 (25 killed) and USAir 427 near Pittsburgh in 1994 (132 killed), 157 in all. The NTSB investigation took years because the failure was rare and unrecoverable. It traced the cause to a thermal-shock condition in the rudder's power-control-unit servo valve — cold-soaked hydraulic fluid hitting a hot valve — that could jam the valve and reverse the rudder. The pilots had no procedure to recognize or recover the failure; the manuals and training never anticipated it, and in that state no available control input could save the aircraft. The 737 rudder cases are the book's case for an unrecoverable failure mode hidden inside a working aircraft.

BACKGROUND

The Boeing 737, the most-produced jetliner in history, steers in yaw with a rudder driven by a hydraulic power control unit (PCU). The PCU's servo valve was a dual-concentric design intended to be fault-tolerant — a reassuring assumption that had passed certification.[0] Because the design was believed to be fault-tolerant, a single jam was treated as something the redundant geometry would contain; that confidence is precisely what let the failure mode hide in plain sight across a fleet flying millions of hours without incident.

WHAT HAPPENED

Twice, 737s rolled and dove out of level flight unrecoverably: United Airlines Flight 585 near Colorado Springs in 1991 (25 killed) and USAir Flight 427 near Pittsburgh in 1994 (132 killed) — 157 deaths between them. In each, the rudder had swung hard over, opposite to what the crew commanded.[1] A control that moves opposite to its input is the cruelest failure a pilot can face: every corrective action deepens the upset, and the three years separating the two losses meant the first crash yielded no usable answer in time to protect the second.

THE INVESTIGATION

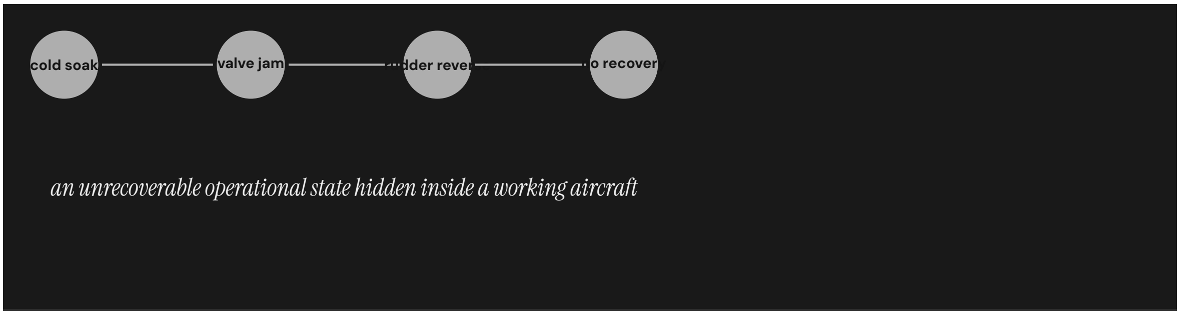
The NTSB investigation took years, because the failure was both rare and unrecoverable, leaving little to reconstruct.[2] An unrecoverable event tends to destroy its own evidence, and a rare one offers no pattern to work from, so the board had to reason toward a mechanism the wreckage could only hint at. It eventually identified a thermal-shock condition in the rudder servo valve — cold-soaked hydraulic fluid striking a hot valve under specific conditions — that could jam the secondary valve and let the rudder move opposite to commanded input, a convergence of cold, heat, and timing so narrow it had eluded every test the design had been put through.[3]

THE CAPABILITY GAP

The gap was not in the pilots: in the failure mode no control input available to them could recover the aircraft, and they had no procedure even to recognize a rudder reversal, because the manuals and training never anticipated it. The missing capability sat upstream — in a certification process that had not surfaced the failure mode, and maintenance procedures that did not test for it. When no human action is recoverable, capability engineering must move to the design and certification, not the cockpit — no amount of pilot skill or training can close a gap that lives in a part the crew cannot reach and a state no procedure names, which is why the only real fix was upstream of the flight deck entirely.[4]

AFTERMATH & REFORM

Boeing redesigned the rudder PCU and the fleet was retrofitted, and the cases reshaped how rare, unrecoverable failure modes are hunted in certification and flight test.[5] Retrofitting the entire fleet conceded that the original fault-tolerant assumption had been wrong, and the changed approach to certification testing accepted that a part passing its tests is not the same as a part proven safe across every condition it will meet. The 737 rudder sits at the intersection of mechanical design, certification testing, and crew capability — all three would have had to fail together for the accidents to occur, and all three did.



737 RUDDER — THE FAILURE MODE CERTIFICATION DID NOT ANTICIPATE

REFERENCES

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2. NTSB, Aircraft Accident Report: United Airlines Flight 585, NTSB-AAR-92-06 (1992) — the initial (undetermined) Colorado Springs investigation, later reopened.
3. NTSB-AAR-99-01 (1999) — the servo-valve reversal mechanism and the absence of any recoverable crew action.
4. Boeing 737 rudder PCU service bulletins and redesign program (1996–2002).
5. W. Langewiesche, Inside the Sky (1998) — the multi-year investigation narrative.
6. NTSB-AAR-99-01 (1999) — the certification and maintenance gaps that left the failure mode unsurfaced.

> *The aircraft was operated for years with a design feature that, under specific conditions, was unrecoverable.*
>
> PARAPHRASING NTSB AIRCRAFT ACCIDENT REPORT NTSB-AAR-99-01 ON THE 737 RUDDER PCU, 1999

LE INSIGHT

The 737 rudder cases are the case for an unrecoverable operational state hidden inside an apparently working system. The capability gap was not in the pilots — there was no capability that would have recovered the aircraft. The gap was in the certification process that had not surfaced the failure mode and in the maintenance procedures that did not test for it.

LENS APPROACH

LENS uses the 737 rudder cases in LEN 5 as a capability-limits case (when no human action is recoverable, capability engineering must move upstream) and in LEN 7 for certification governance.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Test components against the full envelope of conditions they will meet in service — including rare combinations like thermal shock — rather than only the nominal cases certification asks for.
- Treat any "fault-tolerant" claim as a hypothesis to be falsified in flight test, not an assumption that lets a single jam go unexamined.
- Provide crews a procedure to recognize and respond to control reversals, while accepting that for unrecoverable modes the real fix must sit in the design.

IN OPERATION / AFTER

- Reopen and reconcile investigations of rare, unexplained events instead of leaving them undetermined, since a buried first case left the second crew unwarned.
- Build maintenance procedures that actively test for the surfaced failure mode across the fielded fleet, not just the redesigned units.
- Hunt for unrecoverable failure modes proactively in service data, treating their absence in the record as unproven rather than as evidence of safety.

REFLECTION QUESTIONS

1. Where in your domain might an unrecoverable failure mode exist that has not yet manifested? How would you find it?
2. Design the certification-test specification that should have caught the 737 rudder thermal-shock failure in development.
3. The 737 rudder passed certification yet was not safe across every condition it met. What is a component in your domain certified as fault-tolerant on an assumption no one re-tests — and how would you challenge that assumption before it fails?

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CASE 13 GENERAL AVIATION AVIONICS PILOT TRAINING 2010

Glass-Cockpit Transition in Light Aircraft — Technology Outran Training

H Human-System Interface N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT An NTSB safety study of ~8,000 piston aircraft over 2002–2006 found that glass-cockpit aircraft had no better overall safety record — and a higher fatal accident rate — than comparable conventional-instrument aircraft over the period studied; the Board attributed this to inadequate equipment-specific training and issued recommendations A-10-36 through A-10-40

IN BRIEF

Glass cockpits — integrated digital primary flight displays and multifunction displays replacing the inherited six-pack of analog instruments — were introduced into light piston aircraft over the 2000s as a fleet-wide modernization. The NTSB safety study NTSB/SS-10/01 examined approximately 8,000 small piston aircraft over 2002–2006 and reached a finding the technology's own advocates did not expect: glass-cockpit aircraft had no better overall safety record than comparable conventional-instrument aircraft over the period studied, and in fact had a higher fatal accident rate. The NTSB attributed this not to the technology but to inadequate equipment-specific training. The Board issued recommendations A-10-36 through A-10-40 on knowledge-testing standards, simulator availability, and training requirements. The study is explicit that advanced avionics "can increase the safety potential" of light aircraft but that the potential "was not yet realized in the period studied" — an open, not closed, verdict. The case is the canonical 7.1 failure of an inherited capability regime (pilot proficiency) not being re-verified against the new envelope the technology transition introduced. Pair with the aging-system transitions in Cases 22–76 (drafted in parallel).

BACKGROUND

Glass-cockpit avionics — integrated digital primary flight displays and multifunction displays replacing the inherited six-pack of analog round-dial instruments — were introduced into light piston aircraft over the 2000s. The fleet conversion was largely industry-driven; new-build aircraft came with glass as standard, retrofits became increasingly common, and the FAA's certification framework treated the transition as a positive safety move on the strength of the capability the displays delivered to pilots: integrated situational awareness, moving-map navigation, terrain and traffic display, system-status integration. The operationally critical assumption was that pilots transitioning from analog to glass would carry their proficiency across the change.[0]

WHAT HAPPENED

The NTSB safety study NTSB/SS-10/01 examined that assumption directly. The Board studied approximately 8,000 small piston airplanes registered in the United States over the period 2002–2006, comparing accident rates between conventional-instrument and glass-cockpit fleets matched on model and operational class. The headline finding ran against expectation. Glass-cockpit aircraft had no better overall safety record than comparable conventional-instrument aircraft over the period, and the fatal accident rate was higher for the glass fleet. The expected fleet-wide safety gain from the displays had not yet materialized in the accident record.[1]

THE INVESTIGATION

The NTSB's attribution is precise. The technology was not the cause; inadequate equipment-specific training was. The transitioning pilot population was certificated and current under the inherited training regime that assumed analog instruments, and the glass displays — for all their situational-awareness advantages — introduced new failure modes (mode confusion, automation surprises, attention capture by integrated displays, degraded scan in degraded modes) that the inherited training did not address. The Board issued recommendations A-10-36 through A-10-40 to the FAA: equipment-specific knowledge-testing standards, simulator and training-device availability for transitioning pilots, and structured training requirements before operating glass-cockpit aircraft.[2]

THE CAPABILITY GAP

The study's language is open rather than closed. Advanced avionics "can increase the safety potential" of light aircraft, but that potential "was not yet realized in the period studied." The verdict is not that glass is unsafe; it is that the inherited certification of pilot proficiency, built around analog instruments, did not transfer to the new envelope without re-engineered training. The case is the canonical 7.1 failure: a capability regime (pilot proficiency) was not re-verified when the cockpit's interface envelope changed, and the inherited training assumptions silently became inadequate to the new operational reality.[3]

AFTERMATH & REFORM

Drafted alongside the aging-system transitions in Cases 22–76, the glass-cockpit GA study carries the capability-under-system-change pattern at the consumer-aviation scale where the analog cases run at the defense and commercial-aviation scale. The structural lesson is the same: when a system's interface envelope changes — and even when the change is a capability-enhancing one — the inherited certification of operator proficiency has to be re-verified against the new envelope, or the transition outruns the training and the accident record moves in the wrong direction. The NTSB recommendations are the engineering response to that pattern, and their implementation is what closes (or fails to close) the open verdict the study left.

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THE LEARNING ENGINEERING LENS

> *Advanced avionics can increase the safety potential of light aircraft, but the potential was not yet realized in the period studied.*
>
> NTSB SAFETY STUDY NTSB/SS-10/01 (2010).

LE INSIGHT

The NTSB's glass-cockpit GA study is the canonical capability-under-system-change failure at the consumer-aviation scale: a positive technology transition that nevertheless outran the inherited certification of operator proficiency. The Board's attribution is to inadequate equipment-specific training, and the verdict is open — the safety potential is there, and the transition has not yet realized it.

LENS APPROACH

Glass-cockpit GA is the small-fleet capability-under- transition failure (induced 7.1; LENS D3/PT1). LENS uses it in Domain 1 (Systems Analysis) for the inherited certification of operator proficiency that has to be re-verified against the new envelope, and in Domain 3 (Human-System Collaboration) for the new automation-induced failure modes (mode confusion, automation surprises, attention capture) the transition introduced. Pair with the aging-system transitions in Cases 22–76 at the cross-scale capability-under-change layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the inherited certification of operator proficiency in advance of a technology transition, and design the re-verification against the new envelope as part of the transition's deliverable, not as a downstream training catch-up.
- Treat capability-enhancing transitions with the same re-verification discipline as capability-degrading ones; the glass-cockpit transition was a positive technology move that nevertheless required training re-verification the field skipped.
- Name the new failure modes the transition introduces (mode confusion, automation surprises, attention capture) at the design stage of the equipment-specific training, rather than waiting for the accident record to surface them.

IN OPERATION / AFTER

- Preserve the NTSB's open-verdict language ("can increase the safety potential … not yet realized in the period studied") in any report on the transition's outcome; the study did not say glass is unsafe, and the precise language is what makes the recommendation set actionable.
- Track implementation of the NTSB recommendation set (A-10-36 through A-10-40) as the engineering response to the open verdict; the verdict closes when the recommendations are implemented and the next round of evidence is collected.
- Carry the structural lesson into adjacent transitions — aging-system transitions in Cases 22–76 — as evidence that the inherited certification of operator proficiency has to be re-verified against the new envelope across consumer, commercial, and defense scales.

REFLECTION QUESTIONS

1. Identify a technology transition in your domain that is positive on its capability claim and that nevertheless puts the inherited certification of operator proficiency under question. What re-verification of operator proficiency would the transition require — and is it currently part of the deliverable, or assumed away?

2. The NTSB attribution is to inadequate equipment-specific training, not to the technology. What is the analog distinction in your context — between the engineered artifact and the operator-proficiency regime that has to track it?

3. The NTSB verdict is open: advanced avionics "can increase the safety potential" but the potential "was not yet realized in the period studied." What would the next round of evidence in your domain have to look like to close the open verdict, and who would have to commission it?

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CASE 14 GOVERNMENT 2013

Healthcare.gov Launch

K Knowledge & Institutional Memory T Training Gap G Governance & Trust

IMPACT ~27,000 federal-marketplace enrollments in the first month vs. a 7M first-year target; hundreds of millions in remediation

IN BRIEF

When Healthcare.gov launched on 1 October 2013 it collapsed under load it had never been tested for — about 27,000 enrollments through the federal marketplace in its first month against a seven-million first-year target. The people assembled to build it knew insurance markets and government programs but not technology product launches; key technical roles went unfilled, no office clearly owned the integration (CMS thought the contractor CGI was the lead integrator; CGI did not), and no end-to-end test was run before launch. The fix-it team that rescued the site became the U.S. Digital Service. At root it was a capability mismatch at scale: the organization lacked the human capabilities the system required, and the governance chain surfaced no signal of the gap before launch. It is the rare governance failure that produced lasting institutional reform.

BACKGROUND

Healthcare.gov was the federal insurance marketplace at the center of the Affordable Care Act — a high-visibility online system that millions would hit on day one, with a political deadline that could not slip. The people assembled to build it understood insurance markets and large government programs, but not the launch of a consumer technology product at that scale, and key technical positions went unfilled — so the very expertise the launch most needed was the expertise the team most lacked.[0]

WHAT HAPPENED

There was no clear division of responsibility among the many government offices involved; CMS believed the contractor CGI was the lead system integrator, and CGI did not share that understanding — so the single most important role on the program was one no party believed it held. No end-to-end test was run before launch, meaning the assembled pieces were never exercised together as a user would exercise them. The site went live on 1 October 2013 and immediately collapsed under load it had never been validated for: about 27,000 enrollments through the federal marketplace in its first month against a seven-million first-year target.[1]

THE INVESTIGATION

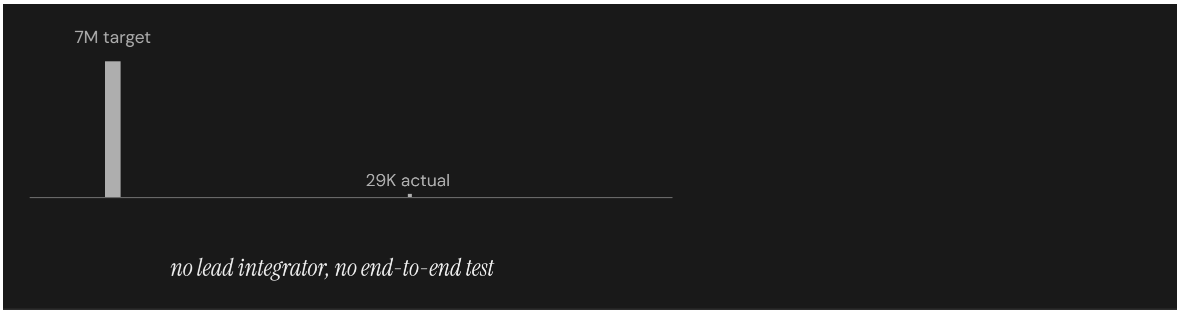
Reviews by the GAO and the HHS Inspector General found that no single person had a clear understanding of the project's status, and that the governance chain had no signal that would surface the readiness gap before launch. With ownership diffused across offices and no integration test to fail, the chain could report progress at every level while the whole remained untested — a structure in which bad news had nowhere to enter and no one positioned to act on it.[2]

THE CAPABILITY GAP

Healthcare.gov is a capability failure wearing a technology costume. The site was salvageable in weeks once the right people arrived, which is the clearest proof that the code was never the binding constraint; the original failure was that the wrong people had been assembled, and that the governance chain meant to catch the mismatch had no mechanism to see it. The missing capability was the matching of human capability to system requirement — and the institutional signal that would have flagged its absence before a deadline locked the launch in place.[3]

AFTERMATH & REFORM

The rescue effort pulled together the team that became the U.S. Digital Service — a permanent institution born from a failure visible on the news every night, its mandate built directly from what the launch had lacked.[4] It is the rare case in this book of a governance failure that produced durable organizational reform, and a reminder that the technical narrative ("the website crashed") often hides the real one: the wrong capability was assembled, unnoticed, by a chain that had no way to notice.



29,000 ENROLLMENTS VS. 7,000,000 TARGET — FIRST TWO MONTHS

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THE LEARNING ENGINEERING LENS

> *No single person had a clear understanding of the project's status.*
>
> PARAPHRASING THE HHS OFFICE OF INSPECTOR GENERAL REVIEW OF HEALTHCARE.GOV, 2016

LE INSIGHT

Healthcare.gov is a capability case wearing a technology costume. The technology was salvageable in weeks once the right people arrived. The original failure was that the wrong people had been assembled in the first place, and the governance chain that should have noticed the mismatch had no signal to surface it. The aftermath — USDS — is the rare case in this book of a governance failure that produced permanent institutional reform.

LENS APPROACH

LENS uses Healthcare.gov in LEN 1 as a problem-framing case (the technical narrative obscures the capability narrative), in LEN 5 to teach capability-requirements analysis for a large government program, and in LEN 8 to discuss the founding of USDS as organizational learning that survived.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Match assembled human capability to the system's real requirements at staffing time — fill the technical roles a consumer-scale launch demands before committing to the deadline.
- Name a single accountable integration owner and confirm every party shares that understanding, so no critical role is one each office assumes someone else holds.
- Gate launch on a full end-to-end test that exercises the assembled system the way users will, not on per-component sign-offs.

IN OPERATION / AFTER

- Build a governance signal that can surface a readiness gap upward, with someone holding the authority to delay a politically fixed launch date.
- Independently audit project status against demonstrated capability, since a diffuse chain can report green at every level while the whole remains untested.
- Institutionalize the rescue capability — as USDS did — so the talent that fixes a launch outlives the crisis that summoned it.

REFLECTION QUESTIONS

1. Healthcare.gov shipped without end-to-end testing. What is the equivalent missing deliverable in a current high-stakes deployment in your domain?
2. USDS was born from the Healthcare.gov failure. What is the institutional capability your domain still lacks, and what failure would have to occur to produce it?
3. No single office believed it owned integration, so the gap stayed invisible until launch day. In a project you know, who actually owns the seam between components — and what signal would let them, rather than the news, learn the system is not ready?

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CASE 15 SPACE / AEROSPACE 1996

Ariane 5 Flight 501

D Designed Out K Knowledge & Institutional Memory H Human-System Interface

IMPACT Maiden flight destroyed itself 37 seconds after launch; ~$500M payloads lost; reused Ariane 4 code never re-verified for Ariane 5

IN BRIEF

On its 1996 maiden flight, the Ariane 5 rocket destroyed itself 37 seconds after launch, losing around half a billion dollars in payloads. The cause was reused software: the inertial reference system inherited code from Ariane 4, where a horizontal-velocity value never exceeded a 16-bit integer's range. Ariane 5's steeper, faster trajectory pushed it past that range; the integer overflow crashed both redundant reference systems almost simultaneously, the vehicle lost guidance and broke up. The inquiry found the offending code path had been irrelevant on Ariane 4 and was never removed or re-verified for Ariane 5 — software reuse had been treated as risk-free. Ariane 5 is the foundational safety-critical-software case for the hazard of reusing code without re-verifying it against the new system's operating envelope.

BACKGROUND

The Ariane 5 heavy-lift rocket reused proven components from its successful predecessor, Ariane 4 — including, in its inertial reference system, software that had flown reliably for years. Reuse of flight-proven code was regarded as a way to reduce risk and cost.[0] The reasoning was seductive precisely because the code's flight record was genuine: software that had worked on Ariane 4 looked like the safest possible choice, and that confidence is what allowed it to cross onto a different vehicle without being re-examined against the new vehicle's conditions.

WHAT HAPPENED

On 4 June 1996, Ariane 5 Flight 501 — its maiden flight — veered off course and broke up under aerodynamic stress 37 seconds after launch; the range-safety system destroyed it, and roughly half a billion dollars of payloads were lost.[1] The vehicle was destroyed in well under a minute of flight, before it had done anything but climb — the whole half-billion-dollar loss flowing from a fault that triggered almost immediately at liftoff, when the new trajectory first pushed the inherited code outside the range it had been built to handle.

THE INVESTIGATION

The inquiry board traced the cause to the reused code. A horizontal-velocity value that never exceeded a 16-bit signed integer on Ariane 4's trajectory did exceed it on Ariane 5's faster one; the resulting overflow shut down both redundant inertial reference systems almost simultaneously, and the vehicle lost guidance.[2] Redundancy gave no protection here because both units ran the identical inherited code and met the identical out-of-range value at the same instant, so the backup failed in lockstep with the primary — duplicating a system defends against a part breaking, not against a shared assumption being wrong. The offending code path had become irrelevant to Ariane 5′s flight after liftoff, yet had neither been removed nor re-verified against the new vehicle's envelope — reuse had been treated as risk-free.[3]

THE CAPABILITY GAP

Ariane 5 is the canonical software-engineering case for the fallacy of risk-free reuse. Code is fit for the envelope it was written and verified against; reusing it in a different envelope is a new design decision and must be re-verified as one. The institutional capability that was missing was the requirement that reused safety-critical code be re-verified, not merely trusted because it had worked before.[4] The flaw was never in the Ariane 4 code, which was correct for the trajectory it was written against; the flaw was the institutional assumption that a record of working in one envelope carried over to another, an assumption no test was required to challenge.

AFTERMATH & REFORM

The fix and the lesson reshaped safety-critical software practice: reuse became a verification event, with explicit checking of every inherited assumption against the new system's operating conditions.[5] The case rhymes with the Patriot (Case 8) and the Mars Climate Orbiter (Case 9) — small, unexamined assumptions inherited across a boundary, fatal when the boundary's conditions changed. What unites the three is that each inherited a quantity sound in its original setting and lethal in the new one, so the reform is the same in every case: make the boundary crossing the moment the assumption is re-checked, rather than the moment it is silently trusted.

37s

AFTER LAUNCH · 16-BIT INTEGER OVERFLOW

code path disabled by the previous vehicle's profile; re-enabled by the new one

ARIANE 5 — THE FALLACY OF RISK-FREE CODE REUSE

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4. Lions Report (1996) — the reused code path neither removed nor re-verified for Ariane 5′s envelope.
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6. Cf. Patriot (Case 8) and Mars Climate Orbiter (Case 9).

THE LEARNING ENGINEERING LENS

> *The internal SRI software exception was caused during execution of a data conversion from 64-bit floating point to 16-bit signed integer value.*
>
> ARIANE 5 FLIGHT 501 FAILURE INQUIRY BOARD, 1996

LE INSIGHT

Ariane 5 is the canonical software-engineering case for the fallacy of risk-free code reuse. Code is fit for its envelope of operation. Reusing it in a different envelope is a new design decision and must be re-verified as one. The institutional capability that was missing was the requirement to re-verify.

LENS APPROACH

LENS uses Ariane 5 in LEN 5 for software-engineering capability deliverables and in LEN 8 for the institutional discipline that treats reuse as a verification event rather than as a savings.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat every reuse of safety-critical code as a new design decision: re-verify each inherited assumption against the new system's operating envelope before flight.
- Document the original operating envelope a component was verified against, so a later reuse can be checked against it rather than blindly trusted.
- Design redundancy to fail independently — diverse implementations or inputs — so a shared inherited assumption cannot take out primary and backup at once.

IN OPERATION / AFTER

- Audit reused components for code paths that are irrelevant in the new system but still active, and remove or re-verify them rather than leaving them dormant.
- Make boundary crossings — code moving onto a new vehicle, a new envelope — a mandatory re-verification event in the institution's process, not a savings.
- Review past reuse decisions for the Ariane / Patriot / Mars Climate Orbiter pattern: a quantity sound in its original setting and unchecked in the new one.

REFLECTION QUESTIONS

1. Identify a piece of reused infrastructure in your domain whose original operating envelope is not documented. What is the silent assumption?
2. Design the verification deliverable that should accompany every reuse of safety-critical software.
3. Both redundant reference systems failed at the same instant because they shared the same inherited assumption. Identify a place in your domain where redundancy gives false comfort because the duplicated units share a common flaw rather than fail independently.

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CASE 16 NEUROSCIENCE SCIENCE POLICY (EU) 2013 – 2023

EU Human Brain Project — Top-Down Vision That Unraveled

G Governance & Trust N Normalization of Deviance

EVIDENCE TIER journalism-tier — source confidence flagged; future validation ongoing.

IMPACT A €1B EU-flagship initiative to simulate a living brain governed top-down under a single PI; the project unraveled as the field disputed both feasibility and approach, restructured under protest, and concluded in 2023 with the founding framing abandoned

IN BRIEF

The EU Human Brain Project, launched in 2013 as one of the EU's Future and Emerging Technologies Flagship programs, set out to build a computer simulation of a living brain under a single principal investigator (Henry Markram). The governance model was top-down: a central vision, a small leadership group, a decade-long funding commitment of about €1 billion, and a research community asked to align around the simulation goal. The case is drafted as the paired contrast to the BRAIN Initiative (Case 176), not as a standalone study: the same era, the same field-scale ambition, opposite governance models, opposite trajectories. The evidence base is largely journalism — MIT Technology Review retrospective, In Silico documentary, science press, the project's own restructuring records — so the evidence-tier flag is rendered under the case title; future validation will continue. The teaching point survives the flag: the governance choice (top-down single-PI versus distributed working-group) was the variable that materially changed the trajectory of the two programs at field scale.

BACKGROUND

In 2013 the European Commission selected the Human Brain Project as one of two Future and Emerging Technologies Flagship programs, with a ten-year horizon and approximately €1 billion in committed funding. The founding vision, articulated and championed by Henry Markram, was a computer simulation of a living human brain. The governance was top-down: a small leadership group, a unified research framing, and a community of European neuroscience labs asked to organize around the simulation goal.[0]

WHAT HAPPENED

The contestation surfaced quickly. Open letters from a sizable fraction of the European neuroscience community questioned both the feasibility of the simulation goal and the project's governance — the breadth of expertise on the leadership group, the relationship between the cognitive neuroscience the field practiced and the molecular-scale simulation the project proposed, and the absence of a process for scope revision. The EU commissioned a mediation and restructuring exercise that broadened the leadership and re-scoped the research agenda around infrastructure platforms rather than a single simulation. The project ran to its scheduled 2023 conclusion; the founding framing was not what it delivered.[1]

THE INVESTIGATION

The case pairs directly with the BRAIN Initiative (Case 176). The two programs launched within months of each other, at comparable field-scale ambition, in the same decade of neuroscience. The governance models were not comparable: BRAIN organized a working group that visibly contested its own composition and re-baselined its scope on the public record; HBP organized around a single PI and a unified framing, and the community had to mediate the project away from that framing. The trajectory difference is not a verdict on either field; it is evidence about the governance variable at the launch moment.[2]

THE CAPABILITY GAP

The evidence-tier flag rendered under the case title is load-bearing here. The detailed account of the early contestation, the open letters, the mediation, and the restructuring lives largely in MIT Technology Review's retrospective, in the In Silico documentary, and in science-press coverage from Science, Nature, and the European press; the project's own restructuring records add the formal layer. The case is teachable from this material — the structural pattern is consistent across sources — but the specific magnitudes and the exact language of the early contestation should be re-confirmed against primary sources before publication. Future validation will continue.[3]

AFTERMATH & REFORM

What the pair (Cases 176 + 108) teaches is that the governance model — distributed working-group with public-record contestation versus top-down single-PI with unified framing — is itself the variable that explains why one field-scale program survives and adapts while the other unravels and is mediated away from its founding goal. The capability deliverable at program launch is not the vision and not the tools; it is a governance architecture that can absorb contestation, document scope drift, and re-baseline without losing the program. BRAIN had one; HBP had to be retrofitted one.[4]

REFERENCES

1. MIT Technology Review (2021), retrospective on big-science brain projects — the principal critical assessment of HBP alongside BRAIN.
2. In Silico (2020), documentary by Noah Hutton — long-form follow of Markram and the HBP through the contestation period.
3. Science and Nature contemporaneous coverage of the open letters and the mediation/restructuring (2014–2016).
4. Human Brain Project final reports (2023) — the project's own restructuring and concluding documentation.
5. Alivisatos et al. (2012), Neuron — the BRAIN position paper that is the paired contrast (Case 176).

THE LEARNING ENGINEERING LENS

> *A program that cannot re-scope without external mediation is a program designed to fail open.*

EDITORS' SYNTHESIS OF THE HUMAN BRAIN PROJECT RECORD.

LE INSIGHT

The EU Human Brain Project is the paired contrast to the BRAIN Initiative (Case 176): same era, same field-scale ambition, opposite governance models, opposite trajectories. Top-down single-PI governance with a unified framing did not survive community contestation; the program was mediated away from its founding goal and ran to its scheduled conclusion in 2023. Evidence base is journalism-tier and the flag is rendered under the title; future validation will continue.

LENS APPROACH

HBP is the field-scale failure case in the v2 corpus (induced 5.1; LENS D1+D3/PT4) drafted as the contrast to BRAIN (Case 176). LENS uses the pair in Domain 1 (Systems Analysis) for the governance-variable comparison and in Domain 5 (Navigating Sociotechnical Constraints) for the contestation-absorption capability. The evidence-tier flag is binding until primary-source confirmation completes; the structural pattern survives the flag, the specific magnitudes do not.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the scope-revision process before the program launches; a program that cannot re-scope without external mediation is a program designed to fail open.
- Stress-test the leadership group's breadth before launch: does it span the disciplines the program will have to coordinate, or is it narrow enough to be captured by the founding framing?
- Treat early community contestation as governance information, not as friction; absorbing it programmatically is the capability the program needs to demonstrate.

IN OPERATION / AFTER

- When journalism-tier sourcing is what the record provides, flag it under the title and carry the standing "future validation ongoing" language into the printed case — the teaching value survives the flag, but the magnitudes do not.
- Commission a structured retrospective on the governance variable specifically: who held authority, how scope drift was named, how the program re-baselined. This is the documentation a future field-scale launch needs more than another success narrative.
- Cross-reference governance evidence across paired programs: a single program's trajectory is not evidence about governance; a paired comparison at the same era and ambition is.

REFLECTION QUESTIONS

1. Identify a large program in your domain that ran a top-down single-PI governance model. What was the scope-revision process designed in at launch, and what was the process the program actually used when contestation arrived?

2. Specify the paired-program comparison you would commission to learn from a field-scale program's trajectory, rather than from the program alone. What is the matched comparison case and what is the governance variable you are isolating?

3. The evidence base for this case is largely journalism. What primary-source confirmation would you require — interviews, restructuring documents, mediation reports — before relying on this case for a deployment decision?

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Chapter 2

Systems Analysis — What Works

When capability requirements become the deliverable.

The standard you walk by is the standard you accept.

AN AXIOM IN CAPABILITY ENGINEERING.

CASE 17 DEFENSE SUSTAINMENT AUDIT 2022 (GAO-22-104533)

GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data

G Governance & Trust K Knowledge & Institutional Memory

IMPACT GAO-22-104533 reviewed sustainment of selected DoD weapon systems and found that operating-and-support (O&S) costs — the dominant share of a system's lifecycle expense — are not reported at a level of completeness, consistency, or comparability that supports portfolio-level sustainment decisions; the finding recurs across GAO's weapon-system portfolio reporting and is a frontier evidence-architecture problem, not a failure attributable to any single program

IN BRIEF

The Government Accountability Office's recurring portfolio reviews of DoD weapon-system sustainment — including GAO-22-104533 — document a structural evidence-architecture problem at the heart of defense sustainment. Operating-and- support (O&S) costs typically dominate a weapon system's total lifecycle expense, often well above the share of cost accounted for by procurement. Decisions about whether to modify, sustain, replace, or retire fielded systems therefore ride on O&S evidence. GAO has found across multiple reports that O&S data are not reported with the completeness, consistency, or comparability across systems and components that portfolio-level sustainment decisions require. The finding is a frontier evidence-architecture problem rather than a per-program failure: the data deficiencies are structural — across services, across categories, across reporting cycles — and the decisions that should be grounded in those data are being made on evidence the auditor describes as insufficient. The case is the sustainment-side companion to MIL-STD-1472H (Case 21): the standards layer produces engineered design requirements; the sustainment-data layer is where the operating evidence to manage those systems over decades is supposed to live, and at the portfolio scale it does not live there yet.

THE SHIFT

Defense sustainment — the operating, supporting, modifying, and ultimately retiring of fielded weapon systems — typically consumes the majority of a system's total lifecycle cost. The procurement decision that receives the most political attention is the smaller fraction; the sustainment decisions that follow, stretching across decades, dominate the budget. For those decisions to be made on decision-grade evidence, the operating-and-support cost data have to be reported in a form that supports comparison across systems, across services, and across years.[0]

WHAT IS EMERGING

The Government Accountability Office has reviewed DoD weapon-system sustainment for decades through its portfolio-level reporting and through individual program reviews. GAO-22-104533, “Weapon System Sustainment,” examined a set of selected systems and reported the recurring finding: the O&S cost data DoD reports to Congress and uses internally are not complete, consistent, or comparable in the way portfolio-level decisions require. Specific findings have included under-reported cost categories, inconsistent definitions across services for what counts as sustainment vs. modification cost, and gaps in the time series that make trend analysis difficult.[1]

THE CAPABILITY QUESTION

The structural form of the finding is what makes the case a frontier rather than a failure. There is no single program manager whose sustainment data the GAO could have fixed; the deficiency is in the reporting architecture across the enterprise. Each service has legitimate reasons for its specific reporting choices; the lack of cross-service comparability is the emergent property of those choices, not the result of any single error. The decisions that ride on the data — modify, sustain, replace, retire across hundreds of systems — are being made anyway, on the strongest available evidence, which the auditor characterizes as insufficient for the question.[2]

EARLY EVIDENCE

The GAO has made recurring recommendations: standardize O&S cost categories across services, improve the completeness of time-series reporting, make the data comparable across systems at the same point in the lifecycle. DoD has accepted many recommendations and implementation is ongoing. The portfolio-level evidence gap has narrowed in specific areas (e.g., depot maintenance reporting) and persists in others. The case is teachable not because the GAO is right and DoD is wrong — the situation is more structural than that — but because the gap between the evidence the decisions require and the evidence the system can currently supply is the canonical frontier instance of the evidence-architecture problem at portfolio scale.[3]

OPEN PROBLEMS

What LENS takes from the case is the requirement-for- requirements form. MIL-STD-1472H (Case 21) is the requirements-discipline mechanism for the design phase; SUBSAFE (Case 19) is the certification-discipline mechanism for the sustainment phase of a specific capability boundary. The GAO finding identifies the gap that those mechanisms do not fill: the portfolio- level evidence architecture for sustainment decisions across the enterprise. The case is the worked example of why “decision-grade evidence” has to be reframed at the portfolio scale — the decision-maker is not the program manager but the budget authority — and is one of the corpus’s strongest grounds for the CLO **Judgment under inadequate evidence**. The portfolio decision-maker does not get to wait for the evidence architecture to be fixed before deciding; the case is about how to make the decision with the evidence acknowledged as insufficient.

REFERENCES

1. Government Accountability Office (2022), “Weapon System Sustainment: Selected Aircraft and Combat Vehicles Sustainment Costs,” GAO-22-104533.
2. Government Accountability Office (recurring), annual “Weapon Systems Annual Assessment” portfolio reports — context for the structural finding.

3. Department of Defense (2022), DoD response to GAO-22-104533 — acceptance and implementation status of recommendations.
4. Defense Acquisition University (ongoing), Operating and Support Cost-Estimating Guide — DoD reference on the cost categories whose comparability GAO addresses.

THE LEARNING ENGINEERING LENS

> *The portfolio decision-maker does not get to wait for the evidence architecture to be fixed. The case is about how to act with the evidence acknowledged as insufficient.*
>
> EDITORS' SYNTHESIS OF GAO-22-104533 AND THE RECURRING SUSTAINMENT PORTFOLIO FINDINGS.

LE INSIGHT

GAO-22-104533 is the frontier evidence-architecture case at portfolio scale: the O&S data that dominate weapon-system lifecycle decisions are not reported in a form the decisions require, and the gap is structural across the enterprise. The case is the worked example of the CLO **Judgment under inadequate evidence** at the budget- authority scale.

LENS APPROACH

GAO weapon-system sustainment is the portfolio-evidence- architecture frontier case (induced 1.4; LENS D1/PT4) — used in Domain 4 for the CLO **Judgment under inadequate evidence** at portfolio scale. Pair with Case 21 and Case 19.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the evidence architecture for portfolio-level decisions as a deliverable in its own right. The GAO finding is that the deliverable does not yet exist at the scale the decisions require, and that gap is the case.
- Specify the cost categories, time-series cadence, and cross-service comparability requirements before the data start flowing, not after — the cost of converging definitions retroactively is high enough to be the recurring GAO finding.
- Design the reporting architecture for the question the decision-maker has to answer (modify, sustain, replace, retire) rather than for the workflow the program offices already run; the question shapes the architecture, not the other way around.

IN OPERATION / AFTER

- Carry the "judgment under inadequate evidence" framing into portfolio-decision documentation. The decision-maker does not get to wait for the evidence architecture to be fixed; the case asks what minimum evidence quality is required to act and what hedges should be documented.
- Treat the GAO recommendation set as the structural-fix roadmap, with progress measured by the narrowing of the comparability gap in specific cost categories rather than by claims of overall improvement.
- Pair the sustainment-evidence frontier with the design-side standards mechanism (Case 21) so the program understands both halves of the requirements architecture: design criteria for the equipment, evidence architecture for the portfolio.

REFLECTION QUESTIONS

1. Identify a portfolio-level decision in your domain that rides on data the auditor would describe as insufficient. What is the minimum evidence-architecture improvement that would convert the data from sub-decision-grade to decision-grade for the specific decision class?

2. Specify the cost or outcome categories whose comparability across units, services, or time would most change the decision quality. The case's lesson is that comparability has to be specified before the data flow, not after.

3. The decision-maker has to act on the evidence available. Design the documentation discipline that would record what hedges the decision was made under, so future audits can distinguish "decision was wrong" from "evidence was inadequate at the time."

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CASE 18 DEFENSE 1954 – PRESENT

U.S. Nuclear Navy / Rickover Training Model

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Zero reactor accidents in 60+ years of U.S. Naval nuclear operations; the most demanding nuclear operator training program in the world

IN BRIEF

Admiral Hyman Rickover built a training and qualification culture for the Naval Nuclear Propulsion Program that has produced zero reactor accidents across more than 60 years and thousands of reactor-years of operation, often in extreme conditions. Every nuclear-trained sailor must pass rigorous qualification to zero-defect standards and demonstrate competence in oral examination by senior nuclear-qualified officers; the culture demands personal accountability, deep technical mastery, and a questioning attitude that obliges operators to challenge assumptions, including superiors'. The sharpest contrast in this book is internal: the same Navy that ran the nuclear program to this standard let surface-warfare training decay to CD-ROMs and paid for it at Fitzgerald and McCain (Case 1). Same institution, two philosophies — and the choice shows up in the casualty columns.

BACKGROUND

In the early 1950s the U.S. Navy set out to put nuclear reactors aboard ships and submarines — machines whose failure could be catastrophic and irreversible, operated by young sailors far from any help. The capability problem was absolute: there was no acceptable accident rate, so the human operating system had to be engineered to an extraordinary standard from inception. Because a reactor at sea could not be evacuated or handed to outside experts, the operators themselves had to be the entire margin of safety — a constraint that forced training to a level no ordinary program would justify.[0]

THE INTERVENTION

Admiral Hyman Rickover established a training and qualification regime requiring every nuclear-trained officer and enlisted sailor to pass demanding programs held to zero-defect standards. Operators must demonstrate competence through oral examination by senior nuclear-qualified officers, and the program embeds continuous re-qualification rather than one-time certification. The oral board tested understanding rather than recall, and the continuous re-qualification meant competence had to be sustained rather than banked once — so the standard governed the operator's whole career, not just entry to it.[1]

HOW IT WORKED

The culture pairs technical mastery with a deliberately engineered attitude: personal accountability and a mandatory questioning posture in which operators are trained to challenge assumptions, including those of superiors. Rickover's premise — that people, not organizations or management systems, get things done — made the qualification ladder, not paperwork, the load-bearing element of safety. The questioning posture is the cultural half of the pair: deep technical mastery alone could still defer to a mistaken superior, so the obligation to challenge assumptions is what keeps competence from being silenced by rank.[2]

THE EVIDENCE

The result is the longest-running continuous capability-engineering record in any high-consequence domain: zero reactor accidents across more than six decades and thousands of reactor-years. The cost — the qualification ladder, the zero-defect oral boards, the continuous re-qualification — is the visible budget-line price of that record. The duration is the key evidence: a record sustained across six decades and many generations of sailors shows the safety came from the engineered system rather than from any one cohort's talent or luck.[3]

WHAT TRANSFERRED

The cleanest test of the model is internal. The same Navy that engineered the nuclear program to this standard let surface-warfare training decay to CD-ROM self-study and paid the price at Fitzgerald and McCain (Cases 1 and 37). Same institution, same era, opposite philosophies, radically different outcomes — the strongest available demonstration of capability treated as a system parameter versus deferred as a cost. The internal comparison controls for nearly everything that usually confounds such claims — one service, one manpower system, one budget process — leaving the training philosophy itself as the variable that diverged.[4]

NUCLEAR	SURFACE (PRE-2017)
0	17
reactor accidents	sailors killed
60+ years	two collisions

SAME NAVY, OPPOSITE PHILOSOPHIES — RADICALLY DIFFERENT OUTCOMES

REFERENCES

1. Polmar, N. & Allen, T. (2007), Rickover: Father of the Nuclear Navy — the program and Rickover's philosophy (paraphrased).
2. Naval Nuclear Propulsion Program documentation (NRC/DOE) — qualification standards and the accident record.
3. Admiral Hyman G. Rickover, "Doing a Job" (Columbia University commencement address, 1982) — "people, not organizations… get things done" (quoted).
4. GAO-21-168, comparison of nuclear and surface Navy training — the internal contrast.
5. Duncan, F. (1990), Rickover and the Nuclear Navy — the qualification culture.

> *Human experience shows that people, not organizations or management systems, get things done.*
>
> ADMIRAL HYMAN G. RICKOVER, "DOING A JOB," COLUMBIA UNIVERSITY, 1982

LE INSIGHT

The Nuclear Navy is the longest-running continuous capability- engineering program in any high-consequence domain. The choice to treat training as a system parameter rather than as a cost center has produced sixty-plus years of zero reactor accidents. The contrast with the Surface Navy is the cleanest available test of what happens when capability is engineered versus when it is deferred — and the price of that engineering (the qualification ladder, the zero-defect oral boards, the continuous re-qualification) is visible on the budget line.

LENS APPROACH

LENS treats the Rickover model in LEN 8 as the foundational organizational-learning case and in LEN 5 as a worked example of capability requirements traceable from operational analysis through qualification standards. The internal Navy comparison anchors the program's core argument about capability as a system parameter.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Engineer the human operating system to the standard the consequences demand from inception — where there is no acceptable failure rate, make the operators themselves the margin of safety.
- Gate qualification on demonstrated understanding through oral examination by senior qualified people, testing comprehension rather than recall.
- Pair technical mastery with a mandatory questioning posture so competence is obliged to challenge assumptions, including superiors', rather than be silenced by rank.

IN OPERATION / AFTER

- Embed continuous re-qualification rather than one-time certification, so competence must be sustained across a career and cannot be banked once and assumed.
- Protect the training as a durable, visible budget line, since the qualification ladder and oral boards are the price of the safety record and the first thing tempo will erode.
- Sustain the standard across generations of operators, treating a multi-decade record as the evidence the safety comes from the engineered system rather than any one cohort.

REFLECTION QUESTIONS

1. Identify an institution that operates two divisions under different capability philosophies. What does the comparison reveal that neither division could see alone?
2. Rickover's standard was zero-defect oral examination. What is the equivalent in your domain — and would you survive it?
3. The Nuclear Navy's safety record is paid for by a durable, visible training budget. What is the equivalent line-item investment in your domain that a comparable safety claim would require — and is it being made?

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CASE 19 NAVAL ENGINEERING DEFENSE SAFETY CERT 1963 – PRESENT

Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable

G Governance & Trust K Knowledge & Institutional Memory H Human-System Interface

IMPACT From 1915 to 1963 the US Navy lost 16 submarines to non-combat causes; since 1963 it has lost one — USS Scorpion, the only post-1963 loss that was not SUBSAFE-certified. The Columbia Accident Investigation Board cited SUBSAFE as a model NASA should emulate

IN BRIEF

USS Thresher was lost with 129 aboard on April 10, 1963. Within fifty-four days the US Navy created SUBSAFE — a program that certifies design, material, fabrication, and testing for every component inside the submarine's watertight-integrity boundary and its safe-recovery systems. The requirements were issued by December 20 of that same year. The program demands what it calls "Objective Quality Evidence" for every step — verifiable fact, not probabilistic assessment — and pairs that with annual training and recurring audits across the entire fleet lifecycle. The documented result is a step-change in non-combat submarine loss rates: 16 losses across the 48 years before SUBSAFE; one loss (USS Scorpion, not SUBSAFE-certified) across the 62 years since. The Columbia Accident Investigation Board cited SUBSAFE in 2003 as the model NASA should emulate. The honest hedge survives: the zero-loss record is correlational across decades with many co-varying factors — submarine design, reactor maturity, operating procedures, intelligence environment — and SUBSAFE's own program literature notes that the requirements can look "excessive." The case is the archetype of treating capability requirements as a recurring, auditable, non-waiverable deliverable across the entire system lifecycle, with the hedges that decades-long capability-engineering claims have to carry.

BACKGROUND

USS Thresher, the lead boat of a new class of US nuclear attack submarines, was lost on April 10, 1963 during post-overhaul sea trials off Cape Cod. All 129 aboard died. The investigation identified a likely sequence — silver-brazed piping joint failure, flooding, electrical fault that scrammed the reactor, inability to blow ballast tanks fast enough to recover — that pointed not to a single defective component but to a gap in the whole way the fleet certified the watertight-integrity boundary and the systems for recovering from a flooding casualty.[0]

THE INTERVENTION

The Navy's response was institutional speed of an unusual order. SUBSAFE was created within fifty-four days of the loss; formal requirements were issued by December 20, 1963. The program scopes itself to two things: the watertight-integrity boundary (every component that holds back seawater at depth), and the safe-recovery systems (the ballast-blow and propulsion systems that get a boat to the surface if flooding starts). Inside that scope it demands certification of design, material, fabrication, testing, and configuration control — with what the program calls "Objective Quality Evidence" attached at every step. That phrase is the load-bearing cultural artifact of the program: verifiable fact, not probabilistic assessment, is what the certification rests on.[1]

HOW IT WORKED

What makes SUBSAFE a sustainment intervention rather than a design intervention is the lifecycle discipline. Annual training across the fleet, recurring audits at shipyards and tenders, change-control on every modification, and a culture that treats requirements as non-waiverable artifacts the program-management chain cannot trade away under schedule pressure. The certification is not done at launch; it is the condition of being allowed to dive. Each overhaul re-engages the certification process. The cost is real — the program's own histories concede the requirements can look "excessive" — and the program treats that cost as the price of the capability the certification produces.[2]

THE EVIDENCE

The documented outcome is one of the cleanest before/after records in the safety-engineering literature. From 1915 to 1963 the Navy lost 16 submarines to non-combat causes (collision, flooding, equipment failure, fire). Since 1963 it has lost one — USS Scorpion in 1968, the only post-1963 loss that was not SUBSAFE-certified. In 2003 the Columbia Accident Investigation Board, examining the loss of the Space Shuttle Columbia, cited SUBSAFE in its final report as the capability-certification model NASA should adopt for human spaceflight. The endorsement is from an investigation body with no Navy institutional stake, examining a different catastrophic-system domain.[3]

WHAT TRANSFERRED

The hedge has to survive into the case. The zero-loss record since 1963 is correlational across more than six decades and many co-varying factors: submarine design generations, reactor maturity, training systems, operating procedures, intelligence environment, the absence of certain operational stressors that the Cold War sometimes produced. Attributing the entire outcome to SUBSAFE alone overstates what the evidence can support. What the evidence does support is that the **program** — the requirements discipline, the Objective Quality Evidence standard, the lifecycle audit cycle, the non-waiverable culture — has been a defining feature of the capability since 1963, and has survived endorsement by an independent investigation in a different domain. The case teaches the requirements-as-sustainment-deliverable form at its strongest, with the honest hedge that decades-long capability claims have to carry.[4]

REFERENCES

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3. NASA SMA (2006), "USS Thresher Lessons Learned" briefing — safety-message archive.
4. Columbia Accident Investigation Board (2003), final report — Volume I, endorsement of SUBSAFE as a model for NASA.
5. US Navy NAVSEA, SUBSAFE program documentation — operating program publications.

THE LEARNING ENGINEERING LENS

> *Objective Quality Evidence — verifiable fact, not probabilistic assessment — is the cultural artifact the program is built on.*
>
> EDITORS' SYNTHESIS OF THE SUBSAFE PROGRAM LITERATURE.

LE INSIGHT

SUBSAFE is the archetype of treating capability requirements as a recurring, auditable, non-waiverable deliverable across the entire system lifecycle. The before/after non-combat-loss record is one of the cleanest in safety engineering — and correlational across many co-varying factors over six decades. The hedge is part of the case.

LENS APPROACH

SUBSAFE is the canonical sustainment-engineering case (induced 1.4; LENS D1/PT3). LENS uses it in Domain 1 (Systems Analysis) for the requirements-as-deliverable discipline; in Domain 4 (Test and Evaluation) for the Objective-Quality-Evidence standard and the recurring-audit cycle; and in Domain 5 (Navigating Sociotechnical Constraints) for the non-waiverable culture that resists schedule pressure. Adjacent to the nurse-ratios case (Case 20) at the requirements-becomes- engineered layer, and to the WHO Surgical Checklist (Case 109) at the mandatory-mechanism layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Scope the certification boundary tightly — for SUBSAFE, the watertight-integrity boundary and the safe-recovery systems — so the discipline is enforceable, not aspirational across an undifferentiated whole.
- Make “Objective Quality Evidence” the cultural standard: verifiable fact at every certification step, not probabilistic assessment, and not signature-without-evidence.
- Treat the requirements as non-waiverable artifacts the program-management chain cannot trade away under schedule pressure; the program’s resistance to waivers is the program.

IN OPERATION / AFTER

- Operate the lifecycle discipline as the program: annual training, recurring audits, change-control on every modification, certification re-engaged at each overhaul.
- Carry the correlational hedge in any communication of the outcome record; a decades-long zero-loss record across co-varying factors is the strongest available evidence the program works, not closed proof.
- Treat external endorsement (Columbia Accident Investigation Board) as a teaching artifact about the **form** of the program, transferable to other catastrophic-system domains under their own scope discipline.

REFLECTION QUESTIONS

1. Identify a capability in your domain where the certification is done at launch and not re-engaged across the lifecycle. What is the analog of SUBSAFE’s annual training and recurring audit — and what is the resistance to it?

2. Specify what “Objective Quality Evidence” would mean in your context: verifiable fact at every step rather than signature-without-evidence. Which steps in your current process would not survive that standard?

3. SUBSAFE’s outcome record is correlational across many co-varying factors. What is the minimum additional evidence you would require before treating a similar long-run record in your domain as evidence the program is what produced the outcome?

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CASE 20 HEALTHCARE WORKFORCE POLICY PATIENT SAFETY 2004 – 2010

California Nurse Staffing Ratios — Legislating a Capability Requirement

G Governance & Trust K Knowledge & Institutional Memory

IMPACT California's mandated minimum nurse-to-patient ratios reduced nurse workload by 1–2 patients per nurse and were modeled to imply 10–14% fewer surgical patient deaths in comparison states if matched — observational, cross-sectional, no California baseline

IN BRIEF

California in 2004 became the first US state to mandate minimum nurse-to-patient ratios in acute-care hospitals — unit-level minimums written into law and enforced through inspection. The Aiken et al. (Health Services Research, 2010) study surveyed 22,336 nurses across California, Pennsylvania, and New Jersey (the latter two with no mandate), and found California nurses cared for 1–2 fewer patients each. Modeling implied that if the two comparison states had matched California's medical-surgical ratios, New Jersey would have seen 13.9% and Pennsylvania 10.6% fewer surgical patient deaths. The study is observational and cross-sectional, and the authors are explicit that the **absence of California baseline measures** fueled a contested stakeholder debate over whether the ratios themselves caused the improvement. The case is the canonical recent instance of a capability requirement (adequate staffing) being converted from a stated aspiration into an engineered, enforced specification by law — and the canonical instance of the methodological hedge such a conversion carries.

BACKGROUND

Nurse staffing is the load-bearing variable in the capability-of-care literature: a substantial body of evidence links nurse workload to medical errors, mortality, and rescue from complications. The professional consensus on the direction of the effect has been durable since the late 1990s. What had been contested is whether a mandated minimum, written into law and enforced through inspection, would convert the stated requirement (adequate staffing) into an engineered one (a specific ratio actually present at the bedside).[0]

THE INTERVENTION

California in 2004 became the first US state to do it. The law specifies unit-by-unit minimum nurse-to-patient ratios — 1:5 in medical-surgical, 1:4 in step-down, 1:2 in ICU, and so on — enforceable through state inspection. The political process took five years from statutory enactment (1999) to regulatory implementation, with industry argument that the ratios would close hospitals and worsen access. The 2004 implementation went ahead substantially as written.[1]

HOW IT WORKED

Aiken et al. (Health Services Research, 2010) is the strongest published evaluation. The study surveyed 22,336 nurses across California (mandated), Pennsylvania, and New Jersey (no mandate) on workload, work environment, and patient-outcome measures. The headline finding is that California nurses cared for 1–2 fewer patients each across comparable unit types. The modeled mortality implication — computed by applying the workload-mortality relationship from the broader Aiken et al. corpus to the workload difference — was that if Pennsylvania and New Jersey had matched California's medical-surgical ratios, surgical mortality would have been 10.6% and 13.9% lower respectively.[2]

THE EVIDENCE

The hedge survives into the case verbatim. The Aiken et al. study is observational and cross-sectional, not a controlled experiment. There was no California baseline measurement to provide a pre/post within-state comparison. The modeled mortality estimates rest on the workload-mortality relationship from prior studies, and a contested stakeholder debate followed publication over whether the ratios themselves caused the observed improvement, or whether California hospitals differed in other ways the cross-section did not capture. The authors do not claim closure; they say the evidence is the strongest available and consistent with the broader literature.[3]

WHAT TRANSFERRED

What the case teaches is the form of converting a stated capability requirement into an engineered, enforced one by law, and the methodological hedge that conversion has to carry. The political path is part of the deliverable: five years from enactment to implementation, sustained industry opposition, contested evidence, and a study design constrained by the absence of a pre-implementation California baseline. In pair with SUBSAFE (Case 19), the case shows that engineered capability requirements at scale require both the requirements-discipline mechanism and an evidence architecture that survives the political process around the requirement.

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THE LEARNING ENGINEERING LENS

> *The headline finding is direct; the modeled mortality estimate is the strongest available evidence and not closed proof.*

EDITORS' SYNTHESIS OF AIKEN ET AL. (2010).

LE INSIGHT

California's nurse-ratio mandate is the canonical recent instance of converting a stated capability requirement (adequate staffing) into an engineered, enforced one by law. The workload effect is direct; the modeled mortality estimate is observational and cross-sectional with no California baseline. The hedge is the case.

LENS APPROACH

Nurse ratios is the legislated-requirement case (induced 1.1; LENS D1/PT3). LENS uses it in Domain 1 (Systems Analysis) for the conversion of stated requirement to engineered specification by law, and in Domain 4 (Test and Evaluation) for the CLO **Judgment under inadequate evidence** — the evaluation evidence is observational cross-sectional, the strongest available and not closed proof. Pair with Case 19 SUBSAFE at the requirements-as-deliverable layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- When converting a stated capability requirement into an engineered specification by law, design the evidence architecture **before** implementation: pre-registered measurement, a baseline window, and a comparison strategy that survives the political process.
- Specify the unit-level requirement at the granularity the work actually has — for nurse staffing, by unit type and acuity — rather than at an aggregate level that lets the requirement be averaged away.
- Anticipate sustained industry argument as part of the implementation cost; a five-year path from enactment to implementation is the political price of the engineered requirement.

IN OPERATION / AFTER

- Report the workload effect (1–2 fewer patients per nurse) separately from the modeled mortality effect (10.6% and 13.9%); the workload finding is direct, the mortality estimate is modeled.
- Preserve the no-baseline hedge in any communication of the evidence; the strongest available evidence is not closed proof of causation.
- Treat the contested stakeholder debate as program documentation, not noise; the debate is part of why the case is teachable.

REFLECTION QUESTIONS

1. Identify a capability requirement in your domain that is currently stated but not engineered. What would a legislated minimum look like at unit-level granularity, and what evidence architecture would you put in place **before** implementation to make the conversion auditable?
2. Specify the political-process cost you would budget for: California's path was five years from enactment to implementation under sustained industry argument. What is the analog in your context, and is the engineered requirement worth that cost?
3. The Aiken evidence is observational cross-sectional with no California baseline. What pre/post or controlled comparison would you build into the next state's implementation to give the next round of evaluation a stronger foundation?

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CASE 21 DEFENSE HUMAN FACTORS STANDARDS 2020 REVISION (SERIES SINCE 1968)

MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

G Governance & Trust K Knowledge & Institutional Memory

IMPACT DoD Design Criteria Standard MIL-STD-1472H, the 2020 revision of a series dating to 1968, converts human-factors and human-engineering findings into binding design criteria across DoD acquisition — controls, displays, anthropometry, workspace, environment, hazards — invoked by acquisition programs as a mandatory or tailored design specification

IN BRIEF

MIL-STD-1472 is the US Department of Defense's design-criteria standard for human engineering — the discipline of making equipment usable, safe, and effective for the human operator. The series originated in 1968 and has been revised through versions A, B, C, D, E, F, G, and most recently H, released September 15, 2020. The standard codifies established human- factors and ergonomic findings into specific binding design criteria: control and display design, anthropometric accommodation ranges, workspace dimensions and access, labeling and signage, environmental limits (noise, vibration, illumination, thermal), and hazard mitigation. Acquisition programs invoke MIL-STD-1472H either as a mandatory standard or with tailored deviation, making human-factors evidence binding rather than advisory. The 2020 revision integrated accumulated findings since 2012 (the prior G revision), including updates to anthropometric data, environmental criteria, and human-system-integration practices. The case is the structural archetype of converting a body of human- factors research into engineered design requirements; it works at the requirement-discipline layer rather than the per-platform layer, and is the standard that programs referenced when they specify human-engineering deliverables.

BACKGROUND

Human factors engineering as a defense discipline emerged from World War II — the documented mismatch between cockpit control layouts and pilot perception (Fitts & Jones's "designed errors" work) showed that platforms could be made un-flyable by avoidable design choices. Through the 1950s and 60s the discipline accumulated a body of findings on control / display design, anthropometry, environmental tolerance, and workspace layout that lived across academic publications, military handbooks, and service-specific guidance. What was missing was a single binding design-criteria standard that an acquisition program could specify in a contract.[0]

THE INTERVENTION

The DoD response was MIL-STD-1472, first issued in 1968, and revised through letter suffixes A (1969), B (1974), C (1981), D (1989), E (1996), F (1999), G (2012), and H (September

2020). Each revision incorporates accumulated findings since the prior version, updated anthropometric data (the surveyed populations evolve), and refinements to specific criteria informed by service experience. The standard is not a list of recommendations; it is a Design Criteria Standard, structured so that an acquisition program can invoke it as a mandatory specification, or tailor specific paragraphs with documented deviation.[1]

HOW IT WORKED

The structural mechanism is the conversion of human- factors findings into engineered requirements. Where the research literature might find that controls of a given size and force are operable across a defined population percentile, MIL-STD-1472H carries that finding as a binding design criterion that the program's controls must meet. Where the literature finds that illumination below a certain level degrades reading performance for given character sizes, the standard carries that as an environmental limit. Anthropometric accommodation — the body-size ranges the equipment must fit — is specified to defined population percentiles. Hazard analysis is required as part of the design process, with mitigation criteria for identified hazards.[2]

THE EVIDENCE

The case's value to LENS is the requirement-discipline form. MIL-STD-1472H is the human-engineering analog of what SUBSAFE (Case 19) does for submarine watertight-integrity: a binding, recurring, auditable requirement set that the program-management chain cannot trade away without a documented deviation. The acquisition contract invokes the standard, the acquirer's human-factors discipline verifies compliance against it, and the engineered design carries the research findings as binding criteria rather than as advice the program might or might not adopt. The 2020 H revision is notable for the cadence — the eight-year gap from G to H — and for the integration of human- systems-integration practices reflecting the rise of complex software-intensive systems.[3]

WHAT TRANSFERRED

The honest framing the case requires is that the standard is necessary but not sufficient. MIL-STD-1472H is the requirements-discipline mechanism; whether the requirements are met in any specific program depends on the program's human-factors engineering capacity, the acquirer's verification rigor, and the tailoring decisions made up front. The standard does not by itself guarantee usable equipment; it makes usability a contractable, auditable deliverable. Cases of operator-platform mismatch in fielded DoD systems (e.g., the F-22 OBOGS instrumentation gap in Case 88) are not failures of the standard's existence but of the tailoring and verification around it. The case teaches the requirements-as-deliverable form at the human-engineering scale, with the qualification that the standard is the mechanism, not the outcome.

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THE LEARNING ENGINEERING LENS

> *The criterion that survives into a binding standard is the criterion the program is held to.*
>
> EDITORS' SYNTHESIS OF MIL-STD-1472 REVISION HISTORY.

LE INSIGHT

MIL-STD-1472H is the structural archetype of converting a body of human-factors research into binding engineered requirements an acquisition contract can specify. The standard is the mechanism by which usability becomes a contractable, auditable deliverable rather than design advice. It is necessary but not sufficient; tailoring and verification determine whether the contract is honored.

LENS APPROACH

MIL-STD-1472H is the binding-standard requirements case (induced 1.1; LENS D1/PT3). LENS uses it in Domain 1 (Systems Analysis) for the conversion of research findings into engineered requirements, and in Domain 5 (Navigating Sociotechnical Constraints) for the tailoring and verification disciplines around the standard. Direct pair with Case 19 (SUBSAFE) at the requirements-as-deliverable layer and with Case 88 (F-22 OBOGS) at the standard-versus-tailoring layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the human-factors literature in your domain as a source of engineered requirements, not as design advice. The criterion that survives into a binding standard is the criterion the program is held to.
- Specify anthropometric and environmental criteria to defined population percentiles, not to “typical users.” The percentile framing is what makes accommodation auditable.
- Design the tailoring discipline around the standard so that any deviation from a binding criterion is documented with rationale, and the program retains a record of which criteria it chose not to meet and why.

IN OPERATION / AFTER

- Maintain the revision cadence the field requires — anthropometric data ages, environmental tolerances drift, human-system-integration practices evolve — so the standard does not become an obsolete reference.
- Verify compliance as a discipline distinct from the standard itself; the standard is the requirement, the verification process is the assurance that the requirement is met.
- Carry the necessary-but-not-sufficient framing: a binding human-engineering standard is the mechanism by which research findings become contractable; whether the contract is honored is the program’s responsibility, not the standard’s.

REFLECTION QUESTIONS

1. Identify a body of research findings in your domain that programs treat as advisory. What would it take to convert those findings into a binding design-criteria standard an acquirer could specify in a contract?
2. Specify the tailoring discipline you would put around a binding standard: which paragraphs may be tailored, with what documented rationale, and what is the cadence at which the standard itself is revised against accumulated experience?
3. The standard is necessary but not sufficient. What verification capacity does your program need to know that the binding criteria are actually met in the delivered system, and where is the gap between standard and verification visible today?

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CASE 22

AVIATION SAFETY | STRUCTURAL ENGINEERING | REGULATORY RULEMAKING

1988 – 2010S

FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule

G Governance & Trust D Designed Out K Knowledge & Institutional Memory

IMPACT

After Aloha Airlines 243 (1988) exposed widespread fatigue cracking in an aging Boeing 737-200, the FAA's Aging Aircraft Program and the Airworthiness Assurance Working Group produced two decades of structural inspection programs culminating in 14 CFR Part 26 (2010), which requires manufacturers to establish a Limit of Validity for each model and embed widespread fatigue damage protection into the maintenance program

IN BRIEF

On April 28, 1988, Aloha Airlines Flight 243 lost an 18-foot section of upper fuselage in flight; the aircraft, a Boeing 737-200, had accumulated 89,680 flight cycles and was operating well past the design assumptions of the original certification. The accident — one flight attendant killed, 65 injured — exposed the structural-engineering category of widespread fatigue damage (WFD): multiple small cracks across many similar structural details, simultaneous enough that established single-crack inspection assumptions did not apply. The FAA stood up the Aging Aircraft Program almost immediately, and the Airworthiness Assurance Working Group (AAWG) operated through the 1990s and 2000s producing structural inspection programs across the transport-airplane fleet. The work culminated in 14 CFR Part 26 (Continued Airworthiness and Safety Improvements; effective 2010), which requires manufacturers to establish a Limit of Validity (LOV) for each type — the operational service goal below which the maintenance program protects against WFD — and to embed WFD prevention into the structural maintenance program itself. Part 26 is one of the most concrete recent examples of legacy assets aging past their original oversight regime being pulled back under structured airworthiness governance, and closes a long-standing zero-state in the induced framework's C7 competency.

BACKGROUND

Aloha Airlines Flight 243 on April 28, 1988, was the sentinel event. The Boeing 737-200 had accumulated 89,680 flight cycles — well above the design service goal — and was operating in a high-cycle short-haul environment that stressed the fuselage skin and lap joints at a rate the original certification analysis had not anticipated. In cruise at 24,000 feet, an 18-foot section of upper fuselage separated. The crew recovered the aircraft and landed it; one flight attendant was lost overboard and 65 occupants were injured. The NTSB investigation identified multiple small fatigue cracks across many lap-joint rivet holes, linking up catastrophically — the textbook presentation of widespread fatigue damage rather than a single-crack failure.[0]

THE INTERVENTION

The structural-engineering category WFD names is specifically the regime where established inspection assumptions break down. A single-crack model assumes one crack initiates, propagates, and is caught by scheduled inspection before reaching critical length. WFD assumes many small cracks initiate at similar structural details across the fleet at similar times, and that link-up between adjacent cracks becomes the dominant failure mode. The inspection cadence, detection threshold, and replacement program a designer would build under the single-crack assumption do not adequately cover the WFD regime, and the airworthiness oversight regime Aloha exposed had not made the distinction.[1]

HOW IT WORKED

The FAA's response was to stand up the Aging Aircraft Program and convene the Airworthiness Assurance Working Group, which operated through the 1990s and 2000s. The AAWG produced mandatory structural inspection documents and supplemental structural inspection programs for transport-category aircraft model-by-model, working through the older fleet systematically. The work was technically substantial — fleet- survey data, fatigue-test campaigns, inspection-program revisions for in-service aircraft — and institutionally sustained across more than two decades. The intermediate deliverables were per-model inspection programs; the capstone was rulemaking.[2]

THE EVIDENCE

14 CFR Part 26 — Continued Airworthiness and Safety Improvements — was finalized in 2007 with the WFD provisions effective in 2010. Two structural elements are load-bearing. First, every transport-category type must have a Limit of Validity established by the manufacturer: the operational service goal, in flight cycles or flight hours, below which the maintenance program is shown to protect against WFD. Operation beyond the LOV requires either the manufacturer establishing an extended LOV with supporting analysis, or operator-specific evidence approved by the regulator. Second, WFD prevention is embedded in the structural maintenance program itself, not handled as a separate one-time inspection event. The maintenance program becomes the carrier of WFD protection across the type's service life.[3]

WHAT TRANSFERRED

What the case teaches at the LENS framing layer is the structural form of pulling legacy assets back under structured airworthiness governance. The original 737-200 certification did not anticipate the cycle count and the WFD regime that high-cycle short-haul operation produced; the regulatory response was not to retire the type but to engineer the oversight regime forward — fleet survey, AAWG, model-by- model inspection programs, and finally Part 26 codifying the LOV concept across the transport-category fleet. The case is one of the v2 sweep's clearest closes of the C7 (Capability under system change, transition, and aging assumptions) zero-state in the induced framework: a sustained two-decade regulatory program that turned an aging-fleet structural surprise into a governed sustainment discipline.[4]

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5. Swift, "Widespread Fatigue Damage Monitoring — Issues and Concerns," International Conference on Aging Aircraft — technical synthesis of the WFD inspection regime.

THE LEARNING ENGINEERING LENS

> *The deliverable was not a single rule. It was the two-decade per-model technical record that made the rule a codification of established practice rather than top-down regulation.*
>
> EDITORS' SYNTHESIS OF THE FAA AGING AIRCRAFT PROGRAM AND THE PART 26 RULEMAKING RECORD.

LE INSIGHT

The FAA Aging Aircraft Program and Part 26 are one of the v2 sweep's clearest closes of the C7 zero-state. Aloha 243 exposed a regime the original certification did not cover; the AAWG operated for two decades; Part 26 codified Limit of Validity and embedded WFD prevention into the maintenance program itself. The sustained two-decade rulemaking is the deliverable, not the rule alone.

LENS APPROACH

FAA aging aircraft is the canonical aging-asset-governance case (induced 7.3; LENS D1/PT3) — Domain 1 for LOV-as- operational-service-goal; Domain 5 for the AAWG institutional discipline. Closes the C7 zero-state. Pair with Cases 23, 24, 76 as the v2 aging-system quartet.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Name the structural regime the original certification analysis did not cover — WFD vs. single-crack here — explicitly, so the inspection program can be designed around the actual failure mode rather than retrofitted to the original assumption.
- Operate the working group across a long horizon (AAWG: two decades): the per-model deliverables build the technical record that supports the eventual rulemaking, and short-cycle deliverables alone do not.
- Make the operational service goal (LOV) the artifact the regulation rests on; the goal is auditable per type, and operation beyond it requires additional analysis rather than implicit extension.

IN OPERATION / AFTER

- Embed the new protection in the maintenance program rather than as a one-time inspection: the maintenance program is the carrier across the service life, and one-time events are not.
- Treat the model-by-model technical work as the legitimacy basis for the rule; without the AAWG's fleet-survey and inspection-program record, Part 26 would have looked like top-down regulation rather than codification of established practice.
- Carry the case as the C7 instance the induced framework needed: legacy assets aging past their original oversight regime, pulled back under structured airworthiness governance by sustained two-decade rulemaking.

REFLECTION QUESTIONS

1. Identify a legacy asset class in your domain whose original certification or design analysis no longer covers the regime the assets are now operating in. What is the analog of the WFD distinction — the failure mode the original analysis did not anticipate?

2. Specify the analog of the LOV: an operational service goal, auditable per asset, beyond which additional analysis is required. What instrument would you use, what cadence, and what would "operation beyond LOV" trigger in your domain?

3. The AAWG operated for two decades before Part 26 codified the practice. What is the institutional discipline you would build for a similar two-decade horizon — the working-group cadence, the per-asset technical record, the eventual rulemaking — and is there a body in your domain that could plausibly carry that horizon?

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CASE 23 AVIATION INFRASTRUCTURE AIR TRAFFIC MANAGEMENT REGULATORY TRANSITION 2003 – 2020

FAA NextGen and the ADS-B Out Transition

G Governance & Trust D Designed Out K Knowledge & Institutional Memory

IMPACT The FAA's Next Generation Air Transportation System (NextGen) shifted US air-traffic management from radar-based surveillance to a satellite-based architecture; the ADS-B Out mandate effective January 1, 2020 required equipage across the controlled-airspace fleet, achieving substantial compliance — with documented schedule slippage and benefit-realization gaps preserved as load-bearing hedges

IN BRIEF

The FAA's Next Generation Air Transportation System (NextGen) is the multi-decade transition of US air-traffic management from a radar-based surveillance architecture to a satellite- based architecture built on Automatic Dependent Surveillance – Broadcast (ADS-B). The ADS-B Out final rule, published in 2010, set the equipage mandate for January 1, 2020 — aircraft operating in most controlled airspace must broadcast their GPS-derived position and identity, replacing the radar-only secondary-surveillance model that defined the era prior. The mandate was substantially met at the deadline; equipage compliance across the affected fleet was high, and ADS-B-based surveillance is now the operational backbone in much of US airspace. The case is one of the largest planned aging- infrastructure transitions in the recent regulatory record — closing a long-standing C7 zero-state in the induced framework — and it carries the load-bearing hedges that GAO and DOT Inspector General reviews have repeatedly documented: significant schedule slippage across the program, benefit- realization gaps relative to original projections, and contested cost-benefit accounting. The transition happened; the transition was harder, slower, and more partial than the original NextGen plan implied.

BACKGROUND

The pre-NextGen US air-traffic management architecture rested on ground-based primary and secondary radar surveillance, with voice communication, paper or quasi-paper flight progress strips at many facilities, and traffic-flow management built around scheduled equipage upgrades that lagged the broader civil-aviation hardware base. By the early 2000s the structural problem was named clearly in the policy literature: a 20th-century surveillance architecture was being asked to handle 21st-century traffic, and incremental modernization inside the radar-paradigm was approaching its limits.[0]

THE INTERVENTION

NextGen, formally launched in 2003 under the Vision 100 – Century of Aviation Reauthorization Act, was the FAA's multi-decade response. The technical core was the transition from radar-based surveillance to ADS-B: aircraft broadcast their GPS-derived position once per second, ground stations and other aircraft receive the broadcast, and the resulting surveillance picture is more precise, lower-latency, and less expensive to operate at scale than secondary radar. The broader NextGen program included additional elements (Data Communications, NAS Voice System, System-Wide Information Management) but ADS-B Out was the load-bearing equipage-mandate piece.[1]

HOW IT WORKED

The ADS-B Out final rule was published in 2010 with the January 1, 2020 compliance deadline — a decade of lead time for operators to equip. At the deadline, substantial compliance was reported across the affected fleet; the FAA's surveillance picture in controlled airspace moved substantially onto the ADS-B architecture, and the operational transition was completed in the sense the rule intended. The transition is one of the largest infrastructure-replacement programs in the recent FAA record, and it executed.[2]

THE EVIDENCE

The hedges have to survive. GAO and DOT Inspector General reports across the 2010s repeatedly documented significant schedule slippage across NextGen — multiple elements missing original delivery dates, the broader program's realized benefits running below original projections, and contested cost-benefit accounting between FAA program reporting and external review. The benefit categories NextGen's business case rested on — fuel-burn reduction, delay reduction, increased airspace capacity — have materialized in some respects and not in others, and the attribution to NextGen specifically (vs. broader operational and weather variation) is contested in the published audit literature. Equipage happened; benefit realization is mixed and ongoing.[3]

WHAT TRANSFERRED

What the case teaches at the LENS framing layer is the sustainment-engineering form of a planned aging- infrastructure transition at continental scale. The transition happened: the regulatory architecture, the decade of lead time, the equipage mandate at the deadline, and the operational migration to ADS-B-based surveillance are real and large. The transition was harder, slower, and more partial than the original NextGen plan implied; the schedule slippage and benefit-realization gaps are not anomalies in the case but part of what infrastructure transitions at this scale look like in practice. Together with the FAA Aging Aircraft program (Case 22), Y2K (Case 24), and LWRS (Case 76), NextGen is part of the v2 quartet that closes the C7 zero-state in the induced framework — and is the instance where the hedges are largest.[4]

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4. Department of Transportation Office of Inspector General, NextGen audit report series — independent program review documenting cost growth and partial benefit realization.
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> *The transition happened. The transition was harder, slower, and more partial than the original plan implied. Both are part of the case.*
>
> EDITORS' SYNTHESIS OF THE GAO NEXTGEN AUDIT SERIES AND FAA PROGRAM REPORTING.

LE INSIGHT

NextGen / ADS-B is one of the largest planned aging- infrastructure transitions in the recent regulatory record. The equipage mandate executed at the January 2020 deadline. The broader NextGen program has documented significant schedule slippage and benefit-realization gaps in sustained external audit. The case closes a C7 zero-state with the largest acknowledged hedges in the v2 aging-system quartet.

LENS APPROACH

NextGen / ADS-B is the planned infrastructure-transition case (induced 7.1; LENS D1/PT4). LENS uses it in Domain 1 (Systems Analysis) for the separation of the auditable equipage mandate from the multifactorial benefit- realization business case, and in Domain 5 (Navigating Sociotechnical Constraints) for the decade of lead time and the sustained external-audit discipline. Closes the C7 zero-state alongside Cases 22 (FAA aging aircraft), 136 (Y2K), and 137 (LWRS) as the v2 aging-system quartet.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the equipage mandate with the lead time the transition actually needs — a decade for ADS-B Out — so the deadline arrives with realistic compliance pathways rather than as a forcing function operators cannot meet.
- Separate the equipage-mandate piece (auditable, has a deadline) from the broader benefit-realization business case (multifactorial, harder to attribute); the equipage piece will deliver, the benefit piece will deliver partially, and conflating them sets the program up to look like a failure where it succeeded.
- Plan for sustained external audit (GAO, DOT IG) as part of the program's operating environment — the schedule slippage and benefit-realization gap reporting is institutional discipline, not a sign the program is unworkable.

IN OPERATION / AFTER

- Report the equipage transition as the deliverable that executed and the benefit realization as the deliverable that is ongoing and mixed; both are real, and the load-bearing hedge is that the original NextGen plan understated the difficulty of the latter.
- Sustain the audit-and-rebaseline cadence after the headline equipage mandate; the benefit-realization picture continues to develop, and the institutional record of slippage and partial realization is part of how transitions at this scale are documented honestly.
- Treat the NextGen case as the instance where the C7 transition delivered with the largest acknowledged hedge — useful precisely because the transition is real and the hedges are also real, and conflating either with the other misreads the lesson.

REFLECTION QUESTIONS

1. Identify an aging-infrastructure transition in your domain whose original business case rested on benefit categories that are multifactorial and contested in attribution. What is the equivalent of the auditable equipage mandate — the piece of the transition that delivers cleanly — and how would you separate its reporting from the broader benefit case?

2. Specify the lead time the transition actually needs from mandate to deadline; ADS-B Out used a decade. What is the analog in your context, and what does the operator-equipage trajectory look like across that horizon?

3. The NextGen case is the instance where the hedges are largest in the v2 aging-system quartet. What does it mean to be honest about a transition that executed and also slipped and underperformed its business case — and what is the institutional discipline that keeps both claims accurate?

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CASE 24

SOFTWARE SUSTAINMENT | CRITICAL INFRASTRUCTURE | FEDERAL PROGRAM MANAGEMENT

1996 – 2000

Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed

G Governance & Trust D Designed Out K Knowledge & Institutional Memory

IMPACT The US federal government and the broader public and private sectors invested an estimated 100 billion dollars (US) over four years remediating two-digit-year date handling in legacy systems; the January 1, 2000 rollover passed with minimal disruption to critical infrastructure — the success contributed to the durable counterfactual debate about whether the threat justified the spending

IN BRIEF

The "Year 2000 problem" — Y2K — was the structural consequence of decades of legacy software representing year fields as two digits, ambiguous between 1900 and 2000 at the rollover. The affected code ran banking systems, embedded controllers in industrial infrastructure, federal benefit-payment systems, air traffic and rail systems, and the broader public and private software base. From 1996 through December 1999 the US federal government, under sustained Office of Management and Budget reporting and GAO audit, drove an inventory-and-remediation program across mission-critical federal systems, while the private sector executed a parallel multi-year effort. Estimates of the total US investment range around $100 billion. The January 1, 2000 rollover passed with minimal disruption to critical infrastructure. The case is the canonical instance of a believed-and-treated aging-system transition in the recent regulatory record: the threat was specific, the program was inventoried at line-item granularity, the deadline was immovable, and the institutional discipline was sustained over four years. The hedge survives in the durable counterfactual debate the success itself produced — would the rollover have passed similarly with less spending? — and the published GAO record characterizes the program as a major federal management success without claiming the counterfactual is closed.

BACKGROUND

The Y2K problem was an artifact of decades of software development in which year fields had been stored as two digits to save memory and database space when memory and database space were expensive. By the mid-1990s the code base had aged into a regime where the two-digit representation was a foreseeable failure: date arithmetic across the January 1, 2000 boundary could return wildly incorrect results — interest calculations, scheduling logic, embedded controller timestamps, federal benefit eligibility checks. The structural form is the C7 case par excellence: legacy code aging past the assumptions of its original design, with a hard, foreseeable deadline.[0]

THE INTERVENTION

The federal-program-management response began in earnest in 1996 and accelerated through 1997 with the creation of the President's Council on Year 2000 Conversion and the OMB quarterly reporting cycle. Federal mission-critical systems were inventoried at line-item granularity — the GAO report series tracked the proportion of federal mission-critical systems Y2K-compliant on a quarterly cadence, agency by agency. The discipline of the program rested on three institutional features: line-item inventory at the level of the actual systems, OMB-enforced quarterly status reporting, and GAO-sustained external audit that named agencies whose remediation lagged.[1]

HOW IT WORKED

The parallel private-sector effort was as large and is harder to characterize precisely. Total US investment estimates cluster around the $100 billion figure; global estimates are larger. Major financial institutions, utilities, telecommunications providers, and industrial operators ran their own multi-year inventory-and-remediation programs. Industry-coordination bodies (banking, electric utility, telecom) operated parallel structures to the federal program's co-ordination role. The deadline forced parallel execution across the public and private sectors at a scale and pace that legacy-software remediation does not usually see.[2]

THE EVIDENCE

January 1, 2000 passed with minimal disruption to critical infrastructure. A small number of localized incidents occurred and were absorbed; the catastrophic-scenario possibilities the program had been built to prevent — major financial system failure, power-grid cascade, air-traffic disruption, federal benefit payment failure — did not materialize. The GAO published a sustained record of the federal program through and past the rollover; the broader retro-spective literature characterizes the four-year effort as one of the major federal program-management successes of the era.[3]

WHAT TRANSFERRED

The hedge that survives the case is the counterfactual debate the success itself produced. The rollover passed; the counterfactual — whether the same outcome would have obtained with substantially less spending — is structurally unobservable, because the program ran. The post-2000 retrospective literature includes serious arguments on both sides: that the threat was real and the investment was the reason the rollover passed quietly, and that the investment was substantially overestimated relative to the actual latent failure population. The case is teachable on the institutional discipline — line-item inventory, OMB-enforced reporting, GAO audit, sustained four-year cadence, immovable deadline — and on the structural feature that the C7 transition succeeded because it was believed and treated. The counterfactual debate is preserved as part of the case rather than smoothed away.[4]

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5. National Research Council, Continued Review of the Tax Systems Modernization of the Internal Revenue Service — Y2K-related sustainment-engineering record.

THE LEARNING ENGINEERING LENS

> *The C7 transition succeeded because it was believed and treated. The counterfactual — whether the threat was as large as the response implies — the program structurally cannot answer.*
>
> EDITORS' SYNTHESIS OF THE GAO Y2K REPORT SERIES AND THE POST-2000 RETROSPECTIVE LITERATURE.

LE INSIGHT

Y2K remediation is the canonical case of a believed-and- treated aging-system transition. Line-item inventory, OMB quarterly reporting, GAO sustained audit, an immovable deadline, and a sustained four-year program-management cadence converted a foreseeable legacy-software failure into a transition the rollover passed quietly. The counterfactual debate the success produced is part of the case.

LENS APPROACH

Y2K is the legacy-software-sustainment case (induced 7.1; LENS D1/PT4) — Domain 1 for line-item inventory + immovable deadline; Domain 5 for OMB-reporting + GAO-audit. Closes C7 with Cases 22, 23, 76. Counterfactual-debate hedge preserved.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Inventory the legacy at line-item granularity — actual systems, not categories — so the remediation status can be reported and audited against a denominator the program can defend.
- Build the immovable deadline into the program's operating discipline; the Y2K deadline could not be rebaselined, and the program's discipline came from the deadline's hardness rather than from management exhortation.
- Pair internal OMB reporting (the program's own status discipline) with external GAO audit (independent verification); the combination is what produced the institutional record the retrospective rests on.

IN OPERATION / AFTER

- Preserve the counterfactual hedge: a transition that succeeds because it was treated cannot prove the threat was as large as the response implies. The case is teachable on the institutional discipline, not on the closed answer to "was it worth it."
- Carry the public/private parallel-execution lesson: the federal program and the broader private-sector effort moved together because the deadline was external to both, and the coordination mechanisms operated alongside each other rather than depending on each other.
- Treat the Y2K case as part of the v2 aging-system quartet (Cases 22, 23, 76) — the instance where the transition was a software-sustainment problem with the largest counterfactual-debate hedge.

REFLECTION QUESTIONS

1. Identify a legacy-software or aging-system transition in your domain whose deadline is foreseeable but rebaselinable. What would the program look like if the deadline were treated as immovable — line-item inventory, quarterly reporting, sustained external audit — and what is the institutional cost of treating it that way?

2. Specify the public/private parallel-execution structure you would build if the transition reached beyond a single organization. The Y2K coordination did not depend on one body controlling the others; it depended on the external deadline being equally hard for everyone.

3. The Y2K success produces a counterfactual debate. Identify a similar success in your domain whose justification rests on the absence of the failure it was built to prevent. What institutional discipline keeps the historical record honest about the counterfactual without diminishing the discipline that produced the outcome?

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CASE 25 AIR TRAFFIC MANAGEMENT LEGACY MODERNIZATION SOFTWARE TRANSFORMATION 2005

Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change

G Governance & Trust K Knowledge & Institutional Memory H Human-System Interface

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT A 17,000-line pilot modernization of the Eurocat air-traffic-management system was scoped explicitly to generate safety evidence that an automated transformation was non-distortive of original functionality; that pilot-tier evidence was used to convince customers to accept a system-wide architecture-driven modernization with 100% automated code transformation

IN BRIEF

The Eurocat Air Traffic Management System was the kind of safety-critical legacy software whose customers cannot accept a big-bang rewrite: the operational system in production cannot tolerate the discontinuity. The 2005 Thales-led pilot modernization was scoped narrowly — 17,000 lines of code — with the deliberate goal of generating safety-equivalence evidence that an automated transformation did not distort the original functionality. The pilot succeeded in producing that evidence, and the evidence was used to convince customers to accept a full architecture-driven modernization with 100% automated code transformation. The teaching pattern is the small-as-gateway-to-big move: the small-tier verification artifact dissolved the customer objection to the large-tier change. The case is documented in a vendor-authored Elsevier technical chapter rather than an independent academic evaluation; the evidence-tier flag is rendered under the title. The case is the small-tier C7 transition-verification companion to the big-tier failures (Patriot/Dhahran, Ariane 5, Knight Capital, CrowdStrike, TSB) already in the corpus. Future validation will continue on the long-run operational behavior of the modernized system.

BACKGROUND

Air traffic management software is a paradigmatic capability-system-misalignment-at-transition problem: the legacy system carries decades of accreted operational logic, the safety case the regulator and the airspace user expect rests on the legacy's documented behavior, and a big-bang rewrite is unacceptable because the operational discontinuity is itself the safety hazard. The Eurocat problem in 2005 was the standard one — modernize without breaking the safety case the legacy already carries.[0]

THE INTERVENTION

The design move was to scope a pilot narrowly enough that the verification could be exhaustive. Seventeen thousand lines of code is small as a fraction of the full system; it is large enough that a working automated transformation pipeline can be exercised end-to-end and the safety-equivalence evidence can be generated against a realistic subset. The pilot was not a feasibility check. It was a deliberately designed evidence artifact: prove, at small scale and on the actual legacy, that the transformation does not distort the function the customer cares about.[1]

HOW IT WORKED

The pilot succeeded in producing the evidence. Customers were then willing to accept a system-wide architecture-driven modernization with 100% automated code transformation. The causal claim the case rests on is not “the modernization worked” — that is a long-run operational question — but “the customer objection to the large-tier change was dissolved by the small-tier verification artifact.” That is the C7 teaching point at the small tier the corpus has not had: the same governance-objection-dissolver move that Case 183 / Case 184 show in the AV regulatory regime, transposed into a legacy-software transition.[2]

THE EVIDENCE

The evidence-tier flag matters. The case is documented in a practitioner-authored chapter in an Elsevier book on information-systems transformation. The authors are from the vendor side; there is no independent academic evaluation of the pilot’s safety-equivalence claims or of the post-modernization operational behavior. The pattern the case teaches — verification at the small tier as the gateway to the large-tier change — is robust across the practitioner literature on legacy modernization, but the Eurocat-specific magnitudes and the long-run system outcomes have not been audited in a peer-reviewed source. Future validation will continue.[3]

WHAT TRANSFERRED

What the case adds at the LENS layer is the small-tier C7 intervention that the big-tier failures already in the corpus do not teach: when the inherited design has to move to a new envelope, the verification work itself can be the capability deliverable that earns the change its legitimacy. The Domain-1 frame applies: systems analysis of the legacy’s safety case as an artifact, and design of the transformation pipeline as another artifact, are the capability-engineering moves the program made. The case is the missing positive example for induced 7.1 and 7.2 at the small tier.

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THE LEARNING ENGINEERING LENS

> *The verification at the small tier is the artifact that earns the large-tier change its legitimacy. The pilot is not a feasibility check. It is the evidence.*
>
> EDITORS' SYNTHESIS OF THE EUROCAT MODERNIZATION CHAPTER.

LE INSIGHT

Eurocat is the small-tier C7 transition-verification intervention the corpus needed: a narrowly scoped pilot designed to produce the safety-equivalence evidence that dissolves customer objection to a system-wide modernization. Evidence is vendor-authored practitioner literature; the long-run operational record of the modernized system is the open question. Future validation ongoing.

LENS APPROACH

Eurocat is the C7 small-tier transition case (induced 7.1 and 7.2; LENS D1/PT1). LENS uses it in Domain 1 (Systems Analysis) for CLO-1 — the safety case and the transformation pipeline are both engineered artifacts — and in Domain 2 (Iterative Development) for CLO-2, since the pilot is the iteration that earns the rollout. Pair with the big-tier C7 failures (Patriot/Dhahran, Ariane 5, Knight Capital, CrowdStrike, TSB) for the failure-and- intervention contrast.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Scope the pilot to be small enough that the verification can be exhaustive against the actual legacy, not just a feasibility check on a synthetic subset.
- Design the safety-equivalence evidence as a deliberate deliverable from the pilot, not a byproduct: specify in advance what the customer needs to see to accept the large-tier change.
- Document the transformation pipeline itself as a verifiable artifact, so the customer can audit the transformation, not just inspect the transformed code.

IN OPERATION / AFTER

- Carry the practitioner-tier flag into any downstream citation; the case is a vendor account and the long-run operational behavior of the modernized system is the open question.
- Treat the small-as-gateway-to-big pattern as the teaching point; the magnitudes and the system-wide rollout are open and depend on the long-run operational record.
- Pair with the big-tier C7 failures (Patriot/Dhahran, Ariane 5, Knight Capital, TSB) when teaching; the pair shows the verification-as-deliverable principle is what separates the failure and intervention threads.

REFLECTION QUESTIONS

1. Identify a legacy system in your context whose customer or operator will not accept a big-bang rewrite. What is the small-tier pilot whose evidence would dissolve the objection to the large-tier change, and what specifically would the customer need to see?

2. Specify the safety-equivalence evidence the pilot would have to generate as a deliberate deliverable. The Eurocat pattern teaches because the pilot was scoped to produce the artifact the customer needed, not just to demonstrate feasibility.

3. The case is vendor-authored practitioner literature. What independent evidence (academic evaluation, multi-customer replication, long-run operational behavior) would you require before treating the small-as-gateway-to-big pattern as a settled engineering practice in your domain?

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CASE 26 NUCLEAR POWER CONTROL-ROOM MODERNIZATION HUMAN FACTORS 2014

INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding

G Governance & Trust K Knowledge & Institutional Memory H Human-System Interface

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT An INL-affiliated study reported that operators were able to use the new digital turbine-control system without extensive additional training or rewriting of operating procedures — i.e., the human-factors verification-and-validation evidence supported a low-burden cutover from the legacy analog control

IN BRIEF

Nuclear-plant control-room modernization is one of the canonical C7 problems — legacy analog systems that have to be replaced with digital equivalents inside a regulatory regime that demands the safety case survive the transition. The Idaho National Laboratory's Light Water Reactor Sustainability (LWRS) program has produced a body of technical reports and conference papers documenting the verification-and-validation work behind specific upgrades. The case here is a specific finding: in a study of turbine-control-system upgrade work, operators were able to use the new digital system without extensive additional training or rewriting of operating procedures. The human-factors V&V evidence supported a low-burden cutover. That finding is the small-tier complement to the big-tier LWRS program case the corpus also tracks: the program produces specific design-V&V case studies, not only program-level claims. The evidence base is INL technical reporting and OSTI-hosted conference papers, not independent academic evaluation; the tier flag is rendered under the title. Future validation will continue on whether the low-retraining finding generalizes beyond the studied subsystem and holds at multi-plant scale.

BACKGROUND

The nuclear-plant control room is the C7 transition problem with the safety-case constraint at the front: the regulator and the licensee both need to know that the upgraded digital control behaves equivalently to the legacy analog under the operating envelopes the safety case covers, and that the operator does not lose capability in the cutover. The LWRS program at INL exists to produce that evidence across the US fleet's modernization needs. The specific study here is one of the program's small-tier deliverables: a human-factors verification of a turbine-control-system upgrade.[0]

THE INTERVENTION

The relevant finding the case rests on is that operators were able to use the new digital turbine-control system without extensive additional training or rewriting of operating procedures. That is a substantive human-factors-V&V claim: the cutover did not impose a retraining burden on the operator workforce that the schedule and the operational logic could not absorb. The study is a per-subsystem evaluation rather than a whole-control-room finding, and the LWRS program treats these small-tier evaluations as the building blocks of the larger fleet-modernization safety case.[1]

HOW IT WORKED

The case is the small-tier complement to the big-tier LWRS program case the corpus also tracks. The big-tier case states the program-level claim: that nuclear plants can be modernized into digital control rooms with the safety case maintained. The small-tier case is the per-subsystem evidence that the program-level claim has to rest on — the individual design-V&V deliverables that, in aggregate, give the regulator and the licensee a reason to accept the modernization. Without the small-tier evidence the program-level claim is unfounded; without the program-level coordination the small-tier evidence does not roll up.[2]

THE EVIDENCE

The evidence-tier flag is load-bearing. The study sits in INL technical reporting and OSTI-hosted conference papers, not in independent academic evaluation. The low-burden-cutover finding is conditional on the specific subsystem, plant, and operator population studied; whether the finding generalizes to other subsystems in the same plant, to other plants in the fleet, or to other digital control products is the open question that the LWRS program itself acknowledges and continues to study. Future validation will continue as the program publishes follow-on reports.[3]

WHAT TRANSFERRED

What the case adds at the LENS layer is the small-tier example of verification-as-deliverable inside a regulated transition. The Domain-1 frame applies: the human-factors analysis of the operator's task under the digital cutover is a systems-analysis artifact that the regulatory regime consumes as part of its acceptance decision. The Domain-3 frame applies because the V&V evidence is itself the measurement instrument the program rests on. The case is a paired small-tier companion to Case 25 (Eurocat ATM) and sits inside the C7 thread alongside the big-tier failures.

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4. O'Hara et al. (2008), "Human Factors Considerations with Respect to Emerging Technology in Nuclear Power Plants," NUREG/CR-6947 — peer-adjacent framing.

> *The program-level claim cannot rest on un-rolled-up small-tier evidence. The per-subsystem V&V is the building block of the modernization safety case.*
>
> EDITORS' SYNTHESIS OF INL LWRS REPORTING.

LE INSIGHT

The INL turbine-control finding is a small-tier C7 verification-as-deliverable case inside the LWRS nuclear-modernization program: the human-factors V&V evidence supports a low-burden cutover. Evidence is INL technical reporting and OSTI conference papers, not independent academic evaluation; the generalization beyond the studied subsystem is the open question. Future validation ongoing.

LENS APPROACH

INL turbine-control upgrade is the C7 small-tier transition case (induced 7.1 and 3.1; LENS D1/PT1). LENS uses it in Domain 1 (Systems Analysis) for CLO-1 — the operator-task analysis under the digital cutover is the systems-analysis artifact the regulator consumes — and in Domain 4 (Test and Evaluation) for CLO-4, since the V&V evidence is itself the measurement instrument the program rests on. Pair with Case 25 (Eurocat) for the small-tier C7 thread and with the LWRS big-tier program case already in the corpus.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat each subsystem upgrade as a per-subsystem human-factors V&V deliverable, not as a sub-task of the larger control-room modernization claim.
- Specify the operator-task analysis the cutover has to satisfy before the upgrade ships, so the V&V evidence has a target the regulator can audit.
- Document the conditioning factors — plant, operator population, subsystem boundary — explicitly, so the generalization question is honestly framed in the report.

IN OPERATION / AFTER

- Roll the per-subsystem V&V studies into the program-level fleet-modernization safety case deliberately; the program-level claim cannot rest on un-rolled-up small-tier evidence.
- Carry the practitioner-tier flag into any downstream citation; the small-tier finding is conditional on the studied subsystem and population.
- Track durability of the low-retraining finding under personnel rotation, procedure updates, and subsequent upgrades; the cutover is not finished when the digital system is in service.

REFLECTION QUESTIONS

1. Identify a control-system upgrade in your context whose cutover burden on the operator is the decisive constraint. What is the per-subsystem V&V deliverable that would settle the burden question, and what is its scope?

2. Specify the rollup logic from per-subsystem V&V evidence to a program-level modernization claim. The INL pattern teaches because the small-tier studies are the building blocks of a fleet-level claim — what would the equivalent be in your context?

3. The case is INL technical reporting, not independent academic evaluation. What additional evidence (independent audit, multi-plant replication, durability tracking under personnel rotation) would you require before treating the low-retraining finding as generalizable?

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CASE 27

E-GOVERNMENT DISTRIBUTED DATA EXCHANGE NATIONAL IT INFRASTRUCTURE

2001 – PRESENT

Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox)

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT By December 2024 Estonia reported effectively 100% digitalization of government services across the X-Road data-exchange layer, with sub-five-minute tax filing for >95% of the population; the load-bearing self-critique is that the country has now created its own legacy system — the very thing the program set out to avoid

IN BRIEF

Estonia launched X-Road in 2001 as the answer to fragmented government databases: a distributed data-exchange layer that lets services interoperate without forcing a central monolith. By December 2024 the country reported effectively 100% digitalization of government services across X-Road, with sub-five-minute tax filing for more than 95% of the population. The case sits inside C7 not because of the headline outcomes but because of the load-bearing self-critique surfaced in the peer-reviewed analysis: by committing the country to X-Road as the data-exchange layer, Estonia has effectively created its own legacy system — the very thing its founders set out to avoid. The "no-legacy paradox" is the C7 teaching the corpus does not get from any other case: success-as-aging is the failure mode, and the modernization regime must contemplate its own future obsolescence as part of its current design discipline. The evidence-tier flag is rendered under the title: the program is well-documented in peer-reviewed and program-report sources; the self-critical framing rests partly on practitioner reflection that the corpus must carry honestly. Future validation will continue on whether the deliberate generational-replacement plans materialize. Cross-listed with Gap 5 (non-US/UK/EU coverage, Estonia).

BACKGROUND

Estonia's pre-2001 problem was the standard one for a small state with a recent administrative inheritance: government databases were fragmented, agency-by-agency, with no common exchange layer. X-Road was the architectural answer — a distributed data-exchange protocol that lets each agency keep its own system but interoperate via a shared, authenticated channel. The design choice avoided the trap of a central monolith and made cross-agency services possible without requiring agencies to surrender their data.[0]

THE INTERVENTION

The deployment trajectory across two decades has been the defining national IT case study in the EU region. By December 2024 the country reported effectively 100% digitalization of government services, with sub-five-minute tax filing for more than 95% of the population. The Mission Mystique and the Hiding Hand chapter (Oxford 2024) and the ICEGOV 2021 historical analysis are the peer-reviewed treatments; the X-Road Global program documentation extends the case to the institutions that have since adopted the protocol elsewhere.[1]

HOW IT WORKED

The load-bearing self-critique is what makes this a C7 case and not just a digital-government success story. The analysis surfaces what practitioners and researchers have begun to call the no-legacy paradox: by committing the country to X-Road as the data-exchange layer, Estonia has effectively created its own legacy system. The protocol is now twenty-plus years old; the agency systems that hang off it have accreted their own logic; the generational replacement of the platform itself is now the modernization problem the founders thought they had designed out. Success-as-aging is the failure mode.[2]

THE EVIDENCE

The evidence-tier flag matters. The peer-reviewed sources cover the program's design and trajectory and the self-critical framing; the program-report and practitioner reflections supply the day-to-day operational picture and the no-legacy paradox in its sharpest form. The country's generational-replacement plans for X-Road exist; whether they materialize at the pace the analysis implies is necessary is the open empirical question. Future validation will continue as the next-generation data-exchange architecture is designed and deployed.[3]

WHAT TRANSFERRED

What the case adds at the LENS layer is the C7 teaching that no other corpus case supplies: a modernization regime must contemplate its own future obsolescence as part of its current design discipline. The Domain-1 frame applies: systems analysis of the data-exchange layer as an artifact whose own aging trajectory is part of the system's behavior. The Domain-4 frame applies because the sociotechnical commitment of a state to a single data-exchange protocol is itself the governance choice that determines what the future modernization will have to undo. The case is also a Gap-5 non-US/UK case from a small EU state, which the corpus needs.

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> *Success creates its own aging problem. The modernization regime must contemplate its own future obsolescence as part of its current design.*
>
> EDITORS' SYNTHESIS OF THE NO-LEGACY PARADOX IN THE ESTONIA X-ROAD ANALYSIS.

LE INSIGHT

Estonia X-Road is the C7 case the corpus needed for the success-as-aging failure mode: a 100%-digitalization program whose own success has now created the legacy system its founders set out to avoid. Evidence is mixed — peer-reviewed analytical chapters plus program-report and practitioner reflection; the generational-replacement trajectory is the open empirical question. Future validation ongoing. Non-US/UK case, Gap-5 cross-listed.

LENS APPROACH

X-Road is the C7 big-tier transition case with a small-tier self-critique (induced 7.1 and 5.3 alternate; LENS D1/PT1). LENS uses it in Domain 1 (Systems Analysis) for CLO-1 — the data-exchange layer is an artifact whose own aging is part of system behavior — and in Domain 5 (Navigating Sociotechnical Constraints) for CLO-5, since the state's commitment to the protocol is itself the governance choice determining future modernization cost. Pair with Cases 25–26 for the C7 thread and with the big-tier C7 failures the corpus already documents.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the data-exchange or platform layer so that its own future obsolescence is contemplated in the current design — the no-legacy paradox is the warning the case carries.
- Treat the generational-replacement plan as part of the deployment commitment, not as a future problem; the Estonia case is open precisely because the replacement is open.
- Document the sociotechnical commitment of the state or institution to the platform explicitly, so the cost of future replacement is in the record from the start.

IN OPERATION / AFTER

- Track the platform's aging trajectory as a system variable — accreted agency logic, protocol drift, dependency depth — not only its operational uptime.
- Carry the self-critical framing honestly; the load-bearing teaching is that success creates its own aging problem, not that the program failed.
- Pair with other C7 cases (Patriot/Dhahran, Ariane 5, Knight Capital, CrowdStrike, TSB) for the failure-and-modernization contrast, and with Cases 25–26 for the small-tier transition thread.

REFLECTION QUESTIONS

1. Identify a platform or data-exchange layer in your context whose success has created a generational-replacement problem the original design did not contemplate. What would the no-legacy-paradox-aware redesign look like?

2. Specify the aging-trajectory variables you would track on the platform (accreted agency logic, protocol drift, dependency depth) so the future-obsolescence question is in the operational record, not a future surprise.

3. The case is peer-reviewed analysis plus program-report plus practitioner reflection. What additional evidence would you require before committing your own institution to a platform that may itself become the legacy system in twenty years?

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Chapter 3

Iterative Development — The Iteration Gap

When the discipline of running an iteration through goes unowned.

The iteration cycle is the field's core method. The casebook documents what iteration produced more often than how iteration was done.

EDITOR'S NOTE ON THE V2 CORPUS.

EDITOR'S NOTE · THE ITERATION GAP

Iterative development is what learning engineering does most often. It is also the capability the casebook documents least directly. The cases that follow in this chapter share a shape: an organization received an unambiguous signal, held the engineering or product position to act on it, and failed to run the iteration cycle through product, business model, and operations until the opportunity had passed. The signal was sometimes a field report (Therac-25, the v2 v1-era cases on adoption and sustainment), sometimes an external launch (BlackBerry meeting the iPhone), sometimes an internal invention the organization could not build a business around (Kodak's 1975 digital camera). In each case the organization had the engineering capability and the institutional position. What it lacked was the discipline of running the iteration through.

The casebook's broader corpus shows what iteration produced — the successful interventions in Chapter 4 (Iterative Development — What Works), the failures whose recovery iterations followed in later chapters. It does not yet show iteration **being done** in detail: the design moves, the dead ends, the reframes. That gap is exactly what the v2.1 amendment to D2 names — "narrate and defend the design iteration in first person" — and what the editor-commissioned first-person Practice Flywheel accounts will fill in the next edition. This chapter names the gap honestly so that the reader knows what to expect from the rest of the volume, and what the discipline is still working to make legible.

CASE 28 AVIATION 2009

Air France Flight 447

T Training Gap H Human-System Interface

IMPACT 228 killed; the deadliest accident in Air France's history

IN BRIEF

Air France Flight 447, an Airbus A330, fell into the South Atlantic on 1 June 2009, killing all 228 aboard — the deadliest accident in the airline's history. At cruise altitude the pitot probes iced over, the airspeed readings failed, and the autopilot handed the jet to a crew that had never trained to fly it by hand in that regime. The pilot flying held the nose up into a full aerodynamic stall and never recognized it; the aircraft descended for nearly four and a half minutes. The BEA traced the loss to the airspeed failure, the crew's inappropriate inputs, and a training system that taught stall prevention at low altitude but never stall recovery at altitude — a gap between the trained envelope and the operational one that reshaped global pilot-training rules.

BACKGROUND

AF447 left Rio for Paris on 31 May 2009 with 228 aboard an Airbus A330 — a highly automated jet with an excellent safety record. Its route crossed the equatorial storm band, and it carried a known vulnerability: Thales AA pitot probes prone to brief icing at high altitude in heavy precipitation, a pattern documented across the A330/A340 fleet in the years before the accident. A replacement program with newer probes was underway, but the accident aircraft had not yet been modified, so the vulnerability the program existed to close was still live on the very jet that crossed the storm band — the retrofit recognized as necessary but not yet fitted where it mattered.[0] The A330's fly-by-wire architecture also carried a design choice that would matter later: the two sidesticks were not mechanically linked, so an input on one was not felt on the other, a philosophy that traded the visual cue of yoke movement for sidestick lightness and independence.

WHAT HAPPENED

At 35,000 feet the probes iced, the airspeed readings went invalid, and the autopilot and autothrust disconnected into Alternate Law — a degraded control regime in which stall protection no longer held. The pilot flying responded with sustained nose-up input; the jet climbed, stalled, and never recovered, falling some 38,000 feet into the ocean in about four and a half minutes. The stall warning sounded, then cut out at extreme angle of attack and resumed when the nose dropped — warning against the one input that would have begun a recovery, so that the cue meant to guide the crew instead punished the correct action and rewarded the fatal one, an inversion no amount of hand-flying instinct could resolve in the time available.[1] The other pilot's correcting inputs on his own sidestick went unfelt on the flying pilot's: the design's clean independence became, under stress, an inability to see what the other crew member was doing.

THE INVESTIGATION

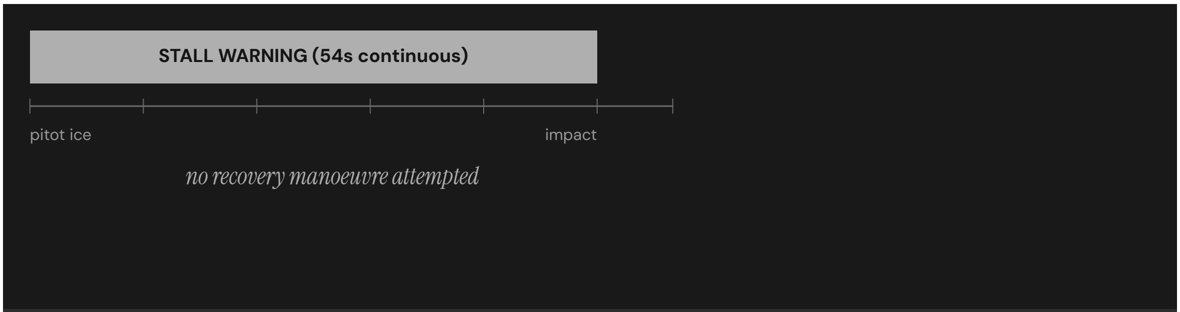
The BEA's three-year investigation, concluded in the 2012 final report, named a chain: airspeed loss from pitot icing, inappropriate crew inputs, and the crew's failure to recognize and recover from the stall.[2] The pilots were not incompetent — they were outside anything their training had prepared them for, and the BEA said so: "the conditions under which pilots are trained and exposed to stalls… did not result in reasonably reliable expected behaviour patterns."[3] The finding reframed the loss from a question of individual airmanship to one of training design: the crew had been drilled in a regime that never produced the responses the emergency demanded. The BEA report also surfaced the longer industry debate over fly-by-wire philosophies — the Airbus convention of independent sidesticks and protected envelopes versus the Boeing convention of linked yokes and trim feedback — not to adjudicate which was safer in steady state but to argue that whichever a manufacturer chose, the training had to make the degraded-mode behavior of that choice reflexive rather than novel.

THE CAPABILITY GAP

The gap was precise. Airlines trained stall recovery at low altitude — the regime certification required and the one simulators could reproduce — while the simulators of the era could not faithfully reproduce a high-altitude stall, so crews never practiced the situation that arrived. Years of reliable automation had also let hand-flying skills atrophy, so when the autopilot handed back control the crew met an unfamiliar regime with rusty manual skills at the worst possible moment. The trained envelope and the operational envelope had quietly diverged, and AF447 fell into the gap between them — a gap no one had been positioned to see widen, because the training record showed full compliance and the operational record showed an excellent safety performance, right up until the regime that neither had covered arrived together with degraded control law.[4]

AFTERMATH & REFORM

The BEA's recommendations reached far beyond one airline: the pitot probes were replaced fleet-wide, and the report pressed for training in manual high-altitude flight, stall recovery from cruise altitude, unreliable-airspeed handling, and angle-of-attack indication.[5] Regulators then made Upset Prevention and Recovery Training mandatory for airline pilots — FAA Part 121 adopted UPRT-aligned stall-recovery training in 2014; EASA phased UPRT into ATPL training by 2019; ICAO codified it in Doc 10011 — closing at the regulatory level the gap that had been invisible at the airline level, and moving the fix from a single carrier's discretion to a binding standard every airline had to meet.[6] Simulator manufacturers were pushed to extend their aerodynamic models into the post-stall regime so the training could actually take place. The crew performed exactly as trained; the training was the wrong training, and only a system-wide mandate could keep that mismatch from recurring elsewhere.



54 SECONDS: STALL WARNING SOUNDED, RESPONSE NEVER CAME

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THE LEARNING ENGINEERING LENS

> *The crew never understood that they were stalling and consequently never applied a recovery manoeuvre.*
>
> BEA, FINAL REPORT ON AIR FRANCE FLIGHT 447, JULY 2012

LE INSIGHT

AF447 is the canonical case of training that matched one envelope of operation perfectly and another not at all. Stall **prevention** training was excellent. Stall **recovery from cruise altitude** training did not exist because the simulators of the era could not faithfully produce it. Layered on top was an automation-to-human handoff that arrived in a degraded control law the crew had never flown and a sidestick architecture that hid one pilot's inputs from the other. The pilots performed exactly as trained. The training was the wrong training, and the reform had to reach the regulator, because no single airline could close the gap unilaterally.

LENS APPROACH

AF447 is the worked example of induced sub-competency 1.2 (capability envelope at the edge of training) and the LENS D2/PT4 pairing — Iterative Development applied to training-program updates that must keep pace with operational regimes. Students use the case to specify training requirements from degraded-mode operational analysis (LENS D1), to design the evidence the BEA used to identify a training-design rather than airmanship failure (LENS D3), and to examine the human-AI handoff as a capability problem (LENS D5): the autopilot disconnection was a transition crews were trained to avoid rather than to handle. The case pairs with Kegworth (Case 55) as the canonical pair on transition-training failure across a changed system, and with the Boeing 737 MAX MCAS sequence as the near-current echo. CLO mapping: CLO-2 (Iterative Development) primary; CLO-4 (Test and Evaluation) for the BEA investigation framing; CLO-3 (Human-System Collaboration) for the automation-handoff dimension.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Define the operational capability envelope explicitly — including degraded-automation and high-altitude regimes — and train to its edges, not only to the routine center.
- Engineer warning logic so that cues never punish the correct recovery input; validate stall-warning behavior across the full angle-of-attack range before fielding.
- Treat manual-flying proficiency in degraded modes as a measured deliverable, not an assumed residual of automated operation.

IN OPERATION / AFTER

- Audit recurrent training against the actual operational record so the trained envelope cannot silently diverge from the regimes crews encounter.
- Monitor for skill atrophy where reliable automation reduces hands-on exposure, and refresh manual competence before it erodes.
- Sustain the reform at the regulatory level (mandatory upset recovery) so a fix proven in one carrier propagates to all rather than lapsing.

REFLECTION QUESTIONS

1. The simulators of 2009 could not produce high-altitude stall behavior. What is the equivalent gap in your domain — the operational regime your training environment cannot reproduce?
2. Design the recurrent-training curriculum that would have caught the AF447 gap. Be specific about cost, evidence, and what makes the curriculum falsifiable against the operational record.
3. The autopilot handed control to a crew at the one moment it was least prepared to take it. Identify an automation-to-human handoff in your domain that occurs precisely when the human is least ready, and design the trigger or warning that would change that.

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CASE 29 TECHNOLOGY HIRING ALGORITHMS MACHINE LEARNING 2014 – 2018

Amazon Hiring AI — Trained Bias, Deprecated 2018

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER journalism-tier — source confidence flagged; future validation ongoing.

IMPACT Amazon internal recruiting-algorithm project initiated 2014, deprecated 2017 – 2018 after engineers determined the model could not be debiased; trained on ten years of historical resume data in which men predominated in technical roles; the model downgraded resumes containing the word "women's" and resumes from all-women's colleges; Reuters single-source investigation Oct 10 2018

IN BRIEF

Jeffrey Dastin's Reuters investigation, published October 10, 2018, reported that Amazon had initiated an internal recruiting-algorithm project in 2014 with the goal of automating resume screening for technical roles, and had deprecated the project in 2017 – 2018 after engineers determined the model could not be debiased. The model was trained on ten years of historical resume data in which men predominated in technical roles. The training-data composition encoded an association between gender-correlated features and role suitability; engineers found that the model downgraded resumes containing the word "women's" (as in "women's chess club captain") and downgraded resumes from all-women's colleges. Attempts to remove the offending features surfaced additional features carrying the same signal — the trained bias could not be debiased through feature engineering inside the model. The case rests on Reuters single-source reporting; Amazon never published the technical detail. The journalism-tier evidence-flag under the title carries the standing language. The load-bearing teaching point — that trained bias cannot in general be debiased through downstream feature manipulation — is the case's curricular value even as the specific case details remain limited to journalism-tier sourcing.

BACKGROUND

The recruiting-algorithm project that Reuters reported on was an internal Amazon effort initiated in 2014, with the stated goal of automating the early-stage screening of resumes for technical roles. The architectural premise was to train a model on historical resume data — applicants who had been hired and applicants who had not, across the preceding decade — and to score new resumes against the learned pattern. The premise was operationally appealing to a company hiring at Amazon's scale and was consistent with contemporary practice in algorithmic hiring across the technology sector. The seam the project's deprecation surfaced is structural to the premise itself.[0]

WHAT HAPPENED

The training data composition encoded the gender imbalance of historical Amazon technical hiring. Men predominated in technical roles across the ten-year window, and the resumes that had been hired carried gender-correlated features — wording, extracurricular activities, college affiliations, vocabulary choice — that the model learned to associate with hire-suitability. The Reuters investigation reports that engineers found the model downgrading resumes that contained the word "women's" (in contexts like "women's chess club captain") and downgrading resumes from all-women's colleges. Attempts to remove the offending features did not eliminate the pattern; the remaining features carried correlated signal that reproduced the same downgrade. The trained bias was structural to the training-data composition and could not be debiased through downstream feature engineering inside the model architecture.[1]

THE INVESTIGATION

Amazon deprecated the project in 2017 – 2018 and did not deploy the model at production scale for hiring decisions. The deprecation is the load-bearing decision in the case: the engineering team determined that the model could not be made fair, and the organization withdrew the project rather than deploying it. The case's evidentiary structure rests on Reuters' single-source reporting — the company never published the technical detail, and the specific mechanism by which engineers verified the irreducibility of the bias is not available in the public record at the level of a peer-reviewed study or a regulator's audit. The evidence-flag under the title is binding: the case is journalism-tier, and future validation of the specific technical detail remains ongoing in the sense that the public record has not deepened beyond the 2018 investigation.[2]

THE CAPABILITY GAP

The case pairs with Case 94 (Bartlett mortgage) for the fairness-through-unawareness-fails thread: removing the gender feature from the training data does not eliminate the gender signal when the remaining features carry correlated signal. Pair with Case 97 (Johnson school surveillance) for the algorithmic-employment-and- surveillance-decisions parallel at a different population. Pair with Case 135 (LiveHint AI bias across foundation models) for the trained-bias-in-foundation-models thread at contemporary scale. The Amazon case is unusual in that the project was deprecated rather than deployed; most cases in the corpus document deployments that ran for years before withdrawal, and the case's curricular value is partly that the engineering team's verification of irreducibility led to the decision not to deploy.[3]

AFTERMATH & REFORM

The load-bearing teaching point is that trained bias cannot in general be debiased through downstream feature manipulation. When the training-data composition encodes a historical disparity in the outcome the model is being trained to predict, the model will learn to reproduce the disparity through whatever features remain in the input space, and the remediation is not feature engineering. The remediation is at the construct-validity layer — what is the model being asked to predict, and is the historical record from which the prediction is being learned a defensible target — or at the deployment-architecture layer — what is the role of the model in the decision process, and what human-in-the-loop infrastructure surrounds it. The case's evidence-tier hedge is binding: the journalism-tier sourcing limits the specificity of the teaching point's mechanism, but the structural form of the teaching point — choose the construct, then ask whether the historical record supports learning to predict it — is the case's curricular anchor.

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THE LEARNING ENGINEERING LENS

> *Trained bias cannot in general be debiased through downstream feature manipulation; when the training-data composition encodes a historical disparity in the outcome the model is being trained to predict, the disparity will be reproduced through whatever features remain in the input space.*
>
> EDITORS' SYNTHESIS OF THE REUTERS INVESTIGATION (DASTIN, 2018) AND THE SURROUNDING ALGORITHMIC-HIRING LITERATURE.

LE INSIGHT

Amazon Hiring AI is the trained-bias-cannot-be-debiased case at major-technology-company scale. The engineering team determined that the model could not be made fair through feature engineering and the organization deprecated the project rather than deploying it. The journalism-tier evidence-flag is binding for the specific technical detail; the structural teaching point — choose the construct, then ask whether the historical record supports learning to predict it — is the case's curricular anchor.

LENS APPROACH

Amazon Hiring AI is the choose-the-construct case at hiring- algorithm scale (induced 8.1; LENS D2+D5/PT6; CLO-4 and CLO-5). LENS uses it in Domain 2 (Iterative Development) for the irreducibility-verification-as-deployment-gate discipline and in Domain 4 (Test and Evaluation) for the construct-validity-at-the-training-data-layer anchor. Pair with Case 94 (Bartlett mortgage), Case 97 (Johnson school surveillance), and Case 135 (LiveHint AI bias across foundation models). The journalism-tier evidence-flag is binding; the structural teaching point is the case's curricular value.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Choose the construct the model is being asked to predict with construct validity in mind before training

IN OPERATION / AFTER

- Carry the journalism-tier evidence-flag under the title without softening; the case rests on single-source begins; the Amazon case demonstrates that when the historical record encodes a disparity in the outcome, the disparity will be learned regardless of feature engineering at the input layer.

- Verify the irreducibility of bias as a deployment gate; the engineering team's determination that the model could not be debiased is what enabled the deprecation decision, and the verification discipline is the curricular target.
- Treat the deprecation decision as the operational artifact that the verification supports; an organization able to deprecate a project at the verification finding is operating with a different decision architecture than one that defaults to deployment.

reporting and the future-validation-ongoing language is binding for the specific technical detail.

- Pair in syllabi with Case 94 (Bartlett) so the fairness-through-unawareness-fails thread is taught at adjacent scales and in adjacent decision domains.
- Use the case to anchor the construct-choice CLO; the curricular target is the discipline of refusing to deploy when the historical record from which the model would be learned does not defensibly support the prediction the deployment requires.

REFLECTION QUESTIONS

1. Identify a deployment in your domain whose training-data composition encodes a historical disparity in the outcome the model is being trained to predict. What construct-validity question would have to be answered before training proceeds, and who has authority to refuse training when the answer is "no"?

2. Specify the irreducibility-verification protocol your team would run before deploying a model whose training data is known to carry historical bias. What is the operational threshold for "cannot be debiased," and what is the decision rule when the threshold is met?

3. Amazon deprecated the project rather than deploying it. Pick a deployment in your domain that proceeded despite known bias-irreducibility findings, and ask: what would have had to be different in the decision architecture for the verification finding to have led to deprecation instead of deployment?

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CASE 30 INDUSTRIAL 1975–2012

Kodak Digital Camera Stagnation

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Inventor of the digital camera (1975) and holder of foundational digital-imaging patents; filed for Chapter 11 bankruptcy in January 2012 with the digital transition essentially un-run

IN BRIEF

In 1975 a young Kodak engineer, Steven Sasson, built the first self-contained digital camera in a Rochester lab: an 8-pound, 0.01-megapixel device that wrote a black-and-white image to cassette tape in 23 seconds. By the 1990s Kodak held foundational digital-imaging patents and had a working line of digital products. Yet the company that invented the category never ran the iteration cycle through to a digital-first business: film and the high-margin consumables it sold remained the protected core, and digital was treated as a threat to be managed rather than a product to be developed against itself. By 2003 digital camera sales overtook film in the United States; by 2007 Kodak had begun mass layoffs; in January 2012 the company filed for Chapter 11. The capability gap was not invention. It was the organization's failure to iterate the business around the technology it already owned.

BACKGROUND

Kodak's twentieth-century position was anchored on a single durable insight: the camera was a loss-leader and the film, paper, and processing chemistry were the recurring high-margin revenue. The business model — what later analysts would call the "razor-and-blades" of imaging — funded a research apparatus deep and well-resourced enough that in December 1975 a 24-year-old Eastman Kodak engineer, Steven Sasson, was able to assemble the first all-electronic still camera from a Fairchild CCD, a cassette-tape recorder, sixteen nickel-cadmium batteries, and a lens cannibalized from a Super 8 movie camera. The prototype wrote a 0.01-megapixel black-and-white image to tape in 23 seconds and played it back on a television.[0]

WHAT HAPPENED

Sasson's later published recollection (IEEE Spectrum, 2007) is that internal reception was polite but unmoved: the demonstration raised the question of when the company would have to compete against its own film business, and the answer settled into a long institutional hedge. Through the 1980s and 1990s Kodak filed foundational digital-imaging patents, built digital products including the 1991 DCS-100 professional digital SLR, and partnered with Apple on the early QuickTake consumer line; the technology was understood and the engineering position was real. What was missing was an iteration cycle that ran the digital signal all the way through product, distribution, pricing, and the corporate identity, rather than holding digital at the perimeter of a film company.[1]

THE INVESTIGATION

Paraphrasing Lucas & Goh (Journal of Strategic Information Systems, 2009): Kodak's leadership read digital primarily as a cannibalization threat to film rather than as an opportunity to be engineered. The retrospective record describes successive strategies — digital as a complement to film, digital kiosks at retail, Ofoto / Kodak Gallery for online printing, the EasyShare consumer line — each of which was a move toward digital but none of which committed the operating company to a digital-first product roadmap with the cadence and capital allocation the transition required. The cycle that a healthy iteration would have run — ship, measure, learn, ship again at higher ambition — was repeatedly truncated by the gravitational pull of the film P&L it would have had to displace.[2]

THE CAPABILITY GAP

By 2003 digital camera sales overtook film camera sales in the United States; by the mid-2000s smartphone cameras began collapsing the standalone-camera category Kodak had partially entered. Kodak's revenues fell from roughly USD 16 billion in 1996 to under USD 6 billion by 2011. The company sold its Health Imaging division in 2007, began mass layoffs the same year, and on 19 January 2012 filed for Chapter 11 bankruptcy protection in the Southern District of New York. In the bankruptcy proceedings Kodak monetized roughly 1,100 of its digital-imaging patents — many traceable to the engineering lineage Sasson's 1975 prototype had started — for approximately USD 525 million, a sale that drew explicit press commentary on what it meant for a company to monetize patents on a category it had failed to commercialize.[3]

AFTERMATH & REFORM

Kodak is the canonical case of an organization that held the technological position and lost on the iteration discipline. The capability gap was not in the lab and was not in the patent portfolio; it was in the leadership system that could not bring itself to run the iteration cycle all the way through the business model it would have replaced. The instructive contrast is Fujifilm, which over the same period diversified into chemicals, materials, and pharmaceuticals on the back of its film-coating expertise — a deliberate organizational iteration against the same signal. The lesson the v2.1 framing names is that "iterative development" at the organizational level is not a product practice but a leadership practice, and the cycle has to run on the business model and the operations as well as on the artifact.[4]

1975

FIRST DIGITAL CAMERA PROTOTYPE · 0.01 MEGAPIXEL

37 years from prototype to Chapter 11

KODAK — THE LONGEST UNACTED SIGNAL IN CONSUMER ELECTRONICS

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THE LEARNING ENGINEERING LENS

> *You don't get to keep the future just because you invented it.*
>
> EDITORS' SYNTHESIS OF THE SASSON (IEEE SPECTRUM 2007) AND LUCAS & GOH (JSIS 2009) RECORD

LE INSIGHT

Kodak is the cleanest available evidence that invention is not iteration. The company held the prototype, the patents, and the engineering depth and still lost the category. The gap was not technical; it was the leadership system's inability to run the iteration cycle through the business model it would have had to displace. v2.1 D2 names this directly: iteration-against-opportunity is an organizational discipline, not a lab discipline, and the cycle has to close on the P&L as well as on the artifact.

LENS APPROACH

Kodak is the v2.1 D2 stagnation exemplar (induced 2.3 transfer to high-consequence settings; LENS D2/PT4 adoption and sustainment; CLO-2 Iterative Development). LENS uses it to make the organizational-iteration claim concrete: 2.2 (run the cycle), 2.3 (transfer the cycle to the high-stakes business decision), and 2.4 (sustain adoption against the gravitational pull of the legacy P&L) all sat un-owned for a generation. Pair directly with BlackBerry (the next case) for the contemporary consumer-electronics analog, and with Fujifilm as the in-category counter-iteration that ran the same signal differently.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat an internally demonstrated category-changing prototype as a signal to run the full iteration cycle on the business, not only on the artifact — including pricing, distribution, sales force, and identity.
- Name explicitly the existing high-margin P&L the new category will have to displace, and assign an executive sponsor whose job is to iterate against that displacement on a schedule rather than around it.
- Set a measured cadence — ship, instrument, learn, ship again at higher ambition — and protect the cadence against the organizational pull of the legacy business at each review cycle.

IN OPERATION / AFTER

- Audit, periodically and without sentimentality, whether the iteration is running through to the operating company or has been parked at the perimeter as research or partnership; treat parking as the failure mode it is.
- Compare against an in-category peer running the same iteration with a different leadership posture (Fujifilm), and use the divergence as evidence rather than letting the legacy P&L be the only reference point.
- Read the late monetization of patents on a category the organization never commercialized as the indictment it is — a signal that the engineering position was held and the iteration discipline was not.

REFLECTION QUESTIONS

1. Identify a category-changing prototype or signal that your organization has internally demonstrated but parked at the perimeter. What would the full iteration cycle through the operating company actually require, and which P&L is the gravitational pull?

2. Kodak iterated on the artifact (digital products shipped) but not on the business model. Distinguish the two in a current initiative in your domain and specify which one is being held still.

3. Fujifilm ran the same signal differently. Identify an in-category peer running an iteration your organization is not, and construct the honest comparison the leadership review would have to engage with.

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CASE 31 TECHNOLOGY 2007–2013

BlackBerry Touchscreen Inertia

D Designed Out N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT From roughly 50% U.S. smartphone share in 2009 to under 1% by 2014; the Z10 full-touch device shipped in January 2013 — six years after the iPhone — and failed commercially; company restructured and effectively exited the consumer handset business

IN BRIEF

On 9 January 2007 Steve Jobs unveiled the iPhone. Research In Motion — then dominant in the keyboard-smartphone market with BlackBerry — read the device, in its co-CEOs' contemporary public statements, as constrained by short battery life, poor typing, and a network burden carriers would resist. The McNish and Silcoff history (Losing the Signal, 2015) documents that RIM had internal touchscreen and full-browser work underway, and that the organization nonetheless did not pivot product, sales, operating-system, or developer strategy quickly enough. The 2008 Storm — RIM's first full-touch device, built to a Verizon timeline — shipped with significant defects. The BB10 platform and the Z10 full-touch handset did not ship until January 2013, six years after the iPhone. By that point U.S. share had collapsed from roughly 50% in 2009 to under 5%, en route to under 1% by 2014, and BlackBerry effectively exited the consumer handset business.

BACKGROUND

Through the mid-2000s Research In Motion had built one of the strongest enterprise-mobile franchises in the industry: a keyboard handset, push email, a secure operating system, and a deep relationship with corporate IT departments and carriers. The BlackBerry Pearl (2006) had extended the product into the consumer market without disturbing the keyboard identity, and by early 2007 RIM was the dominant U.S. smartphone vendor. The product, the carrier relationships, and the developer ecosystem were all built around the keyboard, the network-efficient messaging stack, and the assumption that the smartphone was an enterprise-class communications device first.[0]

WHAT HAPPENED

On 9 January 2007, the iPhone unveiling reframed the smartphone as a full-browser, full-touch, app-centric consumer device. McNish and Silcoff (Losing the Signal, 2015) document RIM's internal response in detail: co-CEOs Mike Lazaridis and Jim Balsillie publicly emphasized what the iPhone could not yet do — battery life, typing accuracy, the network load — and the early carrier-side response was indeed cautious. Internally the engineering organization understood the implication faster than the executive system did; touchscreen and full-browser prototypes existed. The capability gap the case names is not that RIM did not see what was happening; it is that the organization did not run the iteration cycle on product, operating system, developer ecosystem, and go-to-market with the speed and ambition the signal required.[1]

THE INVESTIGATION

Paraphrasing McNish & Silcoff (2015): the 2008 BlackBerry Storm was launched to a Verizon-driven schedule against the iPhone, with a click-screen touch mechanism intended to preserve the keyboard's tactile feedback. The device shipped with reliability and software defects, and large volumes were returned; the iteration cadence the moment called for collided with an organization still optimizing a different category. Successive devices through 2010 and 2011 — the Torch, the PlayBook tablet running QNX — were partial bets that did not consolidate into a single, committed platform transition. The QNX-based BB10 operating system that would eventually power the Z10 took years to harden, and in the interim the developer ecosystem and the consumer-share decline compounded each other.[2]

THE CAPABILITY GAP

The Z10 — RIM's first fully committed full-touch BB10 handset — launched on 30 January 2013, six years after the iPhone. Initial reviews were respectful but the developer ecosystem was thin, carrier promotional support was muted, and consumer demand was weak. In September 2013 BlackBerry (renamed from RIM earlier that year) announced a USD 934 million pre-tax inventory and supply-commitment charge (Q2 FY2014 net loss approximately USD 965 million) and 4,500 layoffs, most of the charge Z10-related. U.S. smartphone share — roughly 50% in 2009 — had fallen below 5% by 2013 and toward 1% by 2014. BlackBerry's subsequent strategy explicitly pivoted to enterprise software, security services, and the QNX automotive-software business, in effect exiting the consumer handset category the iPhone had defined.[3]

AFTERMATH & REFORM

BlackBerry, like Kodak, did not fail on the question of whether the relevant technology existed inside the company. Touchscreen work, full-browser work, and a modern operating system (QNX) were all in the building. The failure was the iteration cycle at the organizational level: the leadership system did not run the cycle on product, OS, developer ecosystem, and go-to-market on a cadence that matched the speed at which the category was being redefined. The v2.1 D2 framing — iteration-against-opportunity as an organizational discipline — names what the case shows: the discipline is to take a signal, run a cycle, ship, measure, and ship again at higher ambition, with conviction allocated to the new category rather than to the defense of the old one.[4]

6 yrs

IPHONE LAUNCH (2007) · Z10 SHIP (2013)

the iteration cycle that did not close

BLACKBERRY — THE GAP BETWEEN THE SIGNAL AND THE RESPONSE

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THE LEARNING ENGINEERING LENS

> *The capability they were missing was not the technology. It was the iteration discipline.*
>
> EDITORS' SYNTHESIS OF MCNISH & SILCOFF, LOSING THE SIGNAL (2015)

LE INSIGHT

BlackBerry shows that a strong engineering position and a dominant market share do not, on their own, supply the iteration discipline a category redefinition requires. The capability gap was at the leadership level: the cycle on product, operating system, developer ecosystem, and go-to-market did not close on the cadence the signal called for. The Z10 launch in 2013, six years after the iPhone, is the legible verdict on the prior six years of partial bets and protected legacy.

LENS APPROACH

BlackBerry is the contemporary consumer-electronics pair to Kodak (induced 2.3; LENS D2/PT4 adoption and sustainment; CLO-2 Iterative Development). LENS uses the case to teach the v2.1 D2 subobjectives at the organizational level: 2.2 — run the iteration cycle on product, OS, ecosystem, and go-to-market together; 2.3 — transfer the cycle to high-stakes decisions (the platform transition itself); 2.4 — sustain adoption against the pull of the legacy enterprise franchise. Pair with Kodak (Case 30) as the longer-time-horizon analog and with the broader v2.1 D2 chapter on what organizational iteration discipline looks like when it works.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Read an external category redefinition as a signal that the iteration cycle has to run at a new cadence — on operating system, developer ecosystem, and go-to-market simultaneously — not on the current product line alone.
- Avoid shipping a transitional device to a partner's schedule when the platform underneath it is not yet ready; partial bets at full marketing weight burn the credibility a real transition will need.
- Allocate executive conviction to the new category explicitly, with named owners for the OS transition, the developer ecosystem, and the consumer go-to-market, and protect that conviction against the gravitational pull of the existing enterprise franchise.

IN OPERATION / AFTER

- Audit honestly whether the iteration cycle is closing on product, OS, and developer ecosystem together, or whether one of the three is being held back by the legacy organization; partial transitions compound rather than buy time.
- Treat consumer share decline and developer-ecosystem thinning as coupled rather than independent — the feedback loop between them is what makes a delayed transition unrecoverable, not either signal alone.
- If the cycle has not closed by the time a committed device ships, treat the launch as the verdict on the prior iteration discipline, not as the start of a new one — the Z10 reception is the case study.

REFLECTION QUESTIONS

1. Identify an external category redefinition currently underway in your domain. Which of product, platform, developer/partner ecosystem, and go-to-market is your organization iterating on, and which is being held still by the legacy P&L?

2. BlackBerry shipped the Storm to a partner's timeline before the platform underneath it was ready. Identify a current initiative in your domain at risk of the same mistake and specify what readiness would actually require.

3. Six years separated the iPhone launch from the Z10. Construct the leadership-review cadence that would have shortened that gap, and identify the specific decisions that cadence would have forced at each review.

WHO BUILDS THIS systems & human-factors engineering · safety science & organizational analysis · knowledge management & data engineering — plus domain experts and a learning engineer to integrate. TOOLS task analysis · design prototyping · incident analysis · safety dashboards · knowledge bases · data pipelines

Chapter 4

Iterative Development — What Works

When design, instrument, evaluate, refine runs through to the next problem framing.

Design capability interventions through iterative engineering cycles that survive contact with real operational environments.

CLO-2, LEARNING ENGINEERING DESIGN AND IMPLEMENTATION.

CASE 32 HEALTHCARE 2003–2017

ACGME 80-Hour Resident Duty-Hour Reform

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT ACGME capped U.S. resident physician work hours at 80/week to reduce fatigue-related errors; subsequent RCTs (FIRST, iCOMPARE) showed mixed effects on patient outcomes and increased hand-off-related errors

IN BRIEF

After the 1984 death of Libby Zion focused decades of concern on resident-physician fatigue, the ACGME capped resident work at 80 hours a week in 2003 and limited first-year shifts to 16 hours in 2011. The logic was intuitive: tired doctors err, so cut the hours. But hours were one input to a many-variable system; cutting them multiplied error-prone hand-offs and cost residents continuity and procedural repetitions. Two large randomized trials — FIRST (2016) and iCOMPARE (2019) — found flexible schedules non-inferior to the strict caps on patient outcomes, so the promised safety gain never appeared, and in 2017 the ACGME relaxed the intern cap. The case is the book's clearest example of a single-variable intervention into a capability system — and of evidence catching up with a well-meant policy.

BACKGROUND

After the 1984 death of Libby Zion — blamed on overworked, under-supervised residents — resident fatigue became a decades-long argument. New York's Bell Commission produced the first hours limits in 1989, and pressure for a national standard built from there — the reform's intuition, that exhausted physicians make errors, was strong enough that the simplest lever, capping the hours, became the obvious answer long before anyone tested what the long shift was producing.[0]

THE INTERVENTION

In 2003 the Accreditation Council for Graduate Medical Education capped resident work at 80 hours a week; in 2011 it limited first-year residents to 16-hour shifts. The logic was clean and intuitive — fatigue causes error, so reduce the hours. It was, in capability terms, a single-variable intervention: change one input and expect the outcome to move — a model that holds only if the rest of the system stays fixed, which in a teaching hospital it never does, since the hours removed had to be absorbed somewhere else.[1]

HOW IT WORKED

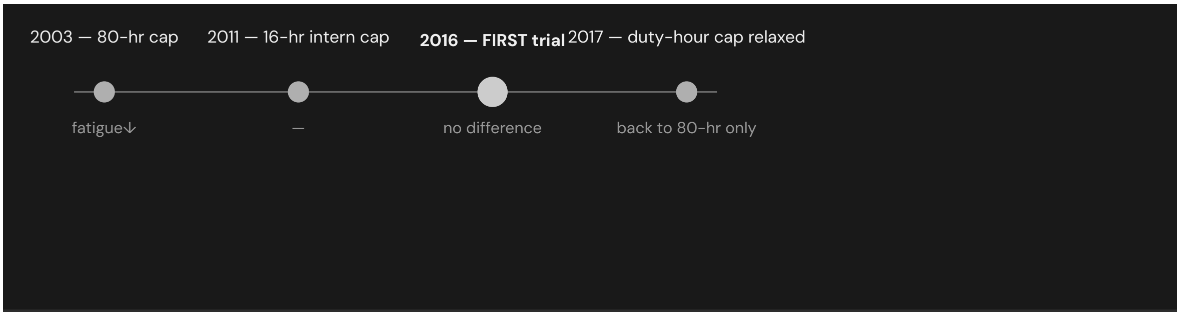
But hours were one input to a system with many. Cutting them redistributed the work and multiplied patient hand-offs — themselves a documented site of error — while reducing residents' continuity and the procedural repetitions that build skill; many reported feeling less prepared, not better rested. The long shift had quietly been doing work no one accounted for — sustaining a patient's care through one set of hands and accumulating the repetitions that turn a trainee into a clinician — and nothing was put in its place when it was cut.[2]

THE EVIDENCE

Two large randomized trials tested the policy. FIRST (Bilimoria et al., NEJM 2016), in surgery, found flexible duty hours non-inferior to the strict caps on patient outcomes and no worse for resident well-being — putting a randomized result against an intuition that had driven policy for years.[3] iCOMPARE (Silber et al., NEJM 2019), in internal medicine, reached a parallel result in a second specialty, making the finding harder to dismiss as an artifact of surgery.[4] Neither found the safety gain the cap had promised, and in 2017 the ACGME relaxed the 16-hour intern limit. The trials did not show fatigue is harmless — only that cutting one input, without rebuilding supervision and hand-offs, did not produce a safer system.[5]

WHAT TRANSFERRED

Duty-hour reform is the book's clearest case of a single-variable intervention into a multi-variable system. Read it against the successes here — the Keystone ICU project, crew resource management, the surgical safety checklist — which worked because they engineered supervision, hand-offs, and measurement **together with** the behavioral change, redesigning the surrounding architecture rather than pulling a single lever and hoping the rest would hold.[6] The lesson is not that fatigue does not matter; it is that capability is a property of the whole system, and a reform that moves one variable while leaving the others untouched will be judged, in the end, by what it actually produced rather than by the plausibility of its intuition.



ACGME DUTY-HOUR REFORM — CAPPING HOURS DID NOT, ON ITS OWN, ENGINEER THE CAPABILITY

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THE LEARNING ENGINEERING LENS

> *Flexible, less-restrictive duty hour policies for first-year residents were associated with non-inferior patient outcomes and no significant difference in residents' satisfaction with overall well-being and education quality.*
>
> PARAPHRASING ICOMPARE TRIAL (SILBER ET AL., NEJM 2019)

LE INSIGHT

The clearest healthcare case in the dataset of a single-variable intervention into a multi-variable capability system. Pairs with Case 4 (fratricide) and Case 141 (V-22). The success cases — Keystone (14), CRM (12), Korean Air (23) — engineered supervisory, hand-off, and measurement architecture **together with** the behavioral change.

LENS APPROACH

LENS uses duty-hour reform in LEN 5 as the foundational capability-system case (what was the long-hour regime **producing** that was lost when the input was capped?), in LEN 4 to discuss measurement architecture (what FIRST and iCOMPARE measured, and what they did not), and in LEN 10 as a studio prompt for the integrated resident-training redesign that the reforms did not deliver.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Map the full set of variables the targeted input is coupled to — continuity, supervision, hand-offs, procedural reps — before changing one of them.
- Design the substitute for whatever the changed input was producing (e.g., structured hand-offs and supervision) into the same reform, not as a follow-on.
- Build the measurement that will test the reform's promised gain into the rollout, so the policy is falsifiable against the operational record.

IN OPERATION / AFTER

- Audit the intervention against patient and trainee outcomes with a controlled comparison, as FIRST and iCOMPARE did, rather than trusting the intuition.
- Monitor the variables that absorbed the change (hand-off frequency, procedural exposure) for the harms a single-lever fix can displace.
- Sustain a willingness to revise the policy when evidence catches up, as the 2017 relaxation did, rather than defending the original lever.

REFLECTION QUESTIONS

1. What capability is the long-hours / heavy-workload regime in your domain currently producing — supervisory exposure, continuity, procedural reps, tacit-knowledge transfer — that a simple cap would lose?
2. Design the integrated redesign — supervision, hand-off, measurement, exposure — that would substitute for the capability the input cap removes.
3. The reform was intuitive enough to set national policy years before FIRST and iCOMPARE tested it. Design the randomized or quasi-experimental check you would build into a future single-variable reform so its promised gain is measured before it is mandated, not after.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 33 DEFENSE EDUCATION AT SCALE 2012 – PRESENT

GIFT and the Adoption Gap

K Knowledge & Institutional Memory G Governance & Trust N Normalization of Deviance

IMPACT **Active open-source ITS framework with demonstrated learning effectiveness; ubiquitous fielded adoption in routine military training has not been achieved**

IN BRIEF

The Generalized Intelligent Framework for Tutoring (GIFT) is an open-source framework, originated at the U.S. Army Research Laboratory, for authoring and delivering intelligent tutoring systems. Computer-based tutoring has been shown to be about as effective as expert human tutors in well-defined domains, and GIFT exists to lower the barrier to building it; the framework is actively developed, with regular releases and a peer-reviewed annual symposium. The puzzle is not that GIFT failed — it did not — but that ubiquitous fielded adoption across routine military training remains limited despite decades of supporting research. That gap is the canonical learning-engineering problem: the science is settled, the platform exists, the studies are positive — and the institutional pathway to scaled adoption is still being built.

THE SHIFT

Five decades of research established that computer-based tutoring can be about as effective as an expert human tutor in well-defined domains, and significantly better than traditional classroom instruction. The open question shifted from “does adaptive tutoring work?” to “why isn’t it everywhere?”[0] That shift matters because it moves the problem out of the laboratory: once the efficacy question is answered, every remaining obstacle to scaled use is institutional rather than scientific, and the field’s research strength stops being the binding constraint.

WHAT IS EMERGING

The Generalized Intelligent Framework for Tutoring (GIFT), originated under the U.S. Army Research Laboratory and now developed at DEVCOM, is an open-source framework for authoring, delivering, and evaluating intelligent tutoring systems — an effort to lower the authoring barrier that has historically made ITS expensive to build. It is actively maintained, with regular releases and a peer-reviewed annual symposium.[1] Open-sourcing the framework and sustaining a research community around it directly attacks the cost-to-build problem, since the expense of authoring a tutor from scratch had long been the practical reason adaptive tutoring stayed confined to well-funded demonstrations.

THE CAPABILITY QUESTION

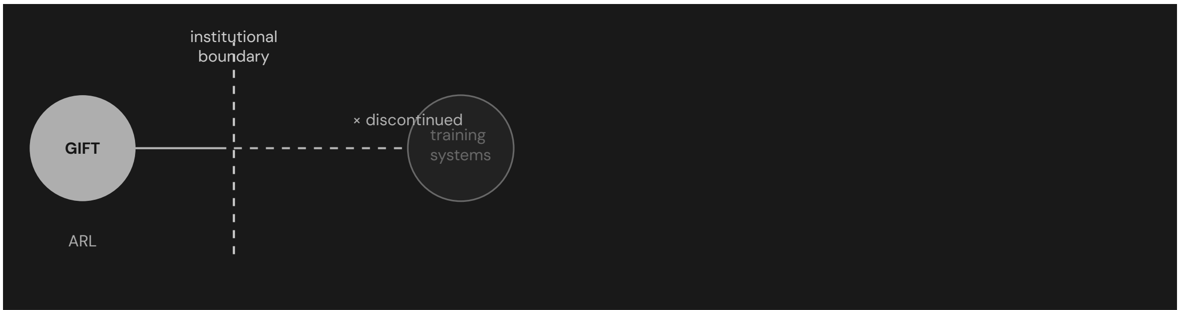
So the puzzle is not failure — GIFT was not discontinued. It is that ubiquitous fielded adoption across routine military training remains limited despite a working framework and decades of positive research. The science is settled and the platform exists; what is missing is the institutional pathway to scaled use.[2] This is the diagnostic feature of the case: the gap is not between idea and prototype but between a maintained, evidence-backed framework and the routine training pipelines that would have to adopt it, and that latter distance is the one no amount of further research closes.

EARLY EVIDENCE

The effectiveness evidence is strong — the tutoring-effectiveness literature is among the more robust in education — and GIFT-based studies continue to demonstrate learning gains. The bottleneck is not efficacy.[3] Because the supporting literature is among the more robust in the field, a decision-maker hesitating to field adaptive tutoring cannot honestly point to weak evidence as the reason; the hesitation traces instead to the missing pathway that would let the proven approach be bought, integrated, and made routine.

OPEN PROBLEMS

What has not been built is the adoption pathway: procurement that can buy adaptive tutoring, integration into existing training pipelines, instructor-workflow redesign, and the authority structure to make adaptive tutoring a default rather than an experiment.[4] GIFT is the case in this book closest to the LENS discipline itself — proof that a working technology and settled science do not adopt themselves, and that the adoption pathway is an engineering deliverable in its own right. Each of those pieces — a contracting vehicle, a pipeline integration, a redesigned instructor workflow, an owner with authority — is a concrete artifact someone must build, and their absence, not any technical shortfall, is what keeps the framework experimental.



ADAPTIVE INSTRUCTION WORKED; ADOPTION DID NOT CROSS THE BOUNDARY

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THE LEARNING ENGINEERING LENS

> *The technology works. The institutional pathway to ubiquitous fielded use does not yet.*
>
> EDITORS' SYNTHESIS OF THE GIFT ADOPTION RECORD

LE INSIGHT

GIFT is the most directly relevant case in this book to the learning-engineering discipline itself. The technology works, the learning science works, the framework is active and supported. What has not been built is the institutional adoption pathway — procurement, training-pipeline integration, instructor workflow redesign, the authority structure — that would make adaptive tutoring a default rather than an experiment.

LENS APPROACH

LENS treats GIFT in LEN 1 as the problem-framing case for the discipline, in LEN 8 as the foundational adoption-pathway case, and in LEN 10 (capstone) as a prompt for designing the institutional deliverables that would close an adoption gap of this shape.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the adoption pathway — procurement, pipeline integration, instructor-workflow redesign, authority — as a named deliverable of the program, not a follow-on hope.
- Build the tutoring capability against an existing training pipeline so integration is designed in, rather than fielding a framework and expecting pipelines to bend to it.
- Specify who holds the authority to make adaptive tutoring a default, so the decision to scale is owned rather than left to volunteers.

IN OPERATION / AFTER

- Measure fielded routine use, not just study-level learning gains, so the institution can see whether adoption is actually happening.
- Sustain the open-source framework and its community so the authoring-cost barrier it lowered does not quietly rise again as releases age.
- Audit each stalled pipeline to find which adoption artifact is missing — contracting vehicle, integration, workflow, owner — and treat that as the engineering gap to close.

REFLECTION QUESTIONS

1. GIFT exists, is supported, and works. Adoption at scale does not. What is the equivalent in your domain — an effective intervention whose adoption pathway has not been engineered?

2. Design the institutional adoption deliverable that would move adaptive tutoring from “available framework” to “default routine practice” in one operational training pipeline.

3. GIFT’s bottleneck is procurement, pipeline integration, instructor-workflow redesign, and authority — not efficacy. For an effective tool in your domain, which of those four is the binding constraint, and who would have to own it for adoption to become the default?

WHO BUILDS THIS knowledge management & data engineering · governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS knowledge bases · data pipelines · governance frameworks · bias & evidence audits · incident analysis · safety dashboards

CASE 34 HEALTHCARE ONGOING

Implementation Science in Healthcare — The 17-Year Gap

K Knowledge & Institutional Memory G Governance & Trust N Normalization of Deviance

IMPACT **Average time from research finding to clinical practice: 17 years; only ~14% of research findings ever reach practice**

IN BRIEF

Implementation science has a canonical finding: it takes an average of about seventeen years for research evidence to reach clinical practice, and only roughly 14% of research findings ever make it at all. This is not a single incident but a systemic condition — effective interventions exist; the system to adopt, sustain, adapt, and measure them at scale does not. Frameworks like the Active Implementation Frameworks (Fixsen et al., 2005) and EPIS (Aarons et al., 2011) were built specifically to attack this gap, and LENS threads implementation science throughout its curriculum in direct response. The seventeen-year gap is the meta-case for the whole book: every success case is a closure of this gap in one domain, every failure case the gap left open. The discipline exists to make seventeen years shorter.

THE SHIFT

Medicine generates more validated knowledge than it can absorb. Implementation science arose to study a stubborn fact: knowing what works and having it practiced are different problems, separated by years.[0] Treating the two as one problem is the error the field formed to correct: a validated finding is not yet a changed practice, and the distance between them is itself a phenomenon to be studied, measured, and engineered rather than waited out.

WHAT IS EMERGING

The canonical figures are stark: it takes an average of about seventeen years for research evidence to be integrated into clinical practice, and only roughly 14% of research findings ever make it at all.[1] Read together, the two figures describe a pipeline that is both slow and leaky: most of what is learned never reaches the bedside at all, and the fraction that does arrives long after the patients who first needed it, so the delay is compounded by sheer attrition.

THE CAPABILITY QUESTION

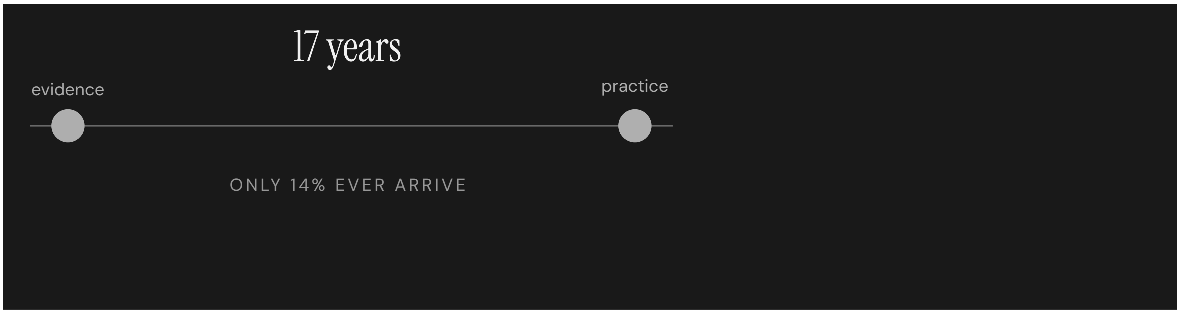
This is not a single case but a systemic condition — the same structural problem the medical-error data (Case 102) describes from the outcome side. Effective interventions exist; the institutional system to adopt, sustain, adapt, and measure them at scale does not.[2] Where the medical-error data counts the harm at the far end of the pipeline, the translation figures name the mechanism that produces it: the same missing adoption-and-measurement system shows up as a delay from one vantage and as a body count from the other.

EARLY EVIDENCE

Frameworks built to attack the gap — the Active Implementation Frameworks (Fixsen et al., 2005) and the EPIS framework (Aarons et al., 2011) — show that implementation can be engineered rather than left to chance, and they inform LENS's choice to thread implementation science through every course rather than isolate it in a module.[3] Threading the discipline through every course rather than confining it to a single module is itself a claim these frameworks support: if implementation is an engineerable property of any intervention, it cannot be quarantined as a specialty and must inform how every design is taught.

OPEN PROBLEMS

The seventeen-year gap is the meta-case for this book. Every success case in the chapters ahead is a closure of the gap in one domain; every failure case is the gap left open.[4] The open problem is general and unglamorous: building, funding, and owning the adoption-and-measurement pathway that turns a proven intervention into routine practice — which is, in one sentence, what the LENS discipline exists to do. Because the problem is general rather than domain-specific, no single clinical result closes it; what closes it is the repeatable, owned, and funded pathway that any proven finding can be run through, again and again.



RESEARCH TO PRACTICE — THE CANONICAL IMPLEMENTATION GAP

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4. G. Aarons et al. (2011), the EPIS framework; L. Damschroder et al. (2009), CFIR.
5. Cf. medical error as systemic failure (Case 102); Goodell & Kolodner, Learning Engineering Toolkit (2022).

THE LEARNING ENGINEERING LENS

> *The answer is 17 years. What is the question?*
>
> MORRIS, WOODING & GRANT, J ROYAL SOC MED, 2011

LE INSIGHT

The 17-year gap is the structural problem that LENS exists to address. It is the difference between knowing what works and having it deployed. Every case in this book is a sample from a distribution governed by that gap. The success cases shorten it; the failure cases let it run.

LENS APPROACH

LENS uses this case in LEN 1 as the foundational problem statement of the discipline, in LEN 10 as a studio constraint (designs must consider implementation, not just efficacy), and in LEN 8 as the central knowledge-transfer challenge. The case is referenced at least once in every required course.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design every intervention with its adoption-and-measurement pathway attached, so implementation is engineered in rather than left to chance after the evidence is published.
- Use an implementation framework (Active Implementation, EPIS) from the outset to plan adoption, sustainment, and adaptation as deliverables of the project.
- Name an owner and a funding line for the pathway, since a proven finding with no one accountable for fielding it is exactly what the gap is made of.

IN OPERATION / AFTER

- Measure both reach and speed — what fraction of practice has adopted the intervention and how long it took — to see the slow-and-leaky pipeline rather than assume publication equals uptake.
- Sustain and adapt fielded interventions, treating drift back to old practice as a measurable failure mode, not a one-time rollout that holds itself.
- Track the gap as a standing metric across the institution, so closing it in one domain becomes a repeatable pathway rather than a one-off success.

REFLECTION QUESTIONS

1. Pick an evidence-based intervention in your domain. Estimate the gap between when the evidence became conclusive and when the intervention reached majority of practice. What did the gap cost?

2. Design the deliverable that would shorten that gap by half in your domain. Be specific about who funds it, who owns it, and what evidence demonstrates the reduction.

3. The translation pipeline is both slow and leaky — most findings never reach practice, and those that do arrive late. For your domain, is the binding problem the delay or the attrition, and what would you measure to tell which one to attack first?

WHO BUILDS THIS knowledge management & data engineering · governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS knowledge bases · data pipelines · governance frameworks · bias & evidence audits · incident analysis · safety dashboards

CASE 35 CORPORATE L&D TRAINING TRANSFER ORGANIZATIONAL BEHAVIOR 2010

Training Transfer — The Gap Between Learning and Doing

K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT A meta-analysis of 89 empirical studies finds training transfer is positively related to cognitive ability, conscientiousness, motivation — and decisively to a supportive work environment; the literature carries explicit hedges about inconsistent measurement and significant variability

IN BRIEF

Blume, Ford, Baldwin, and Huang (Journal of Management 2010) synthesized 89 empirical studies on training transfer — the extent to which what is learned in training produces meaningful change in on-the-job performance. The headline finding is that transfer is positively related to four categories of variable: cognitive ability, conscientiousness, motivation, and the work environment. Of these, the work environment — particularly supervisor and peer support — is among the strongest predictors, and the most decisive at the **system** layer rather than the individual layer. The authors are explicit, and the load-bearing hedge survives into the case: the literature is characterized by "inconsistent measurement of transfer and significant variability in findings," and downstream practitioner summaries note that organizations frequently see limited return because learning fails to transfer to the workplace. The case is the paired peer-reviewed half of the corporate-L&D pair with Kirkpatrick (Case 108): together they close the gap the v1 corpus had open at the workforce-L&D layer, and they motivate the LENS framing that the human is the biggest variable at the interface — here, the return-to-work interface.

THE SHIFT

Training transfer is the structural question corporate L&D sits inside: does what learners did in training produce observable change on the job, and what predicts whether it does? The Kirkpatrick chain of evidence (Case 108) frames the question; the Blume et al. meta-analysis is the strongest single peer-reviewed answer the field has consolidated.[0]

WHAT IS EMERGING

Blume, Ford, Baldwin, and Huang synthesized 89 empirical studies, with several thousand learners in aggregate, and decomposed transfer predictors into four categories: cognitive ability, conscientiousness, motivation, and the work environment (particularly supervisor and peer support). All four categories were positively related to transfer; of them, the work environment was among the strongest predictors, and was the only one that is decisively a **system** variable rather than a learner variable.[1]

THE CAPABILITY QUESTION

The load-bearing hedge survives. The authors are explicit that the literature is "characterized by inconsistent measurement of transfer and significant variability in findings." Downstream practitioner summaries note that organizations frequently see limited training return because learning fails to transfer to the workplace — not because the training did not work in the classroom but because the return-to-work environment did not support application. The meta-analysis is the strongest current synthesis, and it also names what the field has not yet measured well.[2]

EARLY EVIDENCE

The teaching point at the LENS-framing layer is precise. The capability deliverable is not the training event; it is the transfer. And the decisive variable at the transfer layer is the work environment — supervisor support, peer support, the opportunity to practice on actual tasks, the absence of countervailing pressures that punish trying the new behavior. The human is the biggest variable at the interface, and the interface is the return-to-work boundary. Training that ignores this boundary is training that delivers Level 2 evidence (Kirkpatrick, Case 108) and does not deliver Level 3 capability.[3]

OPEN PROBLEMS

In pair with Kirkpatrick, the case closes the corporate-L&D gap the v1 corpus had open and is a worked example of the revised "decision-grade evidence" framing in the v2 research backbone: the strongest synthesis the field has is a meta-analysis with explicit hedges about measurement inconsistency. The practitioner must decide on what to build — training design, supervisor support, practice opportunity — knowing the evidence is the strongest available but not closed. The CLO **Judgment under inadequate evidence** is exactly the capability this meta-analysis pattern asks practitioners to bring.

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THE LEARNING ENGINEERING LENS

> *The training event is not the capability deliverable. The transfer is. And the decisive variable at transfer is the work environment.*
>
> EDITORS' SYNTHESIS OF BLUME ET AL. (2010).

LE INSIGHT

Blume et al. is the strongest current peer-reviewed synthesis on training transfer: cognitive ability, conscientiousness, motivation, and decisively the work environment predict whether training produces on-job behavior change. The literature carries explicit load-bearing hedges about inconsistent measurement and significant variability; the case is included with the hedges intact.

LENS APPROACH

Blume is the corporate-L&D transfer case (induced 2.3; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development) for the design-iteration sub-competency (the work environment is the design variable) and in Domain 4 (Test and Evaluation) for the CLO **Judgment under inadequate evidence** — the meta-analysis is the strongest synthesis the field has, and it explicitly names what it cannot settle. Direct pair with Case 108 (Kirkpatrick); together they close the corporate-L&D gap the v1 corpus had open.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the training intervention with the work environment as a design variable — supervisor briefing, peer-support structures, practice opportunity — not only the classroom content.
- Specify the on-job behavior the training is supposed to enable and the conditions under which it will be observed; the meta-analysis identifies environment as decisive, so the environment must be instrumented.
- Carry the meta-analytic hedge into the deployment documentation: the literature is the strongest current synthesis, and it explicitly notes inconsistent measurement and significant variability.

IN OPERATION / AFTER

- Track the work-environment variables — supervisor support, peer-support structures, practice opportunity — separately from the training event, and report transfer outcomes conditional on environment.
- Use the meta-analytic finding to argue for the work-environment investment in the program design, not to declare the question settled; future replications and better-measured studies will move the magnitudes.
- When transfer is low, attribute the gap with the meta-analysis in hand: was the trainee under-prepared, under-motivated, or returning to an environment that did not support application? The remediation depends on the attribution.

REFLECTION QUESTIONS

1. Identify a training program in your organization that produced strong Level 2 evidence (learning) but unclear Level 3 evidence (behavior change). What does the meta-analysis predict the decisive variable was, and how would you re-design the program with the work environment as a first-class design variable?

2. Specify the work-environment instrumentation you would put in place for a new training deployment — supervisor briefing artifacts, peer-support structures, on-job practice opportunity — so transfer can be measured conditional on environment, not just on training fidelity.

3. The meta-analysis is the strongest current synthesis and carries explicit hedges about measurement inconsistency. What is the minimum additional evidence you would require before treating any specific magnitude from this literature as settled in your decision-making?

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CASE 36 K-12 EDUCATION SCALABLE INTERVENTIONS EQUITY 2019

Growth-Mindset National Experiment — Heterogeneous Effects

D Designed Out N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT A nationally representative RCT of US 9th-graders found a less-than-1-hour online growth-mindset intervention improved grades among lower-achieving students and increased advanced-math enrollment, but the effect was conditional on peer norms — the intervention changed grades only where peer norms aligned with the intervention's message

IN BRIEF

Growth-mindset interventions — short, scalable psychological interventions that teach students that intellectual ability is malleable rather than fixed — accumulated a substantial laboratory and small-school evidence base across the 2000s and 2010s. The open question, framed in the broader scalable- interventions literature, was whether the effects survived at population scale and what the structural moderators were. Yeager et al. (Nature, 2019) ran the test that became the field's reference point. A nationally representative RCT of US 9th-graders received a less-than-1-hour online growth-mindset intervention; the trial documented improved grades among lower-achieving students and increased advanced-math enrollment relative to control. The headline finding for the case is the conditional: the effect was conditional on peer norms. The intervention changed grades only where peer norms aligned with the intervention's message, and the study is notable for treating treatment-effect heterogeneity as the finding rather than as a nuisance. The intervention is not universally effective; naming where it does and does not work is the contribution. The "conditional on peer norms" language survives verbatim into the case. Pair with Case 133 (Gándara) at the scalable-equity-intervention layer.

THE SHIFT

Growth-mindset interventions teach students that intellectual ability is malleable — improvable through effort, strategy, and help-seeking — rather than fixed. The pedagogical claim, developed across two decades of research (Dweck and colleagues), is that students who hold the malleable view respond more constructively to academic challenge: they treat difficulty as informative rather than as a signal of fixed ability, and they persist on problems that the fixed-view student would interpret as evidence to stop. The laboratory and small-school evidence base accumulated substantially across the 2000s and 2010s; the open question, by the late 2010s, was whether the effects survived at population scale and what the structural moderators were.[0]

WHAT IS EMERGING

Yeager et al. (Nature, 2019) ran the trial that became the field's reference point. The design was a nationally representative RCT of US 9th-graders — drawn from a sample stratified to reflect the US ninth-grade population — with the intervention delivered as a less-than-1-hour online module. Outcomes included grades, course-taking, and contextual measures of the classroom and peer environment. The trial pre-registered the moderator analysis the case now anchors on: the effect of the intervention was hypothesized in advance to depend on the peer-norm environment the student returned to after the module.[1]

THE CAPABILITY QUESTION

The headline outcome was a positive finding with structure. The growth-mindset intervention improved grades among lower-achieving students and increased advanced-math enrollment, relative to the active-control condition. The structural finding — the one the case is built around — is the conditional: the effect was conditional on peer norms. The intervention changed grades only where peer norms aligned with the intervention's message — where the peers treated the malleable view as legitimate and the help-seeking behavior the intervention encouraged as socially acceptable. In peer environments where the fixed view was the local norm, the intervention's effect on grades was substantially smaller or absent.[2]

EARLY EVIDENCE

What makes the case methodologically important is the authors' explicit treatment of treatment-effect heterogeneity as the finding rather than as a nuisance to be averaged away. The trial's structural answer is that the intervention is not universally effective, and naming where it does and does not work is the scientific contribution. The "conditional on peer norms" language survives verbatim into the case because it is the load-bearing hedge: a headline-only reading ("growth mindset works at population scale") misses the substance, and an opposite-headline reading ("growth mindset doesn't work") misses it equally. The contribution is the conditional, and the design — a pre-registered moderator analysis with a nationally representative sample — is what makes the conditional defensible.[3]

OPEN PROBLEMS

Drafted alongside Case 133 (Gándara) at the scalable-equity-intervention layer, the case carries the design-predictions-to-trigger-support pattern (induced 8.3, LENS D2/PT5). The intervention targets the students who benefit (lower-achieving students in peer environments where the norm permits the change), rather than applying a blanket treatment that the average effect would credit and the heterogeneity would conceal. The equity-relevant design commitment is to surface the heterogeneity as part of the deliverable, not to launder it through an average-effect headline. The case is the methodologically clean model of how a population-scale RCT can earn the heterogeneity-as-finding stance, and the "conditional on peer norms" qualification is what makes the result useful for the next adaptation.

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THE LEARNING ENGINEERING LENS

> *The effect was conditional on peer norms.*
>
> YEAGER ET AL., NATURE 2019.

LE INSIGHT

The growth-mindset national experiment is the methodologically clean model of how a population-scale RCT can earn the heterogeneity-as-finding stance. The intervention improved grades among lower-achieving students and increased advanced-math enrollment — and the effect was conditional on peer norms. The qualifying language is the load-bearing hedge; headline-only readings in either direction miss the substance.

LENS APPROACH

Yeager et al. 2019 is the designing-predictions-to-trigger- support case (induced 8.3; LENS D2/PT5) — Domain 2 for pre-registered moderator analysis; Domain 5 for the equity-relevant commitment to target support to those who benefit. Pair with Case 133 (Gándara).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pre-register the moderator analysis at the design stage; the heterogeneity-as-finding stance depends on the moderator being a planned analysis rather than a post-hoc inspection.
- Sample so that the moderator can be estimated — a nationally representative sample of US 9th-graders, in this case — so the heterogeneity is observed across the population the headline claim would otherwise speak to.
- Treat the intervention as scalable on the headline finding and conditional on the moderator at the same time; designing predictions to trigger support means targeting the support to those who benefit, in the environments where the support can land.

IN OPERATION / AFTER

- Carry the conditional language ("effect was conditional on peer norms") verbatim into every downstream communication; a headline-only reading and an opposite-headline reading both miss the substance.
- Treat the treatment-effect heterogeneity as program documentation, not as a nuisance to laundered through an average effect; the equity-relevant design commitment is to surface the heterogeneity, not to conceal it.
- Build the next adaptation against the moderator: where would the intervention work, what peer-norm context would it land in, and what is the institutional path to creating that context where it does not yet exist?

REFLECTION QUESTIONS

1. Identify a scalable intervention in your domain whose published evidence runs at the average-effect level. What pre-registered moderator analysis — and what sampling design — would let you treat treatment-effect heterogeneity as the finding rather than as a nuisance?

2. The Yeager finding is "effect was conditional on peer norms." What is the analog conditional in your context — the structural moderator the intervention's effect depends on — and how would you instrument the moderator at scale?

3. The equity-relevant design commitment is to target support to those who benefit rather than to apply a blanket treatment. What would the targeting mechanism look like in your context, and how would you guard against the targeting collapsing into a gatekeeping pattern the prediction was supposed to avoid?

WHO BUILDS THIS systems & human-factors engineering · safety science & organizational analysis · knowledge management & data engineering — plus domain experts and a learning engineer to integrate. TOOLS task analysis · design prototyping · incident analysis · safety dashboards · knowledge bases · data pipelines

CASE 37 DEFENSE 2018 – PRESENT

Navy Surface Warfare Readiness Reform

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Tripled ship-driving training hours; 10 pass-or-fail career assessments; Ready-for-Sea Assessments — 3 of 18 forward-deployed ships immediately sidelined

IN BRIEF

After the fatal 2017 Fitzgerald and McCain collisions (Case 1) exposed gutted seamanship training, the U.S. Navy overhauled how it builds surface-warfare competence. It restored the Surface Warfare Officers School from CD-ROM self-study to classroom and simulator instruction, stood up Maritime Skills Training Centers on both coasts, created ten pass-or-fail career assessments — three of them no-go gates — and adopted aviation-style debriefing. New Ready-for-Sea Assessments evaluated forward-deployed ships against a deliverable standard; three of the first eighteen were immediately sidelined. The structural change is real and the investment substantial. What is missing is the third half: the GAO has noted the Navy still lacks systematic evaluation of whether the reforms work — a live success in structure, with evidence of effect still outstanding.

BACKGROUND

In 2017, the destroyers USS Fitzgerald and USS John S. McCain suffered fatal collisions (Case 1) that investigations traced in part to seamanship and navigation training degraded to CD-ROM self-study. The Navy faced a clear capability gap: officers were going to sea without the hands-on ship-driving competence the job required. The diagnosis was unusually unambiguous — two avoidable collisions in one year pointing at the same eroded fundamentals — which is what converted a long-tolerated shortfall into a mandate for structural change.[0]

THE INTERVENTION

Beginning in 2018, the Navy restored the Surface Warfare Officers School from self-study to classroom-plus-simulator instruction, established Maritime Skills Training Centers on both coasts, roughly tripled ship-driving training hours, and created ten pass-or-fail assessments across an officer's career path — three of them no-go gates that can halt advancement. The no-go gates were the structural teeth: by tying advancement to demonstrated competence rather than time served, they made the qualification something the system would stop on, not merely something it recorded.[1]

HOW IT WORKED

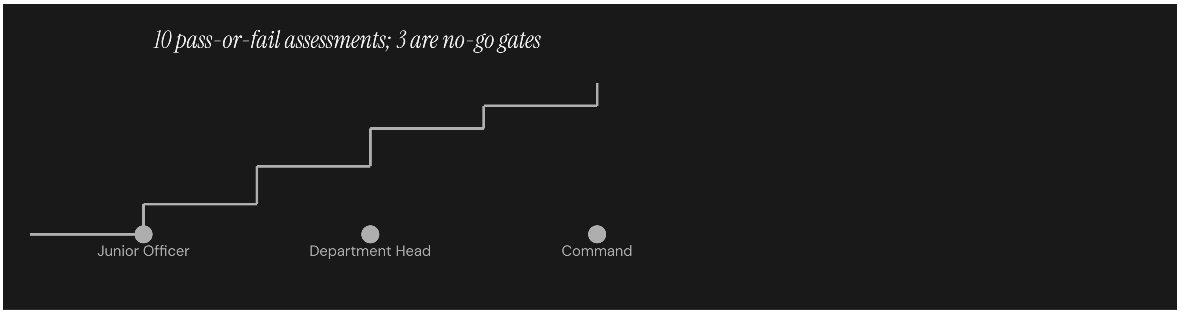
The reform paired technical investment — simulators, restored curricula, qualification gates — with a cultural change borrowed from aviation: structured debriefing and explicit gate ownership. New Ready-for-Sea Assessments evaluated forward-deployed ships against a deliverable standard rather than a paper one; three of the first eighteen ships assessed were immediately sidelined as not ready. That a sixth of the first cohort failed against the new standard showed the assessment had teeth — it measured demonstrated readiness rather than accepting the paper certifications that had masked the pre-collision gap.[2]

THE EVIDENCE

Here the case turns instructive. The Government Accountability Office has noted that the Navy still lacks systematic evaluation of whether the reforms actually improve readiness. The structural intervention happened and the investment was large, but decision-grade evidence on outcomes is incomplete — the measurement to confirm the effect has lagged the change itself. Without an outcome time-series, the Navy cannot yet distinguish a real capability gain from activity that looks like one, which is the same blindness, in milder form, that let the pre-collision gap go unseen.[3]

WHAT TRANSFERRED

The reform is a live, in-progress success: structural change real, evidence of effect outstanding. As a teaching case it argues that mature capability engineering must build the measurement infrastructure from the start — the time-series that lets an institution know whether the capability it bought is materializing — rather than treating evaluation as an afterthought. The two halves the reform did deliver, technical investment and a cultural change, are necessary but not sufficient; the missing third half is the evidence regime that would let the Navy prove, not assume, the gap has closed.[4]



POST-FITZGERALD REFORM: TRIPLED TRAINING; PASS-OR-FAIL GATES

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THE LEARNING ENGINEERING LENS

> *The Navy still lacks systematic evaluation of whether the reforms work.*

PARAPHRASING GAO-21-168 ON NAVY READINESS REFORM, 2021

LE INSIGHT

The Navy reform is a paired intervention in progress: technical (training restored, simulators procured, assessments created) plus cultural (debriefing, gate ownership). What it lacks is the third half: evidence that the intervention has worked. LENS treats this as a teachable case for what mature capability engineering should include from the start — the measurement infrastructure that lets the institution know whether its investment is producing the capability it bought.

LENS APPROACH

LENS uses Navy SWO reform in LEN 10 (capstone) as a worked exercise in capability intervention at scale, and in LEN 4 as a case where the measurement infrastructure has lagged the intervention. Students design the evidence regime that should accompany the reform.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pair the technical investment — restored curricula, simulators, training hours — with a cultural change such as structured debriefing and explicit gate ownership, rather than buying tools alone.
- Install no-go gates that tie advancement to demonstrated competence, so qualification is something the system will stop on, not merely something it records.
- Assess units against a deliverable standard from the first cohort, accepting that a real test will sideline some — that failure rate is the evidence the gate has teeth.

IN OPERATION / AFTER

- Build the measurement infrastructure from the start — the outcome time-series that lets the institution distinguish a real capability gain from activity that merely looks like one.
- Give the senior commander a readiness dashboard that reports demonstrated capability over time, so the same blindness that masked the pre-collision gap cannot recur.
- Treat evidence of effect as the third half of the intervention, not an afterthought, so the reform can be proven rather than assumed to have closed the gap.

REFLECTION QUESTIONS

1. Navy SWO reform happened without an evidence regime to confirm it worked. Identify a current reform in your domain and the evidence regime that should accompany it.
2. Design the dashboard the Chief of Naval Operations should have to know whether SWO capability is improving in time-series.
3. The Ready-for-Sea Assessment sidelined three of the first eighteen ships because it measured readiness against a deliverable standard. Identify a certification in your domain that ratifies rather than tests, and design the gate that would be willing to fail a sixth of its first cohort.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 38 EDUCATION AT SCALE 1990S – PRESENT

Cognitive Tutor / Carnegie Learning

T Training Gap

IMPACT Randomized controlled trials showed significant learning gains; RAND study found positive effects on Algebra I achievement; adopted across 3,000+ schools

IN BRIEF

Carnegie Learning's Cognitive Tutor, built from John Anderson's ACT-R cognitive architecture at Carnegie Mellon, is the most rigorously evaluated intelligent tutoring system in education. It uses Bayesian knowledge tracing to model each student's mastery and adapts instruction accordingly, and a RAND Corporation evaluation found statistically significant positive effects on Algebra I achievement. The system is a learning-engineering success in the discipline's own terms — learning science to engineered software to randomized-trial evidence to deployment across 3,000-plus schools. Its limitations are instructive: it works best in well-defined domains like algebra and less well in ill-structured ones, making it the canonical evidence that the pipeline delivers for problems that fit it — leaving open whether the same discipline can deliver where problems do not.

BACKGROUND

For decades, intelligent tutoring systems promised individualized instruction at scale, but few were grounded in a validated theory of how people learn or rigorously tested for effect. The opportunity was to build a tutor from a real cognitive model and prove its impact with the methods of experimental science. The field's recurring weakness was that promising systems rested on intuition about learning rather than a validated theory, and were rarely subjected to the kind of controlled trial that could separate genuine effect from novelty.[0]

THE INTERVENTION

Carnegie Learning's Cognitive Tutor, developed from John Anderson's ACT-R cognitive architecture at Carnegie Mellon, models the specific skills underlying a subject and tracks each student's mastery using Bayesian knowledge tracing. It adapts problem selection to the individual learner and provides step-level feedback, embodying a full learning-science theory in software. Grounding the tutor in ACT-R rather than designer intuition is what made the system testable: a theory that decomposes a subject into specific skills can be turned into a measurement of mastery and an instrument that responds to it.[1]

HOW IT WORKED

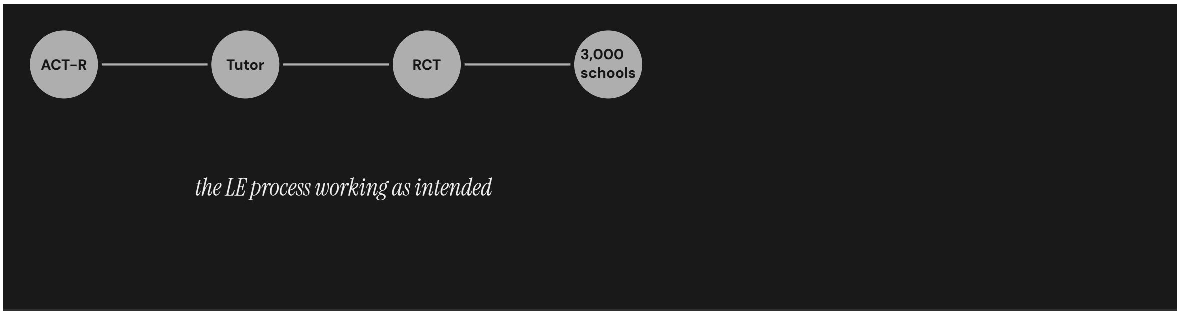
The tutor's effectiveness rests on the chain from theory to instrument: a decomposable skill model, a measurement method (knowledge tracing) that estimates mastery from student actions, and an instrumentable interface that adapts in response. Instruction is targeted where the model detects weakness, so practice concentrates on skills not yet mastered rather than on a fixed sequence. Each link in that chain has to hold for the next to work — the skill model makes mastery measurable, the measurement makes adaptation possible, and the adaptation is what concentrates practice where it pays off.[2]

THE EVIDENCE

The RAND Corporation's multi-site evaluation found statistically significant positive effects on Algebra I achievement, and the program scaled to more than 3,000 schools. The case demonstrates the learning-engineering process working end to end: learning science, to engineered software, to randomized-controlled-trial evidence, to scaled implementation. A multi-site randomized evaluation is the strong form of the claim — it shows the effect survived contact with many real classrooms rather than a single favorable setting, which is what the field's earlier, untested systems had lacked.[3]

WHAT TRANSFERRED

The limitations are as instructive as the success. Cognitive Tutor performs best in well-defined domains like algebra and less well in ill-structured ones, making it the canonical evidence that the learning-engineering pipeline works for problems that fit the pipeline. The frontier question — whether the same discipline can deliver in operational, ill-structured domains where capability matters most — remains open. The dependence on a decomposable skill model is the boundary condition: where a subject cannot be cleanly broken into trackable skills, the very chain that made algebra tractable has nothing to attach to.[4]



THEORY → ENGINEERING → EVIDENCE → SCALE

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> *Cognitive Tutors demonstrate the LE process working: theory → engineering → evidence → scale.*
>
> EDITORS' SYNTHESIS OF ANDERSON ET AL. (1995) AND KOEDINGER & CORBETT (2006)

LE INSIGHT

Cognitive Tutor is the canonical evidence that the learning- engineering process exists, works, and produces measurable benefits at scale — when the problem is tractable. It is also the canonical case for what tractability looks like: well-defined domain, decomposable skill model, instrumentable interface, rigorous evaluation. The frontier for the discipline is whether the same pipeline can deliver in the operational, ill-structured domains where capability matters most.

LENS APPROACH

LENS uses Cognitive Tutor in LEN 1 as the foundational LE-process exemplar, in LEN 4 as the canonical case for Bayesian knowledge tracing as a measurement instrument, and in LEN 9 as a technical case for model-based adaptive instruction.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Ground the intervention in a validated theory that decomposes the subject into specific skills, rather than building on designer intuition about how people learn.
- Build the full chain — a skill model, a measurement method that estimates mastery from learner actions, and an interface that adapts in response — so each link supports the next.
- Target instruction where the model detects weakness, concentrating practice on unmastered skills rather than marching through a fixed sequence.

IN OPERATION / AFTER

- Prove the effect with a multi-site randomized evaluation, so the gain is shown to survive many real settings rather than one favorable classroom.
- Scale only where the problem fits the pipeline — well-defined, decomposable domains — and treat the decomposability of a skill model as the boundary condition for applicability.
- Keep the frontier question explicit when extending the method, testing whether the same theory-to-instrument chain can be built in less-structured domains before assuming it transfers.

REFLECTION QUESTIONS

1. Cognitive Tutor works best in well-defined domains. Identify a problem in your domain that is currently ill-structured. What would have to be true to make the LE pipeline applicable?
2. The case is the strongest evidence the LE process works. What is the equivalent piece of evidence required to demonstrate the same pipeline in a non-cognitive domain — for example, surgical skill, or operational watchstanding?
3. Cognitive Tutor's chain runs from a decomposable skill model through measurement to adaptation. Take a capability in your domain and attempt the decomposition — where it resists being broken into trackable skills, what does that tell you about whether the pipeline can apply?

WHO BUILDS THIS learning science & instructional design — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models

CASE 39 WORKFORCE DEV STEM TRAINING EQUITY 2017 – 2023 (SIX CYCLES)

CIRCUIT — A Scalable, Equity-Centered Research Workforce Model

T Training Gap K Knowledge & Institutional Memory

DISCLOSURE An editor of this volume is the senior author of the underlying study. The case is included on the strength of the published peer-reviewed evidence (ASEE 2023, peer-reviewed full paper with DOI); the editorial framing has been written to maintain a critical distance from the program's own self-presentation, and the open question about external/comparative evaluation is preserved in the case text.

IMPACT An eight-pillar cohort model that in 2022 supported over 100 undergraduate, graduate, and ROTC students from 'trailblazing' backgrounds (first-generation, low-income, limited prior research access); peer-reviewed program description with longitudinal student outcomes across six cycles

IN BRIEF

CIRCUIT is a research workforce-development program at Johns Hopkins APL built on eight explicit pillars — holistic recruiting, mission-driven research, targeted technical training, leadership development, high-resolution assessment, diverse mentorship, university partnerships, and career empowerment. Cervantes, Floryanzia, Sharp, Gray-Roncal, and Johnson ("Empowering Trailblazers toward Scalable, Systematized, Research-Based Workforce Development," ASEE Annual Conference 2023) presents the model and reports longitudinal student outcomes gathered over six program cycles. In 2022 the program supported over 100 undergraduate, graduate, and ROTC students, positioning CIRCUIT as a replicable model for STEM recruitment and retention of underrepresented students. The strongest honest framing — preserved in the case — is that this is a self-authored multi-cycle program evaluation at a single program; an external comparative evaluation would strengthen the causal claim, and the case says so rather than overstate. The COI render under the title (editor is the senior author) is binding. The case is the paired peer-reviewed companion to CIRCUIT proofreading (Case 81) — that case is about deploying capability against automation failure; this case is about building the capability in the first place at the edge of the trainees' prior preparation.

BACKGROUND

The recurring story in STEM workforce-development at the undergraduate level is that the pipeline narrows at every stage, and that the narrowing falls disproportionately on students whose prior preparation has not included the kind of access — to research labs, to mentorship networks, to technical-skills training that meet a research project's immediate needs — that converts undergraduate interest into research capability. CIRCUIT was built to engineer that conversion at the edge of trainees' prior preparation, across a cohort drawn from "trailblazing" backgrounds: first-generation college students, low-income students, and students with limited prior research access.[0]

THE INTERVENTION

The program model is the case's first contribution. Eight pillars are named and operationalized: holistic recruiting that does not screen on credentials a trailblazer's background would not have generated; mission-driven research that lets trainees see why the technical skill they are acquiring matters; targeted technical training built around the project's immediate needs; leadership development; high-resolution assessment (the assessment is the high-resolution version, not a summative pass/fail); diverse mentorship; university partnerships that route the cohort across institutions; and career empowerment that sustains the capability beyond the program. The model is published as a model in the ASEE paper — a peer-reviewed full paper with DOI, not an institutional press release.[1]

HOW IT WORKED

The longitudinal outcome evidence is the case's second contribution. The ASEE paper reports outcomes gathered over the six program cycles from 2017 through 2023 — cohort sizes growing year over year, with over 100 students supported in
2022. The program presents itself as a replicable model,
with the documentation, assessment instruments, and pillar- level operationalization that a replicating institution would need.[2]

THE EVIDENCE

The honest framing preserved in the case is the one the editorial discipline demands. This is a self-authored multi-cycle program evaluation at a single program. The ASEE paper clears the peer-review bar that the build list 02 requires; the model and the operational evidence are auditable. What an external comparative evaluation — by a researcher unaffiliated with the program — would add is the causal-claim half of the evidence: did CIRCUIT produce these outcomes, or did the cohort selection produce them? The case names the open question rather than answering it.[3]

WHAT TRANSFERRED

In pair with Case 81 (CIRCUIT proofreading + MICrONS), the case completes the CIRCUIT picture: building the capability (this case, peer-reviewed) and deploying it against automation failure (Case 81, frontier with evidence-tier flag). The pair also exercises the corpus's COI discipline — both cases carry editor-related COI, both are rendered with the standing gold-bordered "Disclosure" block under the title, and both are anchored to the strongest peer-reviewed evidence available with the institutional and program-evaluation gap honestly named. The book's credibility on these cases rests on plain disclosure, not on hiding the affiliation.

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2. CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional program description.
3. National Academies / NSF research on undergraduate research experience effects on STEM persistence (broader literature against which the case sits).
4. v2 paired case (119) — CIRCUIT proofreading + MICrONS — the deployed-capability companion.

THE LEARNING ENGINEERING LENS

> *The deliverable is the replicable model, not the program brand. The pillars are operationalized for independent evaluation.*
>
> EDITORS' SYNTHESIS OF CERVANTES ET AL. (2023).

LE INSIGHT

CIRCUIT is a peer-reviewed eight-pillar workforce-development program for STEM trainees from trailblazing backgrounds, with longitudinal outcomes over six program cycles. The strongest honest framing is self-authored multi-cycle program evaluation; an external comparative would add the causal half. COI under the title — editor is senior author — is binding.

LENS APPROACH

CIRCUIT workforce model is the equity-engineered workforce-development case (induced 1.2; LENS D2/PT4; with 8.3 alternate). LENS uses it in Domain 2 (Iterative Development) for the operationalized pillar model and the multi-cycle iteration evidence; in Domain 4 (Test and Evaluation) for the high-resolution assessment; and in Domain 5 (Navigating Sociotechnical Constraints) for the holistic-recruiting design choice that converts equity from rhetoric to enrolled cohort. Pair with Case 81 (CIRCUIT proofreading) — building capability vs. deploying it against automation failure. COI render is binding.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Operationalize the program pillars at the same level of detail an external replicator would need; the deliverable is not the program brand, it is the replicable model.
- Build holistic recruiting to actively de-weight credentials a trailblazer background would not have generated; the design choice converts the equity commitment from rhetoric into engineered enrollment.
- Pair high-resolution assessment with the targeted training so feedback can move at the cadence the work demands, not the semester boundary.

IN OPERATION / AFTER

- Commission external comparative evaluation alongside the self-evaluation; the causal-claim half of the evidence is what the program cannot produce on its own, and the case is honest about the gap.
- Publish the model documentation, instruments, and pillar operationalization openly so the model can be replicated and evaluated by other institutions on independent cohorts.
- Render the COI disclosure under the title — the standing language — and preserve the editorial critical distance from the program's own self-presentation in any drafting.

REFLECTION QUESTIONS

1. Identify a STEM workforce-development program in your domain. Which of CIRCUIT's eight pillars are operationalized at the level of detail an external replicator would need, and which are at the level of brand? Where would replication actually be testable?

2. Specify the external comparative evaluation you would commission alongside your own program evaluation. What independent cohort, what instrument, on what cadence — and who is unaffiliated enough to evaluate?

3. The COI render under this case's title is the standing language for editor-author cases. Identify a case in your domain where the program's evaluation is conducted by people with a stake in the program. What is the disclosure architecture that keeps the evaluation honest without hiding the affiliation?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines

CASE 40 EDUCATION AT SCALE MACHINE LEARNING MEMORY AND SKILL 2016

Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale

T Training Gap D Designed Out

IMPACT Settles & Meeder (ACL 2016) introduced Half-Life Regression (HLR), a trainable spaced-repetition model that learns per-item forgetting rates from large-scale learner data; HLR was deployed in Duolingo's production review-scheduling system and the published evaluation reports improvements over heuristic schedulers (Leitner, Pimsleur) on Duolingo's own predictive metric and on a 14-day return-engagement metric

IN BRIEF

Settles and Meeder (Association for Computational Linguistics, 2016) introduced Half-Life Regression (HLR), a trainable statistical model for spaced repetition in language learning. HLR combines the psychological theory of memory half-life (Ebbinghaus's forgetting curve and its descendants) with a learned regression that estimates each item's per-learner half-life from observed practice history and item features. The model was deployed in Duolingo's production review- scheduling system; the published evaluation compares HLR against heuristic schedulers (Leitner spacing, Pimsleur intervals) on Duolingo's predictive recall metric and on a 14-day learner return-engagement metric, and reports improvements on both. The case is one of the few published instances of a spaced-repetition algorithm being deployed and evaluated against meaningful behavioral outcomes at consumer scale (Duolingo had tens of millions of active learners at the time of the study). The hedges that survive into the case verbatim: the evaluation is a single-vendor study, the "learning" outcome is measured by Duolingo's predictive metric and 14-day return rather than independent language-proficiency assessment, and the generalization to other content domains rests on the structural argument rather than a multi-vendor evidence base.

BACKGROUND

The psychological theory of memory has carried, since Ebbinghaus's 1885 forgetting-curve work, the finding that review timed near the point of forgetting produces stronger long-term retention than either massed practice or uniform-interval review. The applied descendants — Leitner boxes (1972), Pimsleur graduated-interval recall (1967), SuperMemo's SM-2 algorithm (1985, and successors) — are all attempts to operationalize the theory as a schedule. Each uses heuristics: items advance through fixed intervals on success and reset on failure. The heuristics work at the population level; they are coarse at the per-learner, per- item level where the actual half-life varies.[0]

THE INTERVENTION

Settles and Meeder's contribution was to treat the half- life as a quantity that can be learned from data. Half- Life Regression (HLR) models the probability that a learner recalls an item at a given delay as an exponential decay whose half-life is a regression on item features (language, part of speech, difficulty proxies) and per-learner practice history. The model is fit on aggregated Duolingo practice data — hundreds of millions of trials — and produces per-item half-life estimates that drive the scheduling: review when the estimated recall probability falls to a target threshold.[1]

HOW IT WORKED

The published evaluation compares HLR against several baselines on two outcomes. The first is Duolingo's predictive metric — how well the scheduler's model of recall matches observed recall on held-out practice data. The second is a downstream behavioral outcome: 14-day learner return rate, which Duolingo treats as a proxy for engagement-driven learning continuation. HLR improves on both axes over Leitner, Pimsleur, and a simpler logistic baseline. The paper deploys HLR into the live review-scheduling system; the evaluation includes the production-deployment outcomes, not just offline validation.[2]

THE EVIDENCE

The case sits in the corpus because it is one of the few published, peer-reviewed deployments of a spaced- repetition algorithm at consumer scale with reported behavioral outcomes. Duolingo's scale at the time of publication — tens of millions of active learners — meant the study had statistical resolution that the academic spaced-repetition literature rarely achieves. The contribution to LENS is the worked example of iterative design in the Domain 2 sense: a learning- engineering model deployed against a measurable behavioral outcome, with the outcome instrumented in the production system rather than inferred from offline validation.[3]

WHAT TRANSFERRED

The hedges have to survive verbatim. This is a single- vendor study published by Duolingo researchers about a Duolingo product; independent replication at comparable scale is not in the literature. The behavioral outcome is Duolingo's 14-day return rate, which is a sensible proxy for engagement but is not a language-proficiency assessment — the case does not say HLR makes learners more proficient than Leitner does; it says HLR does better on the predictive metric and the return rate. Generalization to other content domains (clinical knowledge, technical skill, vocabulary outside language-learning) rests on the structural argument that per-item half-life estimation should outperform heuristic scheduling, not on a multi-domain replication base. The case teaches the trainable- scheduling form, with the qualification that the strongest available evidence for the form is single-vendor.

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4. Leitner, S. (1972), So lernt man lernen — Leitner-box spacing heuristic.

THE LEARNING ENGINEERING LENS

> *HLR does better on the predictive metric and on 14-day learner return. The case does not say it makes learners more proficient than Leitner does.*
>
> EDITORS' SYNTHESIS OF SETTLES & MEEDER (2016).

LE INSIGHT

Half-Life Regression is one of the few peer-reviewed spaced-repetition deployments at consumer scale with reported behavioral outcomes. The structural argument — learn the half-life rather than fix it by heuristic — is strong; the evidence is single-vendor, the outcome is an engagement proxy not a proficiency assessment, and the cross-domain generalization rests on the structural argument rather than replication.

LENS APPROACH

Duolingo HLR is the trainable-scheduling consumer-scale case (induced 2.3; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development) for the production-deployment evaluation form, and in Domain 4 (Test and Evaluation) for the engagement-proxy-vs-proficiency-assessment distinction. Pair with the spaced-education clinical RCTs (Case 45) for the cross-domain replication base, and with the machine-teaming AI-delegation typology where the delegation is to a scheduler optimizing on a proxy metric.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the scheduling parameter (item half-life) as a quantity to learn from data rather than to fix by heuristic. The cost of learning it is the data infrastructure the production system already has; the heuristic was a substitute for missing data, not a principled choice.
- Instrument the production system to record the practice events that the model needs (item, learner, delay, outcome) so the model can be fit and re-fit on observed behavior at consumer scale.
- Choose the deployment outcome metric carefully. Duolingo's choice — predictive recall + 14-day return — is defensible for an engagement-driven product; whether it is the right metric for a proficiency-driven application is a different decision.

IN OPERATION / AFTER

- Carry the single-vendor hedge into communication about the case. HLR is the strongest available evidence for trainable scheduling at consumer scale; it is not multi-vendor evidence and it is not a language-proficiency study.
- Distinguish the structural argument (per-item half-life estimation should outperform heuristic scheduling) from the domain transfer claim (HLR specifically generalizes to other learning content), which the published evidence does not yet support.
- Pair the case with spaced-education clinical RCTs (Case 45) when those are drafted, so the corpus has both a consumer-scale single-vendor deployment and a clinical-domain replication base for the same underlying mechanism.

REFLECTION QUESTIONS

1. Identify a scheduling, dosing, or pacing parameter in your domain that is currently set by heuristic. What data does your production system already record that would let the parameter be learned per-unit rather than fixed?

2. Specify the outcome metric you would use for evaluating a learned scheduler. Duolingo chose engagement proxy + predictive recall; a proficiency-driven application would choose differently, and the choice is part of the evaluation design.

3. The single-vendor hedge is structural to the case. What would a multi-vendor replication base look like in your domain, and what minimum independent evidence would you require before treating a single-vendor production result as the basis for a curriculum-wide claim?

WHO BUILDS THIS learning science & instructional design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · task analysis · design prototyping

CASE 41 MEDICAL EDUCATION PATIENT SAFETY WORKFORCE DISSEMINATION 2009 – 2020S

Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA

T Training Gap K Knowledge & Institutional Memory

IMPACT Barsuk et al. (Northwestern/Feinberg) demonstrated that simulation-based mastery learning for central venous catheter (CVC) insertion reduced catheter-related bloodstream infection rates and procedural complications, with cost-effectiveness shown in a single-hospital evaluation; the program was subsequently disseminated nationally to Department of Veterans Affairs medical centers

IN BRIEF

Central venous catheter (CVC) insertion is a high-volume, high-consequence procedure whose complications — pneumothorax, arterial puncture, catheter-related bloodstream infection (CRBSI) — are well-characterized and partly attributable to procedural skill at the bedside. Barsuk and colleagues at Northwestern/ Feinberg published a series of studies through the late 2000s and 2010s establishing simulation-based mastery learning (SBML) — practice on a simulator to a defined performance standard, not to a clock — as an effective approach for trainee CVC insertion. The single-center evidence reported reduced CRBSI rates, reduced procedural complications, and favorable cost- effectiveness. The program was subsequently disseminated to Department of Veterans Affairs medical centers as a national rollout — the deliberate cross-institutional transfer that distinguishes the case from a successful single-site intervention. The case is the canonical small-tier instance of cross-domain adaptation of a proven intervention (C6.4), and pairs with the multidisciplinary-translation trio (Cases 178 team science, 122 IPE, 123 implementation-science training) as cross-domain workforce evidence. The honest hedge: the dissemination outcome literature is thinner than the original single-center evidence; the case is teachable on the SBML method and the documented dissemination effort, with measurement at the multi-site level a live evidence frontier.

BACKGROUND

Central venous catheter insertion is a procedure trainees in medicine, surgery, and critical care perform routinely; the complication profile — pneumothorax, arterial puncture, catheter-related bloodstream infection — is well-characterized and attributable in part to procedural skill at the bedside. The historical training model relied on graduated exposure on live patients, with supervision but without a defined performance standard before unsupervised practice. The Barsuk group at Northwestern's Feinberg School of Medicine, working across critical care and medical education, set out to replace exposure-based training with simulation-based mastery learning.[0]

THE INTERVENTION

Simulation-based mastery learning is the specific structural form: trainees practice the procedure on a simulator to a defined performance standard, not to a clock. Practice continues until the trainee demonstrates competence; the standard, not the calendar, governs progression. Barsuk et al. (Archives of Internal Medicine, 2009; Critical Care Medicine, Journal of Hospital Medicine, and a series of subsequent papers) reported that the SBML cohort, compared with historical and concurrent non-SBML cohorts at the same hospital, performed CVC insertions with fewer needle passes, fewer arterial punctures, and lower CRBSI rates on the patients they subsequently catheterized.[1]

HOW IT WORKED

The single-center economic evidence was the second pillar. Cohen et al. (Critical Care Medicine, 2010) reported that the SBML program was cost-saving at the single hospital, driven principally by the reduction in CRBSI cases — each bloodstream infection averted is expensive enough that even a modest reduction recovers the simulation-training investment. The combination of the procedural-outcome evidence and the cost-effectiveness evidence made the program transferable beyond the original site.[2]

THE EVIDENCE

The dissemination step is what distinguishes the case from a successful single-site intervention. The SBML CVC program was subsequently disseminated to Department of Veterans Affairs medical centers as a national rollout, supported by the VA's investment in simulation infrastructure and clinical training architecture. The deliberate cross-institutional transfer — from an academic medical center to a federated health system — is the structural feature the C6.4 cross-domain-adaptation competency exists to name. The dissemination outcome literature is thinner than the single-center evidence; multi-site CRBSI tracking is harder to attribute, and the published evidence at the national-VA scale is at the program-report and small-evaluation tier rather than the controlled-comparison tier the single-center papers established.[3]

WHAT TRANSFERRED

In pair with the multidisciplinary-translation trio (Cases 178 team science, 122 IPE, 123 implementation-science training), the Barsuk SBML case completes the workforce- evidence picture: an intervention with strong single-center controlled-comparison evidence (this case) sits alongside a structured team-science training program with validated measurement (121), the field-scale enthusiasm-evidence gap in interprofessional education (122), and the operational- practice gap inside implementation-science training programs (123). The four cases together stage the cross-domain workforce-evidence pattern: the mechanism that works at single-program scale is the demonstrator; the field-scale measurement and the multi-site dissemination evidence are what the discipline is still building.[4]

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THE LEARNING ENGINEERING LENS

> *The standard, not the calendar, governs progression. The trainee practices until they demonstrate competence; the simulator absorbs the cost of the practice.*
>
> EDITORS' SYNTHESIS OF BARSUK ET AL. (2009).

LE INSIGHT

Barsuk SBML for CVC insertion is the canonical small-tier intervention case for cross-institutional dissemination of a proven mechanism. The single-center evidence is controlled- comparison and cost-effective; the VA national dissemination is the cross-institutional transfer step. The multi-site outcome evidence is thinner than the single-center evidence — the live frontier of the dissemination case.

LENS APPROACH

Barsuk SBML is the cross-institutional dissemination case (induced 6.4; LENS D2/PT4) — Domain 2 for mastery-learning replacing exposure-based progression; Domain 4 for the procedural-outcome + cost-effectiveness pairing. Pair with the translation trio (Cases 178, 117, 179).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Replace exposure-based training with a defined performance standard on a simulator; the SBML deliverable is competence demonstrated, not hours accumulated.
- Design the simulator and the standard around the specific complication modes the procedure produces — pneumothorax, arterial puncture, CRBSI for CVC — so the training closes the gaps the outcome data identifies.
- Pair the procedural-outcome evidence with cost-effectiveness evidence at single-site scale; the combination is what makes the program disseminable beyond the original institution.

IN OPERATION / AFTER

- Plan the dissemination as a deliverable of the original program, not as a separate downstream activity; the VA rollout is what makes Barsuk SBML a cross-institutional intervention rather than a successful single-site study.
- Build multi-site outcome measurement into the dissemination from the start; the published evidence at national-VA scale is thinner than the single-center evidence, and the gap is a live frontier the case names.
- Carry the cross-domain pairing with team-science training (Case 178), IPE (Case 117), and implementation-science training (Case 179) into the curricular framing — the four cases together stage what cross-domain workforce dissemination looks like with measurement and what it looks like without.

REFLECTION QUESTIONS

1. Identify a high-volume, high-consequence procedural skill in your domain where training is currently exposure-based rather than performance-standard-based. What is the analog of SBML, and what is the simulator that would absorb the cost of the practice?

2. Specify the procedural-outcome and cost-effectiveness pairing you would design at single-site scale; both halves are what made the Barsuk program disseminable, and the pairing is what distinguishes a teachable single-site case from a national-rollout candidate.

3. The VA dissemination evidence is thinner than the single-center evidence. Design the multi-site outcome-tracking architecture you would build into the next dissemination — what instrument, what comparison, what cadence — so the multi-site evidence catches up to the demonstration evidence.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines

CASE 42 DEFENSE WORKFORCE L&D INTELLIGENT TUTORING 2009 – 2014

DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks

H Human-System Interface K Knowledge & Institutional Memory D Designed Out

IMPACT An IDA independent assessment found that, after 16 weeks of Digital Tutor instruction, US Navy IT graduates with no prior IT experience outscored fleet Information Technology Specialists with an average 7.2 years of experience on a knowledge test, with an effect size of 4.30, and outperformed them on most troubleshooting and design tasks

IN BRIEF

DARPA's Digital Tutor program asked whether a one-on-one intelligent tutoring system, modelled on expert human tutoring, could compress years of operational IT expertise into a 16-week pipeline. The independent evaluation by the Institute for Defense Analyses (Fletcher and Morrison, IDA Document D-4686) compared Digital Tutor graduates — US Navy enlistees with no prior IT experience — against fleet Information Technology Specialists with an average 7.2 years of experience. The Digital Tutor cohort outscored fleet ITs on a knowledge test with an effect size of 4.30 and outperformed them on most troubleshooting and design tasks; only the Security exercise produced a fleet advantage. The IDA report concludes the program "appears to have achieved its goals." Two hedges are load-bearing and survive into the case: knowledge "accounts for about 40 percent of practical-exercise performance variance" and is "an enabler of performance rather than a direct measure of performance itself," and the system-architecture detail in the available documentation is too scant to fully reproduce. The case is the canonical small-tier instance of compressing the capability envelope at the edge of training, paired with CIRCUIT (Cases 81 and 39) on the workforce-capability-at-the-edge axis.

BACKGROUND

The US Navy's Information Technology Specialist rating has a conventional pipeline: an A-school of several months, followed by years of fleet experience that turn the rated sailor into an operational troubleshooter. The capability that matters at the operational end — diagnose a networking failure under time pressure, recover a degraded system, design a workable configuration for an unfamiliar requirement — is conventionally treated as a thing seat time produces. DARPA's Digital Tutor program asked whether an intelligent tutoring system, modelled on the discipline of expert one-on-one human tutoring, could compress that pipeline into 16 weeks of structured instruction.[0]

THE INTERVENTION

The program's design choice was to model the system on what expert human tutors actually do: a continuous dialogue around authentic problems, with the tutor pulling the trainee toward the conceptual move that resolves the situation. The instructional sequence was built around troubleshooting and design problems drawn from the operational domain, not around a syllabus reconstructed from the existing course. The working hypothesis was that the tutorial discipline — not the content coverage — was what produced operational expertise, and that a sufficiently capable system could deliver enough of that discipline at scale to be useful as a training pipeline.[1]

HOW IT WORKED

The Institute for Defense Analyses (Fletcher and Morrison, IDA Document D-4686 / DTIC AD1002362) ran the independent evaluation that the case rests on. Digital Tutor graduates — Navy enlistees with no prior IT background, 16 weeks in — were compared against a sample of fleet Information Technology Specialists with an average 7.2 years of operational experience. On a knowledge test, the Digital Tutor cohort outscored fleet ITs with an effect size of 4.30 — a magnitude that is unusual in workforce L&D research and that the report treats as the headline finding. On troubleshooting and design tasks the Digital Tutor cohort outperformed the fleet sample on most exercises; the Security exercise was the exception where fleet experience showed.[2]

THE EVIDENCE

The IDA report concludes the effort “appears to have achieved its goals,” and the language is deliberate. Two hedges are load-bearing and survive into the case verbatim. First, knowledge “accounts for about 40 percent of practical-exercise performance variance” and is “an enabler of performance rather than a direct measure of performance itself” — so the knowledge-test effect size, as large as it is, is not the same as the operational capability the Navy actually buys. Second, the available program documentation is too scant in system-architecture detail to fully reproduce. The result is teachable; the engineering recipe is not yet open.[3]

WHAT TRANSFERRED

What the case carries for the corpus is the capability-envelope argument at the edge of training (induced 1.2, LENS D2/PT4). The conventional pipeline assumes operational expertise is a function of seat time and exposure. The Digital Tutor evaluation is evidence that the envelope is reachable substantially faster than the inherited course assumed — under one rating, one program, one evaluation — and the hedges name what the evidence does and does not close. Paired with CIRCUIT (Cases 81, 39), the case anchors the workforce-capability-at-the-edge-of-training axis that connectomics proofreading and submarine-system troubleshooting share at the structural level.

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THE LEARNING ENGINEERING LENS

> *The Digital Tutor cohort outscored fleet ITs with 7.2 years' experience on the knowledge test at an effect size of 4.30; the hedge is that knowledge accounts for about 40 percent of practical-exercise variance.*
>
> EDITORS' SYNTHESIS OF FLETCHER & MORRISON (2014), IDA DOCUMENT D-4686.

LE INSIGHT

DARPA's Digital Tutor is the cleanest available evidence that the capability envelope of a training pipeline can be re-specified — from years of seat time to 16 weeks of tutorial-discipline instruction — against an operational comparison the program is built to compete with. The hedges (knowledge as enabler, architecture detail scant) are part of what makes the result interpretable.

LENS APPROACH

Digital Tutor is the canonical workforce-capability-at-the- edge-of-training case (induced 1.2; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development) for the tutorial-discipline-as-instructional-artifact design move, and in Domain 4 (Test and Evaluation) for the operational-comparison evaluation against fleet ITs with 7.2 years of experience. Pair with CIRCUIT (Cases 81, 39) at the workforce-capability-at-the-edge axis — connectomics proofreading and Navy IT troubleshooting share the structural pattern of compressing operational expertise through tutorial discipline.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the operational capability the pipeline must produce in the language of the work — troubleshoot under time pressure, design a workable configuration — not in the language of the existing course's content coverage.
- Treat the tutorial discipline (continuous dialogue around authentic problems, pulling toward the resolving conceptual move) as the load-bearing instructional artifact, rather than the content sequence the legacy course inherited.
- Design the evaluation against the operational comparison the program is built to compete with — for Digital Tutor, fleet ITs with 7.2 years of experience — so the result speaks to the capability envelope, not to a within-program improvement.

IN OPERATION / AFTER

- Report the knowledge-test effect (4.30) and the practical-exercise variance hedge (knowledge accounts for 40%) together; the headline number is real, and the qualification that knowledge is an enabler rather than a direct measure is part of what makes the result interpretable.
- Treat the absence of reproducible architecture detail as program documentation, not as polish: future builders need the engineering recipe, and the next iteration of evidence is conditional on that recipe being available.
- Carry the result into adjacent capability-envelope debates — CIRCUIT proofreading, submarine-system troubleshooting — as evidence that the envelope is reachable faster than the inherited training assumption, under one program and one evaluation.

REFLECTION QUESTIONS

1. Identify a capability in your domain where the inherited training assumption is that operational expertise is a function of seat time. What would a Digital-Tutor-class re-specification — tutorial discipline around authentic problems — look like, and what is the operational comparison your evaluation would have to beat?
2. The Digital Tutor knowledge-test effect size is 4.30 and the report still hedges that knowledge accounts for only 40% of practical-exercise variance. What is the analog hedge in your context: which part of the capability does your evaluation measure directly, and which part is enabler rather than direct measure?
3. The architecture detail is too scant for outside builders to reproduce the system. What is the minimum engineering recipe you would publish alongside a similar result, so that the next iteration of evidence rests on a reproducible base rather than a one-off program?

WHO BUILDS THIS human factors & interaction design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · knowledge bases · data pipelines · task analysis · design prototyping

CASE 43 CLINICAL CARE PATIENT SAFETY TEAM COMMUNICATION 2014

I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer

H Human-System Interface K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Across nine pediatric residency programs, implementation of the I-PASS handoff bundle (mnemonic + training + faculty development + sustainability campaign) was associated with a 23% relative reduction in preventable adverse events and a drop in medical errors, without negatively affecting resident workflow — the study design 'precludes definitively establishing a causal link'

IN BRIEF

The shift-change handoff is the moment in inpatient care where patient state has to be transferred accurately across a human boundary under time pressure. Loss of safety-critical information at handoff is a documented failure mode, and the cognitive demand on the outgoing resident — synthesize, prioritize, and convey — exceeds what unaided improvisation can reliably deliver. I-PASS is a structured handoff bundle: a mnemonic (Illness severity, Patient summary, Action list, Situation awareness and contingency planning, Synthesis by receiver), paired with formal trainee education, faculty development, and a sustainability campaign. Starmer et al. (NEJM, 2014) studied its implementation at nine pediatric residency programs and reported a 23% relative reduction in preventable adverse events and a drop in medical errors, with no negative effect on resident workflow. The hedge survives verbatim: the authors state plainly that "our study design precludes definitively establishing a causal link." Published correspondence cautions against implementing the mnemonic alone without the full bundle. The case is the structured-transfer companion to Case 177 (CIRAS) at the cultural-half-of-capability layer, and the small-tier intervention spine for state-transparency under stress.

BACKGROUND

The shift-change handoff is a structural risk point in inpatient care. The outgoing resident — who has the most detailed mental model of each patient's state and trajectory — has to synthesize, prioritize, and convey enough of that model to the incoming team that the patient's care is not disrupted by the boundary. Loss of safety-critical information at handoff is a documented failure mode in the patient-safety literature; preventable adverse events that trace to communication breakdown are a substantial fraction of inpatient harm. The cognitive demand exceeds what unaided improvisation can reliably deliver, and the institutional tolerance for that improvisation has been falling.[0]

THE INTERVENTION

I-PASS is an explicitly bundled intervention. The mnemonic — Illness severity, Patient summary, Action list, Situation awareness and contingency planning, Synthesis by receiver — structures the handoff conversation around the information classes the receiver needs in order to take over safely. The mnemonic alone is the visible piece; the bundle around it is what makes the mnemonic land. The published implementation pairs the mnemonic with formal trainee education on its use, faculty development so that supervisors model and reinforce it, and a sustainability campaign so that the discipline does not erode after the rollout. The published correspondence cautions explicitly against implementing the mnemonic alone without the full bundle.[1]

HOW IT WORKED

Starmer et al. (New England Journal of Medicine, 2014) studied the I-PASS bundle's implementation at nine pediatric residency programs in the United States and Canada. The design was a pre/post evaluation across the participating sites, with the bundle deployed sequentially and outcomes tracked through chart review and observation. The headline finding was a 23% relative reduction in preventable adverse events from the pre-implementation period to the post-implementation period, alongside a drop in medical errors more broadly, with no observed negative effect on resident workflow — the bundle did not impose net friction that displaced other clinical work.[2]

THE EVIDENCE

The hedge is load-bearing and survives into the case verbatim. The authors state in the published paper that "our study design precludes definitively establishing a causal link" — pre/post evaluation across multiple sites is the strongest practical design in this setting, and it cannot rule out secular trends in patient-safety culture, simultaneous quality-improvement work, or the sites' own selection into the trial. The 23% reduction is the strongest available evidence; it is not closed proof that the bundle alone produced the effect. Treating it as either is a misreading of what the study design supports.[3]

WHAT TRANSFERRED

What the case carries for the corpus is the state-transparency-under-stress pattern at the human–human boundary (induced 3.3, LENS D2/PT5). The handoff is the moment of cognitive transfer at the team boundary; the bundle is the workflow artifact that makes the patient state legible across that boundary. Paired with Case 177 (CIRAS) at the cultural-half-of-capability layer, the case shows that the mnemonic alone is not the intervention — the institutional commitment to the bundle (trainee education, faculty development, sustainability) is what makes the mnemonic real in operation. The pattern is partly borrowed from aviation discipline, so the case also seeds the cross-domain-adaptation conversation that v1 Cases 70 (CRM) and 89 (ASRS) anchor in aviation evidence.

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THE LEARNING ENGINEERING LENS

> *Our study design precludes definitively establishing a causal link.*
>
> STARMER ET AL., NEJM 2014.

LE INSIGHT

I-PASS is the structured-handoff intervention the patient-safety literature anchors on — a 23% relative reduction in preventable adverse events across nine residency programs, with no negative effect on resident workflow. The hedge that survives verbatim is the authors' own: the study design "precludes definitively establishing a causal link," and the bundle, not the mnemonic alone, is what the evidence is about.

LENS APPROACH

I-PASS is the state-transparency-under-stress case at the human–human boundary (induced 3.3; LENS D2/PT5) — Domain 2 for bundle-as-intervention; Domain 5 for the institutional faculty-development commitment. Pair with Case 177 (CIRAS) and v1 Cases 70 (CRM) and 89 (ASRS).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the information classes the receiver needs in order to take over safely (illness severity, patient summary, action list, situation awareness, synthesis by receiver) and structure the handoff conversation around them, rather than around the outgoing operator's narrative preference.
- Treat the mnemonic, the trainee education, the faculty development, and the sustainability campaign as one bundle; the published correspondence is explicit that the mnemonic alone does not carry the result.
- Design the evaluation to track preventable adverse events and resident workflow together, so that the intervention's effect on patient harm and on the work it adds are visible in the same evidence record.

IN OPERATION / AFTER

- Report the 23% relative reduction together with the verbatim hedge that "our study design precludes definitively establishing a causal link"; the strongest available evidence is not closed proof.
- Carry the bundle-not-mnemonic warning into any adaptation: implementations that drop the faculty-development and sustainability components are not implementations of the intervention the published evidence is about.
- Build the cross-domain adaptation conversation deliberately — aviation CRM and ASRS evidence (v1 Cases 70, 111) is part of the I-PASS lineage and CIRAS (Case 177) is the non-aviation companion at the cultural-commitment layer.

REFLECTION QUESTIONS

1. Identify a moment in your domain's workflow where state has to be transferred accurately across a human boundary under time pressure. What are the information classes (the analog of I-PASS's five) the receiver needs in order to take over safely, and what bundle around the structuring artifact would make it land?

2. The 23% relative reduction is the strongest available evidence and the authors are explicit it is not closed proof. What additional evidence — a cluster-randomized rollout, a residual-error analysis at the handoff point — would you require before treating the figure as closure rather than strong signal?

3. The published correspondence cautions against implementing the mnemonic alone. What is the bundle-vs-mnemonic distinction in your context, and what institutional commitment would be required to keep the bundle intact across years and turnover?

WHO BUILDS THIS human factors & interaction design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 44 NURSING EDUCATION CLINICAL SIMULATION REGULATION 2014

NCSBN National Simulation Study — Licensing the 50% Substitution Rule

G Governance & Trust K Knowledge & Institutional Memory D Designed Out

IMPACT A longitudinal RCT randomized students across multiple US nursing programs to control, 25%, or 50% simulation substitution for traditional clinical hours; 660+ took the NCLEX with no statistically significant differences in clinical competency, nursing knowledge, or NCLEX pass rates — the number of nursing regulatory boards permitting up to 50% simulation substitution increased more than 20-fold from 2014 to 2022

IN BRIEF

Pre-licensure nursing education had long rested on a regulatory requirement for traditional clinical hours — time spent caring for patients in a real clinical setting under the supervision of a clinical faculty member. As clinical-placement capacity came under pressure, the question facing nursing regulators was whether high-quality simulation could substitute for some fraction of those hours without degrading the clinical capability of new nurses. The National Council of State Boards of Nursing (NCSBN) ran the study that the field then built on: a longitudinal RCT randomized students across multiple US nursing programs to control, 25%, or 50% simulation substitution. Hayden et al. (Journal of Nursing Regulation, 2014) reported no statistically significant differences in clinical competency, nursing knowledge, or NCLEX pass rates across the three groups; 660+ participants took the NCLEX. The follow-up (2023) documents the institutional transfer: the number of nursing regulatory boards permitting up to 50% substitution increased more than 20-fold from 2014 to 2022 — an unusually clean case of a single evidence base propagating across an entire regulatory field. The hedge survives: the result holds only "under conditions comparable to those described in the study" (high-quality simulation, trained faculty). Pair with Case 178 (Colorado CTSA team-science training) at the cross-domain workforce-intervention layer.

BACKGROUND

Pre-licensure nursing education in the United States has rested on a regulatory requirement, set by state nursing boards, that students complete a defined number of traditional clinical hours — time spent caring for patients in a real clinical setting under the supervision of a clinical faculty member. As nursing programs grew and clinical-placement capacity in hospitals came under pressure, the field was asking whether high-quality simulation — manikin-based or standardized-patient scenarios run in a controlled environment — could substitute for some fraction of traditional clinical hours without degrading the clinical capability of new nurses.[0]

THE INTERVENTION

The National Council of State Boards of Nursing — the umbrella body for the state nursing-regulatory boards that together control nursing licensure — commissioned the definitive study. Hayden et al. (Journal of Nursing Regulation, 2014) reported the NCSBN National Simulation Study: a longitudinal randomized controlled trial that assigned pre-licensure nursing students at multiple US nursing programs to a control group, a 25%-substitution group, or a 50%-substitution group. The substitution condition replaced an equivalent fraction of traditional clinical hours with simulation hours run to specified quality standards (validated scenarios, trained simulation faculty, structured debriefing).[1]

HOW IT WORKED

The headline finding was a null result on the outcome the regulators most cared about: no statistically significant differences in clinical competency, nursing knowledge, or NCLEX pass rates across the three groups. Six hundred and sixty-plus participants took the NCLEX licensure examination, providing the comparison the regulatory decision rested on. The longitudinal design followed cohorts through their first nursing positions, with comparable outcome measures at multiple time points. The study's qualifying language is precise: the result holds only "under conditions comparable to those described in the study," and those conditions are named — high-quality simulation, trained simulation faculty, structured debriefing.[2]

THE EVIDENCE

What followed is the unusual part. Nursing regulation in the US is genuinely decentralized: each state board sets its own pre-licensure requirements, and there is no federal forcing mechanism. The 2023 follow-up study documents that the number of nursing regulatory boards permitting up to 50% simulation substitution increased more than 20-fold from 2014 to 2022. A single evidence base — one study, run by the regulators' own collaborative — propagated across an entire regulatory field, on the strength of the published design and the null result on the licensure examination the boards control.[3]

WHAT TRANSFERRED

The case carries the cross-organization knowledge-transfer pattern at the regulatory-institution layer (induced 6.1, LENS D2/PT4). The mechanism that made the propagation work was the credibility of the study's design (longitudinal RCT, multi-site, blinded outcomes), the alignment of its outcome measure with the boards' own licensure mechanism (NCLEX), and the institutional ownership of the evidence (the regulators commissioned it and own it). Pair with Case 178 (Colorado CTSA team-science training) at the cross-domain workforce-intervention layer — both are small-tier interventions with measurable workforce-capability claims, and both depend on the legitimacy of the assessment instrument the institution then has to defend.

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THE LEARNING ENGINEERING LENS

> *The result holds under conditions comparable to those described in the study — high-quality simulation, trained faculty, structured debriefing.*
>
> EDITORS' SYNTHESIS OF HAYDEN ET AL. (2014).

LE INSIGHT

The NCSBN National Simulation Study is the unusual case of a single, well-designed, regulator-commissioned study propagating a substantial workforce-capability change across an entire decentralized regulatory field. The null result on NCLEX is conditional on the quality conditions the study names — high-quality simulation, trained faculty, structured debriefing — and the qualifying language has to travel with the result.

LENS APPROACH

NCSBN is the regulator-owned cross-organization knowledge-transfer case (induced 6.1; LENS D2/PT4) — Domain 2 for the RCT-design discipline; Domain 5 for the institutional-ownership move that propagated evidence across decentralized regulators. Pair with Case 178 (Colorado CTSA).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Commission the study from inside the regulatory body that will then have to act on it; the institutional ownership of the evidence is what makes the propagation across decentralized regulators possible.
- Specify the quality conditions the result is conditional on (high-quality simulation, trained faculty, structured debriefing) at design time, so the qualifying language travels with the result and is not stripped at the citation stage.
- Align the study's outcome measure with the licensure mechanism the boards control (NCLEX pass rate); the credibility of the propagation depends on the outcome being the one the regulators already use.

IN OPERATION / AFTER

- Carry the qualifying language ("under conditions comparable to those described in the study") into every downstream adoption; the study's null result is conditional on quality standards, not unconditional.
- Document the propagation as a transferable institutional pattern: a decentralized regulatory field can move on the strength of a single, well-designed, body-owned study when the outcome measure is the one the regulators already use.
- Treat the 20-fold expansion of board adoption from 2014 to 2022 as evidence of the mechanism of regulatory transfer, not as evidence that 50% substitution is safe outside the study's quality conditions.

REFLECTION QUESTIONS

1. Identify a decentralized regulatory field in your domain where a substitution or adoption decision rests on whether the new modality preserves capability. What would a NCSBN-class RCT, commissioned from inside the regulatory body, look like — and what would the licensure-aligned outcome measure be?

2. The result is conditional on quality conditions (high-quality simulation, trained faculty, structured debriefing). What is the analog conditional in your context, and what mechanism would carry the qualifying language into the regulatory adoptions that follow?

3. The 20-fold expansion of board adoption is evidence of regulatory transfer, not of safety outside the study's quality conditions. What would you do operationally to ensure the adoptions in your field actually meet the conditions the propagation rests on?

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CASE 45 MEDICAL EDUCATION RETENTION WORKFORCE L&D 2007 – 2009

Spaced Education RCTs in Medical Training

H Human-System Interface K Knowledge & Institutional Memory D Designed Out

IMPACT A multi-institution RCT of urology residents across the US and Canada randomized participants to bolus versus spaced-pattern email delivery of validated study questions; spaced education improved acquisition and retention of medical knowledge, and a follow-up showed the learning benefit persisting for two years

IN BRIEF

Spacing — distributing study sessions over time rather than massing them — is one of the most robust findings in basic learning-science research, with effects across age, population, and content domain. Whether the basic finding transfers into the workplace-L&D context of practicing medical trainees is a separate empirical question. Kerfoot et al. (Journal of Urology, 2007) ran the test that closed the loop. A multi-institution RCT randomized urology residents across US and Canadian programs to receive validated study questions in either a bolus pattern (concentrated delivery) or a spaced pattern (distributed delivery), both via email. The spaced-education condition improved acquisition and retention of medical knowledge against the bolus comparison. The 2009 follow-up (Kerfoot, Journal of Urology, 181:2671) documented that the learning benefit persisted for two years. The case is a methodologically clean small-tier intervention with replication, with standard RCT scope: the result speaks to the delivery-pattern manipulation in the email-delivered validated-question context, not to spacing in general or to other modalities. The strongest randomized strength in the small-tier batch. Pair with Case 40 (Duolingo half-life regression) at the spacing-mechanism-in-deployment layer.

BACKGROUND

Spacing — distributing study sessions over time rather than massing them in a single concentrated block — is among the most robust findings in basic learning-science research. The spacing effect appears across age groups, populations, content domains, and outcome measures, and the broader retrieval-practice literature complements it. What is less obvious in the basic literature is whether the spacing effect transfers to the workplace-L&D context, where the learner is a practicing trainee, the content is the trainee's own clinical specialty, and the delivery mechanism is one the trainee will tolerate as part of their work week.[0]

THE INTERVENTION

Kerfoot et al. (Journal of Urology, 2007) ran the test that closed the loop. The study was a multi-institution randomized controlled trial of urology residents at multiple US and Canadian residency programs. Participants were randomized to receive validated study questions drawn from the urology in-service examination question pool, delivered by email, in either a bolus pattern (concentrated delivery over a short window) or a spaced pattern (distributed delivery across a longer window). Outcomes were knowledge acquisition (immediate post-test) and retention (delayed post-test), measured against the same validated question pool the delivery drew from.[1]

HOW IT WORKED

The result was direct. The spaced-education condition improved acquisition and retention of medical knowledge against the bolus comparison. The intervention and the measurement formed a tight closed loop: the validated questions delivered are the same content the post-test drew from, the email delivery is the manipulation, and the knowledge-outcome measure is the institution's own in-service question pool. The 2009 follow-up (Kerfoot, Journal of Urology, 181:2671) reported a separate analysis with the same residents and documented that the learning benefit persisted for two years — a duration that is rare in the workplace-L&D RCT literature.[2]

THE EVIDENCE

The case's scope is what makes it teachable rather than over-extended. The result speaks to the delivery-pattern manipulation (bolus vs. spaced) in the email-delivered validated-question context, with urology residents at multiple programs. It does not speak to spacing in general, to other modalities (mobile, in-clinic, simulation), or to content classes substantially different from in-service-examination knowledge. The strength of the design — multi-institution RCT, two-year persistence — is the strongest randomized strength in the small-tier batch, and the scope discipline is the qualifying language that makes the strength interpretable. The case is the methodologically clean small-tier closed-loop intervention the LENS Iterative Development domain anchors on.[3]

WHAT TRANSFERRED

Pair with Case 40 (Duolingo half-life regression) at the spacing-mechanism-in-deployment layer. Duolingo is the large-scale operational deployment of a spacing-and- retrieval-practice system in a consumer language-learning product; Kerfoot is the small-scale randomized evidence in the workplace-L&D medical-training context. Together the two cases anchor the spacing-mechanism conversation across laboratory, deployed-product, and randomized-workplace evidence. The Iterative Development discipline LENS teaches depends on having both the controlled-randomized closed loop (Kerfoot) and the operational-mechanism evidence (Duolingo) available in the corpus.

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THE LEARNING ENGINEERING LENS

> *The learning benefit persisted for two years.*
>
> KERFOOT, JOURNAL OF UROLOGY 2009 FOLLOW-UP.

LE INSIGHT

Kerfoot et al. is the methodologically clean small-tier closed-loop spaced-education RCT in workplace medical training, with two-year persistence in the follow-up. It is the strongest randomized strength in the small-tier batch. The scope discipline is what makes it interpretable: the result speaks to the email-delivered validated-question delivery pattern, not to spacing in general.

LENS APPROACH

Kerfoot is the canonical small-tier closed-loop spacing-in-workplace-L&D case (induced 2.3; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development) for the intervention-and-measurement-as-single-design discipline, and in Domain 4 (Test and Evaluation) for the persistence- follow-up that makes the result a workplace-L&D headline. Pair with Case 40 (Duolingo half-life regression) at the spacing-mechanism-in-deployment layer — controlled- randomized evidence with deployed-product evidence together.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the closed loop in advance: the intervention manipulation, the outcome measure, and the validated-question pool together, so the intervention and the measurement form a single design rather than two separately published artifacts.
- Choose the delivery modality (email, in this case) that the trainees will tolerate as part of their work week; the workplace-L&D context is unforgiving of modalities that compete with clinical work.
- Pre-register the persistence follow-up at the design stage; the two-year retention figure in Kerfoot 2009 is what gives the intervention its workplace-L&D strength, and follow-up has to be designed into the original RCT.

IN OPERATION / AFTER

- Report the result in the scope it speaks to: email-delivered validated-question delivery pattern with urology residents; the spacing-in-general claim sits in the basic literature, and the case's contribution is the workplace-L&D closed-loop evidence.
- Treat the two-year persistence figure as the workplace-L&D headline; persistence over years is rare in the workplace-L&D RCT literature, and the figure is what makes the result actionable as a training design.
- Pair the result with Duolingo (Case 40) at the spacing-mechanism layer in any communication of the broader conversation, so the controlled-randomized evidence and the deployed-product evidence are read together.

REFLECTION QUESTIONS

1. Identify a workplace-L&D context in your domain where the basic-literature evidence (spacing, retrieval practice, deliberate practice) is robust and the workplace-transfer evidence is thinner. What would a Kerfoot-class closed-loop RCT look like — the intervention manipulation, the outcome measure, the validated-content pool together?

2. The two-year persistence figure is the workplace-L&D headline. What follow-up cadence would you design into your study at the start, so that persistence over years is part of the original RCT rather than a separately commissioned downstream check?

3. The result's scope is the email-delivered validated-question delivery pattern. What is the analog scope discipline in your context — the qualifying language that has to travel with the headline so the result is read as evidence about a specific mechanism rather than as evidence about a general principle?

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CASE 46 CORPORATE L&D PERFORMANCE CONSULTING LEARNING TRANSFER 2001 – PRESENT

High-Impact Learning System — Engineering the Environment, Not Just the Event

K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT Brinkerhoff & Apking's HILS reframes corporate L&D as a system spanning pre-training (line-manager alignment, work-context preparation), the event itself, and post-training (supervisor support, on-the-job practice) — translating Blume's meta-analytic finding that the work environment dominates transfer into a deployable program model

IN BRIEF

The High-Impact Learning System (HILS), introduced by Brinkerhoff and Apking in 2001, reframes L&D as a system spanning pre-training, the event itself, and post-training. The design principle is that the training event alone explains a small fraction of transfer variance — Blume's 2010 meta-analysis (Case 35) identifies the work environment as the decisive variable — and so the program has to engineer the environment alongside the event. HILS deployments include pre-training line-manager alignment and work-context preparation; the event itself; and post-training supervisor support and on-the-job practice opportunities. Corporate deployments report transfer rates rising from the 10–20% baseline cited in the L&D literature to substantially higher figures, but the per-firm numbers live in vendor whitepapers and conference talks rather than peer-reviewed audits. The case is the deployed-program counterpart to Case 124 (SCM as evaluation instrument): SCM measures whether the program worked at the tails; HILS designs the program so that the conditions for transfer are engineered. Evidence-tier flag is practice-synthesis: the model is documented in practitioner publications and in Watershed and L-TEN summaries, the corporate effect sizes are self-reported, and future validation will continue.

BACKGROUND

Blume et al.'s 2010 meta-analysis (Case 35) names the work environment — supervisor support, peer support, practice opportunity — as the decisive transfer variable at the system layer rather than the learner layer. The L&D practitioner question that follows is what to **deploy** in response to the finding: if the training event is not the capability deliverable, what is the surrounding architecture that has to be built so the transfer actually happens?[0]

THE INTERVENTION

Brinkerhoff and Apking's **High Impact Learning** (2001) is one of the answers in practitioner circulation. HILS reframes the L&D pipeline around three phases. **Before the event**: line-manager alignment, work-context preparation, learner readiness. **The event itself**: content delivery designed against the on-job task it is intended to enable. **After the event**: supervisor support, peer-support structures, deliberately engineered practice opportunities on the actual job tasks. The training event is one component, not the whole program.[1]

HOW IT WORKED

The corporate adoption story — across firms identified in Brinkerhoff materials and in Watershed LRS and L-TEN summaries — reports transfer rates rising from the 10–20% baseline cited in the L&D literature to substantially higher figures. The per-firm numbers live in vendor whitepapers and conference talks, not in peer-reviewed evaluations. The pattern is consistent across the practitioner sources; the magnitudes are self-reported and have not been independently audited.[2]

THE EVIDENCE

HILS is the deployed-program counterpart to Case 124 (SCM as evaluation instrument). Where SCM measures whether the program worked at the tails of the outcome distribution, HILS designs the program so that the surrounding conditions for transfer are engineered. The pair is the operational answer to the Kirkpatrick chain-of-evidence problem (Case 108): design the environment so transfer can happen, and measure the tails to confirm it did. Neither component on its own crosses the Level-2 / Level-3 seam — together they structure the cross.[3]

WHAT TRANSFERRED

The LENS teaching point is the framework-level claim that capability lives at the operator-system interface, not in the individual. HILS is a documented practice pattern that names the environment-as-intervention move; it makes Blume's meta-analytic finding (Case 35) operational. The evidence-tier flag is practice-synthesis — the model and the deployment pattern are durable in practitioner literature, the per-firm effect magnitudes are self-reported, future validation will continue. The CLO **Judgment under inadequate evidence** is exactly the capability HILS asks the practitioner to bring: act on the strongest synthesis the field has produced while naming what is and is not independently audited.[4]

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> *If the work environment is the decisive transfer variable, the work environment has to be a design variable. HILS makes it one.*
>
> EDITORS' SYNTHESIS OF BRINKERHOFF & APKING (2001) AND THE BLUME META-ANALYTIC FINDING.

LE INSIGHT

HILS is the deployed-program operationalization of Blume's environment-as-decisive finding (Case 35) and the design-side counterpart of SCM (Case 124). Evidence is practice-synthesis: the model is durable in practitioner literature, per-firm effect sizes are self-reported, future validation continues. The CLO **Judgment under inadequate evidence** is the capability the case asks for.

LENS APPROACH

HILS is the L&D environment-as-design-variable case (induced 2.4; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development / Learning Engineering Design) for the amended sub-competency that narrates the design iteration explicitly across pre / event / post phases, and exercises the CLO **Judgment under inadequate evidence** because the practitioner must decide on practice-synthesis-tier evidence. Pairs with Case 124 (SCM) and Cases 108 / 112 as the corporate-L&D cluster.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Plan the pre-training phase as a first-class design deliverable: line-manager briefings, work-context preparation, learner readiness assessment, instrumented before the event runs.
- Design the post-training phase before delivery: supervisor support artifacts, peer-support structures, scheduled on-job practice opportunities on the specific tasks the training targets.
- Treat the training event as one component of a three-phase architecture, not as the whole program — Blume's finding (Case 35) is the load-bearing reason the architecture has to span the boundary.

IN OPERATION / AFTER

- Pair with Case 124 (SCM) to measure whether the deployed program worked at the tails — HILS designs the environment, SCM samples the outcomes; together they structure the chain-of-evidence cross (Case 108).
- Carry the practice-synthesis flag honestly: the model is documented and durable, the per-firm effect sizes are self-reported, and any specific magnitude cited from vendor whitepapers should be flagged as such in program documentation.
- Use the CLO **Judgment under inadequate evidence**: the synthesis is the strongest the field has, and the practitioner has to decide whether to deploy HILS at organizational scale on practitioner-tier evidence while naming the validation that remains open.

REFLECTION QUESTIONS

1. Identify a training deployment in your context that currently invests heavily in the event and lightly in the surrounding environment. What pre-training and post-training artifacts would you build to convert the deployment from a single-component program into a three-phase HILS-style architecture?

2. Specify the line-manager and supervisor briefing materials, the peer-support structures, and the on-job practice opportunities you would design — and identify who in your organization would be accountable for each. The accountability map is part of the design.

3. The HILS corporate effect sizes are self-reported. What independent evidence — third-party audit, peer-reviewed evaluation, cross-firm comparison — would you require before treating any specific magnitude as decision-grade, and how would you act on practitioner-tier evidence in the meantime?

WHO BUILDS THIS knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 47 GLOBAL HEALTH HIV CARE TRAINING-MODALITY COMPARISON SUB-SAHARAN AFRICA 2023

PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption

K Knowledge & Institutional Memory N Normalization of Deviance D Designed Out

EVIDENCE TIER preprint-tier — source confidence flagged; future validation ongoing.

IMPACT Across 16 PEPFAR-supported Sub-Saharan African countries, pre/post knowledge and self-confidence assessments showed mean increases across all training modalities — in-person (pre-COVID), virtual, and blended — when pandemic disruption forced a delivery-mode switch; the L1–L2 limitation is explicit and the L3/L4 question remains open

IN BRIEF

Across 16 PEPFAR-supported Sub-Saharan African countries, a 2023 real-world program evaluation compared in-person (pre-COVID), virtual, and blended HIV-care training delivery. Across all assessed knowledge domains and self-perceived confidence dimensions, pre/post assessments showed mean increases regardless of modality. The honest framing the case carries into print is that the outcome metric is knowledge and self-rated confidence — Kirkpatrick L1–L2 territory (Case 108) — not L3 on-job behavior change or L4 patient outcomes. Even so, the study is one of the few real-world cross-country modality comparisons at meaningful scale, and it is the L&D evaluation pattern playing out in global health at multi-country scope. The case is cross-listed with the corporate / workforce L&D cluster (Cases 108, 35, 124, 46) and with the non-US/UK/EU geographic-coverage gap (Cases 185, and the cases in the later supplemental batches). Evidence-tier flag is preprint-tier: the medRxiv version is preprint and the PMC version is journal-published — the editor's citation choice should be carried explicitly. Future validation on whether L1–L2 knowledge gains translate to L3 behavior change or L4 patient outcomes remains ongoing.

BACKGROUND

PEPFAR (the US President's Emergency Plan for AIDS Relief) has been one of the largest sustained global health programs of the past two decades. Workforce capability — clinicians and frontline workers trained on current HIV diagnosis, treatment, and prevention protocols — is one of its load-bearing deliverables. COVID disruption forced a delivery-mode switch in 2020 across countries, from in-person training to virtual and blended modalities, before any of the evidence base on modality equivalence had been built for this workforce.[0]

THE INTERVENTION

The 2023 study compares pre/post-assessment outcomes across 16 PEPFAR-supported Sub-Saharan African countries and across the three modalities. The headline finding is that across all knowledge domains and self-perceived confidence dimensions assessed, mean increases were observed regardless of modality. The pragmatic interpretation in the global-health literature has been that the virtual / blended modalities are not measurably worse on the L1–L2 outcomes the study measured, a finding that has policy implications for program design under future disruption.[1]

HOW IT WORKED

The honest framing the case must carry into print is the Kirkpatrick limitation. The study's outcomes are knowledge and self-rated confidence — Level 1 (reaction and confidence) and Level 2 (learning) in the Kirkpatrick framework (Case 108). It is not Level 3 (behavior change on the job — whether the clinicians actually changed practice) and it is not Level 4 (results — whether patient outcomes improved). The structural limitation Case 108 names is exactly the limitation this study sits inside.[2]

THE EVIDENCE

Even so, the study is one of the few real-world cross-country modality comparisons at meaningful scale, and the multi-country scope means the comparison is not trivially confounded by single-country labor-market or health-system factors. It is the L&D evaluation pattern playing out in global health, and the modality-comparison finding is policy-relevant for PEPFAR program design across the deployment region.[3]

WHAT TRANSFERRED

The evidence-tier flag is preprint-tier: the medRxiv version is the preprint; the PMC version is the journal-published article. The editor's citation choice between the two should be carried explicitly in the printed case. The LENS teaching point is the cross-listed pair with the corporate / workforce L&D cluster (Cases 108, 35, 124, 46) and the non-US/UK/EU cluster (Cases 185). The CLO **Judgment under inadequate evidence** is exercised: the study is the strongest evidence the field has for modality equivalence in this context, and it does not establish L3 / L4 outcomes. Future validation is ongoing.

REFERENCES

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2. PEPFAR program documentation — workforce-capability training as a load-bearing deliverable across Sub-Saharan African deployment countries.
3. Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the chain-of-evidence framework the L1–L2 limitation references (paired Case 108).
4. Blume et al. (2010), Journal of Management 36(4):1065–1105 — the meta-analytic transfer finding the L3 question references (paired Case 35).

> *The outcomes are knowledge and self-rated confidence. The L3 and L4 questions — did practice change, did patient outcomes improve — remain open.*
>
> EDITORS' SYNTHESIS OF THE PEPFAR 16-COUNTRY MODALITY COMPARISON.

LE INSIGHT

PEPFAR's 16-country modality comparison is the L&D evaluation pattern in global health: mean L1–L2 gains across all modalities, with the Kirkpatrick limitation (Case 108) explicit. Evidence-tier flag is preprint-tier — both the medRxiv preprint and the PMC published version are citable — and the L3 / L4 questions remain open. Future validation is ongoing.

LENS APPROACH

PEPFAR is the global-health workforce-capability case (induced 2.3; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development) for the modality-design iteration COVID forced and in Domain 5 (Navigating Sociotechnical Constraints) for the multi-country program scope. The CLO **Judgment under inadequate evidence** is exercised because the study is the strongest available at L1–L2 and does not settle L3 / L4. Pairs with Case 185 (SkillsFuture) as the workforce L&D cross-listing.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the modality comparison around what the assessment instrument can actually establish — knowledge and confidence at L1–L2 are tractable; L3 and L4 require longitudinal data and patient-outcome linkage the study did not have.
- Build the multi-country sampling so the comparison is not trivially confounded by single-country factors — the 16-country scope is part of the study's evidential weight.
- Carry the Kirkpatrick limitation into the program documentation honestly — Case 108's stop-at-L2 pattern is exactly the limitation this study sits inside, and naming it preserves the case's teaching value.

IN OPERATION / AFTER

- Pair with Case 185 (SkillsFuture) as the workforce-capability counterparts at national- and multi-country scale; together they teach what L&D measurement looks like in non-US/UK/EU settings with the evidence-tier honesty intact.
- Use the CLO **Judgment under inadequate evidence**: the study is the strongest evidence the field has for modality equivalence in this context, and the practitioner has to decide modality policy on L1–L2 evidence while L3/L4 evidence develops.
- Carry the preprint-tier flag honestly: the medRxiv preprint and the PMC published article are both citable; future validation requires confirmatory replication and L3/L4 outcome measurement.

REFLECTION QUESTIONS

1. Identify a training program in your context that was forced to switch modality under disruption (COVID, budget constraint, geographic dispersion). What evidence would you have needed at the time to make the modality decision well, and at which Kirkpatrick level was the available evidence?

2. Specify the L3 / L4 measurement architecture you would build to extend the PEPFAR study into behavior-change and patient-outcome evidence — what data sources, what cadence, what linkage between training cohort and clinical outcomes.

3. The case is preprint-tier: both the medRxiv preprint and the PMC published version are citable. What additional confirmatory evidence — replication in different country sets, L3 / L4 follow-up studies, peer-reviewed meta-analysis — would you require before treating modality equivalence as a settled question for program-design decisions?

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CASE 48 SOFTWARE ENGINEERING EDUCATION WORK-BASED LEARNING REFLECTIVE PRACTICE 2025 (PREPRINT)

Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development

K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER preprint-tier — source confidence flagged; future validation ongoing.

IMPACT A 2025 longitudinal preprint study of reflective-practice development across a multi-year work-based software-engineering program — one of the few published instruments aimed at measuring the development of reflective capability rather than only its presence; preprint-tier flag is load-bearing

IN BRIEF

A 2025 arXiv preprint ("The Development of Reflective Practice on a Work-Based Software Engineering Program: A Longitudinal Study," arXiv:2504.20956) reports a longitudinal study of how reflective-practice capability itself develops across a multi-year work-based software-engineering program. The signal the v2 corpus needs from this case is precise: it is one of the few published instruments aimed at measuring the development of reflective capability over time, not merely its presence at a single point. That is the LENS-revised CLO-2′s evaluation problem in miniature — if the program asks the learner to narrate and defend the design iteration in first person, the program also has to be able to evidence that the capability to do so is developing. The preprint-tier flag is load-bearing: not yet peer-reviewed at the time of this drafting, and the case carries the standing "future validation ongoing" language into print. It is the v2 corpus's reference instance of an instrument built to measure the **development** of reflective capability over time, not only its presence at a single point — the prior art the editor- commissioned first-person Practice Flywheel accounts will sit alongside.

THE SHIFT

The amended CLO-2 in v2 asks the learner to narrate and defend the design iteration in first person. The evaluation problem this creates is real: most existing reflective-practice instruments measure whether a learner is reflecting at a given moment, not whether the capability to reflect well is developing across a program. A multi-year work-based software-engineering program is a useful site for this question because the work itself provides successive design iterations the learner can reflect on.[0]

WHAT IS EMERGING

The 2025 preprint reports a longitudinal study of reflective-practice development across such a program. The methodological move worth naming is the focus on development rather than snapshot: the instrument is designed to detect change in the depth and structure of reflection over time, and the study reports the trajectory across a cohort. That is the LENS-revised evaluation problem in miniature — and one of the few published instances of an instrument built for the purpose.[1]

THE CAPABILITY QUESTION

The teaching point is the construct boundary. Measuring the **presence** of reflection is well-trodden ground (rubrics, codings, reflective-essay scoring schemes). Measuring the **development** of reflective capability is a different construct: the comparison is intra-learner across time, the instrument has to be sensitive to depth change rather than presence change, and the study design has to handle the cohort-level variability that arises when learners enter the program with different reflective baselines.[2]

EARLY EVIDENCE

Why the case sits in the v2 supplemental tier rather than in the verified primary list: it is a preprint. The arXiv version has not been through peer review at the time of this drafting, and the case is included with the preprint-tier flag rendered under the title and the standing “future validation ongoing” language preserved. The signal the corpus extracts is structural — the instrument-design move and the construct distinction — rather than the specific magnitudes the preprint reports.[3]

OPEN PROBLEMS

What the case supplies the editor-commissioned first-person Practice Flywheel accounts (CIRCUIT, ERKS-class) is the evaluation pathway: evidence that reflective-practice capability can be measured as it develops, not only observed once. That is the prior art a Flywheel account’s evaluation design can build on. Future validation ongoing — both peer-review pipeline and replication across other work-based engineering programs.

REFERENCES

1. “The Development of Reflective Practice on a Work-Based Software Engineering Program: A Longitudinal Study,” arXiv:2504.20956 (2025) — preprint.
2. D. Schön, The Reflective Practitioner (1983) — the foundational account of reflection-in-action the genre rests on.
3. Boud, Keogh & Walker (eds.), Reflection: Turning Experience into Learning (1985) — reflection as a learning process, and the measurement problem it raises.
4. the proposed revisions — the amended CLO-2 (first-person narration of design iteration) the case evaluates.

THE LEARNING ENGINEERING LENS

> *The construct is the development of reflective capability, not its presence. The instrument has to be sensitive to depth change.*
>
> EDITORS' PARAPHRASE OF THE ARXIV PREPRINT'S CONSTRUCT DISTINCTION.

LE INSIGHT

The 2025 preprint is one of the few published instruments aimed at measuring the development of reflective capability across a multi-year work-based program, not only its presence at a snapshot. Preprint-tier flag load-bearing — not yet peer-reviewed at time of drafting; the case is included on the structural contribution (construct distinction, instrument-design move) rather than specific magnitudes. Future validation ongoing.

LENS APPROACH

The longitudinal SE-program reflective-practice study is the evaluation-pathway case for first-person practice accounts (induced 2.3; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development) for the amended CLO-2 and in Domain 4 (Test and Evaluation) for the CLO **Judgment under inadequate evidence** — the preprint supplies the construct distinction (development vs. presence) without supplying peer-reviewed magnitudes.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- When designing a program-level evaluation of reflective-practice capability, distinguish the development construct from the presence construct at the instrument-design stage; the preprint is the worked example of why the distinction matters.
- Specify the temporal cadence of the instrument — when in the program reflections are collected, against what successive iterations of design work — so the development trajectory can be evidenced rather than inferred.
- Carry the preprint-tier flag through any pedagogical decision the case informs; the structural contribution is the construct distinction, and the specific magnitudes await peer review.

IN OPERATION / AFTER

- Track intra-learner depth change in reflective output across the program as the primary measurement target; cross-learner presence comparisons are a secondary construct and should be reported as such.
- Use the preprint as the prior-art reference for any first-person Flywheel evaluation pathway you propose for CIRCUIT, an ERKS-class effort, or a LENS-graduate program; the construct distinction is portable even where the specific instrument is not.
- When the preprint's peer-reviewed version appears, update the tier flag and re-evaluate the magnitudes; the case is included on the structural contribution, not on the preprint's specific figures.

REFLECTION QUESTIONS

1. Identify a multi-year program in your context where reflective-practice capability is supposed to develop. What instrument would you build to measure development (intra-learner depth change over time) rather than presence (snapshot)? At what temporal cadence?

2. The case is a preprint not yet peer-reviewed. What is the minimum additional evidence — peer-review pipeline outcome, replication across other work-based programs, comparison with snapshot-based instruments — you would require before treating any specific magnitude from this work as settled in your program design?

3. Specify the prior art you would assemble around a first-person Flywheel evaluation pathway for an engineering-practice account; which construct distinctions (development vs. presence) would you carry forward, and which would you supplement with locally produced evidence?

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CASE 49 K-12 MATHEMATICS HOMEWORK SUPPORT FORMATIVE ASSESSMENT 2014 – PRESENT

ASSISTments — National Replication and Long-Term Follow-Through

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT Cluster RCT across 46 schools and 3,035 students: 7th-graders assigned to ASSISTments outperformed controls on end-of-year math; largest gains for lower-performing students; minority students benefited more from the intervention; effect persisted into 8th-grade outcomes in 2020 follow-up

IN BRIEF

The Roschelle, Feng, Murphy, and Mason cluster RCT (AERA Open 2016), conducted across 46 schools and 3,035 students, found that 7th-graders assigned to ASSISTments outperformed controls on end-of-year mathematics, with the largest gains for lower-performing students and a heterogeneous-effect finding that minority students benefited more from the intervention. Murphy et al. (2020) reported that the 7th-grade effect persisted into 8th-grade outcomes. A subsequent Arnold Ventures-funded extension tested a lower-cost virtual-training adaptation in predominantly rural areas, with longitudinal follow-through extended through end of 8th grade. The case is one of the few EdTech tools in the corpus with replicated multi-state RCT evidence at meaningful effect sizes and with deliberate attention to the heterogeneity that matters most for equity outcomes. Pair with Case 45 (spaced education RCTs) for the replication-discipline thread. Open questions the authors carry: whether the virtual-training adaptation matches the in-person-training arm; whether the effect persists post-grade-8.

BACKGROUND

ASSISTments is structurally different from the intelligent tutoring systems that dominate the K-12 EdTech evidence base. It augments homework rather than replacing curriculum; it does not require the institutional commitment to a new instructional system that Cognitive Tutor and its peers require; and the research team behind it (Heffernan and collaborators) has deliberately designed the platform around an evidence-generation loop with classroom teachers. The cluster RCT Roschelle et al. published in 2016 is the first national-scale evaluation of the platform, and it was designed to support claims a single-site trial could not support.[o]

THE INTERVENTION

The trial cluster-randomized 46 schools across multiple states and assigned 3,035 7th-grade students to either an ASSISTments condition or a business-as-usual homework condition. The outcome instrument was end-of-year mathematics achievement. The headline finding is positive: ASSISTments-assigned students outperformed controls. The effect is meaningful in size, and the cluster randomization supports an inference at the school level rather than only at the student level. The teacher-side change required for the intervention was deliberately minimized — the platform is built around teacher-assigned homework problems, with the formative-assessment and feedback loops automated — so the trial estimates an effect achievable under realistic adoption conditions.[1]

HOW IT WORKED

The heterogeneity finding is what makes the case equity-relevant. Effect-size estimates were largest for lower-performing students, and minority students benefited more from the intervention than the average effect would suggest. The pattern is the one Case 157 (Engler / enrollment algorithms) names as the inversion target: prediction and adaptive feedback used to trigger support rather than to gatekeep aid. The heterogeneity finding is not an artifact of subgroup analysis chosen post hoc; it is the pre-specified equity-relevant outcome the trial was designed to estimate, and it is the load-bearing result for the case's pedagogical placement.[2]

THE EVIDENCE

Murphy et al. (2020) extended the evaluation into a longitudinal follow-through. The 7th-grade effect persisted into 8th-grade outcomes — a persistence finding that the EdTech evidence base does not consistently report, and that converts the case from a single-year effect-size study into a multi-year follow-through case. A subsequent Arnold Ventures-funded extension tested a lower-cost virtual-training adaptation in predominantly rural areas with longitudinal follow-through extended through end of 8th grade. The replication structure — trial, replication, longitudinal follow-through, adaptation tested under different deployment conditions — is the closed-loop evidence architecture the corpus's EdTech cases mostly aspire to and rarely report.[3]

WHAT TRANSFERRED

The honest open questions the case carries are the ones the research team itself names. Whether the lower-cost virtual-training adaptation matches the in-person-training arm's effect size is still under analysis. Whether the effect persists past grade 8 is the longer-horizon question that the corpus's evaluation-horizon discipline (Case 127) directly applies to. Pair the case with Case 127 (Cognitive Tutor at-scale evaluation) for the evaluation-horizon thread, with Case 45 (spaced education RCTs) for the replication-discipline thread, and with Case 157 (Engler enrollment algorithms) for the prediction-triggers-support inversion — the equity- relevant heterogeneity finding here is the structural complement to Engler's gatekeeping critique.

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2. Murphy, R. et al. (2020), follow-up evaluation extending the 7th-grade effect into 8th-grade outcomes.
3. Heffernan, N. T., & Heffernan, C. L. (2014), "The ASSISTments ecosystem," International Journal of AI in Education 24:470–497 — platform design and research-loop description.
4. Arnold Ventures RCT documentation of the virtual-training-adaptation arm — longitudinal follow-through through end of 8th grade.

THE LEARNING ENGINEERING LENS

> *The heterogeneity finding is pre-specified, not post hoc. The largest gains are for lower-performing students; minority students benefit more. The case's equity-relevant result is the load-bearing one.*
>
> EDITORS' SYNTHESIS OF ROSCHELLE ET AL. (2016) AND MURPHY ET AL. (2020).

LE INSIGHT

ASSISTments is the case in the corpus with the cleanest closed-loop evidence architecture for EdTech: cluster RCT, longitudinal follow-through into the next grade, adaptation arm under different deployment conditions, and a pre-specified equity-relevant heterogeneity finding. The case grounds the closed-loop evaluation anchor in EdTech the same way Case 178 grounds it in team-science training.

LENS APPROACH

ASSISTments is the closed-loop EdTech evaluation case (induced 2.3; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development) for the teacher-side minimum-change design discipline and in Domain 4 (Test and Evaluation) for the heterogeneity-pre-specification and longitudinal- follow-through structure. Pair with Case 127 (Cognitive Tutor at-scale evaluation horizon), Case 45 (spaced education RCTs), and Case 157 (Engler enrollment algorithms inversion — prediction-triggers-support).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pre-specify the equity-relevant heterogeneity outcomes the trial will estimate; the case's load-bearing finding is pre-specified, not post-hoc, and the pre-specification is the methodological discipline that makes the finding credible.
- Design the teacher-side change to the minimum required for the intervention to operate; the case's external-validity strength depends on its having estimated an effect achievable under realistic adoption conditions.
- Build the longitudinal follow-through into the trial's data infrastructure from the start; the 7th-to-8th-grade persistence finding required data structures that single-year trials do not necessarily provide.

IN OPERATION / AFTER

- Publish the heterogeneity result with the aggregate result; the case's equity-relevant pedagogical value depends on the heterogeneity finding being on the same page as the average effect.
- Track the adaptation arm — the lower-cost virtual-training condition — as a separate replication; the closed-loop evidence architecture the case demonstrates includes adaptation-under-different-conditions as a distinct evidence layer.
- Carry the case in syllabi alongside Case 127 so the evaluation-horizon discipline and the heterogeneity-pre-specification discipline are taught together; the two methodological lessons are independent and both load-bearing for EdTech-evaluation design.

REFLECTION QUESTIONS

1. Identify an EdTech intervention in your domain whose equity-relevant heterogeneity outcome was not pre-specified in the trial design. What pre-specification would the next replication require, and what is the data infrastructure that would support it?

2. Specify the longitudinal-follow-through design you would build into the next at-scale EdTech evaluation. What grade-to-grade or year-to-year outcome would you track, and what data infrastructure does the tracking require?

3. The case sits as the prediction-triggers-support inversion of Case 157. Pick a deployed adaptive system in your domain and ask: in which direction does the adaptation point — toward more support for the lower-performing student or toward higher throughput for the average — and where is that design choice documented?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · task analysis · design prototyping

CASE 50 LEARNING ANALYTICS ONLINE LEARNING MOBILE/DISTANCE LEARNING 2016 – 2025

The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT Van Campenhout et al. (LAK 2023) replicated the doer-effect causal claim across seven courses with 15.2 million data points; L@S 2025 replication held with AI-generated practice questions; LAK 2025 non-WEIRD radio/phone extension found weaker effect for learners with higher prior educational attainment — the load-bearing heterogeneity finding

IN BRIEF

The original “doer effect” causal claim — Koedinger and colleagues at LAK 2016 — held that students who interact with embedded practice activities learn more than students who only read, even after controlling for prior achievement and engagement, and that the effect appears causal. Van Campenhout et al.’s LAK 2023 paper replicated the claim across seven courses with 15.2 million data points; the L@S 2025 follow-up reported the effect held with AI- generated practice questions; the Butler et al. LAK 2025 non-WEIRD extension tested the effect for learners receiving lecture content via community radio and practice via basic mobile phones, and reported that the doer-effect relationship was weaker for learners with higher prior educational attainment — the load-bearing heterogeneity finding the corpus most needs. The case sits with Case 40 (Duolingo half-life) and Case 45 (spaced education RCTs) as the replication-arc thread. The closed loop is closed not by a single trial but by replication; the effect generalizes but not uniformly, and the heterogeneity is itself the teachable result.

BACKGROUND

The doer-effect causal claim is one of the cleanest published claims in the online-learning literature. Koedinger and colleagues, at LAK 2016, drew on student- level data from large online courses and isolated the effect of doing — interacting with embedded practice activities — from the effect of reading, controlling for prior achievement and engagement. The conclusion: doing causally improves learning more than reading does, by an effect size that has stood up across the field’s subsequent replications. The case Van Campenhout et al. open at LAK 2023 is what happens when the claim is treated not as a single result but as a replication target — what does the doer effect look like at scale, across courses, content domains, and delivery modalities?[0]

THE INTERVENTION

The LAK 2023 replication is the broadest, by data volume. Van Campenhout et al. assembled data from seven courses and 15.2 million practice and reading interactions, and reported the doer-effect relationship held in the direction and approximate magnitude the original claim established. The 2025 L@S follow-up addressed a question that did not exist when the original claim was published: what happens to the doer effect when the practice questions are AI-generated rather than human-authored? The L@S 2025 replication reported the effect held with auto-generated content — a meaningful generalization given the rising deployment of LLM-generated practice across educational platforms.[1]

HOW IT WORKED

The 2025 LAK non-WEIRD extension by Butler and collaborators is the case's load-bearing methodological contribution. The extension tested the effect for learners receiving lecture content via community radio and practice via basic mobile phones — a delivery substrate structurally different from the platform-mediated online courses the original claim and the LAK 2023 replication operated on. The doer-effect relationship was weaker for learners with higher prior educational attainment. The heterogeneity is not noise the analysis should adjust away; it is the result. The original causal claim generalizes across delivery modalities, but the effect size is conditioned on prior attainment in a way the WEIRD-population evidence base did not surface.[2]

THE EVIDENCE

The case is closed-loop in a sense the corpus most often aspires to. The loop is closed not by a single trial but by replication: the original claim, the seven-course large-N replication, the AI-generated-content replication, and the non-WEIRD-modality extension are four converging pieces of evidence. The closed loop is also honest about what it closes and what it does not. Long-term retention across multi-year intervals is not yet in the replication record; transfer beyond the included content domains is not yet in the replication record; the non-WEIRD-modality heterogeneity is documented but not yet decomposed into the components — content familiarity, modality affordances, attention conditions — that would let designers act on it. The case carries those open questions rather than collapsing them.[3]

WHAT TRANSFERRED

The case anchors with Case 40 (Duolingo half-life) and Case 45 (spaced education RCTs) as the replication-arc thread. All three demonstrate the closed-loop discipline at field scale: a design principle established as a single result, replicated across contexts, and surfaced as conditional on population and modality. The CLO on judgment under inadequate evidence is operative in a productive sense: the original claim was adequate evidence for the WEIRD-platform-online-course context, and the non-WEIRD-modality evidence the LAK 2025 extension adds is what extends the principle's actionable scope. The case completes the replication-arc thread the corpus needs to teach design-principle generalization honestly.

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THE LEARNING ENGINEERING LENS

> *The doer effect generalizes across delivery modalities, but the effect size is conditioned on prior attainment in a way the WEIRD-population evidence base did not surface. The heterogeneity is the result.*
>
> EDITORS' SYNTHESIS OF VAN CAMPENHOUT ET AL. (2023, 2025) AND BUTLER ET AL. (2025).

LE INSIGHT

The doer-effect replication arc is the closed-loop-via- replication case in the corpus. Original claim, seven-course large-N replication, AI-generated-content replication, non- WEIRD-modality extension — four converging pieces of evidence with the prior-attainment heterogeneity finding as the load- bearing result. The case completes the replication-arc thread alongside Cases 40 and 45.

LENS APPROACH

Doer-effect replication arc is the closed-loop-by- replication case (induced 2.3; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development) for the replication- arc discipline and in Domain 4 (Test and Evaluation) for the cross-population generalization-with-heterogeneity structure. Pair with Case 40 (Duolingo half-life) and Case 45 (spaced education RCTs) — the replication-arc thread teaches the closed-loop discipline at field scale.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the design principle as the replication target, not the original trial; the doer-effect case's structure is built on four converging pieces of evidence rather than on the original claim alone.
- Pre-specify the cross-context replication conditions before launching the replication: course scope, content domain, delivery modality, and population characteristics are all conditions the original effect was estimated under and that the replication should vary.
- Test design-principle generalization at the modality boundary before claiming generalization across modalities; the LAK 2025 non-WEIRD extension is the methodological model for that boundary test.

IN OPERATION / AFTER

- Publish the heterogeneity as the result, not the adjusted-away noise; the non-WEIRD prior-attainment finding is what extends the principle's actionable scope and is the load-bearing pedagogical content of the case.
- Build the long-term-retention and cross-content-transfer studies the replication arc names as the next open questions; the closed loop the case completes is honest about what it does and does not close.
- Carry the case in syllabi alongside Cases 40 and 45 so the replication-arc thread is taught as a thread, not as three independent examples; the closed-loop discipline at field scale is more visible across the three cases than within any one.

REFLECTION QUESTIONS

1. Identify a design principle in your domain that has been replicated within WEIRD platform conditions but not across delivery modalities. What is the modality-boundary test you would design, and what would the analog of the radio-lecture-plus-mobile-phone substrate look like in your domain?

2. Specify the pre-specified heterogeneity outcomes you would build into the next replication of a design principle in your domain. What population characteristic do you expect would condition the effect size, and what would constitute disconfirming evidence?

3. The replication arc's load-bearing pedagogical content is the heterogeneity finding, not the average effect. Pick a published replication in your domain and ask: what heterogeneity did the replication surface, and how does the field's current teaching of the design principle handle the heterogeneity?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · task analysis · design prototyping

CASE 51 K-12 SCIENCE COLLABORATIVE LEARNING DESIGN-BASED RESEARCH 2009

Zhang/Scardamalia — Knowledge Building Across Three Cohorts

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT Three-year design study in a single Grade 4 classroom (Institute of Child Study, Toronto) using Knowledge Forum across an optics curriculum; documented progression from fixed small-group to opportunistic collaboration across cohorts, with associated gains in the depth and distribution of scientific explanations across the class community

IN BRIEF

Zhang, Scardamalia, Reeve, and Messina's 2009 Journal of the Learning Sciences paper reports a three-year design study in a single Grade 4 classroom at the Institute of Child Study in Toronto, working through an optics curriculum supported by the Knowledge Forum platform. Across three successive cohorts, the classroom's collaborative structure progressed from a fixed small-group organization to an opportunistic one in which students convened around emerging questions and dispersed when those questions were addressed. Outcomes on the depth and distribution of scientific explanations across the class community improved in step with the structural progression. The hedges that travel with the case from the source are binding: single teacher, single school, design study not causal in the trial sense; transferability to non-ICS contexts and to teachers without knowledge-building expertise is open. The case is the classroom-scale longitudinal counterpart in the small-tier evidence base; the LE Lens uses it as the JLS- anchored small-tier complement to v1 Case 38.

BACKGROUND

Knowledge Building is the pedagogical program Marlene Scardamalia and Carl Bereiter developed across the 1980s and 1990s, organized around the idea that classrooms can operate as knowledge-creating communities rather than as receivers of established content. The Knowledge Forum platform is the supporting infrastructure: a shared database in which students post conjectures, build on one another's notes, mark rise-above syntheses, and trace the evolution of community ideas. The 2009 Journal of the Learning Sciences paper is the systematic three-year study of how the collaborative structure inside a single Grade 4 classroom developed across cohorts as the teacher and students worked the program through.[o]

THE INTERVENTION

The setting is deliberately constrained. A single Grade 4 classroom at the Institute of Child Study in Toronto, a laboratory school with a Knowledge Building tradition; a single teacher across the three years; an optics curriculum delivered to three successive cohorts; Knowledge Forum as the supporting platform. The design study's interventions across cohorts were not randomized treatment arms but iterated redesigns of how collaboration was organized in the classroom — what kind of grouping the students worked in, how questions were surfaced and addressed, how the classroom community decided what counted as a productive direction. The design-based research method treats this iterative redesign as the unit of inquiry.[1]

HOW IT WORKED

The structural progression Zhang and colleagues document is from fixed small-group collaboration to opportunistic collaboration. In the first cohort, students worked in stable small groups that took on assigned subtopics within optics. By the third cohort, the classroom had moved to an opportunistic structure: students convened around emerging questions as those questions surfaced from the community's shared work in Knowledge Forum, and dispersed when the questions had been addressed. The depth of scientific explanations in the community's shared notes improved in step with the structural change, and the distribution of explanatory contribution across students became less concentrated — more students were authoring substantive contributions, and the substantive contributions were reaching further into the underlying physics.[2]

THE EVIDENCE

The hedges that travel with the case from the source are binding and load-bearing. The study covers a single teacher, a single school, and a single curriculum domain; the design-based research method does not produce a causal estimate in the trial sense, and the authors are explicit that what they report is a developmental account of how the classroom's collaborative practice changed across cohorts, not a controlled comparison. Transferability to non-ICS contexts is open. Transferability to teachers without Knowledge Building expertise is open. The case is not the evidence that the Knowledge Building program works at scale across heterogeneous classrooms; it is the evidence that in a specific well-supported setting, a classroom's collaborative structure can be iteratively redesigned across cohorts in a direction that improves the depth and distribution of scientific explanation.[3]

WHAT TRANSFERRED

The case sits as the classroom-scale longitudinal counterpart in the small-tier evidence base. Pair with Case 178 (Colorado CTSA team science) for the collaboration-measurement thread at a different scale and domain; with Case 77 (Hybrid Human-AI Tutoring) for the small-tier deployment-success counterpart; with Case 115 and Case 128 (OU Analyse) for the distance-higher-education governance frame at a different population. The LE Lens uses the case as the JLS-anchored small-tier complement to v1 Case 38 (Cognitive Tutor's single-site arc); the two cases together teach that the longitudinal classroom record is the substrate that learning-engineering iterations operate on, and that the iteration unit is the cohort, not the lesson.

REFERENCES

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THE LEARNING ENGINEERING LENS

> *The cohort is the iteration unit. The collaborative structure of a single Grade 4 classroom moved from fixed small-group to opportunistic across three years, and the depth and distribution of scientific explanation moved with it.*
>
> EDITORS' SYNTHESIS OF ZHANG, SCARDAMALIA, REEVE, & MESSINA (2009).

LE INSIGHT

Zhang and colleagues' three-year design study is the classroom-scale longitudinal record of how a collaborative structure can be iteratively redesigned across cohorts in a direction that improves the depth and distribution of scientific explanation. The hedges are binding — single teacher, single school, design study not causal — and the case's pedagogical value depends on the hedges being preserved. The cohort is the iteration unit.

LENS APPROACH

Zhang/Scardamalia is the cohort-as-iteration-unit case at classroom scale (induced 2.2; LENS D2/PT4; CLO-2 and CLO-4). LENS uses it in Domain 2 (Iterative Development) for the multi-cohort design-based-research discipline and in Domain 4 (Test and Evaluation) for the depth-and-distribution paired outcome measure. Pair with Case 178 (Colorado CTSA team science collaboration measurement), Case 77 (Hybrid Human-AI Tutoring small-tier deployment), and Cases 115 and 128 (OU Analyse — distance higher-education governance at a different scale). The LE Lens uses it as the small-tier complement to v1 Case 38 (Cognitive Tutor single-site arc).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the cohort as the iteration unit when the classroom collaborative structure is the design target; the case demonstrates that a year is the right horizon for a meaningful redesign cycle and a multi-year horizon is required to surface the progression.
- Design the infrastructure — Knowledge Forum's shared note database — to make the community's evolving ideas inspectable; the redesign across cohorts depended on the teacher and students having a record of how prior cohorts' ideas had moved.
- Track depth and distribution of contribution as paired outcome measures; reporting depth alone hides the equity-relevant distribution change, and reporting distribution alone hides the substantive-progress signal.

IN OPERATION / AFTER

- Carry the binding hedges into print without softening; the case's value to the corpus rests on its specificity, and the transferability questions are open and disclosable as open.
- Pair with Case 178 (collaboration measurement at team-science scale) so the collaboration-as-design-target thread is taught at both the classroom and team scales.
- Use the case as the JLS-anchored small-tier complement to v1 Case 38 (Cognitive Tutor's single-site arc); the cohort-as-iteration-unit lesson is the bridge between the classroom-design literature and the at-scale evaluation literature.

REFLECTION QUESTIONS

1. Identify a classroom or team in your domain whose collaborative structure has not been redesigned across cohorts. What would a cohort-as-iteration-unit redesign cycle look like, and what shared-infrastructure artifact would make the prior cohort's evolution inspectable for the next?

2. Specify the paired outcome measures you would track when collaborative structure is the design target. Depth and distribution travel together in the Zhang case; what is the analogous pair in your setting, and what would reporting one without the other obscure?

3. The case's transferability hedges are open — non-ICS contexts, non-Knowledge-Building-expert teachers. Pick a feature of the ICS setting that the case depends on, and ask: what would have to be true in your setting for the redesign discipline to travel, and what wouldn't?

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CASE 52 K-12 EDUCATION AI-SUPPORTED INSTRUCTION EDUCATIONAL EQUITY 2025

Chen et al. — Rural China AI Devices and the Equity-Direction Finding

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT Quasi-experimental study across 12 schools (4 urban, 8 rural) and 268 teachers, September to November 2024; rural experimental classes gained 17.93% on mathematics and 13.46% on history, while urban experimental classes gained 10.96% on mathematics and 9.55% on history — the rural gain exceeded the urban gain across both subjects

IN BRIEF

Chen, Wu, Chen, and Zhou's 2025 paper in Frontiers in Psychology reports a three-month quasi-experimental study of AI-supported instructional devices across 12 schools — 4 urban and 8 rural — in China, involving 268 teachers from September to November 2024. The headline result is the equity-direction finding: rural experimental classes gained 17.93% on mathematics and 13.46% on history, while urban experimental classes gained 10.96% on mathematics and 9.55% on history. The rural gain exceeded the urban gain across both subjects. The honest hedges that travel with the case from the source are load-bearing and preserved in the prose: the horizon is three months; the study covers 12 schools; assignment is non-randomized; the authors acknowledge self-report and observation biases in the measurement instruments. All four authors are at Chinese institutions — Hangzhou City University, Lingnan University, and Huazhong University of Science and Technology — and the case stands as the first-author-from-deployment-country evidence the corpus's non-WEIRD thread most needs.

BACKGROUND

The corpus's non-WEIRD deployment record is structurally thin. The published learning-engineering evidence base is built largely on WEIRD-population studies — Western, educated, industrialized, rich, democratic — and the deployments that operate at meaningful scale outside that population are rarely written up in peer-reviewed journals by authors based in the deployment country. Chen and colleagues' 2025 Frontiers in Psychology paper is one of the strongest documented exceptions: a deployment of AI- supported instructional devices across rural and urban Chinese schools, evaluated quantitatively, reported by a four-author team based at Chinese institutions, and published in a peer-reviewed journal with the load-bearing methodological hedges named.[0]

THE INTERVENTION

The deployment covers 12 schools — 4 urban and 8 rural — and 268 teachers across the September to November 2024 window. The intervention is AI-supported instructional devices integrated into mathematics and history classroom instruction. The comparison structure is between experimental and control classes within each school setting across both subjects. The headline outcome the paper reports is the equity-direction finding: rural experimental classes gained 17.93% on mathematics and 13.46% on history, while urban experimental classes gained 10.96% on mathematics and 9.55% on history. The rural-over-urban gain pattern is the result the case teaches; the equity direction is unusual in the published EdTech literature, where deployments more often surface as widening the gap between higher- and lower-resourced settings rather than narrowing it.[1]

HOW IT WORKED

The honest hedges the case carries are load-bearing and explicit in the source. The horizon is three months — a September to November 2024 window — and the case's pattern is conditioned on that horizon, not on a school-year or multi-year follow-through. The sample is 12 schools, which is right for the within-subject within-school comparison structure the study runs but is not right for prevalence claims about Chinese rural schooling at population scale. Assignment to experimental and control classes is non- randomized; the authors are explicit that they cannot rule out teacher- or school-level selection effects driving the heterogeneity. Self-report and observation biases in the measurement instruments are acknowledged by the authors. All four hedges travel into the prose without softening; the case is published in a peer-reviewed mid-tier journal, and the evidence-tier flag rendering is intentionally not set — the hedges are carried in the case's own argument rather than in the title-bar render.[2]

THE EVIDENCE

The case's structural placement in the corpus is the equity-direction-finding case at the non-WEIRD deployment seam. Pair with Case 133 (Gándara / community-college predictive equity in AERA Open) for the equity-direction thread at the higher-education scale; with Case 135 (LiveHint AI bias across foundation models) for the bias- surfacing thread in AI-supported instruction; with Case 50 (Doer Effect non-WEIRD radio-and-phone extension) for the non-WEIRD methodological discipline at the heterogeneity- finding axis. The first-author-from-deployment-country structure of Chen et al. is itself the methodological contribution the corpus's non-WEIRD thread needs: the strongest reading of a deployment is the one written by authors with access to the deployment context and published in the journal architecture the field can read.[3]

WHAT TRANSFERRED

The pedagogical seam the case opens is the equity-direction reading of an AI-supported instructional intervention. The conventional reading — that deployments widen the gap between higher- and lower-resourced settings — is not what Chen and colleagues report on the three-month horizon they studied. The rural gain exceeded the urban gain across both subjects, and the result is the load-bearing teaching point the case anchors. The case does not claim that the pattern generalizes past three months; it does not claim that the pattern holds outside the 12 schools the study covers; it does not claim causal identification past the bounds the non-randomized assignment supports. What it reports is the equity-direction finding in the deployment window the study describes, and the case uses that finding to anchor the CLO on fairness beyond omission — the finding is the rare published example where an AI deployment in education narrowed rather than widened a between-setting gap on its evaluation horizon, and the methodological structure of the study supports treating the result as the case's load-bearing observation rather than as a discounted outlier.

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2. Paired Case 133 (Gándara et al., AERA Open) — community-college predictive equity at the higher-education scale.
3. Paired Case 135 (LiveHint AI bias across foundation models, AIED 2025) — bias-surfacing in AI-supported instruction.
4. Paired Case 50 (Doer Effect non-WEIRD LAK 2025 radio-and-phone extension) — non-WEIRD methodological discipline at the heterogeneity-finding axis.

THE LEARNING ENGINEERING LENS

> *Rural experimental classes gained 17.93 percent on mathematics and 13.46 percent on history; urban experimental classes gained 10.96 percent on mathematics and 9.55 percent on history. The rural gain exceeded the urban gain across both subjects — and the load-bearing hedges are three-month horizon, twelve schools, non-randomized assignment, self-report and observation bias acknowledged by the authors.*
>
> EDITORS' SYNTHESIS OF CHEN, WU, CHEN, & ZHOU (2025).

LE INSIGHT

Chen and colleagues' 12-school three-month deployment of AI- supported instructional devices across rural and urban Chinese classrooms reported an equity-direction finding — rural gain exceeded urban gain across both mathematics and history. The load-bearing hedges are explicit and carried in the prose: three-month horizon, twelve schools, non-randomized assignment, self-report and observation bias acknowledged by the authors. The case anchors the CLO on fairness beyond omission with a rare published equity-narrowing result.

LENS APPROACH

Chen et al. is the equity-direction-finding case at the non-WEIRD deployment seam (induced 8.3; LENS D2/PT4; CLO-4 and fairness beyond omission). LENS uses it in Domain 2 (Iterative Development) for the deployment-on-a-defined- horizon discipline and in Domain 5 (Navigating Sociotechnical Constraints) for the equity-direction reading that anchors the fairness-beyond-omission CLO. Pair with Case 133 (Gándara community-college predictive equity), Case 135 (LiveHint AI bias across foundation models), and Case 50 (Doer Effect non-WEIRD radio-and-phone extension). The case's pedagogical value depends on the four binding hedges traveling with the result into print.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pre-specify the equity-direction outcome as the load-bearing reading; the case demonstrates that a deployment study can credibly report a rural-over-

IN OPERATION / AFTER

- Commission the longer-horizon and larger-sample replication the three-month 12-school study cannot perform; the equity-direction finding's persistence urban gain on a defined horizon when the methodological hedges are named.

- Carry the four binding hedges — horizon, sample, assignment structure, measurement bias — in the deployment write-up itself; the case's value to the corpus depends on the hedges being internal to the argument rather than buried in a limitations footnote.
- Treat first-author-from-deployment-country authorship as a methodological feature; the non-WEIRD evidence base is structurally improved by deployments written up by teams with access to the deployment context.

past three months and across more schools is the validation question the case names.

- Pair the case in syllabi with Case 133 (Gándara community-college equity) and Case 50 (Doer Effect non-WEIRD extension) so the equity-direction reading is taught across deployment scales and population settings.
- Use the case to anchor the CLO on fairness beyond omission with a rare published equity-direction finding; the corpus needs equity-narrowing results as visible as equity-widening ones, and the methodological discipline that lets the result be reported credibly is the curricular target.

REFLECTION QUESTIONS

1. Identify a deployment in your domain whose equity-direction finding — narrowing or widening a between-setting gap — has not been pre-specified as a primary outcome. What would the pre-specification require, and what data infrastructure would support it?

2. Specify the four load-bearing hedges you would carry in the prose of a non-WEIRD deployment write-up — horizon, sample, assignment structure, measurement bias. Which of the four is hardest to name credibly in your domain, and what would naming it cost?

3. The case's structural feature is first-author-from-deployment-country authorship. Pick a deployment in your domain that has been written up primarily by external authors, and ask: what would change in the methodological reading if the write-up were led by authors with access to the deployment context?

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CASE 53 MULTIMODAL LEARNING ANALYTICS CLASSROOM DEPLOYMENT DESIGN-BASED RESEARCH 2023

Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER preprint-tier — source confidence flagged; future validation ongoing.

IMPACT First-person practitioner reflection on multiple in-the-wild multimodal learning analytics (MMLA) deployments — eye-tracking, audio capture, spatial positioning in classroom contexts; documents what worked, what failed, what the team would have done differently

IN BRIEF

Martinez-Maldonado et al.'s 2023 arXiv paper, "Lessons Learnt from a Multimodal Learning Analytics Deployment In-the-Wild," is structured as a first-person practitioner reflection on lessons from multiple in-the-wild MMLA deployments — eye-tracking, audio capture, spatial positioning in classroom contexts. The paper documents what worked, what failed, and what the team would have done differently. The case is offered not as a deployment- results case (the deployment outcomes live in adjacent peer-reviewed papers) but as a published-first-person Practice Flywheel exemplar — the genre the front-matter "unpacking is the method" reframing calls for. The case pairs structurally with the reflective-practice cases elsewhere in the v2 supplemental tier and grounds the practitioner-reflection-as-evidence-tier discipline at the LE-specific layer. Preprint-tier evidence-flag is binding on the framing — the arXiv version is preprint, with sections published in adjacent peer-reviewed work, and the standing "future validation ongoing" language applies to both the peer-review pipeline for this version and the broader question of whether the genre takes hold across the LE community.

THE SHIFT

Multimodal learning analytics is the strand of the learning-analytics field where the data substrate moves beyond click-stream and assessment-response logs to include eye-tracking, audio capture, video, and spatial positioning. The lab-deployment record for MMLA is substantial; the in-the-wild classroom-deployment record is structurally different and structurally sparse. What happens when the sensor stack moves from the controlled lab environment into the classroom — where lighting varies, students move, audio overlaps, and the consent architecture has to accommodate the school's operational norms — is a question the published deployment-results papers can answer only partially.[0]

WHAT IS EMERGING

Martinez-Maldonado et al.'s 2023 arXiv paper is structured as a first-person practitioner reflection on lessons from multiple in-the-wild MMLA deployments. The paper's content is what the deployment-results papers do not contain: what worked, what failed, what the team would have done differently. The structure is reflective rather than hypothesis-testing — the team narrates the deployments, identifies the recurring patterns, and surfaces the operational knowledge that does not fit into a methods section of a results paper. The honest framing the paper preserves is that the reflective genre is the right vehicle for the kind of operational knowledge the case contains, and that the peer-review structures of the LE field have not consistently supported the genre.[1]

THE CAPABILITY QUESTION

The case is offered in the corpus not as a deployment- results case but as a published-first-person Practice Flywheel exemplar. The genre the editor's memo (B1) anticipates — first-person practitioner accounts of deployment iterations, intended to be paired with the front-matter "unpacking is the method" reframing — has structural analogs in adjacent fields (Lutz on reflective journaling, CBE-LSE on reflective-practice primers, SE work-based reflective-practice longitudinals) but has historically been under-published in LE. The Martinez-Maldonado paper is the LE-specific instance of the genre at deployment scale; the case carries it on that basis.[2]

EARLY EVIDENCE

The structural anchor the case grounds is the sustaining- tacit-capability-across-generations anchor. Practitioner knowledge — what to do when the eye-tracker calibration drifts mid-session, how to design the consent architecture for an audio capture in a classroom of twenty-five students, what the spatial-positioning sensor placement looks like when the classroom layout shifts — walks out the door if it is not narrated. The paper's first-person genre is the narration vehicle, and the case grounds the curriculum's response to the question of how operational knowledge accumulates and propagates across practitioner cohorts. The pair with the CIRCUIT cases (119, 120) is the workforce-and-capability layer; the pair with Case 77 (hybrid human-AI tutoring) is the design-iteration layer.[3]

OPEN PROBLEMS

The preprint-tier evidence-flag is binding on the framing. The arXiv version is preprint; sections have been published in adjacent peer-reviewed work, but the consolidated lessons-learned synthesis the case carries sits at the preprint tier. The standing "future validation ongoing" language applies along two dimensions. The peer- review pipeline for this specific consolidated synthesis is one dimension. The broader question — whether the first-person practitioner-reflection genre takes hold in the LE community at sufficient scale to function as the Practice Flywheel exemplar the framework names — is the other dimension. The case is included not despite the preprint-tier framing but with it; the framing is part of what the case teaches.

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2. Worsley, M., & Blikstein, P. (2018), "A multimodal multisensor framework for examining how students engage in design," Journal of Learning Analytics — broader MMLA literature backdrop.

3. Schon, D. (1983), The Reflective Practitioner — the genre's theoretical underpinning, referenced across the reflective-practice case tier.
4. Editors' memo (B1) — Practice Flywheel commissioning structure that the case is offered as a published-first-person exemplar within.

THE LEARNING ENGINEERING LENS

> *Practitioner knowledge walks out the door if it is not narrated. The first-person reflective genre is the narration vehicle, and the field has not consistently supported it.*
>
> EDITORS' SYNTHESIS OF MARTINEZ-MALDONADO ET AL. (2023).

LE INSIGHT

Martinez-Maldonado et al. is the LE-specific published- first-person Practice Flywheel exemplar at MMLA in-the-wild deployment scale. The case is offered not as a deployment- results case but as a genre exemplar — the reflective- practice account at the field's preprint tier. Preprint-tier flag binding; future validation ongoing on peer-review pipeline and on whether the genre takes hold across the LE community.

LENS APPROACH

MMLA in-the-wild is the sustaining-tacit-capability case in the LE-conferences tier (induced 6.3; LENS D2/PT4). LENS uses it in Domain 2 (Iterative Development) for the reflective-narration-of-design-iteration discipline and in Domain 4 (Test and Evaluation) for the evidence-tier discipline binding the preprint-tier framing to the genre's pedagogical role. Pair with Cases 81 and 39 (CIRCUIT workforce-and-capability layer) and Case 77 (hybrid human- AI tutoring design-iteration layer). Preprint-tier flag binding under the title.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Narrate the deployment in first person while it is still operating; the operational knowledge the case names is contemporaneous, and the post-hoc reconstruction loses the texture the first-person genre preserves.
- Treat the reflective paper as a deliverable on par with the results paper; the case demonstrates that the operational knowledge has the same evidentiary status as the methods-section content of a results paper, and the publication structure should support it.
- Build the consent and ethics architecture around the in-the-wild deployment substrate from the start; the lessons-learned content includes consent-architecture failures that the lab-deployment record does not surface.

IN OPERATION / AFTER

- Move the consolidated synthesis through the peer-review pipeline; the preprint-tier evidence-flag is binding now, and converting the synthesis to peer-reviewed publication is the validation step the standing language anticipates.
- Carry the Practice Flywheel exemplar designation into the curriculum's first-person-account commissioning structure; the case is the LE-specific anchor for the genre the editor's memo (B1) calls for, and the curriculum is the vehicle that institutionalizes the genre.
- Pair the case with the front-matter "unpacking is the method" reframing so the genre's role in the casebook's pedagogical architecture is visible; the case is offered as an exemplar of the genre, and the genre is offered as the curriculum's response to the sustaining-tacit-capability question.

REFLECTION QUESTIONS

1. Identify a deployment in your domain whose operational knowledge — what worked, what failed, what the team would have done differently — has not been narrated outside the team. What would the first-person reflective account look like, and what publication venue would carry it?

2. Specify the consent and ethics architecture you would build into an in-the-wild deployment in your domain that the lab-deployment record would not have surfaced. The case's lessons-learned content includes consent-architecture failures; what would the deployment-substrate-specific architecture look like?

3. The case is offered as a published-first-person Practice Flywheel exemplar. Identify a practitioner in your domain whose operational knowledge you would commission a first-person account from. What would the commissioning structure look like, and what would the account contain that the published-results papers do not?

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Chapter 5

Human-System Collaboration — What Fails

When the human-system boundary is designed against the human.

The most loaded the human is, the less help the interface gave.

AN OBSERVATION RECURRING ACROSS THE CHAPTER'S CASES.

CASE 54 DEFENSE 1988

USS Vincennes & Iran Air Flight 655

[H] Human-System Interface [T] Training Gap

IMPACT 290 civilians killed — the deadliest shootdown of a commercial airliner by a military force; precipitating case for Aegis CIC display and decision-aid doctrine reform, and a central reference in the human-AI teaming literature

IN BRIEF

On 3 July 1988, during a surface skirmish with Iranian gunboats in the Persian Gulf, the USS Vincennes shot down Iran Air Flight 655 — a civilian Airbus A300 climbing on a scheduled route — killing all 290 aboard, the deadliest airliner shootdown by a military force. The Aegis combat system worked: its radar correctly showed the aircraft ascending on the published civilian-traffic corridor. Yet operators in the Combat Information Center, primed by a simultaneous surface fight to expect a hostile inbound, read the contact as a descending F-14 and fired. The Fogarty Report (DoD, 1988) attributed the tragedy to human error under extreme stress — confirmation bias, “scenario fulfillment,” and the unconscious distortion of data — not equipment failure. The contemporaneous Newsweek “Sea of Lies” investigation and a 2018 US Naval Institute Proceedings retrospective reopened both the operational record and the interface-design lessons. Vincennes is the book’s foundational human-AI-teaming case: the most advanced surface combatant afloat failed because its CIC interface and the operational framing made the correct reading possible in principle and unsustainable in practice under combat stress.

BACKGROUND

In the closing weeks of the Iran-Iraq War, US warships patrolled a tense Persian Gulf where civilian airliners and hostile military aircraft shared the same crowded sky. The USS Vincennes, a Ticonderoga-class cruiser with the Navy’s most advanced Aegis combat system, was in a surface fight with Iranian gunboats — itself a contested engagement that the later Newsweek “Sea of Lies” investigation argued had taken the ship into Iranian territorial waters — when an aircraft took off from Bandar Abbas, an airfield used by both civil and military traffic.[0] The dual-use field meant a single track could be either threat or scheduled flight, so the crew’s reading of the contact carried the whole burden of identification — and the Combat Information Center was already managing divided attention against the surface engagement and against an interface whose track display did not directly show altitude trend, the kind of conditions under which a wrong call becomes far easier to make than a right one.

WHAT HAPPENED

The aircraft was Iran Air Flight 655, a civilian Airbus A300 climbing through 12,000 feet on its scheduled route to Dubai. The Vincennes’ crew identified it as a descending, hostile F-14 and fired two SM-2 surface-to-air missiles; all 290 people aboard, including 66 children, were killed — the deadliest shootdown of a commercial airliner by a military force.[1] An ascending airliner and a diving attack jet are opposite behaviors, yet the crew converged on the second while the Aegis radar reported the first. The engagement consumed only seven minutes from radar detection to launch, collapsing identification, IFF interrogation, and decision into a window too narrow for anyone to slow down and reconcile the contradiction.

THE INVESTIGATION

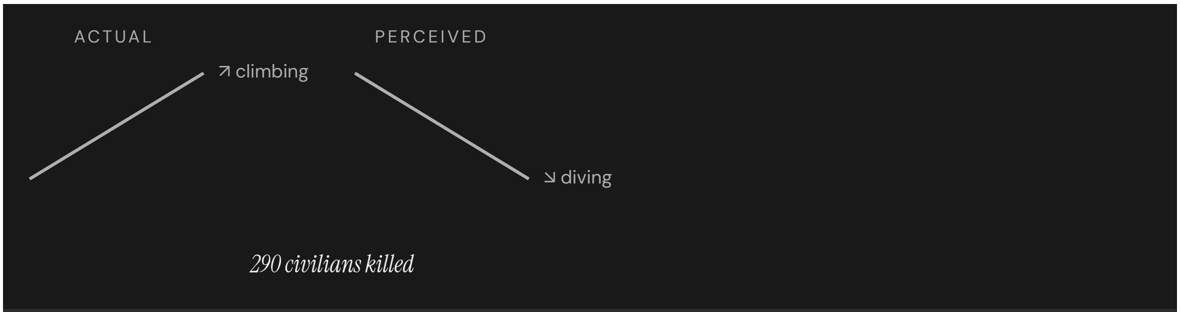
The Aegis SPY-1A radar had not malfunctioned: it correctly showed the aircraft ascending. The IFF interrogation system added a structural confusion of its own: the Vincennes' IFF returns reflected both the airliner's Mode III civilian transponder and, apparently, a Mode II military code from another aircraft on the ground at Bandar Abbas — a mode-confusion failure that left the crew with ambiguous identification at the worst possible moment.[2] The Fogarty Report (Rear Adm. William Fogarty, DoD, August 1988) attributed the tragedy to human error under extreme stress — confirmation bias and the "stress and unconscious distortion of data" — naming "scenario fulfillment" as the psychological mechanism by which operators read every indication through the lens of a presumed hostile attack. Several crew members in the CIC independently believed the aircraft was descending,[3] a shared error that is more damning than a single misread because it shows the interface offered no cross-check strong enough to break a wrong reading once the team had settled into it. The framing arrived before the data did, so each new return was fitted to the expected threat rather than weighed against it.[4]

THE CAPABILITY GAP

The system did not lie, and the operators did not act in bad faith. The interface and the operational framing combined to make a particular misreading not merely possible but likely. Correct performance was possible in principle and unsustainable in practice — and the gap between those two is the engineering problem. An Aegis CIC display that surfaced altitude as a numeric readout but not as a trend, that gave equal weight to civilian and military IFF returns under mode confusion, and that offered no mandatory disconfirmation step before weapons release, had not been designed for the stress it would actually meet. The burden of overriding a presumed hostile attack was left entirely to the operator's discipline, exactly when combat had stripped that discipline of the time and calm it needed to work.[5]

AFTERMATH & REFORM

The Navy's post-incident response included revised tactical doctrine for civilian-traffic deconfliction in the Gulf, updated Aegis training scenarios that explicitly rehearsed the scenario-fulfillment failure mode, and changes to the CIC display sequence and crew procedures around IFF interpretation under mode confusion. The case has remained a standing reference in the human-AI-teaming literature: a 2018 US Naval Proceedings retrospective placed it at the heart of how operators should be teamed with automated decision aids under stress, and the situation-awareness, naturalistic-decision-making, and automation-trust literatures (Klein, Endsley, Cummings) treat it as the canonical worked example.[6] Its lesson is that interface design is a capability deliverable, not an aesthetic one — and that the most advanced system afloat is only as good as the human reading it under fire. The retrospective reframed the loss not as a one-off error but as a predictable outcome of teaming a person with a decision aid that displayed truth without defending it, a pattern that recurs wherever automation is fast and the human is the last check.



AEGIS RADAR DATA CORRECTLY DISPLAYED; FIVE OPERATORS READ IT WRONG

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THE LEARNING ENGINEERING LENS

> *The shootdown of Flight 655 ... reveals lessons for technology adoption and its use in stressful situations.*
>
> PARAPHRASING U.S. NAVAL PROCEEDINGS RETROSPECTIVE ON THE VINCENNES INCIDENT, 2018

LE INSIGHT

Vincennes is the canonical case for human-AI teaming under stress. The system did not lie. The operators did not act in bad faith. The interface and the operational framing combined to make a particular misreading not just possible but likely. Correct performance was possible in principle and unsustainable in practice — and that gap is the engineering problem.

LENS APPROACH

Vincennes is the foundational mode-and-state-transparency case under combat stress (induced 3.3; LENS D2/PT6 human-AI teaming). LENS uses it in Domain 2 (Iterative Development; CLO-2) for interface-as-deliverable: a CIC display tested against the worst case it would actually meet — confirmation bias during a simultaneous surface engagement — not against the calm of acceptance testing. LENS uses it in

Domain 3 (Human-System Collaboration Adaptation; CLO-3) for the human-AI teaming problem itself: a decision aid that displays truth without defending it predictably fails wherever automation is fast and the human is the last check. The case sits at the heart of the program's argument that interface design is a capability deliverable, not an aesthetic one. Pair with EHR/CPOE (Case 61) at the interface-built-to-wrong-specification layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the contact display to make ascent-versus-descent unmistakable at a glance, so the cue that contradicts a presumed threat cannot be passed over under time pressure.
- Test the interface against the worst case it will actually meet — confirmation bias during a simultaneous surface engagement — not against the calm conditions of acceptance testing.
- Build a mandatory disconfirmation step into the engagement sequence so identifying a hostile track requires actively ruling out the civilian one.

IN OPERATION / AFTER

- Audit live engagements and exercises for cases where the crew's reading diverged from sensor data, treating each as a near-miss that exposes an interface gap.
- Train and drill operators specifically on the stress regime — divided attention, presumed-hostile framing — rather than only on nominal track identification.
- Track whether decision aids are trusted past their evidence, and feed that signal back into interface and procedure revisions.

REFLECTION QUESTIONS

1. Identify a high-stress interface in your domain. What framing arrives before the data, and how does it shape what operators see?
2. Vincennes' operators acted under tunnel vision. Design the procedural intervention that would have forced one of the five to call out the contradiction.
3. The Aegis radar reported the truth and the crew still read its opposite. What would an interface have to make impossible — not merely visible — to keep a pre-formed expectation from overriding correct data under stress?

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CASE 55 AVIATION 1989

Kegworth / British Midland 92

T Training Gap H Human-System Interface K Knowledge & Institutional Memory

IMPACT 47 killed and 74 seriously injured when the crew shut down the wrong engine

IN BRIEF

On 8 January 1989 British Midland Flight 92, a nearly new Boeing 737-400, crashed on the M1 motorway embankment near Kegworth, killing 47 and seriously injuring 74. After a fan blade fractured in the left engine, the crew shut down the right engine — the one still working. Reasoning from older 737s, on which the right engine fed the cabin air, they misattributed the smoke; new, harder-to-read electronic engine displays did not correct them, and the cabin crew who saw flames never told the flight deck. The pilots' mental model was right for the airplane they had flown the week before, but the brief conversion course never overwrote it. The AAIB issued 31 recommendations; Kegworth became the textbook case of capability degrading under system change.

BACKGROUND

On 8 January 1989 British Midland Flight 92, a nearly new Boeing 737-400 (G-OBME), left Heathrow for Belfast with 126 aboard. The -400 was a recent variant — bigger engines, a partly redesigned cockpit with new electronic engine instruments. The crew were experienced 737 pilots, but their conversion onto the variant had been brief and largely self-directed, the differences communicated as reading rather than drilled retraining — so the experience that made them confident on the type also seeded the prior model that would mislead them.[0]

WHAT HAPPENED

Climbing through 28,300 feet, a fan blade fractured in the left engine, filling the cabin with vibration and smoke. Reasoning from older 737s — on which the air-conditioning bleed drew from the right engine — the crew concluded the right engine was the source and throttled it back. The symptoms eased (partly because the autothrottle had also cut power on the failing left engine), seeming to confirm it, and they shut the right engine down. On final approach the damaged left engine failed completely; with the good engine off, the aircraft struck the M1 embankment short of the runway — 47 killed, 74 seriously injured. The brief calm after the shutdown was the trap: it appeared to confirm a diagnosis that was wrong, removing the doubt that might have prompted a reassessment.[1]

THE INVESTIGATION

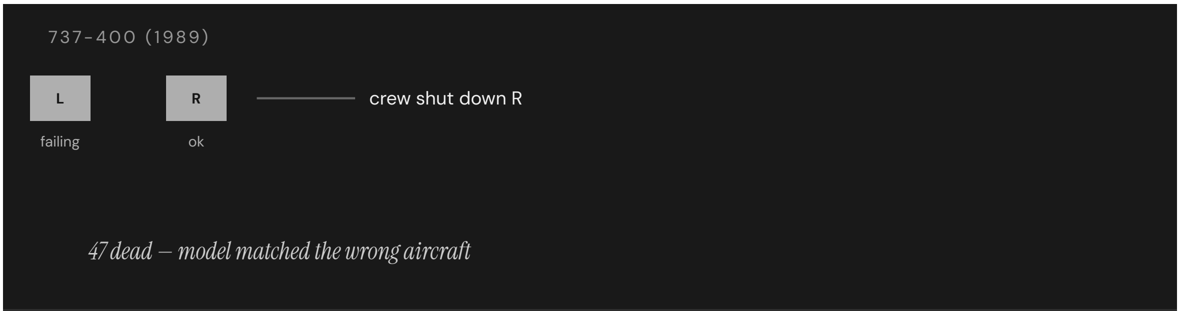
The Air Accidents Investigation Branch traced the disaster to the shutdown of the serviceable engine, and to a mental model carried from earlier 737s and applied to a -400 whose bleed-air configuration Boeing had changed — a difference the conversion training never disturbed, so the crew reasoned correctly from a configuration that no longer applied to the airframe they were flying.[2] The new electronic displays did not help: the vibration indicator that would have pointed at the failing engine was harder to read at a glance than the dials it replaced, so the one instrument that could have corrected the diagnosis was the least likely to be consulted under pressure.[3] And though cabin crew and passengers could see flames from the left engine, that never reached the flight deck — a crew-resource-management gap aviation would later train hard against, because those with the decisive evidence had no path to those making the decision.[4]

THE CAPABILITY GAP

The crew were skilled, and their mental model was correct — for the airplane they had flown the week before. The -400 was treated, for training, as an incremental change to a familiar type; under emergency pressure it behaved as a categorical one. Reading a manual note about revised bleed air is not the same as drilling a new reflex that fires before the old one, and the cockpit redesign had quietly removed perceptual cues a startled crew relies on. The change was filed as incremental precisely because nothing tested it until an emergency did. Capability had degraded under system change with no one noticing, because nothing failed until the day it did.[5]

AFTERMATH & REFORM

The AAIB issued thirty-one safety recommendations spanning conversion and difference training, engine-instrument and vibration-display design, cabin-to-flight-deck communication, and crashworthiness.[6] Kegworth became a standard human-factors teaching case and a reference for how a transition program should be built — so the differences between an old system and a new one are made hard to overlook, and the people closest to the evidence have a path to those making the decision. Each recommendation traced back to a failure the accident exposed — the unforced mental model, the unreadable instrument, the silent cabin — treated together rather than as isolated faults.



CREW MENTAL MODEL FROM OLDER VARIANT; CONFIGURATION CHANGED

REFERENCES

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THE LEARNING ENGINEERING LENS

> *The crew's mental model of the older 737 was inappropriately applied to the 737-400 on which they were operating.*
>
> PARAPHRASING AAIB AIRCRAFT ACCIDENT REPORT 4/90 ON BRITISH MIDLAND 92, 1990

LE INSIGHT

Kegworth is the textbook example of Capability Degradation Under System Change. The pilots were excellent. Their mental model was excellent — for the airframe they had flown the previous week. The difference training had not been engineered to disturb the prior model with enough force to keep it from being misapplied. The new airframe was treated as an incremental change. The crew's response revealed it was a categorical one.

LENS APPROACH

LENS treats Kegworth in LEN 5 as the canonical **system-change** capability problem and in LEN 8 as the institutional version: how knowledge about what has changed travels from engineering to operations and what gets lost in transit. The case sits alongside Patriot (Case 8) in the canonical problem-type pair for system transition.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Classify any change that alters operator mental models — even a "minor" variant — as categorical, and gate it on drilled retraining rather than reading.
- Test new interface designs against startled-crew performance before fielding, so a redesign cannot quietly remove the perceptual cues operators rely on.
- Engineer a communication path from those closest to the evidence (cabin crew, passengers, sensors) to those making the decision, so decisive observations reach the flight deck.

IN OPERATION / AFTER

- Audit transition programs for capability that degraded silently — where nothing failed until an emergency tested the change — using simulator scenarios that force the new reflex.
- Monitor whether difference training actually overwrites the prior model, treating un-disturbed mental models as a measurable latent risk.
- Track that interface and communication recommendations are implemented together, not as isolated fixes, so the integrated failure mode does not recur.

REFLECTION QUESTIONS

1. The Kegworth crew's mental model was right for the previous variant. What change in your domain currently risks an analogous misapplication?

2. Difference training is a generic deliverable in transition programs. Design the artifact that would make differences hard to overlook rather than easy.

3. The cabin crew and passengers saw flames the flight deck never learned of. Identify a path in your domain by which the people closest to the evidence cannot reach the people making the decision, and design the channel that would carry it under emergency conditions.

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CASE 56 AVIATION 2013

Asiana Airlines Flight 214

T Training Gap H Human-System Interface

IMPACT 3 killed, 187 injured; Boeing 777 struck the seawall short of SFO runway 28L

IN BRIEF

On a clear afternoon at San Francisco in July 2013, the crew of Asiana 214 allowed their Boeing 777 to slow far below approach speed without recognizing that the autothrottle was not maintaining it. The aircraft struck the seawall short of runway 28L, killing three of the 307 aboard and injuring nearly two hundred. The NTSB found the crew had mismanaged the approach and inadequately monitored airspeed — but also that the complexity of Boeing's autoflight systems contributed: after the pilots disconnected the autopilot and pulled thrust to idle, the autothrottle reverted to a HOLD mode in which it would not wake to hold the selected speed. The captain believed it would. Asiana 214 is the most prominent recent case of automation surprise on a Western wide-body airliner.

BACKGROUND

Asiana 214 was a scheduled Boeing 777 flight from Seoul to San Francisco. The captain was experienced on other types but new to the 777 and flying it under the supervision of a training captain; the approach was a visual one, with the airport's instrument glideslope out of service for construction — so a pilot still learning the type's automation was hand-flying an approach stripped of the vertical guidance that would normally have anchored it, exactly the conditions under which an unfamiliar mode is most likely to bite.[0]

WHAT HAPPENED

On 6 July 2013, on final approach to runway 28L, the aircraft descended below the proper path and slowed. The crew had set an autopilot mode that left the autothrottle in HOLD, where it would not advance thrust to hold the selected speed. Airspeed decayed to about 103 knots against a target of 137; the 777 struck the seawall short of the runway, killing three and injuring nearly two hundred of the 307 aboard. The thirty-four-knot shortfall built up silently while the crew believed the automation was holding the speed, so the first unambiguous signal of trouble was the airframe itself running out of energy on short final.[1]

THE INVESTIGATION

The NTSB found the probable cause to be the crew's mismanagement of the approach and inadequate monitoring of airspeed, complicated by the complexity of the autothrottle and autopilot flight-director systems and by the crew's faulty mental model of the automation. The captain told investigators he had assumed the autothrottle would maintain speed; in that configuration, it does not — a gap between the system's documented behavior and the mental model the training had left him with, the kind of mismatch that surfaces only when the automation is asked to do what the operator wrongly believes it is already doing.[2]

THE CAPABILITY GAP

Asiana 214 is the textbook automation surprise on a Western wide-body: the mode the autothrottle was actually in and the mode the crew believed it was in diverged silently. The missing capability was not raw pilot skill but transparency — a system that made its own state, and the fact that the crew now owned the airspeed, salient at the moment it mattered, rather than leaving the handoff of responsibility to be inferred from a silent mode change the interface did nothing to announce.[3]

AFTERMATH & REFORM

The NTSB issued recommendations on autoflight training, energy-state monitoring, and the design of mode annunciation, and emphasized training that preserves manual-flying proficiency to counter automation dependence. The accident became a standard reference in airline automation-policy reviews and in the push toward "automation airmanship" curricula — shifting the lesson from one crew's error to a recurring design problem about how modes are annunciated and how monitoring is taught when the automation is doing most of the flying.[4]

EXPECTED	ACTUAL
137 kt	103 kt

the autothrottle was not maintaining the speed the crew believed it was

ASIANA 214 — APPROACH AIRSPEED AT THE SEAWALL

REFERENCES

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THE LEARNING ENGINEERING LENS

> *Complexities of the autothrottle and autopilot flight director systems contributed to the accident.*
>
> NTSB AIRCRAFT ACCIDENT REPORT AAR-14/01, CONTRIBUTING FACTORS, 2014

LE INSIGHT

Asiana 214 is the aviation case for the LENS Human-AI Teaming proposition that automation transparency is a capability deliverable. The mode the autothrottle was actually in and the mode the crew believed it was in diverged silently — a textbook automation surprise. The training, the documentation, and the interface together did not surface the divergence.

LENS APPROACH

LENS treats Asiana 214 in LEN 2 as a worked case for automation transparency requirements. Students design the interface and training intervention that would have made the mode disconnect salient to the crew at the moment the aircraft slowed.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make automation transparency a design requirement: the system must announce its active mode and any silent reversion, not leave it to be inferred.
- Annunciate transfers of responsibility (e.g., when the crew now owns airspeed) explicitly at the moment they occur, so no handoff is implicit.
- Design and validate mode logic against operator mental models during development, so configurations that "do not do what pilots assume" are caught before fielding.

IN OPERATION / AFTER

- Train and audit energy-state monitoring so a slow, silent airspeed decay is caught by the crew before the airframe signals it.
- Preserve manual-flying proficiency through recurrent practice to counter the automation dependence that hid the decay.
- Monitor operations for mode-confusion events and feed them into automation-policy and annunciation-design reviews.

REFLECTION QUESTIONS

1. Identify a mode in an automated system you work with where the operator's mental model and the system's actual behavior can silently diverge. How would the operator know?
2. Design the interface change that would have alerted the Asiana crew to the mode mismatch before the airspeed decayed.
3. The autothrottle's HOLD reversion handed airspeed back to a crew that did not know it now owned it. Design the annunciation that would make a transfer of responsibility between automation and operator unmissable at the instant it occurs.

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CASE 57 AVIATION 2005

Helios Airways Flight 522

T Training Gap H Human-System Interface

IMPACT 121 killed; cabin failed to pressurize after a maintenance setting was left on "manual" and the crew misread the warning

IN BRIEF

Shortly after takeoff from Larnaca in August 2005, the crew of Helios 522 misread a warning horn. The same intermittent horn that signals an incorrect takeoff configuration on the ground sounds, in flight, when cabin altitude climbs past about 10,000 feet — and the pressurization selector had been left in “manual” after a leak check the night before. The cabin never pressurized. The crew, troubleshooting the wrong fault, lost consciousness to hypoxia; the Boeing 737 flew on autopilot until it ran out of fuel and crashed into a hillside near Athens, killing all 121 aboard. The investigation found a single cue carried two meanings with no differentiation, and the airline’s training had drilled only the ground meaning.

BACKGROUND

Helios 522 was a Boeing 737 departing Larnaca, Cyprus, for Athens. The night before, engineers had run a pressurization leak check and left the cabin-pressurization selector in the “manual” position. The pre-flight checks did not catch it, and the aircraft departed with its pressurization system not set to maintain cabin altitude automatically — a maintenance setting left in an abnormal position the following morning, the kind of latent configuration error that a pre-flight verification exists precisely to catch and here did not.[0]

WHAT HAPPENED

As the aircraft climbed on 14 August 2005, the cabin failed to pressurize and a warning horn sounded. On the 737 the same horn indicates an incorrect takeoff configuration on the ground and excessive cabin altitude in flight. The crew interpreted it as a configuration warning and spent minutes troubleshooting the wrong problem while hypoxia set in. By the time the aircraft neared Athens the flight crew was incapacitated; it circled on autopilot until fuel exhaustion and crashed, killing all 121 aboard. The crew spent their few useful minutes diagnosing a configuration fault that did not exist, the ambiguous horn having steered them away from the one problem — climbing cabin altitude — that was quietly disabling them.[1]

THE INVESTIGATION

The Hellenic Air Accident Investigation Board found the crew’s misidentification of the cabin-altitude warning as a takeoff-configuration warning to be a critical, irrecoverable error. The single horn carried two meanings with no differentiation between them, and the airline’s training had emphasized only the ground meaning; the in-flight case was an edge condition the crew had not been prepared to recognize — so an ambiguous cue and a one-sided training regime combined to make the wrong interpretation the natural one, with hypoxia closing the window before the error could be caught.[2]

THE CAPABILITY GAP

Helios 522 is the textbook example of a single cue carrying two meanings without differentiation — an interface that lied by ambiguity while the training filled in only one of the two meanings. The missing capability was a cue, or a training regime, that distinguished the deadly in-flight meaning from the routine ground one before the operators had to guess under hypoxia — a way to resolve the ambiguity at the interface or in training, rather than leaving it to be resolved by a crew already losing the cognition the decision required.[3]

AFTERMATH & REFORM

The investigation drove changes to Boeing's warning logic and to pressurization-related training and checklists across operators; later 737 variants distinguish the cabin-altitude warning from the takeoff-configuration warning. The accident is a standard case study in cue ambiguity, pre-flight verification, and the physiology of hypoxia in human-factors curricula — its remedies attacking the failure on both fronts at once, differentiating the warning at the interface and reinforcing the in-flight meaning in training so neither gap could be papered over by the other.[4]

ACTUAL	INTERPRETED
cabin alt.	T/O config.

one horn, two meanings · the training had drilled only one

HELIOS 522 — A CUE CARRYING TWO MEANINGS WITHOUT DIFFERENTIATION

REFERENCES

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THE LEARNING ENGINEERING LENS

> *The crew's interpretation of the cabin-altitude warning horn as a takeoff-configuration warning was a critical and irrecoverable error.*

PARAPHRASING THE HELLENIC AAIB FINAL REPORT ON HELIOS 522, 2006

LE INSIGHT

Helios 522 is the textbook example of a single cue carrying two meanings without differentiation. The interface lied by ambiguity. The training filled in only one meaning. The combination produced a twenty-minute window in which the crew was solving the wrong problem.

LENS APPROACH

LENS uses Helios in LEN 5 as a case in cue ambiguity and in LEN 2 as a Human-AI Teaming case where the interface failed to communicate which of two possible system states it was warning about. Students redesign the cue.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Differentiate cues that carry distinct meanings in distinct contexts, so a single signal cannot be read as the wrong one under stress.
- Train both meanings of any dual-purpose cue, not just the routine one, so the rare-but-deadly interpretation is recognized first.
- Design pre-flight verification to catch abnormal maintenance configurations, so a setting left in “manual” cannot survive into operation.

IN OPERATION / AFTER

- Audit checklists and warning logic for cues whose meaning depends on phase or context, and resolve the ambiguity before an operator must guess.
- Monitor maintenance-to-operations handoffs for latent configuration errors that pre-flight checks have historically missed.
- Sustain training on the physiology that degrades the operator (e.g., hypoxia), so the window for self-correction is understood and guarded.

REFLECTION QUESTIONS

1. Identify a cue in your domain that carries two meanings in different operational contexts. How would the operator know which?

2. Redesign the Helios configuration warning to distinguish ground- and in-flight meaning without adding cognitive load.

3. A maintenance setting left in “manual” survived the pre-flight checks and reached the air. Design the pre-flight verification step in your domain that would make an abnormal maintenance configuration impossible to depart with unnoticed.

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CASE 58 AVIATION 2015

TransAsia Airways Flight 235

T Training Gap H Human-System Interface

IMPACT 43 killed in Taiwan; crew shut down the only working engine after the other failed

IN BRIEF

Forty-three seconds after takeoff from Taipei Songshan in February 2015, the right engine of a TransAsia ATR 72 auto-feathered following a sensor fault. Working from memory under acute time pressure, the crew shut down the left engine — the one still producing thrust — leaving the aircraft with no power and too little altitude to recover. It clipped a viaduct, struck a taxi, and crashed into the Keelung River, killing 43 of the 58 aboard. The Aviation Safety Council found the captain had failed proficiency checks in the year before the accident and that the crew had not used the published engine-shutdown checklist, responding instead from a flawed memory of the procedure. The wrong-engine error mirrors Kegworth (Case 55) twenty-six years earlier.

BACKGROUND

TransAsia 235 was an ATR 72-600 twin-turboprop departing Taipei Songshan. The captain had failed simulator proficiency checks in the year before the accident and had been noted for weak handling of abnormal procedures. The published response to an engine failure was a checklist-driven procedure; under stress, crews tend to revert to memory — so a captain already flagged for weak abnormal-procedure handling was exactly the operator most likely to skip the checklist precisely when it mattered, a known weakness meeting a known failure mode.[0]

WHAT HAPPENED

On 4 February 2015, forty-three seconds after takeoff, the right engine's autofeather system activated following a faulty sensor signal — the engine itself was capable of producing power. The crew, identifying an engine problem, shut down the left engine, which was operating normally. With neither engine now producing useful thrust and the aircraft low and slow, it stalled, clipped a viaduct and a taxi, and crashed into the Keelung River, killing 43 of the 58 aboard. A recoverable single-engine event became unrecoverable the moment the working engine was shut down, leaving an aircraft with no thrust and no altitude in the seconds just after takeoff when both are least forgiving.[1]

THE INVESTIGATION

The Taiwan Aviation Safety Council found the crew failed to identify which engine had actually failed and shut down the wrong one. They did not follow the published engine-flameout and shutdown checklist, acting from memory under time pressure; the misidentification took roughly fifteen seconds. The captain's documented proficiency deficiencies and the airline's training and crew-resource-management shortfalls were contributing factors — the individual lapse sitting atop an organizational one, the carrier having left the very weaknesses its own proficiency checks had recorded uncorrected before the flight.[2]

THE CAPABILITY GAP

TransAsia 235 is the modern recurrence of the wrong-engine pattern that destroyed British Midland 92 at Kegworth in 1989 (Case 55). Crews under acute time pressure default to memory; if the memory is incomplete, the action follows the flawed memory rather than the checklist. The missing capability is a procedure design and training regime that keeps the verification step in the loop precisely when stress pushes operators to skip it — engineering the confirm-which-engine step so it fires under startle, rather than trusting it to a memory that time pressure is actively eroding.[3]

AFTERMATH & REFORM

The ASC issued recommendations on TransAsia's pilot training, proficiency-check standards, and crew-resource management; the airline ceased operations in 2016, after a second fatal accident the year before. The case is used to argue that the persistence of the wrong-engine pattern across a quarter-century reflects an un-engineered intervention, not merely individual error — the same failure recurring across crews, airframes, and decades points to a design and training gap that no amount of blaming the individual pilot has ever closed.[4]

RIGHT	LEFT
auto-feathered	shut down by crew

the working engine; misidentified in ~15 seconds under stress

TRANSASIA 235 — THE SAME WRONG-ENGINE PATTERN AS KEGWORTH (CASE 55)

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4. AAIB (UK), British Midland Flight 92 (Kegworth) report (1990) — the antecedent wrong-engine case.
5. Dismukes, Berman & Loukopoulos (2007), The Limits of Expertise — memory-driven response under time pressure.

> *The flight crew was unable to identify which engine had failed and shut down the operating engine in error.*
>
> PARAPHRASING THE AVIATION SAFETY COUNCIL (TAIWAN) FINAL REPORT ON TRANSASIA 235, 2016

LE INSIGHT

TransAsia 235 is the modern recurrence of the Kegworth pattern under stress. Crews under acute time pressure default to memory; if the memory is incomplete or wrong, the action follows the memory. The capability gap was the same as in 1989. The intervention pattern needed is also the same.

LENS APPROACH

LENS uses TransAsia 235 in LEN 5 as a recurrence case for well-known capability problems (wrong-engine shutdown) and in LEN 2 to teach the difference between checklist-driven response and memory-driven response under stress.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the abnormal-procedure response so the confirm-which-unit verification step is forced under startle, not left to memory.
- Build proficiency checks whose red signals have the authority to ground an operator before the predicted failure, not merely to document a weakness.
- Train the checklist-driven response under realistic time pressure, so the trained reflex survives the stress that pushes crews to memory.

IN OPERATION / AFTER

- Audit whether documented proficiency deficiencies are actually acted on, treating an uncorrected red check as an organizational, not individual, failure.
- Monitor for recurrence of well-known patterns (wrong-engine shutdown) across crews and airframes as evidence the intervention is un-engineered, not merely human error.
- Sustain crew-resource-management and checklist discipline so the verification step does not erode back to memory under operational pressure.

REFLECTION QUESTIONS

1. Identify a procedure in your domain that is supposed to be checklist-driven but is actually memory-driven under stress. What is the gap?
2. Why has the wrong-engine shutdown pattern persisted across 25 years of aviation reform? What intervention has not yet been engineered?
3. The captain had failed proficiency checks in the year before the accident, yet kept flying. Design the proficiency-check regime in your domain whose red signal would actually remove an operator from duty before, not after, the failure it predicts.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

CASE 59 HEALTHCARE TECHNOLOGY 1985 – 1987

Therac-25

H Human-System Interface D Designed Out G Governance & Trust

IMPACT **Six known massive radiation overdoses; at least three deaths; canonical case in software safety engineering**

IN BRIEF

The Therac-25, a radiation-therapy machine, massively overdosed at least six patients between 1985 and 1987, killing at least three. Its predecessors used hardware interlocks to stop the high-energy beam from firing without its target in place; to save cost, the Therac-25 removed them and trusted software — adapted from the older machines and never engineered for safety — to keep the beam modes separated. A race condition let a fast operator trigger the full beam with no target, while the console showed only a meaningless "Malfunction 54." Leveson and Turner's 1993 investigation, the founding case of software-safety engineering, found a systemic failure: a safeguard removed with nothing put in its place. The lesson — a safeguard you delete does not remove the hazard; it relocates it to whatever you failed to build.

BACKGROUND

The Therac-25 was a radiation-therapy linear accelerator. Its predecessors used hardware interlocks that physically blocked the high-energy beam unless the spreading target was confirmed in place — a mechanical backstop that could not be talked out of stopping the beam. To save cost and simplify the machine, the Therac-25 removed them and trusted software — adapted from the older machines and never engineered from the ground up for safety-critical use — to keep its two beam modes, a hundredfold apart in energy, safely separated. The safety case migrated silently from steel to code.[0]

WHAT HAPPENED

Between 1985 and 1987 the machine massively overdosed at least six patients. A race condition the engineers never knew about meant that if an operator entered a prescription, caught a mistake, and corrected it within about eight seconds — the speed an experienced operator naturally reaches — the full-power beam could fire with no target in place, delivering up to a hundred times the intended dose. The console showed only "MALFUNCTION 54," a code with no documented meaning, and offered to proceed; operators, assured the machine was safe and long accustomed to its cryptic faults, did. At least three patients died of the radiation burns.[1]

THE INVESTIGATION

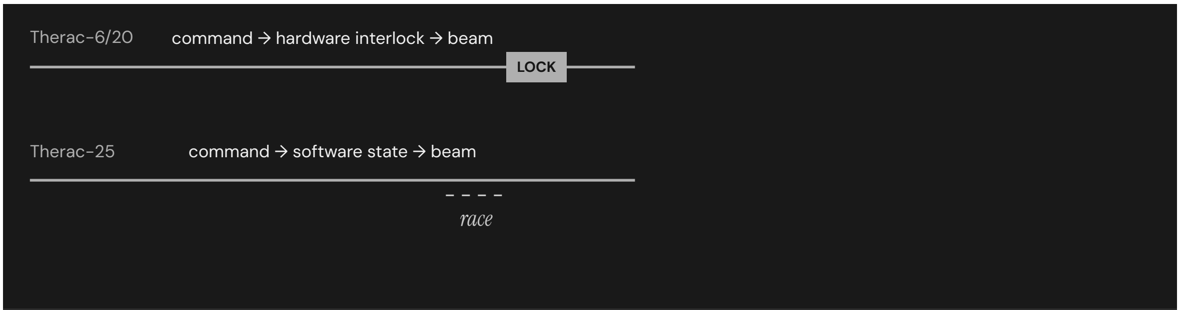
When patients reported searing pain, the manufacturer first insisted an overdose was impossible and treated each report as an isolated complaint rather than a signal. Nancy Leveson and Clark Turner's 1993 investigation — the founding case study of software-safety engineering — found the fault was systemic rather than a single bug: overconfidence in software, removal of the hardware safeguards without replacement, meaningless error messages, reused code never audited for safety, no independent review of the safety-critical logic, and an incident-reporting posture that dismissed the early warnings instead of compounding them into evidence.[2]

THE CAPABILITY GAP

The hardware interlock had not been redundant; it **was** the safety case, the one thing standing between a typing error and a lethal dose. Removing it put nothing in its place — no verified software check, no informed operator action, no independent monitor watching the beam. The operator stayed nominally in the loop but lost any means to detect what the machine was doing wrong, since the only feedback was a code that told them nothing, which made the human presence a formality rather than a safeguard. The question capability engineering would have forced — **what function now carries the interlock's load?** — was never asked of the redesign.[3]

AFTERMATH & REFORM

Therac-25 reshaped safety-critical software practice, making the case by counterexample for independent hazard analysis, safeguards that do not all rest on the same software, error messages that say what is wrong so the operator can act, and treating field reports as evidence to be aggregated rather than complaints to be closed. It remains the canonical teaching case across software, medical-device, and systems-safety curricula.[4] Its lesson is exact and portable: a safeguard you remove does not remove the hazard it guarded — it relocates the hazard to whatever you failed to put in its place, and waits there.



SOFTWARE RACE CONDITION REPLACED HARDWARE INTERLOCK

REFERENCES

1. N. G. Leveson & C. S. Turner, "An Investigation of the Therac-25 Accidents," IEEE Computer 26(7): 18–41 (1993). doi:10.1109/MC.1993.274940 — the removed hardware interlocks and the adapted software.
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3. Leveson & Turner (1993); N. G. Leveson, Safeware: System Safety and Computers (Addison-Wesley, 1995) — the manufacturer's denial and the systemic findings.
4. N. G. Leveson, Engineering a Safer World: Systems Thinking Applied to Safety (MIT Press, 2011) — why removing a safeguard requires explicitly reassigning its safety function.
5. The case's role in medical-device software regulation and software-safety practice (FDA software guidance; IEC 62304 lineage); see also Ethics Unwrapped, UT Austin.

THE LEARNING ENGINEERING LENS

> *Therac-25 illustrates the dangers of relying on software safety controls without rigorous engineering practices.*
>
> PARAPHRASING LEVESON & TURNER, IEEE COMPUTER, 1993

LE INSIGHT

Therac-25 is the moment when removing the human safety margin became visible as a design choice. The hardware interlock was not redundant — it was the safety case. When it was removed, no equivalent capability was put in its place. The operator was retained in the system but stripped of the ability to detect what it was doing wrong. Capability engineering would have asked, before removing the interlock, **what function takes its load?**

LENS APPROACH

LENS frames Therac-25 as a **Design-Out** failure made visible through Interface and Governance pathologies. Studio projects in LEN 5 ask students to produce capability-load diagrams tracing every safety function to a hardware backstop, a software check, or a trained operator action with the information needed to perform it. LEN 7 examines incident reporting as governance.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Before deleting any hardware interlock, write down the safety function it performs and explicitly reassign that function to a verified software check, an independent monitor, or an informed operator action.
- Do not let safety rest entirely on one software path; require an independent channel that does not share the same code, so a single defect cannot defeat the whole safety case.
- Specify error messages as a safety deliverable — each fault code must say what is wrong and what the operator should do, never offer to proceed past an undiagnosed condition.

IN OPERATION / AFTER

- Treat every field report of unexpected harm as evidence to be aggregated, not a complaint to be closed, with a standing path to halt the device when a pattern emerges.
- Audit any safety-critical code reused from a prior system against the new hazard set, since assumptions safe in the old context may be lethal in the new one.
- Instrument the machine so an independent monitor can detect a beam fired without its target and stop it, restoring the interlock's function even where the operator cannot see the fault.

REFLECTION QUESTIONS

1. Identify a system in your domain that migrated a safety function from hardware to software. Where did the human-capability load go, and who is accountable for sustaining it?

2. Therac-25 operators saw "MALFUNCTION 54" and continued treatment. Redesign that interface using LEN 2 principles so that the operator's correct action is the easiest action.

3. The Therac-25's safety-critical software was inherited from earlier machines and never re-audited. Identify reused code in a system you build that now carries a load it was never written for, and specify the review it should have received.

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CASE 60 ENERGY 2003

Northeast Blackout

H Human-System Interface K Knowledge & Institutional Memory

IMPACT 50 million people without power across eight U.S. states and Ontario; $6B+ economic loss; FERC Order 693 followed

IN BRIEF

On 14 August 2003 a high-voltage transmission line in Ohio sagged into a tree and tripped — an event the grid should have absorbed. But FirstEnergy's control-room alarm system had been silently failing for over an hour, so operators did not know the line was gone. Further lines tripped, and a cascade swept across the Eastern Interconnection; within minutes 50 million people across eight U.S. states and Ontario lost power, at a cost above $6 billion. The U.S.-Canada task force found FirstEnergy lacked situational awareness, its alarm system had failed without notice, vegetation management was poor, and the regional coordinator could not intervene. The reforms made reliability standards mandatory and enforceable (FERC Order 693). The gap sat at the automation-operator boundary: a silent failure left operators blind.

BACKGROUND

The Eastern Interconnection is built to ride through the loss of a single transmission line, and control rooms watch the grid through software and alarms. FirstEnergy, the Ohio utility at the center of the story, ran a control room whose alarm system had — unknown to anyone — been silently failing for over an hour.[0] The interconnection's single-line resilience assumes the operators can see which line is gone; a silent alarm broke that assumption at its root, leaving a grid designed to tolerate one fault blind to the fault it was tolerating.

WHAT HAPPENED

On 14 August 2003 a 345 kV line in Ohio sagged into a tree and tripped. With the alarms silent, operators did not know the line was gone and so did not take the corrective steps that would have contained it. Further lines tripped and a cascade swept across the interconnection; within minutes 50 million people across eight states and Ontario lost power, at a cost above $6 billion.[1] The first trip was the routine single-line loss the grid was built to absorb; what turned it into a cascade was not the tree but the hour in which the operators acted on a picture of the grid that no longer matched the grid.

THE INVESTIGATION

The U.S.-Canada Power System Outage Task Force found FirstEnergy operators "did not have adequate situational awareness," that the alarm system had failed without notice, that vegetation management was inadequate, and that the regional reliability coordinator lacked the authority and information to intervene.[2] The reforms produced FERC Order 693, which for the first time made compliance with reliability standards mandatory and enforceable rather than voluntary.[3] Making the standards mandatory addressed the deeper finding that a voluntary regime had let vegetation management and operator awareness drift: when compliance is optional, the practices that prevent a cascade are exactly the ones a cost-pressured utility lets slip.

THE CAPABILITY GAP

The gap sat at the boundary between automation and operator. When the alarm system stopped doing its job, it did so silently — and the operators had no independent signal that they were losing control of their grid segment. The missing capability was the meta-monitor: the watch on the watchman, the check that tells you when the thing that is supposed to tell you has itself stopped.[4] Silence is the most dangerous failure mode because it is indistinguishable from a quiet, healthy grid; the operators were not ignoring a warning but trusting an absence of warning that the broken system could no longer guarantee.

AFTERMATH & REFORM

FERC Order 693 and the new mandatory reliability regime — enforced by an Electric Reliability Organization with audits and penalties — treated grid reliability as a deliverable rather than a best practice. The 2003 blackout endures as the canonical case of silent automation failure: a system that quits without announcing it, leaving the humans nominally in charge with nothing to act on.[5] Backing the standards with audits and penalties conceded that reliability could not be left to good intentions: the practices that would have kept the operators sighted had to be enforceable obligations, not best practices a utility could quietly defer.

NORTHEAST BLACKOUT — SILENT ALARM FAILURE ABSORBED THE OPERATOR'S AWARENESS

REFERENCES

1. U.S.-Canada Power System Outage Task Force, Final Report on the August 14, 2003 Blackout in the United States and Canada (2004) — the tree contact, the silent alarm, and the cascade.
2. Task Force (2004) — 50 million people affected across eight states and Ontario; the minute-by-minute sequence.
3. Task Force (2004) — FirstEnergy "did not have adequate situational awareness," plus the vegetation-management and reliability-coordinator findings (quoted).
4. FERC Order No. 693, Mandatory Reliability Standards for the Bulk-Power System (2007) — enforceable standards.
5. North American Electric Reliability Council reports (2004) and the creation of the Electric Reliability Organization.
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THE LEARNING ENGINEERING LENS

> *FirstEnergy … did not have adequate situational awareness of conditions on its system.*
>
> U.S.-CANADA POWER SYSTEM OUTAGE TASK FORCE, FINAL REPORT ON THE AUGUST 14 2003 BLACKOUT, APRIL 2004

LE INSIGHT

The 2003 blackout is the canonical case for silent automation failure: a system that stops doing its job without telling its operators. The capability that was missing was the meta-monitor — the system that watches the monitor. The reform that followed treated grid-reliability compliance as a deliverable rather than as a best practice.

LENS APPROACH

LENS uses the 2003 blackout in LEN 2 as a Human-AI Teaming case for silent-automation-failure handling and in LEN 8 for the legislative-reform arc that produced enforceable reliability standards.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design monitoring with a positive heartbeat so a silent system is distinguishable from a healthy one, never assuming no alarm means no problem.
- Build a meta-monitor that watches the alarm system itself and announces when the watchman has stopped.
- Give the regional reliability coordinator the authority and information to intervene across utility boundaries before a cascade forms.

IN OPERATION / AFTER

- Audit vegetation management and operator situational awareness against enforceable standards, since voluntary practice drifts under cost pressure.
- Test alarm and monitoring systems for detectable failure, verifying that an outage of the monitor itself raises an alert.
- Back reliability obligations with audits and penalties so the practices that keep operators sighted cannot be quietly deferred.

REFLECTION QUESTIONS

1. Identify an automated monitoring system in your domain whose silent failure would not be detected by its operators. How would they know?
2. Design the meta-monitor that should have been watching FirstEnergy's alarm system in 2003.
3. Silence was indistinguishable from a healthy grid. What positive heartbeat signal in your domain would let operators tell a quiet system from a dead one?

WHO BUILDS THIS human factors & interaction design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · knowledge bases · data pipelines

CASE 61 HEALTHCARE 2005 – PRESENT

EHR / CPOE Implementation

H Human-System Interface D Designed Out G Governance & Trust

IMPACT ~$30B federal investment under HITECH; documented increase in pediatric ICU mortality post-CPOE at one institution; ongoing usability harm at scale

IN BRIEF

The 2009 HITECH Act poured roughly $30 billion into accelerating electronic health records, and adoption surged — but the systems had been built to billing and administrative specifications, not clinical workflow. New categories of harm followed. A disputed but canonical 2005 study found that deploying a commercial order-entry (CPOE) system at one pediatric ICU was associated with a near-doubling of mortality; later deployments showed mixed or improved results, yet the warning stuck. A 2023 survey across 200-plus hospitals found EHR usability is clinicians' top complaint, and that usability scores track patient-safety outcomes. Specific harms recur — a missed abnormal result hidden behind a default, a fatal overdose an unactivated alert would have caught. There is still no regulatory framework monitoring EHR safety. It is the book's live interface-mismatch case.

BACKGROUND

The 2009 HITECH Act authorized roughly $30 billion in incentives to accelerate electronic health record adoption, and adoption surged. But the systems clinicians now had to use had been built to administrative and billing specifications, not to clinical workflow or human-factors specifications — a mismatch baked in before the first order was entered.[0] Because the incentives rewarded adoption rather than usability, hospitals raced to install whatever cleared the bar, and the specification gap propagated to the bedside at national scale before anyone had to demonstrate that the tools supported the work of care rather than merely the work of billing.

WHAT HAPPENED

New categories of harm appeared alongside the new tools. A 2005 study reported that deploying a commercial computerized order-entry (CPOE) system at one pediatric ICU was associated with a near-doubling of mortality — a single-institution result that provoked debate and that later deployments elsewhere did not reproduce, but that became canonical as a warning that powerful tools without workflow integration can disrupt care at the moment of greatest acuity, when the seconds the interface adds to an order are the seconds a critical patient does not have.[1] Specific harms recur: a cancer treatment delayed for years because a default surfaced an old normal result instead of a recent abnormal one; a baby killed by an overdose an alert would have caught had it been switched on — each a case where the interface's defaults and toggles, not the clinician's competence, determined whether the right information reached the decision.[2]

THE INVESTIGATION

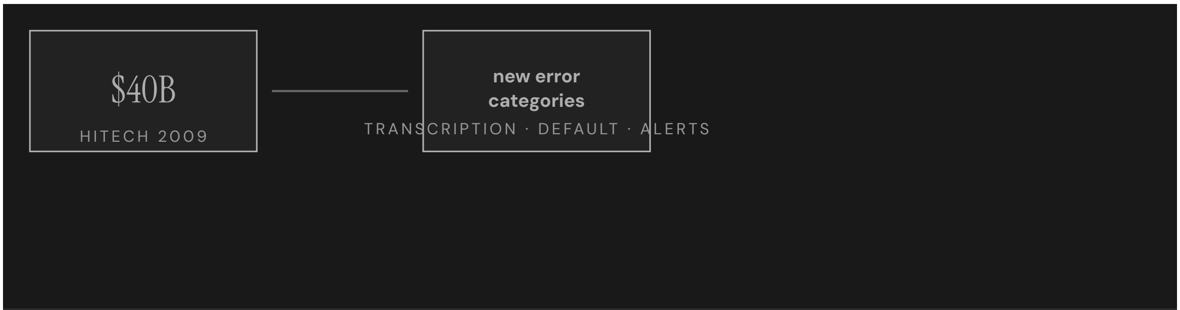
A 2023 KLAS Arch Collaborative survey across more than two hundred hospitals found EHR usability is consistently the top complaint of physicians, nurses, and pharmacists, and that end-user experience scores correlate with patient-safety outcomes — turning a stream of anecdotes into longitudinal evidence that usability is now itself a patient-safety variable. Spanning some two hundred hospitals, the survey converts what could be dismissed as one site's grievance into a reproducible signal, the kind of measurement that makes an interface problem legible as a safety problem rather than a satisfaction one.[3]

THE CAPABILITY GAP

EHR/CPOE is the canonical ongoing case of an interface designed for one set of requirements and deployed against another. The systems work for billing and administration because that was the design constraint; they fail for clinical workflow because clinical workflow was not. Decades and tens of billions of dollars later, usability remains among the largest contributors to in-system harm in U.S. healthcare — a cost that persists precisely because it was never a design requirement and so was never engineered out, only worked around by clinicians absorbing the friction shift after shift.[4]

AFTERMATH & REFORM

Usability and safety have risen on the agenda — AMA/Pew/MedStar usability work, KLAS benchmarking, safety-focused design guidance — but there is still no regulatory framework that monitors deployed EHR safety the way drugs or devices are monitored. The lesson is that buying the tool is not the same as engineering the capability: an interface deployed against the wrong specification keeps extracting a cost no one is formally counting, and without a monitoring regime the harm stays diffuse — spread across millions of encounters, attributable to no single failure, and therefore easy for the system to keep paying indefinitely.[5]



$40B INVESTED; NEW ERROR PATHWAYS EMERGED FROM INTERFACE DESIGN

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THE LEARNING ENGINEERING LENS

> *Reports of new types of errors directly related to EHR implementation — errors that can compromise quality of care and patient safety — have emerged.*
>
> PARAPHRASING SITTIG & SINGH, EHR-RELATED SAFETY RISKS, 2013

LE INSIGHT

EHR/CPOE is the canonical ongoing case of an interface designed for one set of requirements and deployed against another. The systems work for billing and administration. They fail for clinical workflow because clinical workflow was not the design constraint. Forty billion dollars later, usability remains the single largest contributor to in-system harm in U.S. healthcare.

LENS APPROACH

LENS treats EHR/CPOE in LEN 7 as the live, ongoing example of Design-Out and Interface failure under governance opacity, in LEN 2 as a human-AI teaming problem (alert fatigue, default surfacing), and in LEN 9 as a computational systems problem (the alerting architecture itself is a learnable model). The Han 2005 pediatric ICU finding is the durable warning; the KLAS 2023 surveys across two hundred hospitals are the contemporary, longitudinal evidence that usability is now itself a patient-safety variable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Write clinical workflow and human-factors performance into the procurement specification, so a system that serves billing but not bedside care cannot clear the buy.
- Validate defaults, alerts, and result-surfacing against real clinical scenarios before deployment, since it was the default and the unactivated alert — not clinician skill — that determined the harm.
- Tie any adoption incentive to demonstrated usability, so the money rewards a tool that supports care rather than merely one that is installed.

IN OPERATION / AFTER

- Stand up a post-deployment safety-monitoring regime for fielded EHRs comparable to how drugs and devices are surveilled, with thresholds and authority to force changes.
- Track end-user experience scores as a patient-safety variable, using the cross-institution evidence that usability correlates with outcomes.
- Audit recurring interface-driven harms — surfaced defaults, disabled alerts — and feed them back into vendor design requirements rather than absorbing them as clinician burden.

REFLECTION QUESTIONS

1. What is the equivalent system in your domain that was designed for one specification and deployed against another? How would you measure the harm?
2. Design the regulatory architecture that would surface EHR safety harms at scale. Be specific about signal, threshold, and authority.
3. The Han 2005 mortality result was disputed; the 2023 KLAS surveys are consistent across two hundred hospitals. What ongoing measurement architecture would have to exist for the equivalent emerging clinical-AI deployment in your domain to be evaluated honestly while in use?

WHO BUILDS THIS human factors & interaction design · systems & human-factors engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · task analysis · design prototyping · governance frameworks · bias & evidence audits

CASE 62 AUTONOMOUS SYSTEMS 2018

Uber ATG / Tempe Fatality

T Training Gap N Normalization of Deviance G Governance & Trust H Human-System Interface

IMPACT One pedestrian killed — the first fatality involving a self-driving vehicle striking a pedestrian

IN BRIEF

On the night of 18 March 2018 a modified Volvo running Uber's self-driving system struck and killed Elaine Herzberg as she crossed a road in Tempe, Arizona — the first pedestrian killed by an autonomous vehicle. The NTSB found the safety operator had been watching a video on her phone, but placed heavy blame on Uber: it had not recognized the risk of automation complacency, trained for it, or enforced its own no-phone policy. The system itself was programmed not to brake when a crash was deemed unavoidable, and could not classify a pedestrian who was not near a crosswalk. The human was kept in the loop as a passive monitor — a role the NTSB noted is chronically unperformable — with no infrastructure to make it work. Uber ATG is the book's defining human-AI-teaming case.

BACKGROUND

Uber's Advanced Technologies Group tested self-driving cars on public roads with a safety operator behind the wheel, present to take over if the automation failed. The role was passive surveillance: watch a system that drove itself well almost all of the time, and intervene in the rare moment it did not.[0] That structure asks a person to stay vigilant for an event that almost never comes, the precise condition under which human attention is known to lapse — so the role was set up to demand exactly the kind of sustained monitoring that people are least able to deliver.

WHAT HAPPENED

On 18 March 2018 a modified Volvo SUV in autonomous mode struck and killed Elaine Herzberg as she crossed a road at night in Tempe, Arizona — the first pedestrian killed by a self-driving vehicle. The safety operator was looking down at a video on her phone in the seconds before impact.[1] The phone was not an aberration but the predictable filling of an attention vacuum: a role with nothing to do for hours invites exactly that drift, and nothing in the car's design or the monitoring around the seat pulled the operator's eyes back to the road when it finally mattered.

THE INVESTIGATION

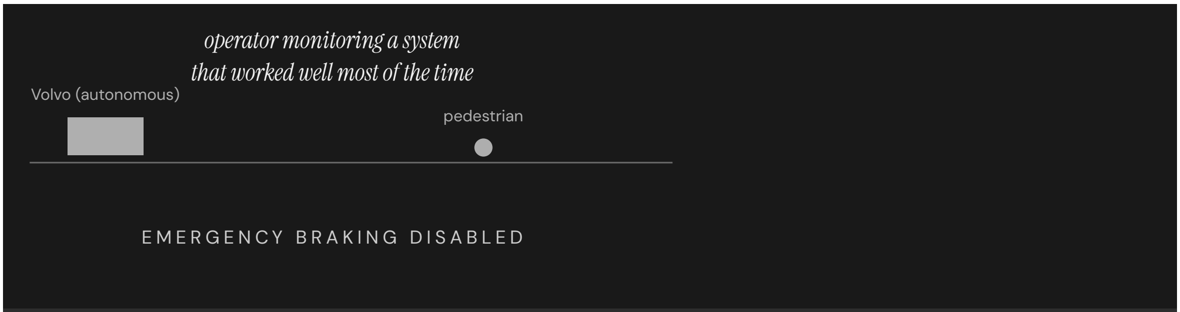
The NTSB's probable cause centered on the operator's failure to monitor the road, but it placed heavy blame on Uber, which "did not adequately recognize the risk of automation complacency and develop effective countermeasures": training was inadequate and the no-phone policy unenforced.[2] Naming the company alongside the operator was the board's way of locating the fault in the design of the role rather than the lapse of the person filling it — the policy existed on paper but had no mechanism behind it to make compliance the default. The system compounded the human gap. It was programmed not to apply emergency braking when a crash was judged unavoidable — removing the automated backstop — and it could not classify an object as a pedestrian unless it was near a crosswalk, so the very situation on the road that night fell into a blind spot the software was not built to see.[3]

THE CAPABILITY GAP

A human was retained not because the designers believed a person could meaningfully catch the failure, but because the regulatory and public posture required one present. The role of "monitor" was assigned without the interface, training, or authority to make it performable — a placeholder for safety rather than an instrument of it. As the NTSB chairman put it, "humans tend to tune out when tasked with monitoring automated systems that work well most of the time." The design was safe only on the assumption that the failure case would not arrive — until it did, and the assumption that had quietly held the whole arrangement together was paid for with a life.[4]

AFTERMATH & REFORM

Uber suspended testing, later exited self-driving, and the case reshaped how the industry and regulators treat safety drivers — toward two-operator teams, driver-monitoring systems, and honest accounting of what a monitor can and cannot do.[5] Each of those reforms is a concession that the single passive observer had been an unsupported role all along: a second operator shares the vigilance burden, and driver-monitoring closes the attention vacuum the original design left open. Its lasting contribution is the reframing of passive monitoring as a role that must be engineered to be performable — or not assigned at all.



DESIGNED AS PASSIVE MONITOR; EQUIPPED ONLY TO FAIL

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5. NTSB Chairman R. Sumwalt, Tempe hearing remarks (2019) — "humans tend to tune out…" (quoted).
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THE LEARNING ENGINEERING LENS

> *Repeatedly, humans tend to tune out when tasked with monitoring automated systems that work well most of the time.*
>
> NTSB CHAIRMAN ROBERT SUMWALT, UBER TEMPE HEARING REMARKS, 2019

LE INSIGHT

Uber ATG is the defining case for the LENS Human-AI Teaming competency. A human was retained in the system not because the designers believed a human could meaningfully act, but because the regulatory architecture required a human be present. The role of "monitor" was assigned without the capability infrastructure to support it. The system that resulted was performable only on the assumption that the failure case would not arrive — until it did.

LENS APPROACH

LENS treats this case in LEN 2 as the live exemplar of monitoring as an unsupportable role. Students reconstruct the capability requirements for the safety operator and design the interface, training, and authority structure that would have made the role performable — or made the case for not retaining the role at all.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Engineer the monitoring role to be performable — give the operator an active task, a usable interface, and the authority to act — or do not assign a human backstop you do not expect to work.
- Keep the automated emergency-braking backstop active rather than suppressing it, so the system does not silently remove its own last line of defense.
- Validate object classification against the real operating environment, including pedestrians away from crosswalks, before fielding the system on public roads.

IN OPERATION / AFTER

- Enforce the no-phone and attention policies with driver-monitoring that detects and corrects drift in real time, not a written rule alone.
- Audit safety-operator attention data continuously, treating sustained lapses as a design failure of the role rather than a fault of the individual.
- Deploy two-operator teams or equivalent redundancy so the vigilance burden does not rest on a single person doing an unsupportable job.

REFLECTION QUESTIONS

1. Identify a passive-monitor role in your domain. What evidence would tell you the role is or is not performable as designed?
2. The Tempe vehicle was programmed not to brake when a crash was unavoidable. Reconstruct the design rationale and propose the deliverable that should have prevented that decision.
3. Uber had a no-phone policy with nothing to enforce it. What is a rule in your domain that exists on paper but lacks the mechanism to make compliance the default — and how would you engineer that mechanism?

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CASE 63 AVIATION 1972

Eastern Air Lines Flight 401

H Human-System Interface T Training Gap

IMPACT 101 killed in the Everglades; the entire flight crew fixated on a landing-gear indicator bulb while the autopilot silently disengaged

IN BRIEF

On 29 December 1972 an Eastern Air Lines L-1011 descended into the Florida Everglades on a night approach to Miami, killing 101 of the 176 aboard. The cause was attention, not mechanics: all three flight-deck crew became absorbed in a landing-gear indicator bulb that had failed to light, and while they worked it, the autopilot's altitude hold was inadvertently disengaged. The aircraft sank slowly; an altitude alert chimed but went unheeded in the cockpit clutter, and no one was monitoring the flight path. The NTSB findings launched decades of research into attention, monitoring, and cockpit-alert design, and the accident is a formative event behind Crew Resource Management. Eastern 401 is the canonical case of a low-priority task crowding out a life-critical one.

BACKGROUND

Eastern Air Lines Flight 401, a wide-body Lockheed L-1011, was on a night approach to Miami on 29 December 1972 with a full flight deck — captain, first officer, and flight engineer.[0] Three crew was the era's safeguard against any one person being overloaded, the assumption being that more eyes meant more coverage; the night would show that without a designed division of those eyes, three people could converge on a single trivial problem as easily as one.

WHAT HAPPENED

The crew noticed that the landing-gear indicator light had not illuminated, and all three focused on the bulb. While they worked the problem, one of them inadvertently nudged the controls and disengaged the autopilot's altitude hold. The TriStar began a slow descent into the Everglades; an altitude-warning chime sounded but, in the auditory clutter of the cockpit, no one registered it. The aircraft hit the swamp, killing 101 of the 176 aboard.[1] The descent was gentle enough to go unfelt, and the single chime carried no more urgency than the routine sounds around it — so the one cue that could have broken the fixation was indistinguishable, by design, from the background it sounded against.

THE INVESTIGATION

The NTSB found the crew had become so engrossed in the landing-gear difficulty that they failed to monitor the flight path.[2] A burned-out indicator bulb had absorbed the attention of an entire qualified crew while a wide-body airliner descended unwatched — the dispro-portion between the trigger and the outcome is exactly what made the case so durable a teaching example. The findings inaugurated decades of research into attention, monitoring, and the design of cockpit alerts — the accident is cited in nearly every introductory cognitive-engineering course as the example of how a low-priority task can crowd out a life-critical one when attention is undivided among the available channels.[3]

THE CAPABILITY GAP

The missing capability was a designed division of attention across the crew, and an alert that carried the priority its message deserved. Three competent people attended to a burned-out bulb while no one watched the altitude; the chime existed but did not cut through. The flight deck had the people and the information — it lacked the design that would have kept one channel of attention on the thing that could kill them. Attention had been treated as something the crew would naturally allocate well, rather than as a resource the cockpit had to be built to protect, and that unexamined assumption was the gap.[4]

AFTERMATH & REFORM

Eastern 401 was one of the formative events behind Crew Resource Management (Case 70) — the explicit allocation of monitoring and cross-checking roles — and behind modern standards for prioritized cockpit alerting.[5] Both reforms encode the same correction: CRM assigns someone to keep flying the aircraft while others troubleshoot, and prioritized alerting ensures the most dangerous condition is the loudest one — turning the night's two failures into permanent structural defenses. Its lesson is that attention is a designable parameter, not a personal virtue: a system that lets all eyes converge on one task must guarantee the critical channel is still watched.

EASTERN 401 — THREE CREW, ONE TASK, NO MONITOR

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THE LEARNING ENGINEERING LENS

> *The crew became so engrossed in the landing-gear difficulty that they failed to maintain altitude.*
>
> PARAPHRASING NTSB AIRCRAFT ACCIDENT REPORT NTSB-AAR-73-14 ON EASTERN 401, 1973

LE INSIGHT

Eastern 401 is the canonical attention-failure case. The capability that was missing was a designed division of attention across the crew. CRM exists because Eastern 401 happened, and the discipline of cockpit alert design exists because the altitude warning chime did not carry the priority weight it needed to.

LENS APPROACH

LENS uses Eastern 401 in LEN 5 to teach attention as a designable parameter and in LEN 2 to introduce alert prioritization as a capability deliverable. The case anchors the CRM origin story.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build an explicit division of attention into the operating procedure, so at least one role is always assigned to the life-critical channel while others troubleshoot.
- Design alerts with a priority hierarchy, ensuring the most dangerous condition is also the most salient rather than one indistinguishable chime among many.
- Treat attention as a resource the system must protect by design, not a virtue the operators are assumed to supply on their own.

IN OPERATION / AFTER

- Drill crews on scenarios where a trivial problem competes with a critical one, and verify the monitoring role actually holds under that pressure.
- Audit whether alerts in service are heard and acted on, retiring or redesigning any that get lost in routine clutter.
- Track distraction-related near-misses so attention-displacement failures are visible before they cause a loss.

REFLECTION QUESTIONS

1. Identify a low-priority task in your domain that could plausibly absorb all of an operator's attention. What is the life-critical task that would be displaced?
2. Redesign the altitude warning chime of an L-1011 so that it cuts through a focused troubleshooting conversation.
3. Eastern 401 had three qualified people and still left the flight path unwatched. How would you assign and verify a "someone is always watching the critical channel" role so it cannot collapse when the whole team is drawn to one problem?

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CASE 64 DEFENSE 1983

Stanislav Petrov / 1983 False Alert

H Human-System Interface T Training Gap

IMPACT Soviet early-warning system reported five incoming U.S. ICBMs; Lt. Col. Petrov correctly assessed the signal as false; retaliation averted

IN BRIEF

On the night of 26 September 1983 the Soviet Oko early-warning system reported five U.S. intercontinental missiles inbound. The duty officer, Lt. Col. Stanislav Petrov, judged it a false alarm — a real first strike would involve hundreds of missiles, not five — and reported it as such rather than up the launch-decision chain. He was right: sunlight glinting off high-altitude clouds had fooled the satellite's infrared sensors. Petrov's judgment is widely credited with averting nuclear war. The case is the book's strongest positive evidence for human-in-the-loop capability: the automation failed in a mode its designers never anticipated, and a person with contextual judgment and the latitude to override it was the recoverability the system had. Keeping humans in some loops is not nostalgia — it is capability engineering.

BACKGROUND

Soviet nuclear command rested partly on Oko, a satellite early-warning system meant to detect U.S. missile launches and feed a launch-on-warning posture. The duty officer at the Serpukhov-15 bunker was responsible for verifying any alert and passing it up the chain.[0] A launch-on-warning posture compresses the time between detection and decision to almost nothing, so the duty officer's verification was not a formality but the single human checkpoint standing between a satellite reading and the machinery of retaliation.

WHAT HAPPENED

On the night of 26 September 1983 Oko reported a U.S. intercontinental ballistic missile launch — then, minutes later, four more, for five in all. Lt. Col. Stanislav Petrov, the duty officer, assessed the signal as a false alarm — a genuine first strike, he reasoned, would involve hundreds of missiles, not five — and reported it as such. He was correct.[1] The reasoning that saved the situation came from outside the system entirely: Oko could report what it saw, but it could not weigh five launches against the doctrine of a real first strike, and that mismatch between the alert and the strategic picture was exactly the judgment the machine had no way to make.

THE INVESTIGATION

The cause was an unanticipated automation failure: sunlight reflecting off high-altitude clouds at a particular geometry had fooled the Oko satellite's infrared sensors into reading launches that were not there.[2] It was a failure of the rarest kind — a benign natural phenomenon the sensors had never been designed to discount — which is precisely why no automated check existed to catch it. Later investigation identified the satellite-geometry failure mode and modified the algorithm; the scenario is now permanently archived in early-warning training as the canonical false positive, a permanent reminder that the system's worst error was one it could not have flagged for itself.[3]

THE CAPABILITY GAP

The automation had failed in a mode its designers had not imagined, and no automated check could catch it. What caught it was a human with the contextual judgment to weigh the alert against what a real attack would look like, and the institutional latitude to override the system rather than simply relay it. Petrov's "funny feeling" was a career's worth of judgment doing the work the automation could not — and crucially, the chain of command had left him room to act on it rather than forcing him to pass the alert upward unfiltered, so the recoverability lived as much in the authority structure as in the man.[4]

AFTERMATH & REFORM

Petrov's decision is widely credited with averting nuclear war, and the case is permanently studied in command-and-control training.[5] Preserving the episode in training is itself a design choice: it keeps the failure mode and the human override alive as institutional memory, so the lesson does not erode the way the original sensor's blind spot had. It is the book's strongest argument that retaining a human in some loops is capability engineering, not nostalgia: the human role is the recoverability of an automation failure the designers did not anticipate — which is exactly why fully automating a strategic decision removes the one element that can catch the unimagined error.

SYSTEM	OPERATOR
5 ICBMs	false alarm

the human in the loop was the recoverability of an automation failure

PETROV, 1983 — HUMAN-AI TEAMING AT THE HIGHEST POSSIBLE STAKES

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THE LEARNING ENGINEERING LENS

> *I had a funny feeling in my gut.*
>
> STANISLAV PETROV, QUOTED IN THE WASHINGTON POST, 1999

LE INSIGHT

Petrov is the canonical positive case for human-in-the-loop nuclear command-and-control. The automation failed in a mode its designers had not anticipated. The human's contextual judgment was the backstop that allowed the failure to be recovered before it became catastrophic. The case is the strongest argument in this book that the design choice to retain humans in some loops is not nostalgia but capability engineering.

LENS APPROACH

LENS uses Petrov in LEN 2 as the positive Human-AI Teaming case at the highest possible stakes: the human role is the recoverability of an automation failure that the designers did not anticipate. The case anchors arguments against full-automation of strategic decisions.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Where automation feeds an irreversible decision, design the human role to be a genuine override — with the context, judgment cues, and authority to refuse a faulty alert, not merely relay it.
- Give operators a strategic picture against which to weigh an alert, so an anomalous signal can be tested against what a real event would look like.
- Treat unimagined failure modes as a design assumption: keep a human in the loop precisely where no automated check can cover the unanticipated.

IN OPERATION / AFTER

- When a failure mode surfaces, modify the algorithm and archive the scenario in training so the lesson persists as institutional memory.
- Audit whether human-in-the-loop roles still carry real override authority over time, or have decayed into ceremonial relays.
- Preserve and refresh the contextual judgment override depends on through deliberate training, so the capability does not erode between rare events.

REFLECTION QUESTIONS

1. Identify an automated system in your domain where retaining a human-in-the-loop is genuinely capability-engineering rather than ceremonial. How would you tell the difference?
2. Petrov's "funny feeling" was contextual judgment built across a career. Design the training that produces it deliberately.
3. Petrov's recoverability lived as much in his latitude to override as in his judgment. Map an automated decision in your domain where the operator has the knowledge to catch a failure but lacks the authority to act on it — and redesign the authority structure.

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CASE 65 SPACE / AEROSPACE HUMAN SPACEFLIGHT SAFETY ENGINEERING 2013

NASA EVA-23 — Water Intrusion Inside a Spacesuit

H Human-System Interface N Normalization of Deviance D Designed Out

IMPACT During the second of two ISS spacewalks, ESA astronaut Luca Parmitano's helmet filled with up to 1.5 liters of water from the suit's cooling-and-ventilation loop, blinding him, fouling his communications, and threatening drowning in vacuum; the EVA was terminated and Parmitano recovered to the airlock guided by tether feel — the NASA Mishap Investigation Board called the outcome a near-fatality

IN BRIEF

On July 16, 2013, during EVA-23 outside the International Space Station, water began collecting inside the helmet of European Space Agency astronaut Luca Parmitano. The contamination grew until the water covered his eyes, ears, and nose, fouled his communications microphone, and would have drowned him in vacuum had the spacewalk continued. The crew terminated the EVA and Parmitano made his way back to the airlock partly by feel along the tether. The NASA Mishap Investigation Board (MIB) report issued later that year identified the proximate cause as blocked separator holes in the suit’s Fan/Pump/Separator assembly that allowed cooling-loop water into the ventilation loop, and the deeper cause as a missed opportunity at the previous EVA-22 a week earlier when a similar (but smaller) water event had been mis-attributed to in-suit drink-bag leakage. The case is the canonical instance in modern human spaceflight of a failure mode the suit was not instrumented to detect, surfacing first as a near-fatal event because the earlier weak signal was rationalized away. The MIB’s recommendations span hardware redesign, in-suit water-detection instrumentation, and the safety-culture response to anomalous EVA telemetry.

BACKGROUND

EVA-23 was the second of two scheduled spacewalks for ISS Expedition 36. Parmitano and NASA astronaut Chris Cassidy exited the Quest airlock to continue tasks begun during EVA-22 a week earlier. Roughly 44 minutes into the EVA, Parmitano reported water at the back of his head inside the helmet. The crew continued briefly while ground controllers assessed; the water continued accumulating, eventually reaching the front of the helmet where it pooled across Parmitano’s eyes, blocked his nose, and fouled the communications microphone. The EVA was terminated and Parmitano made the return traverse to the airlock with degraded vision and communication, partly guided by his sense of the tether.[0]

WHAT HAPPENED

The proximate hardware fault, identified by the NASA Mishap Investigation Board, was in the suit's Fan/Pump/Separator assembly — the component that separates cooling water from the breathing-gas ventilation loop. Contamination had blocked drum-holes in the water separator. Once the separator could not perform its function, water from the cooling loop migrated into the ventilation loop and was pushed up into the helmet by the suit's fan. The hardware had no in-suit instrumentation to detect free water in the ventilation path; the crew detected the failure only when Parmitano felt liquid on his skin.[1]

THE INVESTIGATION

The MIB's deeper finding was about EVA-22, the prior spacewalk one week earlier. Parmitano had reported a smaller volume of water in his helmet during EVA-22 as well. At the time, the event was attributed to a drink-bag leak — a plausible but unverified hypothesis. No suit teardown or anomaly investigation was completed before EVA-23 was approved. The MIB characterized this as a missed opportunity: the EVA-22 event was the weak signal of the same failure mode that nearly killed Parmitano a week later, and the institutional response treated an unverified benign explanation as adequate clearance to proceed.[2]

THE CAPABILITY GAP

The corrective actions span three layers. At the hardware layer the suit's water-separator and ventilation-loop hygiene were re-engineered, and a Helmet Absorption Pad and a snorkel were added so a future water event would not immediately threaten the airway. At the instrumentation layer in-suit water-detection capability was added so the failure mode is no longer detectable only by skin contact. At the safety-culture layer the MIB pressed the "anomaly is anomaly" discipline: an unexplained EVA event requires investigation rather than the most convenient explanation, particularly when the next planned EVA reuses the same hardware.[3]

AFTERMATH & REFORM

The hedge that survives into the case is about attribution depth. The MIB was clear that the in-suit water-detection and hardware fixes address the specific failure mode that produced EVA-23. The harder, lifecycle question — how a suit certified for decades of use developed a contamination pathway not represented in its anomaly catalog, and how future suits will avoid the analogous category of uninstrumented failure — is left as an open sustainment-engineering problem. The case is the canonical modern human-spaceflight instance of a recovery from a near-fatal event whose proximate cause was hardware and whose deeper cause was the institutional decision to accept the most convenient explanation for an anomalous prior event.

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THE LEARNING ENGINEERING LENS

> *The previous occurrence is the strongest available signal of the next one; the institutional decision to accept the convenient explanation is what made the second event near-fatal.*
>
> EDITORS' SYNTHESIS OF THE NASA MIB EVA-23 REPORT.

LE INSIGHT

EVA-23 is the canonical modern human-spaceflight case of a failure mode the suit was not instrumented to detect, surfacing first as a near-fatality because the prior weak signal was rationalized away. The hardware fix and the new in-suit instrumentation are necessary; the safety-culture half — anomaly is anomaly — is what generalizes.

LENS APPROACH

EVA-23 is the human-spaceflight uninstrumented-failure case (induced 3.4; LENS D3/PT5). LENS uses it in Domain 3 (Human-System Collaboration) for the operator's detection-by-skin-contact at the edge of the engineered envelope, and in Domain 4 (Test and Evaluation) for the anomaly-investigation discipline the MIB pressed. Pair with Case 116 (anesthesia monitoring) at the cue/alert-as- capability layer and with Case 88 (F-22 OBOGS) as the aerospace companion in instrumentation-gap failure.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Instrument the failure modes the system was not designed for. EVA-23 was detectable only by skin contact because the ventilation loop had no free-water sensor; instrumentation is the precondition for recovery.
- Treat the previous occurrence of an anomaly as the strongest available signal of the next one. EVA-22's smaller water event was the warning EVA-23 ignored; the institutional decision to accept the convenient explanation is what made the second event near-fatal.
- Engineer the recovery margin into the suit itself — the post-EVA-23 Helmet Absorption Pad and snorkel — so a contamination event has a survivable buffer before it becomes an airway emergency.

IN OPERATION / AFTER

- Require an anomaly investigation, not a hypothesis, before reusing the same hardware for a subsequent high-consequence operation; the cost of a teardown is small against the cost of the failure mode.
- Build the in-suit instrumentation expectation into the certification process for future suits so the lesson generalizes beyond the specific Fan/Pump/Separator failure.
- Carry the sustainment-engineering hedge into program communication: hardware certified for decades can still develop pathways not represented in its anomaly catalog, and the certification process has to keep pace.

REFLECTION QUESTIONS

1. Identify an operational system in your domain where a failure mode is detectable only by direct human sensing (skin contact, smell, the operator noticing). What instrumentation would convert that failure mode from sensed-late to sensed-early?

2. Specify the anomaly-investigation threshold you would set in your program: an unexplained prior event of the same class requires what level of teardown before the next operation is approved? EVA-22 to EVA-23 is the worked example of the cost of getting that threshold wrong.

3. The post-EVA-23 fixes addressed the specific failure mode. What is the lifecycle question your program owes for decades-certified hardware: how does the certification process keep pace with contamination pathways the original anomaly catalog did not represent?

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CASE 66 TECHNOLOGY GOVERNMENT 1999 – 2015

UK Post Office Horizon Scandal

G Governance & Trust H Human-System Interface K Knowledge & Institutional Memory

IMPACT ~900 sub-postmasters wrongfully prosecuted; many imprisoned; documented suicides; described as the most widespread miscarriage of justice in UK history

IN BRIEF

The UK Post Office's Horizon accounting system, built by Fujitsu and rolled out in 1999, generated phantom shortfalls in sub-postmasters' branch ledgers. Rather than accept the software was at fault, the Post Office prosecuted them — around 900 over two decades — for theft and false accounting; people were imprisoned, bankrupted, and driven to suicide, in what is now called the most widespread miscarriage of justice in UK history. Internal documents showed engineers had known about Horizon bugs throughout. Convictions began to be quashed in 2021, and a public inquiry continues. The failure ran through the prosecutor and the courts: each accepted "the computer said so" as authoritative because no actor had the standing or expertise to challenge it. Horizon is the book's case for institutional deference to a flawed algorithm.

BACKGROUND

The UK Post Office ran thousands of branches through sub-postmasters — local operators personally liable for any shortfall in their accounts, a liability that put each operator's livelihood behind the numbers the system reported. In 1999 it deployed Horizon, an accounting system built by Fujitsu, to track every branch's ledger, making the software the single authority on whether a branch's books balanced.[0]

WHAT HAPPENED

Horizon produced systematic accounting errors — phantom shortfalls that appeared in branch ledgers where no money was actually missing. The Post Office treated the shortfalls as real and the sub-postmasters as thieves: over two decades it prosecuted around 900 for theft and false accounting, refusing to accept the system itself was at fault even as the same pattern recurred branch after branch. People were imprisoned, lost homes, went bankrupt, and some died by suicide — the human cost of trusting the ledger over the person.[1]

THE INVESTIGATION

Documents later released through litigation showed Fujitsu and Post Office engineers had known about Horizon bugs throughout the period — the knowledge of fallibility existed inside the institution even as it prosecuted people for the system's errors.[2] The courts began quashing convictions in 2021, and the public inquiry under Sir Wyn Williams found that senior employees "knew, or at the very least should have known, that Legacy Horizon was capable of error" — establishing it as the most widespread miscarriage of justice in UK history, sustained precisely because that internal knowledge never reached the people on trial.[3]

THE CAPABILITY GAP

The gap was at the regulator, the prosecutor, and the courts: each accepted Fujitsu's representation that Horizon was reliable, despite documentation to the contrary, because no institutional actor had the standing or expertise to interrogate it, so the claim of reliability passed unchallenged through every layer that could have tested it. "The computer said so" became, for two decades, a sufficient basis for criminal conviction — the governance hazard of treating automated output as authoritative rather than as evidence to be challenged, with a person's account on the other side of the scale.[4]

AFTERMATH & REFORM

Convictions have been overturned — some by an exceptional act of Parliament, a measure of how far the ordinary appeal routes had failed — compensation schemes established, and Fujitsu and the Post Office called to account before the continuing inquiry.[5] Horizon's lesson is the chapter's in its bluntest form: an automated system's output is not testimony, and any institution that lets "the computer said so" stand unchallenged against a human's account has built a machine for manufacturing injustice, one that runs for as long as no one is empowered to switch it off.

900

WRONGFUL PROSECUTIONS ACROSS 20+ YEARS

"the computer said so" was an institutionally sufficient basis for conviction

HORIZON — INSTITUTIONAL DEFERENCE TO AN ALGORITHM KNOWN TO BE FLAWED

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> *A number of senior, and not so senior, employees of the Post Office knew, or at the very least should have known, that Legacy Horizon was capable of error.*
>
> SIR WYN WILLIAMS, POST OFFICE HORIZON IT INQUIRY, VOLUME 1, JULY 2025

LE INSIGHT

Horizon is the canonical case for institutional deference to automated systems whose internal evidence was already known to be flawed. The capability gap was at every layer that took the software's output as authoritative — including the courts. "The computer said so" became, for two decades, an institutionally sufficient basis for criminal prosecution.

LENS APPROACH

LENS uses Horizon in LEN 7 as the canonical example of institutional deference to algorithmic output and in LEN 2 for the most extensive multi-decade automation-bias case in the dataset. Studio projects examine what evidentiary architecture would require **interrogating** automated output before acting on it.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design automated output to be treated as challengeable evidence, not authoritative testimony, especially where a person's liability rides on it.
- Build a route by which engineers' knowledge of system bugs reaches anyone acting on the output, so internal fallibility cannot stay hidden.
- Give some institutional actor the standing and expertise to interrogate the system's reliability before its output is used against a person.

IN OPERATION / AFTER

- Audit the recurring-error pattern across branches or cases, treating the same fault appearing repeatedly as evidence of the system, not the operators.
- Maintain an appeal path that can challenge automated output without requiring an act of Parliament to overturn a wrong decision.
- Sustain independent review of the system's accuracy throughout its operating life, so a claim of reliability cannot pass unexamined for decades.

REFLECTION QUESTIONS

1. Identify a decision in your domain currently made on the strength of "the computer said so." What evidentiary architecture should sit beside the output?
2. Design the institutional check that would have made Horizon's reliability subject to genuine challenge in 2005.
3. Engineers inside the Post Office and Fujitsu knew Horizon was capable of error, yet that knowledge never reached the courtroom. What pathway in your domain carries — or fails to carry — known system fallibility to the people relying on the output?

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CASE 67 HEALTHCARE/ONCOLOGY COMMERCIAL AI PROCUREMENT 2013 – 2018

Watson for Oncology — Delegation Marketed Ahead of Capability

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER journalism-tier — source confidence flagged; future validation ongoing.

IMPACT IBM marketed Watson for Oncology with tumor-board concordance rates as high as 96%; STAT News investigation (Ross & Swetlitz, 2017–2018) and internal IBM documents documented unsafe and incorrect cancer-treatment recommendations and that the system was trained on a small number of synthetic cases rather than real patient outcomes; hospitals worldwide had purchased the tool on the marketed concordance claim

IN BRIEF

Watson for Oncology was IBM's heavily marketed clinical-decision- support system, sold to hospitals worldwide as a cancer- treatment recommendation engine whose concordance with expert tumor boards was advertised at rates as high as 96%. The capability the system was sold under was AI-grade treatment recommendation across cancer types, delivered with the institutional credibility of IBM and MD Anderson partnerships. The capability the system actually carried — documented in the investigative reporting by Casey Ross and Ike Swetlitz at STAT News through 2017 and 2018, drawing on leaked IBM internal documents — was substantially smaller. Watson was trained on a small number of synthetic cases curated by Memorial Sloan Kettering oncologists rather than on real patient outcomes, generated unsafe and incorrect recommendations in some documented cases, and the internal IBM record showed engineers raising concerns about the gap between marketed concordance claims and what the system actually delivered. Hospitals around the world had purchased Watson on the marketed capability; major deployments at MD Anderson and elsewhere were later wound down. The case is the canonical instance in the AI-delegation typology of capability marketed ahead of capability validated. The evidence-tier flag rendered under the title is binding: STAT News journalism is the primary public source, and the academic record cites it secondarily. "Future validation ongoing" is the standing language; independent peer-reviewed evaluation of Watson for Oncology's operating accuracy at the level the marketing claimed was never produced.

BACKGROUND

Watson for Oncology was the most publicly visible commercial clinical-AI deployment of the mid-2010s. IBM had positioned Watson — originally the Jeopardy-winning question-answering system — as a healthcare engine after the 2011 Jeopardy performance, and oncology was the flagship domain. The product was sold as a cancer-treatment recommendation system: a clinician would enter the patient's case, Watson would query its knowledge base against the training data provided by Memorial Sloan Kettering oncologists, and the system would return ranked treatment recommendations annotated with evidence citations. The marketed capability was treatment recommendation that concorded with expert tumor boards at rates as high as 96%. Hospitals worldwide purchased the system on that claim.[0]

WHAT HAPPENED

Casey Ross and Ike Swetlitz at STAT News produced the investigation of record across 2017 and 2018. Drawing on leaked IBM internal documents and interviews with engineers, oncologists, and former IBM staff, the reporting documented a substantially different operating picture. Watson had been trained not on real patient outcomes — the standard a treatment-recommendation engine would have to clear — but on a small number of synthetic cases curated by MSK oncologists to reflect what those oncologists would have done. The training corpus was a model of expert opinion at one institution, not an outcome-anchored learning record. In documented cases the system generated unsafe and incorrect treatment recommendations. Internal IBM documents showed engineers raising concerns about the gap between marketed capability and operating reality.[1]

THE INVESTIGATION

The structural failure pattern the case carries is delegation of consequential authority — cancer-treatment recommendation — to a tool whose validated capability was substantially smaller than its marketed capability. The procurement mechanism was the carrier: hospitals purchased Watson through capital and IT procurement processes that evaluated the marketed concordance claims and the IBM/MSK institutional credentials, but did not independently validate operating accuracy on the receiving institution's patient population. The capability the procurement was buying — accurate, real-world treatment recommendation at the marketed concordance rate — was not the capability the system carried. Major deployments at MD Anderson and elsewhere were wound down as the gap surfaced.[2]

THE CAPABILITY GAP

The evidence-tier flag rendered under the title is binding and survives into the case verbatim. The STAT News investigation by Ross and Swetlitz is the primary public source, and the academic record cites it secondarily. No independent peer-reviewed evaluation of Watson for Oncology's operating accuracy at the level the marketing claimed was ever produced — partly because the system was proprietary and partly because the deployments were wound down before the kind of prospective study that would have established the operating record could be conducted. The case is teachable on the structural failure pattern at journalism-grade evidence; the standing "source confidence flagged; future validation ongoing" language is the honest tier the case rests on.[3]

AFTERMATH & REFORM

In the AI-delegation typology (Cases 74 TREWS, 102 Epic Sepsis Model, 111 SyRI, and this case), Watson is the marketing-ahead-of-capability failure mode: the procurement decision rode on marketed concordance rates that were not validated at the operating institutions, and the institutional credentials did not substitute for the validation. TREWS shows delegation with strong outcome evidence works; Epic Sepsis shows delegation without validation fails by accuracy disconfirmation; SyRI shows delegation halted on rights grounds; Watson shows delegation driven by procurement and marketing rather than by validated capability. The four together teach the typology of when AI delegation is and is not legitimate. Watson is the case where the marketing finance — the institutional credibility and the headline concordance number — substituted for the evidence the procurement should have required, and the capability the hospitals thought they were buying was not the capability they got.

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THE LEARNING ENGINEERING LENS

> *The capability the procurement was buying — accurate, real-world treatment recommendation at the marketed concordance rate — was not the capability the system carried.*
>
> EDITORS' SYNTHESIS OF ROSS & SWETLITZ (STAT NEWS 2017–2018).

LE INSIGHT

Watson for Oncology is the canonical instance of clinical-AI delegation marketed ahead of validated capability. The operating capability was substantially smaller than the marketed concordance claims; the procurement mechanism evaluated the marketing rather than the validation; the deployments were wound down as the gap surfaced. The evidence-tier flag is binding: journalism-grade reporting is the load-bearing source, and future validation ongoing remains the honest qualifier.

LENS APPROACH

Watson for Oncology is the marketing-ahead-of-capability AI-delegation failure (induced 1.1; LENS D3/PT6) — Domain 3 for **Delegation with revocation** (MD Anderson wind-downs are the revocation half); Domain 4 for requirements-vs- marketing. Pair with Cases 74, 93, 155. Evidence-tier flag binds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Separate marketed capability from validated capability in any procurement decision involving consequential delegation; the marketed concordance number is the carrier of the procurement story, not the evidence the procurement should rest on.
- Require independent operating-accuracy validation on the receiving institution's patient population before delegation; the AI-delegation literature now treats this as the procurement floor, and Watson is one of the reasons.
- Anchor the procurement evidence to outcome-anchored training, not to expert-curated synthetic cases; expert opinion at one institution is not outcome data the procurement can rely on.

IN OPERATION / AFTER

- Render the evidence tier honestly when the primary public source is journalism; the STAT investigation is the load-bearing source, and the academic record's secondary citation is the accurate description, not "peer-reviewed evidence."
- Carry the AI-delegation typology — TREWS, Epic Sepsis, SyRI, Watson — as a unit in any curricular discussion of when delegation is legitimate; the four-case set teaches the typology more clearly than any single case.
- Treat the wind-down at MD Anderson and elsewhere as the case's own correction signal; the institutional decisions to discontinue are themselves evidence

about what the procurement should have required up front.

REFLECTION QUESTIONS

1. Identify a procurement decision in your domain that delegated consequential authority to an automated system. Did the procurement evaluate marketed capability claims or validated capability on the receiving institution's data? Where is the gap most likely to have been smoothed by institutional credentials standing in for evidence?

2. Specify what operating-accuracy validation you would require before delegating: which prospective study, on which population, with which comparison and which outcome metric. The AI-delegation typology has established this as the procurement floor; the question is whether your domain's procurement processes meet it.

3. The evidence-tier flag under this case's title is binding because the primary public source is journalism. Identify a case in your domain where the strongest available evidence is journalism-grade. What is the responsible drafting discipline that surfaces the tier rather than implying peer-reviewed validation?

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CASE 68 AVIATION CUSTOMER SERVICE AI AGENTS 2022 – 2024

Air Canada Chatbot Liability — Delegation Without Revocation

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT British Columbia Civil Resolution Tribunal ruled February 14 2024 in Moffatt v. Air Canada, 2024 BCCRT 149, that Air Canada was liable for bereavement-fare-policy misinformation provided to passenger Jake Moffatt by the airline's website chatbot; tribunal rejected Air Canada's argument that the chatbot was a "separate legal entity" responsible for its own outputs; small-claims-tribunal ruling with limited precedential weight outside BC but cited widely as articulating the principle that organizations are liable for representations made by their AI agents

IN BRIEF

On February 14, 2024, the British Columbia Civil Resolution Tribunal issued its decision in Moffatt v. Air Canada, 2024 BCCRT 149. Passenger Jake Moffatt had consulted Air Canada's website chatbot about the airline's bereavement-fare policy in November 2022, following the death of his grandmother. The chatbot represented that the bereavement fare could be claimed retroactively, after travel. Moffatt booked a full- fare flight in reliance on the chatbot's representation, then submitted a retroactive bereavement-fare claim. Air Canada refused the claim on the ground that the actual policy required pre-booking application. Moffatt sued in the BC Civil Resolution Tribunal — Canada's online small-claims forum — and the tribunal awarded $650.88 in damages. Air Canada had argued that the chatbot was a "separate legal entity" responsible for its own outputs; the tribunal rejected the argument and held that Air Canada was liable for representations made by its chatbot. The ruling has limited precedential weight outside BC but has been cited widely as articulating the delegation-without-revocation principle. The case pairs with Case 93 (Epic Sepsis), Case 67 (Watson for Oncology), and Case 77 (Hybrid Human-AI Tutoring).

BACKGROUND

In November 2022, Jake Moffatt visited Air Canada's website shortly after his grandmother's death and consulted the airline's chatbot — at the time, a customer-service AI agent embedded in the airline's customer-facing web property — about the bereavement-fare policy. The chatbot represented that the bereavement fare could be applied retroactively, after travel, by submitting a claim with supporting documentation. Moffatt booked a full-fare round trip to Toronto in reliance on the representation. After travel, he submitted a retroactive bereavement-fare claim with the documentation the chatbot had described. Air Canada's response was that the actual bereavement-fare policy required pre-booking application — that is, the reduced fare had to be applied for at the time of booking, not claimed retroactively.[0]

WHAT HAPPENED

The structural seam the case opens is that the airline's chatbot was producing representations that diverged from the airline's actual policy. The seam is straightforward operationally — the chatbot's outputs were not constrained to the airline's policy text in a way that would have prevented the misrepresentation — but it is structurally significant in legal terms. When the airline directed a customer to its website for policy information and the website's AI agent produced a representation that the customer relied on to his detriment, the question is whether the airline is liable for the AI agent's output. Air Canada's response in the tribunal was that the chatbot was a "separate legal entity" — the company argued, in effect, that it could delegate customer information to an AI agent without assuming legal responsibility for the agent's representations.[1]

THE INVESTIGATION

The tribunal rejected the argument unambiguously. The decision, written by Tribunal Member Christopher Rivers, found that Air Canada was responsible for "all the information on its website" and that the chatbot was part of the website. The argument that the chatbot was a separate legal entity was found to have no support in law. The tribunal awarded Moffatt $650.88 in damages — the difference between the full fare he paid and the bereavement fare he had been led to believe he could claim. The dollar amount is small; the principle the ruling articulates is what has carried the case into widespread citation. Organizations that deploy AI agents to interact with customers are responsible for the representations the agents make, and the agents are not separate legal persons. The delegation-without-revocation form — the organization delegates customer interaction to the AI agent but cannot revoke responsibility for what the agent says — is the load-bearing structural finding.[2]

THE CAPABILITY GAP

The case pairs with Case 93 (Epic Sepsis) for the delegation-without-validation thread in healthcare AI; the structural form is the same — the organization deploys an AI agent that produces representations or assertions consequential for the affected person, and the organization's accountability for the agent's outputs is the load-bearing governance question. Pair with Case 67 (Watson for Oncology) for the AI-agent-recommendations-in-practice thread. Pair with Case 77 (Hybrid Human-AI Tutoring) for the educational-AI-agent thread at adjacent scale. The Air Canada ruling is a small-claims-tribunal decision with limited precedential weight outside BC, but its principle has been cited in subsequent academic and practitioner writing as the first clear judicial articulation of the delegation-without-revocation form for AI agents.[3]

AFTERMATH & REFORM

The hedges the case carries are load-bearing. The tribunal's ruling has limited precedential weight outside BC and has not been litigated to a higher court; the principle has been cited but not adopted in binding form across Canadian or U.S. jurisdictions. The case teaches the form — organizations are liable for the representations of their AI agents — more than it establishes settled law. The structural reading is the load-bearing one: the case names a delegation structure and the legal question that the delegation surfaces, and it does so in a forum whose decision is operationally consequential for the parties and pedagogically clear for the field. The human-in-the-loop CLO at the customer- interaction-AI-agent seam is anchored by the case in the form the deployment architecture must support — the organization's accountability for the agent's outputs is the architecture's load-bearing constraint.

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THE LEARNING ENGINEERING LENS

> *The chatbot is part of the website; the airline is responsible for all the information on its website; there is no support in law for the argument that the chatbot is a separate legal entity responsible for its own outputs.*
>
> TRIBUNAL MEMBER CHRISTOPHER RIVERS, MOFFATT V. AIR CANADA, 2024 BCCRT 149 (FEB 14, 2024), EDITORS' PARAPHRASE.

LE INSIGHT

Air Canada chatbot is the delegation-without-revocation case at customer-interaction-AI-agent scale. The BC Civil Resolution Tribunal's ruling holds that organizations are liable for representations made by their AI agents and that the agents are not separate legal persons; the small-claims venue limits the precedential weight, but the principle has been cited widely as the first clear judicial articulation of the form. The case teaches the form more than it establishes settled law.

LENS APPROACH

Air Canada chatbot is the human-in-the-loop-at-the-customer- interaction-agent-seam case (induced 5.2; LENS D3/PT6; CLO-3 and CLO-5). LENS uses it in Domain 3 (Machine Teaming and Adaptation) for the organization-is-liable-for-agent-representations principle. Pair with Case 93 (Epic Sepsis delegation-without- validation), Case 67 (Watson for Oncology), and Case 77 (Hybrid Human-AI Tutoring). The small-claims-tribunal venue limits precedential weight; the structural reading is the load-bearing one.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Constrain customer-facing AI agents to representations the deploying organization will stand behind; the Air Canada case demonstrates that the deployment surface of an AI agent's output is the

IN OPERATION / AFTER

- Carry the precedential-weight hedge into print without softening; the ruling is a small-claims-tribunal decision and the precedential limits are part of what the case teaches alongside the structural form it names.

same legal surface as the organization's own representations.

- Build the policy-text-to-agent-output integrity check as part of the deployment, not as a customer-service-recovery process; the divergence between the airline's policy text and the chatbot's representation was the deployment seam the tribunal found dispositive.
- Specify the revocation-and-recovery mechanism the deployment carries when the agent produces a misrepresentation; the organization's accountability for the agent's outputs requires a documented process for honoring the agent's representation or for compensating the affected party.
- Pair in syllabi with Case 93 (Epic Sepsis) so the delegation-without-validation form is taught at both the healthcare and the customer-interaction-agent scales.
- Use the case to anchor the human-in-the-loop CLO at the customer-interaction-AI-agent seam; the curricular target is the discipline of treating the agent's outputs as the organization's representations, and of building the deployment architecture to that constraint.

REFLECTION QUESTIONS

1. Identify a customer-interaction AI agent in your domain whose outputs have not been integrity-checked against the organization's policy text. What divergence between agent representation and policy text would produce a Moffatt-style reliance harm, and what mechanism would close the divergence?

2. Specify the revocation-and-recovery process your deployment carries when the agent produces a misrepresentation. What is the documented decision rule for honoring the representation versus refusing it, and who has authority to decide?

3. The Moffatt ruling has limited precedential weight outside BC. Pick a deployment in your domain and ask: what would have to be true for the delegation-without-revocation principle to apply in your jurisdiction, and what is the deployment architecture that would honor the principle whether or not the law has settled it?

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CASE 69 ENERGY 1979

Three Mile Island

T Training Gap H Human-System Interface

IMPACT Partial meltdown of a Babcock & Wilcox PWR; minimal off-site dose; most serious accident in U.S. commercial nuclear history; catalyst for industry-wide reform

IN BRIEF

On 28 March 1979 a small cooling fault at Three Mile Island's Unit 2, a Babcock & Wilcox reactor near Harrisburg, escalated into a partial core meltdown — the most serious accident in U.S. commercial nuclear history. A relief valve stuck open while a control-room light reported it closed, and operators trained for dramatic design-basis ruptures misread the slow, ambiguous cascade and cut the cooling the core needed. A nearly identical near-miss at Davis-Besse eighteen months earlier had never been propagated to the fleet. The Kemeny Commission concluded the fundamental problems were people-related, not equipment. Off-site radiation was minimal, but the accident produced a system of reform — most enduringly INPO — making it the book's paired example of a failure that engineered lasting capability.

BACKGROUND

Three Mile Island's Unit 2, a Babcock & Wilcox reactor near Harrisburg, was run by operators trained for large, fast, design-basis ruptures — not the slow, ambiguous small-break sequence that actually came.[0] It should not have surprised anyone: in September 1977 a nearly identical stuck-open relief valve had occurred at Davis-Besse, but neither the utility, the vendor, nor the NRC grasped its significance or pushed the lesson out to the fleet — so the precise sequence that would later threaten the core had already been demonstrated and then quietly filed away rather than turned into a drill or a warning.[1] The industry's posture rested on a probabilistic argument that catastrophic accidents in light-water reactors were vanishingly unlikely; that argument had become institutional doctrine, shaping what was trained for, what was instrumented, and what near-misses were treated as.

WHAT HAPPENED

On 28 March 1979 a minor secondary-cooling upset tripped the reactor; a power-operated relief valve (PORV) opened and then stuck open, draining coolant from the primary loop. The control-room light reported the **command** to close it, not its actual position, so the panel read "closed" while the valve stayed open. Misreading the rising pressurizer level, the operators throttled back the high-pressure injection the starving core depended on — taking the one action that turned a recoverable upset into a meltdown, precisely because the instrument they trusted was telling them a state that was not true. About half the core melted; off-site radiation, as it turned out, was minimal, though the public-communication failure that accompanied the accident — the contradictory updates from utility, state, and federal officials — would shape every nuclear emergency-response standard built after.[2]

THE INVESTIGATION

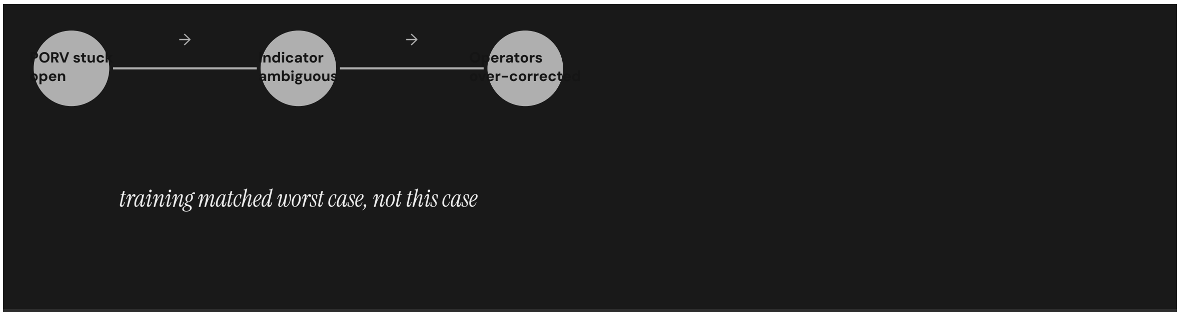
The Kemeny Commission (October 1979) inverted the industry's assumptions: "the fundamental problems are people-related problems and not equipment problems." The equipment was good enough that, but for the human failures, the accident would have been minor. The criticism reached past the operators to management, the utility, and the NRC — an institution-wide belief that serious accidents were effectively impossible, a complacency that had shaped training priorities, staffing, and oversight long before the relief valve ever stuck, so that the operators inherited a posture they had not chosen.[3] The companion Rogovin report for the NRC documented the Davis-Besse precursor and the institutional failure to disseminate it, and the decades-long literature that followed — Walker's historical synthesis, Rees's institutional study — treated TMI as the moment U.S. nuclear stopped being able to assume away its own human factors.[4]

THE CAPABILITY GAP

The gap was not intelligence but the right capability for the event that arrived. Operators drilled on design-basis ruptures had no model for an ambiguous cascade, and a control room reporting commands rather than states made correct diagnosis nearly impossible — the interface and the training were mismatched to the failure that actually arrived. Beneath that lay a second failure: the capacity to **learn** — to turn Davis-Besse into fleet knowledge — had itself broken down, so the industry kept making the same diagnosis blind, which is why the case resists any single-cause reading.[5] The third layer was institutional: there was no organization whose job was to force operating experience from one utility into the training and procedures of every other, and no body the NRC trusted to do it on the industry's behalf.

AFTERMATH & REFORM

TMI produced not another accident but a system of reform. Within nine months the industry created the Institute of Nuclear Power Operations (INPO, December 1979) to set standards, run plant evaluations, and force the sharing of operating experience the Davis-Besse failure had lacked.[6] The NRC overhauled licensing and inspection: it required plant-referenced simulators, placed resident inspectors at every commercial reactor, tied operator training and re-qualification to a national standard, and in Regulatory Guide 1.97 specified the post-accident instrumentation needed to give crews state — not command — information across the regimes accidents actually produce, with subsequent control-room human-factors standards (NUREG-0700) carrying the principle into every new and modernized plant.[7] Each measure attacked one of the failures the accident had exposed: the training gap, the broken learning channel, the gulf between what indicators displayed and what crews inferred, and the institutional absence of a body responsible for cross-fleet learning. TMI is paired later with INPO (Case 172) as the book's strongest argument that failure can engineer durable industry-level reform.



A MINOR COOLING FAULT, AN AMBIGUOUS INDICATOR, A MELTDOWN

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THE LEARNING ENGINEERING LENS

> *The fundamental problems are people-related problems and not equipment problems.*
>
> KEMENY COMMISSION REPORT ON THREE MILE ISLAND, 1979

LE INSIGHT

TMI is the textbook moment when an industry discovered that its capability assumptions did not survive contact with operational reality. Training to design-basis events left the operators blind to the genuinely ambiguous failures that complex systems actually produce; a control room that reported commands rather than states made correct diagnosis structurally hard; and the channel that should have carried Davis-Besse out to every plant did not exist. The reform — INPO, plant-referenced simulators, resident inspectors, RG

1.97 instrumentation, NUREG-0700 human-factors standards — built the missing infrastructure together rather than one piece at a time, which is the case's load-bearing teaching.

LENS APPROACH

TMI is the worked example of induced sub-competency 6.1 (industry-level institution building after catastrophe) and the LENS D5/PT4 pairing — Navigating Sociotechnical Constraints applied to the institutional architecture a catastrophic-system industry needs to learn at scale. Students reconstruct the capability requirements that the design-basis training framework missed (LENS D1), examine the control-room interface and post-accident instrumentation as evidence-architecture problems (LENS D3), and design the cross-fleet learning channel whose absence at Davis-Besse let TMI happen. The case pairs with INPO (Case 172) as the institution-building counterpart to the failure that produced it, and with Fitzgerald/McCain (Case 1) as the contrasting failure where the missing learning channel was internal to one service rather than across a civilian industry. CLO mapping: CLO-5 (Sociotechnical Constraints) primary for the INPO/NRC institutional architecture; CLO-1 (Systems Analysis) for the interface-and-training requirements analysis the accident exposed.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design control-room indicators to report actual system state, not the command issued, so operators diagnose from truth rather than intent.
- Build training around ambiguous, slow-onset cascades — not only dramatic design-basis ruptures — using simulators that mirror the operators' own plant.
- Engineer a learning channel that propagates precursor events across the fleet as drills, so a near-miss at one site becomes practiced knowledge everywhere.

IN OPERATION / AFTER

- Audit whether near-misses are actually reaching operators as changed procedure and training, and force-share operating experience through an independent standards body (the INPO model).
- Sustain plant-referenced simulators and resident oversight so diagnostic skill on ambiguous transients does not decay back to textbook drills.
- Monitor the institutional assumption that serious accidents are impossible, and treat its persistence as a measurable readiness risk.

REFLECTION QUESTIONS

1. TMI operators were trained for worst-case scenarios but failed in an ambiguous one. What is the equivalent training gap in your domain between the trained case and the messy case?
2. The Kemeny Commission called the human element the dominant risk. What evidence would you need to demonstrate the same conclusion in your own domain?
3. The Davis-Besse precursor occurred eighteen months earlier but never reached the operators who needed it. Design the channel in your domain that would turn a near-miss at one site into a drill at every other before the second event arrives.

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Chapter 6

Human-System Collaboration — What Works

When role, interface, and authority are engineered together.

Any operator can stop the line.

TOYOTA PRODUCTION SYSTEM; WIDELY PARAPHRASED.

CASE 70 AVIATION 1981 – PRESENT

Crew Resource Management & CAST

[T] Training Gap [H] Human-System Interface [N] Normalization of Deviance

IMPACT 83% reduction in U.S. commercial aviation fatality risk (1998–2008); 95% reduction in fatalities per 100M passengers over 20 years

IN BRIEF

In March 1977, two 747s collided in fog at Tenerife and 583 people died — in part because a KLM flight engineer's twice-voiced doubt about the runway was overridden by his captain. The failure was not of skill but of the system by which skill in one seat reached another. Crew Resource Management, formalized by United Airlines in 1981, re-engineered the cockpit as a coordinated team: explicit communication protocols, named authority gradients, structured briefings. Twenty years later the Commercial Aviation Safety Team (CAST) added closed-loop analysis of operational data to find and fix hazards before they caused accidents. The pairing — cultural redesign plus continuous evidence — drove an 83% reduction in U.S. commercial-aviation fatality risk between 1998 and 2008, earning the 2008 Collier Trophy.

BACKGROUND

By the 1970s, accident investigations were repeatedly finding that crashes stemmed not from a lack of individual flying skill but from breakdowns in how a crew worked together. The 1977 Tenerife disaster — 583 dead — was the starkest example: a KLM flight engineer questioned, twice and indirectly, whether the runway was clear, and the senior captain dismissed him and continued the takeoff. The pattern that emerged was consistent across investigations — skilled crews failing not because anyone lacked competence, but because the crew's most senior voice could close off information held by a more junior one before it reached the decision.[0]

THE INTERVENTION

Crew Resource Management, formalized by United Airlines in 1981, treated crew coordination as an engineerable property of the system rather than a matter of personality. It introduced explicit communication protocols, named and trained against authority gradients, and instituted structured briefings and debriefings — rebuilding the cockpit as a team in which information from any seat could reach the decision. Rather than exhorting captains to listen better, it built coordination into the standard procedure itself, so the behavior that Tenerife had lacked became the trained default rather than a matter of individual temperament.[1]

HOW IT WORKED

CRM did not teach individual airmanship; it engineered the system of interaction in which airmanship operates. By making it legitimate and expected for a junior officer to challenge a captain, and by giving crews a shared protocol for doing so, it closed the path by which

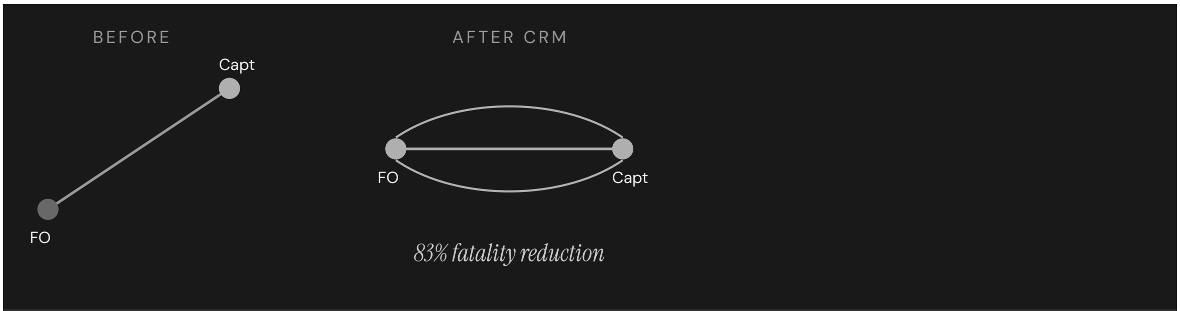
Tenerife-style deference had absorbed safety-critical information instead of transmitting it. The change was structural rather than attitudinal: a challenge that had once depended on a junior officer's nerve now had a named procedure behind it, so the same doubt that went unheard at Tenerife had an authorized route to the decision.[2]

THE EVIDENCE

Twenty years on, the Commercial Aviation Safety Team added the missing second half: closed-loop hazard identification on operational data, prioritized enhancements, tracked implementation, and measured outcomes. CAST set an 80% fatality-reduction target and exceeded it, reaching 83% by 2007 — work recognized with the 2008 Collier Trophy. Over twenty years, fatalities per 100 million passengers fell roughly 95%. The loop closed on itself: data surfaced the next hazard, enhancements were prioritized against it, and the measured outcome fed back into the priorities, so improvement continued rather than plateauing once the cultural change had taken hold.[3]

WHAT TRANSFERRED

CRM and CAST together define what a mature capability-engineering apparatus looks like: a cultural redesign paired with a continuous-evidence loop, where neither half works alone. The model has been exported to surgery, firefighting, and other high-consequence domains, and is now the template for redesigning human roles in AI-augmented systems. What transferred was not the specific protocols but the design logic itself — that coordination is an engineerable system property and that the cultural change needs a measurement loop to keep it honest over time.[4]



AUTHORITY GRADIENT FLATTENED; COMMUNICATION CLOSES THE LOOP

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> *CRM succeeded because it treated crew coordination as an engineerable property of the system.*
>
> PARAPHRASING HELMREICH & FOUSHEE, IN KANKI ET AL., CREW RESOURCE MANAGEMENT (2010)

LE INSIGHT

CRM is the canonical evidence that capability is engineerable at the system level, not just the individual. Tenerife was not solvable by hiring better pilots — only by changing the authority structure inside the cockpit. The intervention worked because it was **paired**: a procedural change plus a cultural change in how the procedure was authorized. Either alone fails.

LENS APPROACH

LENS treats CRM/CAST as the foundational success case across the curriculum. LEN 1 uses it as the problem-framing exemplar; LEN 4 uses CAST as the model closed-loop evidence system; LEN 2 uses CRM as the template for redesigning human roles in automated environments — the logic now being applied to AI-augmented systems.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build the coordination behavior into standard procedure — explicit communication protocols, named authority gradients, and structured briefings — so it is the trained default, not a matter of temperament.
- Authorize the junior-to-senior challenge explicitly and drill it, so safety-critical information has a procedural route to the decision rather than depending on individual nerve.
- Stand up the evidence half from the start: instrument operational data so hazards can be found and prioritized before they cause an accident.

IN OPERATION / AFTER

- Run the closed loop continuously — surface hazards from data, prioritize enhancements, track implementation, and measure outcomes — so improvement does not plateau once the culture shifts.
- Set and exceed an explicit reduction target (the CAST model) so the intervention is judged against a measured outcome, not against its own activity.
- Export the design logic, not just the protocols, when transferring to new domains, adapting the paired cultural-plus-evidence structure to each setting.

REFLECTION QUESTIONS

1. Identify a high-consequence domain whose authority gradient absorbs information rather than transmitting it. What would the CRM equivalent intervention look like there?
2. CRM is paired with CAST. What is the closed-loop evidence system in your domain — and if there is not one, what would it cost to build?
3. CRM made a junior officer's challenge a named procedure rather than an act of nerve. Identify a moment in your domain where the right information depends on someone's courage, and design the protocol that would make raising it the expected default instead.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · incident analysis · safety dashboards

CASE 71 HEALTHCARE 2004 – PRESENT

Keystone ICU / Pronovost Checklist

T Training Gap N Normalization of Deviance

IMPACT Central-line-associated bloodstream infections (CLABSI) reduced to near zero across 103 Michigan ICUs; ~1,500 lives saved in 18 months; ~$100M saved; sustained at ten years

IN BRIEF

Peter Pronovost's central-line checklist has five items — wash hands, clean the skin with chlorhexidine, drape the patient, use full barrier precautions, apply a sterile dressing — and not one of them was unknown to the physicians it governed. The question was never what to do, but whether it would be done every time. The Keystone project, launched across 103 Michigan ICUs in 2004, paired the checklist with a cultural change: nurses were not merely permitted but required to stop the procedure if a step was skipped. That authorization was the load-bearing element. Central-line-associated bloodstream infections fell to near zero, an estimated 1,500 lives and $100 million were saved in eighteen months, and the effect was sustained at ten years.

BACKGROUND

Central-line-associated bloodstream infections were a common, often fatal complication of intensive care, and the steps to prevent them were well established and uncontroversial. The problem was reliability: in the existing culture, a nurse who saw a physician skip a sterile step had no procedural path to intervene without crossing the hospital's authority gradient. The knowledge was universal and the steps cheap; what was missing was any mechanism that made the right action happen every time rather than most of the time, which is precisely where a fatal infection finds its opening.[0]

THE INTERVENTION

In 2004, Peter Pronovost's team launched the Keystone ICU project across 103 Michigan units. It combined a simple five-item central-line checklist — hand hygiene, chlorhexidine skin prep, full-barrier draping, sterile gown-mask-gloves, and a sterile dressing — with an explicit authorization: nurses were required, not merely permitted, to stop any procedure in which a step was missed. The distinction between permitted and required was deliberate — a permission a nurse might decline to exercise against a senior physician became an obligation the institution stood behind, removing the personal risk of intervening.[1]

HOW IT WORKED

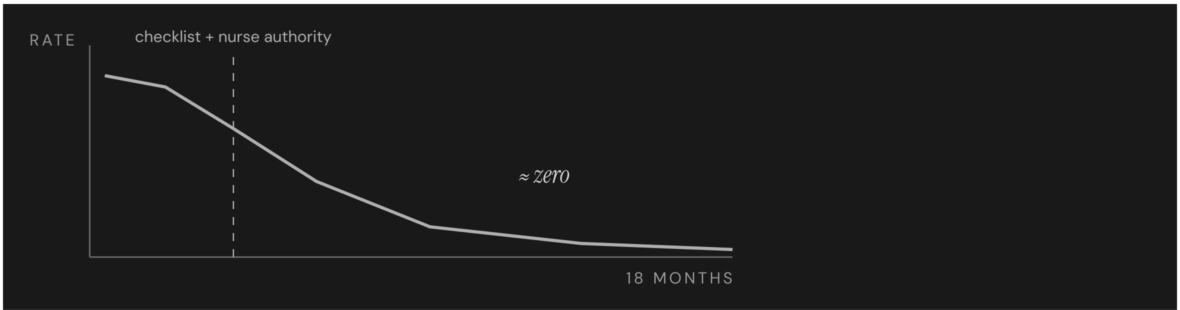
The checklist was the technical half; the nurses' enforcement authority was the cultural half, and it was the load-bearing one. Before Keystone, the path to intervene did not exist; after it, the path was institutional and expected. The pairing converted knowledge everyone already had into behavior that happened every time. The checklist gave the nurse an objective, impersonal basis for the stop — a missed item, not a judgment about the physician — which is what made the authority usable in practice rather than merely on paper.[2]

THE EVIDENCE

Across the Michigan ICUs, the median quarterly CLABSI rate fell to zero, and the state's units outperformed 90% of ICUs nationwide. The program was estimated to have saved roughly 1,500 lives and $100 million within eighteen months. Results were published in the New England Journal of Medicine in 2006, and follow-up showed the effect sustained at ten years. The durability mattered as much as the size of the drop: an improvement that survives a decade is evidence the change was built into the system's structure rather than riding on the initial enthusiasm of a single project.[3]

WHAT TRANSFERRED

Keystone became the clearest evidence in healthcare that a technical intervention without an authority change produces no durable improvement — and vice versa. The model was packaged as the AHRQ CUSP toolkit, adopted in more than forty states, and replicated internationally, establishing the design principle of intervening in matched technical-and-cultural pairs. Packaging the approach as a reusable toolkit was itself part of what transferred — it turned a single project's success into something other institutions could adopt without rediscovering the load-bearing role of the authority change.[4]



BLOODSTREAM-INFECTION RATE ACROSS 103 MICHIGAN ICUS

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THE LEARNING ENGINEERING LENS

> *The checklist was the technical intervention. The nurses' authority to enforce it was the cultural intervention. Neither worked without the other.*
>
> PARAPHRASING PRONOVOST & VOHR, SAFE PATIENTS, SMART HOSPITALS, 2010

LE INSIGHT

Keystone is the clearest evidence in healthcare that a technical intervention without authority intervention produces no durable change, and authority intervention without a technical artifact produces no measurable change. Both are necessary. The empirical record of Keystone is the strongest available argument for designing interventions as **pairs** — and treating the cultural half as engineering, not aspiration.

LENS APPROACH

LENS uses Keystone in LEN 4 and LEN 10 as the canonical worked example of measurement linked to intervention. Studio projects require students to specify both halves of the pair, name the authority that authorizes the cultural half, and identify the measurement signal that will confirm or refute the effect. The course treats "is the cultural change actually authorized?" as falsifiable, not stated.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pair the technical artifact with the authority change from the outset — a checklist plus the explicit, institution-backed right of a junior member to stop the procedure when a step is missed.
- Make the intervention an obligation rather than a permission, so the corrective action does not depend on an individual's willingness to challenge a senior colleague.
- Anchor the stop to an objective, impersonal trigger (a missed checklist item) so the authority is usable without it reading as a judgment of the more senior operator.

IN OPERATION / AFTER

- Measure the outcome directly (the CLABSI rate) and publish it, so the cultural half can be confirmed as authorized in practice rather than merely declared.
- Track the effect over years, not months, to confirm the change is built into the system's structure rather than riding on a project's initial enthusiasm.
- Package the paired design as a reusable toolkit so other institutions can adopt it without rediscovering the load-bearing role of the authority change.

REFLECTION QUESTIONS

1. Identify an evidence-based protocol in your domain that is **known** to work but is not consistently performed. What is the authority change required to pair with it?
2. Design the measurement signal that would confirm the cultural half of a Keystone-style intervention is actually being authorized — not merely declared.
3. Keystone made the nurse's stop an obligation the institution stood behind, not a personal risk. Identify a corrective action in your domain that currently costs the person who takes it, and design the institutional backing that would remove that cost.

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CASE 72 AVIATION 2000 – PRESENT

Korean Air Safety Transformation

T Training Gap N Normalization of Deviance

IMPACT From industry pariah (16 aircraft written off, 700+ lives lost, 1970–1999) to spotless passenger safety record since 1999

IN BRIEF

Between 1970 and 1999, Korean Air had one of the worst safety records in commercial aviation — a loss rate roughly seventeen times United's, more than 700 lives lost — and after the 1997 Guam crash of Flight 801 killed 228, the president of South Korea called it "an embarrassment to the nation." The NTSB traced root causes to cockpit authority gradients: junior officers were culturally unable to challenge a captain's errors. In 2000, Korean Air brought in Delta's David Greenberg to rebuild flight operations — mandating English as the cockpit language, adapting CRM for a high-power-distance culture, and bringing in outside consulting from Boeing and Delta. The airline has had no fatal passenger accident since, winning the 2006 Phoenix Award. Cultural legacy is not destiny when it is deliberately redesigned.

BACKGROUND

From 1970 to 1999, Korean Air suffered repeated fatal crashes — a loss rate roughly seventeen times United Airlines', with over 700 lives lost. The NTSB's investigation of the 1997 Guam crash of Flight 801, which killed 228, identified steep cockpit authority gradients as a root cause: junior officers were culturally unable to challenge a captain's erroneous decisions in time. The pattern echoed Tenerife's lesson but ran deeper, rooted in a national hierarchy the crew carried into the cockpit, so the silence in the right seat was not a lapse of training but an artifact of how rank was expressed in speech.[0]

THE INTERVENTION

After foreign carriers suspended code-shares and the FAA downgraded South Korea's safety rating, Korean Air in 2000 hired David Greenberg from Delta to overhaul flight operations. The interventions were deliberate and cultural: mandated English fluency for all pilots, CRM training adapted for a high-power-distance setting, external consulting from Boeing and Delta, and fleet modernization. The external commercial pressure — suspended code-shares and a downgraded rating — supplied the leverage to act, turning a cultural problem the airline had long lived with into one it could no longer afford to ignore.[1]

HOW IT WORKED

The load-bearing move operated on language. Making English the cockpit language stripped out the Korean honorific hierarchy that had silenced first officers, because English has no honorifics to enforce rank. The CRM adaptation then gave crews an explicit, culturally workable protocol for raising concerns — converting deference into communication. The choice was elegant precisely because it changed the medium rather than asking pilots to override their own culture: with no honorifics to encode, the language itself flattened the gradient that had absorbed the first officer's warnings.[2]

THE EVIDENCE

Korean Air has recorded no fatal passenger accident since the reforms took hold. Air Transport World recognized the turnaround with its 2006 Phoenix Award, and the carrier moved from international pariah to a safety record indistinguishable from the best global operators. The categorical nature of the shift — from a loss rate many times the industry's to none — is what makes the case persuasive: the same crews and the same national culture produced opposite outcomes once the cockpit's communication structure was redesigned.[3]

WHAT TRANSFERRED

Korean Air is the strongest aviation evidence that cultural legacy is not destiny: a specific cultural feature — high power distance in the cockpit — was the binding constraint, and once it was redesigned, the safety record changed categorically. The case also shows that the interface for cultural redesign can itself be engineered, in this instance through language. The transferable lesson is to locate the single cultural feature that actually binds the capability rather than attempting to remake a culture wholesale, then change the medium through which that feature operates.[4]

CULTURAL HIERARCHY REMOVED FROM THE COCKPIT

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THE LEARNING ENGINEERING LENS

> *Korean Air's record before 2000 was changed by an external intervention into cockpit culture, not by criticism of it.*

EDITORS' SYNTHESIS OF NTSB FINDINGS ON KAL 801 AND THE KOREAN AIR TRANSFORMATION

LE INSIGHT

Korean Air is the strongest aviation evidence that cultural legacy is not destiny. A specific cultural feature — high power distance in the cockpit — was the binding capability constraint. Once it was redesigned, the safety record changed categorically. The intervention is also methodologically interesting because it operated on language: English became the cockpit language because English has no honorifics to enforce. The interface for cultural redesign was a linguistic one.

LENS APPROACH

LENS uses Korean Air in LEN 8 as the canonical organizational- learning case under cultural constraint and in LEN 2 as a CRM- extension case for high-power-distance contexts. The case is paired in this book with the Toyota Andon Cord (Case 73) as cultural intervention success stories.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Locate the single cultural feature that actually binds the capability — here, high power distance expressed through honorifics — rather than attempting to remake the whole culture.
- Change the medium through which that feature operates (making English the cockpit language) so the constraint is removed structurally rather than by asking people to override their own culture.
- Pair the structural change with an adapted protocol — CRM tuned for a high-power-distance setting — that gives crews a culturally workable way to raise concerns.

IN OPERATION / AFTER

- Use external pressure (suspended code-shares, a downgraded rating) as durable leverage to hold the reform in place against any drift back to the prior norm.
- Sustain the change with continued outside benchmarking and consulting so the new communication structure is reinforced rather than quietly eroding.
- Track the safety record over years to confirm the categorical shift holds, treating a maintained accident-free record as the evidence the redesign took.

REFLECTION QUESTIONS

1. Identify a cultural feature in an institution you work with that constrains capability. Is it engineerable? What would the redesign look like?
2. Korean Air's intervention operated on language. What surface of culture is engineerable in your domain that you have not yet considered?
3. Korean Air acted only once suspended code-shares and a downgraded rating made the status quo unaffordable. What external pressure exists in your domain that could supply the leverage to redesign a long-tolerated cultural constraint — and how would you use it?

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CASE 73 INDUSTRIAL 1950S – PRESENT

Toyota Production System / Andon Cord

N Normalization of Deviance G Governance & Trust

IMPACT Front-line authority to stop the line resolves most defects quickly at the source; defect-propagation cost minimized; the system adopted globally

IN BRIEF

The andon cord lets any assembly-line worker stop Toyota's entire production line on detecting a defect — handing the lowest-ranking person on the floor the power to halt operations worth millions per hour. The cord is trivially cheap; the authority it confers is the design. The case is decisive for capability engineering because when American automakers copied the cord in the 1980s and 1990s, workers were too afraid to pull it: the tool was present, the empowerment was not. Toyota's system works because it pairs the mechanism with a culture of psychological safety, fast supervisor response, no-blame problem-solving, and the codified "Five Whys" method. When the line stops at Toyota, the team treats it as a learning opportunity. The Andon Cord is the manufacturing twin of the Keystone nurse-authority intervention (Case 71).

BACKGROUND

In high-volume manufacturing, a defect that passes undetected propagates downstream, multiplying the cost of every later correction. Catching problems at the source requires the person who sees them — usually the lowest-ranking worker on the line — to be able to act, in an environment where stopping a line running at millions of dollars per hour is otherwise unthinkable. The economics and the authority structure point in opposite directions: the cheapest moment to fix a defect is the one at which the person who sees it has the least standing to halt production.[0]

THE INTERVENTION

As part of the Toyota Production System, Toyota installed the andon cord: a physical pull-cord that lets any worker signal a problem and, if unresolved, stop the entire line. The inversion of authority is the entire point — the cord itself costs almost nothing, while the protected authority it confers on a front-line worker is the actual design. Handing the lowest-ranking person the power to halt operations worth millions per hour deliberately resolves the contradiction the background poses: it puts the authority to stop exactly where the defect is first visible.[1]

HOW IT WORKED

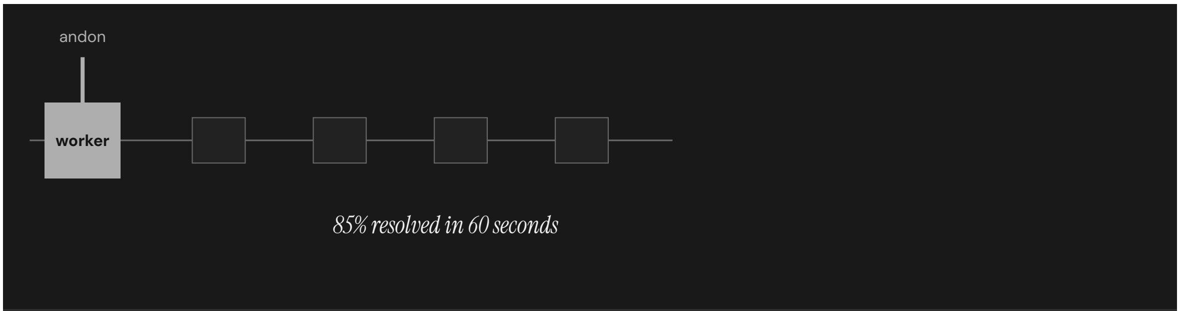
The cord works because Toyota pairs the mechanism with a cultural system: psychological safety, a rapid supervisor-response protocol when the cord is pulled, no-blame root-cause analysis, and the codified "Five Whys" method. A stop is treated as a learning opportunity rather than a failure, so workers actually use it — the technical artifact and the protected authority are inseparable. The rapid supervisor response is what makes the protection credible in practice: a worker who pulls the cord sees help arrive rather than blame, so the no-blame norm is demonstrated each time, not just asserted.[2]

THE EVIDENCE

The proof of the pairing is negative as well as positive. When American manufacturers copied the andon cord in the 1980s and 1990s, workers were too afraid to pull it; the artifact without the authority produced nothing. At Toyota, where the authority is protected, the great majority of activations are resolved within a minute and the system has been sustained and exported for decades. The natural experiment is unusually clean — the same physical cord produced opposite results across two settings, isolating the protected authority, not the hardware, as the variable that mattered.[3]

WHAT TRANSFERRED

The Andon Cord is the foundational evidence that authority interventions and technical artifacts are inseparable — the cord means nothing without the protected authority to pull it, and vice versa. It is the manufacturing counterpart of the Keystone nurse-stop authority (Case 71): same logic, different industry, same load-bearing element, and the same failure mode when only the artifact is copied. That the identical pattern recurs across manufacturing and medicine is what elevates it from a Toyota practice to a design principle: wherever the person who sees the problem lacks the standing to stop, copying the tool alone reproduces the failure.[4]



LOWEST-RANKING WORKER AUTHORIZED TO STOP THE LINE

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> *When American manufacturers copied the andon cord, workers were too afraid to pull it.*
>
> PARAPHRASING LIKER, THE TOYOTA WAY (2ND ED., 2020)

LE INSIGHT

The Andon Cord is the foundational evidence that authority interventions and technical artifacts are inseparable. The cord means nothing without the protected authority to pull it. The protected authority means nothing without the cord to act through. The pair is irreducible — and that irreducibility is the LENS co-optimization commitment in physical form.

LENS APPROACH

LENS uses the Andon Cord in LEN 2 as the foundational example of paired technical-cultural intervention, in LEN 8 to discuss cross-domain transfer (and why imitation without the cultural half fails), and in LEN 10 as a studio exemplar of minimal-artifact design.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Place the authority to stop exactly where the problem is first visible — with the front-line worker — resolving the contradiction that the cheapest moment to fix a defect is when the seer has the least standing.
- Pair the cheap artifact with the protected authority deliberately, recognizing the cord costs almost nothing while the authority it confers is the actual design.
- Stand up a rapid supervisor-response protocol so that pulling the cord brings help, not blame, making the no-blame norm credible from the first activation.

IN OPERATION / AFTER

- Sustain the pairing with no-blame root-cause analysis and a codified method (the Five Whys) so each stop becomes embedded learning rather than a one-off interruption.
- Demonstrate the protected authority repeatedly — help arriving within a minute — so the norm is shown each time rather than merely asserted in policy.
- When transferring the model, export the protected authority and response culture, not just the artifact, since copying the tool alone reproduces the failure mode.

REFLECTION QUESTIONS

1. Identify a technical artifact in your domain whose effectiveness depends entirely on a protected authority. Is the authority protected, or only declared?
2. American manufacturers copied the cord without the authority. What is the equivalent surface-level imitation you have observed in your domain, and what was missing?
3. Toyota's no-blame norm is demonstrated each time help arrives within a minute of a pull rather than blame. Design the visible, repeated response that would prove a protected authority is real in your domain rather than merely written into policy.

WHO BUILDS THIS safety science & organizational analysis · governance, ethics & measurement — plus domain experts and a learning engineer to integrate. TOOLS incident analysis · safety dashboards · governance frameworks · bias & evidence audits

CASE 74 HEALTHCARE CLINICAL AI 2018 – 2022

TREWS / Sepsis Watch

H Human-System Interface K Knowledge & Institutional Memory G Governance & Trust

IMPACT Prospective multi-site evidence of reduced mortality, organ failure, and length of stay when clinicians engaged with ML sepsis alerts in deployed care

IN BRIEF

The Targeted Real-time Early Warning System (TREWS) is a machine-learning sepsis-detection tool deployed at five Johns Hopkins hospitals; Duke's Sepsis Watch follows the same pattern. The Adams et al. prospective multi-site study (Nature Medicine 2022) reported reduced in-hospital mortality, reduced organ failure, shorter length of stay, and improved antibiotic timeliness associated with deployment — conditional on clinicians acting on alerts within a defined window. The benefit was not the model in isolation. It was the model plus a deliberately engineered alert, workflow, and clinician-confirmation layer that made the alert actionable at the bedside. The honest hedge in the literature is that the evidence is prospective and observational, not RCT; the field notes RCTs are still pending. The case is the positive counter to the Epic Sepsis Model (Case 93): same delegation task, opposite outcome, and the difference is in the engineering of the human–machine boundary and the discipline of the surrounding evidence work.

BACKGROUND

Sepsis is among the most consequential time-dependent diagnoses in hospital medicine: every hour of delayed antibiotics is associated with increased mortality, and the disease is heterogeneous enough that bedside clinicians frequently miss the earliest signal in a patient already being treated for something else. The promise of machine learning has been to surface that earliest signal from the continuously updated EHR trace — vitals, labs, medications — and route it to a clinician who can act in time.[0]

THE INTERVENTION

TREWS, deployed across five Johns Hopkins hospitals, and Sepsis Watch at Duke are the two best-documented instances of this approach. The Adams et al. prospective study of 590,000 patient encounters reported that when clinicians confirmed an alert within three hours, in-hospital mortality, organ failure, and length of stay were lower than for matched controls, and antibiotics were given sooner. The evidence is observational rather than randomized, but it is multi-site, pre-registered, and outcome-grounded.[1]

HOW IT WORKED

The capability the deployment supplied was not "the model." It was the alert designed to fit into a specific clinical workflow, the confirmation step that put a clinician between the model and the action, and the institutional commitment to measure outcomes — not adoption —

as the metric of success. The reported benefit collapsed when clinicians did not engage with the alert: the model on its own did nothing. The deliverable was the interface, the role design, and the surrounding evidence loop, not the prediction.[2]

THE EVIDENCE

The honest hedge survives into the literature. The Adams et al. paper, and the broader sepsis-AI field, explicitly note that the outcome inference is conditional on the population, the workflow, and the engagement pattern measured at these sites — and that randomized trials remain pending. The benefit is real on the evidence presented, but it is not a closed proof; it is the strongest available evidence that delegation of early detection to ML can be done as a paired intervention with measurable outcome improvement.[3]

WHAT TRANSFERRED

What TREWS teaches is that the failure pattern of clinical AI (Case 130) is not inevitable. When the model is treated as one component of a deliberately designed human–machine system — with an alert that fits the workflow, a clinician role that retains authority, a deployment that is observed prospectively, and a willingness to report null effects in non-engaged subgroups — the delegation can produce capability rather than alert fatigue. The case is the engineering counter to Watson for Oncology (the model marketed ahead of its capability) and the Epic Sepsis Model (the model deployed ahead of its validation).

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THE LEARNING ENGINEERING LENS

> *The benefit is not the model. It is the model used as part of a system clinicians can act on.*
>
> EDITORS' SYNTHESIS OF ADAMS ET AL. (2022) AND HENRY ET AL. (2022).

LE INSIGHT

TREWS is the strongest current evidence that delegating early sepsis detection to a machine-learning system can improve patient outcomes — when the delegation is engineered as a paired intervention. The capability deliverable is the alert design, the clinician role, and the outcome-grounded evidence loop, not the model.

LENS APPROACH

TREWS is the positive Domain 3 / Problem Type 6 case the corpus needed: a documented delegation to AI that worked, with the explanation locatable in the human–machine interface and the governance discipline rather than the model. LENS uses it in Domain 3 (Human-System Collaboration) for the delegation-with-revocation pattern, in Domain 4 (Test and Evaluation) for outcome-grounded prospective evidence under the judgment-under-inadequate-evidence CLO, and in Domain 5 (Navigating Sociotechnical Constraints) for the workflow-fit discipline. It is the foil drafted directly against the Epic Sepsis Model (Case 93).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the alert to fit a specific workflow, not the average workflow — including the bedside action it should provoke and the timeline within which it must be confirmed.
- Keep a clinician between the model and the action: model flags, human confirms, the model's authority is to surface, not to decide.
- Instrument the deployment for outcomes (mortality, organ failure, antibiotic timeliness) before the first alert fires — adoption is not the metric of success.

IN OPERATION / AFTER

- Report engagement-stratified outcomes honestly — including the subgroups where the alert was not acted on and the benefit was not observed.
- Treat the prospective/observational design as a constraint to be replaced by RCT evidence when feasible, not as a result that ends the evidence work.
- Carry the model's hedges into the deployment documentation so the next site adopts the model and the discipline that produced it.

REFLECTION QUESTIONS

1. Identify a delegation of detection or screening in your domain that succeeded. What were the components of the human–machine interface that made the model actionable, and what would happen to the outcome metric if those components were removed?

2. Specify the engagement-stratified outcome you would report from a deployment at your site — including the subgroup where the alert was not acted on. What would you need to instrument before the first alert fires?

3. The TREWS evidence is prospective and observational, not RCT. What is the minimum additional evidence you would require before recommending the same model deployment at a new site that differs from the validation sites in population, workflow, or EHR configuration?

WHO BUILDS THIS human factors & interaction design · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

CASE 75 HEALTHCARE PATIENT SAFETY HUMAN FACTORS 2024 – 2025

Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff

H Human-System Interface K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT A clinical adaptation of the aviation 'sterile cockpit' rule — no non-essential communication or interruptions during high-workload phases — applied to ward rounds and handoffs; the published evaluation reports reductions in interruption frequency and improvements in information-transfer measures during the protected window, on a single-unit single-study evidence base whose generalization the authors are explicit about

IN BRIEF

The “sterile cockpit” rule in commercial aviation (FAR 121.542, codified 1981) prohibits non-essential communication among flight crew during taxi, takeoff, landing, and any flight phase below 10,000 feet — the high-workload, high- consequence windows when distraction is most likely to produce error. The principle has been adapted across other high-consequence domains; recent work in clinical care has extended the principle to ward rounds and clinical handoff, where interruption-driven information loss is a documented contributor to patient-safety events. Treloar et al. (2025), in World Journal of Surgery, describe a structured intervention on a single surgical ward: a defined window during the handoff or round during which non-essential pages, conversations, and interruptions are prohibited and information transfer is the protected workflow. The evaluation reports reductions in interruption frequency and improvements in information-transfer measures during the protected window. The hedges that survive into the case verbatim: this is a single-unit single-study evidence base, the interruption-reduction effect is direct while the patient- outcome inference rests on the established link between handoff quality and downstream events rather than on direct outcome measurement in this study, and the sustainability of the protected-window discipline beyond the study period is not yet established. The case is the cross-domain adaptation case: an aviation safety-culture rule, adapted with explicit attribution and adapted again for a clinical context, with the evidence at the tier the adaptation has reached.

BACKGROUND

The aviation sterile-cockpit rule (FAR 121.542) was codified in 1981 in response to a documented pattern of accidents in which flight-crew distraction during high- workload phases — non-operational conversation, cabin coordination, administrative communication — contributed to the error chain. The rule is operationally simple: below 10,000 feet, in taxi, takeoff, approach, and landing, only operationally necessary communication is permitted in the cockpit. The principle has been extended in safety-culture practice to other high-workload phases and operations.[0]

THE INTERVENTION

Clinical care has long carried an analogous problem at the handoff and ward-round boundary. Interruption-driven information loss during handoff is a documented contributor to patient-safety events; I-PASS and SIGNOUT? and similar structured-handoff interventions address the information content, but the workflow context — the interruptions that fragment the handoff — has historically been treated as ambient noise rather than as a design variable. The structural argument the clinical adaptation makes is that the workflow context is a design variable, and that the sterile-cockpit principle provides a worked template for engineering it.[1]

HOW IT WORKED

Treloar et al. (2025) describe a structured intervention on a surgical ward. A defined window during the handoff or round is designated as protected; non-essential pages are deferred, side conversations are prohibited, and the staff conducting the handoff are made unavailable for non-emergent interruption during the window. The intervention includes the workflow design (who is responsible for triaging pages, how emergent interruptions are preserved, how the window is signaled and ended) as well as the cultural commitment to honor it. The evaluation measured interruption frequency during the protected window and information-transfer quality against baseline.[2]

THE EVIDENCE

The published outcomes report reductions in interruption frequency during the protected window and improvements in information-transfer measures during that window. The structural learning the case carries is the adaptation discipline: an aviation safety- culture rule, with its specific operational scoping (below 10,000 feet, defined permitted communication), was carried into a clinical context with the operational scoping translated (a defined window, defined permitted interruption classes, defined triage responsibility) rather than the principle imported without translation. Pair the case with I-PASS (a structured-handoff intervention; Case 43 when drafted) at the handoff-as-capability layer, and with CRM (Case 70) at the aviation-safety-culture-to- clinical-care translation layer.[3]

WHAT TRANSFERRED

The hedges have to survive verbatim. This is a single- unit single-study evidence base; replication across units and settings is the natural next step the authors identify, and the case should not be generalized beyond a strong structural argument until that replication is in the literature. The interruption-reduction effect is direct; the inference that fewer interruptions and better information transfer reduce downstream patient-safety events depends on the established link in the handoff- quality literature rather than on direct patient- outcome measurement in this study. The sustainability of the protected-window discipline beyond the study period is not yet established — the operational cost of maintaining the workflow design over the long term is the open question. The case is teachable at the adaptation-discipline level today; the multi-site multi-cycle evidence base is the next deliverable.

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THE LEARNING ENGINEERING LENS

> *The scoping is the adaptation. Sterile cockpit is “below 10,000 feet”; the clinical analog is a defined window with defined permitted interruption classes.*
>
> EDITORS' SYNTHESIS OF PMC12515027 AND THE STERILE-COCKPIT ADAPTATION LITERATURE.

LE INSIGHT

The sterile-cockpit ward-rounds case is the worked example of cross-domain adaptation discipline at small scale: an aviation safety-culture rule, carried into clinical care with its operational scoping translated and its cultural half preserved. The single-unit evidence is direct on interruption frequency and information transfer; the patient-outcome inference rests on the established handoff-quality link, and the multi-site replication is the explicit next deliverable.

LENS APPROACH

Sterile-cockpit ward rounds is the cross-domain adaptation case (induced 3.2; LENS D3/PT5) — Domain 3 for workflow- context-as-design-variable; Domain 5 for the cultural- commitment half. Pair with I-PASS (Case 43), CRM (Case 70), and Case 116.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Adapt the principle with its operational scoping translated, not imported. Sterile cockpit is “below 10,000 feet, defined permitted communication”; the clinical analog is a defined window, defined permitted interruption classes, defined triage responsibility — the scoping is the adaptation.
- Design the triage workflow that handles emergent interruptions during the protected window; the rule cannot be “no interruptions” without a path for true emergencies, and the design of that path is part of the deliverable.
- Build the cultural-commitment half alongside the workflow half; the rule depends on the unit’s willingness to honor it under workload pressure, and the cultural commitment is what makes the rule operative.

IN OPERATION / AFTER

- Carry the single-unit single-study hedge into communication; the case is teachable at the adaptation-discipline level today, with the multi-site replication as the explicit next deliverable.
- Distinguish the direct measure (interruption-frequency and information-transfer-quality reductions during the protected window) from the inferred outcome (downstream patient-safety events), and report them at their respective evidence tiers.
- Plan the long-run sustainability measurement: does the protected-window discipline survive six months, twelve months, leadership turnover? The operational-cost half of the intervention is what the multi-cycle evidence base will eventually answer.

REFLECTION QUESTIONS

1. Identify a high-workload, high-consequence window in your domain that is currently treated as ambient workflow rather than as a protected period. What would the operational scoping of a sterile-window adaptation look like — permitted communication classes, triage responsibility, signaling?

2. Specify the cultural-commitment half of the rule. The workflow design is necessary but not sufficient; honor under workload pressure is what makes the rule operative, and that depends on leadership and unit culture more than on policy.

3. The case's evidence is single-unit single-study. Design the multi-site replication you would want to see before treating the adaptation as established, and the long-run sustainability measurement you would use to know whether the protected-window discipline survives leadership turnover.

WHO BUILDS THIS human factors & interaction design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 76 NUCLEAR ENGINEERING CONTROL-ROOM HUMAN FACTORS RESEARCH AND DEVELOPMENT 2010 – PRESENT

INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet

D Designed Out H Human-System Interface K Knowledge & Institutional Memory

IMPACT The US Department of Energy's Light Water Reactor Sustainability (LWRS) program, executed at Idaho National Laboratory in partnership with utilities, has produced a structured research-and-pilot record for modernizing analog control-room instrumentation in the existing US nuclear fleet — pilot-scale evidence covering hybrid digital/analog operator interfaces, human-factors validation, and qualification pathways for digital instrumentation

IN BRIEF

The US commercial nuclear fleet is dominated by reactors originally licensed in the 1970s and 1980s with analog instrumentation and control-room layouts of that vintage. Aging analog components, vendor obsolescence, and the operational case for digital instrumentation make modernization a foreseeable sustainment requirement; the regulatory environment (Nuclear Regulatory Commission Regulatory Guide 1.180, Branch Technical Position 7-19, and related guidance) makes the qualification path for digital instrumentation deliberately stringent. The US Department of Energy's Light Water Reactor Sustainability (LWRS) program, executed at Idaho National Laboratory in partnership with US utilities, runs the federally-funded research-and-pilot work that supports the fleet's modernization decisions. The program has produced a structured record covering hybrid digital/ analog operator interfaces, human-factors validation studies in full-scope simulators, and qualification-pathway research for safety-related digital instrumentation. The honest hedge survives clearly: LWRS observations are pilot-scale, the sample of plants that have implemented major modernizations is small, and the operational-fleet evidence at scale is forward-looking. The case is teachable on the structured sustainment-research form — a federally-funded research program operating across decades to support an industry's aging-fleet decisions — and closes a C7 zero-state in the induced framework, paired with Cases 22, 23, 24 as the v2 aging-system quartet.

BACKGROUND

The US commercial nuclear fleet operates roughly 90 reactors originally licensed in the 1970s and 1980s. The control rooms of that era were built around analog instrumentation — strip-chart recorders, hardwired indicators, control-panel meters — designed for the operational envelope and the human-factors assumptions of the period. Aging analog components, vendor obsolescence, and the operational case for digital instrumentation (better diagnostics, lower maintenance burden, integration with plant computer systems) make modernization a foreseeable sustainment requirement across the fleet.[0]

THE INTERVENTION

The regulatory environment is deliberately stringent. Nuclear Regulatory Commission guidance — Regulatory Guide 1.180 on electromagnetic and radio-frequency compatibility, Branch Technical Position 7-19 on common-cause failure analysis for digital instrumentation and control, and the broader Standard Review Plan Chapter 7 — sets the qualification path for safety-related digital instrumentation. The stringency is intentional: digital systems can introduce failure modes (common-cause software faults, cyber exposure) that analog systems do not, and the regulatory regime is designed to keep those failure modes bounded as plants modernize.[1]

HOW IT WORKED

The US Department of Energy's Light Water Reactor Sustainability (LWRS) program is the federally-funded research-and-pilot work that supports the fleet's modernization decisions. Executed at Idaho National Laboratory in partnership with US utilities and EPRI, LWRS operates across a multi-decade horizon with annual research- report deliverables. The control-room modernization research line — including the Human Systems Simulation Laboratory at INL with full-scope plant simulators — covers hybrid digital/analog operator-interface designs, human-factors validation studies, and qualification-pathway research that supports utility-level licensing submissions.[2]

THE EVIDENCE

The honest hedge has to be visible. LWRS observations are pilot-scale: full-scope simulator studies, single-plant pilot implementations, and structured human-factors experiments with operator participants from utility partners. The sample of US plants that have implemented major control-room modernizations is small relative to the fleet, and the operational-fleet evidence at scale — fleet-wide reliability, fleet-wide human-error rate, fleet-wide maintenance burden under hybrid digital/analog control rooms — is forward-looking rather than retrospective. The program's own reports characterize the research at this evidence tier, and the case carries the same honesty.[3]

WHAT TRANSFERRED

What the case teaches at the LENS framing layer is the structured sustainment-research form — a federally-funded research program operating across decades in partnership with industry and a regulator, producing the research-and-pilot record that supports licensing decisions on aging-fleet modernization. The form is the sustainment-engineering analog of the FAA Aging Aircraft Program (Case 22): a long-horizon institutional discipline that produces the technical record that aging-fleet decisions can rest on. With NextGen/ADS-B (Case 23) and Y2K (Case 24), LWRS completes the v2 aging-system quartet that closes the C7 zero-state in the induced framework. The LWRS instance is where the evidence is most explicitly pilot-scale, and the case carries that as the program's tier acknowledgement rather than as a weakness to smooth.[4]

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THE LEARNING ENGINEERING LENS

> *The institutional form is what transfers: federal research program plus industry partnership plus regulator engagement, operating across decades to produce a technical record aging-fleet decisions can rest on.*
>
> EDITORS' SYNTHESIS OF THE LWRS ANNUAL REPORT SERIES.

LE INSIGHT

LWRS is the structured sustainment-research case — federally- funded research at INL in partnership with utilities and the regulator, producing the research-and-pilot record that supports aging-fleet modernization decisions across a multi- decade horizon. The observations are pilot-scale; the operational-fleet evidence at scale is forward-looking. The hedge is part of the case.

LENS APPROACH

LWRS control-room modernization is the structured sustainment-research case (induced 7.1; LENS D3/PT4) — Domain 1 for the long-horizon partnership; Domain 5 for the federal-research + industry + regulator triple structure; Domain 3 for the hybrid digital/analog research line. Closes C7 with Cases 22, 23, 24.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pair the research program with full-scope simulator infrastructure (the INL Human Systems Simulation Laboratory) so the human-factors validation work runs in the operational envelope rather than in abstracted laboratory conditions.
- Structure the partnership across federal research (DOE/INL), industry (utilities, EPRI), and regulator (NRC) so the research record is recognized by all three parties — the licensing submissions ultimately rest on that recognition.
- Carry the pilot-scale evidence tier honestly in the program's reports; LWRS characterizes its research at that tier, and downstream practitioners reading the work should read it the same way.

IN OPERATION / AFTER

- Track fleet-scale outcomes as utility implementations proceed: fleet-wide reliability, human-error rate, maintenance burden under modernized control rooms. The forward-looking evidence will accumulate over the next two decades; the program is the institutional carrier of that accumulation.
- Carry the LWRS case in pair with FAA Aging Aircraft (Case 22) as the sustainment-engineering analog at multi-decade horizon, and with Y2K (Case 24) and NextGen (Case 23) as the broader aging-system quartet.
- Treat the structured-sustainment-research institutional form as the transferable artifact: federal research program plus industry partnership plus regulator engagement, operating across decades to produce the technical record aging-fleet decisions can rest on.

REFLECTION QUESTIONS

1. Identify an aging fleet of long-lived assets in your domain whose modernization decisions rest on research that does not yet exist at fleet scale. What is the analog of a federally-funded research program — institutional partner, industry partner, regulator partner — that could operate across the horizon the decisions actually need?

2. Specify the full-scope simulator (or equivalent in-envelope test infrastructure) the program would need so its human-factors validation work runs in operational conditions rather than abstracted laboratory ones. What is the cost of building it, and what does its absence mean for the evidence the research produces?

3. LWRS evidence is pilot-scale and the program reports it as such. Identify a program in your domain whose evidence tier should be acknowledged in its reporting but is not. What institutional discipline keeps the evidence-tier honesty visible — to the program, to industry partners, and to the regulator?

WHO BUILDS THIS systems & human-factors engineering · human factors & interaction design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate. TOOLS task analysis · design prototyping · usability testing · cognitive walkthroughs · knowledge bases · data pipelines

CASE 77 TUTORING HYBRID HUMAN-AI SYSTEMS K-12 EDUCATION 2024

Hybrid Human-AI Tutoring — Augmentation, Not Delegation

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT Three quasi-experimental studies of hybrid human-AI tutoring deployments reported improvements in student learning relative to comparison conditions; the AI is positioned as augmentation, not delegation; the human tutor retains authorization to override and re-direct

IN BRIEF

Thomas et al.'s LAK 2024 best paper, "Improving Student Learning with Hybrid Human-AI Tutoring: A Three-Study Quasi-Experimental Investigation," reports three quasi- experimental studies of hybrid deployments where AI augmentation is added to human tutoring rather than used to replace it. The headline finding is that learning outcomes improved relative to comparison conditions in each of the three studies. The contribution the case carries for the LENS framework is the design positioning: the AI is augmentation, the human tutor retains the authorization to override and re-direct, and the measured outcome is student learning rather than AI-system fidelity. The case is the small-tier intervention-side counterpart to Case 74 (TREWS, the clinician-AI teaming case that worked) translated into education. Pair also with Cases 81 and 39 (CIRCUIT human- AI workforce) and Case 93 (Epic Sepsis, the delegation case that did not work). Open questions: longitudinal persistence; transfer to lower-resource tutoring contexts where human-tutor availability is the binding constraint.

BACKGROUND

The deployment record for AI in tutoring has been pulled in two directions. The fully-automated tutoring track — from Cognitive Tutor through LLM-based tutoring (Case 135) — has tested whether AI alone can replace or substantially reduce the human-tutor role. The augmentation track has tested whether AI can extend the reach and effectiveness of human tutors, with the AI positioned as a tool the tutor uses rather than as a substitute for the tutor. Thomas et al.'s LAK 2024 paper is the strongest published evaluation of the augmentation track to date, and it reports three quasi-experimental studies that converge on a positive finding.[o]

THE INTERVENTION

The three studies test variants of the augmentation pattern in K-12 tutoring deployments. The AI supports the human tutor with information surfacing, problem recommendation, and student-progress visibility; the human tutor retains the conversational and pedagogical lead. Outcome measures are student learning relative to comparison conditions —

control conditions that vary by study but consistently anchor against either tutor-only or AI-only baselines. Across the three studies, the hybrid condition produced measurable improvements in student learning. The replication structure across the three studies — same authorship team, varying institutional context, converging direction of effect — is the methodological backbone of the case.[1]

HOW IT WORKED

The design positioning the case carries for the LENS framework is the augmentation-not-delegation frame. The AI is positioned as augmentation, the human tutor retains the authorization to override and re-direct, and the measured outcome is student learning, not AI-system fidelity. This is the design pattern that worked in clinical-decision-support at Case 74 (TREWS) and that did not work at Case 93 (Epic Sepsis) — where TREWS preserved clinician authorization and built the explanation structure that supported it, the Epic Sepsis deployment pattern collapsed clinician judgment into alert compliance. Hybrid human-AI tutoring is the educational analog of the TREWS pattern, and the LAK 2024 paper is the evidence base that grounds the analog.[2]

THE EVIDENCE

The case anchors with the CIRCUIT pair (Cases 81 and 39) at the workforce-augmentation layer. CIRCUIT proofreading positions human capability as the recovery mechanism for automation failure at petabyte scale; the CIRCUIT workforce model builds the capability in the first place; hybrid human-AI tutoring positions AI as augmentation of an already-capable human tutor. The three cases together teach the augmentation-and-correction pattern across three deployment substrates — connectomics proofreading, neuroscience-workforce development, and tutoring — and they ground the curriculum's machine- teaming and delegation-with-revocation anchors at each substrate.[3]

WHAT TRANSFERRED

The open questions the case carries are the ones the authors name. Longitudinal effect persistence is not yet in the evidence base — the three quasi-experimental studies report end-of-intervention outcomes, not multi-year follow-through. Whether the design transfers to lower-resource tutoring contexts, where human-tutor availability is the binding constraint and the augmentation-of-a-tutor frame may not apply, is the open generalization question. The quasi-experimental design is honest about its causal-inference limits — randomization is not at the level a cluster RCT would provide — and the case carries the qualification. Future validation ongoing on persistence, transfer, and the tutor-scarce-context generalization.

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> *The AI is positioned as augmentation, not delegation. The human tutor retains the authorization to override and re-direct. The measured outcome is student learning, not AI-system fidelity.*
>
> EDITORS' SYNTHESIS OF THOMAS ET AL. (2024).

LE INSIGHT

Hybrid human-AI tutoring is the educational analog of the TREWS clinician-AI teaming pattern. AI is positioned as augmentation; human tutor retains override authorization; student-learning outcome is the measure. Three quasi- experimental studies converge on positive learning effects. The case pairs with Cases 74 / 101 / 118 / 119 in the human-AI teaming thread and grounds the delegation-with- revocation CLO at the educational deployment.

LENS APPROACH

Hybrid human-AI tutoring is the augmentation-not-delegation case in education (induced 6.4; LENS D3/PT6). LENS uses it in Domain 3 (Human-System Collaboration) for the augmentation pattern and the override-authorization frame, and in Domain 2 (Iterative Development) for the three-study converging-design replication. Pair with Cases 74 (TREWS) and 102 (Epic Sepsis) at the clinical analog, and with Cases 81 and 39 (CIRCUIT) at the workforce-augmentation analog.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Position the AI as augmentation explicitly in the design documentation, not implicitly in the deployment pattern; the augmentation-vs-delegation distinction is the load-bearing design choice and should be the named design choice.
- Preserve human-tutor authorization to override and re-direct as a system-design requirement, not as a discretionary affordance; the comparison with Case 93 (Epic Sepsis) is that override authorization collapses when the system pattern does not actively preserve it.
- Measure the student-learning outcome, not the AI-system-fidelity outcome; the case's pedagogical framing depends on the outcome instrument being the educationally relevant one, not the AI-development-internal one.

IN OPERATION / AFTER

- Commission longitudinal follow-through that extends the evidence base past the end-of-intervention horizon; the open persistence question is testable against the same deployment with additional data infrastructure.
- Test the augmentation design in tutor-scarce contexts; the open generalization question is whether the pattern transfers to settings where the binding constraint is human-tutor availability rather than human-tutor effectiveness.
- Pair the case with Case 74 (TREWS) in the curriculum so the augmentation-and-override pattern is taught across clinical and educational substrates; the two cases together ground the delegation-with-revocation CLO with two converging instances.

REFLECTION QUESTIONS

1. Identify an AI deployment in your domain where the design choice between augmentation and delegation has been implicit rather than explicit. What would change in the system design if the choice were named explicitly, and what comparison condition would you build to test the difference?

2. Specify the override-authorization preservation mechanism in your domain's analog deployment. Is the human operator's authority to override and re-direct a system-design requirement, a discretionary affordance, or an implicit assumption? Which of the three is honest about what the system currently supports?

3. The case pairs with Case 74 (TREWS) at the clinical layer and Case 93 (Epic Sepsis) as the delegation-collapse contrast. Pick an AI deployment in your domain and ask: which of the two patterns does your deployment look most like, and what design changes would shift it toward the augmentation pattern?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · task analysis · design prototyping

CASE 78 AUTONOMOUS SYSTEMS 2016 – PRESENT

Tesla Autopilot — Recurring Fatalities

T Training Gap N Normalization of Deviance G Governance & Trust H Human-System Interface

IMPACT Dozens of fatalities documented in NHTSA's Standing General Order data; first U.S. cases of Level-2 automation contributing to fatal injury

IN BRIEF

Tesla's Autopilot and Full Self-Driving Beta are Level-2 driver-assistance systems: the human driver remains legally and operationally responsible at all times. NHTSA's Standing General Order data has documented dozens of fatal crashes involving Autopilot since the 2016 death of Joshua Brown in Florida, and the pattern is consistent — the system performs capably for long stretches, the driver's monitoring attention attenuates, and an edge case (a stationary object, a faded lane line, a crossing vehicle) produces a collision the inattentive driver fails to catch. NHTSA's investigation has found Tesla's driver-engagement design inadequate to sustain the attention safe operation requires. At consumer scale, Autopilot is the live test of whether passive monitoring of good-enough automation is a role a human can actually perform.

THE SHIFT

Partial driving automation has moved from research vehicles into millions of consumer cars. Level-2 systems like Tesla's Autopilot can steer, accelerate, and brake within their operational design domain, but they require the human driver to monitor continuously and take over instantly — a fundamentally new and demanding role assigned to ordinary, untrained consumers. Where a research program could screen, brief, and instrument its safety drivers, a consumer product reaches everyone who buys the car, with no curriculum and no qualification gate standing between purchase and the monitoring task itself.[0]

WHAT IS EMERGING

Since the first fatal Autopilot crash — Joshua Brown, Florida, 2016 — NHTSA's Standing General Order data has documented dozens of fatal crashes involving the system. The pattern is consistent: long periods of capable operation, attenuating driver attention, and then an edge case — a stationary fire truck, a faded lane marking, a perpendicular crossing — that the disengaged driver fails to catch in time. The very reliability that makes the system attractive is what erodes the vigilance it depends on, so each uneventful mile quietly raises the odds that the next intervention will come too late.[1]

THE CAPABILITY QUESTION

The proximate cause in each case is the driver, who was legally responsible. But the deeper question is whether sustained vigilant monitoring of an automation that works well most of the time is a role a human can perform at all. Naming the feature "Autopilot" and designing weak engagement checks shaped the very inattention the system then blamed on the operator — so the architecture both invited the disengagement and reserved the liability for the person least positioned to resist it.[2]

EARLY EVIDENCE

NHTSA's open investigation (ODI EA22-002) has identified Tesla's driver-engagement design as inadequate to maintain the attention safe operation requires, and the recurring fatality pattern across NTSB reports suggests passive monitoring is not a sustainable role as currently engineered. Decades of automation-complacency research point the same way — the finding is not that any one driver failed but that the role asks a human to stay alert to a system precisely because it almost never needs them, a demand the evidence keeps showing is not reliably met.[3]

OPEN PROBLEMS

Tesla Autopilot is the consumer-scale version of the Uber ATG problem (Case 62): a passive-monitoring role deployed without the capability infrastructure — training, engagement design, attention measurement — to make it performable. The open problem is what driver-engagement architecture, if any, could make Level-2 monitoring sustainable for an average driver over years of use, and whether the answer is a better attention check or a concession that the role itself has to be redesigned out of the human's hands.[4]

L2

DRIVER RETAINED · ATTENTION NOT ENGINEERED

the system works well most of the time — and then it does not

TESLA AUTOPILOT — LEVEL-2 MONITORING AS A SUSTAINABLE ROLE

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THE LEARNING ENGINEERING LENS

> *The operational design ... permitted his prolonged disengagement from the driving task.*
>
> NTSB HIGHWAY ACCIDENT REPORT HAR-17/02 (WILLISTON, FLORIDA CRASH), 2017

LE INSIGHT

Tesla Autopilot at consumer scale is the largest live test of Level-2 monitoring as a sustainable role. The early evidence is that it is not. The case is the strongest currently available test of whether consumer-side training and engagement design can produce sustained attention to automation. The recurring fatality pattern suggests the answer.

LENS APPROACH

LENS uses Tesla Autopilot in LEN 2 as the live consumer-scale test of monitoring as a sustainable role and in LEN 7 for the governance dynamics of a Level-2 system marketed at the boundary of higher autonomy. Studio projects examine the driver-engagement design that would make the role performable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Engineer driver-engagement monitoring to the demonstrated limits of human vigilance — verify attention against operational evidence before fielding, not against an assumption that warnings suffice.
- Name and present the feature so its capability boundary is unmistakable to an untrained consumer, rather than implying autonomy the system does not deliver.
- Constrain operation to the design domain the system can actually handle, so the human is not silently relied on as the backstop for edge cases.

IN OPERATION / AFTER

- Monitor the standing-order crash data for the disengagement pattern and treat a recurring signature as evidence the role, not the driver, needs redesign.
- Track attention and takeover performance across years of ownership, since vigilance erodes with the very reliability that accumulates over time.
- Hold the engagement design accountable to an independent regulator with authority to require changes when in-use evidence shows it is inadequate.

REFLECTION QUESTIONS

1. Identify a passive-monitoring role in your domain. What evidence would tell you whether attention is sustainable over years of operation?
2. Design the driver-engagement architecture that would make Level-2 monitoring sustainable for an average consumer.
3. Autopilot assigns full legal responsibility to the operator while engineering the conditions that erode their attention. Where in your domain does liability rest with the person an automated system has made least able to intervene — and how would you realign the two?

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis · governance, ethics & measurement · human factors & interaction design — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · incident analysis · safety dashboards · governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs

CASE 79 HEALTHCARE TECHNOLOGY 2023 – PRESENT

ChatGPT in Healthcare — Hallucination Cases

H Human-System Interface D Designed Out

IMPACT Documented cases of clinicians using LLMs that produce hallucinated citations, fabricated dosages, and fictitious clinical guidelines

IN BRIEF

Since ChatGPT's public release in late 2022, the clinical and peer-reviewed literatures have documented a recurring pattern: clinicians use large language models to draft patient education, summaries, or treatment guidance, and the output contains fabricated citations, hallucinated drug dosages, or fictitious clinical guidelines. The failures range from cosmetic — invented references in academic submissions — to potentially clinical, such as unsafe dosing or fabricated contraindications. The capability gap is at the human-verification interface: the model presents hallucinated content with the same fluent confidence as accurate content, and early reports suggest clinicians accept LLM output less critically than a colleague's. The case is the live, foundational case for LLM integration into clinical workflow — the discipline must specify what verification looks like at the moment of use, while deployment is already happening.

THE SHIFT

Large language models became broadly available with ChatGPT's release in late 2022, and clinicians began using them almost immediately — to draft patient-education materials, summarize records, and look up guidance. For the first time, a tool that produces fluent, authoritative-sounding medical text is in routine, informal use at the point of care — adopted ahead of any guideline, credential, or institutional sign-off, so the practice spread faster than any structure to govern it could be put in place.[0]

WHAT IS EMERGING

A recurring failure pattern has been documented across the clinical and peer-reviewed literatures: LLM output containing fabricated citations, hallucinated drug dosages, or fictitious clinical guidelines. The failures range from cosmetic — invented references in academic submissions — to potentially clinical, such as unsafe medication doses or invented contraindications — a span that matters because the same tool, used the same way, can produce a harmless error and a dangerous one with no change in how confident it sounds.[1]

THE CAPABILITY QUESTION

The capability gap is at the human-verification interface. The model presents hallucinated content with exactly the same fluent confidence as accurate content; the interface does not distinguish the two. The question is whether clinicians can — and will — develop the routine verification practice that the tool's fluency actively discourages, since the very smoothness that makes the output easy to accept is what removes the friction a reader would normally use as a warning.[2]

EARLY EVIDENCE

Early case reports suggest that clinicians who would carefully check a colleague's recommendation accept LLM output less critically, precisely because it reads so authoritatively. JAMA editorials and reviews through 2023–2024 have repeatedly flagged the absence of an established verification practice as the central risk of clinical LLM use — the concern is not that the model errs but that the practice for catching its errors at the moment of use has not yet been defined or taught.[3]

OPEN PROBLEMS

This is the live frontier case for human-AI teaming when the AI is fluent across both accurate and hallucinated content. The capability that does not yet exist is a routine clinical verification practice — an analog to the bibliographic discipline of academic writing — specified at the moment of use rather than after harm. The discipline is being asked to define what good looks like while deployment is already underway, so the verification standard has to be built around a tool already in millions of hands rather than gated in front of it.[4]

TONE	CONTENT
identical	accurate / not

the interface does not distinguish; the clinician must

LLMS IN CLINICAL USE — FLUENCY WITHOUT WARRANTY

REFERENCES

1. JAMA editorials on LLM integration into clinical practice (2023–2024) — the hallucination/verification problem (synthesized).
2. Sallam (2023), "ChatGPT Utility in Healthcare Education, Research, and Practice" — documented benefits and risks.
3. Case reports of LLM hallucinated citations and dosages in clinical and academic use (2023–2024).
4. WHO ethical guidance on AI for health (2024) — verification and oversight requirements.
5. Wachter & Brynjolfsson (2023), JAMA — generative AI in health care.

THE LEARNING ENGINEERING LENS

> *LLMs produce hallucinations indistinguishable in tone from accurate information, and clinicians have not yet developed the practice of routine verification.*
>
> EDITORS' SYNTHESIS OF JAMA EDITORIALS ON LLM CLINICAL USE (2023–2024)

LE INSIGHT

LLM use in clinical practice is the live frontier case for human-AI teaming when the AI's output is fluent across both accurate and hallucinated content. The capability that does not yet exist is the routine verification practice — a clinical analog to the bibliographic discipline of academic writing.

LENS APPROACH

The learning engineer's deliverable here is a specific artifact, not a posture: a verification-at-point-of-use guardrail that counters cognitive offloading. Concretely, a confirm-before-act gate wrapped around the LLM output — every dosage, contraindication, or guideline citation the clinician would rely on must clear a forced source-attribution step that surfaces the underlying reference (or its absence) and requires an explicit confirmation against it before the output can be acted on. The gate is designed so the model's fluency cannot satisfy it: a smooth, confident answer with no attributable source fails the check and is held back from patient-facing use. LENS builds and critiques exactly this guardrail in LEN 2 (the human-AI verification interface), stresses it in LEN 7 (governing the gate as deployment policy), and in LEN 10 (capstone) asks the student to design the confirm-before-act artifact for a clinical workflow already in use.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify a routine verification practice at the moment of use — what a clinician must independently confirm before acting on LLM output — before the tool enters clinical workflow.
- Design the interface to mark provenance and uncertainty, so fluency alone cannot stand in for warranty of accuracy.
- Restrict the sanctioned uses to those where verification is feasible and cheap, keeping unverifiable high-stakes outputs out of patient-facing work.

IN OPERATION / AFTER

- Monitor the documented failure pattern — fabricated citations, dosages, guidelines — across in-use cases to keep the risk profile current as models change.
- Track whether the verification practice is actually being performed, since the tool's fluency discourages exactly the checking it requires.
- Govern adoption against emerging guidance so the standard for verification keeps pace with a tool already in widespread informal use.

REFLECTION QUESTIONS

1. Identify a workflow in your domain currently being augmented by LLMs. What is the verification practice — and does it exist at the moment of use, or only after?
2. Design the verification deliverable that should accompany every clinician's adoption of an LLM tool for patient-facing work.
3. The same fluency that makes LLM output easy to accept is what removes the cues a reader normally uses to doubt it. What interface signal would restore that friction at the moment of use without making the tool unusable?

WHO BUILDS THIS human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · task analysis · design prototyping

CASE 80 TECHNOLOGY 2021 – PRESENT

AI-Augmented Coding Tools

T Training Gap H Human-System Interface

IMPACT Tens of millions of developers using GitHub Copilot, Cursor, and peers; productivity gains documented; security and correctness implications still being characterized

IN BRIEF

AI-augmented coding tools — GitHub Copilot, Cursor, Codeium, and peers — represent the largest real-time experiment in human-AI collaboration in this book, with tens of millions of developers using them daily. Published studies (Peng et al. 2023) document real short-term productivity gains; other work (Pearce et al. 2022) finds a substantial share of AI-generated completions in security-sensitive settings contain vulnerabilities, though a controlled study (Sandoval et al. 2023) found AI assistance did not significantly raise the rate of critical security bugs. The capability question is open: are developers becoming more capable, or more dependent? The short-term gains are real; the long-term consequences — especially for those who learn the craft with these tools always available — are not yet known. The discipline is being asked to define good before the longitudinal evidence is in.

THE SHIFT

AI coding assistants moved from novelty to infrastructure in a few years. GitHub Copilot, Cursor, Codeium, and similar tools now suggest, complete, and generate code for tens of millions of developers daily — the largest real-time experiment in human-AI collaboration in professional knowledge work to date, conducted not in a study design but in the live practice of an entire profession, with no control group and no agreed measure of what it is doing to the underlying craft.[0]

WHAT IS EMERGING

Two findings are accumulating in parallel. Controlled studies (Peng et al. 2023) document real short-term productivity gains. At the same time, the security picture is unsettled: Pearce et al. (2022) found a substantial fraction of Copilot completions in security-relevant scenarios contained vulnerabilities, while a controlled study by Sandoval et al. (2023) found AI assistance did not significantly increase the rate of critical security bugs. The two results do not cancel so much as mark how unsettled the picture is — output clearly rises, but the quality and safety of that output resist a single verdict.[1]

THE CAPABILITY QUESTION

The capability question is open and consequential: are developers using these tools becoming more capable, or more dependent? Short-term output rises, but whether the underlying skill grows or erodes — especially for those who learn the craft with the tools always present

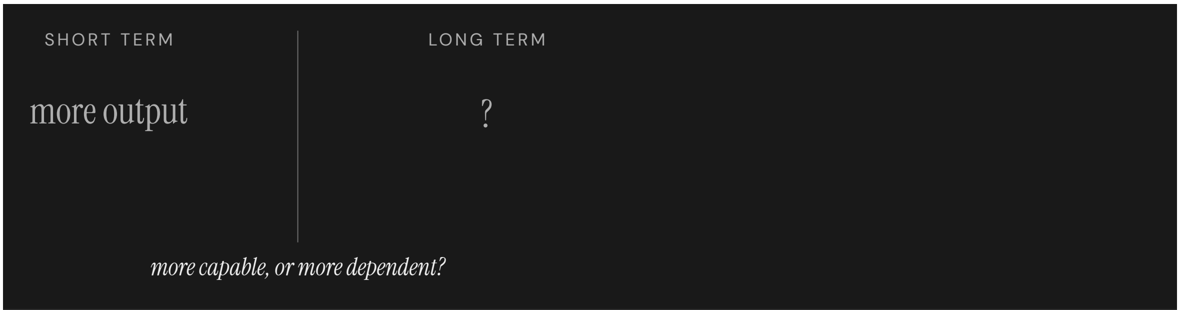
— is precisely what the productivity metrics cannot tell us, because a measure of how much code ships says nothing about whether the person shipping it could still produce or judge it without the assistant.[2]

EARLY EVIDENCE

The longitudinal evidence is not yet sufficient to answer. Dell'Acqua et al. (2023) found a "jagged frontier" in professional LLM use: performance improves on tasks inside the tool's competence and degrades on tasks just outside it, where users over-trust the output. The short-term gains are real; the long-term capability consequences remain uncharacterized — and the jagged frontier is hard to navigate precisely because its edge is invisible from inside the task, so the user cannot tell when they have crossed from where the tool helps to where it misleads.[3]

OPEN PROBLEMS

AI-augmented coding is the live frontier for human-AI teaming in professional knowledge work, and the discipline LENS represents is being asked to specify what good looks like before the long-term evidence is in. The open problem is the longitudinal study that could distinguish capability growth from capability erosion — and a training practice that keeps the human's skill on the growing side, built and adopted while a generation is already learning the craft with the tools always within reach.[4]



AI-AUGMENTED CODING — THE CAPABILITY QUESTION IS OPEN

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1. Peng et al. (2023), "The Impact of AI on Developer Productivity" — short-term productivity gains.
2. Pearce et al. (2022), "Asleep at the Keyboard? Assessing the Security of GitHub Copilot's Code Contributions."
3. Sandoval et al. (2023), "Lost at C," USENIX Security — AI assistance did not significantly increase critical security-bug rates.
4. Dell'Acqua et al. (2023), "Navigating the Jagged Technological Frontier" (HBS / BCG) — professional LLM use.
5. L. Bainbridge, "Ironies of Automation," Automatica 19(6) (1983) — the classic deskilling and over-reliance problem, applied to AI-augmented work.

THE LEARNING ENGINEERING LENS

> *AI assistance changes what developers can do; it may also change what they need to know.*

EDITORS' SYNTHESIS

LE INSIGHT

AI-augmented coding is the live foundational case for human-AI teaming in professional knowledge work. The short-term gains are real. The long-term capability question — does the tool make the operator more capable, or more dependent — is the question the discipline must learn to ask and answer.

LENS APPROACH

The teaching point is a measurement-design problem, and it is the load-bearing one: productivity metrics count output and so cannot distinguish a developer whose skill is growing from one whose skill is quietly eroding under augmentation. The learning engineer's task is to build the instrument that separates those two. The design is a longitudinal, tool-removed probe — periodically measure each developer on representative tasks with the assistant withheld, scoring unaided correctness, debugging, and the ability to judge generated code, and track that aided-minus-unaided gap over time. A widening gap (rising aided output, flat-or-falling unaided competence) reads as skill-atrophy; a narrowing one reads as skill-growth. LENS uses this case in LEN 2 (human-AI teaming), in LEN 8 to build the capability-development instrument itself, and in LEN 10 (capstone) to have the student design the atrophy-versus-growth measurement for an augmented practice in their own domain.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Define what competence the human must retain independently of the tool, and design the workflow so that skill is exercised rather than quietly handed off.
- Engineer the assistant to surface the jagged-frontier edge — flagging where a task sits outside its reliable competence — so users do not over-trust output just beyond it.
- Keep verification of generated code, especially in security-relevant settings, a required step rather than an optional one, given the unsettled quality picture.

IN OPERATION / AFTER

- Run the longitudinal study that productivity metrics cannot substitute for, measuring whether underlying skill is growing or eroding over years of use.
- Monitor for over-reliance at the competence boundary, where the evidence shows performance degrading as users trust the tool past its reliable range.
- Track outcomes for practitioners who learned the craft with the tools always present, the cohort whose long-term capability is most uncertain.

REFLECTION QUESTIONS

1. In your domain, identify a class of practitioners whose work is currently being augmented by AI tools. What evidence would tell you whether their capability is growing or eroding?
2. Design the longitudinal study that would distinguish capability growth from capability erosion in an AI-augmented professional practice.
3. The "jagged frontier" is hard to navigate because its edge is invisible from inside the task. Design a signal or practice that would tell a practitioner in your domain when they have crossed from where the tool helps to where it misleads.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

CASE 81 NEUROSCIENCE/CONNECTOMICS HUMAN-AI TEAMING WORKFORCE TRAINING 2017 – PRESENT

CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale

H Human-System Interface K Knowledge & Institutional Memory N Normalization of Deviance

DISCLOSURE An editor of this volume has research adjacency to connectomics programs discussed in this case and the case originates at the editors' home institution (Johns Hopkins / APL). The connectomics method and infrastructure are anchored to peer-reviewed work; the program training-outcome evidence is institutional documentation rather than independent peer-reviewed evaluation, and that gap is rendered as the evidence-tier flag above.

EVIDENCE TIER internal-pm — source confidence flagged; future validation ongoing.

IMPACT Under IARPA's MICrONS program, automated electron-microscopy segmentation produces brain maps too large and too error-prone to deploy without human correction; CIRCUIT trains undergraduate cohorts as the proofreading workforce that is the recovery mechanism for automation failure at petabyte scale

IN BRIEF

Under IARPA's Machine Intelligence from Cortical Networks (MICrONS) program, automated electron-microscopy segmentation produces petabyte-scale brain maps that are too large and too error-prone to be deployed without human verification. APL's CIRCUIT program trains cohorts of undergraduates to proofread and correct these maps; APL's BossDB stores and serves the petabyte-scale data. The learning-engineering content the case carries is the human-in-the-loop correction layer: where automation fails at the petabyte boundary, a trained human capability is the recovery mechanism that makes the data scientifically usable. The honest evidentiary state — preserved as the evidence-tier flag rendered under the title — is that the connectomics method and infrastructure are documented in the peer-reviewed literature, but CIRCUIT's program training outcomes (did the cohorts reliably produce proofreading capability, at what error rates, with what retention, with what transfer to other tasks) sit in institutional documentation rather than peer-reviewed program evaluation. The case is included as a frontier — the structural pattern is the increasingly central question of how to design a human correction layer for generative and automated systems at scale, and future validation will continue as the program-evaluation literature builds out. The COI render under the title is binding: home institution, research adjacency, evidence-tier flag, all visible at the point of reading.

THE SHIFT

Connectomics — the reconstruction of detailed wiring diagrams of brain tissue from electron-microscopy imaging — is the modern incarnation of a problem with the same structural shape as the early radiology-AI cases (Case 130): automation produces a primary output at scale that no human team could produce manually, and the output is too important and too error-prone to be deployed without verification. The MICrONS program, funded by IARPA, set out to produce reconstructions of cortical microcircuits at the volume scale required to make computational neuroscience comparisons possible.[0]

WHAT IS EMERGING

The automation produces petabytes of data. Automated segmentation labels each voxel with the neuron it belongs to, but at petabyte scale the per-voxel error rate, however low, compounds into a reconstruction that contains many incorrect neuron boundaries — exactly the boundaries the downstream science depends on. The structural form is familiar: the model is good enough to make the project tractable, and not good enough to deploy without a verification layer. The verification layer in this case is human: trained proofreaders work through the segmentation and correct it, neuron by neuron, with tools designed for the task.[1]

THE CAPABILITY QUESTION

The CIRCUIT program at APL trained cohorts of undergraduates to be that proofreading workforce. The program is documented across institutional news, program description, and the MICrONS program literature; CIRCUIT trainees and APL infrastructure (BossDB) together produced the human and computational substrate of the verification layer. The learning-engineering content is the human capability as the recovery mechanism for automation failure at the petabyte boundary: where the model fails, the trained human is the design decision that makes the system's output usable.[2]

EARLY EVIDENCE

The honest evidentiary state is what the evidence-tier flag under the title encodes. The connectomics method, the automated-segmentation literature, and the BossDB infrastructure are documented in peer-reviewed connectomics publications. CIRCUIT's program **training outcomes** — did the cohorts reliably produce proofreading capability at the error rates and retention required, did the trainees transfer to other tasks, what is the program's measured effect on the downstream science — are documented in institutional news and program description rather than in peer-reviewed program evaluation. The case is teachable on the structural pattern; the operating outcomes are at the institutional-program-management tier and should be read at that tier. The COI under the title — research adjacency, home institution — makes the institutional-tier evidence claim auditable.[3]

OPEN PROBLEMS

The frontier note the case carries is the most forward-looking idea from the pass-4 sweep. The human correction layer for generative and automated systems at scale is going to be a defining capability across an expanding number of domains — automated transcription, AI-generated code, model-extracted structured data, document segmentation, scientific imaging reconstruction. CIRCUIT is one instance of the pattern in neuroscience; the structural question — how to design, staff, train, and govern a human correction layer for the gap between what models produce and what is operationally usable — recurs and is not well-named in the existing curriculum. The frontier flag on this case is the case-grounded basis for proposing a sub-competency in this area, pairing with the broader v2 AI-delegation typology and the Domain 5 **Delegation with revocation** CLO.[4]

REFERENCES

1. MICrONS program literature (IARPA) — connectomics method and automated segmentation evidence base.
2. APL BossDB documentation — petabyte-scale connectomics data infrastructure.
3. CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional/program description; the training-outcome evidence is at this tier and the evidence-tier flag is binding.
4. Cervantes et al. (2023), ASEE Annual Conference — the paired peer-reviewed CIRCUIT case (Case 39).
5. Wachter & Brynjolfsson (2023), JAMA — generative AI verification framing, applicable across domains.

THE LEARNING ENGINEERING LENS

> *The recovery mechanism for automation failure is a designed human capability, not an emergent workforce.*
>
> EDITORS' SYNTHESIS OF THE CIRCUIT / MICRONS PROGRAM RECORD.

LE INSIGHT

CIRCUIT / MICrONS is the frontier instance of designing the human correction layer for automated systems at scale. The connectomics method is peer-reviewed; the program training outcomes are institutional documentation, and the gap is rendered as the evidence-tier flag under the title. The forward-looking question the case names — how to design the human correction layer for generative and automated systems at scale — recurs across domains and is not well-named in the existing curriculum.

LENS APPROACH

CIRCUIT proofreading is the human-correction-layer frontier case (induced 3.4; LENS D3/PT6) — Domain 3 for **Delegation with revocation**; Domain 4 for the evidence- tier split. Pair with Case 39 and Cases 74, 93, 155,
67. COI binds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the human correction layer as a designed capability deliverable, not an emergent workforce: define the error rate to achieve, the verification protocol, the trainee qualification, and the tooling.
- Identify in advance which errors the automation will make at scale and design the correction tooling around those error classes — automation failure is structured, and the correction layer should be too.
- Carry the evidence-tier honesty: peer-reviewed sourcing for the method and infrastructure, institutional-program documentation for the program training outcomes, with the gap visible in the case rather than smoothed.

IN OPERATION / AFTER

- Commission peer-reviewed program-evaluation work on the training-outcome side so the institutional record can converge toward independent evidence over time, while the program is operating.
- Track the human correction layer across automation upgrades: as the model improves, what changes about the kind of correction the trainees do, the error classes, the qualification requirements?
- Carry the frontier note into the curriculum conversation: the design of the human correction layer for generative and automated systems at scale is a sub-competency the existing framework does not yet name, and CIRCUIT is one of several emerging instances.

REFLECTION QUESTIONS

1. Identify a domain in your work where automation produces a primary output at a scale that exceeds manual review. What is the human correction layer's design — error classes targeted, verification protocol, trainee qualification, tooling — and which of these decisions are designed vs. emergent?

2. Specify the peer-reviewed vs. institutional-evidence split for a human-correction-layer program you would propose. The connectomics method can be cited from the literature; the program training outcomes will not be. What evidence tier is honest for each layer?

3. The case's frontier note — designing the human correction layer for generative/automated systems at scale — is a sub-competency the existing curriculum does not yet name. What instance from your domain (automated transcription, AI code, structured-data extraction, document segmentation) would be a paired case to anchor this sub-competency against?

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Chapter 7

Test and Evaluation — What Fails

When the institution cannot see what its own evidence shows.

What gets measured gets gamed when the operator controls the measurement.

A PATTERN RECURRING ACROSS THE CHAPTER'S CASES.

CASE 82 AVIATION 2009

Colgan Air Flight 3407

T Training Gap

IMPACT 50 killed near Buffalo; precipitated the FAA's 1,500-hour rule and the Pilot Records Database

IN BRIEF

On approach to Buffalo in February 2009, the captain of Colgan Air 3407 responded to a stall warning by pulling back on the controls instead of lowering the nose; the Bombardier Q400 stalled and crashed into a house in Clarence Center, killing all 49 aboard and one person on the ground. The NTSB found the captain had a documented history of training failures, and the first officer was ill, fatigued, and paid about $16,000 a year. The airline and the hiring pipeline knew the training history; the regulator did not. Victims' families mounted one of the most effective aviation-safety campaigns in a generation, producing the 2010 law that raised first-officer experience requirements to 1,500 hours and created the Pilot Records Database. The capability gap was visible to everyone except the system that licensed the pilots.

BACKGROUND

Colgan Air 3407, a Bombardier Q400 flying a regional route to Buffalo, was crewed by a captain with a documented history of training failures and a first officer who was ill, fatigued, and paid roughly $16,000 a year. The airline and the hiring pipeline knew the captain's record; the regulator did not — the information that should have shaped who sat in that seat lived inside the carrier's files and never crossed into the licensing system that was supposed to vouch for the crew.[0]

WHAT HAPPENED

On 12 February 2009, on approach to Buffalo, the aircraft slowed and its stall-warning stick shaker activated. The captain responded by pulling back on the control column — the opposite of stall recovery — and the Q400 stalled and crashed into a house in Clarence Center, killing all 49 aboard and one person on the ground. The reflex he reached for was exactly wrong for the situation, the kind of error a pattern of failed training events should have predicted long before he was handed the controls.[1]

THE INVESTIGATION

The NTSB found the probable cause to be the captain's inappropriate response to the stick shaker, against a backdrop of fatigue, weak airline training, and a hiring system blind to the captain's history. The data that should have flagged him existed — multiple failed training events — but did not flow to the decision that put him in the seat, so the hiring choice was made on a record that omitted the very facts that mattered most, an absence that looked like a clean slate.[2]

THE CAPABILITY GAP

Colgan is the canonical case for the gap between the information an institution holds about its operators and the information that reaches the decisions about those operators. The capability that was missing was not pilot skill in the abstract but a data flow: a way to make a pilot's documented training history actionable at the hiring and licensing decision, rather than buried in files no one consulted — a record can exist in full and still fail completely if it never reaches the moment and the person whose choice it was meant to inform.[3]

AFTERMATH & REFORM

The victims' families organized one of the most effective aviation-safety lobbying efforts in a generation, producing the Airline Safety and FAA Extension Act of 2010 — which raised the minimum experience for airline first officers to 1,500 hours and created the Pilot Records Database to make pilot history flow between carriers.[4] The 1,500-hour rule remains debated; the records database fundamentally restructured how the information that was missing in 2009 now moves, converting a buried file into a record a hiring carrier is required to retrieve — the difference between a fact that exists and one that arrives where the decision is made.

50

KILLED, CLARENCE CENTER NY

the data was in the system; the data flow was not

COLGAN 3407 — AND THE REFORM THAT FOLLOWED

REFERENCES

1. NTSB, Aircraft Accident Report: Colgan Air Flight 3407, NTSB/AAR-10/01 (2010) — probable cause (quoted).
2. NTSB/AAR-10/01 (2010) — the captain's training history, and the first officer's fatigue and pay.
3. NTSB/AAR-10/01 (2010) — the information-flow gap between known pilot history and the hiring decision.
4. Airline Safety and Federal Aviation Administration Extension Act of 2010, Pub. L. 111-216 — the 1,500-hour rule and the Pilot Records Database.
5. GAO-12-203, Pilot Records Database Implementation (2012); the Families of Continental Flight 3407 campaign.

THE LEARNING ENGINEERING LENS

> *The captain's inappropriate response to the activation of the stick shaker led to an aerodynamic stall from which the airplane did not recover.*
>
> NTSB AIRCRAFT ACCIDENT REPORT AAR-10/01, PROBABLE CAUSE, 2010

LE INSIGHT

Colgan is the canonical case for the gap between the information an institution holds about its operators and the information that reaches the decisions about whether those operators should be in that seat. The data was already in the system — multiple failed training events. The data flow that would have made that data actionable at the hiring decision did not exist.

LENS APPROACH

LENS uses Colgan in LEN 4 as a case for evidence-system design (the PRD as a designed information flow) and in LEN 8 as a case for advocacy-driven institutional change. Studio projects examine how families of victims became the load-bearing element of a reform the industry had resisted for years.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make a pilot's documented training history a required, structured input to the hiring and licensing decision, not a record held passively in carrier files.
- Design the data flow so that adverse signals (failed training events) are pushed to the decision point rather than waiting to be requested.
- Define record retention and transfer obligations across carriers up front, so a clean-looking application cannot conceal a known history.

IN OPERATION / AFTER

- Audit whether the records actually reach hiring decisions, treating a complete file that never arrives as a live failure, not a solved one.
- Monitor the pipeline for operators whose adverse history is not surfacing, and act before the gap is paid for in an accident.
- Sustain the records database against drift and gaps in coverage, since a data-flow remedy decays as fast as its weakest link.

REFLECTION QUESTIONS

1. What information about operators in your domain exists somewhere in the system but does not flow to the decisions that depend on it?
2. Design the data-flow architecture that would make a Colgan-equivalent visible **before** the accident rather than after.
3. The reform in 2009 was driven by victims' families, not the industry that had resisted it. What load-bearing constituency in your domain could force a stalled data-flow fix into existence, and what evidence would mobilize them?

WHO BUILDS THIS learning science & instructional design — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models

CASE 83 AVIATION 2019

Atlas Air Flight 3591

T Training Gap K Knowledge & Institutional Memory

IMPACT 3 killed; Boeing 767 cargo flight crashed in Texas after the first officer mistakenly activated go-around mode

IN BRIEF

On approach to Houston in February 2019, the first officer of Atlas Air 3591 — an Amazon Prime Air cargo flight — inadvertently triggered the go-around mode during a turbulent descent. Experiencing a somatogravic illusion that the aircraft was pitching up, he pushed the nose down sharply and dove the Boeing 767 freighter into Trinity Bay, killing all three aboard. The NTSB found the first officer had a long, undisclosed history of training failures across multiple carriers — and that Atlas could not see it, because the Pilot Records Database mandated by Congress after Colgan Air 3407 (Case 82) was not yet operational. Atlas relied on the older PRIA system, which surfaced only five years of history. The case is the live recurrence of Colgan: the information existed; the data-flow system was not yet complete.

BACKGROUND

Atlas Air 3591 was a Boeing 767 freighter operating for Amazon Prime Air from Miami to Houston. The first officer had failed training events at several previous employers and had not disclosed them on his Atlas application. At the time of hiring, the Pilot Records Database directed by Congress after Colgan Air 3407 was not yet operational; Atlas used the older PRIA system, which required only five years of records — a window too short to surface a pattern of training failures spread across several prior employers, so the record Atlas could legally obtain was structurally incapable of showing what it most needed to see.[0]

WHAT HAPPENED

On 23 February 2019, during a turbulent descent toward Houston, the first officer inadvertently activated the aircraft's go-around mode, which commanded a pitch-up. Experiencing a somatogravic illusion that the nose was rising into a stall, he pushed forward hard. The 767 entered a steep dive from which the crew did not recover and crashed into Trinity Bay, killing all three aboard. The startle and spatial disorientation that drove the fatal push were the same kind of breakdown his prior training failures had repeatedly recorded, had that history been visible to the carrier that hired him.[1]

THE INVESTIGATION

The NTSB found the probable cause to be the first officer's inappropriate response to the inadvertent go-around activation, driven by a startle and spatial-disorientation response, and identified his pattern of training deficiencies and Atlas's inability to access his full record as contributing factors. The records that would have informed the hiring decision existed but did not reach it — the same information-flow failure named after Colgan a decade earlier, recurring here because the remedy built to fix it was authorized but not yet carrying the full history.[2]

THE CAPABILITY GAP

Atlas Air 3591 is the iteration test of the Colgan reform. The reform partially worked — the Pilot Records Database had been authorized — but it was not yet operational and its coverage was incomplete, so the same information-flow gap that killed fifty people in 2009 was still partly open in 2019. Partial implementation of a remedy leaves a measurable harm inside the aperture — the years between authorization and full coverage are not a neutral transition but a window in which the original failure can recur on the cases the incomplete remedy does not yet reach.[3]

AFTERMATH & REFORM

The PRD final rule was published in 2021 and became effective for Part 121 carriers in 2022, with full historical coverage phasing in through 2024 — closing the gap the case exposed. Atlas 3591 is cited in implementation-fidelity discussions as evidence that a remedy is only as good as its completed coverage — that the date a rule is authorized and the date it actually reaches every carrier are different dates, and the distance between them is where the avoidable harm accumulates.[4]

THE COST OF PARTIAL IMPLEMENTATION BETWEEN COLGAN AND THE PRD'S FINAL COVERAGE

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5. GAO reports on Pilot Records Database implementation (2014–2024).

THE LEARNING ENGINEERING LENS

> *Atlas Air did not have access to portions of the first officer's training record that would have informed its hiring decision.*
>
> PARAPHRASING NTSB AIRCRAFT ACCIDENT REPORT AAR-20/02, 2020

LE INSIGHT

Atlas Air 3591 is the iteration test of the Colgan reform. The reform partially worked: the PRD exists. The reform was incomplete: not all carriers were covered. The case is a cautionary worked example of why partial implementation of a regulatory remedy leaves the original capability gap partially open.

LENS APPROACH

LENS uses Atlas Air 3591 in LEN 4 as a case for measurement-system coverage (the PRD had been built but its mandatory coverage was incomplete), and in LEN 8 for the iteration cycle of reform: an intervention that leaves an aperture creates a measurable harm inside the aperture.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the records system to require a full operator history, not a fixed-length window that can hide a multi-employer pattern.
- Plan the coverage phase-in explicitly, with interim controls protecting the cases the remedy does not yet reach.
- Specify the data the hiring decision needs and make its delivery mandatory before the decision, not optional or self-reported.

IN OPERATION / AFTER

- Audit coverage completeness, not just authorization, and treat the gap between the two as an active source of harm until closed.
- Monitor for adverse events occurring inside the aperture of a partially implemented remedy, and use them to accelerate full coverage.
- Sustain attention through the full phase-in, since a remedy authorized but not yet universal leaves the original gap measurably open.

REFLECTION QUESTIONS

1. Identify a regulatory remedy in your domain whose coverage is partial. What harm is occurring inside the aperture?
2. Design the deliverable that would close the PRD coverage gap in advance of a future Atlas Air 3591.
3. The PRD was authorized in 2010 but not fully effective until 2024, and Atlas 3591 fell inside that gap. For a remedy in your domain, design the interim control that would protect the cases the remedy does not yet reach while its coverage phases in.

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CASE 84 INDUSTRIAL 2008 – 2023

Takata Airbag Inflators

D Designed Out G Governance & Trust

IMPACT More than 30 deaths and hundreds of injuries linked to inflator ruptures; largest automotive recall in history

IN BRIEF

Takata built its airbag inflators around ammonium nitrate — cheap and energetic but unstable, and used by almost no one else without a moisture-absorbing desiccant. Takata's inflators omitted the desiccant. Over years of heat and humidity the propellant degraded and could rupture the metal housing on deployment, spraying shrapnel at the driver; more than two dozen deaths and hundreds of injuries followed. Takata's own tests had shown the ruptures, but it reported them as isolated anomalies and, in places, falsified data; in 2017 it pleaded guilty to wire fraud, paid roughly $1 billion, and went bankrupt. The recall — over 100 million inflators across 19 automakers — is the largest in automotive history. Two capabilities were designed out: the desiccant, and the independent verification regulators never had.

BACKGROUND

An airbag inflates by burning a propellant inches from a person's face, so the propellant must be energetic **and** stable across the years and climates a car will see. To cut cost, Takata built its inflators around ammonium nitrate — cheap and energetic but notoriously unstable, shifting crystalline structure and absorbing moisture as temperature cycles day after day. The few competitors who used it at all added a desiccant to keep it dry over the life of the car; Takata's inflators, for years, shipped with none.[0]

WHAT HAPPENED

Over years of heat and humidity the propellant degraded, and a degraded charge can burn too fast, generating more pressure than the housing was built to contain. A firing inflator could then rupture its own metal housing and spray shrapnel into the cabin — turning the device meant to save a life into a fragmentation hazard aimed at the driver. More than two dozen deaths worldwide and hundreds of injuries followed. The recall grew to more than a hundred million inflators across some nineteen automakers — the largest in automotive history — and Takata went bankrupt; the toll keeps rising as unrepaired vehicles stay on the road and their inflators keep aging.[1]

THE INVESTIGATION

Takata's own testing had shown the ruptures, but internal documents revealed engineers raising the alarm internally while the company reported the failures to automakers and regulators as isolated anomalies rather than a systemic propellant problem — and, in places, manipulated the test data outright.[2] In 2017 Takata pleaded guilty to wire fraud and paid roughly a billion dollars in fine, restitution, and a victims' fund. The legal finding was pointed: not that a part had failed, but that the company had spent years misrepresenting what its own engineers knew.[3]

THE CAPABILITY GAP

Two capabilities were designed out, and the second matters as much as the first. The product capability was the desiccant — a stabilizing component competitors used, omitted to save cost, in plain view of anyone comparing designs side by side. The system capability was independent verification: the recall regime treated the manufacturer's representations about its own safety data as authoritative, with no independent pipeline to test inflator behavior across the fleet as it aged in the field — which let an obvious failure mode hide for years inside a process that kept receiving the evidence and reporting it back as noise.[4]

AFTERMATH & REFORM

The recall is still being worked off, vehicle by vehicle, long after the company that built the inflators ceased to exist — concrete proof that a designed-out capability can outlive the firm that removed it and become someone else's burden. The episode pushed regulators toward more aggressive, coordinated recall management that does not leave pace to each manufacturer. Its central lesson is the pairing: Takata is the modern Pinto in its product failure, and more in its system failure — the evidence pipeline a regulator relies on is itself a safety capability, and omitting it is as consequential as omitting the desiccant from the inflator.[5]

100M+

INFLATORS RECALLED · 19 AUTOMAKERS

the desiccant competitors used was the designed-out capability

TAKATA — THE LARGEST AUTOMOTIVE RECALL ON RECORD

REFERENCES

1. U.S. National Highway Traffic Safety Administration, Takata airbag inflator recall coordination materials — the ammonium-nitrate-without-desiccant design and the propellant-degradation rupture mechanism.
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3. U.S. Department of Justice, settlement and guilty plea with Takata Corporation (2017) — wire fraud, $1B in fine and restitution, and Takata's subsequent bankruptcy.
4. U.S. DOJ (2017) and Takata internal documents released in litigation — the sustained misrepresentation of inflator test data to automakers and regulators.
5. NHTSA Takata recall status reporting — the years-long completion of replacements and the continuing risk in unrepaired vehicles.
6. NHTSA Coordinated Remedy Program for the Takata recalls — the regulator's move to actively prioritize and manage replacement across nineteen automakers rather than leave pace to each manufacturer.

THE LEARNING ENGINEERING LENS

> *Takata engaged in a sustained pattern of misrepresenting inflator safety data to its automaker customers and to regulators.*
>
> PARAPHRASING THE U.S. DEPARTMENT OF JUSTICE SETTLEMENT WITH TAKATA, 2017

LE INSIGHT

Takata is the modern Pinto: an engineering choice to omit a stabilizing component, internal data showing the consequence, and a regulatory architecture that treated the manufacturer's representation as authoritative rather than as one input. The capability gap was at the regulator's evidence pipeline as much as at the manufacturer's bench.

LENS APPROACH

The load-bearing lesson is post-deployment surveillance. Field rupture reports arrived over years but were never aggregated and interpreted as a single class-level safety signal, so an obvious failure mode stayed hidden inside a process that kept receiving the evidence and reporting it back as noise. The ammonium-nitrate-without-desiccant chemistry is the substrate of the hazard, but the capability that should have caught it is the surveillance-aggregation function that turns scattered field reports into one population-level signal with authority to act. LENS uses Takata in LEN 5 to build that aggregate-the-field-signal capability and in LEN 7 as an industrial-governance failure spanning manufacturer, regulator, and customer auditors.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Qualify a safety-critical material against the full service life and climate range it will see, not just its as-delivered state, so slow degradation cannot turn a safe part lethal years later.
- When choosing a cheaper, less stable material, require the stabilizing measure competitors use — here a desiccant — rather than omitting it to save cost.
- Design inflator testing to detect the rupture failure mode as a class across temperature and humidity cycling, so it cannot be dismissed as an isolated anomaly.

IN OPERATION / AFTER

- Stand up an independent verification pipeline that tests fielded inflators across the aging fleet, so a regulator does not depend on the manufacturer's own representation of its safety data.
- Aggregate field rupture reports as a population signal rather than closing them one by one, with authority to trigger recall when a pattern appears.
- Manage the recall as a coordinated program that prioritizes replacement across automakers, since a designed-out hazard outlives the firm and keeps aging in unrepaired cars.

REFLECTION QUESTIONS

1. Where in your domain does a regulator receive manufacturer test data without an independent verification pipeline? What is the load-bearing trust assumption?

2. Design the verification regime that should have surrounded ammonium-nitrate inflator testing. Who funds it, who runs it, and what does it produce?

3. Takata's propellant degraded slowly with heat and humidity, so a part safe at delivery became lethal years later. Identify a component in your domain whose qualification testing does not cover its full service life, and specify the aging test that would close the gap.

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CASE 85 ENERGY 2010

Deepwater Horizon

T Training Gap N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT 11 killed; largest marine oil spill in U.S. history; $65B+ in damages

IN BRIEF

On 20 April 2010 the Macondo well blew out beneath the Deepwater Horizon rig in the Gulf of Mexico, killing 11 workers and releasing roughly 4.9 million barrels of oil — the largest marine spill in U.S. history, and more than $65 billion in eventual costs to BP. The crew had misread the negative-pressure test, the primary check of well integrity, and kept working a well that was no longer sealed. The National Commission found human error, not equipment, the root cause, and judged the failures so systematic they cast doubt on the safety culture of the whole industry. Every defense — procedure, training, supervision, contractor coordination, equipment — had drifted independently until none caught the others. It is the book's canonical multi-layer normalization failure. Four major investigations — the National Commission, the joint Coast Guard / BOEMRE inquiry, the Chemical Safety Board's process-safety review, and the Deepwater Horizon Study Group at Berkeley — each surfaced a distinct facet of the same drift; the disagreements among them are themselves load-bearing for the case.

BACKGROUND

The Deepwater Horizon was drilling BP's Macondo exploratory well in the Gulf of Mexico when, on 20 April 2010, the well blew out. Eleven workers were killed, the rig burned and sank, and roughly 4.9 million barrels of oil flowed into the Gulf over 87 days — the largest marine oil spill in U.S. history, eventually costing BP more than $65 billion.[0] The well sat at the end of a long chain of contractors and schedules, each operating to its own tolerance, so that the 87 days of uncontrolled flow stand as a measure not of one mistake but of how far the accumulated drift had to be unwound before the Gulf was sealed again.

WHAT HAPPENED

The proximate trigger was a misread safety check. The crew ran a negative-pressure test — the primary test of whether the well was sealed — got anomalous readings, accepted an incorrect explanation for them, and proceeded to displace the heavy drilling mud. The well was not sealed; hydrocarbons surged up the riser to the rig and ignited.[1] The anomalous pressure was the well speaking plainly that it remained live, but the explanation the crew accepted let the displacement proceed on schedule, and once the heavy mud was gone nothing stood between the reservoir and the rig but a barrier that had already failed.

THE INVESTIGATION

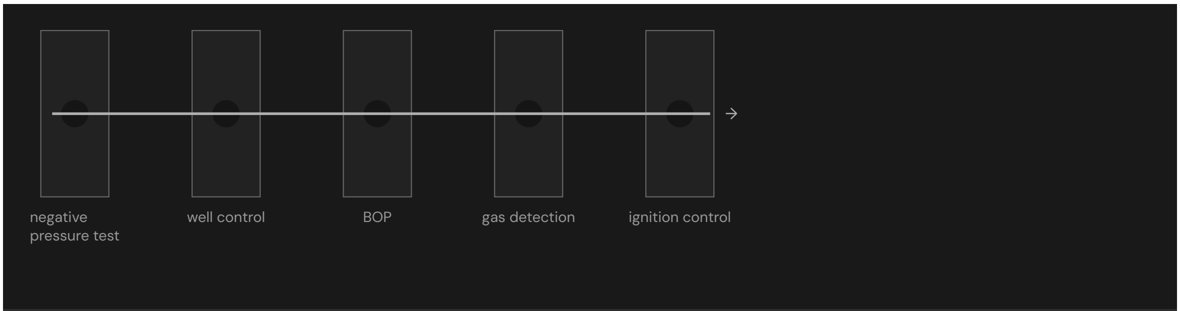
Four major investigations reached the wreck and pulled on different threads. The National Commission concluded that human error, not mechanical failure, was the root cause, and that the mistakes revealed "such systematic failures in risk management that they place in doubt the safety culture of the entire industry."[2] The joint Coast Guard / BOEMRE inquiry traced the blowout-preventer failure to a buckled drill pipe and a maintenance backlog that the leasing-and-safety regulator had not caught. The Chemical Safety Board's separately published process-safety review made the load-bearing distinction explicit: BP's lost-time-injury rate had been industry-leading on the rig, yet personal-safety metrics had no purchase on the catastrophic process-safety risks that produced the blowout — the same inversion the CSB had already documented at Texas City. Government and academic reviews found training that had not covered the well-control situation the crew faced, an unclear chain of command, and a cascade of failed defenses — "a complex and interlinked series of mechanical failures, human judgments, engineering design, operational implementation and team interfaces."[3] That the Commission reached past the rig to the whole industry's safety culture marked the failure as structural rather than local: the same drift, it judged, was latent wherever the same pressures and the same defenses operated unexamined.

THE CAPABILITY GAP

Deepwater Horizon is the canonical multi-layer failure: procedure, training, supervision, contractor coordination, and equipment had each drifted independently until none caught the others. The negative-pressure test was the trigger, but the system had been carrying accumulated procedural debt for years, and the capability to recognize an unsafe well-state was simply not present at the moment it was needed.[4] Because each defense had degraded inside its own tolerance, no single review of any one layer would have flagged a crisis; the danger lived in the correlation between layers, which is precisely the property no individual procedure was designed to watch. The cement job, the blowout preventer's design and maintenance, the well-control culture before the incident, and the regulator's split mandate were each, on their own, a partial story; the failure was the alignment among them, and no defense layer had been instrumented to read that alignment in time.

AFTERMATH & REFORM

The blowout drove a restructuring of offshore regulation — the Minerals Management Service was broken up and replaced by the Bureau of Safety and Environmental Enforcement (BSEE) and the Bureau of Ocean Energy Management (BOEM), separating the leasing function from the safety function so revenue pressure could no longer erode well-integrity enforcement — and drilling-safety and well-integrity rules were tightened, while BP paid tens of billions in penalties and settlements. The deeper lesson is the normalization one: no single layer failed catastrophically on its own; each had quietly drifted within tolerance until the day the tolerances lined up.[5] Splitting the regulator conceded that an agency both leasing acreage and policing safety carried a built-in conflict, and tightening the well-integrity rules conceded that the negative-pressure test had been a real barrier all along — one the system had been quietly permitted to misread. The hedges survive into the case: the process-safety / personal-safety distinction the CSB names is load-bearing and easy to lose; the OSHA-vs-EPA enforcement gap on offshore facilities was structural, not incidental; and the four investigations did not converge on a single cause precisely because the catastrophe had several, all of which had been quietly tolerated.



CASCADING DEFENSES — EACH FAILED IN TURN

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THE LEARNING ENGINEERING LENS

> *The immediate causes of the Macondo well blowout can be traced to a series of identifiable mistakes… that reveal such systematic failures in risk management that they place in doubt the safety culture of the entire industry.*
>
> NATIONAL COMMISSION, DEEP WATER (REPORT TO THE PRESIDENT), 2011

LE INSIGHT

Deepwater Horizon is the canonical multi-layer failure: training, procedure, supervision, contractor-coordination, and equipment all drifted independently until none caught the others. The negative-pressure test was the proximate trigger, but the system as a whole had been operating with accumulated procedural debt for years. The capability to recognize an unsafe well-state simply was not present at the moment it was needed. The CSB's process-safety / personal-safety distinction is load-bearing: BP's industry-leading lost-time-injury record was a measurement of the wrong construct, and reading it as safety was itself a normalization.

LENS APPROACH

Deepwater Horizon anchors the cue-and-alert-design competency (induced 3.1; LENS D4/PT5): the negative-pressure test was the cue, and the cue's ambiguity at the moment of decision was the capability gap. LENS uses it in Domain 4 (Test and Evaluation) for the cue-design failure and the wrong-construct measurement; in Domain 1 (Systems Analysis) for multi-layer drift across procedure, training, supervision, contractor coordination, and equipment; and in Domain 5 (Navigating Sociotechnical Constraints) for the OSHA-vs-EPA enforcement gap and the leasing-vs-safety regulator conflict the BSEE/BOEM split conceded. Pair with Texas City (Case 137) on the process-safety-vs-personal-safety inversion, and with Challenger / Columbia (Case 140) on the multi-layer-drift form.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make the negative-pressure test a hard, instrumented gate: define the pass criteria so an anomalous reading stops work rather than inviting a rationalizing explanation.
- Specify a single, unambiguous well-control chain of command across operator and contractors before drilling begins, so no decision falls between organizations.
- Build well-control training around the ambiguous small-anomaly case the crew actually faced, not only the textbook blowout.

IN OPERATION / AFTER

- Audit each defense layer for drift independently — and audit the correlation between them, since the danger lived where tolerances aligned.
- Track accumulated procedural debt as a standing metric so silent degradation surfaces before tolerances coincide.
- Structurally separate the regulator's leasing and safety roles so revenue pressure cannot erode well-integrity enforcement.

REFLECTION QUESTIONS

1. Identify a multi-defense process in your domain. For each layer, what is the procedural debt that has accumulated since it was designed?
2. The negative-pressure test was the proximate trigger. Design a capability check that would have caught the misinterpretation in real time — and specify what makes the cue unambiguous enough that an anomalous reading stops work rather than inviting a rationalizing explanation.
3. The Commission judged the drift industry-wide, not rig-specific. What measure in your domain would reveal whether a single failure is a local lapse or a sample from a systemic distribution?
4. BP's lost-time-injury rate had been industry-leading on the rig. Name the measurement in your domain that is most at risk of being the wrong construct — a personal-safety analog where the catastrophic process-safety risk lives elsewhere — and design the second instrument that would surface the harm the first one cannot see.

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CASE 86 HEALTHCARE 2005 – 2009

Mid Staffordshire NHS Foundation Trust

G Governance & Trust N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT Excess deaths at Stafford Hospital documented across years; the Francis Inquiry produced 290 recommendations and restructured UK patient-safety governance

IN BRIEF

Between roughly 2005 and 2009, Stafford Hospital, run by the Mid Staffordshire NHS Foundation Trust, subjected patients to appalling neglect — left without food, water, or basic care — while mortality ran above expected. The trust had been chasing financial targets that depended on staffing cuts. Robert Francis QC's public inquiry produced a 2,000-page report and 290 recommendations, identifying the structural problem as the gap between the trust's reported performance and patients' actual experience: every governance layer above the ward received reports that targets were being met, and none checked those reports against what was happening to patients. Mid Staffs is the dataset's strongest case for the harm done when measurement and reality diverge over years.

BACKGROUND

The Mid Staffordshire NHS Foundation Trust ran Stafford Hospital in England. Pursuing Foundation Trust status and the financial targets that came with it, the board cut staffing — and the cuts fell on the wards. The institution's reported performance, the numbers that travelled upward, stayed on target.[0] The targets the board chased were financial and procedural, so cutting ward staff improved the very figures the trust was measured on even as it removed the people on whom patient care directly depended.

WHAT HAPPENED

On the wards the reality was appalling. Patients were left in their own excrement, denied food and water, given the wrong medication or none, for years; mortality ran substantially above expected. The harm was not a single incident but a sustained condition — visible to anyone on the ward and invisible in the reports that left it.[1] Because the suffering was a continuous state rather than a nameable event, it never generated the kind of discrete incident a reporting system is built to catch, and so it accumulated for years beneath numbers that stayed reassuringly on target.

THE INVESTIGATION

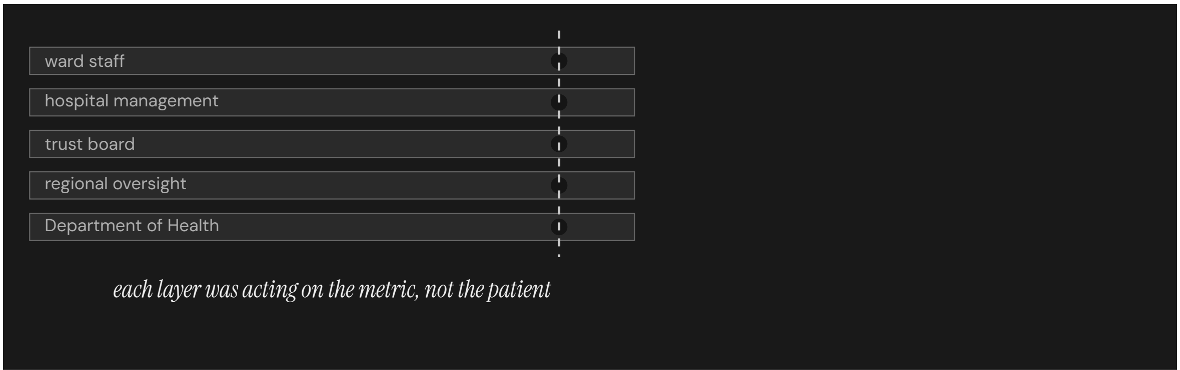
The Healthcare Commission investigated; the layers above did not. Robert Francis QC's public inquiry ran to some 2,000 pages and 290 recommendations.[2] Its structural finding was the divergence between reported performance and patient experience: every governance layer above the ward had received reports that the hospital was meeting its targets, and none had checked them against what was happening to patients. "The system as a whole failed in its most essential duty — to protect patients from unacceptable risks of harm."[3] The phrase "the system as a whole" located the failure deliberately above any single ward or manager: no one layer was solely at fault, because each had trusted the layer below to be reporting reality rather than targets.

THE CAPABILITY GAP

Mid Staffs is the British analog of the VA wait-time scandal (Case 101): measurement and reality diverged over years while every layer of governance acted on the measurement. The capability that was missing was not clinical skill on the ward but the institutional habit of testing whether the numbers corresponded to the patients — a check no layer above the ward performed.[4] Each layer reasonably assumed the check belonged to someone else, so the verification that the report matched the patient fell into the gap between layers — exactly the place a reporting chain built only to pass numbers upward is structurally unequipped to look.

AFTERMATH & REFORM

The Francis Inquiry restructured how the NHS treats patient safety, and the Berwick review that followed pressed for a culture of learning over targets. The lesson is the measurement one in its starkest form: a reporting chain can run clean from ward to Department of Health while, underneath it, the thing being reported on quietly fails.[5] Berwick's framing — learning over targets — named the deeper correction: as long as the target is the thing the institution rewards, the report will describe the target, and only a culture that prizes finding the gap will keep checking the report against the patient.



MID STAFFS — MEASUREMENT AND REALITY DIVERGED ACROSS YEARS

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THE LEARNING ENGINEERING LENS

> *The system as a whole failed in its most essential duty — to protect patients from unacceptable risks of harm.*
>
> ROBERT FRANCIS QC, REPORT OF THE MID STAFFORDSHIRE NHS FOUNDATION TRUST PUBLIC INQUIRY, 2013

LE INSIGHT

Mid Staffordshire is the British analog of the VA wait-time scandal (Case 101). Measurement and reality diverged over years; every layer of governance above the operating environment was acting on the measurement; patients paid the cost of the divergence. The Francis Inquiry recommendations changed how the NHS thinks about patient safety as an institutional capability.

LENS APPROACH

LENS uses Mid Staffs in LEN 4 for the divergence-of-measurement problem and in LEN 7 for the governance failure across multiple layers. Studio projects examine the Francis recommendations as a template for institutional reform deliverables.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design targets so that gaming them (e.g., cutting ward staff) cannot improve the metric while degrading the outcome it stands for.
- Build a direct patient-experience signal — independent of the financial and procedural targets — into the reporting chain from the start.
- Assign explicit ownership of the report-versus-reality check at each layer, so verification cannot fall into the gap between them.

IN OPERATION / AFTER

- Audit sustained conditions, not just discrete incidents, since continuous harm never trips an event-based reporting system.
- Cross-check upward reports against ground truth at the ward periodically, treating a clean report as a hypothesis to be tested.
- Reward a culture of learning over target attainment, so the institution keeps looking for the gap rather than describing the target.

REFLECTION QUESTIONS

1. Identify a multi-layer reporting chain in your domain. What would it take for the top layer to know whether the reports correspond to reality?
2. The Francis Inquiry produced 290 recommendations. Pick five that you think were most load-bearing and explain why.
3. The verification fell into the gap between layers because each assumed it belonged to someone else. In your domain, who explicitly owns the check that a report matches reality — and how would you know if no one does?

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CASE 87 ENERGY INDUSTRIAL 2010

Upper Big Branch Mine Explosion

N Normalization of Deviance G Governance & Trust K Knowledge & Institutional Memory

IMPACT 29 killed in West Virginia coal mine; MSHA found systematic falsification of safety records; first U.S. mining-industry CEO convicted of a federal mine-safety charge (misdemeanor)

IN BRIEF

On 5 April 2010 coal dust and methane ignited at Massey Energy's Upper Big Branch mine in West Virginia, killing 29 miners — the worst U.S. mine disaster in forty years. Investigators found Massey had kept two sets of records: an internal log of actual conditions and a separate, clean log for federal inspectors. Foremen were directed to suppress methane readings, disable monitors, and falsify pre-shift examinations — not as an aberration but as a stable routine across years and management layers. CEO Don Blankenship was convicted of a misdemeanor conspiracy to violate mine-safety standards — the first U.S. mining CEO criminally convicted on such a charge. Upper Big Branch is the dataset's clearest case of a measurement system engineered deliberately to defeat the regulator.

BACKGROUND

Massey Energy's Upper Big Branch mine in West Virginia operated under federal safety rules enforced through inspections and the records the mine kept. Massey kept two: an internal log of actual conditions, and a separate, clean log maintained for the inspectors.[0] The enforcement regime depended entirely on the records reflecting reality, so a second, sanitized set of books did not merely break a rule — it disabled the very mechanism by which the regulator was supposed to see the mine at all.

WHAT HAPPENED

On 5 April 2010 accumulated coal dust and methane ignited and tore through the mine, killing twenty-nine miners — the worst U.S. coal disaster in forty years. The conditions that fed the blast — high methane, inadequate ventilation, dust not rendered inert — were the very ones the clean, inspector-facing records had been built to hide.[1] The records had done their intended work right up to the blast: they kept the conditions that killed twenty-nine men out of the regulator's view precisely while those conditions were building toward ignition.

THE INVESTIGATION

MSHA's investigation and a parallel U.S. Department of Justice probe found foremen instructed to suppress methane readings, disable monitors, and falsify pre-shift examinations.[2] Massey CEO Don Blankenship was eventually convicted of a misdemeanor count of conspiring to willfully violate mine-safety standards — the first U.S. mining-industry CEO criminally convicted on a mine-safety charge — though acquitted on the felony counts.[3]

That the instructions ran down to foremen and the conviction reached up to the CEO marks the practice as vertical, not local: the deception was operated at the working level and conceived above it, spanning the management layers in between.

THE CAPABILITY GAP

The dual-records architecture was not the work of a few bad foremen; it was a stable institutional practice sustained across years and multiple management layers. The capability gap was not in the miners but in the executive ranks that designed and operated a measurement system specifically to defeat the regulator — which makes it the book's clearest case of measurement engineered as deception.[4] Where other cases show measurement drifting or pointed at the wrong dimension, here it was deliberately constructed to mislead, so no improvement in the regulator's reading of the official records could ever have helped — the records were the lie.

AFTERMATH & REFORM

The case anchors the argument that decision-grade evidence needs structural protection from the institution that produces it, when that institution has a stake in what the evidence says. Blankenship's conviction set a marker for corporate-officer accountability — and left open the regulator-side question: what architecture would have detected two sets of books before twenty-nine people died?[5] Holding a CEO criminally accountable raised the cost of designing such a system, but accountability after the fact is not detection; the unanswered question is what independent signal could have exposed the divergence between the two logs while the mine was still running.

UPPER BIG BRANCH — MEASUREMENT ENGINEERED AGAINST THE REGULATOR

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> *Massey kept two sets of books — one for inspectors, one for itself.*
>
> PARAPHRASING FEDERAL INVESTIGATORS, MSHA UPPER BIG BRANCH REVIEWS (2011 – 2012)

LE INSIGHT

Upper Big Branch is the case for deliberately engineered measurement deception. The dual-records practice was not error. It was capability design — by management, against the regulator. The case anchors the LENS argument that decision-grade evidence requires structural protection against the institution that produces it having a stake in what it reports.

LENS APPROACH

LENS uses Upper Big Branch in LEN 4 as the most explicit case for measurement-system protection and in LEN 7 as a corporate-criminal accountability case. Studio projects examine what regulator-side architecture would have detected the dual-records system earlier.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Source safety-critical data from monitors the operator cannot disable or edit, so a clean official log cannot be constructed by hand.
- Design measurement so the producing institution has no unilateral control over the record the regulator relies on.
- Build cross-checks that compare independent streams, making a single sanitized set of books detectably inconsistent.

IN OPERATION / AFTER

- Establish whistleblower and anomaly-detection channels that can surface a divergence between actual and reported conditions while operations continue.
- Audit for the structural conditions that make dual records possible — operator-controlled monitors, no independent verification — not just for violations.
- Pair corporate-officer accountability with detection architecture, since raising the cost of deception does not by itself reveal it in time.

REFLECTION QUESTIONS

1. Where in your domain could two sets of records plausibly be kept? What architectural change would make that impossible?
2. What does it mean for a measurement system to be “structurally protected” from the institution that produces it?
3. Accountability after the fact is not detection. What independent signal could have exposed the divergence between the two logs while the mine was still running?

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CASE 88 AVIATION DEFENSE HUMAN FACTORS 2008 – 2012

F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap

H Human-System Interface K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Between 2008 and 2012 USAF F-22 Raptor pilots reported a cluster of physiological events consistent with hypoxia; one fatal accident (Capt. Jeffrey Haney, Nov 2010) was attributed in part to operator response to a bleed-air shutoff; the fleet was stood down in 2011, and the USAF Scientific Advisory Board found no single root cause but a contributing oxygen-delivery issue rooted in the On-Board Oxygen Generation System (OBOGS) and associated equipment

IN BRIEF

Between 2008 and 2012 the USAF accumulated a cluster of F-22 Raptor "physiological events" — pilot reports consistent with hypoxia, often with persistent post-flight symptoms (chronic cough, disorientation). One fatal class-A accident in November 2010 (Capt. Jeffrey Haney) was attributed in part to operator response after a bleed-air shutoff. In May 2011 the USAF stood the fleet down for several months. The USAF Scientific Advisory Board's investigation, with NASA contribution, found **no single root cause** — the load-bearing finding of the case. The set of contributing factors included the On-Board Oxygen Generation System (OBOGS) and its associated equipment (regulators, hoses, upper-pressure-garment vest), aircrew flight equipment fit, and the absence of in-cockpit physiological monitoring that would have permitted earlier detection. The case is the canonical recent instance of a sustainment-era failure whose diagnosis was protracted because the system as fielded had no instrumentation of the variable that mattered — pilot blood oxygenation — and the failure mode therefore had to be inferred from pilot self-report and incident clustering.

BACKGROUND

The F-22 Raptor entered operational service in 2005. Like most modern fighters it uses an On-Board Oxygen Generation System (OBOGS) rather than carrying liquid oxygen — a molecular sieve separates oxygen from engine bleed air to supply the pilot's breathing mask. Beginning in 2008, USAF Raptor pilots reported a cluster of "physiological events": symptoms consistent with hypoxia during flight, often with persistent post-flight effects — chronic cough, disorientation, exercise intolerance. The cluster did not conform to a single platform or sortie profile, and the reports accumulated faster than the diagnostic process could resolve them.[0]

WHAT HAPPENED

The forcing event was the loss of Capt. Jeffrey Haney and his aircraft in November 2010 over Alaska. The Air Force Accident Investigation Board found that a bleed-air leak had triggered an automatic bleed-air shutoff, which in turn cut OBOGS supply, and that the accident sequence involved pilot response under conditions consistent with hypoxia. The board's findings were contested by the pilot's family and reviewers; what the accident crystallized was that the platform's response to an OBOGS cutoff placed an unreasonable burden on pilot recognition and reaction in a window the platform did not instrument. In May 2011 the USAF stood the entire F-22 fleet down for several months pending investigation.[1]

THE INVESTIGATION

The USAF Scientific Advisory Board investigation, with participation from NASA's review of the hypoxia incidents, reported the load-bearing finding the case turns on: there was **no single root cause**. The contributing factors included the OBOGS system and its associated equipment (regulators, hose configurations, the upper-pressure garment vest that under some conditions inhibited normal breathing), aircrew flight-equipment fit issues, and the absence of in-cockpit physiological monitoring. The investigation made several corrective recommendations — including modifications to the upper-pressure garment, changes to OBOGS components, and the addition of physiological monitoring during flight — and the symptom reporting decreased substantially in the period after implementation.[2]

THE CAPABILITY GAP

What makes this a sustainment-era case rather than a design-era one is the instrumentation gap. The F-22 as fielded had no in-cockpit measurement of the pilot's oxygenation or of the actual gas composition reaching the mask. The variable that mattered most for the failure mode — was the pilot getting enough oxygen? — had to be inferred from pilot self-report after the fact. The diagnostic process took years partly because the population of events had to be characterized statistically from a noisy reporting channel, and partly because the multi-cause structure of the failure made any single intervention only partially testable against the population.[3]

AFTERMATH & REFORM

The hedge survives into the case verbatim. The USAF SAB explicitly declined to identify a single root cause, and the corrective actions were a bundle: garment modification, OBOGS component changes, training emphasis on hypoxia recognition, and the long-overdue addition of physiological monitoring. Symptom reports decreased after the bundle was implemented, but the attribution of the decline to any single component within the bundle is not possible from the available evidence. The case teaches the sustainment-instrumentation-gap form: a high-performance platform fielded without measurement of the operator-physiology variable that determines whether the mission can be completed, and a diagnostic process forced to work from self-report and clustering because the platform did not surface the data.

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THE LEARNING ENGINEERING LENS

> *The variable that mattered most — was the pilot getting enough oxygen? — was the variable the platform did not measure.*
>
> EDITORS' SYNTHESIS OF THE USAF SAB AND NASA REVIEWS.

LE INSIGHT

F-22 OBOGS is the canonical recent sustainment-instrumentation- gap case. A high-performance platform was fielded without in-cockpit measurement of pilot oxygenation, the diagnostic process took years against a noisy self-report channel, and the corrective action was an irreducibly multi-cause bundle. The “no single root cause” finding is the case.

LENS APPROACH

F-22 OBOGS is the aerospace instrumentation-gap case (induced 2.4; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for the no-single-root-cause finding and the bundle-of-corrections evaluation problem, and in Domain 3 (Human-System Collaboration) for the operator burden when an automatic action cuts a life-critical supply. Pair with Case 65 (EVA-23) as the human- spaceflight instrumentation-gap companion and with Case 116 (anesthesia monitoring) at the cue/alert-as- capability layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Instrument the operator-physiology variable that determines mission completion before fielding a high-performance platform. The F-22 OBOGS case is the worked example of the cost of fielding without it: years of self-report-based diagnosis and one fatality.
- When a failure-event cluster does not conform to a single platform or sortie profile, treat the absence of a single root cause as a finding, not as a failure of the investigation. The bundle of contributing factors is the diagnosis.
- Design the platform’s automatic responses (e.g., bleed-air shutoff) with the pilot’s recognition window in mind; an automatic action that cuts a life-critical supply has to be paired with cueing the pilot can act on in time.

IN OPERATION / AFTER

- Implement the corrective actions as a bundle — garment modification, OBOGS components, training, physiological monitoring — and report the population-level outcome decline as bundle evidence, not as evidence about any single component.
- Carry the “no single root cause” hedge into program communication; the multi-cause structure is the case’s teaching point and the basis for the sustainment-instrumentation argument.
- Treat the F-22 OBOGS experience as a fielded-platform sustainment lesson for subsequent platforms (e.g., F-35) — the analogous instrumentation question should be answered before the cluster appears, not after.

REFLECTION QUESTIONS

1. Identify a fielded high-performance platform in your domain where the variable that determines mission completion is not directly measured. What is the analog of in-cockpit physiological monitoring, and what would it cost to add at sustainment vs. having designed it in?
2. Specify what your program would treat as “no single root cause” — the irreducibly multi-factor finding — and how you would report bundle-of-corrections evidence rather than overstating attribution to any single change.
3. The F-22 cluster appeared in 2008 and stand-down came in 2011. What signal-detection threshold would your program use on a noisy self-report channel to act earlier on a population-level pattern?

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CASE 89 EDUCATION AT SCALE 2009–2018

Tennessee Voluntary Pre-K Study

G Governance & Trust D Designed Out

IMPACT Vanderbilt longitudinal RCT of a state-funded universal pre-K program: early gains faded by third grade; sixth-grade outcomes were worse than the control group on several measures

IN BRIEF

Tennessee's Voluntary Pre-K program enrolled some 18,000 four-year-olds a year, and Vanderbilt researchers studied it with a rare large-scale randomized controlled trial — children admitted by lottery against those who applied but didn't enroll. Through kindergarten the pre-K children showed the expected gains in literacy and vocabulary. By third grade the gains had faded, and by sixth grade the pre-K group did somewhat worse than controls on several academic and behavior measures. The unwelcome result was contested, its methods attacked, and the field largely declined to absorb it. The case is in the book not because pre-K is bad — other programs show durable effects — but because the measurement was unusually rigorous and the discipline's capacity to act on an inconvenient finding was tested and largely failed.

BACKGROUND

The Tennessee Voluntary Pre-Kindergarten Program served roughly 18,000 four-year-olds a year. Demand exceeded supply, and that scarcity created an ethical lottery — a fair way to ration scarce seats that doubled as a clean randomizer. It let Vanderbilt's Peabody Research Institute run something rare in education: a randomized controlled trial, following children admitted by lottery against children who applied but did not enroll, the kind of design rarely available in a live policy setting.[0]

WHAT HAPPENED

Through kindergarten the pre-K children showed the expected gains — stronger letter knowledge, vocabulary, early literacy — exactly the early result the program had been funded to produce. By third grade the gains had faded. By sixth grade, the researchers reported, the pre-K children were doing somewhat **worse** than the control group on several state academic measures and on teacher-reported behavior — a reversal that turned an expected success story into an uncomfortable one the longitudinal design had been built to detect.[1]

THE INVESTIGATION

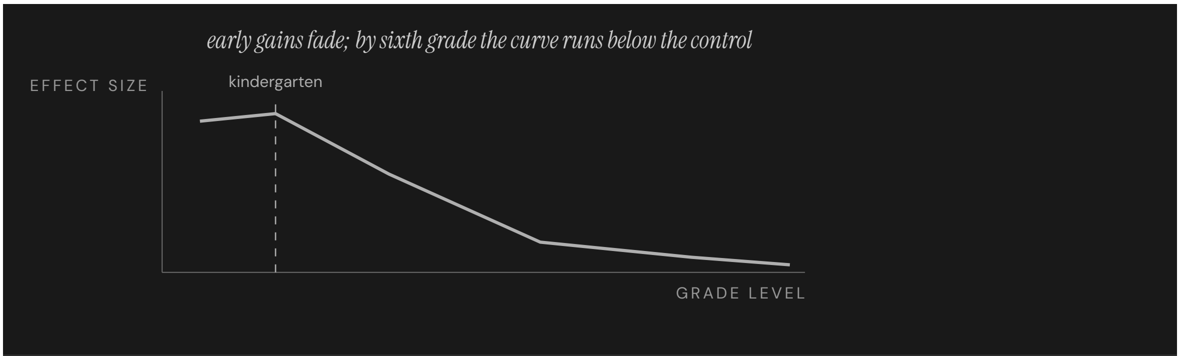
The result contradicted policy consensus and provoked an unusual response: the study was attacked, its methods contested, and the field largely declined to internalize the findings, defending the policy rather than interrogating it. Other rigorous programs — Perry Preschool, Abecedarian — remain durable, so the lesson is not that pre-K fails; it is that a discipline met an unwelcome, well-measured answer and mostly looked away, treating the inconvenient evidence as an attack to repel rather than a finding to absorb.[2]

THE CAPABILITY GAP

Tennessee Pre-K is the cleanest case in the dataset for what happens when a field has not engineered its own capacity to update on contrary evidence. The measurement instrument worked — a real RCT in a real policy setting, the rare study whose design could not easily be waved off. The institutional architecture for acting on what it found did not, so a strong instrument fed a discipline with no pathway to receive its answer. A measurement that returns an inconvenient result is only as valuable as the discipline's willingness to absorb it, and here the willingness was the part that was missing.[3]

AFTERMATH & REFORM

The debate continued for years through follow-up studies and counter-analyses, and the episode became a touchstone in the methodology of early-childhood research — argued over more than acted upon.[4] Its place in this book is as a governance-of-evidence case: the capability that needed engineering was not a better study but an implementation-science pathway that could route an unwelcome finding into program redesign rather than rejection, so the cost of the study bought a course correction instead of a controversy.



TENNESSEE PRE-K — MEASUREMENT ARCHITECTURE WORKED; THE FIELD'S RESPONSE DID NOT

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> *By sixth grade, children who had attended TN-VPK were doing somewhat worse on academic achievement and discipline measures than children in the control group.*
>
> DURKIN, LIPSEY, FARRAN & WIESEN (2022), VANDERBILT PEABODY RESEARCH INSTITUTE

LE INSIGHT

Tennessee Pre-K is the cleanest case in the dataset for what happens when a discipline has not engineered its own capacity to update on contrary evidence. The measurement instrument worked. The institutional architecture for acting on what it found did not. Compare to Case 59 (Makary methodology debate) and Case 101 (VA Wait-Time): in each, a measurement that returned an unwelcome answer was contested rather than absorbed.

LENS APPROACH

LENS uses Tennessee Pre-K in LEN 4 as a measurement-architecture case (longitudinal RCT in a real policy context), in LEN 7 to discuss the institutional politics of unwelcome findings, and in LEN 10 as a studio prompt for designing the implementation-science pathway that would absorb such findings into program redesign rather than rejection.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Commit in advance to a decision rule for how an unwelcome but well-measured result will change the program, so the response is designed before the finding arrives.
- Build the longitudinal measurement to detect fade-out and reversal, not just the early gains a program is funded to show.
- Establish an implementation-science pathway that can route a contrary finding into redesign, giving the evidence somewhere to go.

IN OPERATION / AFTER

- Audit how the discipline actually receives inconvenient evidence, treating reflexive method-attacks as a symptom of a missing absorption pathway.
- Sustain follow-up measurement through later grades so a faded or reversed effect cannot be obscured by an early success.
- Protect the independence of the measurement function so a strong instrument is not dismantled for returning an answer the field did not want.

REFLECTION QUESTIONS

1. What measurement instrument in your domain has returned an unwelcome answer, and how did the discipline respond?
2. Design the implementation-science pathway that would absorb a Tennessee-Pre-K-style finding into program redesign rather than rejection.
3. The Tennessee RCT was strong enough that its methods were attacked rather than its conclusion accepted. What decides, in your field, whether a rigorous but inconvenient result is absorbed or contested — and who holds the authority to act on it?

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CASE 90 EDUCATION AT SCALE ONGOING

Algorithmic Bias in Educational Predictive Analytics

G Governance & Trust H Human-System Interface D Designed Out

IMPACT Predictive "at-risk" models show racial calibration bias that can misdirect support away from Black and Latinx students; a large majority of U.S. public colleges now use predictive analytics

IN BRIEF

Most U.S. public colleges now use predictive analytics to flag “at-risk” students for early support. Research finds these models carry racial calibration bias: they miscalibrate by race in ways that can misclassify Black and Latinx students — and so misdirect the very support the prediction is meant to trigger. The magnitude depends heavily on how “at-risk” is defined, making this partly a construct-definition problem: the choice of what to predict is itself a capability decision with equity consequences. The models inherit historical patterns of discrimination from their training data, and a “flag” can confirm a biased instructor’s low expectations rather than prompt help. It is the educational analog of the UK A-level case — an algorithm built to do good that can allocate help away from the students who need it most.

BACKGROUND

Colleges increasingly use predictive analytics to identify students at risk of failing or dropping out, so advisors can intervene early — a genuinely well-meant aim, to reach struggling students before they vanish. A large majority of U.S. public colleges now use some form of these models, which makes any bias they carry a quiet, system-wide feature of how support is allocated rather than an isolated experiment.[0]

WHAT HAPPENED

The intervention is well-intentioned, but research finds the models carry racial calibration bias: they miscalibrate by race in ways that can misclassify Black and Latinx students relative to their actual outcomes — and so misdirect the support the flag is meant to trigger, turning a tool built to help into one that can steer help away. Crucially, the magnitude of the bias depends on how “at-risk” is defined, which makes it partly a construct-definition problem rather than a coding bug — a flaw in what the model was asked to predict, not in how it computes the prediction.[1]

THE INVESTIGATION

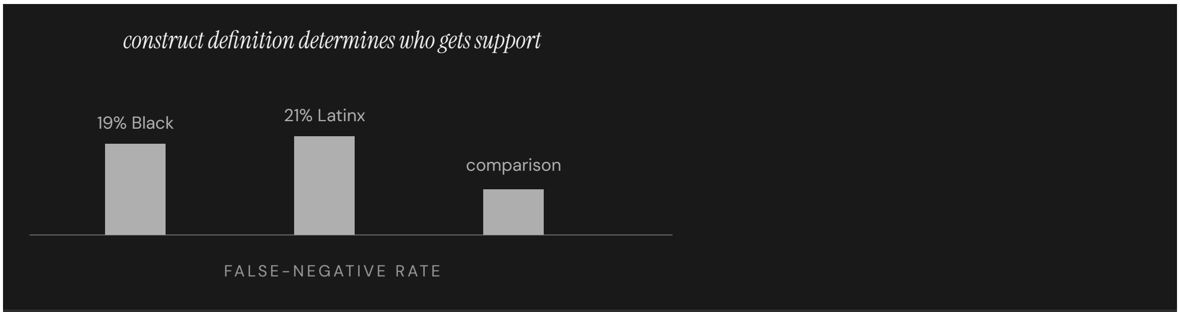
Researchers studying equity in completion-prediction models have documented these calibration gaps and traced them to training data that encodes historical patterns of discrimination — so the model learns the past's inequities and projects them forward as predictions.[2] As Baker and Hawn put it, algorithmic bias in education "poses significant threats to educational equity, potentially amplifying existing social and economic disparities" — and the harm compounds when an instructor with deficit assumptions reads an "at-risk" flag as confirmation rather than a cue to help, letting the prediction become a self-fulfilling label.[3]

THE CAPABILITY GAP

The bias is not in the math; it is in the definition of "at-risk." Define it one way and support flows to one population; define it another way and it flows to another, so the construct silently sets who the system decides to help. The choice of what the model predicts — the construct — is a capability-engineering decision with measurable equity consequences, and it is the part most often made implicitly, by whoever assembles the training labels, rather than governed deliberately by anyone accountable for where help is sent.[4]

AFTERMATH & REFORM

Unlike the discrete failures elsewhere in this chapter, this one is ongoing and quiet — embedded in advising dashboards at hundreds of institutions — which is what makes it dangerous: there is no single collapse to force a reckoning, only a steady misallocation no headline announces.[5] Its lesson, pushed upstream, is the chapter's: governing an algorithm's fairness begins not at deployment but at construct definition and label choice, with an equity audit of what the model is asked to predict and for whom the prediction allocates help — catching the bias where it is introduced rather than where it surfaces.



PREDICTIONS OVER-FLAG THE VERY STUDENTS WHO SUCCEED

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THE LEARNING ENGINEERING LENS

> *Algorithmic bias in educational systems poses significant threats to educational equity, potentially amplifying existing social and economic disparities.*
>
> BAKER & HAWN, 2021

LE INSIGHT

Educational predictive analytics is the ongoing live case for algorithmic bias at the construct level. The bias is not in the sigmoid; it is in the definition of “at-risk.” Defining at-risk in one way allocates support to one population; defining it in another way allocates support to another. The choice of definition is a capability-engineering decision with measurable equity consequences.

LENS APPROACH

LENS treats this case as the positive counterpart to Georgia State (Case 110). LEN 4 examines construct definition as the load-bearing measurement choice. LEN 7 examines the governance architecture that determines whose construct gets adopted. LEN 9 covers the technical bias-mitigation methods.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Govern the construct definition deliberately: have an accountable owner decide and document what “at-risk” means and whom the prediction will route help toward.
- Audit training data for encoded historical discrimination before fitting, so the model does not learn past inequities as future predictions.
- Test calibration across racial groups during development and treat disparate misclassification as a design defect, not a tolerable residual.

IN OPERATION / AFTER

- Monitor deployed dashboards for the quiet, distributed misallocation that has no single failure event to announce it.
- Train advisors and instructors to read an “at-risk” flag as a cue to help rather than confirmation of a deficit assumption.
- Sustain a recurring equity audit of what each model predicts and for whom, since a bias embedded across hundreds of institutions persists until someone is tasked to find it.

REFLECTION QUESTIONS

1. Pick a predictive analytic in your institution. Reconstruct the construct definition behind it. What is the equity consequence of that definition?
2. Design the governance review that a new predictive model should pass before it allocates resources to or away from a population.
3. This failure is ongoing and quiet, with no single collapse to force a reckoning. What would it take to make a slow, distributed misallocation visible enough that someone with authority had to act on it?

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CASE 91 TECHNOLOGY GOVERNMENT 2011 – 2016

Wells Fargo Fake Accounts

G Governance & Trust N Normalization of Deviance

IMPACT ~3.5 million unauthorized accounts opened; ~$3B in penalties; CEO resigned; the Federal Reserve capped the bank's assets

IN BRIEF

To meet aggressive sales quotas, Wells Fargo employees opened roughly 3.5 million unauthorized customer accounts over years. The behavior was widespread and visible to internal risk and compliance functions, but the bank's response was to fire individual "bad apples" while leaving the incentive structure intact — and that structure was the actual cause: sales targets most employees could not meet by ethical means. The 2016 CFPB and OCC actions brought $3 billion in penalties, the CEO resigned, and the Federal Reserve took the unprecedented step of capping the bank's assets. Wells Fargo is the canonical case of an incentive architecture that manufactured the misconduct the institution then prosecuted at the front line, while insisting the misconduct was individual.

BACKGROUND

Wells Fargo built its retail strategy on "cross-selling" — opening many products per customer — and drove it with aggressive sales quotas pushed down to branch employees, whose pay and jobs depended on hitting them. The bank's signature metric, "Gr-eight" (an average of eight products per household), and a daily "solution" scorecard reported up the management chain made cross-sell the single most-watched proxy for branch performance. The metric the bank chose to manage by thus became the thing every front-line employee was structurally compelled to maximize, whatever it took. The controls function carried no comparably visible counter-measure — no widely reported figure for the share of those accounts that customers had actually authorized — so the incentive ran without a designed brake.[0]

WHAT HAPPENED

Unable to meet the quotas honestly, employees opened roughly 3.5 million unauthorized accounts in customers' names — the rational response to a target most could not reach by legitimate means. Practices documented by investigators included opening checking, savings, and credit-card accounts without customer consent, forging signatures, moving funds between accounts to manufacture activity, and enrolling customers in online banking they had not requested. The behavior was widespread and longstanding, and visible to internal risk and compliance functions for years before the 2013 Los Angeles Times reporting made it public; the institutional response was to discipline individual employees as bad apples — Wells Fargo terminated more than 5,300 employees between 2011 and 2016 — while leaving the incentive structure intact, fixing the symptom and preserving the cause.[1]

THE INVESTIGATION

The 2016 CFPB and OCC actions brought roughly $3 billion in penalties, and the bank's own independent-directors investigation (2017) documented how the sales-target architecture drove the misconduct — locating the cause in the design, not the people executing it.[2] Investigators tied the misconduct directly to the bank's incentive-compensation structure, which had made such sales practices a foreseeable result rather than an aberration. The CEO resigned, and the Federal Reserve imposed an unprecedented cap on the bank's assets, reaching past individual penalties to constrain the institution itself.[3]

THE CAPABILITY GAP

Wells Fargo is the strongest evidence in the dataset that incentive architecture is a capability-engineering deliverable. The measurement system used to manage employees produced exactly the behavior the institution then prosecuted — the same structure that demanded the result punished the people who delivered it. The gap was not at the front line but at the governance layer that designed the incentives and then, for years, treated the predictable result as individual moral failure, mistaking a designed outcome for a character flaw.[4]

AFTERMATH & REFORM

Stumpf's successor Tim Sloan, promoted from chief operating officer in the immediate aftermath, was himself forced out in 2019 after the Federal Reserve's asset cap proved more durable than the bank had assumed. The cap — imposed in February 2018 at roughly $1.95 trillion — restricted Wells Fargo's growth pending evidence of governance and risk-management remediation, and as of 2024 it remained in effect, constraining the bank's growth for a sixth consecutive year and making it the longest-running enforcement action of its kind against a major U.S. bank. The case became the standard teaching example of measurement-gaming and incentive design.[5] Its lesson is that any quota becomes a target to be gamed, and that an institution is accountable for the behavior its measurement system makes rational, not just the behavior endorsed in its values statement — because employees respond to the incentive they are paid on, not the value they are told to honor.

3.5M

UNAUTHORIZED ACCOUNTS

the incentive architecture made misconduct rational for the front line

WELLS FARGO — THE MEASUREMENT SYSTEM PRODUCED THE BEHAVIOR THE INSTITUTION THEN PROSECUTED

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THE LEARNING ENGINEERING LENS

> *Wells Fargo's sales practices were a foreseeable consequence of its incentive compensation structure.*
>
> PARAPHRASING THE 2016 REGULATORY AND INDEPENDENT-DIRECTORS FINDINGS ON WELLS FARGO

LE INSIGHT

Wells Fargo is the strongest available evidence that incentive architecture is a capability-engineering deliverable. The measurement system used to manage employees produced the behavior the institution then prosecuted. The gap was not at the front line. It was at the governance layer that had designed the incentives.

LENS APPROACH

Wells Fargo is the canonical "protecting the measurement from gaming" case (induced 2.2; LENS D4/PT5), with measuring-the- failure-mode-you-care-about (induced 2.1) as the alternate anchor. LENS uses it in LEN 4 as the measurement-gaming case and in LEN 7 for the corporate-governance dynamics that allow such incentives to persist for years. Studio projects redesign the incentive architecture and the countervailing audit measure that would have detected the gaming as a structural pattern rather than as individual misconduct. Adjacent to Texas City (Case 137) at the wrong-measurement-reported-as-a-win layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design incentive and quota structures against the behavior they will rationally produce, treating the measurement system as a capability deliverable in its own right.
- Set targets that are achievable by legitimate means, since a quota most employees cannot reach honestly is an instruction to cheat.
- Pair every performance metric with a countervailing measure that detects gaming before it becomes systemic.

IN OPERATION / AFTER

- Audit for the gap between the behavior the measurement system rewards and the conduct the institution claims to value, and treat divergence as a design fault.
- Empower risk and compliance to escalate a pattern of gaming as a structural cause, not to absorb it as isolated misconduct.
- Hold the governance layer that designed the incentives accountable for predictable outcomes, rather than disciplining only the front line.

REFLECTION QUESTIONS

1. Identify a measurement system in your organization that is currently producing the behavior the organization claims to deplore. What is the gap between the measurement and the goal?

2. Design the incentive architecture for a bank's branch sales force that does not produce a Wells-Fargo-equivalent failure.

3. Wells Fargo's misconduct was visible to risk and compliance for years before it was addressed at the structural level. What lets an organization see a pattern of gaming and still treat it as individual bad apples — and who would have to be empowered to name the incentive as the cause?

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CASE 92 INDUSTRIAL GOVERNMENT 2015

Volkswagen Dieselgate

D Designed Out G Governance & Trust

IMPACT ~11 million vehicles equipped with defeat-device software; USD 4.3B U.S. DOJ criminal-and-civil plea (Jan 2017); total global cost above USD 30B across penalties, vehicle buybacks, and settlements; multiple criminal convictions

IN BRIEF

Volkswagen engineered a “defeat device” into its diesel emissions software: code that detected when a car was on a regulatory test bench and switched on full emissions controls only then. On the road the controls were largely disabled, producing nitrogen-oxide emissions up to forty times the legal limit, across roughly 11 million vehicles for years. The deception was uncovered not by regulators but by a small West Virginia university team comparing real-world to lab measurements. Internal documents showed the defeat device was an institutional decision — a deliberate response to a standard VW’s engineers did not believe they could meet within cost — not a rogue act. VW pleaded guilty to a USD 4.3 billion criminal and civil settlement with the U.S. Department of Justice (January 2017); total Dieselgate cost across penalties, vehicle buybacks, and settlements has been estimated above USD 30 billion globally, with multiple criminal convictions. Dieselgate is the book’s case for organized evasion of a measurement system.

BACKGROUND

Volkswagen had staked its U.S. growth on “clean diesel,” promising cars that met strict nitrogen-oxide limits without sacrificing performance or cost — a marketing position that left no room to admit the engineering could not deliver all three at once. Its engineers did not believe they could actually meet the standard within the platform’s cost constraints, putting the company’s public promise on a collision course with its own technical reality.[0]

WHAT HAPPENED

So they cheated by design. VW’s emissions software detected when a vehicle was on a regulatory test bench — by its steering, speed, and duration patterns, the telltale signature of a lab rather than a road — and switched on full emissions controls only during the test. On the road the controls were largely disabled, producing emissions up to forty times the legal limit, across roughly 11 million vehicles for years, so the pollution the standard existed to prevent flowed freely everywhere except where it was measured.[1]

THE INVESTIGATION

The deception was uncovered not by the regulator but by a small university research team in West Virginia comparing real-world emissions to lab results — the gap between road and bench being exactly what the regulator’s own test could never reveal.[2] Internal documents then showed the defeat device had been authorized inside VW’s engineering hierarchy as a deliberate institutional response to a standard the team could not meet — not the work of a rogue engineer, but a decision made up the chain of command. VW pleaded guilty to a USD 4.3 billion U.S. DOJ criminal and civil settlement (January 2017) — total global Dieselgate cost across penalties, vehicle buybacks, and settlements has been estimated above USD 30 billion — and multiple executives were convicted.[3]

THE CAPABILITY GAP

The exploitable gap was at the regulator's instrument: the emissions test ran in a regime the vehicle could detect, so a manufacturer determined to evade could tune its behavior to the test rather than to the road, optimizing for the measurement instead of the goal it stood for. Dieselgate is the canonical case of institutionally engineered evasion of a measurement system — and a reminder that any test the measured party can recognize is a test it can defeat, because a detectable test invites the very gaming it is meant to catch.[4]

AFTERMATH & REFORM

The reform closed the gap at the instrument: the EU introduced real-world driving-emissions testing, moving the measurement off the predictable bench and onto the road the standard actually cared about, making it far harder to game.[5] The pattern parallels Takata (Case 84) — a manufacturer's fraud meeting a regulator whose evidence pipeline trusted the manufacturer's test conditions, and a fix that had to upgrade the measurement itself, not just punish the cheat, because punishing the cheat leaves the exploitable instrument in place for the next one.

TEST BENCH	REAL WORLD
compliant	up to 40×

the protocol operated in a regime the vehicle could detect

DIESELGATE — ENGINEERED EVASION OF A MEASUREMENT SYSTEM

REFERENCES

1. U.S. EPA, Notice of Violation to Volkswagen (2015) — the defeat device and emissions exceedances.
2. West Virginia University / ICCT real-world diesel-emissions study (2014) — the discovery comparing road to lab.
3. U.S. Department of Justice Plea Agreement with Volkswagen AG (January 2017) — the institutional decision, USD 4.3 billion criminal-and-civil settlement, and convictions (quoted).
4. Volkswagen internal documents released through litigation — authorization of the defeat device within the engineering hierarchy.
5. EU Real Driving Emissions (RDE) testing regulation — the post-Dieselgate measurement reform.
6. J. Ewing, Faster, Higher, Farther: The Volkswagen Scandal (2017); cf. Takata (Case 84).

THE LEARNING ENGINEERING LENS

> *The defeat device was the product of a long-standing institutional decision to evade emissions standards.*
>
> PARAPHRASING THE U.S. DEPARTMENT OF JUSTICE PLEA AGREEMENT WITH VOLKSWAGEN AG, 2017

LE INSIGHT

Dieselgate is the canonical case for institutionally engineered evasion of a measurement system. The capability gap was at the regulator's test protocol: it operated in a regime the vehicle could detect. The reform — real-world emissions testing — was a capability deliverable upgrade at the regulator's instrument boundary.

LENS APPROACH

LENS reads Dieselgate at the independent-instrument and operator-inspector boundary (induced 2.2): the defeat device worked by detecting the test condition itself, so the measurement was one the operator could see coming and game. The lesson is designing an emissions-measurement and inspection regime the operator cannot detect and defeat — real-driving emissions, surprise and outside-the-loop testing — rather than the bare observation that fraud occurred. The reform pattern parallels Takata (Case 84).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design measurement protocols the measured party cannot detect and tune to, so behavior is optimized for the goal rather than the test.
- Surface the gap between a public commitment and the engineering reality early, so an unmeetable standard is renegotiated rather than secretly evaded.
- Build a governance check between an impossible target and the team asked to meet it, so the response is escalation, not a defeat device.

IN OPERATION / AFTER

- Audit real-world behavior against test-condition behavior, treating a divergence between road and bench as the signature of engineered evasion.
- Upgrade the measurement instrument itself after a discovered cheat, not just penalize the cheat, since the exploitable test invites the next one.
- Sustain independent, outside-the-loop verification — as the university team provided — to catch fraud the regulator's own protocol cannot see.

REFLECTION QUESTIONS

1. Identify a measurement protocol in your domain that operates in a regime detectable by the entity being measured. What is the evasion potential?

2. Design the upgrade to a regulatory test protocol that makes Dieselgate-style evasion structurally infeasible.

3. VW's defeat device was authorized up the engineering hierarchy, not the act of a rogue engineer. What allows an institution to sanction fraud as a deliberate response to an unmeetable standard — and what governance check should sit between an impossible target and the team asked to meet it?

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CASE 93 HEALTHCARE CLINICAL AI GOVERNANCE 2017 – 2021

Epic Sepsis Model

D Designed Out G Governance & Trust N Normalization of Deviance

IMPACT External validation across 38,455 hospitalizations found AUROC 0.63 versus reported 0.76–0.83, missing ~67% of sepsis at the operational threshold; deployed in hundreds of hospitals without independent validation or FDA clearance

IN BRIEF

The Epic Sepsis Model was a proprietary machine-learning sepsis prediction tool embedded in the Epic EHR and deployed in hundreds of US hospitals — the most widely operational clinical AI in American medicine. Until Wong et al. (JAMA Internal Medicine 2021) ran an external validation across 38,455 hospitalizations, no independent evaluation had been published. The reported AUROC was 0.63, well below the 0.76–0.83 Epic had reported; at the operational threshold, the model missed roughly two-thirds of sepsis cases, with a 12% positive predictive value and substantial alert burden. The case is not principally about the model's performance. It is about the governance seam that let the deployment happen at scale without independent validation: as an EHR-embedded proprietary feature, the model sat outside the FDA's medical-device oversight regime, so the machinery that would have surfaced its limitations at clearance was never engaged. The post-deployment surveillance pattern (Vioxx, Case 105) is the analog: the harm was the absence of a standing system to check the tool once it was in the population's hands.

BACKGROUND

By the late 2010s the Epic Sepsis Model was, by deployment count, the most widely operational clinical AI in American medicine — embedded in hundreds of hospitals as a default feature of the dominant inpatient EHR. The tool was presented as a help to clinicians trying to catch sepsis earlier in a workflow already saturated with alerts. The model's design and validation, however, were proprietary and not externally evaluated.[0]

WHAT HAPPENED

In 2021 Wong et al. published the first large external validation in JAMA Internal Medicine: 38,455 hospitalizations at the University of Michigan, with sepsis diagnosed using two consensus definitions. The model achieved an AUROC of 0.63, materially below Epic's reported 0.76–0.83. At the operational threshold for bedside alerting, the model missed roughly two-thirds of sepsis cases. The 12% positive predictive value meant most alerts were false; the alert burden landed on clinicians who could not distinguish the few real alerts from the many spurious ones.[1]

THE INVESTIGATION

The governance seam is the structural lesson. Because the Epic Sepsis Model was distributed as a feature of an EHR rather than as a stand-alone clinical-decision-support device, it did not require FDA clearance. The machinery that would normally require independent validation, post-market surveillance, and demographic stratification of performance was never engaged. The model's deployment was a regulatory non-event because the regulatory regime treated the EHR layer as out of scope. The clinical AI question and the device-oversight question diverged.[2]

THE CAPABILITY GAP

What surfaced the failure was post-deployment external validation — the exact discipline that the clearance pathway omits. The Wong et al. paper was disconfirmation in the form the system did not otherwise provide. Epic subsequently revised the model and added stratification to its documentation; many hospitals turned the alert off, recalibrated, or replaced it. The corrective action worked through publication, not through governance. That is the gap: a tool can be deployed at hundreds of sites, alert at the bedside for years, and still be disconfirmable only by an academic paper rather than by the surveillance architecture the deployment was supposed to have.[3]

AFTERMATH & REFORM

The Epic case is the negative pair to TREWS (Case 74) and the governance-seam analog to Radiology AI Miscalibration (Case 130). Together they teach a typology: delegation done well as a paired intervention with engineered interface and outcome-grounded evidence (TREWS); delegation done badly as a proprietary EHR-embedded model deployed outside the device-oversight regime without independent validation (Epic); delegation halted on rights grounds because the system was both ineffective and rights-violating (SyRI); delegation marketed ahead of capability (Watson for Oncology). The four together are the canonical AI delegation typology.

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2. Habib et al. (2021), commentary on Wong et al., JAMA Internal Medicine — on the implications for proprietary clinical AI.
3. FDA, Clinical Decision Support Software: Final Guidance (2022) — the post-Wong reframing of the EHR-embedded oversight question.
4. Adams et al. (2022), Nature Medicine — the paired positive case (101).

THE LEARNING ENGINEERING LENS

> *A deployment is not a validation. Deployment without independent validation is delegation without evidence.*
>
> EDITORS' SYNTHESIS OF WONG ET AL. (2021).

LE INSIGHT

The Epic Sepsis Model is the canonical case of consequential clinical-AI delegation at scale without independent validation. The structural lesson is not the model's poor performance; it is the governance seam that let the deployment proceed without the validation and surveillance machinery that the medical-device pathway would have required, surfaced only by post-deployment external work.

LENS APPROACH

Epic is the Domain 4 + Domain 3 / Problem Type 6 failure that motivates the post-deployment-surveillance discipline LENS teaches. Used in Domain 4 (Test and Evaluation) for measurement architecture under proprietary opacity and the gap-attribution CLO; in Domain 3 (Human-System Collaboration) for the delegation-with-revocation CLO — Epic was delegated without a pre-specified revocation criterion; and in Domain 5 (Navigating Sociotechnical Constraints) for the cross-regime / platform governance seam. Pairs directly against TREWS (Case 74) and sits in the AI-delegation typology with SyRI and Watson.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Require independent external validation before deployment of any consequential clinical AI, regardless of whether it ships as a stand-alone device or as a feature of a host platform.
- Specify in advance the disconfirming evidence — population, threshold, PPV, alert burden — that would revoke the delegation, and the channel through which that evidence would surface.
- Identify the regulatory regime the tool falls under, and where the seam between regimes is — proprietary EHR features should not be exempt from clinical-AI oversight by virtue of their packaging.

IN OPERATION / AFTER

- Build post-deployment surveillance as a standing institutional capability — outcome metrics, demographic stratification, alert-burden audit — so disconfirmation does not require a single academic paper to surface.
- Close the cross-regime seam: clinical AI embedded in EHRs should be subject to the same independent validation and surveillance as stand-alone clinical-decision-support devices.
- When disconfirming evidence arrives, treat it as a designed input: revise, recalibrate, or remove on a defined timeline, with the corrective action visible to the clinicians who used the tool.

REFLECTION QUESTIONS

1. Identify a clinical AI tool deployed in your domain. Where in the regulatory architecture would independent validation have been required, and where could it slip the seam? What pre-specified disconfirming evidence would revoke the delegation?
2. Design the post-deployment surveillance deliverable that should accompany every deployment of consequential clinical AI — including embedded-in-EHR features that currently fall outside the device-oversight regime.
3. The disconfirmation in this case came from a single academic paper, not from a standing institutional architecture. What is the minimum surveillance machinery that would have surfaced the model's performance gap at the operational threshold without requiring the Wong et al. paper to exist?

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CASE 94 FINANCE ALGO FAIRNESS 2018 – 2022

Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity

D Designed Out G Governance & Trust N Normalization of Deviance

IMPACT Algorithmic mortgage underwriting reduced face-to-face discrimination but preserved a measured pricing disparity even when race was excluded from the inputs — the variable's omission did not fix the harm

IN BRIEF

Bartlett, Morse, Stanton, and Wallace (Journal of Financial Economics, 2022) analyzed several million US mortgage applications across the fintech transition and documented two patterns that together unsettle a common intuition. Algorithmic underwriting was, on average, less discriminatory than face-to-face underwriting along the **acceptance** margin — fintech lenders accept-rejected Black and Latino applicants more equitably than traditional lenders. But the **pricing** margin persisted: minority borrowers were charged systematically higher rates for equivalent loans even when race was not in the model inputs. The excluded variable did not stay excluded; it re-entered through geography, credit-score history, and other features that correlate with race in the present US population. The case is the canonical instance of why "fairness through unawareness" is not, by itself, fairness. It pairs directly with the Coots et al. fintech fairness audit (Case 107, preprint-tier): the next teaching step is that competing fairness definitions can disagree about what counts as fair even when the inputs are held constant.

BACKGROUND

The US mortgage market is the most consequential consumer-credit market in which algorithmic underwriting now competes with face-to-face underwriting at large scale. Fintech entrants over the 2010s automated significant parts of the application-to-approval pipeline, and the public-policy question that followed was whether automation reduced or preserved the discrimination documented in traditional lending.[0]

WHAT HAPPENED

Bartlett et al. assembled a corpus of several million applications from a period spanning fintech adoption, and decomposed discrimination along two distinct margins: the **acceptance** decision (do you get a loan) and the **pricing** decision (what interest rate you pay if you do). On acceptance, algorithmic lenders did better: Black and Latino applicants with equivalent observable characteristics were accepted at rates closer to those of comparable White applicants than face-to-face lending produced. The result is consistent with the intuition that removing the loan officer removes some of the channel through which bias entered.[1]

THE INVESTIGATION

The pricing margin tells the harder story. Even when race was not among the model inputs, minority borrowers were charged systematically higher rates than equivalent White borrowers — by about 8 basis points on purchase loans and 3 on refinances in the paper's central estimate. The disparity did not vanish because the variable was omitted; the variable returned through features correlated with race in the present US population — geography, credit-history depth, and other proxies that the model is allowed to use and that carry the historical signal of where lending has and has not flowed.[2]

THE CAPABILITY GAP

What the case teaches is the structural form of "fairness through unawareness": when protected attributes are excluded from a model that operates over a population in which other admissible features correlate with the protected attribute, the model can preserve the disparity it was meant to remove. Omission shifts the channel of discrimination but does not close it. The capability deliverable is a measurement architecture that surfaces disparate impact in the **outputs**, not assurance about the **inputs**. The acceptance/pricing asymmetry also reframes the policy question: a model can be more equitable on one decision margin and unchanged on another within the same transaction.[3]

AFTERMATH & REFORM

The case is the headline mortgage-finance instance of a pattern that now surfaces in clinical algorithms (eGFR, Cases 113, 114 and 95), in hiring and proctoring tools (Case 96 small-tier proctoring bias), and in welfare administration (SyRI). It pairs with the Coots et al. fintech fairness audit (Case 107) which shows the next layer: once the practitioner accepts that omission is not the answer, competing fairness definitions disagree about which adjustment is the right one — and the choice has to be made on judgment under irreducible uncertainty, not on a technical optimum.

REFERENCES

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3. Dwork et al. (2012), "Fairness Through Awareness," ITCS 2012 — the foundational technical statement of the limits of unawareness.
4. Mitchell, Potash, Barocas, D'Amour, & Lum (2021), "Algorithmic Fairness: Choices, Assumptions, and Definitions," Annual Review of Statistics and Its Application 8:141–163 — the competing-definitions framing.

THE LEARNING ENGINEERING LENS

> *The omitted variable does not stay omitted. It re-enters through every feature that carries the same signal.*
>
> EDITORS' SYNTHESIS OF BARTLETT ET AL. (2022) AND DWORK ET AL. (2012).

LE INSIGHT

Bartlett et al. is the canonical mortgage-finance instance of why "fairness through unawareness" is not fairness. Algorithmic underwriting reduced the acceptance-margin disparity but the pricing-margin disparity persisted even with race excluded — because correlated admissible features carry the same signal. The capability deliverable is output-level disparate-impact measurement, not input-level assurance.

LENS APPROACH

Bartlett is the headline equity-and-construct case in consumer credit (induced 8.2; LENS D4+D4/PT6). LENS uses it in Domain 4 (Test and Evaluation) for the CLO **Fairness beyond omission**, in Domain 3 (Human-System Collaboration) for delegation to underwriting models, and in Domain 5 (Navigating Sociotechnical Constraints) for the fair-lending regulatory regime. Direct pair with Case 107 (Coots fintech fairness audit). Adjacent to the race-construct trio in clinical medicine (Cases 113, 114 and 95) — same structural lesson at the construct-definition layer rather than the pricing layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify, before deployment, the disparate-impact measurement on outputs (acceptance rate, pricing) stratified by protected attribute, with a pre-registered threshold for what would trigger remediation.
- Audit the model's correlated features for the channel through which an excluded attribute can re-enter — geography, credit-history depth, transaction venue — and decide whether each feature's predictive value justifies its disparate-impact contribution.
- Separate acceptance and pricing as distinct decision margins; do not assume parity on one implies parity on the other.

IN OPERATION / AFTER

- Publish stratified outcome metrics at intervals long enough for selection effects to settle; the central finding required a multi-year panel.
- Treat omission of the protected attribute as a baseline, not a remedy; the test of fairness is the output distribution, not the input set.
- When the measurement surfaces a disparity, name the fairness definition under which it is a problem — group calibration, equalized odds, demographic parity — and the trade-offs of the chosen remediation.

REFLECTION QUESTIONS

1. Identify a model in your domain where a protected attribute is excluded from the inputs. Which admissible features correlate with the excluded attribute in your population? What is the channel through which the excluded variable could re-enter?

2. Design the disparate-impact measurement you would publish at intervals after deployment. Specify the fairness definition, the decision margin (acceptance, pricing, escalation), and the threshold that would trigger remediation.

3. Bartlett's central finding is that the acceptance margin can be more equitable than the face-to-face baseline while the pricing margin remains unchanged. What policy or design intervention would address each margin without assuming parity on one implies parity on the other?

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CASE 95 CLINICAL MEDICINE MEDICAL EDUCATION HEALTH EQUITY 2016

Racial Bias in Pain Assessment — The False-Belief Mechanism

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT **About half of surveyed medical students and residents endorsed at least one false belief about biological differences between Black and White people; those who held more false beliefs rated Black patients' pain as less severe and recommended less accurate treatment**

IN BRIEF

Hoffman, Trawalter, Axt, and Oliver (PNAS 2016) surveyed medical students and residents on a battery of false beliefs about biological differences between Black and White people (e.g., "Black people's skin is thicker," "Black people's nerve endings are less sensitive"). About half endorsed at least one such belief. The paper's experimental layer showed that respondents who endorsed more false beliefs rated the pain of mock Black patients as less severe than the same pain in mock White patients, and made less accurate treatment recommendations. The mechanism the case identifies is specific and unusually precise for a bias study: the pain-assessment gap is traceable to a small set of nameable false biological beliefs, not to diffuse implicit bias or structural-only explanation. That precision is what makes Hoffman the human-development case in the race-construct trio. The capability deliverable is not awareness training; it is curriculum that specifically disconfirms the named false beliefs and instrumentation that surfaces when bedside ratings of pain diverge by patient race.

BACKGROUND

Pain assessment is a clinician's judgment, made repeatedly across a day, on patients whose subjective report of pain has to be translated into a numeric rating and a treatment decision. A documented finding in the medical literature is that Black patients in the United States are systematically under-treated for pain across emergency-department, post-surgical, and end-of-life settings. The bias has been measured at the population level for decades; the mechanism was less precisely named.[o]

WHAT HAPPENED

Hoffman et al. (2016) administered a battery of statements about biological differences between Black and White people to 222 medical students and residents — some true, some false (e.g., "Black people's skin is thicker," "Black people's blood coagulates more quickly," "Black people's nerve endings are less sensitive"). About half of respondents endorsed at least one false belief; a smaller subset endorsed several. The experimental layer of the study presented respondents with two mock patient cases identical except for race, asked them to rate the patients' pain, and asked them to recommend treatment.[1]

THE INVESTIGATION

The pattern was that respondents who endorsed more false beliefs rated the pain of the Black mock patient as less severe than the pain of the White mock patient, and recommended less accurate treatment for the Black mock patient. Respondents who endorsed no false beliefs did not show the rating gap. The case is unusual in identifying a specific cognitive mechanism — a small set of named false biological beliefs — that mediates a documented population-level disparity. Most bias studies leave the mechanism diffuse; Hoffman names it precisely enough that a curriculum or assessment can target it.[2]

THE CAPABILITY GAP

What the case teaches at the construct layer is that the capability deliverable in medical education is not generic "implicit bias" awareness — it is curriculum that specifically disconfirms the named false beliefs, with assessment instruments that test whether the beliefs were actually disconfirmed. Operationally, the deliverable is a bedside instrument or surveillance pattern that surfaces when pain ratings diverge by patient race in ways that survive case-mix adjustment. The Hoffman finding makes both deliverables specifiable in a way that more diffuse bias findings did not.[3]

AFTERMATH & REFORM

Hoffman pairs with pulse oximetry (Case 114) and eGFR (Case 113) in the race-construct trio. The three cases are the same surface harm — minority patients systematically under-served across a clinical decision — attributable to three distinct layers of the system: the construct definition (eGFR), the validation architecture (pulse oximetry), and the human-development mechanism (Hoffman). The trio is the case-grounded basis for the CLO **Gap attribution**: distinguishing capability gaps attributable to human development, system design, and organizational performance, and selecting measurement that isolates the intended cause.

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THE LEARNING ENGINEERING LENS

> *The mechanism the paper names is precise enough to teach against. Awareness training is not a curriculum; a curriculum has to disconfirm something specific.*
>
> EDITORS' SYNTHESIS OF HOFFMAN ET AL. (2016).

LE INSIGHT

Hoffman et al. is the human-development case in the race-construct trio. The pain-assessment disparity in medical settings is mediated, in measurable part, by a nameable set of false biological-difference beliefs held by clinicians in training. The capability deliverable is curriculum that specifically disconfirms the beliefs and instrumentation that surfaces the bedside rating gap when it persists.

LENS APPROACH

Hoffman is the human-development case in the race-construct trio (Cases 113, 114 and 95). LENS uses it in Domain 4 (Test and Evaluation) for the CLO **Gap attribution** — the gap is in the clinician's training, not the construct or the device — and in Domain 2 (Learning Engineering Design) for the curriculum-design implication. The trio together is the case-grounded basis for **Gap attribution**: same surface harm, three distinct layers, three distinct remediations. Adjacent to the lending pair (Cases 94–107) at the construct layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build curriculum that specifically disconfirms the named false biological beliefs identified in the Hoffman survey, with assessment items that test whether the disconfirmation took hold.
- Instrument bedside pain ratings to surface case-mix-adjusted divergence by patient race; the gap is otherwise invisible to the system that produces it.
- Identify the layer of the gap before designing the remediation: construct, validation, or human-development. A construct fix cannot remediate a clinician-belief fix and vice versa.

IN OPERATION / AFTER

- Re-administer the Hoffman survey periodically as a curriculum-evaluation instrument; a curriculum that does not move the false-belief endorsement rate is not closing the mechanism the paper identifies.
- Track whether the bedside pain-rating disparity narrows in cohorts that received the disconfirming curriculum, with reporting at intervals long enough for selection effects to settle.
- Cross-reference the human-development result against the construct (eGFR) and validation-architecture (pulse oximetry) layers, so the overall equity capability of the clinical system is not assessed only at the layer the institution finds easiest to instrument.

REFLECTION QUESTIONS

1. Identify a documented disparity in your domain whose mechanism is treated as diffuse. What would a Hoffman-style survey look like — a battery of named false beliefs or assumptions whose endorsement could be measured and whose presence predicts the operational decision?

2. Design the curriculum-evaluation instrument you would use to test whether a curriculum has actually disconfirmed the false beliefs. What endorsement-rate change would you require before claiming the mechanism has been addressed?

3. Hoffman is the human-development case in the race-construct trio. Pulse oximetry (Case 114) is the validation-architecture case; eGFR (Case 113) is the construct-definition case. Which of the three layers does your domain currently address, and which does it leave invisible?

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CASE 96 ED-TECH ALGO FAIRNESS ASSESSMENT 2022

Algorithmic Bias in Automated Exam Proctoring

D Designed Out N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT The first quantitative study of facial-detection bias in automated exam proctoring software found that students with darker skin tones and Black students were significantly more likely to be flagged for instructor review for potential cheating; at the race–sex intersection, women with the darkest skin tones were far more likely to be flagged than other groups

IN BRIEF

The COVID-era expansion of remote learning produced a rapid deployment of automated exam-proctoring software across higher education: computer-vision systems that monitor the student via webcam during an exam and flag suspicious behavior for instructor review. Yoder-Himes et al. (Frontiers in Education, 2022) ran the first quantitative study of facial-detection bias in this class of software. Students with darker skin tones and Black students were significantly more likely to be flagged for instructor review for potential cheating than students with lighter skin tones; at the race–sex intersection, women with the darkest skin tones were far more likely to be flagged than other groups. The study examined one major proctoring product and concludes the product "may employ biased AI algorithms that unfairly disadvantage students." It documents the disparity but not a remediation, so it is a failure / diagnosis case. The harm comes directly from validation that did not stratify across skin tone, surfacing a group-specific failure in a deployed system. Trio with Cases 113 (eGFR), 106 (pulse oximetry), and 107 (Hoffman pain bias) at the race-construct-and-validation layer.

BACKGROUND

Automated exam proctoring is a recent and rapidly deployed class of ed-tech: a webcam-based monitoring system that observes the student during a remote-administered exam and uses computer vision to flag suspicious behavior — looking off-camera, multiple faces in the frame, the face leaving the frame — for instructor review. The COVID-era pivot to remote instruction expanded the deployment of this software across US higher education at speed. The systems' face detection and face tracking are the load-bearing computer- vision components: a failure in face detection translates directly into a false flag for instructor review, and systematic failures in face detection across demographic groups translate into systematic group-level harm.[o]

WHAT HAPPENED

Yoder-Himes et al. (Frontiers in Education, 2022) ran the first published quantitative study of facial-detection bias in this class of software. The study examined one major proctoring product across a sample of student-submitted sessions, with skin-tone classification on a standard scale and demographic information available. The headline finding ran in the direction the broader face-recognition-bias literature had documented (Buolamwini and Gebru, Raji, and others) and extended that finding into the assessment context. Students with darker skin tones, and Black students specifically, were significantly more likely to be flagged for instructor review for potential cheating than students with lighter skin tones.[1]

THE INVESTIGATION

The intersectional analysis sharpened the finding. Women with the darkest skin tones were far more likely to be flagged than other groups — the race-by-sex intersection produced the largest disparity, consistent with the broader face-recognition literature's intersectional findings. The harm class is not abstract: a flag for instructor review under an academic-integrity process produces real downstream consequence — the student has to defend themself against a suspicion the software generated, and the institutional resolution mechanism is not designed for the case where the suspicion was generated by a software bias rather than by a student behavior.[2]

THE CAPABILITY GAP

The study's claim is calibrated and direct: the product "may employ biased AI algorithms that unfairly disadvantage students." The study documents the disparity in a real deployment of a real product and does not document a remediation, so it is a failure / diagnosis case rather than a failure-to-intervention arc. The harm comes directly from validation that did not stratify across skin tone: the computer-vision face-detection model behind the proctoring system was deployed without demographic validation of its detection rates, and the group-specific failure was therefore present in the deployed system from day one and only surfaced post-hoc by external researchers.[3]

AFTERMATH & REFORM

Drafted alongside the race-construct trio (Cases 113 eGFR, 106 pulse oximetry, 107 Hoffman pain bias), the proctoring case extends the validation-must-be- demographically-stratified lesson into the ed-tech and assessment context. The structural form is the same as pulse oximetry: a deployed system measured on an unrepresentative sample, producing aggregate accuracy that conceals a group-specific failure for years, until external researchers stratify the validation post-hoc. The eGFR cross-reference is the construct-definition counterpart; the Hoffman cross-reference is the human-judgment counterpart. All four sit in the small-and-big-tier conversation about validation discipline as an equity design commitment.

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THE LEARNING ENGINEERING LENS

> *The product may employ biased AI algorithms that unfairly disadvantage students.*
>
> YODER-HIMES ET AL., FRONTIERS IN EDUCATION 2022.

LE INSIGHT

Automated exam proctoring is the assessment-domain counterpart to pulse oximetry's medical-device bias. Yoder-Himes et al. 2022 is the first quantitative study of the disparity in this class of software, with the intersectional finding the broader face-recognition literature predicts. The case documents the disparity in one product; the remediation is not yet documented.

LENS APPROACH

Proctoring bias is the small-tier ed-tech validation-stratification failure (induced 8.2; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for the validation-must-be-demographically-stratified discipline, and in Domain 5 (Navigating Sociotechnical Constraints) for the institutional resolution mechanism that has to handle the case where the flag was generated by software bias. Trio with Cases 113 (eGFR), 106 (pulse oximetry), and 107 (Hoffman pain bias) at the race-construct-and-validation layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Stratify validation across skin tone (and across the race–sex intersection) before deployment, not after, so the group-specific failure surfaces in the engineering record before it surfaces in the harm record.
- Specify the harm class the false-positive flag produces — academic-integrity defense, downstream consequence to the student — and design the institutional resolution mechanism to handle the case where the flag was generated by software bias.
- Require the proctoring vendor to publish demographic stratification of their detection rates, on a standard skin-tone scale, as a deployment condition rather than as a post-hoc disclosure.

IN OPERATION / AFTER

- Treat the Yoder-Himes finding as the diagnosis it is: the disparity is documented in one product; the remediation is not.
- Carry the intersectional reading explicitly (women with the darkest skin tones far more likely to be flagged) in any communication about the case; the broader race-by-sex intersectional finding is consistent with the face-recognition-bias literature and should not be smoothed into a single-dimension finding.
- Build the cross-case reading with eGFR (Case 113), pulse oximetry (Case 114), and Hoffman pain bias (Case 95): the validation-must-be-demographically-stratified lesson runs across clinical, device, and assessment domains, and the proctoring case is the assessment-domain anchor.

REFLECTION QUESTIONS

1. Identify a deployed system in your domain whose validation rests on an aggregate accuracy figure rather than a demographically stratified one. What would the stratified validation actually require, and who would have to commission it before deployment rather than after?
2. The Yoder-Himes finding is intersectional: women with the darkest skin tones were far more likely to be flagged. What is the analog intersectional structure of the harm in your domain, and is it visible in the engineering record before the harm record?
3. The case documents the disparity in one product; the remediation is not yet documented. What would a remediation look like — vendor disclosure, regulatory disclosure requirement, institutional resolution-mechanism redesign — and which of those is in your scope?

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CASE 97 K-12 EDUCATION SCHOOL SAFETY INFRASTRUCTURE RACIAL DISPARITIES 2010S – 2022

School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

DISCLOSURE An editor of this volume shares an institution (Johns Hopkins School of Engineering) with the underlying study's authors. The case is included on the strength of the published peer-reviewed evidence in the Journal of Criminal Justice (2022); editorial framing maintains critical distance.

IMPACT Peer-reviewed study (Journal of Criminal Justice, 2022) examining school-surveillance infrastructure and Black student outcomes; the mechanism is the surveillance infrastructure (cameras, metal detectors, school resource officers), not the students, and the infrastructure drove a measured share of the outcome gap — one of the motivating cases for the CLO Gap attribution

IN BRIEF

Johnson and colleagues, writing in the Journal of Criminal Justice in 2022, analyzed the relationship between school-surveillance infrastructure — cameras, metal detectors, school resource officers, ID-check protocols — and outcomes for Black students across US schools. The study's learning-engineering content is in where it locates the mechanism. The disparity in outcomes between Black students and white students under surveillance-heavy schooling cannot be explained, the analysis shows, by student behavior alone; a measured share of the outcome gap is attributable to the surveillance infrastructure itself — to the institutional posture that schools serving predominantly Black student populations more often adopted. The mechanism, in other words, is the infrastructure, not the students. The case is one of the motivating cases in the v2 sweep for the CLO **Gap attribution** — the discipline of asking, when a disparity in outcomes is observed, what share of the disparity is attributable to the institutional or technical infrastructure rather than to the population the infrastructure is operating on. The standing COI rendered under the title is binding: an editor of this volume shares an institution with the study's authors, and the case is anchored to the peer-reviewed Journal of Criminal Justice evidence rather than to institutional press. The case extends the race-construct trio (Cases 113 eGFR, 106 pulse oximetry, 107 Hoffman) into the K-12 education domain at the infrastructure layer.

BACKGROUND

US public schools have, over the past two decades, deployed a progressively more elaborate surveillance infrastructure: cameras throughout common areas; metal detectors at entry points; school resource officers (sworn law enforcement stationed in schools); ID-check protocols and visitor management systems; behavioral-tracking software. The distribution of that infrastructure across schools has not been uniform. Schools serving predominantly Black student populations have, on average, carried more of it. The institutional rationale has typically been safety — the surveillance infrastructure is described as keeping students safe — and the question the Johnson study confronts is whether the surveillance is operating as safety infrastructure or as something else, measured by the outcomes the infrastructure produces.[0]

WHAT HAPPENED

Johnson and colleagues (Journal of Criminal Justice, 2022) analyzed the relationship between school-surveillance infrastructure and outcomes for Black students across the US public-school sector. The study's design separates student-level variables from school-level infrastructure-deployment variables, so that the share of the outcome gap attributable to each can be estimated. The headline analytic result the case rests on: a measured share of the disparity in outcomes between Black students and white students under surveillance-heavy schooling is attributable to the surveillance infrastructure itself, not to differences in student behavior. The infrastructure is acting as a driver of the outcome gap, not only as a response to it.[1]

THE INVESTIGATION

The mechanism is what makes the case the canonical gap-attribution case. When a disparity in outcomes is observed, the analytical default in many institutional settings is to attribute the disparity to the population the outcomes are measured on — to differences in preparation, behavior, family context. The Johnson study shows that, in the school-surveillance case, that default is wrong in a measurable sense: the institutional infrastructure is itself a driver of the disparity, and attributing the outcome gap to the students rather than to the infrastructure mis-locates the mechanism in a way the data does not support. The case extends the race-construct trio (Cases 113 eGFR, 106 pulse oximetry, 107 Hoffman) into K-12 education at the infrastructure layer: in eGFR and pulse oximetry the mechanism was in the device or the formula; in Hoffman the mechanism was in the practitioner's cognitive baseline; here the mechanism is in the institutional architecture itself.[2]

THE CAPABILITY GAP

The standing COI rendered under the title is binding. An editor of this volume shares an institution (Johns Hopkins School of Engineering) with the study's authors. The case is anchored to the published peer-reviewed evidence in the Journal of Criminal Justice, not to institutional press or to commentary by the editor or the authors outside the paper. The editorial framing has been written to maintain critical distance from the home-institution affiliation; the case's claim is what the published evidence supports, and the disclosure is the safeguard that makes the institutional adjacency visible at the point of reading.[3]

AFTERMATH & REFORM

The case is one of the motivating cases in the v2 sweep for the CLO **Gap attribution** proposed in — the discipline of asking, when a disparity in outcomes is observed, what share of the disparity is attributable to the institutional or technical infrastructure rather than to the population the infrastructure is operating on. The race-construct trio established the pattern at the device/formula/cognitive- baseline layers; this case carries the pattern at the institutional-infrastructure layer. The four-case set (Cases 113, 114, 95, 97) is the case-grounded basis for Gap attribution as a designed competency: practitioners and program designers have to be trained to look for the mechanism in the infrastructure they built, not only in the population they are serving, and the evidence the case carries is what makes the proposed CLO defensible.

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THE LEARNING ENGINEERING LENS

> *The mechanism is the infrastructure, not the students. Attributing the outcome gap to the population mis-locates the mechanism in a measurable sense the data does not support.*
>
> EDITORS' SYNTHESIS OF JOHNSON ET AL. (JOURNAL OF CRIMINAL JUSTICE, 2022).

LE INSIGHT

The Johnson school-surveillance study locates the mechanism of an outcome disparity in the institutional infrastructure, not in the population the infrastructure operates on. The case extends the race-construct trio (eGFR, pulse oximetry, Hoffman) into K-12 education at the infrastructure layer and is one of the motivating cases for the CLO Gap attribution. COI under the title — shared institution — is binding and rendered openly.

LENS APPROACH

Johnson school surveillance is the infrastructure-as-mechanism gap-attribution case (induced 8.1; LENS D4/PT5) — Domain 4 as the case-grounded basis for **Gap attribution**; Domain 5 for the institutional-architecture-as-mechanism framing. Pair with Cases 113, 114, 95. COI binds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the analysis to separate population-level variables from infrastructure-level variables so the share of the disparity attributable to each can be estimated; the Johnson study's design is the worked example.
- Treat the institutional infrastructure as a candidate mechanism, not as a neutral background; the default attribution-to-population analysis cannot test the infrastructure if the infrastructure is not a variable in the model.
- Anchor the editorial framing to the peer-reviewed evidence rather than to institutional press; the COI is rendered openly under the title and the framing maintains critical distance.

IN OPERATION / AFTER

- Carry the Gap-attribution discipline into curriculum design: practitioners have to be trained to look for the mechanism in the infrastructure they built, not only in the population they are serving.
- Pair the case with the race-construct trio (Cases 113, 114, 95) so the gap-attribution pattern is taught at multiple layers — device, formula, cognitive baseline, institutional architecture.
- Preserve the COI render — shared institution, anchored to peer-reviewed evidence, critical editorial distance — as the standing language for home-institution-shared cases across the corpus.

REFLECTION QUESTIONS

1. Identify a disparity in outcomes in your domain that is currently attributed primarily to the population the outcomes are measured on. What infrastructure-level variables would you have to add to the analysis to test whether the institutional architecture is itself a mechanism of the disparity?

2. Specify the design pattern the Johnson study uses: separating student-level from school-level variables so the share of the disparity attributable to each can be estimated. What is the analog in your context, and where is the default analysis most at risk of mis-locating the mechanism?

3. The case is one of the motivating cases for the CLO Gap attribution. What instance from your work — a device, a formula, a cognitive baseline, an institutional architecture — would you carry as the case-grounded basis for training practitioners in your context to look for the mechanism in the infrastructure they built?

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CASE 98 GOVERNMENT EDUCATION AT SCALE HIGH-STAKES ASSESSMENT 2020

UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Ofqual standardisation algorithm applied to summer 2020 A-level grades following examination cancellation downgraded approximately 39.1% of teacher-estimated grades; results released August 13 2020; algorithm withdrawn August 17 2020 after four days of public protest; Centre Assessment Grades (teacher estimates) substituted; UK House of Commons Education Committee report (2021) documented disproportionate downgrade rates for state-school students

IN BRIEF

With summer 2020 examinations cancelled in response to the COVID-19 pandemic, the UK Office of Qualifications and Examinations Regulation (Ofqual) deployed a statistical standardisation model to produce A-level grades from Centre Assessment Grades (teacher estimates) and Centre-level historical performance. Results were released on August 13,
2020. Approximately 39.1 percent of teacher-estimated grades

were downgraded by the algorithm. State-school students in larger cohorts were downgraded at higher rates than independent- school students in smaller cohorts, because the model relied on Centre-level historical performance more heavily where Centre-level cohorts were larger. After four days of public protest, on August 17, 2020, the algorithm was withdrawn and Centre Assessment Grades were substituted. The Ofqual technical report acknowledges that the standardisation goal was incompatible with population-level fairness given individual-student variance and the dependence of the model on cohort size. The case pairs with Case 133 (Gándara / AERA Open community-college fairness), Case 135 (LiveHint AI bias across foundation models), and Case 97 (Johnson school surveillance).

BACKGROUND

The summer 2020 examination cancellation removed the mechanism by which A-level grades had historically been produced. The Department for Education and Ofqual judged that teacher estimates alone would generate grade inflation incompatible with university admissions and higher-education capacity planning. The decision was to combine teacher Centre Assessment Grades with a statistical standardisation model that drew on Centre-level historical performance to adjust the distribution of grades each Centre's cohort received. The intent of the standardisation was to preserve year-on-year comparability of the national grade distribution; the seam that surfaced under deployment was that population-level comparability and individual-student fairness are not reconcilable when the standardisation mechanism depends on cohort size.[0]

WHAT HAPPENED

The model's mechanics carried the seam. Where a Centre's cohort for a subject was small, the model relied more heavily on the teacher-submitted Centre Assessment Grade and rank order, because small-cohort historical performance was not informative enough to standardise against. Where a Centre's cohort was large, the model relied more heavily on the Centre's historical performance, because the larger cohort gave the standardisation more purchase. The distributional consequence was structural: independent schools and selective settings with small cohorts received grades close to teacher estimates; state schools and large- cohort comprehensive settings received grades pulled downward toward the Centre's historical distribution. The headline result was that approximately 39.1 percent of teacher-estimated grades were downgraded and that the downgrade rate was higher for state-school students in large cohorts.[1]

THE INVESTIGATION

The four-day withdrawal arc is the governance record. Grades were released on Thursday, August 13, 2020. Public protest — students gathering with signs reading "the algorithm stole my future," extensive press coverage of named individual cases, and rapid political pressure — built across the weekend. On Monday, August 17, 2020, Ofqual and the Department for Education announced that A-level and GCSE grades would be reissued at the teacher-submitted Centre Assessment Grade level. The withdrawal was structural — it affected the entire 2020 cohort — and it was rapid in a way that few national-scale algorithmic deployments have been. The House of Commons Education Committee's 2021 report adjudicated the governance record and named the consultation-with-stakeholders failure as the load-bearing one; the technical report had been internally honest about the cohort-size dependence, and the failure was that the dependence had not been surfaced to affected schools and students in advance of deployment.[2]

THE CAPABILITY GAP

The case pairs with Case 133 (Gándara / community-college predictive equity in AERA Open) at the higher-education scale: both cases turn on the question of whether a standardisation or prediction mechanism that is statistically defensible at the population level can be deployed in a way that is defensible at the individual-student level. Pair with Case 135 (LiveHint AI bias across foundation models) for the bias-surfacing thread in education-deployed algorithms. Pair with Case 97 (Johnson school surveillance) for the algorithmic-administration-in-education parallel at a different scale. The Ofqual case is unusual in the casebook because it is the rare deployment that was withdrawn within days under public pressure; most cases in the corpus document deployments that ran for years before withdrawal or that were never withdrawn.[3]

AFTERMATH & REFORM

The technical report's hedge is binding and load-bearing. Ofqual's own document acknowledges that the standardisation goal — preserving year-on-year comparability of the national distribution — was incompatible with population- level fairness given the variance in individual-student circumstances and the cohort-size dependence of the model. The case teaches the change-control-and-disclosure-as- governance-artifacts pattern: an algorithm that is internally documented as carrying a distributional seam cannot be deployed at population scale without the distributional seam being surfaced to the affected population in advance of deployment. The four-day withdrawal is the governance evidence that the population had not been consulted; the structural argument the case anchors is that the consultation is the governance artifact whose absence the withdrawal made visible.

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THE LEARNING ENGINEERING LENS

> *Approximately 39.1 percent of teacher-estimated grades were downgraded; the algorithm was withdrawn within four days under public protest; the technical report was internally honest about the cohort-size dependence and the failure was that the honesty did not travel out of the document.*
>
> EDITORS' SYNTHESIS OF THE OFQUAL TECHNICAL REPORT AND THE HOUSE OF COMMONS EDUCATION COMMITTEE REPORT.

LE INSIGHT

The Ofqual A-level case is the rare national-scale algorithmic deployment withdrawn within days under public pressure. The technical report was internally honest about the cohort-size dependence and the incompatibility of population-level standardisation with individual-level fairness; the four-day withdrawal is the evidence that the internal honesty did not travel out of the document to the affected population in advance of deployment.

LENS APPROACH

Ofqual A-level 2020 is the change-control-and-disclosure-as- governance-artifacts case at national scale (induced 5.4; LENS D4+D3/PT5; CLO-4 and CLO-5). LENS uses it in Domain 4 (Test and Evaluation) for the consultation-with-affected- stakeholders process as the test surface and in Domain 5 (Navigating Sociotechnical Constraints) for the cohort-size dependence as the distributional seam the deployment carried. Pair with Case 133 (Gándara community-college predictive equity), Case 135 (LiveHint AI bias across foundation models), and Case 97 (Johnson school surveillance). The technical report's acknowledgement of incompatibility is the load-bearing hedge.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Surface the distributional seam an internally documented standardisation mechanism carries to the affected population in advance of deployment; the Ofqual technical report was internally honest about the cohort-size dependence, and the governance failure was that the honesty did not travel out of the document.
- Treat the consultation-with-affected-stakeholders process as the change-control artifact a national-scale algorithmic deployment requires; the four-day withdrawal under public protest is the evidence that the consultation had not occurred.
- Pre-specify the individual-student fairness criterion against which a standardisation mechanism will be evaluated, and refuse deployment when the criterion is incompatible with the standardisation goal.

IN OPERATION / AFTER

- Carry the technical report's hedge — "standardisation incompatible with population-level fairness given individual-student variance" — into print without softening; the case's pedagogical value depends on the internal documentation of the seam being visible alongside the public withdrawal.
- Pair in syllabi with Case 133 (Gándara) so the population-level-versus-individual-level fairness tension is taught at both the secondary-to-higher-education transition scale and the community-college transition scale.
- Use the case as the rare example of an algorithmic deployment withdrawn at national scale within days; the four-day withdrawal arc is the curricular target for governance-response speed under public pressure.

REFLECTION QUESTIONS

1. Identify a standardisation or prediction mechanism in your domain whose internal documentation flags a distributional seam. What is the consultation process that would surface the seam to the affected population in advance of deployment, and what would deployment without consultation look like?

2. Specify the individual-level fairness criterion against which a population-level standardisation in your setting would be evaluated. Is the criterion compatible with the standardisation goal, and if not, which governs?

3. The Ofqual case is unusual for the speed of withdrawal — four days. Pick a deployment in your domain whose distributional seam has not yet surfaced publicly, and ask: what would have to be true for a four-day withdrawal arc to be possible, and what would have to be true for the seam to have been surfaced in advance instead?

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CASE 99 CRIMINAL JUSTICE PREDICTIVE ANALYTICS ALGO FAIRNESS 2014 – 2018

COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Northpointe COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) risk-assessment instrument used in pretrial, parole, and sentencing decisions across multiple U.S. jurisdictions; ProPublica's May 2016 investigation reported a 2× higher false-positive rate for Black defendants; Chouldechova (2017) and Kleinberg, Mullainathan & Raghavan (2017) independently formalized the impossibility of simultaneously satisfying calibration and equal false-positive/false-negative rates across groups with unequal base rates

IN BRIEF

Northpointe's COMPAS risk-assessment instrument, deployed in pretrial, parole, and sentencing decisions across multiple U.S. jurisdictions, became the central case in the algorithmic-fairness literature after ProPublica's May 2016 investigation reported that the instrument produced false-positive rates approximately twice as high for Black defendants as for white defendants. Northpointe's response argued that COMPAS satisfied predictive parity — that within each risk score, the rate of subsequent recidivism was approximately equal across groups. Both parties were correct by their respective definitions. Chouldechova's 2017 paper and Kleinberg, Mullainathan, and Raghavan's 2017 paper independently formalized the impossibility result: when base rates of the outcome differ across groups, calibration (predictive parity) and equal false-positive and false- negative rates cannot be simultaneously satisfied except in degenerate cases. The case pairs with Case 94 (Bartlett mortgage — fairness through unawareness fails), Case 107 (Coots — competing fairness definitions), and Case 155 (SyRI). The impossibility result is the load-bearing teaching point.

BACKGROUND

COMPAS is a proprietary risk-assessment instrument developed by Northpointe (now Equivant) and used in pretrial release, parole, and sentencing decisions across many U.S. jurisdictions through the 2010s. The instrument scores a defendant on a scale of recidivism risk based on a questionnaire covering criminal history, employment, education, family circumstances, and attitudes. The score is then surfaced to judges, parole boards, and pretrial services as one input among several into consequential decisions about the defendant's liberty. The deployment scale was large enough that the instrument was the central target of the contemporary algorithmic-fairness literature when the first sustained external audit was published.[0]

WHAT HAPPENED

ProPublica's May 2016 investigation, led by Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner, audited COMPAS scores against subsequent recidivism for approximately 7,000 defendants in Broward County, Florida. The headline finding was that among defendants who did not go on to reoffend within two years, Black defendants had been scored as high-risk at roughly twice the rate of white defendants — a false-positive-rate disparity. Northpointe's response, authored by William Dieterich, Christina Mendoza, and Tim Brennan, argued that COMPAS satisfied predictive parity (also called calibration): within each risk score band, the rate of subsequent recidivism was approximately equal across racial groups. The defendant assigned a "high risk" score had approximately the same probability of reoffending whether Black or white. The two findings appear contradictory but are not; they describe two different fairness criteria applied to the same instrument.[1]

THE INVESTIGATION

Chouldechova's 2017 paper in Big Data and Kleinberg, Mullainathan, and Raghavan's 2017 ITCS paper independently formalized the impossibility result. Calibration within groups (predictive parity) and equality of false-positive and false-negative rates across groups cannot be simultaneously satisfied when the base rates of the outcome differ across the groups, except in degenerate cases. The base rate of subsequent recidivism in the Broward County data was higher for Black defendants than for white defendants. Under that base-rate difference, an instrument calibrated equally across groups will produce unequal false-positive rates, and an instrument with equal false- positive rates across groups will produce miscalibration. The mathematics is binding. The choice between fairness criteria is a normative and governance question, not a technical one.[2]

THE CAPABILITY GAP

The case pairs with Case 94 (Bartlett mortgage discrimination) for the fairness-through-unawareness-fails thread: removing protected attributes from training data does not eliminate disparate-impact concerns when the remaining features carry protected-attribute signal. Pair with Case 107 (Coots) for the competing-fairness-definitions thread at a different domain and scale. Pair with Case 155 (SyRI) for the governance-objection-correct-in-advance complement; in COMPAS the objection surfaces in the auditing record, in SyRI the objection succeeded in court before population- scale harm was produced. The COMPAS case is the central reference in the contemporary algorithmic-fairness literature because the impossibility result was formalized against its audit record; the literature's subsequent decade of work on fairness criteria operates inside the constraint the case made legible.[3]

AFTERMATH & REFORM

The hedges the case carries are load-bearing. Both Northpointe and ProPublica are correct by their respective definitions, and the impossibility result formalizes the tension rather than resolving it. The case does not teach that COMPAS is fair or that COMPAS is unfair; it teaches that the choice between fairness criteria is governance and normative work that the deployment did not surface to the affected jurisdictions or to the defendants whose liberty depended on the score. The CLO on fairness beyond omission is anchored by the case in its mature form — the impossibility result requires the deploying institution to choose, document, and disclose which fairness criterion the instrument optimizes, and to make the trade-off legible to the people the criterion does not protect.

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THE LEARNING ENGINEERING LENS

> *Both Northpointe and ProPublica were correct by their respective definitions of fairness; the impossibility result is that calibration and equal false-positive rates cannot be simultaneously satisfied when base rates differ across groups, except in degenerate cases.*
>
> EDITORS' SYNTHESIS OF THE COMPAS AUDIT RECORD AND THE 2017 IMPOSSIBILITY-RESULT PAPERS.

LE INSIGHT

COMPAS is the central reference in the contemporary algorithmic-fairness literature because the impossibility result was formalized against its audit record. Both the ProPublica finding (unequal false-positive rates) and the Northpointe response (predictive parity within risk scores) are correct by their respective definitions; the Chouldechova and Kleinberg/Mullainathan/Raghavan results show that the two cannot be simultaneously satisfied when base rates differ. The choice between fairness criteria is governance work, not technique.

LENS APPROACH

COMPAS is the impossibility-result case at consequential- decision scale (induced 8.4; LENS D4+D3/PT6; CLO-4 and CLO-5). LENS uses it in Domain 4 (Test and Evaluation) for the multi-criterion-audit discipline and in Domain 5 (Navigating Sociotechnical Constraints) for the surfacing-bias-through-governance-not-just-technique anchor. Pair with Case 94 (Bartlett mortgage — fairness through unawareness fails), Case 107 (Coots — competing fairness definitions), and Case 155 (SyRI governance-objection- correct precedent). The impossibility result is the load- bearing teaching point; both Northpointe and ProPublica are correct by their respective definitions.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Choose, document, and disclose the fairness criterion the instrument optimizes for in advance of deployment; the impossibility result requires the deploying institution to make the choice and to make the trade-off legible to the people the criterion does not protect.
- Audit the deployed instrument against multiple fairness criteria simultaneously; the COMPAS record demonstrates that an instrument can satisfy predictive parity and fail equality of false-positive rates at the same time, and the audit must surface both.
- Treat the base-rate difference across groups as a governance fact, not a technical artifact; the difference is what makes the impossibility binding, and

IN OPERATION / AFTER

- Carry the impossibility result into print as the load-bearing teaching point; the case does not teach that COMPAS is fair or that COMPAS is unfair, and the editorial framing must preserve the formal constraint that both audit findings instantiate.
- Pair in syllabi with Case 94 (Bartlett) so the fairness-through-unawareness-fails thread and the impossibility-of-multiple-criteria thread are taught together as complementary structural arguments about disparate impact.
- Use the case to anchor the fairness-beyond-omission CLO; the curricular target is the discipline of choosing and disclosing the fairness criterion when the impossibility result rules out satisfying all of them simultaneously.

pretending it can be eliminated is the rhetorical move the case teaches to refuse.

REFLECTION QUESTIONS

1. Identify a risk-assessment instrument in your domain whose fairness criterion has not been chosen and disclosed in advance of deployment. What are the candidate criteria, and which of them are jointly satisfiable given the base-rate differences across affected groups?

2. Specify the multi-criterion audit you would run against a deployed instrument. What is the format in which the audit surfaces incompatibility findings to the deploying institution, and what is the documented decision rule when incompatibility is found?

3. The impossibility result is mathematical; the choice between criteria is governance and normative work. Pick a setting in your domain and ask: who has authority to make the choice, who is accountable for documenting and disclosing the trade-off, and to whom is the trade-off disclosable?

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CASE 100 EDUCATION AT SCALE PREDICTIVE ANALYTICS ALGORITHMIC EQUITY 2012 – 2024

Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Wisconsin Department of Public Instruction's Dropout Early Warning System (DEWS) deployed since 2012, producing risk scores for approximately 200,000 sixth- through ninth-grade students per year; Perdomo, Britton, Hardt, & Abebe FAccT 2025 regression-discontinuity analysis across ~10 years of data found the analysis cannot rule out zero treatment effect on graduation; DPI's own 2021 internal equity audit and The Markup's 2023 investigation found the system was less accurate for Black, Hispanic, and English-learner students

IN BRIEF

The Wisconsin Department of Public Instruction has operated the Dropout Early Warning System (DEWS) since 2012, producing dropout-risk scores for approximately 200,000 sixth- through ninth-grade students across the state each year. Two evidence streams converge on the deployment. Perdomo, Britton, Hardt, and Abebe's 2025 FAccT paper used a regression-discontinuity design on approximately ten years of DEWS-and-graduation data and found that the analysis cannot rule out zero treatment effect on graduation outcomes for students above versus below the DEWS risk threshold — the prediction did not appear to trigger interventions that affected the predicted outcome. The Wisconsin DPI's own 2021 internal equity audit, titled "Is DEWS Fair?", found that the system was less accurate for Black, Hispanic, and English-learner students, and the agency continued to operate it unchanged. The Markup's 2023 investigation by Todd Feathers documented the disparate-impact finding and the agency's response. Both findings are load-bearing. The case pairs with Case 97 (Johnson school surveillance), Case 133 (Gándara community-college predictive equity), and Case 106 (Purdue Course Signals — reverse causality). This case carries both the peer-reviewed and journalism-investigation evidence streams in one entry rather than parallel ones.

BACKGROUND

The Wisconsin Dropout Early Warning System was developed by the Wisconsin Department of Public Instruction in collaboration with researchers from the University of Wisconsin – Madison and deployed in 2012. The system ingests student-level administrative data — attendance, course performance, behavioral incidents, mobility, and demographics — and produces a dropout-risk score for each student in grades six through nine. The score is surfaced to school administrators, counselors, and teachers, with the operational theory that the early warning enables school-level interventions — additional counseling, tutoring, mentoring, or family contact — that improve the student's graduation trajectory. The system has operated at scale for more than a decade and produces risk scores for approximately 200,000 students each year.[0]

WHAT HAPPENED

Perdomo, Britton, Hardt, and Abebe's 2025 paper at FAccT (also available as arXiv 2304.06205) is the load-bearing causal-inference evidence on the deployment. The researchers used a regression-discontinuity design, exploiting the DEWS risk-threshold cutoff to compare students just above the threshold (classified as high- risk and surfaced to the school's intervention surface) against students just below it (not so surfaced), controlling for the continuous risk score itself. The analysis covered approximately ten years of DEWS data and the corresponding graduation outcomes. The headline finding is that the analysis cannot rule out zero treatment effect of being above the DEWS threshold on subsequent graduation. The confidence interval on the treatment effect includes zero, and the point estimate is small enough that the prediction's operational role — triggering school-level interventions that would change the predicted outcome — is not evidenced in the ten-year record. The hedge the paper preserves is binding: the RDD analysis cannot establish a negative finding either (the interval includes small positive effects), but it can and does establish that the prediction-triggers- intervention-that-changes-outcome theory is not supported in the deployment record.[1]

THE INVESTIGATION

The Wisconsin DPI's own 2021 internal equity audit, titled “Is DEWS Fair?”, is the load-bearing disparate- impact evidence on the deployment. The audit found that DEWS was less accurate for Black, Hispanic, and English- learner students than for white and non-English-learner students — that the false-positive and false-negative rates diverged across student-population subgroups in ways the audit characterized as inconsistent with fairness. The agency's own response to its own audit was to continue operating DEWS unchanged. The Markup's 2023 investigation by Todd Feathers documented both the finding and the agency's response, and the journalism- investigation evidence stream entered the public record in 2023. The case carries both evidence streams — the peer-reviewed causal-inference null and the agency-and- investigation disparate-impact finding — in one entry, rather than as parallel cases, because the deployment is one deployment and the two streams describe complementary structural problems with it.[2]

THE CAPABILITY GAP

The case pairs with Case 97 (Johnson school surveillance) for the algorithmic-public-administration-in-education parallel; both cases involve administrative-data predictions deployed against student populations and both surface disparate-impact concerns at the deployment surface. Pair with Case 133 (Gándara community-college predictive equity) for the predictive-equity-in-education thread at adjacent population scale. Pair with Case 106 (Purdue Course Signals) for the reverse-causality and null-effect thread in education predictive analytics; the Purdue case named the same conceptual problem that the Perdomo et al. analysis evidences at population scale. DEWS is the rare deployment in the corpus that is evidenced from both the peer-reviewed causal-inference direction and the disparate-impact-investigation direction, and the editorial decision to carry both streams in one entry reflects the structural unity of the deployment.[3]

AFTERMATH & REFORM

The hedges the case carries are load-bearing and both streams are preserved. The Perdomo et al. RDD analysis cannot rule out zero treatment effect on graduation across approximately ten years of data — the prediction- triggers-intervention-that-changes-outcome theory is not supported in the deployment record. The DPI's own equity audit found that DEWS is less accurate for Black, Hispanic, and English-learner students, and the agency continued operating the system unchanged after the audit's release. Both findings are binding and travel together. The CLO on designing predictions to trigger support rather than gatekeeping is anchored by the case at the deployment-with-null-causal-effect-and-disparate-accuracy seam: a prediction system operating at population scale for more than a decade, without evidence that the prediction triggers outcome-changing intervention and with evidence that the prediction's accuracy varies across protected-attribute subgroups, is the deployment form whose persistence the case asks the reader to account for.

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THE LEARNING ENGINEERING LENS

> *The RDD analysis cannot rule out zero treatment effect on graduation across approximately ten years of data; the agency's own equity audit found DEWS is less accurate for Black, Hispanic, and English-learner students; the agency continued operating the system unchanged.*
>
> EDITORS' SYNTHESIS OF PERDOMO ET AL. (2025, FACCT), THE WISCONSIN DPI INTERNAL EQUITY AUDIT (2021), AND THE MARKUP INVESTIGATION (FEATHERS, 2023).

LE INSIGHT

Wisconsin DEWS is the load-bearing case for a prediction system operating at population scale for more than a decade without evidence that the prediction triggers outcome- changing intervention and with evidence that the prediction's accuracy varies across protected-attribute subgroups. Both the peer-reviewed null and the agency-audit disparate-impact finding travel together; the case carries both evidence streams in one entry rather than parallel ones.

LENS APPROACH

Wisconsin DEWS is the designing-predictions-to-trigger- support-not-gatekeeping case at population scale (induced 8.3; LENS D4+D3/PT6; CLO-4 and CLO-5). LENS uses it in Domain 4 (Test and Evaluation) for the causal-inference- on-multi-year-deployment discipline and in Domain 5 (Navigating Sociotechnical

Constraints) for the equity- audit-as-binding-input anchor. Pair with Case 97 (Johnson school surveillance), Case 133 (Gándara community-college predictive equity), and Case 106 (Purdue Course Signals reverse causality). Both the peer-reviewed null and the journalism-and-agency-audit disparate-impact finding are load-bearing.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pre-specify the causal-inference design that will evaluate whether the prediction triggers outcome-changing intervention; the Perdomo et al. RDD is the deployment-evidence form that the prediction-triggers-support theory requires for verification at multi-year horizons.
- Treat the agency's own equity audit as a binding governance input, not as an internal document; the DPI's "Is DEWS Fair?" audit was internally honest about the disparate-accuracy finding, and the case's load-bearing observation is the agency's continued operation of the system unchanged.
- Design predictions to trigger support, not gatekeeping; the case's central CLO anchor is the discipline of building the intervention surface that the prediction is meant to trigger, and of verifying — not assuming — that the prediction's operational role produces the outcome change the deployment theory requires.

IN OPERATION / AFTER

- Carry both evidence streams into print as load-bearing and complementary; the case's pedagogical value depends on the peer-reviewed null and the agency-audit disparate-impact finding traveling together rather than being separated into parallel entries.
- Pair in syllabi with Case 97 (Johnson) and Case 106 (Purdue) so the algorithmic-public-administration-in-education and reverse-causality threads are taught alongside the multi-year-deployment-with-null-causal-effect finding.
- Use the case as the anchor for the designing-predictions-to-trigger-support CLO; the curricular target is the multi-year deployment record that demonstrates the gap between the prediction's operational theory and the evidence the deployment produces.

REFLECTION QUESTIONS

1. Identify a prediction system in your domain that has operated at scale for years without RDD-style causal evaluation of whether the prediction triggers outcome-changing intervention. What would the causal-evaluation design look like in your setting, and what would a null finding require of the deployment?

2. Specify the equity-audit-as-binding-input discipline you would apply when an internal audit finds disparate accuracy across protected-attribute subgroups. What is the documented decision rule for modifying or withdrawing the system, and what is the rule when the agency continues operation unchanged?

3. The case carries both the peer-reviewed null and the disparate-impact-investigation finding in one entry. Pick a deployment in your domain that has evidence streams in tension, and ask: what is the editorial discipline that carries the streams together rather than separating them, and what would separating them lose?

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CASE 101 GOVERNMENT 2014

VA Wait-Time Scandal

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Veterans died waiting for care; 300,000+ on waiting lists or waiting 6+ months; staff falsified records

IN BRIEF

In 2014 the VA Inspector General found that staff at the Phoenix VA — and then nationwide — had created secret waiting lists and falsified appointment data to hide that veterans were waiting weeks or months for care; some died waiting while the system reported success. A 14-day access target, unrealistic given staffing, was met by hiding reality rather than surfacing it. The warning signs ran back fifteen years: GAO had flagged data-reliability problems since 2000, the IG since 2005, with no systemic fix — and schedulers, the staff operating the measurement, are among the VA's highest-turnover roles, so the institution kept losing the knowledge even to see it was lying to itself. The VA case is the book's canonical example of normalization of deviance applied to measurement itself.

BACKGROUND

The Veterans Health Administration measured access to care against a 14-day appointment target — a target that, given staffing, was often unrealistic to meet honestly. Schedulers, the staff who operate that measurement, are among the VA's highest-turnover positions.[0] The target functioned as the headline number leadership watched, so the pressure to show 14-day compliance bore down hardest on the very front-line role least equipped, through constant churn, to record the access data accurately or to question what the number was leaving out.

WHAT HAPPENED

In 2014 the VA Office of Inspector General found that staff at the Phoenix VA — and then across the system — had created secret waiting lists and falsified appointment data to hide that veterans were waiting weeks or months for care. Some veterans died waiting, inside a system that, by its own metrics, was succeeding.[1] The secret lists were the mechanism by which an impossible target was reconciled with reality: official records showed appointments inside 14 days while the true wait accumulated off the books, so the more the metric was gamed, the more confidently the reporting line above it declared the access problem solved.

THE INVESTIGATION

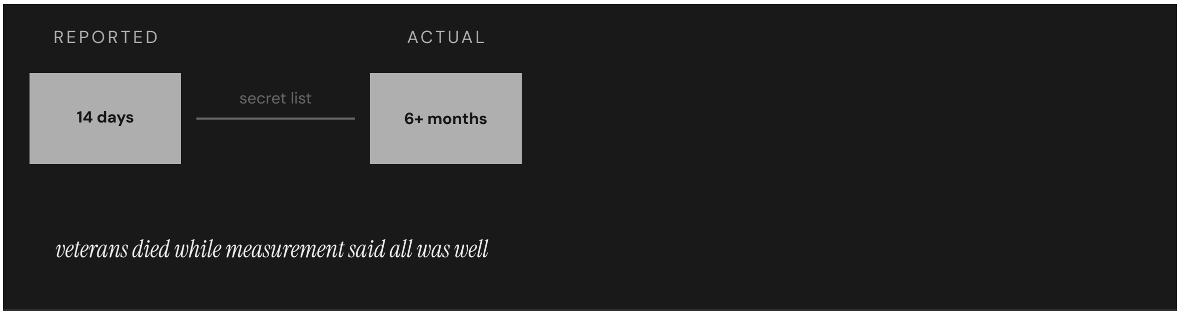
The warning signs went back fifteen years: GAO had flagged data-reliability concerns since 2000, and the VA IG had identified problems in 2005, 2007, and 2008, with the incoming administration warned in 2008 — none of which produced systemic change.[2] GAO named scheduler training as a root cause: with schedulers among the top-ten highest-turnover roles, the institution perpetually lost the knowledge required even to run the measurement honestly, and five years on still reported data-reliability concerns.[3] Each warning had named a real defect in how access was recorded, yet because the schedulers who held the practical knowledge of the system kept turning over, every wave of findings landed on a workforce that had to relearn the instrument from scratch, so the same defect resurfaced report after report.

THE CAPABILITY GAP

This is normalization of deviance applied to measurement itself. When the measurement system cannot capture reality — and the people operating it churn before the gaming can be unlearned — the gap between reported and actual performance becomes invisible. Veterans died inside a system whose numbers said it was fine, which is the lethal form of the evidence problem this book treats as a design failure, not a reporting one.[4] An institution that cannot retain the staff who run its measurement loses not only accuracy but the memory that the number was ever wrong, so the gaming stops registering as deviance at all and hardens into the ordinary way the work is done.

AFTERMATH & REFORM

The scandal forced resignations, the Veterans Access, Choice, and Accountability Act (2014), and a long, uneven effort to rebuild scheduling and data integrity.[5] Its lesson is that decision-grade evidence is a design requirement: an institution must be able to surface its own failures without relying on the very people incentivized — and too transient — to hide them. The Choice Act bought access by routing care outside the VA, but the underlying capability — a measurement the institution could trust even as its schedulers churned — was the harder thing to rebuild, and the data-reliability concerns that persisted for years afterward show why.



SCHEDULERS FALSIFIED RECORDS; THE MEASUREMENT SYSTEM HID THE GAP

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THE LEARNING ENGINEERING LENS

> *Schedulers are among the top ten highest-turnover positions in the VA.*
>
> DEBRA DRAPER (GAO DIRECTOR OF HEALTH CARE), HOUSE VA COMMITTEE TESTIMONY, GAO-19-687T, JULY 2019

LE INSIGHT

The VA case is the canonical example of measurement failure as a capability failure. The system that should have surfaced the gap instead generated reports that hid it. The turnover among schedulers — the human capability operating the measurement instrument — meant the institution lost the knowledge to even notice it was lying to itself. The case stands as the strongest argument in this book for treating decision-grade evidence as a design requirement, not a reporting requirement.

LENS APPROACH

LENS treats the VA case in LEN 4 as the canonical evidence-gap case (the measurement system itself was the source of harm), in LEN 7 as a governance failure (multiple warnings unactioned over fifteen years), and in LEN 8 as a knowledge-loss case via turnover.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Set access targets against actual staffing capacity, so the metric cannot be met only by falsifying it.
- Design the scheduling instrument so correct data entry is the path of least resistance, not a discipline that churning staff must be retrained into.
- Build an independent read on real wait times — separate from the staff incentivized to report them — into the measurement architecture from the start.

IN OPERATION / AFTER

- Audit reported access against an out-of-band signal (direct veteran survey, third-party booking records) with authority to act when the two diverge.
- Treat scheduler turnover as a measurement risk: monitor it, and protect the knowledge of how to run the instrument honestly against constant churn.
- Track whether old data-reliability findings keep recurring — a repeat finding is the signal that the institution is relearning the same gap rather than closing it.

REFLECTION QUESTIONS

1. Identify a measurement system in your domain that is also operated by a high-turnover role. What is the institutional risk that the system stops measuring reality?
2. Design the evidence pipeline that would have surfaced the Phoenix VA gap without relying on the people who were gaming the metrics.
3. The 14-day target was unrealistic given staffing, so it was met by hiding reality. Identify a target in your organization that the people measured against it cannot honestly meet — and design the correction that surfaces the gap rather than burying it.

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CASE 102 HEALTHCARE 1999 – PRESENT

Medical Errors as Systemic Failure

T Training Gap H Human-System Interface N Normalization of Deviance
K Knowledge & Institutional Memory G Governance & Trust

IMPACT IOM 1999 estimate of 44,000–98,000 US deaths/year from medical error; Makary & Daniel (2016) estimate of ~250,000 deaths/year — substantively contested on methodological grounds; 2023 NEJM inpatient-harm study confirms persistence; 25-year reform arc with bounded successes and an unmoved population count

IN BRIEF

Medical error in the United States is not a single incident but a systemic condition that the system's own measurement instruments cannot see. The Institute of Medicine's To Err Is Human (1999) raised the alarm with a 44,000–98,000 annual-deaths estimate and reframed harm as a systems problem rather than an individual one; Crossing the Quality Chasm (2001) and the 2015 Improving Diagnosis in Health Care sequel extended the program. In 2016, Makary and Daniel of Johns Hopkins published a BMJ analysis estimating 250,000 deaths a year — which would make medical error the third leading cause of death behind heart disease and cancer. Death certificates do not record medical error as a cause, so the problem is structurally invisible to the systems meant to track it. The 250,000 figure has been substantively contested on methodological grounds (Shojania & Dixon-Woods, BMJ Quality & Safety, 2017; Carr in Health Affairs), a dispute that is itself a worked example of the gap-attribution problem. The field's 25-year arc shows bounded interventions that work — computerized order entry, handoff protocols, the WHO surgical checklist (Case 109), the Keystone ICU project (Case 71), TeamSTEPPS (Case 173) — alongside a population-scale mortality count that has resisted both intervention and precise estimation.

BACKGROUND

The Institute of Medicine's To Err Is Human (1999) was the field-defining moment: it estimated 44,000–98,000 deaths annually from medical error in US hospitals — at the lower bound, more Americans than die in motor-vehicle accidents — and made the explicit case that the problem was a systems problem, not an individual-clinician problem. The 2001 sequel Crossing the Quality Chasm set six aims for the redesign of care, and the 2015 sequel Improving Diagnosis in Health Care extended the framing to diagnostic error. The framing of harm as systemic mattered: it created the cultural permission for non-punitive incident reporting, root-cause analysis, and the broader patient-safety movement that followed.[0] Yet modern medicine's core mortality-measurement instrument — the death certificate — records a proximate physiological cause and has no field for medical error. Because the certificate is keyed to an ICD billing taxonomy built for disease, a death set in motion by a care-process breakdown is recorded under whatever organ ultimately failed, and the causal role of the system disappears into the physiology.

WHAT HAPPENED

In 2016, surgeon Martin Makary and Michael Daniel published an analysis in the BMJ estimating that medical errors cause more than 250,000 deaths a year in the United States — which would rank third behind heart disease and cancer. Their core claim was that "people don't just die from heart attacks and bacteria, they die from system-wide failings and poorly coordinated care." By relocating the cause from the individual clinician to the coordination of care, the framing recast a ledger of isolated mistakes as a single population-scale failure mode the existing statistics were never built to count. The number itself was computed by extrapolating from four prior studies — the IOM 1999 estimate, the 2010 OIG Medicare adverse-events study, the Landrigan 2010 NEJM study, and the Classen 2011 Global Trigger Tool study — to the contemporary inpatient population, an extrapolation the authors acknowledged as approximate.[1]

THE INVESTIGATION

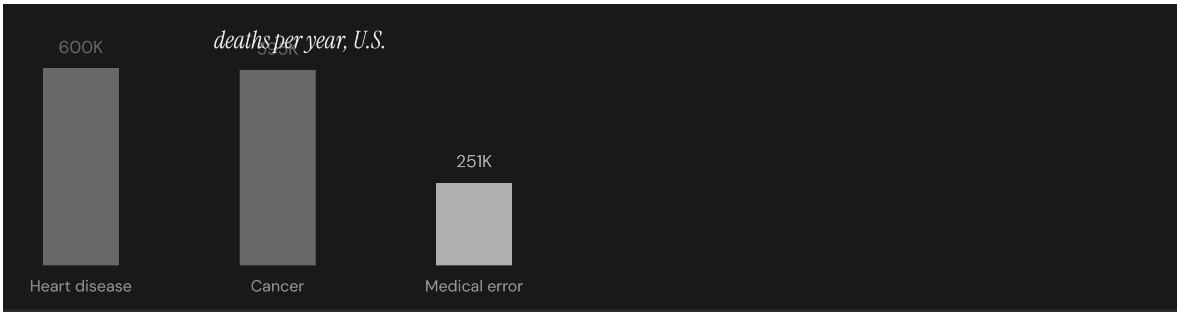
The 250,000 estimate was substantively contested almost immediately. Shojania and Dixon-Woods, writing in BMJ Quality & Safety in 2017, challenged both the extrapolation and the attribution method, arguing that counting deaths "due to" error is far harder than a single headline number implies. A Health Affairs commentary by Carr and a series of letters in the BMJ pressed the same point: the source studies use different definitions of "preventable death," the extrapolation rests on prior estimates already known to be unstable, and the ranking against CDC cause-of-death categories is methodologically incompatible (CDC counts use ICD codes; the Makary estimate does not).[2] The objection turned on counterfactual attribution — how confidently one can say a frail, already-dying patient would have survived but for the error — a judgment that resists the clean tallying a headline number demands. The dispute is itself a worked example of the gap-attribution problem: how much of a counted death is the learning system, how much the system design, how much the underlying disease.[3]

THE CAPABILITY GAP

The deeper failure is one of measurement: a system that cannot see its own failure modes cannot manage them. With no field on the certificate, no reliable count, and no agreed attribution method, every safety program competes for resources against a harm that the official record renders invisible, so even effective interventions struggle to prove their worth at population scale. The bounded interventions that have worked — computerized physician order entry, structured handoff protocols (I-PASS), the WHO Surgical Checklist (Case 109), the Keystone ICU central-line bundle (Case 71), TeamSTEPPS (Case 173) — each move a specific harm in a specific setting, and each can prove it. What none can prove is movement on the population count, because the population count does not exist in a form precise enough to be moved. The missing capability is an instrument that captures medical error as a tracked cause of harm — and an attribution method robust enough that the resulting number can guide intervention rather than fuel a methodological stalemate.[4]

AFTERMATH & REFORM

The 25-year arc since To Err Is Human is the case's most important teaching artifact. The IOM framing catalyzed a patient-safety movement, an Agency for Healthcare Research and Quality patient-safety program, and the bounded interventions noted above. Yet later work, including the 2023 NEJM study by Bates et al. of inpatient harm across eleven Massachusetts hospitals, confirms that the problem persists at scale: about a quarter of admissions involved an adverse event, and roughly a quarter of those events were preventable. The system has not built the instrument the original report implied it would need — no national active-surveillance system for inpatient harm, no death-certificate field for care-process failure, no agreed attribution method for "deaths due to error." The interventions exist; the measurement and the implementation still lag, and the headline mortality kept escaping measurement even as bounded harms fell. The case is the canonical worked example of the gap-attribution problem at population scale, and a standing reminder that an instrument the system cannot see through is one it cannot govern.[5]



THIRD LEADING CAUSE OF DEATH — AND STRUCTURALLY INVISIBLE

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> *People don't just die from heart attacks and bacteria, they die from system-wide failings and poorly coordinated care.*
>
> MARTIN MAKARY, PRESS STATEMENTS ACCOMPANYING MAKARY & DANIEL, BMJ (2016)

LE INSIGHT

The Makary data is the anchor evidence for the LENS argument because of its scale, its provenance (Johns Hopkins), and its framing — system failure rather than individual error. The seventeen years between **To Err Is Human** and Makary's reassessment is also the implementation gap of Case 34 in another guise — and the cost of leaving it open is measured in lives at population scale.

LENS APPROACH

Medical error is the central evidence anchor of the curriculum (induced 2.1; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation; CLO-4) for the foundational measurement problem: the death certificate cannot record the failure mode it is supposed to govern, and the contested mortality estimates show what happens when an instrument's gap is filled by extrapolation. LENS uses it in Domain 1 (Systems Analysis) as the foundational problem statement of the program and in Domain 5 (Navigating Sociotechnical Constraints) for industry-level institution building. The 25-year arc pairs the case with the WHO Surgical Checklist (Case 109), Keystone ICU (Case 71), and TeamSTEPPS (Case 173) as the bounded interventions that worked; pair with Vioxx (Case 105) at the population-scale-surveillance layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Give the mortality-recording instrument an explicit field for care-process failure, so a system-caused death is captured, not absorbed into the proximate cause.
- Specify the attribution method up front — the counterfactual test for "due to" error — so the count guides action rather than collapsing into dispute.
- Pair each safety intervention with the population measure it should move, gating reforms on demonstrated effect.

IN OPERATION / AFTER

- Audit reported mortality against an independent count of care-related harm.
- Use active surveillance of inpatient harm rather than waiting for the death certificate.
- Hold the count and the intervention to the same measurement discipline, closing the implementation gap.

REFLECTION QUESTIONS

1. Identify a measurement instrument in your domain that systematically fails to capture the failure modes it should be designed to surface. What would it cost to fix?
2. Two hundred fifty thousand deaths a year is the third leading cause of death in the U.S. Design the measurement and intervention regime that would shift the curve over a five-year horizon. Estimate the deliverable and the evidence.
3. Makary and Shojania disagreed not on whether error kills but on how to attribute a death to it. Specify an attribution method robust enough to survive that dispute — and name who would hold the authority to act on the number it produces.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · safety science & organizational analysis · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · incident analysis · safety dashboards · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

CASE 103 GOVERNMENT TECHNOLOGY 2003 – 2012

LIBOR Manipulation

G Governance & Trust N Normalization of Deviance

IMPACT Major banks fined more than $9B; the benchmark underlying ~$350T of contracts was manipulable; replaced by SOFR

IN BRIEF

LIBOR — the London Interbank Offered Rate — was the benchmark interest rate referenced in roughly $350 trillion of financial contracts, yet it was set each day not from observed transactions but from a small panel of banks' estimates of what they would pay to borrow. For about a decade, banks systematically shaded those estimates to favor their own derivatives positions; the coordination was later laid bare in subpoenaed trader-and-submitter messages. The capability gap was in the design of the benchmark itself: it asked for declarations, not observations, and the architecture invited gaming. Regulators levied more than $9 billion in fines, and LIBOR was eventually replaced by SOFR, a rate built from actual transactions. The reform changed the unit of measurement, not just the rules.

BACKGROUND

LIBOR was the benchmark interest rate underlying an estimated $350 trillion in loans, mortgages, and derivatives worldwide. It was set daily by a panel of large banks, each submitting an estimate of the rate at which it could borrow — a self-reported figure rather than a record of any transaction that had actually occurred. Because the same banks that set the rate held derivatives whose value moved with it, the instrument asked the parties with the most to gain to supply the very numbers that would decide their gains.[0]

WHAT HAPPENED

For roughly a decade from the early 2000s, traders at multiple banks asked their firms' rate submitters to shade the daily LIBOR figure up or down to benefit the banks' derivatives books. The requests were routine and documented in internal messages later subpoenaed by regulators — casual, repeated, and unconcealed, which is the signature of conduct the participants did not believe could be detected. During the 2008 crisis, some banks also lowballed submissions to appear healthier than they were, bending the rate to mask their own funding stress as well as to profit.[1]

THE INVESTIGATION

Investigations by the U.S. Department of Justice, UK regulators, and others produced settlements with Barclays, UBS, Deutsche Bank, RBS, and others totaling more than $9 billion. The UK Treasury commissioned the Wheatley Review, which concluded that the benchmark's reliance on subjective estimates rather than transactions was the structural flaw that made manipulation possible. The review located the fault in the instrument rather than only in the individuals who exploited it, treating the gaming as the predictable output of a design that left the door open.[2]

THE CAPABILITY GAP

The capability that was missing was the basic architectural choice to measure observations rather than declarations. A benchmark built on what banks said they would pay is gameable by anyone willing to misstate; a benchmark built on transactions that actually occurred is not. The gaming, once it began, was systematic rather than occasional — exactly what the design invited — and because the figure flowed into hundreds of trillions in contracts, a small shading of the submission moved enormous sums while leaving no trace in any executed trade.[3]

AFTERMATH & REFORM

LIBOR was phased out and replaced by SOFR, the Secured Overnight Financing Rate, which is constructed from observed overnight Treasury repo transactions. The transition took roughly a decade and required coordination across regulators in multiple jurisdictions; the reform re-engineered the measurement at the architecture level rather than merely policing the old one. By anchoring the benchmark to trades that actually settle, the redesign removed the discretion that had been the lever for manipulation, rather than asking submitters to exercise that discretion more honestly.[4]

LIBOR	SOFR
estimates	transactions

the reform changed the unit of measurement, not the regulator

LIBOR — AN INSTRUMENT WHOSE DESIGN INVITED THE GAMING

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THE LEARNING ENGINEERING LENS

> *The architecture of LIBOR invited the manipulation it experienced.*

PARAPHRASING THE WHEATLEY REVIEW OF LIBOR, 2012

LE INSIGHT

LIBOR is the canonical financial-system case for an instrument whose design invites the gaming it then experiences. The capability that was missing was the basic architectural choice to measure observations rather than declarations. The reform — SOFR — re-engineered the measurement at the architecture level.

LENS APPROACH

LIBOR is a measurement-gaming exemplar (induced 2.2): the benchmark was a self-reported estimate panel with no independent instrument behind the number, so the parties being measured authored the measurement, and the figure flowed into hundreds of trillions in contracts. LENS binds it to the measurement-gaming cluster alongside Wells Fargo (Case 91) and Volkswagen (Case 92): in each, the reporting party controlled the input it was judged on. The design lesson is to build an instrument the reporting party cannot author — anchor the measure in observations the measured party cannot supply, rather than in declarations they can.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Anchor the benchmark in observed, settled transactions from the outset, so the figure cannot be set by anyone's declaration of what they would pay.
- Separate the parties who supply the input from the parties who profit from its level, eliminating the conflict of interest the panel design built in.
- Stress-test the instrument against an adversary who wants to move it: if a small misstatement shifts large contract value undetectably, redesign before fielding.

IN OPERATION / AFTER

- Audit submissions against independent transaction data so a divergence between what is reported and what actually trades surfaces the gaming early.
- Monitor submitter-trader communications and position-aligned shading as a leading indicator, rather than waiting for a subpoena to reveal it.
- Sustain the reform at the architecture level — keep the benchmark transaction-based — rather than relying on policing discretion that can quietly return.

REFLECTION QUESTIONS

1. Identify a benchmark in your domain that is constructed from declarations rather than observations. What gaming pressure does it experience?
2. Design the observation-based replacement for that benchmark.
3. The same panel banks set LIBOR and held positions that moved with it. Identify a measurement in your domain where the party supplying the figure benefits from its level, and design the separation that would remove the conflict.

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CASE 104 EDUCATION AT SCALE GOVERNMENT 2009 – 2015

Atlanta Public Schools Cheating Scandal

G Governance & Trust N Normalization of Deviance

IMPACT ~180 educators implicated; 35 indicted; 11 convicted under RICO statute; foundational U.S. education measurement-gaming case

IN BRIEF

Under a celebrated superintendent, Atlanta Public Schools was held up nationally as a high-performing urban district. The performance was substantially fabricated. A state special investigation, supported by the Georgia Bureau of Investigation, and reporting by the Atlanta Journal-Constitution found that for several years educators across dozens of schools had systematically erased and corrected students' answers on state standardized tests. The cheating was organized — principals pressured teachers, staff held "erasure parties" after testing days — and the incentive system rewarded the gaming with bonuses and promotions. Around 180 educators were implicated, 35 indicted, and 11 convicted under Georgia's racketeering statute. The capability gap lay in the measurement architecture: the institution being measured operated the instrument that measured it, with no independent audit.

BACKGROUND

During the 2000s, Atlanta Public Schools under superintendent Beverly Hall was celebrated nationally for rapid gains on Georgia's high-stakes standardized tests. Bonuses, public recognition, and job security were tied directly to those scores, and the district administered and reported the very tests on which it was being judged. With the reward attached to the number and the number produced in-house, the people under pressure to improve the score also controlled the answer sheets that determined it.[0]

WHAT HAPPENED

Over several years, educators at dozens of APS elementary and middle schools systematically changed students' answers after testing. The cheating was organized rather than incidental: principals pressured teachers to hit targets, and staff held after-hours "erasure parties" to correct answer sheets. Suspiciously high rates of wrong-to-right erasures flagged the pattern — a statistical fingerprint the gaming left behind precisely because correcting a wrong answer to a right one is far rarer, by chance, than the reverse.[1]

THE INVESTIGATION

A 2009 Atlanta Journal-Constitution analysis of erasure data, followed by the state's Office of Student Achievement and a Governor-ordered special investigation supported by the Georgia Bureau of Investigation, documented the scheme in a 2011 report. Roughly 180 educators were implicated; 35 were indicted and 11 convicted under Georgia's RICO statute, the report finding the administration had emphasized results and praise "to the exclusion of integrity and ethics." That an investigation came from outside the district — a newspaper and state investigators, not the schools — is itself the mark of the missing independent check.[2]

THE CAPABILITY GAP

The capability gap was at the measurement architecture: the institution being measured also operated the instrument that measured it. High-stakes incentives without an independent audit are a textbook setup for Campbell's Law — the more a measure is used for decision-making, the more it will be corrupted. The students were not learning; the system was rewarding the appearance of learning, so the score rose while the underlying capability it was meant to certify went unmeasured and, for the children, unmet.[3]

AFTERMATH & REFORM

The convictions made APS the most prominent U.S. case of high-stakes-testing fraud and fuelled a broader reassessment of accountability-testing regimes. The structural lesson — that a measurement used for consequential decisions needs an audit independent of the institution being measured — is direct, but is still rarely implemented at scale, because the independent audit costs money and political will that the high score, once reported, makes seem unnecessary.[4]

180

EDUCATORS IMPLICATED · DOZENS OF SCHOOLS

the institution being measured operated the instrument that measured it

ATLANTA PUBLIC SCHOOLS — MEASUREMENT GAMING UNDER HIGH-STAKES TESTING

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THE LEARNING ENGINEERING LENS

> *Dr. Hall's administration emphasized test results and public praise to the exclusion of integrity and ethics.*
>
> GEORGIA SPECIAL INVESTIGATORS' REPORT ON ATLANTA PUBLIC SCHOOLS, 2011

LE INSIGHT

The APS cheating scandal is the strongest available case for measurement-gaming under high-stakes incentives in education. The capability gap was at the measurement architecture: the institution being measured operated the instrument that measured it. The reform pattern is direct — independent measurement audit — but rarely implemented at scale.

LENS APPROACH

APS is the protect-the-instrument-from-the-operator-who-controls-it case (induced 2.2): the educators measured on the test also administered and scored it, so the instrument was capturable by the very people it judged. LENS pairs it explicitly with VA wait-times (Case 101), a same-shape instrument-integrity failure where the unit being measured operated the measurement and quietly rewrote the number. The lesson is an instrument-integrity design choice — separate the measured from the measurer, and audit independently of the institution being judged — not the surface reading that people cheated under pressure.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Separate test administration and scoring from the institution being judged, so the party with the incentive to inflate the number never controls the instrument that produces it.
- Engineer a statistical integrity check — such as wrong-to-right erasure-rate monitoring — into the measurement system before scores carry consequences.
- Anticipate Campbell's Law when attaching bonuses and job security to a metric, and design the audit as part of the incentive, not as an afterthought.

IN OPERATION / AFTER

- Run independent, ongoing audits of the high-stakes measure rather than relying on a newspaper or state investigators to surface fraud after years.
- Monitor for the appearance of learning diverging from independent evidence of learning, so the gap is caught while the score still means something.
- Sustain the audit's funding and authority against the complacency a high reported score creates, since the cost of checking looks unnecessary exactly when it is most needed.

REFLECTION QUESTIONS

1. Identify a high-stakes measurement in your domain whose audit is operated by the institution being measured. What would change with independent audit?
2. Apply Campbell's Law to a current high-stakes measurement system. What distortion is predicted?
3. The erasure-rate analysis caught APS only after the fact. Design a continuous integrity signal — a statistical fingerprint of gaming — that would flag the pattern as it emerged rather than years later.

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CASE 105 HEALTHCARE 1999 – 2004

Vioxx Withdrawal

G Governance & Trust D Designed Out

IMPACT Tens of thousands of excess cardiovascular events estimated (FDA's Graham, 2004; Lancet 2005); Merck withdrew Vioxx in September 2004; ~\ $4.85B settlement; FDA Amendments Act of 2007 and the Sentinel Initiative (2008) institutionalized active post-market surveillance

IN BRIEF

Merck’s painkiller Vioxx (rofecoxib) was approved by the FDA in May 1999 and prescribed to an estimated 80 million people worldwide. The 2000 VIGOR trial (Bombardier et al., NEJM) found a roughly five-fold higher rate of myocardial infarction in the Vioxx arm than in the naproxen comparator — a signal Merck and many readers attributed to a hypothesized cardio-protective effect of naproxen rather than a cardiovascular risk of Vioxx, a reading that required the absent placebo arm to be true. Not until the placebo-controlled APPROVe trial (Bresalier et al., NEJM) was halted in September 2004 was the risk established and the drug withdrawn. For nearly four years the disclosure architecture between Merck’s internal data, FDA reviewers, and prescribers failed to surface the magnitude of the danger to a decision point. FDA Office of Drug Safety scientist David Graham testified to the Senate Finance Committee in November 2004 that the risk had been visible well before withdrawal, and estimated 88,000–139,000 excess cardiovascular events attributable to Vioxx; the 2005 Graham et al. Lancet analysis put the figure in similar range. Merck litigation discovery produced internal Merck communications used in the Senate hearings to argue that publication-bias and authorship-by-Merck-employees patterns had suppressed the cardiovascular signal in the published record. The reforms that followed — Risk Evaluation and Mitigation Strategies (REMS), the FDA Amendments Act of 2007, and the Sentinel Initiative (2008) — built the active post-market surveillance architecture that had not previously existed.

BACKGROUND

Vioxx, a selective COX-2 inhibitor for arthritis pain, was approved by the FDA in May 1999 and became one of the most widely prescribed drugs of its era — an estimated 80 million people worldwide were exposed before withdrawal. Detecting rare or delayed harms in a drug at that scale depends on post-market surveillance — aggregating adverse-event data after approval — a function that, at the time, was thin relative to the size of the exposed population. The FDA’s MedWatch system relied on voluntary submission from clinicians; no active query of the underlying claims and electronic-health-record data was operational. A risk too small to surface in a pre-approval trial becomes a large absolute toll once a drug reaches millions, which is precisely the regime that demands strong after-market monitoring — and precisely the regime in which voluntary reporting fails to detect it.[o]

WHAT HAPPENED

The VIGOR trial (Bombardier et al., NEJM 2000) reported about a five-fold higher rate of myocardial infarction in 8,076 patients taking Vioxx than in those taking naproxen. Merck and many readers interpreted the gap as naproxen being cardio-protective rather than Vioxx being harmful. That reading was not absurd — it was the more comfortable of two explanations for the same numbers — but it required the absent placebo arm to be true, and the data could not adjudicate between the readings. The placebo-controlled APPROVe trial (Bresalier et al., NEJM 2005), originally designed to test Vioxx for colorectal-polyp prevention, was terminated early in September 2004 when its data safety monitoring board observed a doubling of cardiovascular events in the Vioxx arm relative to placebo. Merck withdrew the drug worldwide within days.[1]

THE INVESTIGATION

Senate Finance Committee hearings in November 2004 and the subsequent FDA Office of Inspector General review found that signals of cardiovascular harm had been present in the trial record for years before withdrawal. FDA Office of Drug Safety scientist David Graham testified under oath that the cardiovascular risk had been visible to him by 2000, that he had been pressured by FDA management not to publish his estimate, and that he believed Vioxx had caused 88,000–139,000 excess heart attacks and strokes in the United States, of which 30–40% were probably fatal. The Graham et al. Lancet 2005 analysis, using Kaiser Permanente data, produced a population-level estimate in similar range. Merck litigation discovery, made public through New Jersey and federal court filings and reported in the NEJM editorial trail, included internal Merck communications and ghost-authorship patterns in published Vioxx papers; a 2008 JAMA analysis by Ross et al. documented the publication-bias and authorship patterns directly.[2] The harm was not hidden in some unmeasured corner — it sat in the trial record the whole time, waiting for an architecture that would carry it to a decision rather than leave it to interpretation.[3]

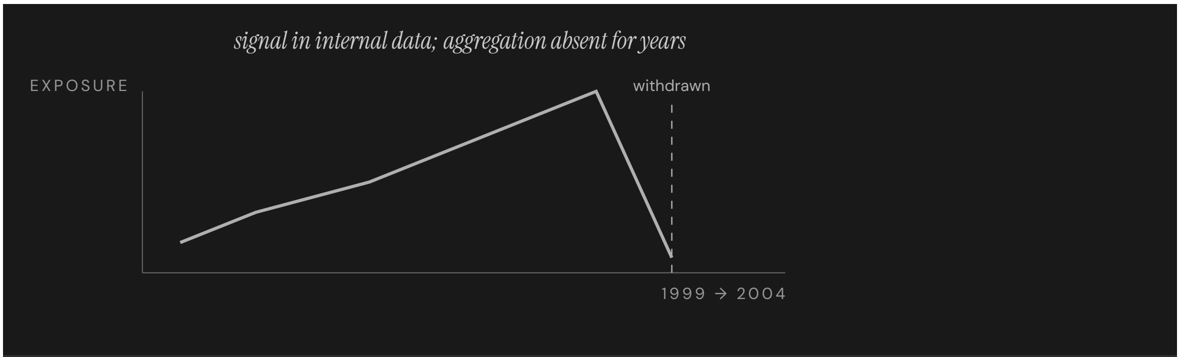
THE CAPABILITY GAP

The capability gap was in post-market surveillance: the disclosure architecture between manufacturer-held data, FDA reviewers, and prescribers did not aggregate the signal to a decision boundary for nearly four years. The FDA's MedWatch adverse-event reporting system relied on voluntary submissions structured to minimize signal, and was not adequate to the volume of the drug's distribution. A system that waits for a clinician to choose to file a report will always lag a harm spread thinly across millions of prescriptions, because no single prescriber sees enough of the pattern to recognize it. The Merck-side gap was structural in a different way: VIGOR was designed to test gastrointestinal safety (the COX-2 selling point), and the cardiovascular signal was a secondary finding that the trial was not powered to adjudicate against the naproxen-protective hypothesis. That trial-design choice compounded with a publication-bias pattern, an FDA adverse-event-reporting architecture insufficient for the population at risk, and a regulator–manufacturer disclosure protocol that had no decision boundary for ambiguous signals.[4]

AFTERMATH & REFORM

Merck eventually settled US litigation for about $4.85 billion across approximately 27,000 plaintiffs. The case drove a coordinated regulatory response. The FDA Amendments Act of 2007 gave the FDA explicit authority to require Risk Evaluation and Mitigation Strategies

(REMS) and post-market study commitments as conditions of approval. In 2008 the FDA launched the Sentinel Initiative — an active post-market surveillance system that queries distributed health-data partners covering hundreds of millions of patient-years, rather than waiting for voluntary reports — a direct response to the Vioxx-era detection failure. The NEJM tightened conflict-of-interest disclosure for trial reports; the JAMA Ross et al. (2008) analysis became the reference point for publication-bias diagnosis in drug safety. By going out to the data instead of waiting for it to arrive, the reform inverted the logic that had let the signal sit unaggregated for years. The reform built the surveillance infrastructure that had not previously existed; the Vioxx record is what made the case for it.[5]



VIOXX — POST-MARKET SURVEILLANCE AS A MISSING DELIVERABLE

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THE LEARNING ENGINEERING LENS

> *The cardiovascular risk was visible in Merck's internal data years before the drug was withdrawn.*

> PARAPHRASING DAVID GRAHAM (FDA OFFICE OF DRUG SAFETY), SENATE FINANCE COMMITTEE TESTIMONY, NOVEMBER 2004

LE INSIGHT

Vioxx is the canonical pharmaceutical case for post-market surveillance as a capability deliverable. The signal existed; the institutional architecture to aggregate it did not. The reform pattern — Sentinel — built the architecture. The case is the drug-industry analog of the EHR case (Case 61): a measurement architecture too thin for the system that depended on it.

LENS APPROACH

Vioxx is the canonical pharmaceutical post-deployment-surveillance case (induced 2.4; LENS D4+D4/PT5). LENS uses it in Domain 4 (Test and Evaluation; CLO-4) for post-market surveillance as a measurement deliverable — the signal existed; the institutional architecture to aggregate it did not. LENS uses it in Domain 3 (Human-System Collaboration; CLO-4) for the change-control and disclosure architecture between manufacturer, regulator, and prescriber that determines whether ambiguous safety signals reach a decision boundary. Pair with Radiology AI (Case 130) as the post-market-surveillance failure pattern at a new technological boundary; pair with EHR/CPOE (Case 61) at the measurement-architecture-too-thin layer; pair with Medical Errors and IOM (Case 102) at the population-scale-instrument layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Field an active post-market surveillance architecture sized to the exposed population before scale-up, so a harm spread thinly across millions can still aggregate to a decision boundary.
- Design trials and data collection to adjudicate between competing readings of a signal, rather than leaving an ambiguous result to the more comfortable interpretation.
- Treat signal-aggregation as a defined deliverable carrying evidence from manufacturer data through reviewers to prescribers, not a byproduct of voluntary reporting.

IN OPERATION / AFTER

- Audit the trial record and adverse-event data for signals already present but never carried to a decision, closing the lag between visibility and action.
- Monitor by querying health-data partners actively rather than waiting for voluntary reports that no single prescriber sees enough of the pattern to file.
- Sustain post-market surveillance funding against the commercial pressure to defer aggregation, since the cost of waiting compounds across the exposed population.

REFLECTION QUESTIONS

1. Identify a post-deployment surveillance architecture in your domain that is too thin for the scale of the system it monitors. What is the missing deliverable?
2. Design the Sentinel-equivalent post-market surveillance system for a new domain.
3. The VIGOR signal admitted two readings — harmful drug or protective comparator — and the data could not decide. Identify a measurement in your domain whose interpretation is underdetermined, and design the study or instrument that would force the question to a decision.

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CASE 106 HIGHER-ED ANALYTICS EARLY-WARNING SYSTEMS EVIDENCE ARCHITECTURE 2012 – 2013

Purdue Course Signals — The Reverse-Causality Retention Claim

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Purdue's widely cited claim that students taking two or more Course Signals classes were 21% more likely to be retained was challenged by Mike Caulfield in 2013, who identified the dose–response curve as an artifact of selection: students persist and therefore take more Signals courses, not the reverse — Alfred Essa reproduced the apparent retention gain by substituting 'received a piece of chocolate' for 'took a Signals class' in a simulation

IN BRIEF

Purdue's Course Signals was one of the most-cited early-warning learning-analytics interventions of the early 2010s. The program's headline outcome claim — students who took two or more Signals courses were 21 percent more likely to be retained — was published by Arnold and Pistilli at LAK 2012 and circulated widely in the learning-analytics community and in vendor materials. In 2013 Mike Caulfield, blogging at e-Literate and in Inside Higher Ed, identified the dose–response curve as a reverse-causality artifact: students who persist in college therefore take more Signals courses, so the apparent retention gain reflects selection, not effect. Alfred Essa built a simulation that substituted "received a piece of chocolate" for "took a Signals class" and reproduced the apparent retention gain, demonstrating the methodological flaw. The original study was never peer-reviewed outside conference proceedings yet became one of the most-referenced learning-analytics studies of its era — which is itself the cautionary point about the field's evidence architecture. The case is the small-tier methodological companion to v1 Cases 90 (educational predictive-analytics bias) and 39 (Georgia State predictive analytics).

BACKGROUND

Course Signals at Purdue was a faculty-facing dashboard that classified students enrolled in a course as green, yellow, or red, based on a predictive model of academic risk built from LMS and grade-book signals. Faculty used the classification to send targeted communications to students flagged at risk. The design and the operational use were not the subject of the later critique; what became contested was the system's headline outcome claim, published by Arnold and Pistilli at Learning Analytics and Knowledge 2012: students who took two or more Signals courses were 21 percent more likely to be retained at the institution than students who took fewer.[0]

WHAT HAPPENED

The figure circulated. It appeared in conference keynotes, vendor materials, accreditor presentations, and in widely shared accounts of what early-warning analytics could deliver. The claim's status in the literature outran its evidentiary base: the LAK 2012 paper was a conference paper, not a peer-reviewed journal article, and the institutional dataset behind the headline number was not made available for independent reanalysis. The field cited the result anyway, because it was the kind of result the field wanted to be true.[1]

THE INVESTIGATION

In 2013 Mike Caulfield, writing at e-Literate and in Inside Higher Ed, asked a specific methodological question: was the dose–response curve — more Signals classes, more retention — what it appeared to be? Caulfield argued the relationship was a reverse-causality artifact. Students who persist at the institution have more semesters in which to take Signals courses; students who depart cannot. The "took two or more Signals courses" group was therefore an inadvertent selection on persistence — not a sample exposed to a different treatment intensity. Alfred Essa then built a simulation that substituted "received a piece of chocolate" for "took a Signals class," with chocolate having no causal effect on anything, and reproduced the apparent retention gain. The reverse-causality reading survived the simulation replication.[2]

THE CAPABILITY GAP

The methodological point is precise: the published analysis did not isolate the treatment from the selection mechanism that determined treatment intensity, and the dose–response curve that looked like the effect was generated by the selection itself. The case is not an argument that Course Signals had no effect on retention. It is an argument that the published evidence could not distinguish effect from selection, and that the institution measured a number which **felt** like the failure mode it cared about (retention) using a design that could not actually answer the causal question. This is the textbook 2.1 failure: measuring the failure mode with a design the institution can deceive itself with.[3]

AFTERMATH & REFORM

Drafted as a deeper-evidence-of v1 Cases 90 and 110, the Purdue case carries a named methodological failure into the corpus's predictive-analytics conversation. The cautionary thread runs through three places at once: the original study's status (conference proceedings, never peer-reviewed outside that venue) outpaced its evidentiary base; the field's citation practice amplified the headline without probing the design; and the correction (Caulfield, Essa) was mounted from outside the original study's institutional network. The case teaches the evidence-architecture failure mode that the LENS Iterative Development domain and the Navigating Sociotechnical Constraints domain both have to protect against — and that v1 Cases 90 and 110 anchor at the bias and the institutional-deployment layers respectively.

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THE LEARNING ENGINEERING LENS

> *Substituting "received a piece of chocolate" for "took a Signals class" in a simulation reproduces the apparent retention gain.*
>
> EDITORS' SYNTHESIS OF ESSA'S 2013 SIMULATION DEMONSTRATION.

LE INSIGHT

Course Signals is the named methodological failure in the predictive-analytics conversation: the institution measured a number that felt like the failure mode it cared about, using a design that could not isolate effect from selection. The original study was never peer-reviewed outside conference proceedings; the correction came from outside the original network. Both are part of the cautionary reading.

LENS APPROACH

Course Signals is the small-tier evidence-architecture failure (induced 2.1; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for measurement design that cannot deceive the institution, and in Domain 5 (Navigating Sociotechnical Constraints) for the citation-practice failure mode in which a headline outcome outpaces the evidence it rests on. Deeper-evidence-of v1 Cases 90 (predictive-analytics bias) and 39 (Georgia State predictive analytics) — distinct because this is a named methodological-validity failure, not a bias finding.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the causal question the analysis is built to answer in advance, and design the comparison group so that the selection mechanism into "treatment intensity" is not itself the variable doing the work.
- Pre-register the dose–response analysis with the threats-to-validity table — selection, censoring, reverse causality — visible at design time, so the published headline cannot outrun the analysis it rests on.
- Choose the publication venue that fits the evidentiary claim: conference proceedings for an early result, peer-reviewed journal with independent reanalysis for the figure the field will cite.

IN OPERATION / AFTER

- Make the dataset available for independent reanalysis; the Caulfield / Essa correction succeeded because the published claim was simulatable on plausible data, not because the original dataset was inspected.
- Treat the simulation-replication finding as program evidence about the analysis design, not as a verdict on the intervention; Course Signals may have had an effect, and the published study could not detect or measure it.
- Carry the cautionary reading into the field's citation practice: a headline outcome circulating widely without peer review outside conference proceedings is itself an evidence-architecture failure, separate from any specific methodological flaw.

REFLECTION QUESTIONS

1. Identify a predictive or early-warning analytics deployment in your domain whose published outcome claim circulates more widely than the peer-reviewed evidence supports. What threats to validity — selection, reverse causality, censoring — would a Caulfield-style external critic name first?

2. Specify the comparison-group design that would isolate effect from selection in your context. What pre-registered analysis plan, with simulated placebo treatment, would let the field check the claim before it circulates?

3. The Course Signals correction came from outside the original institutional network. What is the analog in your domain — the independent reanalysis path that protects the field from citation practice that outruns the evidence?

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Chapter 8

Test and Evaluation — What Works

When evidence is engineered to outlive the incentives to ignore it.

Decision-grade evidence is the evidence we need to justify a decision, on incomplete information.

LENS V2.1 REFRAMING OF DECISION-GRADE EVIDENCE.

CASE 107 FINANCE ALGO FAIRNESS FINTECH 2025

Fintech Lending Fairness Audit — When Including Race Reduces Disparity

D Designed Out G Governance & Trust N Normalization of Deviance

EVIDENCE TIER preprint-tier — source confidence flagged; future validation ongoing.

IMPACT A fintech-lending fairness audit reports that incorporating protected attributes under controlled mitigation reduces measured disparity below the unawareness baseline — surfacing the next teaching point: competing fairness definitions disagree, and the choice is a judgment

IN BRIEF

The Coots et al. (2025) fintech lending fairness audit, posted as a preprint, examines a deployed consumer-lending model under both the unawareness baseline (the protected attribute excluded from inputs) and a mitigation regime that uses the protected attribute in a controlled fashion at training or post-processing time. The audit reports that the mitigation regime produced lower measured disparity on standard group-fairness metrics than the unawareness baseline did — a small-scale instance of the same pattern Bartlett documented at the mortgage market scale. The case is a frontier candidate: it sharpens the teaching point that "fairness through unawareness" is not the conservative or safe choice it is often assumed to be, and it surfaces the next layer — competing fairness definitions disagree about what the same model is doing, and the practitioner has to decide **which definition** the deployment is being held to before the measured disparity can even be interpreted. The evidence-tier flag is preprint: the finding has not yet completed peer review, the metric choices and mitigation specifics may move, and future validation will continue. The teaching point survives those caveats, which is why the case is included with the flag rather than held.

THE SHIFT

The lending pair (Cases 94 and 107) takes the practitioner past the first equity intuition — **just don't use the protected attribute** — and into the harder territory the equity literature now operates in. Bartlett showed that omission preserves the disparity through correlated features. The Coots audit shows the next thing: when the protected attribute is included under a principled mitigation, the disparity on standard fairness metrics drops below the unawareness baseline. Both findings are about the same family of models; they disagree about what makes a model fair.[o]

WHAT IS EMERGING

The audit examines a deployed fintech-lending model under two regimes. In the unawareness regime, the protected attribute is withheld from the model's inputs at training and inference. In the controlled-mitigation regime, the protected attribute enters at training or post-processing time as the basis for an explicit adjustment intended to equalize a chosen fairness criterion. The audit reports lower measured disparity on standard group-fairness metrics under controlled mitigation than under unawareness — small scale, but the pattern matches the mortgage-finance finding (Case 94) and the broader fair-ML literature.[1]

THE CAPABILITY QUESTION

The case is a frontier candidate because it surfaces the layer Bartlett alone does not reach: competing fairness definitions disagree about the same model. A model that is more equitable on group calibration may be less equitable on equalized odds, and vice versa; the impossibility results of the fair-ML literature (Chouldechova; Kleinberg, Mullainathan, & Raghavan) make this explicit. The Coots audit is concrete enough to ground the choice: the practitioner cannot postpone the question of **which definition** the deployment is being held to.[2]

EARLY EVIDENCE

The evidence-tier flag is load-bearing here and is rendered under the case title. The Coots audit is a preprint; the metric choices, the mitigation specifics, the dataset window, and the reported magnitude may move in peer review. The structural pattern — that controlled inclusion of the protected attribute can produce a lower measured disparity than omission — is consistent with the broader fair-ML literature and with Bartlett's mortgage-finance finding, but the specific magnitudes in the audit should be treated as the strongest current preprint claim, not as a settled fact. Future validation will continue, and this case will be updated when peer review lands.[3]

OPEN PROBLEMS

What the pair (Cases 94 + 103) teaches together is the form of the equity capability deliverable: the practitioner must specify, in advance, the fairness definition the deployment will be evaluated against, audit on outputs rather than reasoning about inputs, and decide on judgment that the trade-offs across competing definitions are acceptable for the deployment context. This is the case-grounded basis for the CLO **Fairness beyond omission** and the CLO **Judgment under inadequate evidence** — the audit is itself a worked example of deciding under irreducible disagreement.[4]

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1. Coots, Bartlett, Nyarko, & Goel (2025), "Algorithmic Bias in Lending: Evidence from a Fintech Audit," arXiv:2512.20753 — audit of 80,000 personal loans showing model miscalibration disparities and that controlled inclusion of protected attributes could correct them; the evidence-tier flag is binding until peer review completes.
2. Bartlett, Morse, Stanton, & Wallace (2022), "Consumer-lending discrimination in the FinTech era," Journal of Financial Economics 143(1):30–56 — the paired big-tier case (103).
3. Chouldechova (2017), "Fair Prediction with Disparate Impact," Big Data 5(2):153–163 — the impossibility result for calibration and error-rate parity.
4. Kleinberg, Mullainathan, & Raghavan (2017), "Inherent Trade-Offs in the Fair Determination of Risk Scores," ITCS — competing fairness definitions are not jointly achievable.
5. Mitchell, Potash, Barocas, D'Amour, & Lum (2021), "Algorithmic Fairness: Choices, Assumptions, and Definitions," Annual Review of Statistics and Its Application 8:141–163 — practitioner-facing survey.

THE LEARNING ENGINEERING LENS

> *Once you accept that omission is not the answer, the next question — which fairness definition — is the one the deployment cannot avoid.*
>
> EDITORS' SYNTHESIS OF COOTS ET AL. (2025) AND MITCHELL ET AL. (2021).

LE INSIGHT

The Coots audit is the small-tier frontier instance of the finding that controlled inclusion of a protected attribute can produce lower measured disparity than omission. With Bartlett (Case 94) it forms the canonical lending pair: omission does not fix the harm; competing fairness definitions disagree about what fix is. Evidence is preprint-tier; future validation will continue.

LENS APPROACH

Coots is the small-tier frontier counter-case to Bartlett. LENS uses the pair in Domain 4 (Test and Evaluation) for the CLO **Fairness beyond omission**; in Domain 4 again for the CLO **Judgment under inadequate evidence** (the pair is itself a decision under irreducible disagreement); and in Domain 3 (Human-System Collaboration) for delegation of consequential consumer-finance decisions to a model. The preprint-tier flag is binding until peer review completes.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify, before model selection, which fairness definition the deployment is being held to — group calibration, equalized odds, demographic parity — and the metric values that will count as acceptable.
- If the protected attribute is included under a controlled-mitigation regime, document the inclusion path (training-time, post-processing) and the auditing pipeline that verifies the mitigation does what it claims.
- Distinguish what is preprint-tier evidence from what is settled in the literature you are drawing on, so a deployment decision does not ride on the strongest unreviewed claim.

IN OPERATION / AFTER

- Audit the deployed model on outputs stratified by protected attribute, with the pre-registered fairness definition; publish the metric values and the cases that disconfirm them.
- Track the preprint to publication; when peer review lands, update the auditing pipeline to reflect the settled metric choices and any revised magnitudes.
- When competing fairness definitions disagree, name the trade-off explicitly in the deployment documentation; do not present the chosen definition as the technical optimum.

REFLECTION QUESTIONS

1. Identify a fairness audit in your domain conducted at preprint or unpublished stage. What part of its claim survives if peer review modifies the metric choices? What part is contingent on the specific magnitudes?
2. Specify which fairness definition your deployment is being held to before the audit is run. What trade-off — across calibration, equalized odds, demographic parity — does that choice accept?
3. Coots' finding is consistent with Bartlett (Case 94) and with the broader fair-ML literature, but the specific magnitudes are preprint-tier. What is the minimum follow-up evidence you would require before allowing this case to drive an operational decision in your context?

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CASE 108 CORPORATE L&D TRAINING EVALUATION WORKFORCE DEV 1959 – PRESENT

The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops

K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT **Across a US corporate-training market sized in the >\$125B/year range, the dominant evaluation framework structurally collapses: most teams stop at Levels 1–2 (reaction + learning) and never measure Level 3 (behavior change) or Level 4 (business results) — measuring the variable that flatters the program**

IN BRIEF

Donald Kirkpatrick's four-level model — Reaction, Learning, Behavior, Results — has been the dominant corporate-training evaluation framework for sixty years, increasingly described as a "chain of evidence" from satisfaction through workplace behavior change to business outcomes. The documented systemic pattern is that most organizations stop at Levels 1 and 2. Level 3 (behavior change on the job) and Level 4 (business results) require data that the training organization typically cannot access: longitudinal performance traces, business-unit outcome metrics, line-manager observation. Satisfaction scores do not predict on-job performance, and knowledge retention does not guarantee workplace application. The case is the corporate-scale instance of the enthusiasm-evidence gap and a direct illustration of the revised "decision-grade evidence" framing in the v2 research backbone: the evidence most L&D decisions ride on is structurally sub-decision-grade. The evidence base is practice-synthesis: Devlin Peck, D2L, Valamis, and related evaluation-practice guides documenting the stop-at-L2 pattern, plus the Blume meta-analysis (Case 35) for the transfer half of the chain. The evidence-tier flag is rendered under the title; future validation will continue as the corporate-L&D evaluation literature consolidates.

THE SHIFT

In 1959 Donald Kirkpatrick proposed four levels at which a training program could be evaluated: Reaction (did learners like it), Learning (did they acquire the content), Behavior (did they apply it on the job), and Results (did business outcomes improve). The four levels became the dominant corporate-training evaluation framework and were later described as a "chain of evidence" — each level meant to provide the evidence for the next.[0]

WHAT IS EMERGING

The documented systemic pattern in corporate L&D evaluation practice is that most organizations stop at Levels 1 and 2. Practice-literature synthesis across major evaluation-guidance sources (Devlin Peck, D2L, Valamis, and corporate L&D benchmarking studies) reports that Levels 3 and 4 are attempted by a minority of programs and reported reliably by fewer. The structural reason is that Level 3 (behavior change) requires longitudinal observation that the training organization cannot conduct, and Level 4 (results) requires business outcome data that often sits outside L&D's reach.[1]

THE CAPABILITY QUESTION

The consequence is a field that spends heavily on training — US organizations reported >$125B/year on workplace training, per the ASTD figure cited in Blume et al. (Case 35) — while measuring mostly the variable that flatters the program. Satisfaction scores do not predict on-job performance; knowledge retention measured immediately after training does not guarantee workplace application. The chain of evidence is cited as the framework; in practice the chain is broken at the link between Level 2 and Level 3, and the decisions made on the available data are not decisions about whether training is producing capability change.[2]

EARLY EVIDENCE

The evidence-tier flag rendered under the case title is load-bearing. The case is documented through evaluation-practice guides synthesizing the stop-at-L2 pattern across many organizations rather than through a single peer-reviewed study of the phenomenon. The pattern is consistent across the sources and is the practitioner consensus, but the magnitudes vary by sector and the sectoral breakdown is in flux as the field consolidates its evaluation evidence. Future validation will continue as the empirical synthesis improves; the case is included with the flag because the pattern itself is teachable and the practitioner consensus is durable.[3]

OPEN PROBLEMS

What the case teaches in pair with Blume (Case 35) is the structural form of the enthusiasm-evidence gap at corporate scale, and it is the case-grounded basis for the revised "decision-grade evidence" framing proposed in the v2 research backbone. The capability deliverable is an evaluation architecture that crosses the Level 2 / Level 3 seam — by partnering with line management for behavior observation, by instrumenting the workplace tasks the training targets, and by reporting honestly what evidence is and is not available — rather than declaring the chain satisfied at the Level the training organization can control. Until that crossing happens, most corporate L&D decisions are made on structurally sub-decision-grade evidence, and the CLO **Judgment under inadequate evidence** is exactly the capability the practitioner needs.[4]

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1. Kirkpatrick (1959–1960), original four-level evaluation series in Journal of the ASTD; updated as Kirkpatrick & Kirkpatrick, Evaluating Training Programs (3rd ed., 2006).
2. Devlin Peck, "Kirkpatrick Model: A Guide to the Four Levels of Training Evaluation" — synthesis of the stop-at-L2 pattern in corporate practice.
3. D2L, "Kirkpatrick's 4 Levels of Training Evaluation," practitioner guide documenting the same pattern.
4. Valamis, "Kirkpatrick Model" practitioner guide on evaluation-practice gaps.
5. Blume, Ford, Baldwin, & Huang (2010), "Transfer of Training: A Meta-Analytic Review," Journal of Management 36(4):1065–1105 — the paired peer-reviewed case (113).

THE LEARNING ENGINEERING LENS

> *The chain of evidence is cited as a framework. In practice it breaks at the link between Level 2 and Level 3, and the decisions made on the available data are not decisions about capability.*
>
> EDITORS' SYNTHESIS OF THE KIRKPATRICK MODEL AND CORPORATE L&D EVALUATION PRACTICE.

LE INSIGHT

The Kirkpatrick stop-at-L2 pattern is the corporate-scale instance of the enthusiasm-evidence gap and the direct illustration of the revised "decision-grade evidence" point: the evidence most L&D decisions ride on is structurally sub-decision-grade. Evidence base is practice-synthesis tier; the pattern is consistent across sources, the magnitudes still consolidating; future validation will continue.

LENS APPROACH

Kirkpatrick is the corporate-L&D evaluation case (induced 2.1; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for the CLO **Judgment under inadequate evidence** — the practitioner must decide well on structurally sub-decision-grade evidence — and in Domain 5 (Navigating Sociotechnical Constraints) for the Level-2/Level-3 seam that the training organization cannot cross alone. Direct pair with Case 35 (Blume transfer meta-analysis) for the workplace-environment half of the chain.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the evaluation across the Level-2/Level-3 seam before the training is delivered: partner with line management for behavior observation, identify the workplace tasks the training targets, and instrument them.
- Specify in advance the on-job behavior and business outcome that would count as evidence the training worked, with a reporting cadence long enough for transfer effects to surface.
- Treat the practice-synthesis evidence base honestly: name the pattern, flag the tier, and do not present a stop-at-L2 outcome as if it were a Level-3 or Level-4 finding.

IN OPERATION / AFTER

- When Level 3 and Level 4 data are not available, say so plainly in the evaluation report rather than substituting Level 2 metrics; the CLO **Judgment under inadequate evidence** is what the practitioner needs to do well here.
- Track the work environment as a separate variable — Blume's meta-analysis (Case 35) names it as decisive — and report the training outcome conditional on environment, not as a property of the training alone.
- Use the structural pattern to argue for the evaluation architecture investment, not to abandon evaluation. The gap is a capability gap, not an argument against measurement.

REFLECTION QUESTIONS

1. Identify a recent corporate training program in your organization. At which Kirkpatrick level did evaluation stop? What would the Level 3 and Level 4 measurement have required, and who would have had to provide the data?
2. Specify the evaluation architecture you would build to cross the Level-2/Level-3 seam: which on-job behavior, which business outcome, on what cadence, in partnership with which line-management role.
3. The case is practice-synthesis tier. What is the minimum peer-reviewed or program-evaluation evidence you would require before relying on the stop-at-L2 pattern to justify an evaluation-architecture investment in your context?

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CASE 109 HEALTHCARE 2008 – PRESENT

WHO Surgical Safety Checklist

T Training Gap N Normalization of Deviance

IMPACT **Death rate 1.5% → 0.8% in eight-site pilot; complications down >33%; adopted by the majority of surgical providers worldwide; Ontario population study (2014) found no significant mortality benefit after a mandated rollout**

> IN BRIEF
>
> In 2008, Atul Gawande's team and the WHO introduced a single-page, nineteen-item surgical checklist applied at three junctures — before anesthesia, before incision, and before the patient leaves the operating room. Piloted across eight hospitals in eight countries, from Toronto to Tanzania, it nearly halved the surgical death rate (1.5% to 0.8%) and cut serious complications by more than a third — results published in the NEJM in 2009. The artifact was the checklist; the intervention was the system of forced pauses it created, three moments when a moving team had to stop and confirm shared state. A later Ontario study found no mortality benefit after a mandated rollout, surfacing the fidelity question: the artifact works only when the institution authorizes its honest use.

BACKGROUND

Surgical harm — wrong-site operations, retained instruments, missed allergies, post-operative infection — was widespread across health systems rich and poor, and much of it stemmed not from a lack of skill but from teams that never paused to confirm shared understanding before acting. The knowledge to prevent these events existed; the reliable practice did not. A surgical team in motion rarely stopped to verify that everyone held the same picture of the patient and procedure, so a mismatch any member could have caught passed silently into the operation. The cost was global and patterned — the WHO estimated that, of the roughly 234 million operations performed each year, at least seven million produced major complications and about a million ended in death — and the bulk of that harm sat in failure modes that a brief verbal cross-check would have closed.[0]

THE INTERVENTION

In 2008, the WHO and a Harvard team led by Atul Gawande introduced a single-page, nineteen-item Surgical Safety Checklist applied at three critical junctures: before anesthesia, before skin incision, and before the patient leaves the operating room. The team piloted it across eight hospitals in eight countries spanning very different economic and operational conditions — Toronto, London, Seattle, Auckland, Amman, New Delhi, Manila, and Ifakara — selecting sites deliberately to span the global income gradient. Testing the same artifact from Toronto to Tanzania was deliberate — it had to work in settings with vastly different resources to demonstrate that the gain came from the forced pause itself, not from any one system's wealth. The artifact was intentionally minimal: items the team had tested against the surgical-safety literature for the smallest set that still spanned the highest-risk junctures.[1]

HOW IT WORKED

The checklist was the artifact; the intervention was the system of pauses it imposed. At each juncture a team that would otherwise keep moving had to stop, look at one another, and confirm names, the procedure, allergies, and equipment aloud. The pauses were the load-bearing element — the requirement to halt and speak mattered more than the specific list of items. Saying the confirmations aloud to one another, rather than each member assuming them privately, is what surfaced the mismatches a moving team would otherwise have carried into the incision. The mechanism turned the operating room into a momentarily flat-hierarchy space: a circulating nurse who noticed a missing antibiotic dose at the time-out had institutional cover to say so before the incision, a transaction the pre-checklist culture had not reliably authorized. The artifact's authority — that the team had to stop — was what made the speech act safe.[2]

THE EVIDENCE

The 2009 NEJM study reported the surgical death rate falling from 1.5% to 0.8% and major complications dropping by more than a third across all eight sites — confirming Gawande's framing that "gaps in teamwork and safety practices in surgery are substantial in countries both rich and poor." The checklist was subsequently adopted by the majority of surgical providers worldwide. That the death rate roughly halved across all eight sites, despite their differences in wealth and practice, is what made the result so persuasive — the effect tracked the intervention rather than the setting. The published mortality reduction was contested almost from the moment of publication: observational, pre/post, no concurrent control, susceptible to secular trends and Hawthorne-style attention effects. The pilot was always best read as the formation of a strong hypothesis under pragmatic conditions, not as a closed proof of population-scale mortality benefit.[3]

WHAT TRANSFERRED

Urbach and colleagues, writing in NEJM in 2014, examined more than 200,000 procedures across 101 Ontario hospitals before and after a province-wide mandated rollout and found no statistically significant reduction in operative mortality, in-hospital complications, length of stay, or readmission. The null result did not refute the checklist so much as illustrate its dependence on implementation fidelity: where a mandate replaced genuine authorization of the pause, the measured effect attenuated, making the checklist a paired lesson in both minimal-artifact design and the limits of mere compliance. The contrast between the pilot and the mandated rollout is the lasting teaching point: when the pause was genuinely used it worked, and when it became a box to check before proceeding the same paper produced nothing. Subsequent multi-country replications and the WHO's own follow-up work have been mixed — the institutional uptake outran the closed evidence of population-scale mortality benefit, and the contested mortality reduction is the part of the case that does not resolve.[4]

THREE PAUSE POINTS ACROSS THE SURGICAL TIMELINE

REFERENCES

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2. WHO Safe Surgery Saves Lives campaign documentation — the nineteen-item checklist and the three junctures.
3. Gawande, A. (2009), The Checklist Manifesto — the pause as the active mechanism.
4. Urbach, D. et al. (2014), "Introduction of Surgical Safety Checklists in Ontario, Canada," NEJM — null mortality result after mandate.
5. Bosk, C. et al. (2009), "Reality check for checklists," The Lancet — implementation fidelity.

THE LEARNING ENGINEERING LENS

> *Gaps in teamwork and safety practices in surgery are substantial in countries both rich and poor.*
>
> ATUL GAWANDE, HARVARD GAZETTE, JANUARY 2009

LE INSIGHT

The Surgical Safety Checklist is the canonical evidence that a tiny artifact — one page, nineteen items — can produce population-scale effects when paired with the structural change of **requiring a pause.** The checklist alone is paper. The pause alone is anxiety. Together they constitute the smallest effective capability intervention in the dataset. The Ontario follow-up underscores the secondary requirement: the artifact carries the effect only when the institution actually enforces the pause. Where mandate replaced authorization, the measured effect attenuated.

LENS APPROACH

LENS uses the WHO checklist as the canonical pre-committed institutional response paired with closed-loop evidence (induced 4.4; LENS D4/PT5). It runs in LEN 10 (capstone) as the end-to-end design exemplar — a problem identified, a minimal artifact prototyped, a multi-site pilot, a measurement regime, and global scale-out — and in LEN 4 for the measurement architecture that made the pilot effect provable and the Ontario null interpretable. Adjacent to SUBSAFE (Case 19) at the mandatory-mechanism layer, where the artifact's authority is the intervention.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the minimal artifact — a single-page checklist — around a small number of forced pauses at the

IN OPERATION / AFTER

- Measure outcomes (death and complication rates) directly so the effect can be confirmed and the

highest-risk junctures, keeping the list short enough to be used every time.
- Make the confirmations spoken aloud and shared across the team, so the pause surfaces mismatches rather than letting each member assume them privately.
- Pilot across deliberately varied sites so the measured gain can be shown to track the intervention rather than any one setting's resources.

artifact is not assumed to work merely because it is in use.
- Guard implementation fidelity as the scale grows — a mandate that turns the pause into a box to check reproduces the artifact without the effect.
- Build a governance signal that distinguishes genuine, authorized use from compliance check-off, so the fidelity gap is visible before a rollout is judged a failure.

REFLECTION QUESTIONS

1. What is the smallest possible capability artifact in your domain that, paired with a required pause, would shift outcomes?
2. The WHO checklist was studied across eight countries. Design the multi-site evaluation that would establish whether your candidate intervention generalizes.
3. The Ontario mandated rollout produced no measurable mortality reduction. What governance signal would have surfaced the fidelity gap between authorized use and compliance check-off before the trial was declared a failure?

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CASE 110 EDUCATION AT SCALE 2012 – PRESENT

Georgia State University Predictive Analytics

T Training Gap K Knowledge & Institutional Memory

IMPACT Six-year graduation rate 32% → 54%; equity gaps in graduation eliminated; 2,000+ more graduates per year

IN BRIEF

Georgia State University built a predictive-analytics advising system that tracks some 800 risk factors per student and triggers proactive outreach when early warning signs appear. Crucially, it was designed with equity as a primary constraint — the explicit goal was to eliminate, not reproduce, graduation gaps — and the alerts prompt human advisors rather than making automated decisions. The six-year graduation rate rose from 32% to 54%; Black and Pell-eligible students now graduate at the same rate as their peers, and GSU produces roughly 2,000 more graduates a year. The difference from the algorithmic-bias cases (35–37) is design: GSU built equity in from the start, used predictions to trigger more human support rather than gatekeeping, and tracked outcomes by demographic group as a primary metric.

BACKGROUND

Georgia State was a regional commuter university where only about a third of students finished in six years, with large gaps by race and income. Like many institutions it had predictive data but such systems, where they existed elsewhere, were typically used for triage or gatekeeping — risking the reproduction of existing inequities rather than their repair. A model that flags at-risk students can just as easily steer them away as toward help; the same prediction serves opposite ends depending on what the institution decides to do with it.[0]

THE INTERVENTION

Beginning in 2012, GSU deployed a predictive-analytics advising system that monitors roughly 800 behavioral and academic risk factors per student daily and fires an alert to an advisor when warning signs — a missed assignment, a poor grade in a gateway course — appear. The system was built with equity as a primary design constraint, with the explicit aim of closing graduation gaps. Daily monitoring meant the alert fired while there was still time to act — a slipping student was caught at the missed assignment rather than at the failed semester, when intervention could still change the outcome.[1]

HOW IT WORKED

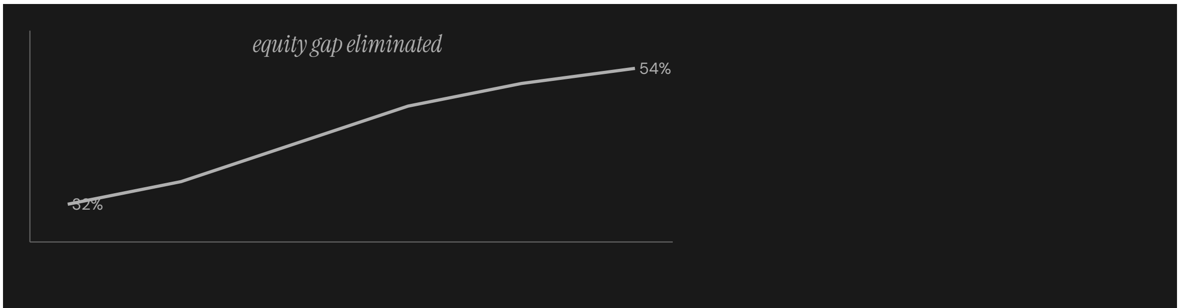
The load-bearing design choice was the human-loop architecture: alerts trigger proactive advising — a phone call, a meeting, a financial-aid check — rather than automated decisions. Predictions are used to deliver more support to at-risk students, not to gatekeep them out. Human judgment stays in the loop, and the model functions as decision support rather than decision-maker. Routing every alert through an advisor rather than an automated action is what kept the prediction in service of the student: the model identified who needed attention, and a person decided what that attention should be.[2]

THE EVIDENCE

GSU's six-year graduation rate rose from 32% to 54%, and the institution now produces some 2,000 additional graduates a year. The graduation rate for Black students rose to match the overall rate, and Pell-eligible students graduate at the same rate as non-Pell students — the equity gap was eliminated rather than merely narrowed. Eliminating the gap rather than narrowing it is the decisive result: the overall rate rose while the disparities by race and income closed, so the gain did not come at the expense of the students the system was most at risk of leaving behind.[3]

WHAT TRANSFERRED

GSU is the positive counterpart to the algorithmic-harm cases of Chapter 5 (A-Level, Robodebt, educational bias). The same technical capability — a predictive model — produced an equity gain rather than an equity harm because of how it was framed and governed. The case is the strongest evidence that construct definition and human-loop architecture, not the model itself, determine whether prediction helps or harms. Holding the technology constant and varying only the design constraint and governance is what isolates the lesson: the same predictive capability that harmed in Chapter 5 helped here, so the framing and the human loop, not the model, are where intent lives.[4]



GRADUATION RATE, WITH EQUITY AS A PRIMARY CONSTRAINT

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THE LEARNING ENGINEERING LENS

> *Predictions trigger support, not gatekeeping.*

EDITORS' SYNTHESIS OF THE GSU ADVISING MODEL, DRAWN FROM RENICK & STROM (2020) AND NEW YORK TIMES COVERAGE (2018)

LE INSIGHT

GSU is the positive counterpart to A-Level (35), Robodebt (36), and educational algorithmic bias (37). The same technical capability — a predictive model — was deployed under a different design constraint and produced an equity outcome rather than an equity harm. The case is the strongest available evidence that the **construct definition** and the **human-loop architecture** determine whether a predictive model produces good or harm, not the model itself.

LENS APPROACH

LENS treats GSU as the canonical positive case for predictive analytics in education. LEN 4 examines the measurement architecture that made equity a primary outcome. LEN 7 examines the governance structure that kept the system as decision support rather than automated decision. LEN 1 uses it as a problem-framing exemplar.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Set equity as a primary design constraint from the start — the explicit aim of closing gaps — rather than discovering disparities after deployment.
- Build a human-loop architecture so alerts trigger proactive support routed through an advisor, with the model as decision support and a person deciding the action.
- Tune the monitoring to fire early — at the missed assignment, not the failed semester — so the intervention reaches the student while it can still change the outcome.

IN OPERATION / AFTER

- Track outcomes by demographic group as a primary metric, so the system is judged on whether it closes gaps rather than merely raises the average.
- Confirm the overall gain does not come at the expense of the most at-risk students — eliminating the gap, not just narrowing it, is the test that the design held.
- Keep human judgment in the loop as the system scales, so prediction continues to deliver more support rather than drifting into automated gatekeeping.

REFLECTION QUESTIONS

1. What is the difference between GSU's predictive analytics and the algorithmic-bias cases of Chapter 5? Be specific about what makes the GSU implementation work.
2. Design the equity-as-primary-constraint version of a predictive system in your domain. What would you measure first?
3. GSU used predictions to deliver more support rather than to gatekeep, with an advisor between the alert and the action. Identify a predictive system in your domain and specify the human-loop architecture that would keep it serving the people it flags rather than screening them out.

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CASE 111 AVIATION 1976 – PRESENT

Aviation Safety Reporting System (ASRS)

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT NASA-administered confidential reporting system; more than 2M reports received; foundational architecture for evidence-driven aviation safety

IN BRIEF

The Aviation Safety Reporting System, run by NASA on behalf of the FAA since 1976, accepts confidential reports from pilots, controllers, mechanics, and cabin crew about incidents, near-misses, and safety concerns. Its load-bearing feature is institutional, not technical: reporting an event to ASRS confers immunity from FAA enforcement for the conduct reported, within specified limits, making honest reporting the rational choice. Over nearly fifty years and more than two million reports, ASRS has become the world's largest repository of aviation-safety information, surfacing patterns — automation surprise, runway incursions, fatigue effects — before they reached formal investigation thresholds. The architecture has been emulated across domains, and is the canonical success case for an evidence system paired with a credible commitment to non-punitive use.

BACKGROUND

The most valuable safety information in any high-consequence domain lives with front-line operators — the near-misses and quiet errors that never reach an accident report. But operators will not surrender that information to an institution that can punish them for it, so the data that could prevent the next accident stays locked in the people who hold it. The incentives run backward: the person best placed to report a near-miss is the same person a punitive system gives the strongest reason to stay silent, so the data that matters most is the data least likely to surface.[0]

THE INTERVENTION

In 1976 the FAA and NASA created the Aviation Safety Reporting System, a confidential channel for pilots, controllers, mechanics, and cabin crew to report incidents, near-misses, and concerns. NASA — not the regulator — administers it, and reporting an event confers immunity from FAA enforcement for the conduct reported, subject to specified limits. Putting a neutral party between the reporter and the enforcer directly addressed the backward incentive — an operator now had a positive reason to report, because doing so converted potential jeopardy into protection.[1]

HOW IT WORKED

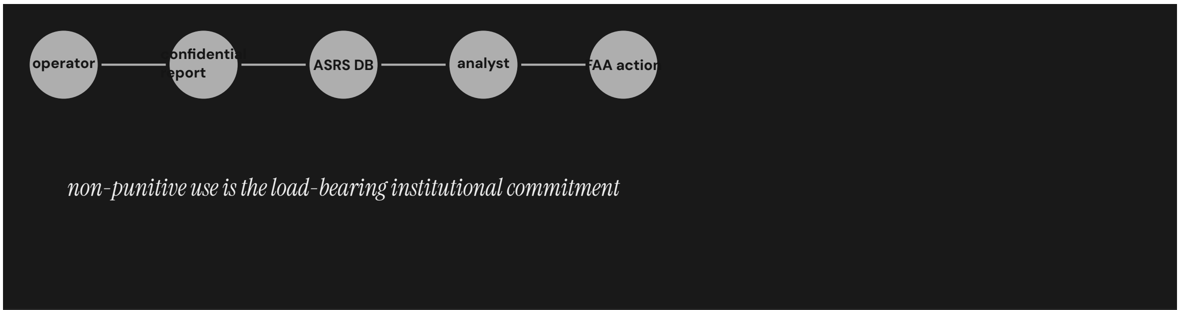
The system pairs a technical artifact (the reporting form and a searchable database) with a cultural commitment (protected, non-punitive use). The immunity provision makes reporting the rational choice for the operator, and NASA's role as a neutral third party makes the protection credible. Either half alone fails; together they make the data flow to the institution that can act on it. The credibility of the protection is what does the work — a promise of non-punishment from the regulator itself would be doubted, so routing it through NASA is what makes operators trust it enough to report.[2]

THE EVIDENCE

Over nearly fifty years ASRS has accumulated more than two million reports — the largest single repository of aviation-safety information in the world. Patterns such as automation surprise, runway incursions, and fatigue effects were first identified at scale through ASRS data before they crossed formal investigation thresholds. Surfacing a pattern before it reaches an accident is the whole point — the value of the system is the hazards it lets the industry act on while they are still near-misses, not the reports it collects after the fact.[3]

WHAT TRANSFERRED

ASRS has been studied and emulated across domains — patient-safety reporting systems, the maritime and aviation CHIRP scheme, and similar systems in rail and nuclear power. It is the canonical positive case for evidence architecture paired with an institutional commitment to non-punitive learning, the defining design pattern of a "just culture." The breadth of emulation underscores that the load-bearing element travels — wherever the most valuable safety data sits with operators who fear punishment, the same protected-reporting design recurs as the way to unlock it.[4]



ASRS — PAIRED TECHNICAL ARTIFACT AND PROTECTED REPORTER

REFERENCES

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2. Reason, J. (1997), Managing the Risks of Organizational Accidents — non-punitive reporting as a model (paraphrased).
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4. Dekker, S. (2012), Just Culture — the cultural commitment to non-punitive use.
5. CHIRP and patient-safety reporting-system documentation — cross-domain emulation.

THE LEARNING ENGINEERING LENS

> *ASRS is the model for confidential, voluntary, non-punitive incident reporting in any high-consequence domain.*

> PARAPHRASING JAMES REASON, MANAGING THE RISKS OF ORGANIZATIONAL ACCIDENTS, 1997

LE INSIGHT

ASRS is the canonical positive case for paired-intervention evidence architecture. The technical artifact (the reporting form and the database) is paired with the cultural commitment to non-punitive use. Either alone fails. Together they have produced the most comprehensive operational-safety dataset in any domain.

LENS APPROACH

LENS uses ASRS in LEN 4 as the foundational positive case for evidence architecture and in LEN 8 for institutional commitment to non-punitive learning. Studio projects design ASRS-equivalents for new domains.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pair a simple reporting artifact (a form and searchable database) with a credible commitment to non-punitive use, since the channel without the protection collects nothing of value.
- Confer immunity for reported conduct so that reporting becomes the rational choice, directly reversing the incentive that otherwise keeps the most valuable data hidden.
- Route the protection through a neutral third party rather than the enforcer, so operators trust the non-punitive promise enough to report against their own interest.

IN OPERATION / AFTER

- Analyze the accumulated reports to surface patterns — automation surprise, runway incursions, fatigue — and act on them while they are still near-misses rather than accidents.
- Protect the immunity provision over time, since a single high-profile punishment of a reporter would collapse the trust the whole system depends on.
- Export the protected-reporting design, not just the database, to new domains, adapting the neutral-party structure wherever valuable safety data sits with operators who fear punishment.

REFLECTION QUESTIONS

1. Identify a domain in your institution that would benefit from an ASRS-equivalent. What cultural commitment would be required for it to function?

2. Design the institutional commitment that makes an ASRS-equivalent operational rather than merely declared.

3. ASRS made the protection credible by routing it through NASA rather than the regulator. Identify a reporting channel in your domain that operators distrust, and specify the neutral party or structural separation that would make its non-punitive promise believable.

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CASE 112 HEALTHCARE 1984 – PRESENT

Bristol Heart Babies Reform

G Governance & Trust N Normalization of Deviance

IMPACT Foundational UK case in clinical outcomes transparency; produced specialty-level performance reporting in UK cardiac surgery

IN BRIEF

Through whistleblowing and a public inquiry, the Bristol Royal Infirmary was found to have had substantially worse pediatric cardiac-surgery outcomes than other UK centers for years. The Kennedy Inquiry — one of the most influential UK healthcare inquiries in modern times — located the capability gap in the absence of routine specialty-level outcomes reporting: surgeons did not know how their results compared with peers, patients did not know, and referrals did not respond to actual outcome data. The reform built national specialty-level outcomes registries, first in cardiac surgery and then across other specialties, making the UK one of the few countries that routinely publishes surgeon-level results — a paired intervention of technical registry plus institutional commitment to publish that ended outcomes data as the private property of clinicians.

BACKGROUND

In pediatric cardiac surgery, small differences in a unit's performance can mean the difference between a child living and dying — yet in the UK of the 1980s and early 1990s, no system routinely compared outcomes between centers. A surgeon, a hospital, or a referring physician had no reliable way to know whether a given unit's results were normal or dangerously poor. Without comparison across centers, a dangerously poor unit and an ordinary one looked alike from inside, so the very gap that cost children's lives was the one no one was positioned to see.[0]

THE INTERVENTION

After whistleblowing and a public inquiry into deaths at the Bristol Royal Infirmary between 1984 and 1995, the Kennedy Inquiry recommended routine, risk-adjusted, specialty-level outcomes reporting. The reform built national registries — beginning with cardiac surgery and extending to other specialties — together with a commitment to publish results, including at the level of individual surgeons. Starting in cardiac surgery and then extending outward was deliberate sequencing — the specialty where the harm had been exposed proved the model, and the registry then spread to fields that had not had their own Bristol.[1]

HOW IT WORKED

The intervention was explicitly paired. The technical half — registry design, statistical risk adjustment so that surgeons taking hard cases are not penalized, and a publication architecture — was necessary but not sufficient. The cultural half — surgeons accepting that their comparative results would be visible — was equally load-bearing, and was the harder of the two to secure. The risk adjustment was what made the cultural half securable: without it, surgeons who took the sickest patients would have been punished by the raw numbers, giving them every reason to resist publication or avoid hard cases.[2]

THE EVIDENCE

The UK became one of the few countries that routinely publishes surgeon- and unit-level outcomes, and the cardiac-surgery registry is among the most mature specialty-outcomes regimes in any country. Outcomes data ceased to be the private property of clinicians and became a shared resource for patients, referrers, and surgeons themselves. That surgeons themselves became users of the data, not just subjects of it, is part of why the regime endured — comparison that had once felt like exposure became a tool the profession relied on to know where it stood.[3]

WHAT TRANSFERRED

Bristol is the foundational UK case for outcomes transparency as a paired-intervention deliverable, and its registry model has been extended across NHS specialties and influenced later national-quality reforms. It pairs with Keystone ICU (Case 71) as healthcare-outcomes interventions operating at different layers — the bedside protocol and the system-level measurement regime. The two layers are complementary rather than competing: Keystone changes what happens at the point of care, while Bristol changes what the system can see about results across centers, and a mature regime needs both.[4]

BEFORE	AFTER
private	published

outcomes data ceased to be the private property of clinicians

BRISTOL — OUTCOMES TRANSPARENCY AS PAIRED INTERVENTION

REFERENCES

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4. Sherlaw-Johnson et al. — risk-adjusted outcome methodology.
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> *Outcomes data must cease to be the private property of clinicians and become a shared institutional resource.*
>
> PARAPHRASING THE KENNEDY INQUIRY FINAL REPORT (LEARNING FROM BRISTOL), 2001

LE INSIGHT

Bristol is the canonical UK case for outcomes-transparency as a paired intervention. The technical capability — registry design, statistical risk adjustment, publication architecture — was necessary. The cultural capability — surgeons accepting that their results would be visible and comparable — was equally necessary. The pair has produced one of the most mature specialty-outcomes regimes in any country.

LENS APPROACH

LENS uses Bristol in LEN 4 for outcomes-transparency as a paired- intervention deliverable and in LEN 7 for the institutional reform that made surgeon-level publication acceptable. The case pairs with Keystone ICU (Case 71) as healthcare-outcomes interventions at different layers.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pair the technical instrument — a registry with statistical risk adjustment and a publication architecture — with the cultural change of practitioners accepting that comparative results will be visible.
- Build risk adjustment in from the start so those who take the hardest cases are not punished by raw numbers, which is what makes the cultural half securable.
- Sequence the rollout to begin where the harm was exposed, proving the model in one specialty before extending it to fields that have not had their own crisis.

IN OPERATION / AFTER

- Make practitioners users of the data, not just subjects of it, so comparison becomes a tool the profession relies on rather than an exposure it resents — which is what sustains the regime.
- Extend the registry model across specialties over time, turning a single reform into a system-wide measurement regime.
- Pair the system-level transparency layer with point-of-care interventions (as with Keystone), since a mature outcomes regime needs both what the system can see and what happens at the bedside.

REFLECTION QUESTIONS

1. What is the equivalent of surgeon-level outcomes transparency in your domain? What cultural change would have to accompany the technical instrument?
2. Design the registry and publication architecture for a specialty in your domain currently operating without outcomes transparency.
3. Bristol's risk adjustment was what let surgeons accept publication, by ensuring those who took the hardest cases were not punished by raw numbers. Identify a transparency measure in your domain that practitioners resist, and design the adjustment that would make the comparison fair enough to accept.

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CASE 113 CLINICAL MEDICINE HEALTH EQUITY 2021

Removing the Race Coefficient from eGFR

D Designed Out G Governance & Trust N Normalization of Deviance

IMPACT A clinical estimating equation that for two decades raised estimated kidney function for Black patients was retired through a governance process; the documented effect on disparities remains unknown

IN BRIEF

For two decades the standard equation used to estimate glomerular filtration rate from serum creatinine — the kidney-function number on routine lab reports — included a coefficient that raised the estimate for patients reported as Black. The downstream effects were documented: later nephrology referral, later wait-listing for transplant. In 2021 the joint NKF-ASN Task Force, after reviewing over twenty candidate approaches with patient and clinician input, recommended immediate national implementation of the race-free 2021 CKD-EPI creatinine equation (Inker et al., NEJM 2021); clinical laboratories moved to adopt it. The case is a governance intervention — a change-control process that retired a construct after three decades of operational use. The honest hedge, preserved from the Task Force report and from follow-up commentaries, is that the new equation introduces a small bias for all groups, the disparities effect **remains unknown**, and the case is the construct-definition act, not a closed outcome proof. It pairs with pulse oximetry (Case 114) and pain assessment (Case 95) as the trio of "what counts as the same patient across race."

BACKGROUND

Glomerular filtration rate (GFR) is the standard measure of kidney function and the threshold variable for nephrology referral, medication dosing, and transplant wait-listing. Because direct GFR measurement is impractical at clinic scale, equations estimate it from serum creatinine and demographic inputs. The dominant equations from the late 1990s onward included a Black-race coefficient that raised the estimate for patients reported as Black — making them appear to have better kidney function than the same creatinine value implied for White patients.[0]

THE INTERVENTION

The downstream effects of that coefficient are documented across the nephrology literature: later referral to specialty care, later transplant wait-listing, and a population-level shift in who was counted as having advanced chronic kidney disease. The defenders of the coefficient pointed to differences in creatinine generation across self-identified populations; the critics pointed to the logical and ethical problems of embedding a population-level adjustment in an individual-patient decision tool, and to the construct-definition question — what is the equation supposed to be measuring, and across whom does it have to mean the same thing?[1]

HOW IT WORKED

The NKF-ASN Task Force, formed in 2020, ran the construct-revision process as a governance exercise: an external panel, patient-clinician input, more than twenty candidate alternatives, a sustained review window, and a published report (Delgado et al., Am J Kidney Dis 2021). The Task Force recommended immediate national adoption of the 2021 CKD-EPI creatinine equation, which eliminates the race coefficient. The replacement equation was published in parallel (Inker et al., NEJM 2021). Major laboratories and health systems moved to adopt the new equation within months.[2]

THE EVIDENCE

The hedge in both the Task Force report and the follow-up commentary is load-bearing and is preserved in the case. The new equation introduces a small bias for all groups relative to measured GFR; the **net effect on documented disparities in nephrology care and transplant access** is not yet measured at population scale, and the literature explicitly states that the disparities outcome **remains unknown**. The case is the construct-definition act — the right kind of governance intervention, run with the right kind of process — and the outcome evidence is the continuing work.[3]

WHAT TRANSFERRED

What the eGFR case teaches is that some equity capability deliverables are construct-definition acts: choosing what gets predicted and what counts as the same patient is the design decision, not a downstream remediation. It pairs with pulse oximetry (Case 114), where the failure was in the validation architecture rather than the equation; and with pain assessment (Case 95), where the failure was in clinician-held false beliefs rather than the instrument. The trio together is the case-grounded basis for the CLO **Gap attribution** — distinguishing capability gaps attributable to human development, system design, and organizational performance — and for the CLO **Fairness beyond omission**, of which eGFR is the construct-removal instance.

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3. Eneanya, Yang, & Reese (2019), "Reconsidering the Consequences of Using Race to Estimate Kidney Function," JAMA 322(2):113–114 — the equity argument that motivated the revision.
4. Vyas, Eisenstein, & Jones (2020), "Hidden in Plain Sight — Reconsidering the Use of Race Correction in Clinical Algorithms," NEJM 383(9):874–882 — broader race-in-algorithms survey.

> *What counts as the same patient across race is a construct-definition decision, not a downstream remediation.*
>
> EDITORS' SYNTHESIS OF DELGADO ET AL. (2021) AND VYAS ET AL. (2020).

LE INSIGHT

eGFR is the canonical recent instance of construct-definition as an equity capability deliverable. A race coefficient was embedded in a continuous estimating equation for two decades, documented downstream effects on referral and transplant access, and was retired through a governance process. The disparities effect of the change remains unknown; the case is the construct-revision act, not the closed outcome.

LENS APPROACH

eGFR is the construct-definition equity intervention in the race-construct trio (Cases 113, 114 and 95). LENS uses it in Domain 4 (Test and Evaluation) for the CLO **Gap attribution** — attributing the gap to construct definition, not to clinicians or patients — and in Domain 5 (Navigating Sociotechnical Constraints) for the governance process that revised the standard. Direct trio with Case 114 (pulse oximetry — the validation-architecture mechanism) and Case 95 (pain assessment — the human-development mechanism). Adjacent to the lending pair (Cases 94–107) at a different layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Reconsider the construct: ask what the equation should predict and across whom it should mean the same thing, separately from optimizing the residual.
- Run construct-revision as a governance process: an external panel, patient and clinician input, candidate alternatives examined openly, and a published report.
- Carry the hedge — the small all-group bias introduced and the unmeasured disparities effect — into the deployment documentation, not the press release.

IN OPERATION / AFTER

- Instrument the downstream outcomes (nephrology referral, transplant wait-listing) demographically so the disparities effect of the construct revision is actually measurable at population scale.
- Treat construct revision as ongoing: a coefficient was removed; a new validation architecture is the longer work.
- Make the construct-definition decision auditable — publish the candidate alternatives that were considered and the basis on which the chosen alternative was selected.

REFLECTION QUESTIONS

1. Identify a clinical or algorithmic construct in your domain that embeds a population-level adjustment in an individual-patient decision. Was the construct revised, examined, or never questioned? What governance process would you run if it had to be revisited?
2. Specify the downstream outcomes you would instrument demographically to make the disparities effect of a construct revision measurable at population scale.
3. The new CKD-EPI equation introduces a small bias for all groups and the disparities effect **remains unknown**. What is the minimum follow-up evidence you would require before concluding the construct revision improved or worsened the equity outcome you care about?

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CASE 114 MEDICAL DEVICES CLINICAL CARE HEALTH EQUITY 1990 – 2020 – 2025

Pulse Oximetry Across Skin Tones

D Designed Out G Governance & Trust N Normalization of Deviance

IMPACT A bedside device validated on a non-representative population systematically under-detected hypoxemia in Black patients for thirty years; the bias persisted because device validation was never demographically stratified

IN BRIEF

Pulse oximetry — the noninvasive bedside measurement of blood oxygen saturation — is one of the most widely used patient-monitoring tools in clinical medicine. Sjoding et al. (NEJM 2020) found that across two large cohorts, Black patients had nearly three times the frequency of **occult hypoxemia** (arterial saturation <88% despite a pulse-ox reading of 92–96%) as White patients. The finding replicated Jubran & Tobin (Chest 1990), published thirty years earlier and overlooked operationally. The bias persisted because device validation was never demographically stratified — the aggregate accuracy number on FDA clearance documentation concealed a group-specific failure mode. The discovery drove FDA review and, in January 2025, a draft guidance recommending that manufacturers evaluate device performance across diverse skin pigmentations during validation. The case is a **failure-to-intervention arc**: the failure sat in the validation architecture for three decades; the intervention is the regulatory change-control on validation, and its measured effect on the disparities outcome is not yet documented.

BACKGROUND

Pulse oximetry replaced repeated arterial blood draws as the bedside standard for monitoring oxygen saturation in the 1980s and 1990s. The device shines two wavelengths of light through tissue and infers saturation from the absorbance ratio. The inference depends, among other things, on the absorbance properties of the intervening tissue — including melanin pigmentation. The clearance documentation reports an aggregate accuracy number against arterial blood gas measurements.[o]

THE INTERVENTION

In 1990 Jubran & Tobin reported, in Chest, that pulse oximeters tended to overestimate true oxygenation in patients with darker skin. The paper was published, cited intermittently, and did not drive a change in validation practice. Thirty years later Sjoding et al. (NEJM 2020) revisited the question in two large modern cohorts and reported that Black patients had nearly three times the frequency of **occult hypoxemia** — true arterial saturation below 88 percent despite a pulse-oximeter reading of 92–96 percent — as White patients. The structural form of the finding was the same as Jubran & Tobin's; the population and the salience were different, and the COVID-19 pandemic, which made pulse oximetry a household-scale triage tool, made it harder to ignore.[1]

HOW IT WORKED

The bias persisted because device validation was never demographically stratified at clearance. The aggregate accuracy number — clinically acceptable on average — concealed a group-specific failure mode. The capability gap was not in the clinician using the device or in the manufacturer's engineering; it was in the regulatory machinery that approved a measurement device without checking whether its measurement held across the patients it would meet. The Sjoding paper's lasting contribution was not the technical finding alone — Jubran & Tobin had supplied that — but the disconfirmation of the validation architecture.[2]

THE EVIDENCE

The FDA's January 2025 draft guidance recommends that pulse oximeter manufacturers evaluate device performance across diverse skin pigmentations during validation, and that the validation protocol explicitly stratify accuracy by skin tone. The guidance is the corrective-action half of the case: an intervention in the validation architecture, not in the device itself. The measured effect on the downstream disparities outcome — under-treated hypoxemia in patients of color — is not yet documented; it is the continuing work the intervention sets up.[3]

WHAT TRANSFERRED

What the case teaches is that a measurement-device failure can persist for three decades inside an aggregate accuracy number, and that the capability deliverable is a validation architecture that surfaces group-specific failure modes by stratifying outcome metrics by relevant demographic characteristics. Pulse oximetry pairs with eGFR (Case 113) and pain assessment (Case 95) in the race-construct trio. The mechanisms are distinct — eGFR is construct definition; pulse oximetry is validation architecture; Hoffman is clinician-held false belief — and the case-grounded lesson is that the diagnosis of the same surface harm has to attribute the gap to the right layer of the system before a remediation can land.

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2. Jubran & Tobin (1990), "Reliability of Pulse Oximetry in Titrating Supplemental Oxygen Therapy in Ventilator-Dependent Patients," Chest 97(6):1420–1425 — original finding, published thirty years earlier.
3. FDA (2025), "Pulse Oximeters for Medical Purposes — Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations: Draft Guidance for Industry and Food and Drug Administration Staff," issued January 7 2025, Docket No. FDA-2023-N-4976; Federal Register notice 2024-31540 — the regulatory corrective-action artifact, language may evolve in final.
4. Fawzy et al. (2022), "Racial and Ethnic Discrepancy in Pulse Oximetry and Delayed Identification of Treatment Eligibility Among Patients With COVID-19," JAMA Internal Medicine — downstream effect during the pandemic.

> *Aggregate accuracy is not group accuracy. A device can be acceptable on average and unsafe for one population.*
>
> EDITORS' SYNTHESIS OF SJODING ET AL. (2020) AND THE FDA 2025 DRAFT GUIDANCE.

LE INSIGHT

Pulse oximetry is the validation-architecture instance of the race-construct trio. The bias was published in 1990, persisted for thirty years inside aggregate clearance accuracy, was re-documented in 2020, and reached the regulatory architecture in 2025. The capability deliverable is demographic stratification at validation, not after deployment.

LENS APPROACH

Pulse oximetry is the validation-architecture intervention in the race-construct trio (Cases 113, 114 and 95). LENS uses it in Domain 4 (Test and Evaluation) for the CLO **Gap attribution** — the gap is in the validation architecture, not the device or the clinician — and in Domain 5 (Navigating Sociotechnical Constraints) for the FDA clearance / device oversight regime. Adjacent to Case 130 (radiology AI miscalibration), which is the same diagnosis at a different technological boundary, and to the Epic Sepsis Model (Case 93) for the post-deployment-surveillance pattern.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Stratify device-validation outcomes by every demographic characteristic that can change the measurement physics, before clearance, not after deployment.
- Specify the group-specific accuracy that would count as acceptable, separately from the aggregate; do not allow the aggregate to substitute.
- Treat replication of an earlier finding (Jubran & Tobin → Sjoding) as a verification trigger, not a duplication — the same finding in a different population is itself evidence.

IN OPERATION / AFTER

- Audit deployed devices on the population that actually uses them, on a schedule that surfaces drift; aggregate accuracy is not a substitute.
- Tie the regulatory clearance update to a downstream-outcome surveillance plan — under-treated hypoxemia, in this case — so the intervention's effect on the harm can be measured.
- When a published finding sits operationally inert for years, ask whether the publication channel reached the operational community; the structural problem may be in how validation evidence diffuses, not in whether it exists.

REFLECTION QUESTIONS

1. Identify a measurement device in your domain whose validation reports an aggregate accuracy number. Across which demographic characteristics could the measurement physics change? What would a stratified validation protocol look like, and who would have to approve it?
2. The Sjoding finding replicated Jubran & Tobin thirty years later. What is the institutional architecture that should have surfaced the original finding to the regulatory regime? Where did the publication-to-operations channel break, and where does it still break in your domain?
3. The FDA 2025 draft guidance corrects the validation architecture. Specify the downstream outcome (under-treated hypoxemia in patients of color) and the surveillance plan you would tie to the guidance so the intervention's effect on harm is measurable.

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CASE 115 HIGHER ED (UK) LEARNING ANALYTICS DATA GOVERNANCE 2014 – 2025

Open University 'Ethical Use of Student Data' and OU Analyse

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT The first institutional 'Ethical Use of Student Data' policy in higher education (2014); an eight-principle consent architecture co-designed with students that unblocked predictive analytics on hundreds of thousands of learners and supported documented intervention work

IN BRIEF

Predictive learning analytics at the Open University faced a real governance objection: large-scale processing of student data for intervention, with pre-GDPR scrutiny and credible surveillance concerns. The OU's response in 2014 was to author the first institutional "Ethical Use of Student Data for Learning Analytics" policy in higher education — eight principles built through wide stakeholder consultation including students, framing students as participants rather than data subjects. The consent-and- transparency architecture was the enabling engineering, not a compliance afterthought; the deployment followed because trust was established first. OU Analyse — a weekly machine-learning at-risk prediction system — operated on top of that architecture, and the Analytics4Action framework (Rienties et al., JIME 2016) paired predictions with tutor judgment and documented interventions on modules of 3,000+ students. A 2019 evaluation (Herodotou et al., BJET) across 559 teachers and 14,000+ students examined how degree of system usage related to outcomes. The honest open question, raised by the OU's own researchers, is whether predictive analytics genuinely serves students versus surveils them — a tension that remains contested, and the policy has since been superseded by a broader Data Ethics Policy. Governance as a living artifact, not a solved problem.

BACKGROUND

The Open University runs higher education at distance scale — cohorts of tens of thousands across a single module, with no classroom signal that a student is falling behind. Predictive learning analytics on the engagement and assessment trace can, in principle, surface that signal in time for a tutor to act. The governance objection in the early 2010s was not abstract: the data volumes were large, the scrutiny pre-GDPR but tightening, and the surveillance concern — that institutional analytics treat students as objects rather than partners — was credible enough that several universities had paused similar programs.[0]

THE INTERVENTION

The OU's intervention in 2014 was to author the "Ethical Use of Student Data for Learning Analytics" policy, the first such policy in higher education. The development process was the teaching point as much as the document was: wide stakeholder consultation including the Students Union, eight principles that framed students as participants in the analytics rather than its subjects, and a published artifact the institution could be held to. The policy was the **enabling engineering** for analytics deployment, not a compliance overlay applied to a deployment that was happening anyway.[1]

HOW IT WORKED

OU Analyse — a weekly machine-learning at-risk prediction system — operated on top of that architecture. The Analytics4Action framework (Herodotou et al., JIME 2016) paired the prediction with tutor judgment rather than treating the model output as a decision: tutors reviewed flagged students, made the intervention call, and documented what action they took. The framework was evaluated on modules of more than 3,000 students. A 2019 evaluation (Herodotou et al., BJET) across 559 teachers and 14,000+ students examined how degree of teacher engagement with the system related to outcomes; the pattern was that engagement, not the raw prediction, was what tracked with intervention success.[2]

THE EVIDENCE

The honest open question, raised by the OU's own researchers, is whether predictive learning analytics genuinely serves students versus surveils them. The OU's policy did not foreclose that question; it made the deployment legible enough to argue about, and it superseded the 2014 policy with a broader Data Ethics Policy as practice and the scrutiny regime evolved. Governance as a living artifact is the case. The capability deliverable is not a final answer to the surveillance question — there isn't one — but a consent-and-transparency architecture that lets the institution decide and the students participate in the deciding.[3]

WHAT TRANSFERRED

What OU teaches in the pair (Cases 115 + 110) is the governance-objection diagnostic: when the objection is about trust and accountability — as it was at the OU — good design can dissolve it, and the deployment can proceed under a credibly co-authored consent architecture. The pair's contrast case is the Dutch SyRI (Case 155), where the governance objection was correct: the system was both rights-violating and operationally ineffective, and the District Court of The Hague stopped it on Article 8 ECHR grounds. The diagnostic capability is to tell those two situations apart before deployment, not after.

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> *The consent architecture is the enabling engineering. The deployment is what follows once trust is established.*
>
> EDITORS' SYNTHESIS OF THE OU 2014 POLICY AND SLADE & PRINSLOO (2013).

LE INSIGHT

The Open University case is the cleanest instance in the sweep of a governance objection dissolved by design. The 2014 ethical-use policy was the enabling engineering; OU Analyse and Analytics4Action operated on top of it; the 2019 evaluation showed teacher engagement tracked with intervention success. The open question — serve vs. surveil — remains contested, and the policy is a living artifact, not a solved problem.

LENS APPROACH

OU is the positive Domain 4 / Problem Type 5 stakeholder governance case (induced 5.1; LENS D4/PT5). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the governance-objection diagnostic — the dissolvable-objection side — and in Domain 4 (Test and Evaluation) for the engagement-tracked-outcome evidence. Direct pair with Case 155 (Dutch SyRI), where the governance objection was correct and design could not have dissolved it. Adjacent to Case 93 (Epic Sepsis Model) as the inverse pattern: OU built consent before deployment; Epic deployed without validation, and the objection that should have been raised wasn't.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build the consent-and-transparency architecture as enabling engineering, not as compliance afterthought; the document is the artifact the deployment can be held to.
- Co-author the policy with the people the analytics will be applied to — students or operators — including their voice in the principles before the system is built.
- Pair predictions with human judgment by design: the prediction surfaces a candidate; the tutor or operator makes the decision and documents what they did.

IN OPERATION / AFTER

- Evaluate the system on engagement and intervention quality, not adoption alone; engagement is what tracks with student outcome at the OU.
- Treat the governance policy as a living artifact: when the scrutiny regime or practice evolves, supersede the policy openly rather than letting it fall into disuse.
- Keep the open question — does this genuinely serve students or surveil them? — visible in the institutional record; the legitimacy of the program depends on the question staying askable.

REFLECTION QUESTIONS

1. Identify a deployment in your domain that faced a governance objection about trust and accountability. Was the objection dissolved by design — a consent or transparency architecture co-authored with the affected parties — or was it managed as compliance overlay? What did the difference cost?

2. Specify the engagement metric you would track, separately from adoption, to know whether your deployment is producing intervention quality. At the OU it was teacher engagement; what is the analog in your context?

3. The OU's open question — does predictive analytics genuinely serve students or surveil them — remains contested, and the policy has been superseded. What is the institutional discipline that keeps the question visible and the policy a living artifact rather than a one-time document?

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CASE 116 ANESTHESIOLOGY PATIENT SAFETY MEDICAL DEVICES 1986 – PRESENT

Anesthesia Monitoring Standards and the APSF — The Mortality Decline

H Human-System Interface K Knowledge & Institutional Memory G Governance & Trust

IMPACT Mandatory pulse oximetry and capnography monitoring (Harvard 1986; ASA 1986–87) converted silent hypoxemia and esophageal intubation from undetectable failures into monitored, recoverable ones; anesthesia-related mortality fell dramatically over subsequent decades — multifactorial decline

IN BRIEF

Through the early 1980s, anesthesia in the United States carried a documented patient-safety crisis: silent intraoperative hypoxemia and unrecognized esophageal intubation produced catastrophic outcomes — brain injury and death — that were structurally undetectable until the harm had occurred. A 1982 ABC news special on anesthesia catastrophes converged with a malpractice-insurance crisis to force institutional attention. In 1986 Eichhorn et al. (JAMA, 1986) published the Harvard Medical School minimum monitoring standards — continuous pulse oximetry and capnography were the load-bearing additions, designed specifically to make hypoxemia and misplaced endotracheal tubes detectable early enough to recover. The ASA adopted essentially the same standards in 1986–87. The Anesthesia Patient Safety Foundation, founded in 1985, institutionalized the broader change-management effort. Anesthesia-related mortality fell dramatically over subsequent decades — one widely cited Brazilian series reports a fall toward zero — and malpractice premiums declined alongside. The field's own histories preserve the hedge: the decline has multiple co-varying causes (training, device design, pharmacology, team composition), and the device standards themselves still fail in documented edge cases.

BACKGROUND

Through the early 1980s, anesthesia in the United States carried a documented patient-safety crisis with two structurally similar failure modes at its center. Silent intraoperative hypoxemia — falling oxygenation in a paralyzed, ventilated patient — produced brain injury and death without the clinical signal a conscious patient would have given. And esophageal intubation — endotracheal tube misplaced in the esophagus instead of the trachea — produced no ventilation at all, often noticed only when the patient was already decompensating. Both failures were detectable in principle and in practice undetectable, because the cue did not reach the anesthetist in time.[0]

THE INTERVENTION

The institutional forcing function arrived from outside the field. A 1982 ABC news special documented a string of catastrophic anesthesia outcomes, and the malpractice insurance environment for anesthesiology had reached a crisis level that made the status quo untenable. In 1985 the Anesthesia Patient Safety Foundation was founded — among the first specialty-organized patient-safety foundations in medicine. In 1986 Eichhorn et al. published in JAMA the Harvard Medical School minimum monitoring standards: a defined minimum set of continuous monitors for every anesthetized patient, with continuous pulse oximetry and capnography as the load-bearing additions.[1]

HOW IT WORKED

Pulse oximetry made oxygenation continuously visible to the anesthetist; capnography made the end-tidal carbon-dioxide trace visible, which immediately distinguishes tracheal from esophageal intubation in the first breaths. Together they converted two structurally undetectable failures into monitored, recoverable ones. The American Society of Anesthesiologists adopted essentially the same minimum standards in 1986–87. Eichhorn's follow-up (Anesthesiology, 1989) reported declines in preventable mishaps in the Harvard hospitals consistent with the mechanism the standards targeted.[2]

THE EVIDENCE

Anesthesia-related mortality fell dramatically over the subsequent decades. A widely cited Brazilian series reported decline toward zero. Malpractice premiums for anesthesiology declined alongside — one of the few examples in American medicine of a specialty's malpractice cost falling as the specialty's safety record improved. The case is regularly cited as a canonical example of the cue/alert design as the capability deliverable: when the right signal is made visible at the right point in the workflow, an undetectable failure becomes recoverable, and the institutional outcome record moves.[3]

WHAT TRANSFERRED

The hedges the field's own histories preserve are load-bearing. The mortality decline has multiple co-varying causes — anesthetist training, device design improvements, pharmacological change (newer agents are inherently safer), team composition with broader monitoring presence, and the monitoring standards themselves — and attribution of the entire effect to the standards overstates what the evidence can support. The device standards themselves still fail in documented edge cases: pulse oximetry is unreliable across skin tones (Case 114 in this same v2 batch), and capnography can mislead in specific physiology. The case teaches the cue/alert-design form of capability engineering at its most durable, with the honest qualification that the institutional decline is consistent with the standards but not isolatable to them alone.[4]

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5. Sjoding et al. (2020), NEJM — the racial-bias edge case the standard still carries (cross-reference Case 114).

> *The deliverable is not the device. It is the standard that makes the device's signal non-waiverable across the specialty.*
>
> EDITORS' SYNTHESIS OF EICHHORN ET AL. (1986) AND APSF FOUNDING DOCUMENTS.

LE INSIGHT

The Harvard / ASA / APSF anesthesia-monitoring intervention is the canonical cue/alert-design success: silent intraoperative hypoxemia and esophageal intubation were converted from undetectable failures into monitored, recoverable ones, and the specialty's mortality and malpractice record moved over subsequent decades. The decline is multifactorial; the device standards still fail in documented edge cases (Case 114).

LENS APPROACH

Anesthesia monitoring is the canonical cue/alert intervention (induced 3.1; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for the cue/alert design as deliverable; in Domain 5 (Navigating Sociotechnical Constraints) for the APSF / ASA institutional discipline that made the standard non-waiverable; and in Domain 3 (Human-System Collaboration) at the human–device boundary. Adjacent to Case 114 (pulse oximetry across skin tones) — same instrument, the canonical edge case the standard still carries. Pair with Case 109 (WHO Surgical Checklist) at the mandatory-standard layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the cue/alert to surface the failure mode that produces harm — silent hypoxemia, misplaced tube — at the point in the workflow where the operator can still recover, not in a post-hoc record.
- Pair the device standard with the institutional commitment (APSF, ASA) that makes the standard non-waiverable across the specialty, so adoption is a profession-level deliverable rather than a per-institution choice.
- Build the standard with edge-case acknowledgment: the device is the strongest available signal, not closed proof; pulse oximetry across skin tones (Case 114) is the canonical edge case the standard has to keep visible.

IN OPERATION / AFTER

- Track outcome metrics (mortality, malpractice premiums) alongside the standards, while attributing carefully — multifactorial declines are the norm, not the exception, in long-window safety interventions.
- Treat the malpractice-cost signal as institutional evidence the standard is recognized to have worked, not as the safety evidence itself; the safety evidence is mortality and harm.
- Maintain the edge-case surveillance the standard sets up: pulse oximetry across skin tones, capnography under specific physiology — each documented edge case is a place where the standard's coverage is incomplete.

REFLECTION QUESTIONS

1. Identify an undetectable failure mode in your domain — one where the cue does not reach the operator in time to recover. What is the analog of pulse oximetry / capnography for that failure: the continuous signal that would convert it from undetectable to monitored?

2. Specify the institutional commitment (specialty foundation, regulatory standard) that would make the new signal non-waiverable. The device alone is not the deliverable; the standard is.

3. The anesthesia mortality decline is multifactorial. What is the minimum decomposition you would publish — training, device, pharmacology, team — to let downstream practitioners learn which components carried the most weight, rather than attributing the outcome to a single intervention?

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CASE 117 HEALTH PROFESSIONS EDUCATION INTERPROFESSIONAL COLLABORATION PATIENT SAFETY 2013 – 2015

Interprofessional Education and the Evidence Gap

K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Decades-long well-funded movement to educate health professionals together for collaborative care; Cochrane 2013 found only 15 studies between 1999 and 2011 met inclusion criteria; IOM 2015 made the gap the central finding — 'paucity of high-quality research' linking IPE to measurable changes in practice and patient outcomes

IN BRIEF

Interprofessional Education (IPE) is a decades-long, well-funded movement premised on the idea that doctors, nurses, pharmacists, and allied professionals should learn together so they can collaborate better in practice. The Cochrane review (Reeves et al., 2013) found only 15 studies between 1999 and 2011 met its inclusion criteria, and while those studies showed some positive outcomes, the evidence base for linking IPE to measurable changes in practice and patient outcomes was thin. The IOM report (2015) made the gap the central finding: there remains a "paucity of high-quality research" connecting IPE interventions to patient outcomes, and it proposed a conceptual model for doing the measurement properly. The case is the canonical instance in the corpus of a large, sincere, multidisciplinary translation effort whose core problem is that the field cannot yet measure whether the intervention works. It is the case-grounded basis for the enthusiasm-evidence gap as a curricular concept and motivates the Domain 3 sub-competency proposed in — the recurring pattern that a field instruments its enthusiasm faster than its outcomes.

THE SHIFT

The premise of Interprofessional Education is straightforward and credible: health-professions practice is collaborative, so health-professions training should be collaborative. Doctors, nurses, pharmacists, dietitians, social workers, and allied professionals should learn alongside each other, ideally in clinical and simulation contexts, so that the coordination capability is built during training rather than improvised on the wards. The movement is decades old, has institutional support across major accreditation bodies, and is well-funded by government and foundation sources.[0]

WHAT IS EMERGING

Reeves et al.'s 2013 Cochrane review applied the standard systematic-review machinery to the IPE outcome literature: what studies, between 1999 and 2011, met the inclusion criteria for evaluating IPE interventions against collaborative-practice or patient outcomes? The answer was

15. Across that small set, the studies showed some positive

outcomes — on clinician self-reported behavior, patient satisfaction, and a small number of clinical metrics — but the evidence base for linking IPE to measurable changes in practice and patient outcomes was thin. The reviewers' own conclusion was that the field had not produced the outcome evidence its scale of investment implied it should have.[1]

THE CAPABILITY QUESTION

The IOM's 2015 report, Measuring the Impact of Interprofessional Education on Collaborative Practice and Patient Outcomes, made the evidence gap the central finding. There remains, the IOM concluded, a "paucity of high-quality research" connecting IPE interventions to patient outcomes. The report proposed a conceptual model for doing the measurement properly — a chain from IPE intervention through learner outcome, collaborative behavior, organizational practice, and patient outcome — and named the methodological work the field had to do to get from the current evidence state to a defensible causal claim. The report is itself the load-bearing artifact: a national-academy synthesis that the central problem of the field is the evidence architecture, not the intervention.[2]

EARLY EVIDENCE

What the case teaches at the LENS framing layer is the structural form of the enthusiasm-evidence gap at the multidisciplinary-translation scale. IPE is a sincere, well-funded, decades-long movement on which a great deal of curricular and operational investment has been made; the strongest synthesis of the outcome literature concludes that the field instrumented its enthusiasm faster than it instrumented outcomes. The case is the case-grounded basis for the Domain 3 sub-competency proposed in — naming the enthusiasm-evidence gap as a canonical anti-pattern — and for the CLO **Judgment under inadequate evidence**: practitioners and program designers in IPE have had to decide for decades on what to build, on evidence that does not establish the causal claim the field's premise rests on.[3]

OPEN PROBLEMS

In pair with the Colorado CTSA team-science training case (121) and implementation-science training (123), IPE is the frontier endpoint of the multidisciplinary-translation thread. Team science showed the measurement can be done at single-program scale with a validated instrument; IPE shows what happens when the field-scale evidence architecture has not been built. The pair-plus-trio teaches the Domain 3 sub-competency in both directions: collaboration as a unit of measurement is possible (Case 178), and the field-scale evidence is structurally absent (this case).

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4. v2 paired cases: Team-science training (121), Implementation-science training (123).

THE LEARNING ENGINEERING LENS

> *The field instrumented its enthusiasm faster than its outcomes. The strongest synthesis names the gap as the central finding, not the intervention as the failure.*
>
> EDITORS' SYNTHESIS OF REEVES ET AL. (2013) AND IOM (2015).

LE INSIGHT

Interprofessional Education is the canonical enthusiasm-evidence-gap case in the corpus. A decades-long, well-funded movement; the strongest synthesis of the outcome literature concludes that the evidence base for the field's central claim — that IPE produces measurable changes in practice and patient outcomes — is structurally thin. The IOM 2015 conceptual model is the proposed evidence architecture; the field's task is to build to it.

LENS APPROACH

IPE is the field-scale enthusiasm-evidence-gap case (induced 2.1; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) as the case-grounded basis for the enthusiasm-evidence-gap sub-competency and for the CLO **Judgment under inadequate evidence** — IPE is the pattern's largest instance. Pair with Case 178 (team- science training, where measurement is possible at program scale) and Case 179 (implementation-science training, where stated goals run ahead of operational practices).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the IPE intervention with an outcome-measurement chain in mind from the start: learner outcome → collaborative behavior → organizational practice → patient outcome. The IOM model is the published reference.
- Specify, before the intervention launches, which links in the chain the evaluation will measure and which it will leave as assumed; the field-scale gap is partly the result of every program leaving the same links unmeasured.
- Treat enthusiasm-as-evidence as a foreseeable failure mode in any field-scale capability movement; the IPE pattern recurs across other interdisciplinary translation efforts (see Case 179).

IN OPERATION / AFTER

- Commission field-level evidence architecture, not only program-level evaluation; the gap the IOM names is at the field level and the program-level studies cannot close it on their own.
- Publish negative or null findings as a contribution to the evidence base, not as a program failure; the field's evidence gap is partly the result of selective publication on the positive side.
- Carry the IOM conceptual model into curriculum design conversations: a sub-competency that names the enthusiasm-evidence gap explicitly is the case-grounded curricular response to this pattern.

REFLECTION QUESTIONS

1. Identify a field-scale capability movement in your domain whose evidence architecture has not kept pace with its operational and curricular investment. What would the IOM-style conceptual model look like for that field's outcome chain?

2. Specify, for the next IPE-style program you would design or evaluate, which links in the outcome chain (learner / collaborative behavior / organizational practice / patient outcome) the evaluation will measure and which it will leave assumed. The field-scale gap is the accumulated result of leaving the same links assumed.

3. The case names enthusiasm-as-evidence as a foreseeable failure mode in any field-scale capability movement. What is the curricular response — a sub-competency that names the gap explicitly — and what cases would anchor it for your trainees?

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CASE 118 AVIATION SAFETY ENGINEERING HUMAN FACTORS 1996 – 2002

EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation

H Human-System Interface K Knowledge & Institutional Memory G Governance & Trust

IMPACT Honeywell's Enhanced Ground Proximity Warning System (EGPWS, 1996), mandated by the FAA as Terrain Awareness and Warning System (TAWS) for US-registered turbine aircraft beginning in 2001 and broadly worldwide by 2002, converted controlled flight into terrain (CFIT) — historically one of the largest categories of commercial-aviation fatalities — into a category whose rate in equipped fleets has fallen sharply; CFIT events on properly equipped and operating airliners are now rare

IN BRIEF

Controlled flight into terrain (CFIT) — a serviceable aircraft under the pilot's control flown unintentionally into the ground, water, or an obstacle — was for decades one of the largest categories of commercial-aviation fatalities. The 1979 Air New Zealand Mt Erebus crash (257 dead) and the 1995 American Airlines 965 Cali crash (159 dead) are canonical examples. The first-generation Ground Proximity Warning System (GPWS) developed by C. Donald Bateman at Sundstrand / Honeywell in the 1970s used radio altimeter and rate-of-descent inputs to warn of imminent terrain contact; it reduced CFIT but produced late warnings and was blind to terrain ahead of the aircraft. Enhanced GPWS (EGPWS), introduced commercially in 1996, added a digital terrain database and aircraft position to the input set, permitting forward-looking terrain-avoidance alerting. The FAA mandated EGPWS-class equipment (formally TAWS) on US-registered turbine aircraft beginning March 2001, with full compliance required by 2005; ICAO and most national regulators followed. The published outcome record is that CFIT in EGPWS-equipped commercial fleets has become rare — NTSB, FAA, and Flight Safety Foundation analyses consistently report a sharp decline. The hedge that survives: residual CFIT events still occur, typically involving disabled or inhibited equipment, deviation from procedure, or terrain outside the database, and the case has to honor the system- in-operation discipline rather than the system-as-installed claim.

BACKGROUND

Through the 1960s and 70s, controlled flight into terrain was one of the highest-fatality categories in commercial aviation. The pattern was structurally consistent: serviceable aircraft, qualified crew, often in IMC (instrument meteorological conditions) or at night, flown into rising terrain the crew had not visualized correctly. The Air New Zealand Mt Erebus crash (1979, 257 dead) and the American Airlines 965 Cali crash (1995, 159 dead) are the canonical examples — competent crews who lost situational awareness about their position relative to terrain in conditions that prevented visual recovery.[0]

THE INTERVENTION

The first engineered intervention was the Ground Proximity Warning System (GPWS), developed in the early 1970s by C. Donald Bateman at Sundstrand (later Honeywell). GPWS used radio altimeter readings and rate-of-descent to generate "pull up" and similar warnings when the aircraft was descending toward terrain directly below it. GPWS reduced CFIT meaningfully through the 1970s and 80s, but had two structural limits: it produced late warnings (the aircraft was already close to terrain when the alert fired), and it was blind to terrain ahead of the aircraft — the Cali accident occurred in a GPWS-equipped aircraft because the rising terrain was ahead of the flight path, not below.[1]

HOW IT WORKED

Enhanced GPWS (EGPWS), introduced by Honeywell in 1996, addressed both limits by adding a digital terrain database and aircraft position (GPS / IRS) to the input set. The system can now compute a forward-looking terrain surface relative to the aircraft's projected flight path and provide alerts well before terrain contact is imminent. The FAA codified the capability in the Terrain Awareness and Warning System (TAWS) regulation, requiring TAWS-class equipment on US-registered turbine aircraft with six or more passenger seats beginning March 29, 2001, with full equipage by 2005. ICAO and most national regulators followed with parallel mandates.[2]

THE EVIDENCE

The published outcome record across NTSB accident statistics, FAA reporting, and Flight Safety Foundation analyses is that CFIT in EGPWS-equipped commercial fleets has fallen sharply. The category that once dominated airliner-fatality statistics is now an uncommon-event category in equipped fleets. The structural claim the case makes is the cue/alert-design one: a failure mode in which the operator's perception of terrain was the limiting variable was converted into a monitored, recoverable mode by surfacing the forward terrain picture as an actionable alert. Pair with anesthesia monitoring (Case 116) at the cue/alert-as- capability layer, and with TCAS (Case 119) at the automated-advisory-system layer.[3]

WHAT TRANSFERRED

The hedge has to survive into the case. CFIT events still occur, typically involving one or more of: EGPWS disabled or inhibited (crew action, MEL release, maintenance), deviation from procedure (e.g., descent below minimum sector altitude under pressure), or terrain or obstacles not represented in the database (rapidly changing wind-turbine and structure environments are a known frontier). The system-in- operation has to be flying and the crew has to act on the alert; a rule-of-thumb in the safety community is that EGPWS is most useful when its warnings are taken seriously enough that they are rare in operation. The case teaches the cue/alert intervention at its most durable, with the qualification that the capability depends on the standard being honored in operation, not on the equipment being installed.[4]

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THE LEARNING ENGINEERING LENS

> *The capability depends on the standard being honored in operation, not on the equipment being installed.*
>
> EDITORS' SYNTHESIS OF FAA TAWS RULE HISTORY AND FSF ALAR ANALYSES.

LE INSIGHT

EGPWS / TAWS is the canonical cue/alert intervention at fleet scale. The forward-looking terrain alert converted a failure mode in which the crew's terrain perception was the limiting variable into a monitored, recoverable mode. CFIT in equipped fleets has become rare; residual events typically involve inhibited equipment or procedure deviation, and the hedge is the case.

LENS APPROACH

EGPWS / TAWS is the aviation cue/alert intervention case (induced 3.1; LENS D4/PT5) — Domain 4 for cue-design-as- deliverable; Domain 3 for the operator-cue boundary. Pair with TCAS (Case 119) and Case 116 (anesthesia monitoring).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Identify the operator-perception variable that is the limiting variable in a failure mode (here: the crew's awareness of terrain ahead of the aircraft) and engineer the system that surfaces that variable as an actionable alert with enough lead time to recover.
- Pair the alert design with a regulatory mandate that makes the equipment non-waiverable across the fleet, so adoption is fleet-level capability rather than per-operator choice. The 1996-to-2001 gap between commercial availability and mandate is the political-process cost.
- Build the cue's lead time around the time the operator needs to act, not the time the equipment can produce the alert; a too-late alert is the GPWS limitation EGPWS was specifically built to address.

IN OPERATION / AFTER

- Carry the system-in-operation hedge into communication: the capability depends on EGPWS being operational, not inhibited, and on the crew acting on the alert. Inhibition discipline is part of the deliverable.
- Maintain the terrain database as a continuously updated artifact; the equipment as installed is only as good as the database it queries, and rapidly changing obstacle environments (wind turbines, structures) are a known frontier.
- Treat residual CFIT events as evidence about the operational discipline, not as evidence against the intervention; the system that has reduced a fatality category to rare uncommon events is doing the work the case claims.

REFLECTION QUESTIONS

1. Identify a failure mode in your domain where operator perception of an external variable is the limiting factor. What is the analog of the digital terrain database — the engineered representation of the variable — and what lead time would the cue need to be actionable?

2. Specify the regulatory or institutional mandate path you would expect: EGPWS reached the market in 1996, was mandated in 2001, and was fully equipaged by 2005. Five years from commercial availability to full equipage is a useful planning datum for a fleet-scale capability mandate.

3. The system-in-operation hedge is binding. What inhibition discipline would your program require so that the engineered recovery layer is operating when the failure mode appears, and how would you instrument that the discipline is being honored?

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CASE 119 AVIATION AUTOMATION HUMAN FACTORS 1981 – 2008 (TCAS II VERSION 7.1)

TCAS — Coordinated Collision Avoidance and the Überlingen Lesson

H Human-System Interface K Knowledge & Institutional Memory G Governance & Trust

IMPACT TCAS II — the Traffic Alert and Collision Avoidance System — provides cockpit traffic display and coordinated Resolution Advisories (RAs) between aircraft on conflicting trajectories; mandated on US air-carrier and on most international turbine aircraft, TCAS converted mid-air collision in commercial aviation from a recurring fatality category to a rare event; the 2002 Überlingen mid-air (71 dead) exposed a specific coordination failure mode and drove the 2008 release of TCAS II Version 7.1 with the 'level off, level off' reversal logic

IN BRIEF

TCAS — the Traffic Alert and Collision Avoidance System, standardized in RTCA DO-185 and successors — is the cockpit automation that monitors transponder returns from nearby aircraft, computes potential conflicts, and issues Traffic Advisories and Resolution Advisories (RAs) to the crew. Operational TCAS II was mandated on US air-carrier aircraft by FAA rule in the early 1990s and on most international turbine aircraft by ICAO. RAs are coordinated: when two TCAS-equipped aircraft are in conflict, one is instructed to climb and the other to descend by negotiated inversion of the data-link. The intervention converted mid-air collision in commercial aviation from a recurring fatality category to a rare event. The case’s load-bearing edge case is the 2002 Überlingen mid-air collision (71 dead), in which one crew followed the TCAS RA and the other followed an ATC instruction in the opposite direction. The BFU investigation identified the human–TCAS–ATC coordination failure mode and drove the 2008 release of TCAS II Version 7.1, which added the “level off, level off” reversal logic and clarified the precedence of TCAS RAs over ATC instructions. The hedge survives into the case: TCAS is among the strongest aviation automation interventions in the outcome record, and the Überlingen failure mode and its V7.1 correction are part of the case rather than smoothed away.

BACKGROUND

Mid-air collision in commercial aviation has been a recurring fatality category since the 1950s. The 1956 Grand Canyon mid-air (128 dead) prompted the modern US air-traffic-control system, but ATC alone cannot always prevent collision when traffic densities or coordination errors exceed the controller’s reach. The FAA and RTCA developed the Traffic Alert and Collision Avoidance System (TCAS) through the 1980s as a cockpit collision-avoidance automation independent of ATC. TCAS II — the operational version with Resolution

Advisories — was mandated on US air-carrier aircraft beginning in the early 1990s, and on most international turbine aircraft by ICAO over the following decade.[0]

THE INTERVENTION

The TCAS II architecture is what the case rests on. Each equipped aircraft interrogates the transponders of nearby aircraft and computes a closest-point-of-approach projection from range, altitude, and rate data. If the projection enters the conflict envelope, TCAS issues a Traffic Advisory (TA) — a cue to the crew to acquire the other aircraft visually if possible. If the conflict persists, TCAS issues a Resolution Advisory (RA): a specific vertical-rate command ("Climb, climb" or "Descend, descend"). When two TCAS-equipped aircraft are in conflict, the two RAs are coordinated via the Mode S data link so the aircraft are instructed to diverge — one climbing, one descending — rather than both maneuvering in the same direction.[1]

HOW IT WORKED

The deployed outcome record across the 1990s and 2000s was strong. Mid-air collision in TCAS-equipped fleets became rare. The case nevertheless turns on Überlingen. On July 1, 2002, a Russian Tu-154 and a DHL Boeing 757 approached on conflicting trajectories at FL360 over southern Germany. Both aircraft received coordinated TCAS RAs — the 757 to descend, the Tu-154 to climb. The Tu-154 crew received the climb RA and, almost simultaneously, an ATC instruction to descend from a Skyguide controller working alone on degraded console configuration. The Tu-154 crew followed the ATC instruction; the 757 followed the TCAS RA; both aircraft descended into each other. 71 people died, including 52 children on a school trip.[2]

THE EVIDENCE

The BFU investigation identified a specific failure mode at the human–automation–controller boundary: TCAS RA precedence over ATC instructions was insufficiently clear in the crew procedures, the data-link coordination between the two TCAS units had performed as designed but could not enforce the result on the crew, and the ATC single-controller / degraded-console context was a systemic failure. The aviation response was operational and technical: ICAO and national regulators clarified that TCAS RAs take precedence over conflicting ATC instructions; ATC procedures were tightened around coordination of conflict-resolution between controllers and TCAS; and RTCA / Eurocontrol developed TCAS II Version 7.1 (released 2008), which added the "level off, level off" reversal logic for the specific scenario where one aircraft does not follow its RA, and clarified RA wording and behavior at the boundary.[3]

WHAT TRANSFERRED

The hedge survives into the case. TCAS is among the strongest aviation automation interventions in the outcome record; the Überlingen failure mode is not a refutation of the system but a documented coordination limit that drove the V7.1 correction. The case teaches the coordinated-automation form at its most durable, with the discipline that the human–automation–human- operator triangle (crew, TCAS, ATC) has to be designed coherently. A single-controller / degraded-console situation and an unclear precedence rule converted a working automation into a fatal outcome; both were addressed in the post-Überlingen response, and both remain part of the case rather than smoothed away.[4]

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THE LEARNING ENGINEERING LENS

> *The intervention is not less effective for having a documented coordination edge case; it is more credible because the edge case is named and addressed.*
>
> EDITORS' SYNTHESIS OF THE BFU ÜBERLINGEN REPORT AND THE V7.1 DEVELOPMENT RECORD.

LE INSIGHT

TCAS is among the strongest aviation automation interventions in the outcome record; the Überlingen coordination failure mode is part of the case rather than smoothed away. The human–automation–ATC triangle has to be designed coherently; V7.1's reversal logic and the clarified precedence rule are the iterative-design response to the documented edge case.

LENS APPROACH

TCAS is the coordinated-cockpit-automation case (induced 3.1; LENS D4/PT5) — Domain 4 for the Überlingen-driven V7.1 iteration; Domain 3 for the crew–TCAS–ATC precedence rule. Pair with EGPWS (Case 118) and Case 116.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the coordinated-automation logic for the case where one of the coordinated agents does not comply — the Überlingen failure mode — not just for the case where both comply. The V7.1 reversal logic is the worked example of that design discipline.
- Specify precedence at the human–automation boundary unambiguously and in advance: TCAS RAs take precedence over conflicting ATC instructions. Leaving precedence to crew judgment under time pressure is the design choice that produced Überlingen.
- Treat the ATC context (single-controller, degraded console) as part of the system the automation operates in, not as a precondition the automation can assume away. Systemic failures at the boundary determine the boundary behavior.

IN OPERATION / AFTER

- Carry the Überlingen failure mode into the case rather than smoothing it away. The intervention is not less effective for having a documented coordination edge case; it is more credible because the edge case is named and addressed.
- Track the post-V7.1 outcome record as evidence about the correction, not just about the original intervention; the lesson is the iterative-design discipline that the human–automation triangle requires after a failure mode is exposed.
- Use the case as the canonical pair to EGPWS (Case 118): two cockpit automations, two outcome categories closed, one with a coordination edge case that drove a version revision and one without.

REFLECTION QUESTIONS

1. Identify a coordinated-automation system in your domain where two agents must comply for the resolution to work. What is the analog of the V7.1 reversal logic — the design for the case where one agent does not comply?

2. Specify the precedence rule at the human–automation boundary in your system. Überlingen turned on an ambiguous precedence rule; the post-2002 clarification is the worked example of why precedence has to be unambiguous in advance.

3. The systemic context (Skyguide single-controller / degraded console) was part of the failure. What contextual preconditions does your automation assume that, if they fail, would convert a working automation into a failure mode?

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CASE 120 CLINICAL CARE PATIENT SAFETY HEALTH IT 2010

Bar-Code Medication Administration — Cue at the Point of Care

H Human-System Interface K Knowledge & Institutional Memory D Designed Out

IMPACT A before-and-after study at an academic medical center associated bar-code electronic medication administration (bar-code eMAR) with a 41.4% reduction in nontiming administration errors and a 50.8% reduction in potential adverse drug events; transcription errors on order documentation fell 80.3%; a later single-site rollout reported a 55.4% reduction in actual patient-harm events

IN BRIEF

Wrong-drug and wrong-patient administration errors are a persistent failure mode in hospital pharmacy: the cue the human operator needs in order to catch the mismatch is structurally absent at the bedside, because the order, the dispensed medication, and the patient are connected only by paper documentation and clinical memory under time pressure. Bar-code medication administration (BCMA) supplies the cue in hardware: a bedside scan of the medication's bar code against the patient's wristband, gated by the electronic medication administration record. Poon et al. (NEJM, 2010) evaluated bar-code eMAR at an academic medical center using a before-and-after observational design and reported a 41.4% reduction in nontiming administration errors, a 50.8% reduction in potential adverse drug events, and an 80.3% reduction in transcription errors on order documentation. A later single-site rollout (PMC6257885) reported a 55.4% reduction in actual patient-harm events. The study is explicit that the design is quasi-experimental — before-and-after / observational — not a randomized trial, and that significant workflow changes were required for the cue to land. The case is the canonical point-of-care cue/alert intervention, paired with Case 116 (anesthesia monitoring / APSF) at the cue-as-deliverable layer and with Case 109 (WHO Surgical Checklist) at the mandatory-mechanism layer.

BACKGROUND

The wrong-drug and wrong-patient administration error is a structural failure mode of the hospital medication chain. A physician's order, a pharmacist's dispense, a nurse's bedside administration, and a patient's wristband identification are connected only by paper documentation and the operator's clinical memory under time pressure. The cue that would let the bedside nurse catch a mismatch — "this medication's identity does not match this patient's record" — is not present in the workflow unaided. The conventional safeguards (the five rights, double-checks, naming protocols) are cognitive and procedural; the cue itself is structurally absent.[o]

THE INTERVENTION

Bar-code medication administration supplies the cue in hardware. The medication unit-of-use carries a bar code, the patient wears a coded wristband, and the electronic medication administration record (eMAR) gates the administration on a bedside scan that matches medication-to-order-to-patient. The cue lands at the moment of administration, before the harm, and the system refuses to acknowledge the administration if any element does not match. The structural form is the same as the cue/alert interventions in Case 116 (continuous pulse oximetry, capnography under anesthesia): the cue at the right point in the workflow converts an undetectable failure into a monitored, recoverable one.[1]

HOW IT WORKED

Poon et al. (New England Journal of Medicine, 2010) evaluated bar-code eMAR at a large academic medical center using a before-and-after observational design. The headline findings were a 41.4% reduction in nontiming administration errors (the failure mode the cue is designed to catch), a 50.8% reduction in potential adverse drug events (the harm the errors would have produced), and an 80.3% reduction in transcription errors on order documentation upstream of administration. A subsequent single-site rollout reported in PMC6257885 documented a 55.4% reduction in actual patient-harm events, supporting the transfer of the result across institutional contexts.[2]

THE EVIDENCE

The study's design is what the case's hedge rests on. The evaluation is quasi-experimental — before-and-after / observational — not a randomized trial, and the authors are explicit that significant workflow changes were required for the cue to land in operation. The cue is not the deliverable alone; the workflow is also the deliverable, and the attribution of the observed reduction to the cue rather than to the workflow change rests on plausibility and on the mechanism the cue is designed against. The 55.4% transfer number from the later rollout supports the mechanism, and that rollout figure also rests on observational data rather than randomized comparison.[3]

WHAT TRANSFERRED

What the case carries for the corpus is the cue-as-deliverable pattern at the point of care, with the explicit before-and-after-design hedge. Paired with Case 116 (anesthesia monitoring / APSF), bar-code eMAR shows the same structural form in a different specialty: the cue at the right point in the workflow converts an undetectable mismatch into a monitored one. Paired with Case 109 (WHO Surgical Checklist), the case sits at the mandatory-mechanism layer: the bar-code scan, like the checklist, is a workflow artifact the operator cannot route around, and the institutional commitment to that non-routability is part of why the cue lands.

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THE LEARNING ENGINEERING LENS

> *The cue is supplied in hardware at the point of administration; the workflow change is part of the deliverable, and the design is observational rather than randomized.*
>
> EDITORS' SYNTHESIS OF POON ET AL. (2010) AND THE BCMA IMPLEMENTATION LITERATURE.

LE INSIGHT

Bar-code medication administration is the canonical small-tier cue/alert intervention at the point of care. The cue lands at the bedside before the harm, the workflow change is part of the deliverable, and the headline reductions (41.4%, 50.8%, 80.3%, and a 55.4% transfer figure) rest on a before-and-after / observational design that the case preserves verbatim.

LENS APPROACH

BCMA is the small-tier point-of-care cue/alert intervention (induced 3.1; LENS D4/PT5) — Domain 4 for cue-as-deliverable; Domain 3 for the workflow-around-the- cue. Pair with Case 116 (anesthesia/APSF) and Case 109 (WHO Surgical Checklist).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the cue to land at the point in the workflow where the operator can still recover — for bar-code eMAR, the bedside administration scan, before the dose reaches the patient — not at a post-hoc reconciliation point.
- Treat the workflow change as part of the deliverable, not as friction to minimize; the cue's effect is conditional on the workflow that surrounds it, and the evaluation has to acknowledge both.
- Pair the device standard with the institutional commitment that makes the scan non-routable around — the equivalent of the APSF / ASA monitoring standard in anesthesia, or the WHO checklist's mandatory-mechanism status.

IN OPERATION / AFTER

- Report the headline reductions (41.4%, 50.8%, 80.3%, 55.4%) together with the design hedge — before-and-after / observational, significant workflow changes required — so the evidence the field cites is the evidence the field can actually use.
- Track the transfer figure (the 55.4% from the later single-site rollout) as evidence the mechanism survives a new institutional context, while preserving the same observational-design qualification on the new figure.
- Build the next round of evaluation against the residual error rate (the failures that survive the cue) so the institution learns where the workflow leaks remain, rather than treating the reduction as the closure of the failure mode.

REFLECTION QUESTIONS

1. Identify a point in your domain's workflow where the cue the operator needs in order to catch a mismatch is structurally absent. What would the hardware-supplied analog of bar-code eMAR look like — and at what point in the workflow does it have to land for the operator to still recover?

2. The headline reductions (41.4%, 50.8%, 80.3%, 55.4%) rest on a before-and-after observational design. What additional evidence — randomized comparison, segmented analysis by unit type, residual-error analysis — would you require before treating the figures as a closure rather than a strong signal?

3. The workflow change is part of the deliverable. Specify the institutional commitment in your context that would make the analog of the bedside scan non-routable around, and name the cost the institution would absorb to keep that commitment.

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CASE 121 SURGERY CLINICAL OUTCOMES ASSESSMENT 2013

Surgical Skill Video Peer-Rating Predicts Complications

H Human-System Interface D Designed Out K Knowledge & Institutional Memory

IMPACT Twenty bariatric surgeons each submitted a representative gastric-bypass video, rated blind by at least 10 peers; skill ratings ranged 2.6–4.8; the bottom skill quartile had a higher complication rate (14.5%) than the top quartile across a registry of 10,343 patients, and greater skill was associated with fewer reoperations, readmissions, and emergency department visits

IN BRIEF

Surgical complications are conventionally attributed to patient factors, hospital factors, and case-mix. Birkmeyer et al. (NEJM, 2013) asked a more direct question: can the surgeon's actual technical capability be measured well enough to predict patient outcomes? Twenty bariatric surgeons each submitted a representative video of a laparoscopic gastric bypass; the videos were rated blind by at least 10 peers on a structured skill scale. Skill ratings ranged from 2.6 to 4.8. Linked to a Michigan registry of 10,343 patients, the bottom skill quartile had a higher complication rate (14.5%) than the top quartile, and greater skill was associated with fewer reoperations, readmissions, and emergency department visits. The authors call the findings preliminary and name the skill-versus-volume confound explicitly: low-skill surgeons also did fewer cases and operated more slowly, so the contribution of skill itself versus the contribution of case volume is partly open. The hedge is part of the case. The proposed primary anchor is 2.1 (measuring the failure mode you care about) with C3 and C1 alternates; the editor may move it. Adjacent to JIGSAWS (Case 122) at the surgical-skill-measurement layer.

BACKGROUND

The conventional accounting for surgical complications attributes outcomes to patient factors (age, comorbidity, severity), hospital factors (volume, ICU support), and case-mix. The surgeon's actual technical capability — what the surgeon does with the instruments in this operation — is conventionally treated as too hard to measure at scale, and so as a hidden variable in the outcome equation. Birkmeyer et al. (New England Journal of Medicine, 2013) asked whether that variable could be brought into the open: can the surgeon's skill be measured well enough, at scale, to predict patient outcomes?[o]

THE INTERVENTION

The study's design was deliberately simple. Twenty bariatric surgeons in a Michigan collaborative each submitted a representative video of a laparoscopic gastric bypass — a standardized, common procedure. The videos were rated blind by at least 10 peers on a structured skill scale derived from the surgical-education literature: tissue handling, time and motion, instrument knowledge, exposure, completion of the operation. Skill ratings ranged from 2.6 to 4.8, with the inter-rater reliability adequate to support the rank-ordering the study then took into the outcomes analysis.[1]

HOW IT WORKED

The outcomes evidence linked the peer-rated skill scores to a Michigan registry of 10,343 patients treated by the same twenty surgeons. The headline finding: the bottom skill quartile had a higher complication rate (14.5%) than the top quartile, and greater skill was associated with fewer reoperations, fewer readmissions, and fewer emergency department visits in the months after surgery. The skill score predicted the complication signal at the cohort scale. In the surgical-outcomes literature, this is the first large-registry evidence that operative skill, measured directly from operative video by peer rating, predicts the patient-outcome signal the institution actually cares about.[2]

THE EVIDENCE

The authors call the findings preliminary and name the load-bearing confound explicitly: low-skill surgeons also did fewer cases and operated more slowly. The skill-versus-volume confound is the central methodological hedge of the case. It is plausible — and consistent with the broader volume-outcome literature — that what the skill rating captured was partly the surgeon's accumulated operative experience, and that the rated skill is downstream of case volume rather than the other axis of the outcome. The published study does not separate the two; the editor may move the primary anchor away from 2.1 toward 1.1 (engineered vs. stated requirements) or 6.2 (operator-to-institution feedback channels) on that basis.[3]

WHAT TRANSFERRED

What the case carries for the corpus is the evidence- architecture move of measuring the operator's actual technical capability against the outcome the institution cares about, with the volume-confound hedge intact. The case is adjacent to JIGSAWS (Case 122) at the surgical-skill-measurement layer — JIGSAWS provides the controlled-task instrumented evidence and Birkmeyer the naturalistic operative-video evidence, and the two together anchor the small-tier C3 conversation about measuring skill in surgery. The preliminary-findings language is part of what the case teaches.

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> *Greater skill was associated with fewer reoperations, readmissions, and emergency department visits; the low-skill surgeons also did fewer cases and operated more slowly.*
>
> EDITORS' SYNTHESIS OF BIRKMEYER ET AL. (2013) AND THE ACCOMPANYING EDITORIAL COMMENT.

LE INSIGHT

The Birkmeyer skill–outcomes study is the first large- registry evidence that operative skill, measured directly from operative video by blind peer rating, predicts the complication signal the institution cares about. The findings are preliminary; the skill-versus-volume confound is explicit; the multi-anchor (2.1 primary, 1.1 and 6.2 alternates) is the editor's call.

LENS APPROACH

The surgical-skill peer-rating study is the small-tier measure-the-failure-mode-you-care-about case (induced 2.1; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for the measure-the-capability-against-the-outcome design move, and in Domain 3 (Human-System Collaboration) for the peer-rating workflow that makes the assessment scalable. Adjacent to JIGSAWS (Case 122) at the surgical-skill-measurement layer — JIGSAWS the controlled-task instrumented evidence, Birkmeyer the naturalistic operative-video evidence. The skill-versus-volume confound is the case.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the capability the assessment is built to measure (operative skill on a standard procedure) and the outcome the institution actually cares about (complication rate, reoperation, readmission), so the evidence architecture connects the measured variable to the institutional signal.
- Use blind peer rating on standardized video to make the assessment scalable and resistant to the gaming patterns of self-report or volume-weighted reputation.
- Pre-register the threats-to-validity table — volume confound, selection of submitted video, inter-rater reliability — so the published headline is reported alongside the qualifications the design carries.

IN OPERATION / AFTER

- Report the 14.5% bottom-quartile complication finding together with the volume-confound hedge; the preliminary-findings language is what makes the result interpretable for the next study.
- Design the follow-up analysis to separate skill from volume — for example, by matching on case volume within skill quartile, or by instrumenting the case-volume time-course — so the next round of evidence resolves the confound the present study leaves open.
- Carry the result into the surgical-education and credentialing conversation as evidence that peer-rated operative-video skill is a measurable, outcome-predictive variable, while preserving the qualifications the authors themselves attach.

REFLECTION QUESTIONS

1. Identify a capability in your domain where the operator's actual technical capability is conventionally treated as too hard to measure at scale and therefore left as a hidden variable in the outcome equation. What would a Birkmeyer-style scalable, blind peer-rated measure look like?
2. The skill-versus-volume confound is the case. Design the follow-up analysis that would separate skill from volume in your context. What matching, what instrumentation of the volume time-course, what controlled-task companion would be required?
3. Birkmeyer is anchored at 2.1 with C3 and C1 alternates. Which anchor would you choose for your own use, and what does that choice say about which conversation in your domain the case is the load-bearing evidence for?

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CASE 122 SURGERY SKILL ASSESSMENT HUMAN-MOTION ANALYSIS 2009 – 2016

Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

T Training Gap K Knowledge & Institutional Memory H Human-System Interface

DISCLOSURE This case originates at the editors' home institution (Johns Hopkins). The case is anchored to the peer-reviewed PLOS One paper and the publicly released JIGSAWS dataset rather than institutional press; the affiliation is disclosed openly.

IMPACT JHU's Language of Surgery project treated surgical motion as language — decomposing tasks into gesture and sub-gesture motion primitives — and released JIGSAWS, a public da Vinci kinematic/video/gesture/skill-rating dataset that became a standard benchmark; experts used fewer gestures (26.29 vs 31.30) and fewer gesture errors than novices for a knot-tying task

IN BRIEF

The Language of Surgery project at Johns Hopkins, led by Gregory Hager and a cross-departmental team of roughly twenty investigators across engineering, computer science, and surgery, treated surgical motion as language — decomposing tasks into gestures (the "surgemes") and sub-gesture motion primitives (the "dexemes") fine enough to distinguish expert from novice. The project released JIGSAWS (JHU-ISI Gesture and Skill Assessment Working Set), a publicly available da Vinci surgical-robot dataset with synchronized kinematic, video, gesture-annotation, and skill-rating tracks for suturing, knot-tying, and needle-passing tasks. JIGSAWS became a standard benchmark in surgical skill-assessment and gesture-recognition research. Vedula et al. (PLOS One, 2016) used the dataset to show that experts used fewer gestures (26.29 vs. 31.30 on a knot-tying task) and made fewer gesture errors than novices, with quantifiable sub-task-level differences. The case establishes that surgical skill is decomposable and machine-measurable; the honest open question — preserved here verbatim — is whether automated motion-level feedback accelerates trainee skill acquisition or improves patient outcomes. The dataset enables the question more than it answers it. The case pairs directly with Case 121 (Birkmeyer's video-rated bariatric-surgical-skill outcome study), which establishes that skill matters; together they form the skill-measurement spine the corpus needed.

BACKGROUND

Surgical training has carried, for decades, a structural measurement gap: skill at the trainee-versus-attending level is universally acknowledged to matter, and is universally measured by rater judgment — case logs, milestone evaluations, OSATS scores — that resolves the construct only at the summary level. What such measures cannot resolve is the unit of skill: the specific motion sequence, the gesture choice, the sub-gesture smoothness, the corrective re-engagement after a near-error. The Language of Surgery project began from the position that skill could be made measurable at the gesture level if surgical activity were decomposed the way language is decomposed into phonemes and words.[0]

THE INTERVENTION

The intervention is dataset-and-method, jointly. The team instrumented the da Vinci surgical robot to capture synchronized kinematic traces of the arms, stereo video of the surgical field, and frame-level human annotation of which gesture was being performed. Onto that base they layered expert skill ratings task by task. The technical decomposition was structured as a hierarchy: a task (e.g., knot-tying) consists of a sequence of gestures or surgemes (e.g., “reach for needle”, “position needle”, “drive needle”), each of which is composed of motion primitives or dexemes at the kinematic-trajectory level. The team released the resulting corpus as JIGSAWS in open form so the broader research community could test methods against a common benchmark, rather than each lab building its own private dataset.[1]

HOW IT WORKED

Vedula, Ishii, and Hager (PLOS One, 2016) used the dataset to analyze the structure of surgical activity for a suturing and knot-tying task and reported that experts used fewer gestures (26.29 on a knot-tying task) than novices (31.30), and that novices made more gesture errors. The differences were not at the gross-task-outcome level — both groups completed the task — but at the gesture-composition level the decomposition made visible. Subsequent work on JIGSAWS by other groups developed automated gesture-recognition methods, automated skill-rating classifiers, and motion-primitive analyses; the dataset’s role was to make these results comparable across labs.[2]

THE EVIDENCE

The honest open question survives into the case verbatim. The project demonstrates that surgical skill is decomposable and machine-measurable at the gesture level. What remains open is whether automated motion-level feedback delivered to trainees actually accelerates skill acquisition, and whether gesture-level skill differences translate into patient-outcome differences for the trainees once they reach the operating room. JIGSAWS enables both questions to be asked rigorously; it does not, by itself, answer either. The case’s learning-engineering content is the construction of the evidence architecture — the measurable unit, the open dataset, the cross-lab benchmark — that makes downstream measurement possible. The home-institution disclosure under the title is the standing safeguard against boosterism.[3]

WHAT TRANSFERRED

In pair with Case 121 (Birkmeyer et al.’s video-rated bariatric surgical-skill study), the case completes a skill-measurement pair the corpus had needed: Birkmeyer shows that rated skill predicts patient outcome at scale (skill matters), and this case shows that skill is decomposable into machine-measurable units (skill is engineerable). Drafted together they operationalize the new framing the v2 induced framework proposes — that capability is engineerable when the unit of capability is named and the measurement instrument follows. JIGSAWS’s continued use as a benchmark, more than a decade after release, is the field-scale evidence that the decomposition was the right resolution for the question.

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THE LEARNING ENGINEERING LENS

> *Surgical motion is language. The unit of skill is the gesture; the corpus is the dataset; the benchmark is the field's continued use of it more than a decade after release.*
>
> EDITORS' SYNTHESIS OF VEDULA ET AL. (2016) AND THE JIGSAWS RELEASE.

LE INSIGHT

Language of Surgery / JIGSAWS is the corpus's worked example of decomposing a tacit capability — surgical skill — into machine-measurable units, and releasing the measurement infrastructure openly so the field can build on it. The decomposition is established; the downstream question — whether motion-level feedback accelerates skill or improves patient outcomes — is open. The case enables the question rather than answering it.

LENS APPROACH

Language of Surgery is the skill-decomposition case (induced 2.1; LENS D4/PT5) — Domain 4 for construct-resolution; Domain 1 for task-decomposition. Pair with Case 121 (Birkmeyer) and Case 116. Home-institution disclosure under the title.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Decompose the capability into units fine enough to distinguish expert from novice but coarse enough to be reliably annotated; the surgeme / dexeme hierarchy is the worked example of the trade-off.
- Release the dataset openly with synchronized signal tracks (kinematic, video, annotation, rating) so that downstream methods can be compared on a common benchmark rather than each lab evaluating itself.
- Pair the dataset with a baseline analysis paper that demonstrates the decomposition resolves real expert/novice differences; the Vedula et al. analysis is what makes JIGSAWS more than a data dump.

IN OPERATION / AFTER

- Carry the open question explicitly — whether automated motion-level feedback accelerates skill acquisition is not what the dataset establishes — so downstream researchers and trainees see the gap as a research target, not a settled claim.
- Preserve the home-institution affiliation disclosure in any communication; the standing language anchors the case to the peer-reviewed paper and the public dataset, not to institutional press.
- Treat the cross-lab benchmark adoption as the case's strongest evidence that the decomposition was correct at the resolution chosen; the field's continued use of JIGSAWS more than a decade after release is itself the test.

REFLECTION QUESTIONS

1. Identify a tacit capability in your domain that is currently rated at the summary level. What is the unit of the capability — the gesture-equivalent — at which the decomposition would resolve expert/novice differences, and what signal tracks would the dataset need to synchronize?

2. Specify the open question your decomposition would not answer on its own. JIGSAWS does not establish that motion-level feedback accelerates skill; it enables the question. What is the analog in your context — the question the dataset enables but does not close?

3. The case's evidence of correctness is field-scale adoption of the benchmark a decade after release. What is the publication, release, and open-license strategy that would let your decomposition be tested by labs that have no stake in the original design?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · human factors & interaction design — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · usability testing · cognitive walkthroughs

CASE 123 HIGHER EDUCATION (BRAZIL) LEARNING ANALYTICS INSTITUTIONAL GOVERNANCE 2024

MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil)

K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT MMALA is a maturity model spanning infrastructure, human resources, ethics, and pedagogy; expert evaluation rated it comprehensive and suitable; three-institution validation exercise in Brazilian universities found it could outline essential practices and support self-assessment for scaling — instrument for responsible adoption, downstream learning outcome open

IN BRIEF

Freitas, Mello, Gasevic, Costa, and Andrade (Journal of Learning Analytics, 2024) developed and validated MMALA — a Maturity Model for Adopting Learning Analytics — designed to let an institution self-assess its readiness across the four dimensions responsible adoption actually depends on: infrastructure (technical capability), human resources (analytical and pedagogical staffing), ethics (governance and consent architecture), and pedagogy (integration with learning design). Experts evaluated the model as comprehensive and suitable; a three-institution validation exercise at Brazilian universities found that MMALA could outline essential practices and support self-assessment for scaling learning analytics responsibly. The case is one of the corpus's clearest worked examples of governance-as-instrument: a structured artifact an institution can use to convert the abstract goal "we should adopt learning analytics responsibly" into specific assessments of where its current capability sits and what it has to build next. The honest limit preserved verbatim: the instrument is validated by expert opinion and a three-institution exercise, not by longitudinal outcomes of institutions that used it to adopt LA — it is an instrument for responsible adoption, with the downstream effect on student learning still to be measured. The case pairs with the OU policy (Case 115) and the LALA CANVAS (Case 181) as the institutional-instrument layer of the non-US LA governance set.

BACKGROUND

Institutional adoption of learning analytics typically collapses into one of two patterns: a top-down deployment that meets resistance because the institution is not yet ready (governance is thin, staff are not analytically trained, the pedagogical integration was not designed), or a stalled aspiration where leadership wants responsible adoption but does not have a structured way to know what "ready" means. MMALA was built to address the second pattern: to give an institution a structured self-assessment instrument across the dimensions readiness actually has, so the adoption decision becomes specific rather than aspirational.[o]

THE INTERVENTION

The instrument is dimensional. Infrastructure asks whether the technical capability — data pipelines, secure storage, analytical platforms — is in place at the level the planned adoption requires. Human resources asks whether the institution has the analytical and pedagogical staffing the tools will need. Ethics asks whether the governance and consent architecture has been built. Pedagogy asks whether the analytics integrate with the institution's learning design, or whether they would be bolted on as a separate track. Each dimension is resolved into maturity levels so the self-assessment yields a structured picture of which dimensions are ready and which are not, rather than a single overall readiness score.[1]

HOW IT WORKED

The Freitas et al. validation has two halves. The first is expert evaluation: subject-matter experts in learning analytics, institutional research, and educational technology evaluated the instrument as comprehensive (covering the dimensions adoption actually depends on) and suitable (resolved at the right level for institutional self-assessment). The second is a three-institution validation exercise at Brazilian universities, which found that MMALA could outline essential practices and support self-assessment for scaling. The instrument was usable in practice and produced actionable structure for the institutions that piloted it.[2]

THE EVIDENCE

The honest limit survives verbatim. MMALA is validated by expert opinion and a three-institution exercise, not by longitudinal outcomes of institutions that used it to adopt LA. The instrument's claim is to be a suitable structure for responsible adoption — what the published evidence supports — and the downstream effect on student learning is the next study, not this one. The case is teachable on the instrument and the validation method; it is honest that the causal chain from "institution adopted via MMALA" to "students learned more" is not yet closed by evidence.[3]

WHAT TRANSFERRED

In the non-US LA governance set — OU (Case 115, policy and operating system), LALA (Case 181, participatory adoption framework), African data privacy (Case 169, cross-regime seam), Norway (Case 182, national commission), and MMALA (this case, institutional maturity model) — the cases together teach the level structure of governance for the same underlying capability. MMALA's specific contribution is the institutional-instrument layer: when a single institution is the unit of adoption and the governance artifact has to be usable by the institution itself, the maturity model is the form. The pentad demonstrates that governance is producible at multiple levels for the same capability, and the choice of level is itself a governance decision.

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3. Hilliger et al. (2020), Internet and Higher Education — LALA participatory adoption companion (Case 181).
4. Norwegian Expert Commission on Learning Analytics, final NOU (2023) — national-scale companion (Case 182).

> *The instrument converts "we should adopt responsibly" into a structured per-dimension self-assessment. The instrument is validated; the downstream learning outcome is the next study.*
>
> EDITORS' SYNTHESIS OF FREITAS ET AL. (2024).

LE INSIGHT

MMALA is the institutional-instrument instance of governance for learning-analytics adoption: a structured maturity model across infrastructure, human resources, ethics, and pedagogy. The validation evidence — expert evaluation and three-institution pilot — is what the case claims; the downstream effect on student learning is the next study, and the case is honest about that gap.

LENS APPROACH

MMALA is the institutional-maturity-model case (induced 5.4; LENS D4/PT4). LENS uses it in Domain 4 (Test and Evaluation) for the structured self-assessment instrument and the two-layer validation discipline; in Domain 5 (Navigating Sociotechnical Constraints) for the governance-as-instrument framing; and on the CLO **Judgment under inadequate evidence** for the instrument-validation-vs-outcome-validation tier distinction. Pair with Cases 115, 181, 169, 182 as the non-US LA governance pentad; MMALA is the institutional-instrument layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Resolve readiness into the dimensions adoption actually depends on — infrastructure, human resources, ethics, pedagogy — rather than a single overall score; the per-dimension structure is what makes the instrument actionable.
- Build maturity levels within each dimension so the self-assessment yields a structured next-step picture rather than a binary ready / not-ready.
- Validate the instrument in two layers: expert evaluation for comprehensiveness and suitability, and institutional pilot for usability — both are necessary, and the case carries both.

IN OPERATION / AFTER

- Report the evidence at its tier: expert opinion and three-institution validation establish the instrument; longitudinal outcome evidence on institutions that used it is the next study, and the case names the gap rather than glossing it.
- Commission follow-up studies on the institutions that pilot MMALA to convert the instrument-validation evidence into adoption-outcome evidence over time.
- Carry the level-of-governance lesson into the broader v2 governance set: institutional-instrument, multi-country participatory, national commission, and judicial review are different governance modes for the same capability; MMALA is the institutional-instrument instance.

REFLECTION QUESTIONS

1. Identify a capability your institution is considering adopting where readiness is currently discussed as a single overall question. What are the dimensions readiness actually has, and what maturity levels within each dimension would yield a structured next-step picture rather than a binary judgment?

2. Specify the two-layer validation you would build for the instrument: expert evaluation for comprehensiveness and suitability, and institutional pilot for usability. Where would each layer be at risk of being skipped, and what would that skip cost the instrument's credibility?

3. The case is honest that instrument-validation evidence is not the same as adoption-outcome evidence. What is the longitudinal study you would commission alongside MMALA-style adoption, on what cadence and against what comparison, to convert the instrument into outcome evidence over time?

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CASE 124 CORPORATE L&D TRAINING EVALUATION WORKFORCE DEV 2005 – PRESENT

Brinkerhoff Success Case Method — Tails as the Evaluation Instrument

K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT When ROI-style evaluation of corporate training is intractable, Brinkerhoff's Success Case Method samples the tails of the outcome distribution — the highest- and lowest-impact participants — and reconstructs the system conditions that made the program work for some and fail for others; deployed at Cargill, Ford, Merck, World Bank, ICRC

IN BRIEF

The Success Case Method (SCM), introduced by Robert Brinkerhoff in 2005, deliberately samples the tails of a training program's outcome distribution. Instead of attempting to derive an average effect that flatters most programs and gives L&D nothing actionable, SCM identifies the highest- and lowest-impact participants, studies them in detail, and reconstructs the system conditions that made the program work for some and fail for others. The method's argument — and the one that places it inside the LENS framework — is that extreme cases reveal whether the program ever produces meaningful work-performance change and what the surrounding system has to provide for transfer to happen. The method is peer-reviewed in **Advances in Developing Human Resources**; the corporate deployments at Cargill, Ford, Merck, World Bank, and the International Committee of the Red Cross are documented in practitioner channels — case-study writeups, conference talks, vendor whitepapers — rather than in peer-reviewed evaluation literature. SCM is the operational answer to the chain-of-evidence problem named in Case 108 (Kirkpatrick): the practitioner instrument that crosses the Level-2 / Level-3 seam by sampling where the evidence is most informative. Evidence-tier flag is practice-synthesis; future validation will continue as more firms publish their SCM outcome data.

BACKGROUND

Corporate L&D evaluation faces a structural problem the Kirkpatrick chain-of-evidence pattern names (Case 108): Level 3 (behavior change on the job) and Level 4 (business results) require longitudinal data the training organization typically cannot access. Average-effect statistics across whole programs flatter most programs, because the distribution of training outcomes is heavily skewed — a minority of participants produce most of the work-performance change, and the average buries that signal under the participants for whom the program produced nothing.[0]

THE INTERVENTION

Robert Brinkerhoff's 2005 **Advances in Developing Human Resources** paper proposes a different sampling logic: do not estimate the average; identify the participants in the tails of the outcome distribution and study them in detail. The Success Case Method identifies the highest-impact participants (where did the program work, and what conditions made the work-performance change possible), the lowest-impact participants (where did the program fail, and what conditions were missing), and reconstructs the system around each.[1]

HOW IT WORKED

The method's argument is that the tails carry the decision-grade information. A program that produces meaningful work-performance change for any participants proves it **can** work; the conditions that distinguish the success cases from the failure cases name what the surrounding system has to provide for transfer. The argument is the practical complement of the Blume meta-analytic finding (Case 35) that work environment is the decisive transfer variable: SCM operationalizes the finding by sampling where the variable's effects are most visible.[2]

THE EVIDENCE

The Brinkerhoff Evaluation Institute lists deployments at Cargill, Ford, Merck, the World Bank, and the International Committee of the Red Cross. These deployments are documented in case-study writeups, conference talks, and vendor whitepapers, not in peer-reviewed program evaluations. The evidence-tier flag is therefore practice-synthesis: the method itself is peer-reviewed, but the per-firm impact data on which the corporate adoption story rests live in practitioner channels. The honest framing in print is that the methodological pattern is teachable and durable, but the per-firm effect sizes are not independently audited.[3]

WHAT TRANSFERRED

The LENS teaching point pairs SCM with the chain-of-evidence problem from Case 108 and the meta-analytic transfer finding from Case 35. SCM is the operational instrument that crosses the Level-2 / Level-3 seam by sampling where the evidence is most informative — and it exercises the CLO **Judgment under inadequate evidence** directly, because the method asks practitioners to act on detailed case reconstructions rather than wait for population-scale causal estimates that corporate L&D often cannot produce. Future validation will continue as more firms publish their SCM outcome data.

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2. Brinkerhoff Evaluation Institute deployment list — Cargill, Ford, Merck, World Bank, International Committee of the Red Cross — practitioner channel.
3. Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the chain-of-evidence framework SCM operationalizes (paired Case 108).
4. Blume, Ford, Baldwin, & Huang (2010), Journal of Management 36(4):1065–1105 — the meta-analytic environment-as-decisive finding SCM samples around (paired Case 35).

> *The tails carry the decision-grade information. The average flatters the program.*
>
> EDITORS' SYNTHESIS OF BRINKERHOFF (2005) AND THE CORPORATE SCM DEPLOYMENTS.

LE INSIGHT

SCM is the practitioner instrument that operationalizes Blume's environment-as-decisive finding (Case 35) by sampling the tails of the outcome distribution. The method is peer-reviewed; the per-firm impact data at Cargill, Ford, Merck, World Bank, ICRC live in practitioner channels. Evidence-tier flag is practice-synthesis; future validation will continue as more firms publish.

LENS APPROACH

SCM is the corporate-L&D tail-sampling case (induced 2.1; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for the CLO **Judgment under inadequate evidence** — the method asks practitioners to act on case-reconstruction evidence rather than population estimates — and in Domain 2 (Iterative Development) by way of the evaluation-feedback loop SCM enables. Pairs with Case 46 (HILS) for the redesign of the surrounding work environment.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Before the program is delivered, plan the SCM sampling: define the outcome metric (specific work-performance change), identify how the success and failure cases will be surfaced (manager nominations, performance data, structured interviews), and pre-commit to the sampling logic.
- Structure the success-case and failure-case interviews around the system conditions Blume's meta-analysis (Case 35) names as decisive — supervisor support, peer support, practice opportunity, environment — so the reconstruction is theory-grounded, not anecdote-grounded.
- Report the tails honestly: the success cases are existence proofs the program **can** work; the failure cases are evidence of what the surrounding system did not provide. Neither is a population effect size.

IN OPERATION / AFTER

- Translate the tail evidence into a redesign of the surrounding system — supervisor briefings, peer-support structures, on-job practice opportunity — rather than into a redesign of the training event alone. Pair with Case 46 (HILS).
- Use the CLO **Judgment under inadequate evidence**: act on the case-reconstruction evidence the method produces while being explicit that it is not a population causal estimate; document the uncertainty.
- Carry the practice-synthesis flag honestly in any program documentation citing SCM corporate deployments — the method is peer-reviewed, but the per-firm effect sizes are not.

REFLECTION QUESTIONS

1. Identify a recent training program in your organization. Define how you would identify the highest- and lowest-impact participants and what structured interview you would conduct with each to reconstruct the system conditions around them.

2. Map the SCM tails to the Blume meta-analytic variables (Case 35): supervisor support, peer support, practice opportunity, environment. What pattern would the tails have to show for you to redesign the surrounding system rather than the training event itself?

3. The corporate SCM deployments at Cargill, Ford, Merck, World Bank, ICRC are documented in practitioner channels. What additional evidence — independent program audit, peer-reviewed evaluation, cross-firm comparison — would you require before treating any specific per-firm effect size as decision-grade?

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CASE 125 HEALTHCARE EHR USABILITY CLINICAL DECISION SUPPORT 2020

Annual-Screening UI Redesign + CDS at University of Missouri Health Care

T Training Gap D Designed Out N Normalization of Deviance

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT A multidisciplinary EHR redesign of ambulatory annual-screening prompts (advance directives, depression, falls risk, alcohol/drug misuse), paired with embedded CDS, reported improvements in task time, error rates, System Usability Scale scores, and the downstream screening-rate outcomes the project was scoped to move

IN BRIEF

A multidisciplinary team at University of Missouri Health Care redesigned the EHR interface clinicians use to prompt and perform annual screening — advance directives, depression, falls risk, alcohol and drug misuse — and embedded clinical decision support inside the redesigned workflow. The team reported gains on the usability metrics (task time, error rate, System Usability Scale) and on the downstream process outcome the project was scoped to move: the actual rate at which guideline-recommended screening was completed. It is a small-tier intervention case for cue-and-alert design as a capability deliverable, with both human-factors and clinical-process outcomes in the same report. The corpus has long needed a small-tier C3 positive example to set against the interface failures already documented at the big tier (Therac-25, CPOE/EHR adoption, the Helios pattern). The evidence base is a HIMSS case-study format rather than a top-tier peer-reviewed journal article; the tier flag renders under the title and is load-bearing. Future validation will continue as the downstream clinical-outcome literature on screening-rate gains matures.

BACKGROUND

Ambulatory annual screening — advance directives, depression, falls risk, alcohol and drug misuse — is the kind of guideline-recommended care that is easy to declare and hard to land. The cue lives in the EHR; the action lives in a time-pressured encounter; and the gap between prompt and completion is where most screening programs lose their numbers. The University of Missouri Health Care project framed the problem squarely as cue-and-alert design: if the prompt cannot be acted on inside the workflow without friction, the screening will not happen.[0]

THE INTERVENTION

The redesign was multidisciplinary by construction — clinicians, informaticists, and usability specialists working against the existing screening interface. The team rebuilt the prompt presentation, added embedded clinical decision support that surfaced the next action at the point of decision, and tightened the path between recognizing a positive screen and placing the appropriate order. The design move is the one the induced framework flags as the C3 deliverable: change the interface so that the desired action is the path of least resistance, not a separate sub-task layered on top of the visit.[1]

HOW IT WORKED

The reported outcomes cross two layers. At the usability layer the team reported reductions in task time and error rate and a gain on the System Usability Scale. At the process layer they reported an increase in the screening-rate metric the project was scoped to move — the clinical process the cue exists to drive. That second layer is what makes the case a C3 small-tier intervention rather than a usability study: the interface gain translated into the downstream behavior, at least over the reported observation window.[2]

THE EVIDENCE

The evidence-tier flag rendered under the title is load-bearing. The case is documented in a HIMSS chapter case-study format rather than a top-tier peer-reviewed journal article. The magnitudes reported sit inside a single-institution quality-improvement project with its own outcome metric; replication at other institutions and durability across EHR upgrades and personnel rotation have not been independently audited. Future validation will continue as the institution and others publish follow-on screening-rate data and as the relationship between screening completion and downstream patient outcomes is tracked.[3]

WHAT TRANSFERRED

What the case teaches at the LENS layer is that the C3 interface-failure pattern documented at the big tier is not a cosmic constraint — it is redressable by deliberate design at the small tier, when usability specialists, clinicians, and CDS authors are seated together. The capability deliverable is the redesigned cue plus the embedded decision support plus the measurement of the downstream process the cue exists to drive. The case is the missing positive example for induced 3.1 at the small tier and a paired teaching companion for the failures already in the corpus.

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3. Office of the National Coordinator for Health IT, SAFER Guides on CDS design — practitioner-tier guidance the redesign instantiates.
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> *The C3 interface failure mode is not a cosmic constraint. It is redressable by deliberate design — when usability, clinicians, and CDS authors sit on the same redesign.*
>
> EDITORS' SYNTHESIS OF THE UMHC HIMSS CASE STUDY.

LE INSIGHT

The UMHC redesign is the small-tier C3 positive example the corpus needed: cue-and-alert design as a capability deliverable, with both usability and downstream screening-rate gains in the same project. The evidence is practice-synthesis tier (HIMSS case-study format), not a top-tier peer-reviewed article; magnitudes await independent replication and durability tracking. Future validation ongoing.

LENS APPROACH

UMHC is the C3 small-tier intervention case (induced 3.1; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for the CLO-4 deliverable that cue redesign must show its downstream process effect, and in Domain 1 (Systems Analysis) for CLO-1 — the multidisciplinary team did the analysis of the screening workflow as the precondition for the redesign. The case is the paired positive example for the big-tier C3 failures (Therac-25, CPOE/EHR adoption, Helios) the corpus already documents.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Frame the screening problem as cue-and-alert design from the start; do not separate usability from the clinical-process outcome the cue exists to drive.
- Seat usability specialists, clinicians, and CDS authors together on the redesign team; the C3 deliverable is the integrated artifact, not a hand-off.
- Specify the downstream process metric (screening-rate completion) before the redesign ships, so the usability-layer and process-layer outcomes are measured against the same scope.

IN OPERATION / AFTER

- Treat the QI report honestly: a single-institution case study is a small-tier intervention, not a settled magnitude. Carry the practice-synthesis-tier flag into any downstream citation.
- Track durability — across EHR upgrades, personnel rotation, and downstream clinical outcomes — as a separate post-deployment commitment, not a footnote to the implementation report.
- Pair with the big-tier C3 failures already in the corpus when teaching; the failure-and-intervention pair is the teaching artifact, not either alone.

REFLECTION QUESTIONS

1. Identify a screening or recommended-care prompt in your EHR. What proportion of prompts result in the completed action? Where does the cue-action gap sit, and what is the redesign that would close it?

2. Specify the downstream process metric you would commit to before any usability redesign ships. The UMHC case is teachable because it reported the screening-rate outcome, not only the SUS gain. What would be the equivalent in your context?

3. The case is practice-synthesis tier (HIMSS case-study format). What is the minimum independent replication evidence you would require before treating the reported magnitudes as a basis for an institutional investment in your own setting?

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CASE 126

HEALTHCARE | CLINICAL DECISION SUPPORT | EHR DESIGN | NURSING WORKFLOW

2019 – 2024

Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

T Training Gap   D Designed Out   N Normalization of Deviance

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT Structured EHR alert redesign — fewer alerts, severity reclassification, interruptive-to-passive conversion, role-tailoring — reduced alert burden in published systematic-review and quality-improvement evidence; the 2024 case studies report alert-rate reduction with the underlying safety signal preserved

IN BRIEF

Alert fatigue is the structural failure mode the C3 thread names at the small tier: an EHR that fires so many alerts the actionable ones are lost in the noise. The 2019 JAMIA systematic review by Co and colleagues aggregates the evidence that structured redesign — interaction design changes and clinical-role tailoring — can reduce alert burden; most optimization studies in the review reported alert-rate reduction after intervention. Two 2024 quality-improvement publications extend the pattern with named maneuvers: a nursing-workflow redesign of four high-firing, low-action alerts using quantitative alert-firing analysis, empathy mapping, and iterative user feedback; and the replacement of an interruptive COVID-precautions alert with passive clinical decision support, targeting both alert burden and the timeliness of precautions orders. The case is the small-tier intervention companion to a C3.2 failure thread that v1 left almost entirely populated by failures (Uber ATG, Robodebt, Northeast Blackout, UK Post Office, Tesla Autopilot). The evidence tier is mixed: the systematic review is peer-reviewed; the per-site QI projects are practice-tier publications. Future validation will continue on whether the redesigns survive EHR upgrades and personnel rotation.

BACKGROUND

Monitoring under alert burden is the unsupportable role the C3.2 induced sub-competency names — the operator asked to keep attention on a stream of low-signal alerts and to spot the consequential one in real time. In the EHR setting the burden becomes structural: high-firing, low-action alerts train the clinician to dismiss alerts as the default, which is where the actionable alert is missed. Alert fatigue is the failure mode; the question for C3 is whether deliberate redesign can reduce the burden without cutting the safety signal.[0]

THE INTERVENTION

The 2019 JAMIA systematic review by Co and colleagues aggregates the published evidence on EHR alert optimization. The headline finding: interaction-design changes and clinical-role tailoring reduce alert burden, and most evaluated optimization studies in the review reported alert-rate reduction post-intervention. The mechanism is not a single intervention but a family of moves — severity reclassification, conversion of interruptive alerts to passive decision support, role-based tailoring so the alert reaches the clinician who can act on it, and removal of alerts whose firing-to-action ratio shows the alert is no longer working.[1]

HOW IT WORKED

The 2024 quality-improvement publications instantiate the moves. One project redesigned four high-firing, low-action alerts in the nursing workflow using mixed methods — quantitative analysis of firing data, empathy mapping of the nursing experience, and iterative user feedback as the redesign was refined. A second project replaced an interruptive COVID-precautions alert with passive CDS, with dual outcomes: reduce alert burden and improve the timeliness of precautions orders. Together the projects show the redesign-pattern is operable at the per-alert level and that the evaluation can report both halves of the trade-off the redesign exists to manage.[2]

THE EVIDENCE

The evidence-tier flag matters. The 2019 systematic review is peer-reviewed, and the 2024 Applied Clinical Informatics paper sits in a peer-reviewed informatics journal. The practical maneuver of treating the per-site QI projects as generalizable, though, rests on practice-synthesis logic across the body of work rather than on a single multi-site randomized evaluation. Magnitudes vary by site, EHR vendor, and alert category; the redesigns must be re-verified after EHR upgrades and personnel rotation. Future validation will continue as the optimization-study literature consolidates.[3]

WHAT TRANSFERRED

The teaching point pairs with Case 125 and with the v1 C3.2 failure thread. C3.2 in v1 is entirely failures; this case is the small-tier intervention that demonstrates the failure mode is redressable by design. The capability deliverable is the redesigned alert architecture itself — severity tiers, interruptive-vs-passive decisions, role-tailoring rules, and an ongoing measurement loop on firing-to-action ratios — not a one-time clean-up. The new CLO around delegation with revocation applies here: when automated decision support oversight is delegated to the bedside, the redesign discipline is part of the delegation.[4]

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THE LEARNING ENGINEERING LENS

> *Alert fatigue is the failure mode the alert architecture trains. The redesign discipline is part of the delegation.*
>
> EDITORS' SYNTHESIS OF CO ET AL. (2019) AND THE 2024 QI LITERATURE.

LE INSIGHT

The 2019 JAMIA review plus the 2024 QI projects are the small-tier C3.2 intervention companion the corpus needed — the failure thread (Uber ATG, Robodebt, UK Post Office, Tesla) is redressable by deliberate alert redesign. The systematic review is peer-reviewed; the per-site QI publications are practice-tier; magnitudes and durability open. Future validation ongoing.

LENS APPROACH

Alert-fatigue redesign is the C3.2 small-tier intervention case (induced 3.1 and 3.2; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for CLO-4 — the redesign must report the safety signal alongside the alert-rate cut — and in Domain 3 (Human-System Collaboration) for CLO-3 oversight of automated decision support delegated to the bedside, with the CLO on delegation with revocation explicit. Pair with Case 125 for the small-tier C3 thread.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Instrument every consequential alert with its firing-to-action ratio; the metric that names the failure mode must be reported alongside the redesign.
- Decide the interruptive-vs-passive call deliberately for each alert; the conversion is the C3 move that reduces burden when the signal does not require the interruption to be actionable.
- Use mixed methods (firing data + clinician experience + iterative feedback) to design the per-alert redesign; quantitative alone misses why the alert is being dismissed.

IN OPERATION / AFTER

- Track the safety signal the alert exists to protect as a separate post-redesign outcome; an alert-rate cut that lost the signal is a failure, not a win.
- Re-audit the redesigned alerts after EHR upgrades and personnel rotation; the redesign is not a one-time clean-up and the v1 thread shows the failure mode returns.
- Treat the per-site QI publications as practice-synthesis evidence when generalizing; the systematic review supports the pattern, the magnitudes require local replication.

REFLECTION QUESTIONS

1. Identify an EHR alert in your context with a high firing-to-action ratio. Which of the redesign moves (severity reclassification, interruptive-to-passive conversion, role-tailoring, removal) would you apply, and what would the measured outcome be on both alert burden and the safety signal?
2. Specify the mixed-methods design (firing data + clinician experience + iterative feedback) you would use to make a per-alert redesign decision. What does the empathy-mapping step add beyond quantitative firing analysis alone?
3. The per-site QI publications are practice-synthesis tier. What is the minimum cross-site or randomized evidence you would require before generalizing the magnitudes from the 2024 reports to your own setting?

WHO BUILDS THIS learning science & instructional design · systems & human-factors engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · task analysis · design prototyping · incident analysis · safety dashboards

CASE 127 K-12 MATHEMATICS INTELLIGENT TUTORING EDUCATION POLICY 2007 – 2014

Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT Cluster-randomized 147 high schools across seven states; year-one posttest scores showed no significant difference between Cognitive Tutor and control conditions; year-two posttest scores showed CTAI schools significantly outperforming controls; a one-year evaluation would have published the wrong answer

IN BRIEF

Pane, Griffin, McCaffrey, and Karam's 2014 paper in Educational Evaluation and Policy Analysis reports the central at-scale evaluation of Cognitive Tutor Algebra I (CTAI), the canonical learning-sciences-to-classroom translation. The RAND team cluster-randomized 147 high schools across seven states to continue their current Algebra I curriculum or to adopt CTAI for two years. Year-one posttest scores: no significant difference between conditions. Year-two posttest scores: CTAI schools significantly outperformed control schools. The honest reading the authors press is that a one-year evaluation would have published a null result and the two-year evaluation surfaced a real effect — and both findings were in the same trial. The case is the deeper-evidence-of update on v1 Case 38 (Cognitive Tutor), translating that case from a system-description case into a methodological- discipline case about evaluation horizons. The timeline of the evaluation is itself a falsifiable design choice; the case grounds the CLO on judgment under inadequate evidence where the inadequacy is the evaluation horizon, not the sample size.

BACKGROUND

Cognitive Tutor is the case the learning-sciences-to- classroom translation thread cannot avoid teaching at v1 Case 38. The system instantiates a model of student cognition, drives a problem-by-problem adaptive curriculum, and is the canonical published instance of an intelligent tutoring system in K-12 mathematics. The question the v1 case left open — and the question the Pane et al. 2014 paper answers — is what happens when the system is deployed at scale, in the classroom and curricular conditions that classroom adoption actually generates, rather than in the developer-supported conditions of an early efficacy trial.[o]

THE INTERVENTION

The RAND team built the evaluation to support that question. 147 high schools across seven states were cluster-randomized to either continue their current Algebra I curriculum or to adopt CTAI for two years. The sample was deliberately wide — multiple states, multiple district contexts, classroom adoption with the support structures the publisher could realistically supply. The outcome instrument was an Algebra I posttest applied at end of school year. The design supports an effect-size estimate at end of year one and a separate estimate at end of year two, with the same students in the same treatment condition for both years.[1]

HOW IT WORKED

The year-one result was a null. Posttest scores in CTAI schools did not differ significantly from posttest scores in control schools, across the cluster-randomized sample. The year-two result was different. Posttest scores in CTAI schools were significantly higher than in control schools, with an effect size that an Algebra I curriculum evaluation would treat as meaningful. The two findings were generated by the same trial, run in the same schools, with the same instrument. The methodological discipline the case teaches turns on what the field would have learned if the evaluation had been designed to a one-year horizon: it would have published a null result against an intervention that the same trial, on its planned horizon, established as positive.[2]

THE EVIDENCE

The authors press the honest reading without softening it. The timeline of the evaluation is itself a falsifiable design choice, not a methodological default. The reasons year one returned a null are recoverable from the trial record — teacher fluency with the system grew across the year, district-level scheduling and pacing adjusted across the year, student familiarity with the adaptive workflow stabilized across the year. None of these are noise; all are part of the deployment substrate the intervention depends on, and all required the second year to stabilize. The two-year horizon was the right horizon because the intervention's effect is conditioned on a deployment-substrate stabilization that takes more than one year.[3]

WHAT TRANSFERRED

The case is the deeper-evidence-of update on v1 Case 38. v1 Case 38 carries Cognitive Tutor as the system description; this case carries the at-scale evaluation and the timeline-of-evidence lesson. The CLO on judgment under inadequate evidence is directly motivated: the one-year evaluation would have been inadequate evidence, and a field that publishes the one-year null without noting the planned second-year horizon publishes a wrong answer on the available data. Pair with Case 49 (ASSISTments) for the multi-year follow-through paired case, and with Case 93 (Epic Sepsis) for the evaluation-horizon discipline in clinical AI. The closed loop the case completes is the two-year-horizon-was-the- right-horizon record that lets a field design the next at-scale evaluation honestly.

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THE LEARNING ENGINEERING LENS

> *A one-year evaluation would have published a null. The two-year evaluation surfaced a real effect. Both findings were in the same trial.*
>
> EDITORS' SYNTHESIS OF PANE ET AL. (2014).

LE INSIGHT

Pane et al. is the at-scale evaluation case for Cognitive Tutor and the worked example of evaluation horizon as a falsifiable design choice. Year one: null. Year two: significantly positive. The case is the deeper-evidence-of update on v1 Case 38 and the curriculum's primary anchor for the CLO on judgment under inadequate evidence where the inadequacy is the horizon, not the sample size.

LENS APPROACH

Pane / CTAI at scale is the closed-loop evaluation-horizon case (induced 2.3; LENS D4/PT4). LENS uses it in Domain 4 (Test and Evaluation) for the horizon-as-design-choice discipline and in Domain 2 (Iterative Development) for the deployment-substrate-stabilization frame — teacher fluency, scheduling, workflow familiarity are designable parts of the intervention substrate, not noise. Pair with Case 49 (ASSISTments follow-through), v1 Case 38 (Cognitive Tutor system), and Case 93 (Epic Sepsis horizon discipline).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Plan the evaluation horizon as a design decision, not a budget default; the case demonstrates that the horizon choice is causally consequential and that a shorter horizon can produce a wrong-direction finding.
- Document the deployment-substrate stabilization that the intervention depends on: teacher fluency, scheduling adjustment, student familiarity with the workflow are all year-over-year stabilization mechanisms the trial should track.
- Design the trial so the year-one and year-two estimates are both interpretable; the case's pedagogical value depends on having both estimates from the same trial, not on having only the longer-horizon estimate.

IN OPERATION / AFTER

- Publish the year-by-year result pattern, not the aggregate effect; the case's teaching power depends on the year-one null being on the same page as the year-two positive.
- Carry the timeline-of-evidence lesson into the curriculum's evaluation-design teaching; the case grounds the CLO on judgment under inadequate evidence with a rare worked example where the inadequacy is the horizon, not the sample size.
- Pair the case with Case 49 (ASSISTments multi-year follow-through) so the field-level discipline is taught with two converging examples; the methodological lesson is more useful as a pattern than as a single instance.

REFLECTION QUESTIONS

1. Identify an intervention in your domain whose effect depends on a deployment-substrate stabilization that takes more than a year (teacher fluency, workflow familiarity, scheduling alignment). What evaluation horizon is honest for that intervention, and what would the year-one estimate, taken alone, falsely conclude?

2. Specify the year-by-year result pattern your next evaluation will report. The case's pedagogical value depends on having both the year-one and year-two estimates from the same trial; what reporting structure would you commit to that supports the pattern view, not only the aggregate?

3. The CLO on judgment under inadequate evidence is grounded in this case where the inadequacy is the evaluation horizon. Identify a published null result in your domain whose horizon was potentially too short; what would the replication-with-longer-horizon look like?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · task analysis · design prototyping

CASE 128 HIGHER EDUCATION LEARNING ANALYTICS DISTANCE LEARNING 2019 – 2023

OU Analyse — Predictive Learning Analytics and Teacher Use at Scale

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT Across 9 courses and 559 teachers (189 with OU Analyse access), teachers' engagement with predictive learning analytics was associated with measurable improvements in student performance for >14,000 students; three-year post-implementation follow-up extends the evidence into sustained adoption and perceptions

IN BRIEF

Herodotou et al.'s 2019 British Journal of Educational Technology paper reports the evaluation of OU Analyse, the Open University UK's predictive-learning-analytics dashboard for online tutors, across 9 courses and 559 teachers (189 of whom had access to OU Analyse) with more than 14,000 students. Teachers' engagement with the predictive learning analytics was associated with measurable improvements in student performance. The LAK 2023 three-year-post-implementation follow-up extends the picture into questions of sustained adoption — how teachers' use of the predictions stabilized, what fraction continued to act on them, how perceptions evolved. The case is distinct from the OU consent-and-ethical- use frame Case 115 covers; this case carries the post- deployment teacher-use evaluation at multi-cohort scale. The authors' hedges are binding: causal attribution to OU Analyse use specifically — versus teacher selection effects — is bounded; the 2019 study uses propensity-style matching rather than RCT randomization. Future validation ongoing on multi-institution transfer as the system is licensed beyond the Open University.

BACKGROUND

Predictive learning analytics dashboards have been deployed across higher-education institutions for more than a decade. The deployment record is largely a collection of pilots and single-cohort studies, with Course Signals (the discontinued Purdue deployment) as the structural cautionary case. OU Analyse is structurally different: deployed across an entire distance-learning institution, used by hundreds of tutors across multiple cohorts and multiple courses, and evaluated at the scale the institutional deployment supports. The 2019 Herodotou et al. paper is the central peer-reviewed evaluation of the teacher-use side of the deployment.[0]

THE INTERVENTION

The evaluation design covers 9 courses, 559 teachers, and more than 14,000 students. 189 of the 559 teachers had access to OU Analyse; the comparison structure is between these teachers and the OU Analyse-naive teachers in the same courses. The outcome is student performance on course assessments. Teachers' engagement with the predictive learning analytics — operationalized as dashboard-usage and acted-upon-prediction proxies — was associated with measurable improvements in student performance for the cohorts those teachers taught. The effect size is meaningful at the scale the institutional deployment supports.[1]

HOW IT WORKED

The 2023 LAK three-year-post-implementation follow-up extends the picture into a question the single-cohort pilots cannot address: what happens to teacher use of the predictions across multiple cohorts and across years? The follow-up paper documents how teachers' use stabilized, what fraction continued to act on the predictions, and how perceptions evolved as the institutional norm around the dashboard solidified. Across the three years, the pattern surfaces as one of sustained adoption with stratification: a fraction of teachers used the predictions actively, a fraction used them as background reference, and a fraction did not engage. The delegation-with-revocation structure is operative: each teacher chose, in each cohort, whether to act on the prediction, and the choice itself is the load-bearing capability the dashboard supports.[2]

THE EVIDENCE

The case is distinct from the OU consent-and-ethical-use frame Case 115 covers. Case 115 carries the governance story — the Policy on Ethical Use of Student Data, the institutional review of analytic deployments, the consent architecture. This case carries the post-deployment teacher-use evaluation at multi-cohort scale. The editor's decision (memo A6) anchors LENS primary, induced secondary, CLO carried; the two cases together teach the deployment across both governance and post-deployment-use frames without collapsing one into the other. The frame the present case adds to the corpus is the rare successful learning-engineering intervention with both deployment scale and longitudinal teacher-use evidence at the journal-graded evidence tier the corpus needs.[3]

WHAT TRANSFERRED

The authors' hedges are binding. The causal attribution to OU Analyse use specifically — as distinct from teacher selection effects — is bounded. The 2019 study uses propensity-style matching but does not randomize at the teacher level; the 2023 follow-up extends the evidence base on adoption and perception but does not close the teacher-selection question. Future validation ongoing on multi-institution transfer — OU Analyse has been licensed beyond the Open University, and the transfer outcomes are not yet in the peer-reviewed literature. Pair with Case 115 (OU consent governance), Case 127 (Cognitive Tutor at-scale evaluation), and the Purdue Course Signals cautionary case from v1 — the OU Analyse evaluation is the methodologically more careful descendant of the Course Signals lineage.

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THE LEARNING ENGINEERING LENS

> *The delegation-with-revocation structure is operative: each teacher chose, in each cohort, whether to act on the prediction. The choice itself is the load-bearing capability the dashboard supports.*
>
> EDITORS' SYNTHESIS OF HERODOTOU ET AL. (2019, 2023).

LE INSIGHT

OU Analyse is the rare successful learning-engineering intervention with both deployment scale and longitudinal teacher-use evidence at journal tier. The 2019 BJET paper establishes the effect on student performance at the institutional deployment; the 2023 LAK follow-up extends the picture into three-year sustained adoption. The delegation-with-revocation structure is operative and teachable. Hedges binding on causal attribution and on multi-institution transfer.

LENS APPROACH

OU Analyse teacher-use is the human-in-the-loop predictive- analytics case at institutional scale (induced 5.2; LENS D4/PT5). LENS uses it in Domain 4 (Test and Evaluation) for the multi-cohort longitudinal-follow-through design and in Domain 3 (Human-System Collaboration) for the delegation-with-revocation structure — the teacher chooses, each time, whether to act on the prediction. Pair with Case 115 (OU consent governance), Case 127 (Cognitive Tutor at-scale), and Case 49 (ASSISTments multi-year).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat teacher use of the prediction as the load-bearing capability the dashboard supports; the delegation-with-revocation structure — each teacher chooses, in each cohort, whether to act — is the design point, not an emergent behavior.
- Build the deployment at multi-cohort and multi-course scale from the start; the case's evidence-tier strength depends on the deployment having been institutional, not pilot, and the institutional deployment supports the longitudinal-follow-through evidence the case provides.
- Design the dashboard's prediction surface around the action the teacher would take — a flagged-student list with an intervention prompt, not a probability score the teacher has to interpret unaided.

IN OPERATION / AFTER

- Commission the next-tier evaluation that addresses the teacher-selection-effect question; an RCT at the teacher level, or an instrumental-variable design exploiting institutional dashboard-rollout variation, would close the residual causal-attribution gap.
- Publish the multi-institution-transfer evaluations as the system is licensed beyond the OU; the case's value as a generalizable instance depends on the transfer evidence the OU-internal evaluation cannot provide.
- Pair the case in the curriculum with Case 115 (OU consent governance) so the deployment is taught across both governance and post-deployment-use frames; the two cases together teach the institutional anchor without collapsing one frame into the other.

REFLECTION QUESTIONS

1. Identify a predictive-analytics dashboard in your domain whose operator-use stratification — active, background reference, non-engagement — has not been longitudinally tracked. What would the multi-cohort follow-up look like, and what data infrastructure would it require?

2. Specify the design choice for the prediction surface in your domain: a flagged-list-with-intervention-prompt vs. a probability-score-the-operator-interprets-unaided. Which design is in place currently, and what would change if the design switched?

3. The case's hedge on teacher selection effects is the open methodological question. What randomized or quasi-experimental design would close the residual attribution gap in your domain's analog deployment — and what institutional support would the design require?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · task analysis · design prototyping

CASE 129 HEALTHCARE CLINICAL DECISION SUPPORT MACHINE LEARNING 2022 – 2024

COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT Boussina et al. NPJ Digital Medicine 2024 prospective multi-site implementation of the COMPOSER (COnformal Multidimensional Prediction Of SEpsis Risk) deep-learning model at UC San Diego Health; reported 1.9 percentage-point absolute decrease in in-hospital sepsis mortality and 5.0 percentage-point absolute decrease in 28-day sepsis-related readmission during the deployment period, evaluated against the pre-deployment baseline period within the same health system

IN BRIEF

Aaron Boussina, Shamim Nemati, and colleagues at UC San Diego Health published the COMPOSER (COnformal Multidimensional Prediction Of SEpsis Risk) prospective implementation study in NPJ Digital Medicine in early 2024. The model is a deep- learning sepsis-risk prediction system that uses conformal prediction to handle uncertainty — alerting only when the model's calibrated confidence threshold is met and abstaining when the input is sufficiently out-of-distribution. The deployment evaluation reported a 1.9 percentage-point absolute decrease in in-hospital sepsis mortality and a 5.0 percentage- point absolute decrease in 28-day sepsis-related readmission during the deployment period at two UC San Diego hospitals, evaluated against the pre-deployment baseline period within the same health system. The case completes the AI-delegation typology in sepsis at three deployments — Case 74 (TREWS at Johns Hopkins), Case 93 (Epic Sepsis Score across multiple health systems), and COMPOSER at UCSD. The honest hedges from the source are binding: the deployment is prospective but not RCT-grade, the mortality reduction is multifactorial (the COMPOSER deployment ran alongside other process improvements at UCSD), and the author team is explicit about the structural attribution question. Pair with Case 74, Case 93, and Case 130 (Radiology AI Miscalibration).

BACKGROUND

The contemporary clinical-AI sepsis-prediction literature has converged on a small number of well-evidenced deployments. Case 74 documents TREWS — the Targeted Real- Time Early Warning System at Johns Hopkins — with its prospective evaluation showing mortality reduction associated with prompt clinician response to alerts. Case 93 documents the Epic Sepsis Score deployed across multiple U.S. health systems, with external validation by Wong and colleagues finding substantially lower sensitivity and higher false-positive rates than the vendor's validation data implied. COMPOSER is the third deployment in the typology and the second prospective-positive case. The deployment is at UC San Diego Health across two hospital sites, integrated into the electronic health record and into emergency-department and inpatient workflows.[0]

THE INTERVENTION

The methodological feature that distinguishes COMPOSER is the conformal-prediction framework. Conformal prediction is a statistical method for producing calibrated uncertainty intervals around individual predictions: rather than producing a single risk score, the model produces a prediction together with a measure of how confident the model is in that prediction given the input. The operational implication is that the model can abstain from alerting when the input is sufficiently out-of-distribution — when the model's calibrated confidence does not support a clinically actionable alert. The abstention structure is what handles one of the load-bearing failure modes of Case 93 (Epic Sepsis), where the model alerted at high rates against patient populations it had not been adequately calibrated against. COMPOSER's abstention structure is one mechanism for refusing to alert in those circumstances.[1]

HOW IT WORKED

The deployment outcomes Boussina and colleagues report are the load-bearing intervention-evidence. Across the deployment period at the two UC San Diego hospital sites, in-hospital sepsis mortality decreased by 1.9 percentage points in absolute terms, and 28-day sepsis-related readmission decreased by 5.0 percentage points, both evaluated against the pre-deployment baseline period within the same health system. The evaluation is prospective implementation, not RCT — the comparison is against the same hospital sites' historical baseline rather than against a contemporaneous randomized control arm — and the authors are explicit about the limitation. The deployment ran alongside other process improvements at UCSD's sepsis-care pathway, and the authors are honest that the mortality reduction is multifactorial: COMPOSER's contribution cannot be cleanly separated from the contribution of the surrounding process changes.[2]

THE EVIDENCE

The case pairs with Case 74 (TREWS) for the prospective- positive sepsis-prediction thread: both deployments report mortality benefits, both are integrated into specific health-system workflows, and the comparison between TREWS and COMPOSER turns on the alerting architecture and on the role of the surrounding process. Pair with Case 93 (Epic Sepsis Score) for the external-validation-revealed- gaps thread: COMPOSER's abstention structure is one response to the calibration-gap failure mode that the Epic Sepsis evidence base named. Pair with Case 130 (Radiology AI Miscalibration) for the broader medical-AI miscalibration-in-deployment thread. The three sepsis cases together — Case 74, Case 93, COMPOSER — define the AI-delegation typology in sepsis: prospective-positive with workflow integration (TREWS), externally validated and found wanting against vendor claims (Epic Sepsis), and prospective-positive with conformal-prediction abstention (COMPOSER).[3]

WHAT TRANSFERRED

The hedges the case carries are load-bearing and preserved in the prose. The deployment is prospective implementation, not an RCT; the mortality and readmission reductions are evaluated against the same health system's historical baseline, and the comparison cannot rule out concurrent improvements in sepsis care that would have produced similar reductions in the absence of COMPOSER. The mortality reduction is multifactorial — COMPOSER was deployed alongside other process improvements at UCSD, and the authors are explicit that the deployment evaluation cannot cleanly separate COMPOSER's contribution from the contribution of the surrounding process. The conformal- prediction abstention structure is the methodological contribution the case anchors, and the cue-and-alert design CLO is anchored at the deployment seam where abstention is a clinical-workflow capability: the model that can refuse to alert when its calibrated confidence does not support an alert is the deployment artifact the case names as the load-bearing one.

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THE LEARNING ENGINEERING LENS

> *The deployment is prospective implementation, not RCT; the mortality reduction is multifactorial — COMPOSER was deployed alongside other process improvements at UCSD, and the contribution cannot be cleanly separated; the conformal-prediction abstention is the methodological contribution the case anchors.*
>
> EDITORS' SYNTHESIS OF BOUSSINA ET AL. (2024) NPJ DIGITAL MEDICINE AND THE SURROUNDING SEPSIS-PREDICTION DEPLOYMENT LITERATURE.

LE INSIGHT

COMPOSER is the third clinical-AI sepsis case in the delegation typology — prospective-positive with conformal- prediction abstention. The deployment reduced in-hospital sepsis mortality by 1.9 percentage points and 28-day readmission by 5.0 percentage points; the load-bearing hedges are that the evaluation is prospective not RCT and that the mortality reduction is multifactorial. The abstention structure is the methodological contribution the case anchors.

LENS APPROACH

COMPOSER is the cue-and-alert-design-as-capability-deliverable case at clinical-deployment scale (induced 3.1; LENS D4+D4/PT6; CLO-4 and CLO-3). LENS uses it in Domain 4 (Test and Evaluation) for the prospective-implementation- evaluation discipline and in Domain 3 (Human-System Collaboration) for the abstention-as-clinical-workflow- capability anchor. Pair with Case 74 (TREWS), Case 93 (Epic Sepsis), and

Case 130 (Radiology AI Miscalibration). The three sepsis cases together — Case 74, Case 93, Case 129 — define the AI-delegation typology in sepsis at the deployment-evidence level.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build the abstention mechanism as part of the deployment, not as a post-hoc filter; the conformal-prediction framework's clinical value is that the model can refuse to alert when its calibrated confidence does not support an alert, and the abstention is a deployment capability rather than an analysis artifact.
- Pre-specify the multifactorial-attribution question before evaluation; the case demonstrates that prospective implementation alongside concurrent process improvement requires the evaluation to acknowledge the attribution limit, and pre-specifying the acknowledgement is the discipline.
- Integrate the model into specific workflow surfaces — ED admission, inpatient deterioration — rather than as a hospital-wide alerting layer; the deployment's specificity is part of what makes the conformal-prediction abstention structure operationally meaningful.

IN OPERATION / AFTER

- Carry the multifactorial-mortality hedge into print without softening; the case's pedagogical value depends on the attribution question being explicit alongside the deployment-outcome reporting.
- Pair in syllabi with Case 74 (TREWS) and Case 93 (Epic Sepsis) so the AI-delegation typology in sepsis is taught as a three-case set at the deployment-evidence level.
- Use the case to anchor the cue-and-alert design CLO; the curricular target is the discipline of building abstention into the alerting architecture as a capability deliverable, and of evaluating the abstention against the calibration-gap failure mode the prior cases named.

REFLECTION QUESTIONS

1. Identify a clinical-AI deployment in your setting whose alerting architecture does not include an abstention mechanism. What is the calibration-gap failure mode the deployment would encounter, and what would the conformal-prediction abstention structure look like in your workflow?

2. Specify the multifactorial-attribution acknowledgement you would carry in a prospective-implementation evaluation. What concurrent process improvements would have to be enumerated, and what is the documented decision rule for what the deployment evaluation can and cannot establish?

3. The three sepsis cases — TREWS, Epic Sepsis, COMPOSER — together define an AI-delegation typology. Pick a clinical-AI category in your setting and ask: what is the analogous typology, and which deployments would have to be evidenced to define it at the same level of resolution?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · task analysis · design prototyping

CASE 130 HEALTHCARE TECHNOLOGY 2018 – PRESENT

Radiology AI Miscalibration

H Human-System Interface K Knowledge & Institutional Memory D Designed Out

IMPACT Recurring documented cases of FDA-cleared radiology AI tools performing worse in deployment than in validation, often along demographic lines; the canonical v1 anchor for clinical-AI deployment without surveillance, cross-referenced by the Epic Sepsis (Case 93) and pulse-oximetry (Case 114) deployment-evidence cases

IN BRIEF

FDA-cleared radiology AI tools — for chest X-ray classification, mammography, CT triage — have been repeatedly documented performing worse in deployment than in their validation studies, often with the degradation concentrated in patient groups under-represented in the training data. Larrazabal et al. (PNAS 2020) showed this structurally for chest-X-ray classifiers across gender imbalance; Seyyed-Kalantari et al. (Nature Medicine 2021) extended the finding to under-served racial and socioeconomic subgroups across three large public chest-X-ray datasets. Obermeyer et al. (Science 2019) showed that bias in the labels used to train clinical AI can under-allocate care to Black patients even when the model looks well-calibrated on its chosen proxy. Wachter and Brynjolfsson (JAMA 2023) raised the generative-AI extension. The FDA's 510(k) clearance pathway — the route most cleared radiology AI tools have taken — does not routinely require demographic stratification of validation metrics or post-market monitoring of how a tool performs in the population using it; the De Novo pathway used for a small number of novel tools imposes more, but is rarely the chosen route. The 2025 FDA draft guidance on AI/ML-based Software as a Medical Device (SaMD), with its Predetermined Change Control Plan and Good Machine Learning Practice provisions, begins to address this gap; the institutional architecture for demographic post-market surveillance is still being built. The capability gap is in the regulatory architecture, not the model: clearance is not the same thing as clinically performable deployment. The case is the v1 anchor for the cross-references in the Epic Sepsis (Case 93) and pulse-oximetry (Case 114) deployment-evidence cases.

THE SHIFT

Machine-learning tools are moving rapidly into radiology and other diagnostic medicine, cleared for market and integrated into clinical workflows that affect real patients. The FDA has now authorized more than 1,000 AI/ML-enabled medical devices, the majority through the 510(k) clearance pathway as devices substantially equivalent to a predicate. Unlike a drug, a model can pass its validation study and still behave very differently once it meets a population that differs from its training data — the same model file that scored well on the clearance set can quietly carry a different error profile into every hospital whose patients do not resemble it. The 510(k) substantial-equivalence framing was built for an era of physical devices whose behavior was largely determined by their design; it was not built for a class of systems whose behavior depends on the distribution of the data they meet.[0]

WHAT IS EMERGING

Multiple FDA-cleared radiology tools — chest-X-ray classifiers, mammography aids, CT triage systems — have been documented in the peer-reviewed literature performing worse in deployment than in validation, with the degradation often concentrated in under-represented patient groups. Larrazabal et al. (PNAS 2020) demonstrated structural sensitivity drops for groups under-represented in chest-X-ray training data, using the NIH ChestX-ray14 and CheXpert datasets to show that classifier sensitivity for a given group tracks that group's prevalence in the training set. Seyyed-Kalantari et al. (Nature Medicine 2021) extended the finding directly: across three large public chest-X-ray datasets, classifiers under-diagnosed Black, Hispanic, female, and lower-socioeconomic patients at higher rates, with the disparity present across model architectures — evidence that the shortfall is not a stray bug but a predictable consequence of which patients the training set did and did not contain.[1]

THE CAPABILITY QUESTION

The problem is not confined to imaging. Obermeyer et al. (Science 2019) showed that a widely deployed care-management algorithm covering an estimated 200 million Americans systematically under-allocated resources to Black patients because it was trained on healthcare cost as a proxy label for need: equally sick Black patients had lower historical costs because they had received less care, so the model rated them as lower-need. The question is how a regulator can certify a model as safe without checking how it behaves across the populations and labels it will actually meet — since a model can look well-calibrated on its chosen proxy while the proxy itself encodes the inequity it then propagates. Wachter and Brynjolfsson (JAMA 2023) raised the generative-AI extension of the same question: the proxy-and-population problem deepens under LLM-class tools whose outputs are harder to validate against any well-defined label at all.[2]

EARLY EVIDENCE

Similar deployment-performance degradation has been reported in mammography AI (validation sets often under-represent dense-breast tissue and Black women), sepsis-prediction tools (the Epic Sepsis Model — Case 93 — is the canonical example of an AI tool deployed at scale whose external validation found it substantially worse than its developer-reported numbers), pulse-oximetry behavior (the structural racial miscalibration documented in Sjoding et al. — Case 114 — sits under several downstream AI sepsis and triage tools), and skin-lesion classifiers (training-set under-representation of darker skin tones produces accuracy gaps in deployment). Yet the FDA's 510(k) clearance pathway does not routinely require demographic stratification of validation metrics, nor post-market monitoring of in-use performance — so the divergence between clearance and deployment is largely invisible while the tool is in use, and a shortfall concentrated in one patient group can persist unmeasured across the entire period the tool is influencing care.[3]

OPEN PROBLEMS

The FDA's 2025 draft guidance on AI/ML-Based Software as a Medical Device begins to address this gap. The proposed Predetermined Change Control Plan would let manufacturers pre-specify the model updates and validation procedures that can be applied without a new clearance; the companion Good Machine Learning Practice principles emphasize representative training data, transparent performance metrics, and lifecycle monitoring. The

De Novo pathway, used for a small number of novel tools, imposes more — but is rarely the chosen route. What the guidance does not yet require, as of its draft form, is mandatory demographic stratification of validation metrics at clearance and mandatory population-level post-market surveillance of in-use performance for the cleared model. Radiology AI thus remains the live, recurring case for the gap between regulatory clearance and clinically performable deployment. The capability gap is at the regulatory architecture, not the model: the institutional machinery to require demographic post-market surveillance has not yet been built. It is the medical-AI analog of the Vioxx post-market-surveillance failure (Case 105) at a new technological boundary — a case where the harm comes not from a hidden defect but from the absence of any standing system to watch the tool once it is in the population's hands.[4]

VALIDATION	DEPLOYMENT
AUC 0.9+	drops

the clearance dataset and the deployment population diverge

RADIOLOGY AI — CLEARANCE IS NOT DEPLOYMENT

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3. Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019), "Dissecting racial bias in an algorithm used to manage the health of populations," Science 366(6464):447–453 — proxy-label bias in care-management algorithm covering 200 million Americans.
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> *Performance metrics on a clearance dataset are not the same as performance metrics in deployment populations.*
>
> EDITORS' SYNTHESIS OF THE FDA AI/ML-BASED SAMD DISCUSSION PAPER (2019)

LE INSIGHT

Radiology AI is the canonical contemporary case for the gap between regulatory clearance and clinical deployment performance in medical AI. The clearance dataset and the deployment population diverge. The institutional architecture to surface the divergence — demographic post-market surveillance — has not yet been built.

LENS APPROACH

Radiology AI is the canonical contemporary clinical-AI deployment-and-surveillance case (induced 7.2; LENS D4+D4/PT5). LENS uses it in Domain 4 (Test and Evaluation; CLO-4) for the clearance-vs-deployment measurement architecture and for demographic stratification of validation as a deliverable. LENS uses it in Domain 3 (Human-System Collaboration; CLO-4) for the human-AI deployment-without-surveillance pattern, and in Domain 5 (Navigating Sociotechnical Constraints) for the FDA AI/ML regulatory trajectory itself. The Obermeyer (2019) finding generalizes the diagnosis: bias enters through the labels and through the population, both of which the 510(k) process currently treats as outside its scope. Pair with Vioxx (Case 105) as the post-market-surveillance-failure pattern at a new technological boundary; cross-references the Epic Sepsis (Case 93) and pulse-oximetry (Case 114) cases at the clinical-AI-without-surveillance layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Require validation metrics stratified by demographic group at clearance, so a tool's performance is established on the populations it will actually meet, not an aggregate.
- Specify the intended deployment population and label definition explicitly, and treat divergence from the training distribution as a known risk to be measured, not assumed away.
- Build the data pipeline for in-use performance capture before deployment, so post-market signals are collectable from the first patient rather than reconstructed after harm.

IN OPERATION / AFTER

- Mandate demographic post-market surveillance of in-use performance, the institutional machinery the clearance pathway currently lacks.
- Monitor for the clearance-to-deployment performance gap continuously, since a shortfall concentrated in one group can otherwise persist invisibly.
- Tie continued authorization to demonstrated in-population performance, so a tool that degrades in deployment can be withdrawn before the divergence compounds.

REFLECTION QUESTIONS

1. Identify a model in your domain whose deployment population diverges from its training population. What is the institutional architecture to surface the divergence?
2. Design the demographic post-market surveillance deliverable that should accompany every FDA clearance of medical AI.
3. FDA 510(k) clearance does not currently require demographic stratification of validation metrics, nor does it require post-market monitoring of how a cleared tool actually performs on the population using it. What is the minimum reporting deliverable a regulator should require so the gap is visible while the tool is in use?

WHO BUILDS THIS human factors & interaction design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · knowledge bases · data pipelines · task analysis · design prototyping

CASE 131 GOVERNMENT TECHNOLOGY 2011 – PRESENT

Predictive Policing — PredPol

G Governance & Trust H Human-System Interface D Designed Out

IMPACT Predictive-policing tools deployed by hundreds of U.S. police departments; several cities have abandoned them after equity analyses

IN BRIEF

PredPol and similar predictive-policing tools use historical crime data to forecast where future crime is likely, directing patrols to those locations. Multiple analyses (Lum & Isaac 2016; Richardson, Schultz & Crawford 2019) found that because the historical data records where police have enforced, not where crime has occurred, the algorithm tends to reinforce existing enforcement patterns rather than predict underlying crime — a feedback loop that concentrates policing on already-over-policed neighborhoods. Cities including Santa Cruz, New Orleans, and Los Angeles have since suspended or abandoned predictive-policing deployments after equity reviews. The capability gap is at the construct definition: "where crime occurs" and "where crime is recorded" are different variables, and the tools treated them as one. It is the canonical algorithmic-governance case in U.S. policing.

THE SHIFT

Police departments have adopted data-driven tools that forecast where crime will occur and direct patrol resources accordingly. PredPol and its peers promised to make policing more objective by replacing officer intuition with statistical prediction — relocating a discretionary judgment into an algorithm trained on historical records, and lending the output a veneer of neutrality that the human judgments embedded in those records did not actually possess.[0]

WHAT IS EMERGING

Researchers examining these systems found a structural flaw. Because the training data records where police have made arrests rather than where crime has actually occurred, the model learns enforcement patterns, not crime patterns. Lum & Isaac (2016) showed the result is a feedback loop: patrols are sent where police already went, generating more recorded incidents that confirm the prediction — a loop that grows more confident the longer it runs, because its own output becomes the next cycle's evidence.[1]

THE CAPABILITY QUESTION

The capability gap is at the construct definition. "Where crime occurs" and "where crime is recorded" are not the same variable, yet predictive-policing tools have treated them as interchangeable. The question is whether a model trained on a record of institutional behavior can ever predict the underlying phenomenon, or only amplify the behavior that produced its data — a question no amount of modeling accuracy can answer, because the gap is in what the data measures, not in how well the model fits it.[2]

EARLY EVIDENCE

Richardson, Schultz & Crawford (2019) documented “dirty data” — records produced during periods of biased or unlawful policing — feeding directly into predictive systems. After equity reviews, cities including Santa Cruz, New Orleans, and Los Angeles suspended or abandoned their predictive-policing deployments — abandonment that came only after the tools were already in service, the construct problem surfaced by external review rather than caught before the systems shaped where officers were sent.[3]

OPEN PROBLEMS

Predictive policing is the canonical algorithmic-governance case in U.S. policing and pairs with COMPAS (Case 99) and educational algorithmic bias (Case 90). The open problem is a construct-validity audit — a way to establish, before deployment, whether a predictive system’s training data is a record of ground truth or merely of institutional behavior — implemented in some jurisdictions and absent in most, so the same construct error remains available to the next department that mistakes a record of enforcement for a map of crime.[4]

TRAINING DATA	GROUND TRUTH
past arrests	past crime

the system learns enforcement patterns, not crime patterns

PREDICTIVE POLICING — CONSTRUCT GAP AT LAW-ENFORCEMENT SCALE

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5. Municipal records on suspension and abandonment of predictive policing (Santa Cruz, New Orleans, Los Angeles).

THE LEARNING ENGINEERING LENS

> *Predictive policing systems learn from a record of past policing, not from a record of past crime.*
>
> PARAPHRASING LUM & ISAAC, SIGNIFICANCE, 2016

LE INSIGHT

Predictive policing is the canonical case for the difference between training data and ground truth at law-enforcement scale. The institution being predicted (where crime occurs) is not the institution producing the training data (where police make arrests). The capability gap is at the construct.

LENS APPROACH

LENS uses predictive policing in LEN 7 as a foundational AI- governance case in policing and in LEN 9 for the technical construct-validity analysis. The case is paired with COMPAS as criminal-justice algorithmic cases of different kinds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Establish a construct-validity audit before deployment that tests whether the training data records the phenomenon to be predicted or merely the institution's own behavior.
- Define the prediction target explicitly as "where crime occurs," not "where arrests are recorded," and reject data that cannot speak to the former.
- Engineer against the feedback loop — for example, decoupling patrol allocation from the data the model then re-ingests — before the system can amplify its own output.

IN OPERATION / AFTER

- Monitor for the enforcement feedback loop in deployment, watching whether predicted areas simply accumulate more recorded incidents that confirm the prediction.
- Require periodic equity review of in-use outcomes, since the construct flaw in these systems has surfaced through external analysis rather than internal metrics.
- Track the provenance of incoming training data and quarantine records produced during periods of biased or unlawful enforcement.

REFLECTION QUESTIONS

1. Identify a predictive system in your domain whose training data is itself a record of institutional behavior rather than ground truth. What is the construct gap?
2. Design the construct-validity audit that should precede deployment of a predictive system in any institutional setting.
3. Predictive policing's feedback loop grows more confident the longer it runs, because its own output becomes the next cycle's training data. Where in your domain does a deployed model shape the data it later learns from — and how would you break the loop?

WHO BUILDS THIS governance, ethics & measurement · human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate. TOOLS governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · task analysis · design prototyping

CASE 132 HEALTHCARE TECHNOLOGY 2020 – PRESENT

AlphaFold — Protein Structure Prediction

T Training Gap

IMPACT Substantially solved a 50-year protein-structure prediction problem; 200M+ structures publicly released; foundational positive AI case

IN BRIEF

DeepMind's AlphaFold (2020) and AlphaFold 2 (2021) substantially solved the half-century-old protein-structure prediction problem in computational biology. The publicly released AlphaFold Protein Structure Database now contains predicted structures for more than 200 million proteins — essentially the entire known protein universe — and has been integrated into structural-biology and drug-discovery workflows worldwide. AlphaFold is the strongest positive AI case in this book, and its lesson is in the conditions that made the benefit possible: a benchmark (CASP) the field had agreed on for decades, high-quality training data, an output biologists could verify against experimental structures, and an open release that let the global community adopt it fast. Each was a precondition for the success — and none of them was the model itself.

THE SHIFT

Predicting a protein's three-dimensional structure from its amino-acid sequence was one of biology's grand challenges for half a century — slow, expensive experimental work that bottlenecked drug discovery and basic research. Deep learning offered, for the first time, the prospect of solving it computationally at scale, turning a problem that had been a years-long experimental undertaking per protein into one that could be approached for the whole proteome at once.[0]

WHAT IS EMERGING

DeepMind's AlphaFold (2020) and AlphaFold 2 (2021) substantially solved the problem, predicting structures at accuracies rivaling experiment. The publicly released AlphaFold Protein Structure Database now holds predicted structures for more than 200 million proteins — close to the entire known protein universe — and has been folded into research workflows worldwide, so the benefit arrived not as a single laboratory's advantage but as a shared resource the wider community could build on immediately.[1]

THE CAPABILITY QUESTION

What made AlphaFold a benefit rather than just a benchmark win? The case poses the question of which conditions allow an AI capability to be safely and widely useful — and the answer turns out to lie around the model, not in it: an agreed benchmark, trustworthy training data, a verifiable output, and a deliberate decision about release. Each of these is an institutional or evidentiary precondition, not an artifact of the architecture, which is why the case reads as a lesson about capability infrastructure rather than about a model.[2]

EARLY EVIDENCE

AlphaFold's success rested on four features, each a precondition rather than the model: the CASP benchmark the field had used for decades, high-quality experimental training data, output that biologists could check against known structures, and an open release that let the global community adopt the tool quickly. Where those conditions hold, AI amplifies capability; the technical model alone does not — and the same architecture dropped into a domain missing any one of those four would not have produced a comparable, trusted, widely adopted result.[3]

OPEN PROBLEMS

AlphaFold is the strongest positive AI case in the dataset for what supports beneficial deployment in a well-defined scientific domain. The open problem is generalization: most consequential problems lack an agreed benchmark, clean training data, or verifiable output. The frontier question is how much of the AlphaFold pattern can be reconstructed where those preconditions are not given for free — that is, whether a field can deliberately build the benchmark, the data, and the verification path that protein structure happened to have accumulated over decades.[4]

200M

PREDICTED STRUCTURES · PUBLICLY RELEASED

agreed benchmark, training data, verifiable output, open release

ALPHAFOLD — THE CONDITIONS FOR BENEFICIAL AI DEPLOYMENT

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5. Hassabis (DeepMind) public commentary — the open-release governance decision.

THE LEARNING ENGINEERING LENS

> *This will be one of the most important achievements in AI in the past decade.*
>
> PARAPHRASING JOHN MOULT (ORGANIZER OF THE CASP BENCHMARK) ON ALPHAFOLD2, 2020

LE INSIGHT

AlphaFold is the strongest positive AI case in the dataset. The technical achievement is real. The conditions that made the benefit possible — agreed benchmark, training data, verifiable output, open release — are the capability infrastructure around the model, not the model itself. The case is the strongest available evidence for what supports beneficial AI deployment.

LENS APPROACH

LENS uses AlphaFold in LEN 1 as a problem-framing case for what productive AI deployment looks like, in LEN 9 as a technical achievement, and in LEN 7 for the open-release governance decision that distributed the benefit globally.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Establish an agreed benchmark and high-quality training data for the target problem before building the model, treating these as preconditions rather than afterthoughts.
- Engineer the output to be verifiable against an independent ground truth, so users can check predictions rather than having to trust them.
- Decide the release and access terms deliberately as a governance choice, since open release is what distributed AlphaFold's benefit globally.

IN OPERATION / AFTER

- Monitor downstream use to confirm the verifiable-output property holds in practice, and that users are in fact checking predictions where stakes are high.
- Track whether the four preconditions still hold as the tool is applied to new protein families or adjacent problems beyond its validated domain.
- Sustain the open resource and benchmark over time, so the community-wide benefit does not erode as the field and the data move on.

REFLECTION QUESTIONS

1. Identify a domain in your work where the conditions that supported AlphaFold's success (benchmark, training data, verifiable output) are present. What is the analogous opportunity?
2. The open release of AlphaFold's predictions was a governance decision. Design the equivalent decision for an AI capability your institution might develop.
3. AlphaFold inherited a benchmark, clean data, and a verification path that protein structure had accumulated over decades. Pick a problem in your domain that lacks one of those preconditions, and lay out how a field would deliberately build it.

WHO BUILDS THIS learning science & instructional design — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models

CASE 133 HIGHER EDUCATION ALGO FAIRNESS POLICY ANALYSIS 2024

Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Gándara, Anahideh, Ison, and Picchiarini (AERA Open, 2024) audited predictive models of college student success and showed that models which look acceptable on overall accuracy are systematically less accurate for Black and Hispanic students and overestimate success for white and Asian students — small-tier frontier evidence that the choice of construct and the stratification used in evaluation, not only model-bias mitigation, determine whether an equity-oriented prediction is fair to the groups the equity commitment is meant to protect

IN BRIEF

Predictive models of college student success — models that score students on predicted graduation, retention, or course completion to drive advising, outreach, and support decisions — have become a routine tool across community colleges and four-year institutions. Gándara, Anahideh, Ison, and Picchiarini, publishing in AERA Open (2024), audited such models across racialized groups and found that a model which looks acceptable on overall accuracy is systematically less accurate for Black and Hispanic students — making more prediction errors for them — while overestimating success for white and Asian students. The apparent fairness of the system depends materially on upstream choices: the construct the model is built to predict (predicted graduation vs. predicted benefit vs. predicted need), the stratification used in evaluation (overall accuracy vs. accuracy by income, race/ethnicity, first-generation status), and the decision context the prediction is consumed in (whether end users are trained to contextualize a flagged prediction or treat it as a verdict). The paper's contribution is the frontier-shaped finding that fairness in equity-oriented prediction is a construct-definition and stratified-evaluation problem before it is a model-bias problem. The case pairs explicitly with the v2 race-construct trio (Cases 113 eGFR, 106 pulse oximetry, 107 Hoffman) and with the broader equity-construct competency C8.2: demographic stratification of validation and outcomes as a design commitment.

THE SHIFT

Predictive modeling of student success in higher education is now routine. Community-college and four-year institutions operate models that score students on predicted graduation likelihood, predicted retention, predicted benefit-from-intervention, or predicted financial need. The scores feed downstream decisions: which students get outreach emails, which get advising appointments, which get need-based aid or enrollment incentives. The structural capability question — whose interest does the model maximize, by what construct, and against what stratification — is the layer where the case operates.[0]

WHAT IS EMERGING

Gándara, Anahideh, Ison, and Picchiarini, publishing in AERA Open, audited predictive models of college student success across racialized groups. The central finding is that models which look acceptable on overall accuracy are systematically less accurate for Black and Hispanic students — making more prediction errors for them — while overestimating success for white and Asian students. The methodological move is to show that the choice of construct the model is built to predict, and the way the model is evaluated, have substantial consequences for which students the model treats accurately. A model optimized and validated on aggregate accuracy is not interchangeable with one whose validation is stratified by racialized group; the institution's choice of construct and evaluation regime is a normative decision the model architecture cannot make on its own.[1]

THE CAPABILITY QUESTION

The second methodological move concerns evaluation stratification. Overall accuracy, the standard summary metric, can mask substantial disparity by income, race/ ethnicity, and first-generation status — populations whose base rates and observed outcomes differ from the modal student the training data over-represents. The paper's stratified evaluation shows the canonical equity-construct finding: a model that looks fair under overall accuracy can be substantially less accurate, or substantially biased in its prediction direction, for the subgroups the equity commitment is supposed to protect. The induced framework's C8.2 sub-competency — demographic stratification of validation and outcomes as a design commitment — is the analytic anchor.[2]

EARLY EVIDENCE

The third move concerns the decision context the prediction is consumed in and the mitigations available. A model is not deployed in isolation; its predictions are read by advisors and administrators who decide what to do with a flagged student. Gándara and colleagues stress training end users on the potential for algorithmic bias — so a prediction is contextualized for the individual student rather than treated as a verdict — and evaluate bias-mitigation techniques that reduce, but do not eliminate, the cross-group accuracy gap. The system's apparent fairness therefore depends not only on the model but on the evaluation regime and the decision practice the institution actually deploys.[3]

OPEN PROBLEMS

What the case teaches at the LENS framing layer is the frontier-shaped finding: fairness in equity-oriented prediction is a construct-definition problem before it is a model-bias problem. The induced framework's C8.2 sub- competency and the equity-construct CLOs proposed find their case-grounded basis here. The case explicitly cross-references the v2 race-construct trio — eGFR (Case 113), pulse oximetry (Case 114), and Hoffman pain bias (Case 95) — at the construct-definition layer: in those cases the construct (race correction in eGFR, single-sensor calibration in pulse oximetry, the pain-perception assumption in

Hoffman) was the design decision that produced the disparate outcome; in this case the construct (predicted enrollment vs. predicted need vs. predicted benefit) is the design decision the algorithmic- targeting system has to make explicitly.

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THE LEARNING ENGINEERING LENS

> *Fairness in equity-oriented prediction is a construct-definition problem before it is a model-bias problem. Which student outcome the model is maximizing is the upstream decision.*
>
> EDITORS' SYNTHESIS OF GÁNDARA, ANAHIDEH, ISON, & PICCHIARINI, _AERA OPEN_ (2024).

LE INSIGHT

Gándara's student-success-prediction audit is the small-tier frontier instance of fairness-as-construct-definition. Which construct the model maximizes (predicted graduation vs. benefit vs. need), which stratification is used in validation, and the decision context the prediction is consumed in each determine the fairness properties of the deployed system; the audit found models systematically less accurate for Black and Hispanic students. Cross-references the v2 race-construct trio at the construct-definition layer.

LENS APPROACH

Gándara student-success-prediction fairness is the equity-construct-definition case (induced 8.2; LENS D4/PT5) — Domain 4 for stratified evaluation; Domain 5 for the decision-context and mitigation question. Cross-reference Cases 113, 114, 95 — case-grounded basis for the equity- construct CLOs.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make the construct choice explicit at design time: predicted enrollment, predicted benefit, predicted need are different constructs with different policy implications, and the institution's normative position on which to optimize for has to be on the record.
- Build stratified evaluation by income, race/ethnicity, first-generation status, and other equity-relevant axes into the model's validation; the C8.2 sub-competency operationalized.
- Pair the targeting model with the specific intervention it will feed; the fairness properties of the system depend on the pairing, and evaluating the model in

IN OPERATION / AFTER

- Report which construct the model maximizes, on which stratifications it was evaluated, and which intervention it is paired with — three pieces of information any deployed targeting system should publish together.
- Treat the fairness conversation as a construct-definition conversation first, and as a model-bias conversation second; the latter is operationally important, the former determines what the model is fair **about**.
- Carry the race-construct trio (Cases 113, 114, 95) as the cross-domain pair in the curriculum: in each, the construct definition is the upstream design decision;

isolation underestimates the variation a deployed system actually exhibits.

the framework's C8.2 sub-competency and the equity-construct CLOs are anchored here.

REFLECTION QUESTIONS

1. Identify a predictive-targeting model in your domain. Which construct does it maximize — observed outcome, predicted benefit, predicted need — and is the choice on the record as a normative institutional decision, or absorbed into the model's training objective without disclosure?

2. Specify the stratifications you would build into the model's validation: which equity-relevant axes (income, race/ethnicity, first-generation, geography, disability) and what comparison structure (per-axis accuracy, per-axis calibration, per-axis intervention effectiveness)?

3. The case cross-references the v2 race-construct trio (Cases 113, 114, 95). In each, the construct definition is the upstream design decision that produced the disparate outcome. Identify a construct decision in your domain that is currently absorbed into the design rather than on the record — and what would it take to make the choice explicit?

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CASE 134 LEARNING ANALYTICS ALGO FAIRNESS ED-TECH RESEARCH 2021 – 2024

Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Yu, Lee, and Kizilcec, publishing in the LAK/EDM literature, examined whether and how protected attributes (race/ethnicity, gender, socioeconomic status) should be included in learning-analytics predictive models, and showed that whether including or excluding the attribute produces fairer outcomes depends on the construct, the model class, the downstream intervention, and the population — small-tier frontier evidence that the include-or-exclude question is the wrong framing

IN BRIEF

A long-running practical question in learning analytics is whether protected attributes — race/ethnicity, gender, socioeconomic status, first-generation status — should be included as features in predictive models. The intuitive "fairness through unawareness" answer is to exclude them. The technical-fairness literature has shown the unawareness answer is incomplete: omitted protected attributes are typically reconstructable from correlated features (zip code, course history, prior achievement), so excluding the attribute does not exclude its predictive footprint and can make discrimination harder to detect and audit. Yu, Lee, and Kizilcec, publishing in the LAK/EDM literature, examined the include-or-exclude question empirically across multiple learning-analytics prediction tasks. The headline finding is the frontier-shaped one: whether inclusion or exclusion produces fairer outcomes depends on the construct being predicted, the model class, the intervention the prediction feeds, and the population. The case is the small-tier frontier instance of "surfacing bias through governance, not just technique" (C8.4 in the induced framework). It cross-references the v2 race-construct trio (Cases 113 eGFR, 106 pulse oximetry, 107 Hoffman) at the model-fairness layer: in those cases the construct was the design decision; in this case the question is whether the attribute is allowed into the model that operationalizes the construct.

THE SHIFT

A practical question that recurs across learning-analytics deployments is whether protected attributes — race/ethnicity, gender, socioeconomic status, first-generation status, and similar — should be included as features in predictive models. The intuitive policy answer ("fairness through unawareness") is to exclude them: a model that does not see race cannot discriminate on race. The intuition is operationally appealing and statutorily aligned in some domains where regulators have read the omission requirement strictly.[0]

WHAT IS EMERGING

The technical-fairness literature has shown the unawareness answer is incomplete. Protected attributes that are omitted from a model are typically reconstructable from correlated features the model does see — zip code, school assignment, prior achievement, course history, language at home, family income proxies. Omitting the attribute removes the label, not the predictive footprint, and can in some cases make discrimination harder to detect and audit precisely because the auditable record no longer carries the attribute that would let an evaluator stratify the model's output.[1]

THE CAPABILITY QUESTION

Yu, Lee, and Kizilcec, publishing in the LAK/EDM (Learning Analytics and Knowledge / Educational Data Mining) literature, examined the include-or-exclude question empirically across multiple learning-analytics prediction tasks — course completion, performance prediction, dropout risk — with multiple model classes (regression, tree-based, neural) and multiple downstream intervention contexts. The headline finding is the frontier-shaped one: whether including or excluding a protected attribute produces fairer outcomes depends on the construct being predicted, the model class, the intervention the prediction feeds, and the population. There is no general answer; the include-or- exclude question is the wrong framing.[2]

EARLY EVIDENCE

The right framing the paper develops is governance and audit. The decision to include or exclude a protected attribute is one of several decisions a learning-analytics deployment makes that determine its fairness properties; the decision has to be made deliberately, recorded with reasoning, and paired with stratified evaluation and audit cadence that catches the consequences of the decision in operation. The case is the induced framework's C8.4 instance at small scale: "surfacing bias through governance, not just technique." The technical-fairness machinery alone does not answer the question; the governance architecture around the model is the carrier of the answer in any specific deployment.[3]

OPEN PROBLEMS

In pair with Case 133 (Gándara on community-college targeting) and with the v2 race-construct trio (Cases 113 eGFR, 106 pulse oximetry, 107 Hoffman), the case completes the v2 equity-construct frontier picture. Case 133 names the construct-definition layer; this case names the protected-attribute-in-the-model layer; the race-construct trio names the construct-encoded-in-the-instrument layer (race correction in eGFR, sensor calibration in pulse oximetry, pain perception in Hoffman). Together, the five cases stage the equity-construct competency across the construct-definition / attribute-handling / instrument-construct axes — the case-grounded basis for the equity-construct sub-competencies proposed in the v2 research backbone.[4]

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THE LEARNING ENGINEERING LENS

> *The include-or-exclude question is the wrong framing. The right framing is governance: a deliberate decision, recorded with reasoning, paired with stratified evaluation and audit cadence.*
>
> EDITORS' SYNTHESIS OF YU, LEE, & KIZILCEC (LAK/EDM).

LE INSIGHT

Yu, Lee, and Kizilcec's protected-attributes work is the frontier instance of surfacing bias through governance, not just technique. Whether including or excluding a protected attribute produces fairer outcomes depends on the construct, the model class, the intervention, and the population. The governance architecture around the model is the carrier of the answer in any specific deployment.

LENS APPROACH

Yu/Lee/Kizilcec protected attributes is the model-fairness- governance case (induced 8.4; LENS D4/PT5) — Domain 4 for stratified-evaluation-by-attribute; Domain 5 for the decision-plus-reasoning-plus-audit architecture. Cross- reference Case 133 and the race-construct trio (105, 106, 107).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the include-or-exclude decision for each protected attribute as a deliberate design choice, recorded with reasoning, rather than absorbed into the data-engineering pipeline.
- Build stratified evaluation by the protected attribute regardless of whether the model itself uses it as a feature; auditing the model's output by attribute is independent of whether the attribute is an input.
- Pair the decision with the downstream intervention context; the include-or-exclude answer that produces fairer outcomes in one intervention context may produce less fair outcomes in another.

IN OPERATION / AFTER

- Publish the protected-attribute handling decisions for any deployed learning-analytics model — included, excluded, and with what reasoning — as part of the model's governance documentation.
- Operate stratified audit on a regular cadence; the include-or-exclude consequences in operation are what the audit catches, and audit absence makes the decision functionally invisible to the institution.
- Carry the five-case equity-construct set into the curriculum: Cases 113 (eGFR), 106 (pulse oximetry), 107 (Hoffman) on the instrument-construct layer; Case 133 (Gándara) on the construct-definition layer; this case on the protected-attribute-in-the-model layer. The set is the case-grounded basis for the equity-construct sub-competencies.

REFLECTION QUESTIONS

1. Identify a learning-analytics or analogous predictive model in your domain. Which protected attributes are inputs to the model, which are not, and is the include-or-exclude decision on the record with reasoning, or absorbed into the data-engineering pipeline?

2. Specify the stratified-audit cadence you would operate for the model regardless of whether protected attributes are inputs; the audit catches the consequences of the include-or-exclude decision in operation, and its absence makes the decision functionally invisible.

3. The case sits in a five-case equity-construct set with Case 133 (Gándara) on construct definition and the v2 race-construct trio (Cases 113, 114, 95) on instrument construct. Identify a deployment in your domain that sits across all three layers — construct definition, attribute handling, instrument construct — and which layer's decisions are currently most invisible.

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CASE 135 AI IN EDUCATION INTELLIGENT TUTORING ALGO FAIRNESS 2025

LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Repeated probing of LiveHint AI (an LLM-based tutor under development at Carnegie Learning) with identity-marked student queries surfaced response differences in tone, detail, and pedagogical appropriateness across identities; choice of foundation model materially affected the level of differentiation

IN BRIEF

The AIED 2025 paper “Evaluating an AI Tutor for Bias Across Different Foundation Models” tests LiveHint AI, an LLM-based tutor under development at Carnegie Learning, against a structured probing protocol. Realistic student queries are modified to include explicit or implicit statements of identity — nationality, dialect markers, demographic cues — and the tutor’s responses are assessed for tone, detail, and pedagogical appropriateness. The choice of foundation model materially affects the level of differentiation in responses. The authors are explicit that this is not a deployment-bias audit (LiveHint is in development); it is a methods- development paper proposing how foundation-model-level fairness evaluation should be done before deployment. The case extends the race-construct trio (Cases 113, 114, 95, 97) into the LLM-tutoring layer where the structurally new variable is the foundation model. Open questions: whether lab-style probing matches deployed-conversation patterns; whether vendor selection across foundation models becomes a routine fairness deliverable.

THE SHIFT

The deployment surface for LLM-based tutoring expanded faster than the fairness-evaluation literature for it. Most pre-deployment fairness work on educational AI was built for classifiers — proctoring systems, recommendation engines, automated essay scoring — where the output is a score or a flag and the fairness question can be formulated as a disparate-impact comparison across groups. LLM tutoring shifts the output to a conversational response, and the fairness question shifts with it: it is no longer whether two equivalently-prepared students receive equivalent flags, but whether they receive equivalent pedagogical engagement.[0]

WHAT IS EMERGING

The AIED 2025 paper builds the evaluation method against that shifted target. LiveHint AI, an LLM-based tutor under development at Carnegie Learning, is the subject system. The probing protocol generates realistic student queries — calibrated to the kinds of mathematics questions a tutor would receive — and produces matched variants of each query that differ only in identity-marker content: explicit statements of nationality, dialect markers, demographic cues. The tutor's responses to the matched pairs are then assessed across three dimensions: tone (is the response respectful and appropriately framed for a tutoring context), detail (does the response give the same quality of pedagogical content), and pedagogical appropriateness (does the response engage the underlying mathematics question with the same instructional intent).[1]

THE CAPABILITY QUESTION

The headline finding is that the choice of foundation model materially affects the level of differentiation in responses. Different foundation models — the underlying LLMs the tutoring layer wraps — produce different patterns of identity-conditioned response variation. The variation is not uniform across foundation models, not uniform across the three response dimensions, and not uniform across identity-marker types. The vendor-selection decision — which foundation model the tutoring product is built on — is itself a fairness-relevant design choice, and the case names it as such. The structurally new variable in the case, by comparison with the race- construct trio in the corpus, is the foundation-model layer, which did not exist as a deployment-side decision when the proctoring-bias and pulse-oximetry cases were studied.[2]

EARLY EVIDENCE

The authors' framing is explicit and binding on the case. LiveHint AI is in development, not deployment; this is a methods-development paper proposing how foundation-model- level fairness evaluation should be done before deployment. The case is not the deployment-bias-audit case the corpus carries at Cases 113 (Hoffman pain assessment), 106 (pulse oximetry), 107 (eGFR), and 156 (Johnson school surveillance). It is the structurally new methods-development case at the layer above those deployments — the foundation-model layer — and it grounds the curriculum's demographic-stratification anchor at that layer.[3]

OPEN PROBLEMS

The open questions the authors preserve are the case's load-bearing hedges. Whether the differentiation patterns documented in lab-style probing match what students encounter in deployed conversations — where session length, follow-up turns, and student adaptation affect the response trajectory — is a question the present study cannot answer. Whether vendor selection across foundation models becomes a routine fairness deliverable for the educational-AI procurement pipeline is a market-evolution question the present study can name but cannot resolve. The case pairs with the race-construct trio for the stratified-validation discipline and with Case 77 (hybrid human-AI tutoring) for the deployment-side complement — the augmentation pattern Case 77 documents depends on the foundation-model-level evaluation Case 135 is methodologically grounding.

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THE LEARNING ENGINEERING LENS

> *The choice of foundation model is itself a fairness-relevant design decision. The evaluation should be a pre-deployment deliverable, not a post-deployment audit.*
>
> EDITORS' SYNTHESIS OF THE AIED 2025 LIVEHINT AI BIAS EVALUATION.

LE INSIGHT

LiveHint AI is the methods-development case at the foundation- model layer of LLM-tutoring fairness evaluation. The matched- pair identity-probing protocol surfaces tone, detail, and pedagogical-appropriateness differences across identities; the foundation-model choice materially affects the differentiation level. The case extends the race-construct trio into the LLM- tutoring layer with a structurally new variable — the foundation model — that did not exist at the deployment-audit layer.

LENS APPROACH

LiveHint AI is the demographic-stratified validation case at the foundation-model layer (induced 8.2; LENS D4/PT6). LENS uses it in Domain 4 (Test and Evaluation) for the matched- pair probing methodology and in Domain 3 (Human-System Collaboration) for the foundation-model-selection-as- fairness-decision frame. Pair with the race-construct trio (Cases 113, 114, 95, 97) at the deployment-audit layer and with Case 77 (hybrid human-AI tutoring) as the augmentation-pattern complement.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build the probing protocol to vary identity markers within matched query pairs; the methods-development contribution depends on the matched-pair design that isolates the identity-conditioned response variation.
- Treat the choice of foundation model as a fairness-relevant design decision, not as an upstream procurement decision; the case demonstrates that the foundation-model layer materially affects the tutoring-layer's identity-conditioned response patterns.
- Conduct the fairness evaluation before deployment, not after; the methods-development framing is that the evaluation should be a pre-deployment deliverable, parallel to the race-construct-trio cases at the deployment-audit layer.

IN OPERATION / AFTER

- Commission the deployed-conversation evaluation that the lab-style probing cannot perform; the open question on whether probing patterns match deployment patterns is testable against deployment logs as the system moves toward release.
- Publish the foundation-model-level fairness findings as part of the procurement record; the case argues that vendor selection across foundation models is a fairness deliverable, and the publication discipline is what would make that deliverable operational.
- Pair the case in the curriculum with the race-construct trio (Cases 113, 114, 95, 97) so the demographic-stratification anchor is taught across both the deployed-system layer and the foundation-model layer above it.

REFLECTION QUESTIONS

1. Identify an LLM-based system in your domain whose foundation-model choice was treated as an upstream procurement decision rather than as a fairness-relevant design decision. What would the matched-pair probing protocol look like for that system, and what response dimensions would you assess?

2. Specify the pre-deployment fairness deliverable you would build into the procurement record for an LLM-based product in your domain. What would the published artifact contain — probe set, response-dimension assessment, foundation-model comparison — and what would the procurement decision turn on?

3. The case's open question is whether lab-style probing matches deployed-conversation patterns. What deployment-log analysis would convert the methods-development evidence into deployment-audit evidence, and what privacy and consent architecture does that analysis require?

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CASE 136 HEALTHCARE MEDICAL IMAGING MACHINE LEARNING 2020

DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT McKinney et al. 2020 Nature paper reported a deep-learning mammography screening system outperforming radiologists on retrospective UK and U.S. screening datasets, with reductions in false-positives (5.7 percentage points in the U.S. set, 1.2 in the UK set) and false-negatives (9.4 and 2.7 percentage points respectively); Haibe-Kains et al. October 2020 Nature comment critiqued the paper for failing to release code and trained models, arguing that reproducibility could not be assessed and that screening-comparison claims required deployment-grade evidence

IN BRIEF

Scott Mayer McKinney and colleagues at Google Health and DeepMind published "International evaluation of an AI system for breast cancer screening" in Nature on January 1, 2020. The paper reported that a deep-learning system outperformed radiologists on retrospective UK and U.S. screening datasets, with reductions in false-positives of 5.7 percentage points (U.S.) and 1.2 percentage points (UK) and reductions in false-negatives of 9.4 and 2.7 percentage points respectively. The paper drew unusual press attention and rapidly entered the policy conversation on AI-assisted screening. Haibe-Kains and colleagues' October 14, 2020 Nature comment titled "Transparency and reproducibility in artificial intelligence" critiqued the McKinney paper for failing to release code, trained models, or sufficient methodological detail to permit independent reproduction. The load-bearing hedge the Haibe-Kains comment delivers is that a large fraction of the methodology was not reproducible, and the screening-comparison framing the original paper offered has been refined by subsequent deployment evidence rather than confirmed at the deployment scale the headline implied. The case pairs with Case 130 (Radiology AI Miscalibration), Case 93 (Epic Sepsis), and Case 114 (Pulse oximetry).

THE SHIFT

The McKinney et al. paper was published on January 1, 2020, in Nature — a top-tier venue and an unusually high- profile publication for a deep-learning medical-imaging study. The work was a collaboration across Google Health, DeepMind, and clinical partners at Cancer Research UK Imperial Centre, Northwestern University, the Royal Surrey County Hospital, and the National Cancer Institute. The retrospective evaluation used UK and U.S. screening datasets and compared the AI system's outputs against single-reader and double-reader radiologist performance. The headline framing was that the AI system reduced both false-positives and false-negatives relative to radiologists, with the U.S. dataset showing larger absolute reductions than the UK dataset.[0]

WHAT IS EMERGING

The press response was substantial. Mainstream coverage framed the result as "AI outperforms radiologists at breast cancer screening," and the framing entered the policy conversation on AI-assisted medical imaging quickly. The framing carried more weight than the underlying retrospective comparison was designed to support: a retrospective evaluation against historical reader performance is informative about model output, but prospective deployment against current radiologists operating in their current workflow involves variables — reader fatigue, screen presentation, integration with reading worklists, recall thresholds — that the retrospective study does not measure. The McKinney paper itself was careful in its claims; the gap between the paper's careful claims and the press's framing of the headline is part of the case.[1]

THE CAPABILITY QUESTION

The October 14, 2020 Nature comment by Benjamin Haibe- Kains, George Adam, Ahmed Hosny, Farnoosh Khodakarami, Massive Analysis Quality Control (MAQC) Society Board of Directors, Levi Waldron, Bo Wang, Chris McIntosh, Anna Goldenberg, Anshul Kundaje, Casey S. Greene, Tamara Broderick, Michael M. Hoffman, Jeffrey T. Leek, Keegan Korthauer, Wolfgang Huber, Alvis Brazma, Joelle Pineau, Robert Tibshirani, Trevor Hastie, John P. A. Ioannidis, John Quackenbush, and Hugo J. W. L. Aerts is the load- bearing reproducibility critique. The comment argued that the McKinney paper had not released code, had not released trained models, and had not provided sufficient methodological detail to permit independent reproduction. The comment was specific: a large fraction of the methodology was not reproducible from the published paper, and the screening-comparison claim could not be independently validated. The comment did not allege error in the paper; it argued that reproducibility had not been established.[2]

EARLY EVIDENCE

The case pairs with Case 130 (Radiology AI Miscalibration) for the medical-imaging-AI-deployment-evidence thread: retrospective evaluation produces one class of evidence; prospective deployment produces another, and the two are not interchangeable. Pair with Case 93 (Epic Sepsis) for the high-profile-result-versus-deployment-evidence thread in healthcare AI; Epic Sepsis is the load-bearing case in the corpus for the gap between vendor or developer claims and external evaluation, and DeepMind Mammography sits in the same conceptual family at a different domain. Pair with Case 114 (Pulse oximetry) for the population- heterogeneity-in-medical-AI thread; the McKinney paper's UK-versus-U.S. effect-size difference (5.7 vs 1.2 pp on false-positives) is itself evidence that the system's performance varies across screening populations, and the variation has implications for deployment.[3]

OPEN PROBLEMS

The honest hedges the case carries are load-bearing. The Haibe-Kains comment is not a finding that the McKinney paper was wrong; it is a finding that the paper as published did not establish reproducibility. The subsequent five years of deployment evidence on AI-assisted breast cancer screening have refined the screening- comparison framing — prospective evaluations have shown benefits in some settings and not in others, and the operational variables the retrospective comparison did not measure have proved load-bearing in deployment. The case teaches the verification-as-deployment-event pattern: a high-profile retrospective result is the starting point of a verification arc, not its endpoint, and the reproducibility infrastructure the Haibe-Kains comment named is the condition for the arc to be possible.

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THE LEARNING ENGINEERING LENS

> *A large fraction of the methodology was not reproducible from the published paper, and the screening-comparison framing has been refined by subsequent deployment evidence rather than confirmed at the deployment scale the headline implied.*
>
> EDITORS' SYNTHESIS OF THE MCKINNEY ET AL. (2020) AND HAIBE-KAINS ET AL. (2020) NATURE EXCHANGE.

LE INSIGHT

DeepMind Mammography is the verification-as-deployment-event case at the high-profile-publication scale. The McKinney paper's retrospective result was the starting point of a verification arc, not its endpoint; the Haibe-Kains comment named the reproducibility infrastructure as the condition for the arc, and subsequent deployment evidence has refined the screening-comparison framing the original paper offered.

LENS APPROACH

DeepMind Mammography is the deployment-and-reuse-as- verification-events case at the high-profile-publication seam (induced 7.2; LENS D4+D4/PT6; CLO-4 and CLO-3). LENS uses it in Domain 4 (Test and Evaluation) for the reproducibility-infrastructure-as-verification-condition discipline and in Domain 3 (Human-System Collaboration) for the retrospective-versus-prospective-evidence distinction. Pair with Case 130 (Radiology AI Miscalibration), Case 93 (Epic Sepsis), and Case 114 (pulse oximetry population heterogeneity). The Haibe-Kains comment is a reproducibility finding, not a finding of error; the distinction is the load-bearing hedge.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Release code, trained models, and sufficient methodological detail to permit independent reproduction as a condition of publishing a high-profile retrospective medical-AI result; the Haibe-Kains comment names the reproducibility

IN OPERATION / AFTER

- Carry the Haibe-Kains comment's specific framing into print without softening; the comment is a finding on reproducibility, not a finding of error, and the case's pedagogical value depends on the distinction being preserved.

infrastructure as the condition for the verification arc that the original paper opens.

- Specify in advance the deployment variables — reader fatigue, recall threshold, worklist integration, screening-population characteristics — that a retrospective comparison does not measure but that a deployment will encounter.
- Treat the gap between the paper's careful claims and the press's framing as a deployment surface, not a communications problem; the framing the field receives is the framing the deployment will operate under in the policy conversation.
- Pair in syllabi with Case 130 and Case 93 so the high-profile-result-versus-deployment-evidence pattern is taught across the medical-imaging-AI deployment seam at multiple instances.
- Use the case to anchor the verification-as-deployment-event frame; the curricular target is the discipline of treating a high-profile retrospective result as the starting point of a multi-year verification arc rather than as a deployment-ready endpoint.

REFLECTION QUESTIONS

1. Identify a high-profile retrospective result in your domain whose reproducibility infrastructure — code, trained models, methodological detail — has not been released. What would the verification arc the result opens require to proceed, and what currently blocks it?

2. Specify the deployment variables a retrospective comparison in your setting does not measure but a deployment will encounter. What is the prospective evaluation design that would surface those variables before deployment scale?

3. The press framing of a result often carries deployment implications the paper's careful claims do not. Pick a result in your domain and ask: what is the gap between the careful claim and the framing the field receives, and what would have to be true for the gap to be narrowed in advance of deployment decisions?

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Chapter 9

Sociotechnical Constraints — What Fails

When the system the deployment must survive in is not the system the deployment was designed for.

Good designs that survive in the real world are designed for the real world.

CLO-5, CONTEXT AND DOMAIN FLUENCY.

CASE 137 ENERGY INDUSTRIAL 2005

Texas City BP Refinery Explosion

N Normalization of Deviance T Training Gap K Knowledge & Institutional Memory G Governance & Trust

IMPACT 15 killed, 180 injured at BP's Texas City refinery; CSB found systemic safety-culture decay

IN BRIEF

On 23 March 2005 a startup at BP's Texas City refinery overfilled a distillation tower — operators were working from instruments that had malfunctioned for years — and venting hydrocarbon vapor found an ignition source and exploded, killing 15 workers in trailers parked beside the unit and injuring 180. The Chemical Safety Board found accumulated safety-culture decay, deferred maintenance, and a celebrated cost-cutting program. Its central finding reshaped U.S. industrial safety: BP's **personal**-safety metrics were among the best in the industry, while its **process**-safety state — the integrity of the hazardous process itself — had been drifting, unmeasured. Texas City is the canonical evidence that the wrong measurement, reported as a win, is worse than no measurement at all.

BACKGROUND

BP's Texas City refinery, one of the largest in the U.S., had absorbed years of deferred maintenance and corporate cost-cutting, with instruments and alarms tolerated in a degraded state. On the surface it looked safe: its personal-injury rates were among the best in the industry.[0] The strong injury numbers actively reassured the organization, because the metric the company trusted most was the one that carried no information about the degrading instruments and alarms on which a safe startup actually depended.

WHAT HAPPENED

On 23 March 2005, during a unit startup, operators overfilled a distillation tower far past its safe level, working from a level indicator that had malfunctioned for years and alarms that did not sound. Hydrocarbon vapor vented, drifted across the site, found an ignition source — an idling truck — and exploded. Fifteen workers in temporary trailers parked beside the unit were killed and about 180 injured.[1] Every contributor had been tolerated as routine for years — the broken indicator, the silent alarms, the trailers parked beside a hazardous unit — so the startup was run blind to a danger the site had long since stopped seeing.

THE INVESTIGATION

The U.S. Chemical Safety Board's investigation became the most influential in the agency's history.[2] It found accumulated safety-culture decay, deferred maintenance, broken instruments tolerated as routine, trailers sited dangerously close to a hazardous unit, and a cost-cutting program — branded internally as "1000 Day" and "Forward" — celebrated as a success while it consumed the process-safety margin. The CSB drew the distinction that would reshape the field: “indicators of personal safety are not indicators of process safety.”[3] The Board’s force came from naming the deeper error as one of measurement: the company had not failed to measure but had measured the wrong dimension and then trusted the reassuring number it produced. BP’s recordable personal-injury rate was excellent at the moment of the disaster, the front-page metric on which management-incentive plans paid out; the process-safety state of the unit that killed fifteen workers had no comparable reporting line at all.

THE CAPABILITY GAP

Texas City is the foundational evidence that an organization can measure the wrong thing and report excellent performance while its real capability gap widens. Personal-safety metrics — slips, trips, recordable injuries — carried no information about the integrity of the hazardous process, so the signal regime was blind in the very dimension that killed people. The wrong measurement, trusted, is worse than none.[4] A blank dashboard at least invites suspicion; a confident green reading on the wrong axis manufactures false assurance, which is why the excellent injury record made the process-safety drift harder to see rather than easier.

AFTERMATH & REFORM

The Baker Panel — chaired by former Secretary of State James Baker, commissioned by BP at the CSB’s recommendation — reported in 2007 that the failure was a corporate one, not a Texas City one: a safety-culture decay extending across BP’s five U.S. refineries, driven by cost pressure and management-of-change failures that the personal-safety metric system was structurally incapable of surfacing. BP invested heavily in process safety afterward, and the process-safety/personal-safety distinction — developed in the CCPS literature and codified in OSHA’s 1992 process-safety-management standard — moved into mainstream U.S. industrial regulation after 2005, with API Recommended Practice 754 (2010) establishing process-safety leading and lagging indicators as an industry standard. The case’s lasting contribution is a measurement lesson: count the thing that can kill you, not the thing that is easy to count.[5] That the distinction had been available in the CCPS literature and the OSHA standard before the explosion underscores the point: the knowledge existed, but the refinery’s reporting had not been built to carry it upward where the hazard actually lived.

TEXAS CITY — THE WRONG MEASUREMENT, REPORTED AS A WIN

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3. CSB (2007) — "indicators of personal safety are not indicators of process safety" (quoted).
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5. A. Hopkins, Failure to Learn: The BP Texas City Refinery Disaster (CCH, 2008).
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THE LEARNING ENGINEERING LENS

> *Indicators of personal safety are not indicators of process safety.*
>
> U.S. CHEMICAL SAFETY BOARD, BP TEXAS CITY FINAL INVESTIGATION REPORT, 2007

LE INSIGHT

Texas City is the foundational evidence that organizations can be measuring the wrong thing and reporting excellent performance while their actual capability gap widens. Personal-safety metrics had no information about process-safety state. The signal regime was blind in the dimension that killed people.

LENS APPROACH

Texas City is the canonical "measuring the wrong failure mode" case (induced 2.1; LENS D4/PT5), with cue/alert design (induced 3.1) and change-control (induced 5.4) as alternate anchors. LENS uses it in LEN 4 as the wrong-measurement case and in LEN 7 for the governance failure that allowed years of cost-cutting to be reported as wins. Studio projects design the process-safety measurement deliverable BP's executives should have been receiving in 2003. Adjacent to Wells Fargo (Case 91) at the measurement-system-blind-to-the-real-failure-mode layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Instrument process-safety state directly — barrier integrity, alarm health, instrument validity — rather than inferring safety from personal-injury rates.
- Treat degraded instruments and silent alarms as startup-blocking conditions, not routine items to defer past the next run.
- Set facility-siting rules that keep occupied trailers away from hazardous units as a design constraint, not a tolerated exception.

IN OPERATION / AFTER

- Audit whether the headline metric actually carries information about the hazard that can kill, and retire reassuring numbers that do not.
- Track deferred maintenance and tolerated-defect counts as process-safety leading indicators reported to executives.
- Verify that the process-vs-personal-safety distinction is wired into the reporting chain so it reaches the layer that funds maintenance.

REFLECTION QUESTIONS

1. Identify a "personal safety vs. process safety" equivalent in your domain. What capability gap is invisible to the metric your institution currently reports?
2. Design the process-safety dashboard that BP Texas City's executives should have been receiving in 2003.
3. A confident green reading on the wrong axis manufactured false assurance. What headline metric in your organization might be reassuring precisely because it measures the wrong dimension?

WHO BUILDS THIS safety science & organizational analysis · learning science & instructional design · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate. TOOLS incident analysis · safety dashboards · scenario simulation · competency models · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

CASE 138 DEFENSE 1941 – 1943

Mark 14 Torpedo Failures

T Training Gap K Knowledge & Institutional Memory G Governance & Trust

IMPACT Persistent torpedo failures across the first ~20 months of the Pacific War; resolved only after fleet-level testing forced acknowledgment of the defects

IN BRIEF

The U.S. Navy entered World War II with the Mark 14 torpedo, so expensive that the Bureau of Ordnance had effectively forbidden live testing in peacetime. Through 1942 submarine crews reported torpedoes running deep, failing to detonate, or exploding prematurely; the Bureau insisted the weapon was sound and blamed the operators. It took eighteen months and the personal intervention of Admiral Charles Lockwood — who ordered fleet-level live-fire tests — to confirm three separate defects: the torpedo ran about ten feet too deep, its magnetic exploder fired erratically, and its contact pin crushed on a square hit. The fixes were simple once the defects were acknowledged. The binding constraint was institutional: a bureau insulated from the operators who knew the weapon did not work.

BACKGROUND

The Mark 14 was the U.S. Navy's standard submarine torpedo at the start of the Pacific War. It had been so expensive to test that the Bureau of Ordnance had effectively forbidden live trials in the 1930s; the weapon went to war essentially unproven against realistic conditions, so the very decision meant to conserve a scarce and costly weapon guaranteed that its defects would first be discovered in combat, by the crews who could least afford them to surface there.[0]

WHAT HAPPENED

From early 1942, submarine crews reported a litany of failures: torpedoes that ran far below their set depth, magnetic exploders that detonated prematurely or not at all, and contact firing pins that crushed on a direct hit. The Bureau of Ordnance insisted the weapon worked and attributed the misses to crew error; captains who reported failures risked their careers — an arrangement that turned every field report into a self-accusation and so suppressed the very evidence that would have isolated the defects the Bureau refused to acknowledge.[1]

THE INVESTIGATION

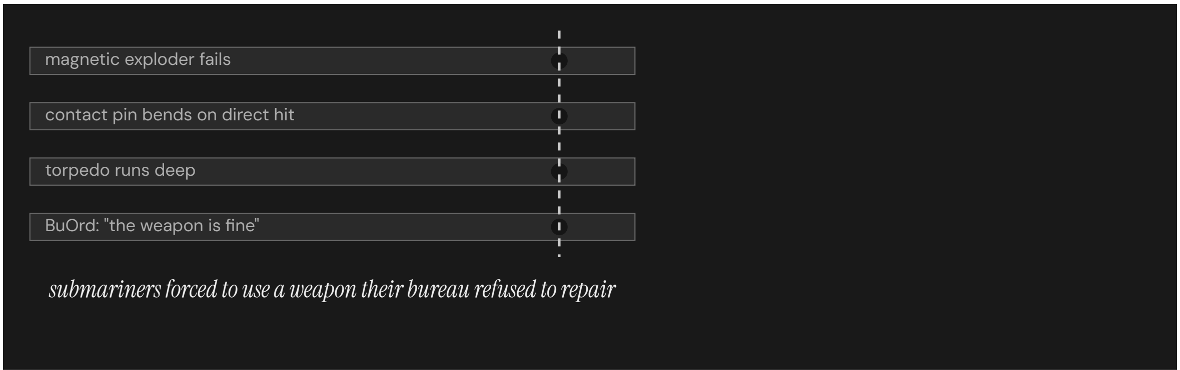
It took eighteen months and the intervention of Admiral Charles Lockwood, commander of the Pacific submarine force, who ordered fleet-level testing. A live-fire trial — and the USS Tinosa's July 1943 attack on the Tonan Maru, in which eleven torpedoes struck the stopped ship squarely and failed to detonate — forced the issue. The tests confirmed the torpedo ran about ten feet too deep, that the Mark 6 magnetic exploder failed routinely, and that the contact pin buckled on perpendicular impact — three independent defects that had been masking one another at sea, which is why a fleet commander's controlled trial, not another combat patrol, was finally able to separate and prove them.[2]

THE CAPABILITY GAP

The defects were real and separate, but the binding constraint was institutional. The capability that was missing was not at the front; it was a channel by which the bureau that owned the weapon could be made to hear what the boats already knew. The fixes — recalibrating depth, deactivating the magnetic exploder, redesigning the contact pin — were small. The refusal to believe the operators was the expensive part, measured not in engineering hours but in the months of patrols and the targets that escaped while a bureau defended a verdict the boats had already disproven.[3]

AFTERMATH & REFORM

By late 1943 the three defects were corrected and the Mark 14 became an effective weapon for the rest of the war. The episode entered U.S. Navy institutional history as the canonical case of a procurement bureau insulated from operator feedback, and is cited in modern organizational-learning literature on the cost of suppressing field reports — a reminder that the structure that decides whose evidence counts is itself a capability, and that its absence can cost more than any single technical defect it conceals.[4]



MARK 14 — MULTIPLE DEFECTS, SINGLE INSTITUTIONAL REFUSAL

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> *The Bureau certified the weapon; field reports of failure were treated as evidence of operator error.*
>
> PARAPHRASING BLAIR, SILENT VICTORY, 1975

LE INSIGHT

The Mark 14 is the canonical Navy case for the institutional refusal to believe operator feedback. The capability that was missing was not at the front. It was at the bureau that owned the weapon. The cost of the refusal was paid by the submariners forced to use it until the bureau yielded. The eventual fixes — resetting depth calibration, replacing the magnetic exploder, redesigning the contact pin — were technical and small. The reform was institutional and slow. The capability that had to be built was the channel by which the bureau could be made to hear what the boats already knew.

LENS APPROACH

LENS uses the Mark 14 in LEN 7 as a governance failure case and in LEN 8 as an organizational-learning case in which the operators had the diagnosis and the institution lacked the structure to receive it. Studio projects (LEN 10) examine what the equivalent operator-to-institution feedback channel should look like.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Require realistic live testing before fielding, even for scarce and costly items, so defects surface in trials rather than in combat.
- Build an operator-feedback channel into the weapon's ownership from the start, with a path that does not put the reporter's career at risk.
- Separate the authority that certifies a system from the authority that investigates its field failures, so a bureau cannot judge its own product.

IN OPERATION / AFTER

- Audit field reports for suppressed or career-risking signals, and treat a pattern of "operator error" verdicts as a warning to test the system, not the crew.
- Empower an operational commander to order controlled trials when field reports conflict with the owner's certification.
- Sustain the feedback channel so multiple masking defects can be separated and proven before the cost compounds.

REFLECTION QUESTIONS

1. Identify a system in your domain whose owners are institutionally insulated from the operators using it. What feedback would they refuse to hear, and what would it cost?

2. Design the operator-feedback channel that the U.S. Navy Bureau of Ordnance should have had in 1941. Who signs, who receives, what triggers action?

3. The USS Tinosa fired a string of torpedoes that struck a stopped ship and failed to detonate before the bureau accepted the diagnosis. What is the operator-evidence threshold in your domain that would force the equivalent institutional acknowledgment — and how would you make sure it is reached before the cost is paid?

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CASE 139 INDUSTRIAL 2002 – 2014

GM Ignition Switch

D Designed Out G Governance & Trust

IMPACT 124 deaths attributed; ~2.6M vehicles recalled for the defective switch; $900M federal forfeiture (DOJ, 2015); the fix existed for eight years before the recall

IN BRIEF

A faulty ignition switch in several GM compact cars (the Chevrolet Cobalt, Saturn Ion) could slip from “run” to “accessory” while driving, cutting power steering and brakes and — fatally — disarming the airbags. GM engineers identified it in 2002. In 2006 an engineer approved a redesigned switch but did not change its part number, so the fix propagated as neither a revision nor a recall, and defective cars kept selling. The recall came only in 2014 — about 2.6 million vehicles, with 124 deaths attributed through GM’s compensation fund. The Valukas report found a culture that absorbed safety concerns, and GM paid a $900 million federal penalty. What was designed out was not a part but the pathway by which a known safety problem reaches the decision to recall.

BACKGROUND

The ignition switches in several GM compact cars — the Chevrolet Cobalt and Saturn Ion among them — had too little detent torque to hold their position, so a jostle, a bump in the road, or a heavy keychain could rotate the switch out of “run” while driving. That cut power steering and braking and, fatally, disarmed the airbags so they would not deploy in the very crash that loss of control often produced. GM engineers identified the problem in 2002, during development, before the cars ever reached customers.[0]

WHAT HAPPENED

In 2006 a GM engineer approved a redesigned switch but did not change its part number — and in an engineering organization the part number is the very mechanism by which the system knows something changed. The fix propagated as neither a revision nor a recall; defective cars kept selling, and crashes with non-deploying airbags were investigated piecemeal over years, no one connecting them to a switch the records said had “never changed.” The recall came only in 2014 — about 2.6 million vehicles, with 124 deaths attributed through the compensation fund. What finally broke the silence was a wrongful-death lawsuit, in which a family’s expert found the part had been quietly changed under the same number — the smoking gun GM’s own records had been built to miss.[1]

THE INVESTIGATION

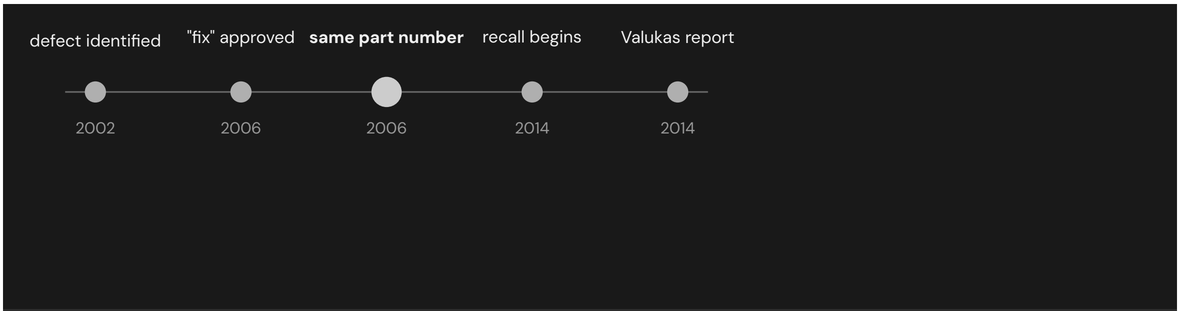
GM’s commissioned investigation by Anton Valukas (2014) found not a single villain but a culture that absorbed safety concerns until they dissipated — the “GM nod,” in which a room agrees a thing should happen and then no one acts, and the “GM salute,” arms crossed and each person pointing elsewhere — and a fundamental failure to use the escalation processes the company already had on the books.[2] In 2015 GM paid a $900 million federal forfeiture for concealing the defect; total penalties and settlements ultimately exceeded $2.6 billion.[3]

THE CAPABILITY GAP

What was designed out of GM was not a part but a pathway — the institutional route by which a known safety problem reaches the decision to recall. The fix existed for eight years; the path from fix to recall did not, so the knowledge sat inert. The mechanism is mundane and all the more instructive for it: the part-number convention was the company’s own way of seeing what had changed, and quietly breaking it blinded the organization to its own action. Suppressing a signal need not be a conspiracy; here it was one engineer taking the path of least resistance through a process no one was assigned to guard.[4]

AFTERMATH & REFORM

Mary Barra, who became CEO as the recall broke, used the Valukas report to restructure GM’s safety decision-making — a global-safety leadership role, consolidated escalation channels, and a “Speak Up for Safety” program to give concerns a route upward that did not depend on one person’s persistence to survive. The case became a standard teaching example in governance and psychological safety: an organization can hold the fix for the better part of a decade and still fail to act if the channel that carries bad news upward has been allowed to fail quietly, with no one accountable for keeping it open.[5]



GM IGNITION SWITCH — EIGHT YEARS BETWEEN THE FIX AND THE RECALL

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THE LEARNING ENGINEERING LENS

> *There was a fundamental failure to use the formal escalation processes that GM had established.*
>
> PARAPHRASING THE VALUKAS REPORT TO THE GM BOARD OF DIRECTORS, 2014

LE INSIGHT

The GM ignition switch case is the canonical example of a corporate organizational structure that suppressed the upward flow of safety information by procedural design. The fix existed for eight years before the recall. The institutional path between the fix and the recall did not.

LENS APPROACH

The load-bearing lesson is change control. Engineers changed the ignition switch but kept the same part number, defeating configuration traceability so the field failures could never be tied back to a design change and the records insisted the part had "never changed." The capability to build is configuration management that forbids altering a part without renumbering it, so the system can always see what changed and connect a field failure to the revision that caused it. LENS uses GM in LEN 4 for that change-control and part-number-as-signal discipline, and secondarily in LEN 7 and LEN 8 for the corporate-cover-up and Valukas-style retrospective accountability framing.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the part number as a safety-critical signal: make changing a part without changing its number impossible by process, so the system cannot be blinded to its own revision.
- Specify the ignition switch's detent torque as a hold-position requirement and verify it, so a known under-torque defect cannot pass into production as a tolerable quirk.
- Build the route from "fix approved" to "recall decision" as an explicit deliverable, not a path that depends on one person choosing to escalate.

IN OPERATION / AFTER

- Correlate field crashes with non-deploying airbags as a population, so piecemeal incidents are connected to a common cause rather than investigated and closed one at a time.
- Run an escalation channel that carries bad news upward independent of any single person's persistence, with a named owner accountable for keeping it open.
- Audit whether approved fixes have actually propagated to a recall or revision, closing the gap in which a known defect keeps shipping under an unchanged record.

REFLECTION QUESTIONS

1. What information channel in your organization carries the same load that GM's part-number system did? Could it be silently bypassed?
2. Design the escalation deliverable that would have moved the GM ignition switch fix to a recall in 2006.
3. A single engineer changed the switch without changing its part number, and no one was assigned to guard that convention. Identify a process in your organization whose integrity depends on an unenforced convention, and name who should own enforcing it.

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CASE 140 SPACE / AEROSPACE 1986 / 2003

Challenger & Columbia

N Normalization of Deviance K Knowledge & Institutional Memory G Governance & Trust

IMPACT 14 astronauts killed (7 per accident); 17 years between identical organizational pathologies

IN BRIEF

NASA lost two Space Shuttle crews to the same organizational pathology seventeen years apart: Challenger in 1986, when O-ring seals failed in cold weather, and Columbia in 2003, when foam debris breached the wing's thermal protection — fourteen astronauts in all. Both flaws had been seen repeatedly and accepted as routine; foam shedding had been documented on at least sixteen prior missions before Columbia, and O-ring erosion in the field joints had been on the engineering record since the early flights. Sociologist Diane Vaughan named the mechanism "normalization of deviance" from the Challenger investigation; the Columbia Accident Investigation Board found the same culture intact seventeen years later and concluded NASA's organizational culture had as much to do with the accident as the foam. The pair is the strongest single-institution evidence that culture is an engineerable property of a system — and that a pathology diagnosed but left unrepaired will recur, at the same cost.

BACKGROUND

The Space Shuttle flew with two flaws its engineers knew about. The solid-rocket booster joints sealed with O-rings that stiffened in cold; the external tank shed foam insulation that struck the orbiter on ascent. Both had appeared on flight after flight without catastrophe, and both were progressively reclassified from hazards to accepted, "routine" features of flying.[0] Each survived flight became evidence that the flaw was tolerable, so the very absence of disaster fed the reclassification — the safety margin redefined downward not by decision but by the accumulating habit of getting away with it.

WHAT HAPPENED

On 28 January 1986 Challenger launched on an unusually cold morning; an O-ring failed to seal, and the vehicle broke apart 73 seconds after liftoff, killing all seven aboard. Seventeen years later, on 1 February 2003, foam that had struck Columbia's wing on ascent had opened a breach in its thermal protection; the orbiter disintegrated on reentry, killing its seven.[1] Both flaws were the ones engineers had already flagged and the institution had already filed as routine, so each accident was less a surprise than the arrival of a bill the system had decided, repeatedly, not to pay.

THE INVESTIGATION

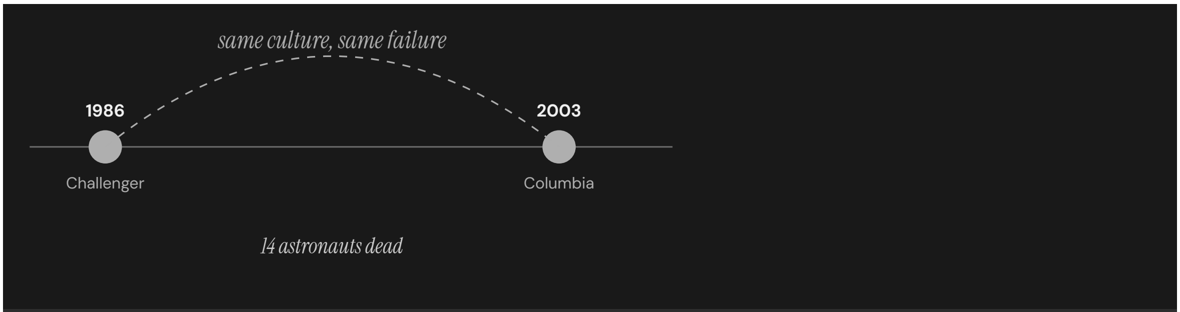
The Rogers Commission traced Challenger to the O-ring and to a launch decision that overrode engineers' cold-weather warnings — the Thiokol teleconference on the eve of launch, at which the engineering position ("do not launch below 53°F") was reversed under management pressure and the launch went ahead at 36°F.[2] Diane Vaughan's The Challenger Launch Decision (1996) re-examined the same teleconference with a decade's documentary access and reframed the decision: not deviant individuals overriding good engineering, but a working group whose rules of evidence had drifted, one accepted anomaly at a time, until a launch at 36°F was inside what the group's own decision rules counted as acceptable risk. Seventeen years later the Columbia Accident Investigation Board found the same patterns intact — flying with flaws defined as routine, foam strikes filed under "in-family" across at least sixteen prior missions, and a structure that suppressed the upward flow of safety concerns — and concluded that "the NASA organizational culture had as much to do with this accident as the foam."[3] That the same board found the same structure seventeen years on is the case's sharpest point: the first reform had fixed the hardware and the schedule pressure but not the mechanism by which a flaw becomes redefined as acceptable.

THE CAPABILITY GAP

Diane Vaughan's "normalization of deviance," developed from Challenger, named the mechanism: deviations from the safety baseline become acceptable through production pressure, weak communication, and habit, one small step at a time. Columbia validated the concept against its author's intent — Vaughan herself was retained by the CAIB and the Board's final report cites the mechanism by name. The pathology had been diagnosed in 1986 and never engineered away — which is the point: culture is a system property, and a diagnosis without a remediation decays.[4] Naming the mechanism did not arrest it, because a name lives in a report while the production pressure and the suppressed warnings live in the daily flow of work, where they kept doing what they had always done. The institutional-learning gap is itself the load-bearing finding: between 1986 and 2003 NASA had reorganized twice, lost an administrator, and adopted multiple safety initiatives, and the structural pathway from a dissenting engineer to a launch decision was substantially unchanged.

AFTERMATH & REFORM

Each accident produced reform — the Rogers Commission's redesign of the booster joint, the CAIB's call to treat culture as a safety variable, the establishment of the NASA Engineering and Safety Center, and the Board's instruction to rebuild independent technical authority — and the Shuttle was retired in 2011 after completing the International Space Station assembly. The pair stands as the book's strongest evidence that organizational culture is engineerable, and that leaving it unengineered is a choice with a recurring, lethal cost.[5] The CAIB's insistence on independent technical authority conceded the deeper lesson the booster-joint redesign alone had missed in 1986: the upward path for a dissenting engineer is itself a piece of safety hardware, and one that has to be rebuilt and defended rather than assumed. The hedge that survives into the case: Vaughan's "normalization of deviance" is the load-bearing analytic claim across both accidents and is never to be smoothed; the seventeen-year institutional-learning gap is the empirical claim, and the CAIB's cross-referencing of Rogers Commission language ("the same decision-making structures") is the documentary anchor.



17 YEARS, SAME PATTERN — CHALLENGER TO COLUMBIA

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THE LEARNING ENGINEERING LENS

> *These repeating patterns mean that flawed practices embedded in NASA's organizational system continued for 20 years and made substantial contributions to both accidents.*
>
> COLUMBIA ACCIDENT INVESTIGATION BOARD, 2003

LE INSIGHT

Challenger/Columbia is the strongest documentary evidence that organizational culture is an engineerable property of a system, and that diagnoses without engineered remediations decay. The same pathology, twice, seventeen years apart, in the same institution, with the same human cost — and with the diagnosis already on the record. Vaughan's "normalization of deviance" names the mechanism and is the load-bearing analytic claim across both accidents. Capability engineering treats culture as a deliverable.

LENS APPROACH

Challenger/Columbia anchors the multi-layer-drift-and-cumulative- inadequacy competency (induced 7.4; LENS D1/PT4): a cascade of marginal-but-tolerable conditions across decision rules, schedule pressure, and communication structure aligned twice, seventeen years apart. LENS uses the pair in Domain 1 (Systems Analysis) for normalization of deviance as a systems concept and the institutional-learning gap as a measurable property; in Domain 5 (Navigating Sociotechnical Constraints) for the governance failure that allowed a diagnosed pathology to persist and for the upward-channel design the CAIB called a piece of safety hardware. Pair with Deepwater Horizon (Case 85) on the multi-layer-drift form, and with Bhopal (Case 147) and Fukushima (Case 143) on cumulative inadequacy in catastrophic-system operations.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Hold known flaws as open hazards with explicit owners, so a clean flight cannot silently reclassify a deviation as routine.
- Build independent technical authority into the launch decision, giving dissenting engineers a path that production pressure cannot override.
- Define the safety baseline quantitatively at design time, so any drift below it is a flagged change rather than an unremarked habit.

IN OPERATION / AFTER

- Re-audit every diagnosed-but-unrepaired pathology on a fixed cadence, verifying remediation in practice rather than on paper.
- Track the upward flow of safety concerns as a measured signal — count and resolve dissents — so suppression becomes visible.
- Treat the cultural mechanism, not just the hardware fix, as the deliverable that must persist between major incidents.

REFLECTION QUESTIONS

1. What is the equivalent "diagnosed but not repaired" pathology in your domain? What evidence would close the loop?
2. The CAIB called culture engineerable. Sketch the engineering deliverable for the cultural intervention you would propose in your domain — including its measurement signal.
3. Seventeen years separated two accidents with the same root pathology. What mechanism in your organization would verify that a past diagnosis has actually been remediated rather than merely documented?
4. Vaughan reframed the Thiokol teleconference not as deviant override but as a working group whose decision rules had drifted. Identify a recurring decision in your domain whose rules of evidence may have drifted incrementally, and specify the audit that would surface the drift before the next high-consequence call.

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CASE 141 DEFENSE 1991 – PRESENT

V-22 Osprey

T Training Gap H Human-System Interface N Normalization of Deviance

IMPACT ~65 killed across ~17 hull-loss accidents since 1991; serious-mishap rate above comparable fleets (GAO-26-108905, 2025); some fixes stretch to the 2030s

IN BRIEF

The V-22 Osprey — the tiltrotor flown by the Marines, Air Force, and Navy — has had about 17 hull-loss accidents and roughly 65 fatalities since 1991, including 19 Marines in a single 2000 test crash and 8 airmen off Yakushima, Japan, in 2023. The Yakushima crash traced to cracks in a transmission gear (a flaw in the X-53 steel alloy) and a pilot's decision to keep flying through warnings. In December 2025, GAO and a NAVAIR review found the V-22's joint program office had failed for years to assess and address mounting safety risks, that serious-mishap rates exceeded comparable fleets, and that some fixes stretch toward
2034. The Osprey is the steady-state form of normalization of deviance:

a documented, reviewed shortfall accepted as the cost of flying the airframe.

BACKGROUND

The V-22 Osprey tilts its rotors to take off like a helicopter and cruise like a turboprop — an ambitious capability shared awkwardly across three services and a joint program office. Since development began in the 1980s it has suffered about 17 hull-loss accidents and roughly 65 fatalities, including 19 Marines in a single crash during 2000 testing.[0] The same configuration that makes the tiltrotor uniquely useful makes it uniquely demanding to sustain, and the joint arrangement diffused ownership of that burden across three services that each carried the airframe but none of which fully owned its safety trajectory.

WHAT HAPPENED

On 29 November 2023 a CV-22B broke up off Yakushima, Japan, killing all eight airmen aboard. The Air Force traced it to cracks in a transmission gear — rooted in metallic inclusions in the X-53 steel alloy used for the gears — and to the pilot's decision to keep flying despite warnings to land. The crash grounded Osprey fleets worldwide for months.[1] The materiel flaw and the human decision to press on through warnings were the same paired factors earlier reviews had named, so Yakushima read less as a novel failure than as one more draw from a pattern the program had already characterized but not closed.

THE INVESTIGATION

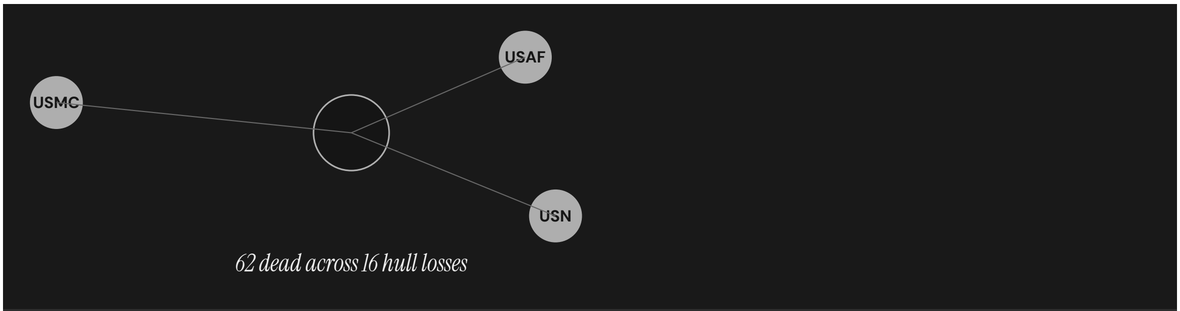
In December 2025 the Government Accountability Office and a NAVAIR review reported together that the joint program office had failed for years to adequately assess and address mounting safety risks, even as service members died.[2] Serious-mishap rates generally exceeded those of comparable Navy and Air Force fixed- and rotary-wing fleets from FY2015 to FY2024 and spiked in 2023–2024; a gearbox flaw dating to 2006 was not evaluated until 2024, and full fixes for some issues are not expected until the 2030s.[3] An eighteen-year gap between a gearbox flaw arising and its being evaluated is the timescale of normalization made literal: the deviation persisted long enough to become the airframe's accepted background condition rather than an open defect demanding action.

THE CAPABILITY GAP

The V-22 is the steady-state form of normalization of deviance. The shortfall is not unknown — it has been documented across successive reviews — but the system built to remediate it has accepted its own incompleteness as the cost of operating the airframe. Each crash carries a familiar set of factors (materiel failure, human error, weak coordination across the three services), and each is followed by adjustments that do not converge.[4] Convergence is the missing property: three services adjusting in parallel, none holding final authority over the whole, produce motion without resolution, so the mishap rate stays elevated while every individual response looks reasonable on its own terms.

AFTERMATH & REFORM

Groundings, gear-inspection regimes, and a redesign program have followed, with full fixes for some gearbox issues stretching toward 2034. The case is instructive precisely because the problem is recognized: a known, reviewed capability gap can persist for decades when each incident is treated as an isolated event rather than a sample from a distribution the program keeps drawing from.[5] Fixes that stretch toward 2034 mean the airframe will keep flying for years inside a margin already judged worse than comparable fleets — the program in effect electing to operate the distribution it has been told it is drawing fatalities from.



PERSISTENT CAPABILITY GAPS ACROSS THREE SERVICES

REFERENCES

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3. U.S. GAO, Osprey Aircraft: Additional Oversight and Information Sharing Would Improve Safety Efforts, GAO-26-108905 (Dec. 2025) — the joint program office's failure to assess and address risks.
4. GAO-26-108905 (2025) — serious-mishap rates exceeding comparable Navy/Air Force fleets (FY2015–FY2024); the 2006 gearbox flaw evaluated only in 2024; fixes into the 2030s.
5. NAVAIR independent review of the V-22 (Dec. 2025) — materiel and cross-service-coordination factors and unresolved catastrophic parts issues.
6. USNI News, V-22 program coverage (2024–2025).

THE LEARNING ENGINEERING LENS

> *Materiel failure and human-error factors were the most frequently cited causal factors in serious Osprey accidents.*
>
> PARAPHRASING THE NAVAIR INDEPENDENT REVIEW OF THE V-22, 2025

LE INSIGHT

V-22 demonstrates the steady-state version of normalization of deviance: a platform whose shortfall has been documented, reviewed, and acted on across multiple administrations without producing sustained improvement. The capability gap has itself been normalized. Each incident is treated as an event rather than as a sample from a distribution the program continues to draw from.

LENS APPROACH

LENS treats V-22 in LEN 5 as a multi-service capability-coordination problem and in LEN 8 as a study in long-cycle organizational learning failure. Students design the evidence system that would distinguish a true reduction in mishap rate from the natural variation of a chronically marginal platform.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assign one accountable owner for the airframe's safety trajectory rather than diffusing it across three services with no final authority.
- Treat each known materiel flaw as an open defect on a clock, so a problem like the gearbox cannot quietly become accepted background condition.
- Define an absolute mishap-rate threshold against comparable fleets that triggers mandatory action, not merely review.

IN OPERATION / AFTER

- Aggregate incidents as samples from one distribution, not isolated events, so the elevated rate itself becomes the thing being managed.
- Audit whether parallel service-level adjustments are actually converging on a lower rate, and escalate when they are not.
- Gate continued operation on remediation progress, so fixes stretching toward 2034 do not silently license years of degraded margin.

REFLECTION QUESTIONS

1. Identify a platform or process in your domain that has been operating in a documented shortfall for years. What measurement would have to change for the shortfall to become unacceptable?
2. The V-22's three services do not converge on remediation. Design the governance structure that would force convergence.
3. A gearbox flaw went eighteen years between arising and being evaluated. What mechanism in your domain converts a long-tolerated defect back into an open item that demands action?

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CASE 142 ENERGY 2002

Davis-Besse Reactor Head Corrosion

N Normalization of Deviance K Knowledge & Institutional Memory G Governance & Trust

IMPACT Football-sized cavity discovered in the reactor pressure-vessel head; near-miss; ~$600M recovery and extended outage

IN BRIEF

During a 2002 refueling outage at the Davis-Besse nuclear plant in Ohio, workers found a football-sized cavity eaten through six inches of the carbon-steel reactor-vessel head by leaking boric acid, leaving only a thin layer of stainless cladding between the cavity and reactor coolant at 2,200 psi — a serious near-miss for a loss-of-coolant accident. The leakage had been visible for years and inspections deferred; FirstEnergy had won an NRC deferral of a mandatory inspection to fit its refueling schedule, and the NRC's Inspector General later found the decision had weighted economics over safety. Davis-Besse is the canonical post-Three-Mile-Island near-miss — evidence that even an industry that built INPO to engineer safety can let regulator-operator dynamics erode it.

BACKGROUND

After Three Mile Island, U.S. commercial nuclear power had built — through INPO and a strengthened NRC — a reputation for engineering safety as a system. Davis-Besse, a pressurized-water reactor in Ohio, operated inside that regime, with a known industry-wide hazard: boric acid leaking from cracked nozzles could corrode the carbon-steel reactor-vessel head.[0] The hazard was generic and understood across the fleet, which is what makes the case sharp: the danger was not a surprise mechanism but a recognized one the post-TMI regime was precisely supposed to keep inspected and contained.

WHAT HAPPENED

During a refueling outage in early 2002, workers found a cavity about the size of a football eaten clean through six inches of the carbon-steel head, leaving only a thin layer of stainless-steel cladding between the corrosion and reactor coolant at 2,200 psi. Had the cladding given way, the result would have been a medium-break loss-of-coolant accident with a badly degraded safety margin.[1] The thin remaining cladding was the entire distance between routine operation and the kind of accident the entire regulatory regime existed to prevent — a margin measured in a fraction of an inch of unintended material rather than in any engineered barrier.

THE INVESTIGATION

The boric-acid leakage had been observable for years, and inspections had been deferred. FirstEnergy had requested — and the NRC had granted — a deferral of a mandatory inspection (NRC Bulletin 2001-01) so it would align with the plant's February 2002 outage.[2] The

NRC's Office of Inspector General later found the agency had inappropriately weighted the utility's economic arguments over safety — its oversight, the OIG concluded, had not been adequate to ensure that safety would not be compromised.[3] The deferral was not a covert lapse but a documented decision both parties signed off on, which is the unsettling part: the erosion happened through the regulator's normal process, in the open, agreed to be safe to wait.

THE CAPABILITY GAP

Davis-Besse is a near-miss in the dimension above operations: the regulator-operator relationship. The same industry that had shown, through INPO, that it knew how to engineer safety let a known corrosion mechanism go uninspected because inspecting it was inconvenient and expensive. Regulatory capture — the regulator adopting the operator's economic frame — is a capability failure at the institutional layer above the plant.[4] The plant's own engineering competence was never the missing piece; what failed was the independence of the layer meant to overrule a utility's schedule when safety required it, and that layer had quietly adopted the schedule as its own.

AFTERMATH & REFORM

INPO and the NRC restructured significant parts of their inspection regimes, and reactor-head inspection requirements were tightened across the fleet. The lesson pairs with the book's TMI / INPO arc: institutional capability is not built once. It erodes if it is not re-engineered — and the erosion is quietest where the regulator and the regulated agree it is safe to wait.[5] Tightening the head-inspection requirement across the fleet conceded that a mandatory inspection had been a real barrier all along — one the deferral process had been allowed to treat as negotiable against a refueling schedule.

1/4″

OF STAINLESS CLADDING REMAINED

between a 6-inch corrosion cavity and reactor coolant at 2,200 psi

DAVIS-BESSE — THE POST-TMI NEAR-MISS

REFERENCES

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6. J. V. Rees, Hostages of Each Other (1994) — INPO and the fragility of institutional safety capability.

THE LEARNING ENGINEERING LENS

> *The NRC's actions were not adequate to ensure that safety would not be compromised.*
>
> PARAPHRASING THE NRC OFFICE OF INSPECTOR GENERAL DAVIS-BESSE REPORT, 2002

LE INSIGHT

Davis-Besse is the case for how regulator-operator dynamics can erode the capability of an industry that had previously demonstrated, through INPO, that it knew how to engineer safety. Regulatory capture is a capability failure at the institutional layer above operations.

LENS APPROACH

LENS uses Davis-Besse in LEN 7 to study regulatory-capture dynamics and in LEN 8 to compare with INPO's earlier success. The case is a reminder that institutional capability requires sustained re-engineering.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make safety-critical inspections of known generic hazards non-deferrable, so a mandatory check cannot be traded against a refueling schedule.
- Require any deferral decision to be argued on safety margin alone, with the utility's economic case explicitly excluded from the record.
- Preserve the independence of the oversight layer so it can overrule an operator's timeline rather than adopt it.

IN OPERATION / AFTER

- Audit granted deferrals for whether the regulator reasoned from safety or from the operator's economics, and flag drift toward the latter.
- Track observable degradation (like boric-acid leakage) against inspection currency, so a known mechanism cannot run uninspected for years.
- Re-engineer institutional safety capability on a cadence, treating post-incident regimes like INPO as maintained, not permanent.

REFLECTION QUESTIONS

1. Identify a regulator-operator relationship in your domain in which the regulator may be at risk of accepting the operator's economic argument over its own safety judgment. What signal would surface it?

2. Design the institutional control that would prevent a Davis-Besse-style deferral from being granted.

3. The erosion here happened openly, through the regulator's normal process. What audit would distinguish a defensible deferral from one in which the regulator has quietly adopted the operator's schedule as its own?

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CASE 143 ENERGY 2011

Fukushima Daiichi

N Normalization of Deviance G Governance & Trust K Knowledge & Institutional Memory

IMPACT Three reactor meltdowns following the Tōhoku earthquake and tsunami; ~160,000 people displaced; cleanup projected at $200B+

IN BRIEF

On 11 March 2011 the Tōhoku earthquake and the tsunami it spawned overwhelmed TEPCO's Fukushima Daiichi plant: the wave topped the seawall, flooded the emergency diesel generators, and cut cooling to the reactors. Three of the six cores melted down and hydrogen explosions spread radioactive material, displacing some 160,000 people; cleanup is projected above $200 billion. The independent Diet commission (NAIIC), chaired by Kiyoshi Kurokawa, called it a disaster "made in Japan" — the product of regulatory capture and a culture reluctant to challenge utility assumptions; evidence of large historical tsunamis had been discussed internally but never forced a change to the seawall. Other reviews stressed under-estimated external hazards. Fukushima is the post-TMI evidence that safety institutions like INPO must be deliberately built, not assumed.

BACKGROUND

Fukushima Daiichi sat on Japan's northeast coast, its reactors and their backup diesel generators protected by a seawall sized to a design-basis tsunami. Evidence of much larger historical waves — back to the ninth-century Jōgan event — had been discussed in TEPCO's own internal assessments, but no institutional path turned that evidence into a higher seawall.[0] The gap was not in the data but in the conversion: the larger-wave evidence lived inside the utility's own assessments, where it could be discussed indefinitely without ever becoming a binding requirement to raise the wall it implicated.

WHAT HAPPENED

On 11 March 2011 the Tōhoku earthquake struck and the tsunami that followed topped the seawall. The plant lost off-site power; the diesel generators meant to keep cooling running were inundated. Cooling failed, three of the six reactor cores melted down, and hydrogen explosions spread radioactive material across the region. Some 160,000 people were displaced, and cleanup is projected above $200 billion.[1] Siting the backup generators where a wave that overtopped the seawall would reach them tied the entire cooling chain to that single design assumption, so once the wall was topped the loss of cooling followed almost mechanically from the layout itself.

THE INVESTIGATION

The independent investigation chaired by Kiyoshi Kurokawa for the National Diet (NAIIC) called the accident "made in Japan" — the product of regulatory capture, deference to authority, and an institutional reluctance to challenge utility assumptions.[2] Other major reviews — the Hatamura government commission (2012) and the IAEA Director General's report (2015) — emphasized the under-estimation of external hazards and defense-in-depth assumptions over the cultural critique; the Kurokawa framing is the most-cited but not the only consensus reading.[3] That serious independent reviews diverged on emphasis — cultural capture versus under-estimated external hazards — is itself part of the record, and the book treats Kurokawa's as the most-cited reading rather than the settled one.

THE CAPABILITY GAP

Fukushima is the post-TMI case showing that the INPO pattern (Case 172) is not self-executing. The U.S. industry built INPO to force operating discipline and shared learning; the Japanese industry did not build an equivalent with the independence to override a utility's optimistic assumptions. The internal tsunami evidence existed; the institutional capability to act on it did not.[4] Evidence without an independent body empowered to act on it is inert: the larger-wave assessments could be acknowledged and shelved indefinitely because no institution stood outside the utility with the standing to convert them into a mandated change.

AFTERMATH & REFORM

The accident drove a restructuring of Japanese nuclear regulation — a new, more independent Nuclear Regulation Authority — and a global re-examination of external-hazard margins and station-blackout protection. Paired with INPO and Davis-Besse, it triangulates the book's claim: sustained nuclear-safety capability is an institution that must be deliberately built and rebuilt, not a property a competent industry simply has.[5] Creating a more independent regulator after the fact conceded the structural diagnosis directly: the missing piece had been an authority sitting outside the utility, and the reform was precisely to build the independence that the pre-2011 arrangement had lacked.

3 of 6

REACTORS MELTED DOWN

tsunami exceeded design basis the institutional evidence had already questioned

FUKUSHIMA DAIICHI — "MADE IN JAPAN," PER THE DIET INQUIRY

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THE LEARNING ENGINEERING LENS

> *What must be admitted — very painfully — is that this was a disaster "Made in Japan."*
>
> NATIONAL DIET OF JAPAN FUKUSHIMA NUCLEAR ACCIDENT INDEPENDENT INVESTIGATION COMMISSION, 2012

LE INSIGHT

Fukushima is the post-TMI case that establishes that the INPO pattern (Case 172) is not self-executing. The U.S. industry built INPO; the Japanese industry did not. The cost of the difference, paid in 2011, is the strongest available evidence that capability institutions must be deliberately built, not assumed.

LENS APPROACH

LENS uses Fukushima in LEN 8 as the cross-cultural comparison to INPO and in LEN 7 for the regulator-utility dynamics study. The case is paired with INPO and Davis-Besse to triangulate what sustained nuclear-safety capability requires.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build an institutional path that converts an acknowledged hazard (like the larger-wave evidence) into a binding design requirement, not a discussable assessment.
- Site critical backups so a single overtopped barrier cannot disable the whole cooling chain at once.
- Establish an oversight body independent of the utility, with standing to override optimistic in-house assumptions about external hazards.

IN OPERATION / AFTER

- Audit whether internally held hazard evidence has actually driven design changes, treating shelved assessments as an open finding.
- Re-examine external-hazard margins on a cadence rather than freezing them at a design-basis figure set once.
- Verify that the regulator retains the independence to act, since institutional safety capability erodes if it is assumed rather than rebuilt.

REFLECTION QUESTIONS

1. Identify a regulatory regime in your domain whose effectiveness depends on a cultural willingness to challenge authority. What if the culture changes?
2. INPO is U.S.-specific. Design the structural features that would have to be in place for a comparable institution to function in a different national context.
3. The tsunami evidence was discussed but never converted into a binding requirement. What institutional path in your domain turns an acknowledged hazard into a mandated change rather than a shelved assessment?

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CASE 144 TECHNOLOGY 2024

CrowdStrike Falcon Outage

D Designed Out K Knowledge & Institutional Memory G Governance & Trust

IMPACT 8.5 million Windows machines crashed; airlines, hospitals, broadcasters, and banks affected simultaneously; largest IT outage on record

IN BRIEF

On 19 July 2024 CrowdStrike pushed a content update to its Falcon endpoint sensor that contained a logic error in a small configuration file. On every Windows machine running the affected sensor, the file drove the kernel driver to crash, looping the blue screen of death and requiring hands-on recovery of each device. Roughly 8.5 million machines went down at once — across hospitals, airlines, banks, broadcasters, and governments worldwide — the largest IT outage on record. CrowdStrike's post-incident review found the content update had not been put through the same automated testing or staged rollout as code: the pipeline treated "content" as a lesser category than "code," though the operating system did not. CrowdStrike is the cybersecurity-vendor analog of Knight Capital, six orders of magnitude larger.

BACKGROUND

CrowdStrike's Falcon sensor runs deep inside Windows — in the kernel — to detect threats, and it updates constantly: not only its code, but "content," the rapid-response detection configuration pushed to customers continuously. The deployment pipeline treated that content as a lighter category than code, with less testing and no staged rollout.[0] The distinction had a logic: content shipped fast and often, precisely so the sensor could keep pace with new threats, and slowing it down with full code-grade testing seemed to defeat its purpose. That speed was the very reason the safety gate was lowered on the artifact that could still crash a kernel.

WHAT HAPPENED

On 19 July 2024 a content update contained a logic error in a small configuration file. On every Windows machine running the affected sensor, the file drove the kernel driver to read out of bounds and crash, looping the blue screen of death and requiring manual recovery of each device. Roughly 8.5 million machines failed at once — hospitals, airlines, banks, broadcasters, and governments worldwide — the largest IT outage on record.[1] Because the file went to every affected sensor simultaneously with no staged rollout, there was no first wave to catch the fault and no blast radius short of the whole install base; the requirement for hands-on recovery of each device turned a single bad push into weeks of physical labor across the world.

THE INVESTIGATION

CrowdStrike's own post-incident review found the content update had not been put through the same depth of testing and staged rollout as its code releases. The fault was not exotic: a category boundary in the deployment pipeline — content treated as safer than code — that did not match the operational reality, in which a bad content file could crash the kernel exactly as bad code could. The boundary was an organizational convenience rather than a technical truth, and the kernel, which executes whatever reaches it, recognized no such distinction at all.[2]

THE CAPABILITY GAP

The missing capability was the recognition that, for deployment safety, content **is** code: anything that can crash the kernel must clear the same testing and staged-rollout gates. CrowdStrike's customers had trusted the vendor's deployment safety, and that trust turned out to be load-bearing for the operation of a large slice of the global economy on a single morning. Each customer had implicitly outsourced a safety gate to the vendor's pipeline, so the one missing gate inside that pipeline was multiplied across every institution that ran the sensor, with no independent check standing between a bad push and their kernels.[3]

AFTERMATH & REFORM

CrowdStrike moved content updates onto staged rollouts and stronger validation, Microsoft revisited kernel-level access for security vendors, and the episode prompted scrutiny of concentration risk in endpoint security.[4] Each response targets a different layer of the same failure: staged rollout limits the blast radius of any one push, reconsidering kernel access limits how much a vendor fault can break, and the concentration-risk scrutiny acknowledges that a single vendor had become a shared point of failure for much of the economy. It is the cybersecurity-vendor analog of Knight Capital (Case 10) — an unverified deployment to a system wired into critical operations — at a scale six orders of magnitude larger.

8.5M

MACHINES · SINGLE CONFIGURATION FILE

content treated differently from code in the deployment pipeline

CROWDSTRIKE — THE COST OF A CATEGORY BOUNDARY IN A DEPLOYMENT

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THE LEARNING ENGINEERING LENS

> *Our content configuration update process did not include the same depth of testing and staged rollout as our code releases.*
>
> PARAPHRASING CROWDSTRIKE PRELIMINARY POST-INCIDENT REVIEW, JULY 2024

LE INSIGHT

The CrowdStrike outage is the live case for what happens when a deployment safety architecture treats categories of artifact differently than the operational system actually does. Content looked operationally identical to code; it was treated as different in the deployment pipeline. The mismatch became the largest IT outage in history.

LENS APPROACH

LENS uses CrowdStrike in LEN 5 as a categories-and-boundaries capability case and in LEN 2 for the vendor-customer trust architecture: customers trusted CrowdStrike's deployment safety; that trust was load-bearing for the operation of the global economy on a single day.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Define deployment-safety gates by operational impact, not artifact category: anything that can crash the kernel clears the same testing and staged rollout as code.
- Make staged rollout mandatory for every update, so a faulty push is caught by a first wave instead of reaching the entire install base at once.
- Design for recovery, not just prevention — assume a bad update will ship and ensure it does not require hands-on intervention at every device.

IN OPERATION / AFTER

- Audit vendor deployment-safety practices your institution depends on, rather than treating the outsourced safety gate as trustworthy by default.
- Map and reduce concentration risk so a single vendor fault cannot take down a large slice of critical operations simultaneously.
- Run post-incident reviews that interrogate category boundaries in the pipeline and feed the findings back into validation gates.

REFLECTION QUESTIONS

1. Identify a vendor relationship in your domain whose deployment-safety practice your institution does not audit. What would the audit reveal?
2. Design the cross-vendor staged-rollout protocol that should be standard for endpoint security software.
3. CrowdStrike's pipeline treated content as safer than code, but the kernel did not. What category boundary in one of your systems is an organizational convenience that the operational reality ignores — and what would it cost if it broke?

WHO BUILDS THIS systems & human-factors engineering · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate. TOOLS task analysis · design prototyping · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

CASE 145 TECHNOLOGY 2018

TSB Bank IT Migration

H Human-System Interface G Governance & Trust

IMPACT 1.9 million UK customers locked out of accounts; £330M+ in compensation and remediation; CEO resigned

IN BRIEF

In April 2018 TSB Bank tried to migrate some five million customer accounts off its former parent Lloyds' systems onto a new platform from its current owner, Sabadell, over a single weekend. When services came back online, nearly every component failed: 1.9 million customers were locked out, some saw strangers' accounts, mortgages vanished, payments bounced. Recovery took months and cost over £330 million; the CEO resigned and the regulator fined the bank. The independent review found the platform had been tested under conditions that did not approximate real load, certified ready by a process that did not challenge the certification, and pushed live against technical recommendations that it was not ready. TSB is the financial-sector analog of Healthcare.gov: a migration shipped without adequate testing because schedule pressure overrode the technical signal.

BACKGROUND

TSB Bank, spun out of Lloyds and acquired by Spain's Sabadell, needed to move some five million customer accounts off Lloyds' systems onto a new Sabadell-built platform. The cutover was scheduled for a single weekend.[0] Compressing a five-million-account migration into one weekend left no room for partial failure: the schedule itself became a forcing function, framing readiness as a date to be hit rather than a condition to be proven, and that framing would later prove decisive when the technical signal said the platform was not ready.

WHAT HAPPENED

When customer-facing services came back online that Sunday evening in April 2018, nearly every component of the new platform had problems. About 1.9 million customers were locked out; some saw other people's accounts, mortgages disappeared, and card payments failed. The recovery took months, cost more than £330 million in compensation and remediation, and the chief executive resigned.[1] That nearly every component failed at once points away from a single defect and toward a platform that had never been exercised under real load — the kind of systemic breakdown that follows when a system is proven only in conditions it will never actually meet.

THE INVESTIGATION

The Slaughter and May independent review found the migration had been tested under conditions that did not approximate real customer load, and that the platform had been certified ready by a process that did not adequately challenge the certification — a certification that confirmed readiness rather than interrogating it, which is how a system that would fail under real conditions could be signed off as fit.[2] Decisively, the executive decision to proceed had been taken against technical recommendations that the platform was not ready; the Financial Conduct Authority later fined TSB, treating the override of a known technical objection as a failure of governance and not merely of engineering.[3]

THE CAPABILITY GAP

The technical signal existed — the platform was not ready, and people knew it. But the decision authority sat at the executive layer, where the signal arrived weakened by passage through intermediate layers, and the institutional architecture gave the technical layer no way to halt the migration. The missing capability was not testing knowledge but a governance structure in which a "not ready" could stop a scheduled go-live. Knowing a system is unready is worthless if the knowledge cannot reach the decision with its force intact and the authority to act on it; here the truth was present but powerless.[4]

AFTERMATH & REFORM

TSB rebuilt its testing and migration governance, paid out and was penalized, and the case entered the literature as a study in technical-decision authority.[5] Rebuilding governance rather than merely the platform was the right diagnosis: the failure had been one of who could say "stop" and be heeded, so the durable fix had to live in the decision structure rather than the code. It is the financial-sector analog of Healthcare.gov (Case 14): a large migration shipped without the testing the institution knew it needed, because schedule pressure overrode a technical signal that had no authority to win.

TSB — READINESS SIGNAL ARRIVED ATTENUATED AT THE DECISION

REFERENCES

1. Slaughter and May, Independent Review of the TSB Migration (2019) — the single-weekend cutover and the testing failures.
2. Slaughter and May (2019) and FCA materials — 1.9 million customers locked out, £330M+ in costs, and the CEO's resignation.
3. Slaughter and May (2019) — inadequate load testing and an unchallenged readiness certification.
4. Financial Conduct Authority, Final Notice on TSB Bank (2022) — the regulatory penalty and proceeding against technical advice.
5. House of Commons Treasury Committee, report on the TSB IT migration (2018).
6. Cf. Healthcare.gov (Case 14) and the migration-safety literature.

> *The migration proceeded notwithstanding clear signals that the platform was not ready.*
>
> PARAPHRASING THE SLAUGHTER AND MAY INDEPENDENT REVIEW OF THE TSB MIGRATION, 2019

LE INSIGHT

TSB is the canonical case for schedule pressure overriding technical signal in a regulated industry. The technical signal existed. The decision authority was at the executive layer where the signal arrived weakened by passage through multiple intermediate layers. The institutional architecture did not allow the technical layer to halt the migration.

LENS APPROACH

LENS uses TSB in LEN 7 as a corporate-governance case and in LEN 8 for the institutional structure of technical-decision authority. Studio projects compare TSB and Healthcare.gov.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Test the platform under conditions that approximate real production load, since a system proven only in unrepresentative conditions will fail when it meets the real ones.
- Make certification adversarial — a process that tries to disprove readiness — rather than a sign-off that confirms it.
- Avoid single-weekend big-bang cutovers where feasible; stage the migration so partial failure is survivable rather than catastrophic.

IN OPERATION / AFTER

- Build a governance structure in which a technical “not ready” can halt a scheduled go-live, so the signal reaches the decision with its force and authority intact.
- Ensure the decision authority hears the technical signal directly rather than through intermediate layers that attenuate it.
- After any failure, fix the decision structure that allowed a known objection to be overridden, not just the system that broke.

REFLECTION QUESTIONS

1. Where in your organization does a technical signal arrive at the executive layer attenuated by intermediate layers? What is the cost of the attenuation?
2. Design the institutional structure that would allow a technical lead to halt a migration like TSB’s without resigning.
3. TSB’s certification confirmed readiness rather than interrogating it. Examine a sign-off process in your domain that rubber-stamps rather than challenges — what would it take to make the certification adversarial enough to catch an unready system?

WHO BUILDS THIS human factors & interaction design · governance, ethics & measurement — plus domain experts and a learning engineer to integrate. TOOLS usability testing · cognitive walkthroughs · governance frameworks · bias & evidence audits

CASE 146 EDUCATION AT SCALE 2014

inBloom

G Governance & Trust

IMPACT $100M initiative collapsed in ~14 months; 9 states withdrew; data infrastructure for education set back years

IN BRIEF

inBloom was a $100-million, Gates-funded shared data infrastructure for U.S. student records — and the technology worked: no bug, no breach, no performance failure. What killed it in about fourteen months was everything around the technology. It launched without consent frameworks, community engagement, transparency about data use, or a way for parents to participate in decisions about their children's data. Parent-privacy groups organized opposition state by state, and nine states withdrew. Analysts read inBloom as the failure of technocratic reform — the assumption that technically sound infrastructure generates its own legitimacy. It is the purest governance failure in the dataset, and the book's clearest argument that in education at scale, stakeholder trust and governance are not optional features but load-bearing structure.

BACKGROUND

inBloom was an ambitious shared data infrastructure for U.S. K–12 student records, backed by $100 million from the Gates Foundation, hosted on commercial cloud, and built by enterprise engineers. The premise was that a common store would spare districts from rebuilding the same plumbing and let applications interoperate across systems that had never spoken to one another. Technically it was sound — no bug, no breach, no performance failure ever undid it, and that very soundness is what makes the case instructive.[0]

WHAT HAPPENED

What undid it was everything around the technology. inBloom launched without adequate consent frameworks, without meaningful community engagement on data governance, without transparency about what was collected and why, and without any way for parents to participate in decisions about their children's data. Each omission read, to a worried parent, as a decision made about their child without them in the room. Parent groups organized opposition state by state, and nine states — among them New York, Louisiana, and Illinois — withdrew as the political cost of staying overtook any promised efficiency. Within about fourteen months the $100-million initiative collapsed.[1]

THE INVESTIGATION

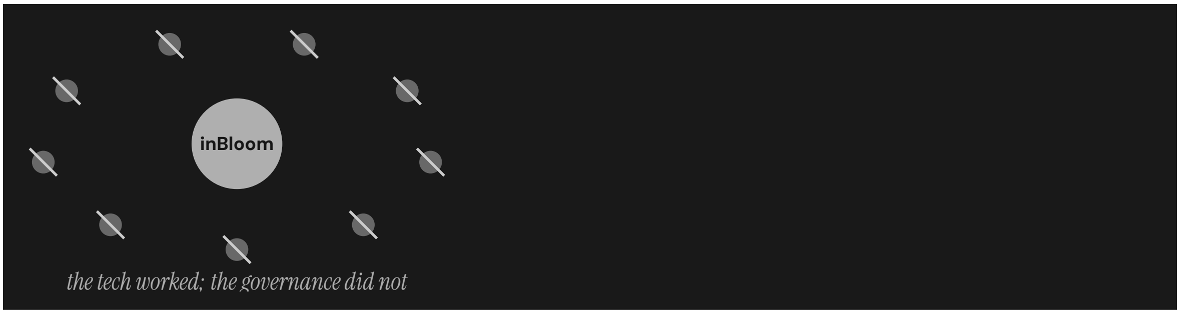
Analysts at Data & Society read inBloom as the failure of technocratic education reform: the assumption that technically sound infrastructure generates its own legitimacy proved catastrophically wrong. Legitimacy, on this reading, is earned from the stakeholders a system acts upon, not conferred by the quality of its engineering. The technology was never the problem; the governance was — the consent, transparency, and trust that had been treated as add-ons rather than as the foundation the whole effort needed before a single record moved.[2]

THE CAPABILITY GAP

inBloom is the purest governance failure in this dataset: nothing technical was wrong, and everything sociotechnical was. The missing capability was the design of stakeholder trust — consent, accountability, and a voice for the families whose data was at stake — treated as a precondition for deployment rather than a feature to add later. Once opposition formed, no patch could retrofit the trust that should have been built in from the start, because trust withheld at launch cannot be engineered back in under fire. In education at scale, those are load-bearing elements, not optional ones.[3]

AFTERMATH & REFORM

inBloom's collapse set back shared education-data infrastructure for years and became a standard cautionary tale; it also helped drive a wave of state student-data-privacy laws that codified, after the fact, the consent and transparency the project had skipped.[4] The lesson the book takes from it is that ethics-as-design-constraint is not ideology but engineering — and inBloom is the $100-million empirical test of what happens when you skip it, paid not in downtime but in a dead initiative and a chilled field.



$100M OF WORKING TECHNOLOGY; NINE STATES WITHDREW

REFERENCES

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2. Education Week and Hechinger Report coverage of the state withdrawals (2013–2014) — nine states exiting.
3. Bulger et al. (2017) — governance, not technology, as the cause; "the technology was not the problem."
4. N. Selwyn, Distrusting Educational Technology (2014); d. boyd & K. Crawford, "Critical Questions for Big Data" (2012).
5. Parent Coalition for Student Privacy archives and the wave of state student-data-privacy legislation that followed inBloom.

THE LEARNING ENGINEERING LENS

> *The technology was not the problem. The governance was the problem.*

PARAPHRASING THE ANALYSIS IN BULGER, MCCORMICK & PITCAN, DATA & SOCIETY, 2017

LE INSIGHT

inBloom is the purest governance failure in this dataset. Nothing technical was wrong. Everything sociotechnical was. The case is the strongest argument in the book for treating ethics-as-design- constraint not as ideology but as engineering: a $100M empirical test of what happens when you do not.

LENS APPROACH

LENS uses inBloom in LEN 7 as the canonical governance failure and in LEN 1 as a stakeholder-analysis case. The case anchors the LENS threading commitment that equity and accountability are design commitments, not modules. Studio projects (LEN 10) require students to produce a stakeholder-trust deliverable as a precondition for deployment.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat consent, transparency, and a parent-facing voice as load-bearing requirements gathered before the data store is built, not features bolted on after launch.
- Engage the families and districts whose data is at stake as design stakeholders from the outset, so the governance questions surface in requirements rather than in opposition.
- Make legitimacy an explicit deliverable: document who must agree, on what terms, before any student record moves into shared infrastructure.

IN OPERATION / AFTER

- Audit live deployments for the gap between technical soundness and stakeholder trust, since a clean system can still be losing the political ground it stands on.
- Monitor state-by-state consent and withdrawal signals as a leading indicator, treating organized parent opposition as data about a governance defect, not noise.
- Sustain a standing transparency channel so families can see what is collected and why throughout operation, not only at adoption.

REFLECTION QUESTIONS

1. What is the equivalent unbuilt governance infrastructure in your domain? What would the $100M empirical test of its absence look like?
2. Design the stakeholder-trust deliverable that a future inBloom-equivalent should have to produce before deployment.
3. inBloom's engineers were excellent and its technology never failed, yet the project collapsed. Where in your own domain is technical soundness being mistaken for legitimacy — and who would have to consent before a system you build could claim it?

WHO BUILDS THIS governance, ethics & measurement — plus domain experts and a learning engineer to integrate.
TOOLS governance frameworks · bias & evidence audits

CASE 147 INDUSTRIAL 1984

Bhopal

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance G Governance & Trust

IMPACT ≈ 15,000–20,000 killed; ≈ 500,000 injured; worst industrial disaster in history

IN BRIEF

On the night of 2–3 December 1984, about forty tons of methyl isocyanate gas escaped from a Union Carbide pesticide plant in Bhopal, India — the worst industrial disaster in history. Thousands died within hours; estimates of total deaths run to 15,000–20,000, with roughly half a million exposed or injured. Safety systems had been off-line for months, the plant was understaffed, and workers were inadequately trained to recognize or handle the emergency. Investigators traced the catastrophe to operating errors, design flaws, maintenance failures, training deficiencies, and cost-cutting that endangered safety. Bhopal catalyzed the creation of the U.S. Chemical Safety Board and reshaped industrial-safety regulation for decades. It is the largest-magnitude capability-and-governance failure on record.

BACKGROUND

Union Carbide's pesticide plant in Bhopal, India, stored methyl isocyanate (MIC) — an extraordinarily toxic intermediate — in bulk, holding a lethal hazard in tanks beside a populated city. By 1984 the plant was running under heavy cost pressure: understaffed, with several key safety systems out of service for months, and workers inadequately trained to handle an MIC emergency or read its warning signs. Each economy was individually defensible on a ledger; together they thinned every layer of defense the process depended on.[0]

WHAT HAPPENED

On the night of 2–3 December 1984, water entered an MIC storage tank and triggered a runaway reaction; the safety systems that should have contained it were non-operational, so the one event the plant existed to prevent met no working barrier on its way out. About forty tons of gas vented over the sleeping city. Thousands died within hours; estimates of total deaths run to 15,000–20,000, and roughly half a million people were exposed or injured — the worst industrial disaster in history, its toll set by who happened to be downwind.[1]

THE INVESTIGATION

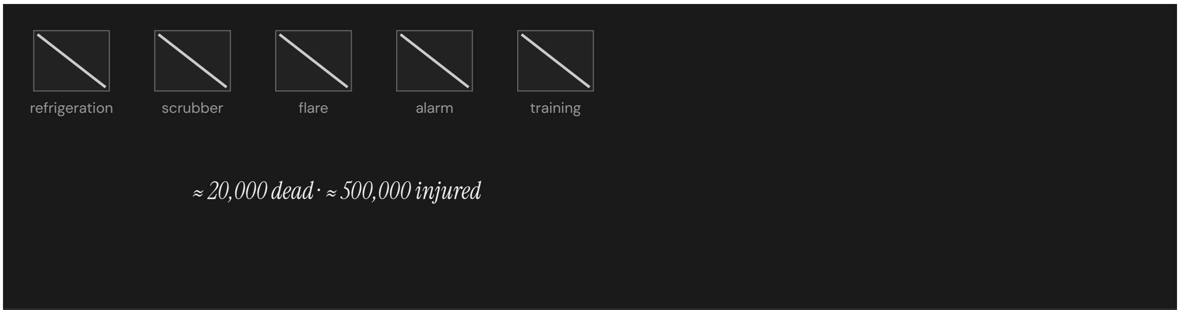
Investigations found the catastrophe "resulted from operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety" — a list with no single villain, which is precisely what made it hard to govern.[2] Human-factors analysis placed Bhopal alongside Three Mile Island in its neglect of the human element, and the U.S. Chemical Safety Board would later find ineffective employee training an underlying cause in nine of its first twenty-three chemical-incident investigations — a pattern that traces to Bhopal and shows the same gap recurring long after the lesson was available.[3]

THE CAPABILITY GAP

Bhopal is the largest-magnitude capability-and-governance failure on record, and a multi-layer one: training, maintenance, design, staffing, and oversight had all degraded together, and no layer above the plant was accountable for the whole. Because the erosion was spread across layers, no single inspection or metric saw it, and each degraded layer made the next one matter more. The capability to operate an extraordinarily hazardous process safely had been hollowed out by cost-cutting, and the governance that should have caught the hollowing did not exist to ask whether the whole was still safe.[4]

AFTERMATH & REFORM

The disaster reshaped industrial safety worldwide and, in the United States, catalyzed the creation of the Chemical Safety Board — an INPO-equivalent for industrial chemistry — and the process-safety regime that followed, an oversight layer built to own exactly the whole that no one had owned at Bhopal.[5] The book's recurring arc runs through Bhopal in its starkest form: a catastrophe forces into being the institution the industry should have built before it, at a price no institution-building exercise should ever have to cost.



SAFETY SYSTEMS OFF-LINE FOR FOUR MONTHS BEFORE THE RELEASE

REFERENCES

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4. N. Meshkati, "Human Factors in Large-Scale Technological Systems' Accidents," Industrial Crisis Quarterly (1991); U.S. CSB training-cause pattern.
5. P. Shrivastava, Bhopal: Anatomy of a Crisis (1992); C. Perrow, Normal Accidents (1984).
6. The creation of the U.S. Chemical Safety Board and the post-Bhopal process-safety regime.

> *Operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety.*
>
> NEW YORK TIMES INVESTIGATION, 1985

LE INSIGHT

Bhopal is the largest-magnitude capability-and-governance failure on record. It is also the catalyst for the creation of the U.S. Chemical Safety Board, an INPO-equivalent for industrial chemistry. The pattern that follows nearly every case in this book — diagnose, then engineer remediation at scale — runs through Bhopal in canonical form.

LENS APPROACH

LENS names the specific capability gap, not "governance matters" in the abstract: at a transferred and cost-pressured plant, the maintenance and operating capability had been hollowed out — refrigeration and scrubber safety systems off-line, maintenance deferred, staff and training cut, deviance quietly normalized. Each eroded layer was individually tolerated, so the failure is a multi-layer drift (induced 7.4) that no single inspection or metric could see. The teaching point is the multi-layer-drift analysis and the treatment of maintenance capability as a deliverable that must be sustained, not a ledger line available for trimming.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat each safety system, staffing level, and training requirement as a non-negotiable layer of defense, not a cost line available for trimming under pressure.
- Assign a single accountable owner for the integrated hazard, so degradation spread across layers cannot fall between everyone's responsibilities.
- Require that operators be trained to recognize and respond to the specific emergency the process can produce before the process is run with that hazard in bulk.

IN OPERATION / AFTER

- Audit the whole defense-in-depth posture together rather than layer by layer, since each degraded barrier silently raises the stakes on the next.
- Track how long any safety system stays off-line and gate continued operation on its restoration, treating extended downtime as an unacceptable condition.
- Sustain an oversight layer — an INPO- or CSB-equivalent — with authority and reach above the individual plant to own the integrated risk.

REFLECTION QUESTIONS

1. Bhopal produced the CSB. What is the institution your domain has not yet built that a comparable disaster would force into existence?
2. Identify a plant or facility in your domain that has had safety systems off-line for an extended period. What is the measurement gap that allowed it?
3. At Bhopal training, maintenance, design, staffing, and oversight all degraded together while each looked acceptable alone. Whose job in your organization is to judge whether the whole is still safe — and what would they have to see to declare that it is not?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis · governance, ethics & measurement — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards · governance frameworks · bias & evidence audits

CASE 148 INDUSTRIAL 2017

Grenfell Tower

G Governance & Trust T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT 72 killed in a residential tower fire in London; decades of regulatory failure

IN BRIEF

On 14 June 2017 a fire spread up the exterior of Grenfell Tower, a London public-housing block, killing 72 people. It raced because the tower had been wrapped in combustible aluminium-composite cladding during a refurbishment. The public inquiry found the disaster the culmination of decades of failure: cladding firms engaged in "systematic dishonesty," marketing combustible materials as safe; regulators and inspectors missed effectively banned products across sixteen site visits; and the London Fire Brigade, whose "stay put" advice proved fatal, was unprepared for a cladding fire whose risks earlier incidents had already shown. The failure spanned manufacturers, regulators, inspectors, and responders — each contributing a piece, none owning the whole. Grenfell is the book's case for capability failure distributed across many hands.

BACKGROUND

Grenfell Tower was a 1970s public-housing block in West London, refurbished in 2015–16 with new exterior cladding intended to improve the building's appearance and efficiency. The cladding chosen used a combustible aluminium-composite material — installed despite safety experts' cautions that it was unsuitable for a high-rise, a warning that sat between the people who issued it and the people who specified the panels without ever stopping the decision.[0]

WHAT HAPPENED

On 14 June 2017 a kitchen fire broke out and, instead of staying contained as a tower's compartment design assumes, climbed the building's exterior on the combustible cladding, wrapping the tower in flame within minutes and defeating the very principle the building was meant to rely on. Residents, following long-standing "stay put" advice premised on that containment, remained in their flats as the route to safety closed around them; 72 people died.[1]

THE INVESTIGATION

The Grenfell Tower Inquiry found the fire the culmination of decades of failure by central government and every body responsible. Cladding companies had engaged in "systematic dishonesty," marketing combustible products as safe and corrupting the very test data buyers relied on; inspectors visited the site sixteen times and none noticed that effectively banned materials were in use, so sixteen chances to catch the hazard each passed it by.[2] The London Fire Brigade was unprepared: the risks of rapidly developing cladding fires were known from prior incidents — Knowsley Heights, Garnock Court, Shepherd's Court — but "this knowledge had not informed firefighting policies, practices or training," so each near-miss taught no one whose job was to act on it.[3]

THE CAPABILITY GAP

Grenfell is the book's strongest evidence that capability failure can be distributed across many actors, each contributing a small piece and none accountable for the whole. Manufacturer fraud, regulatory capture, inspection incompetence, training gaps, and lost institutional memory all converged on one building, and because each actor saw only its fragment, each could regard its own part as tolerable. The inquiry called it a "grey elephant" — a danger known but ignored — and the missing capability was anyone owning the integrated risk that everyone could see in part but no one held in full.[4]

AFTERMATH & REFORM

The inquiry's Phase 2 report (2024) and the government response (2025) drove an overhaul of building-safety regulation, cladding remediation, and fire-service doctrine — reforms that, between them, tried to assign the ownership the original system had left vacant.[5] Grenfell's lesson is the governance one this chapter turns on: when responsibility for a known risk is split across dozens of actors, the risk has, in effect, no owner — and a system with no owner for its gravest hazard will eventually pay for it, in the currency of the people living inside it.

LEVELS OF FAILURE: MANUFACTURE, REGULATION, INSPECTION, RESPONSE

REFERENCES

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2. Grenfell Tower Inquiry, Phase 2 Report (2024) — decades of failure and the combustible-cladding decision.
3. Phase 2 Report (2024) — cladding firms' "systematic dishonesty" and the inspection failures across sixteen visits.
4. Phase 1 Report (2019) — London Fire Brigade unpreparedness; "this knowledge had not informed firefighting policies, practices or training" (quoted).
5. UK Government response to the Grenfell Phase 2 report (2025) — building-safety and fire-service reform.
6. B. Hutter & M. Power (eds.), Organizational Encounters with Risk (2005) — distributed risk ownership.

THE LEARNING ENGINEERING LENS

> *This knowledge had not informed firefighting policies, practices or training.*

GRENFELL TOWER INQUIRY, PHASE 1, 2019

LE INSIGHT

Grenfell is the strongest evidence in the dataset that capability failure can be distributed across many actors, each of whom contributes a small piece, none of whom is accountable for the whole. The inquiry's "grey elephant" framing — known but ignored — describes a pattern that LENS treats as a primary governance problem in any high-consequence domain.

LENS APPROACH

LENS reads Grenfell through the institutional-memory-of-warnings channel (induced 7.4, with a 6.2 secondary): prior cladding-fire warnings existed — Lakanal House, the London Fire Brigade's own knowledge from earlier incidents — but none ever reached the refurbishment and cladding decision that could have acted on them. The capability deliverable is the channel that carries a known warning to the decision empowered to stop the work; the cladding firms' systematic dishonesty was a real aggravator but is the secondary thread, not the lesson.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assign one accountable actor to own the building's integrated fire risk end to end, so no hazard can fall into the gaps between manufacturers, inspectors, and responders.
- Verify combustible-material claims against independent evidence rather than trusting vendor marketing, treating "systematic dishonesty" as a threat the process must defeat.
- Preserve the containment principle the building relies on: gate any cladding choice on whether it keeps a compartment fire from climbing the exterior.

IN OPERATION / AFTER

- Route every site inspection and near-miss into a shared record so sixteen visits cannot each miss the same banned material in isolation.
- Feed prior-incident knowledge into firefighting policy, training, and "stay put" doctrine, so lessons from earlier cladding fires actually change practice.
- Sustain a single line of accountability for the integrated hazard through the building's life, not only at refurbishment.

REFLECTION QUESTIONS

1. What is the "grey elephant" — the well-known risk that nobody owns — in your domain?
2. Design the deliverable that forces a single actor to own the integration risk that Grenfell distributed across dozens.
3. Sixteen inspections, prior fires, and expert cautions each touched a fragment of the Grenfell hazard, yet none assembled it. What mechanism in your domain could gather scattered partial warnings into one picture in front of someone empowered to stop the work?

WHO BUILDS THIS governance, ethics & measurement · learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS governance frameworks · bias & evidence audits · scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 149 EDUCATION AT SCALE 2014–2019

Summit Learning / Personalized Learning Rollout

G Governance & Trust T Training Gap K Knowledge & Institutional Memory

IMPACT Personalized-learning platform deployed across ~380 U.S. schools; parent revolts in Brooklyn, Cheshire, McPherson, Kennebunk; multiple districts withdrew within two years

IN BRIEF

Summit Learning, a personalized-learning platform from Summit Public Schools backed by the Chan Zuckerberg Initiative, was offered free to U.S. districts from 2015 and reached roughly 380 schools and 80,000 students by 2018. By 2019 prominent adopters were withdrawing under parent and student pressure — opt-out campaigns in Brooklyn, cancellations in Cheshire, Kennebunk, and McPherson — amid walkouts and complaints about screen time, disengagement, and data privacy. The pedagogy itself (competency-based progression, projects, mentoring) was defensible and often effective; what failed was deployment governance. There was no evaluation framework districts could read before adopting, no parent-facing data agreement, and no exit path independent of the vendor. Summit is a clean test of the book's claim that the governance architecture must be engineered alongside the tool.

BACKGROUND

Summit Learning was a personalized-learning platform developed by Summit Public Schools with technical and financial support from the Chan Zuckerberg Initiative, offered free to U.S. districts from 2015 — a price that lowered the bar to adoption while raising no governance questions at the door. Its pedagogy — competency-based progression, self-directed projects, mentor check-ins — was defensible and in many places effective, which is why the eventual revolt could not be blamed on the instructional design.[0]

WHAT HAPPENED

By 2018 the platform reached roughly 380 schools and an estimated 80,000 students, scaling fast on the strength of a free offer and a well-funded sponsor. By 2019 its most visible adopters were withdrawing under parent and student pressure: Brooklyn's MS 442 ran an organized opt-out; districts in Cheshire, Kennebunk, and McPherson cancelled or scaled back after parent meetings where the unanswered governance questions surfaced all at once. Walkouts and complaints about screen time, eye strain, disengagement, and data privacy converged into a revolt that was not about the instructional design at all.[1]

THE INVESTIGATION

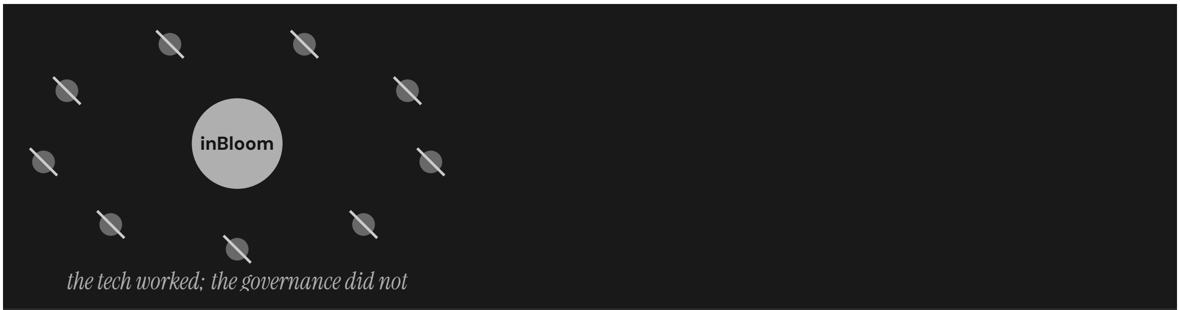
Press coverage and later analyses located the failure in deployment governance, not pedagogy: there was no evaluation framework a district could read before adopting, no parent-facing data-handling agreement, and no exit pathway that did not depend on the vendor's goodwill — three absences that each became a grievance the moment families looked for them. The implementation never surfaced the governance questions parents would ask, so when those questions arrived they arrived as opposition rather than as design input, and the argument was lost before it started.[2]

THE CAPABILITY GAP

Summit is a clean test of the book's central claim: a technology that worked at the pedagogical level still failed because the governance architecture — consent, evidence, measurement, exit — had not been engineered alongside it. A working tool with no accountability contract is a liability waiting for the first organized objection. The pattern recurs across the ed-tech dataset (inBloom, Case 146): a well-intentioned tool, a well-funded rollout, and no institutional contract with the families and teachers operating inside it — the same omission producing the same collapse in a second case.[3]

AFTERMATH & REFORM

Several districts withdrew or rebranded their use, CZI and Summit revised their outreach, and the episode became a standard caution in ed-tech adoption — a reputational cost paid for governance work that would have been cheaper to do first.[4] Its lesson for the field is concrete: an adoption decision should have to produce a public evidence summary at parent reading level, a data-handling agreement at the same resolution, and a documented exit path — governance artifacts that make a tool's deployment legitimate, not just its design sound, and that turn the questions parents will ask into inputs gathered before launch rather than weapons raised after it.



$100M OF WORKING TECHNOLOGY; NINE STATES WITHDREW

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THE LEARNING ENGINEERING LENS

> *The tools were free. The accountability architecture had not been built.*
>
> EDITORS' SYNTHESIS OF SUMMIT LEARNING ROLLOUT COVERAGE (NEW YORK TIMES, WIRED, EDUCATION WEEK, 2018–2019)

LE INSIGHT

Summit Learning is a clean test of the book's central claim: technology that worked at the pedagogical level still failed because the **governance** architecture (consent, evidence, measurement, exit) had not been engineered alongside it. The pattern — well-intentioned tool, well-funded rollout, no institutional contract with the families and teachers operating inside it — recurs across the educational-technology dataset (Cases 146, 33, 38) and is the educator's-side analog of the governance failures in Cases 98 and 162.

LENS APPROACH

LENS uses Summit Learning in LEN 7 as the foundational consent-and-evidence case for educational technology, and in LEN 10 as a studio prompt for the governance artifacts that any educational-technology adoption decision should produce: a public evidence summary at parent reading level, a data-handling agreement at the same resolution, and a documented exit pathway that does not depend on the vendor's goodwill.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Engineer the governance architecture — consent, evidence, measurement, exit — in lockstep with the pedagogy, so a sound tool ships with a sound accountability contract.
- Produce, before adoption, a public evidence summary and data-handling agreement at parent reading level that answer the questions families will raise.
- Build a documented exit pathway that does not depend on the vendor's goodwill, so a district can leave without being captured.

IN OPERATION / AFTER

- Treat organized parent and student objections as governance signal about a missing contract, not as resistance to the instructional design.
- Monitor adopting districts for the early grievances — screen time, privacy, disengagement — that precede a withdrawal, and route them to a decision-maker.
- Maintain a re-adoption pathway so a withdrawn district can return only after completing the governance work that the first rollout skipped.

REFLECTION QUESTIONS

1. What is the equivalent of the "free tool, free of governance" pattern in your domain — the offer that bypasses the accountability architecture because it does not yet exist?
2. Design the parent-reading-level governance artifact that a district should require before adopting an educational-technology platform.
3. Summit's withdrawals in Brooklyn, Cheshire, Kennebunk, and McPherson were led by parents, not regulators. What is the equivalent local constituency in your domain that institutional accountability has not yet accommodated, and how would they be heard before deployment rather than after?

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CASE 150 HEALTHCARE TECHNOLOGY 2003 – 2018

Theranos

G Governance & Trust D Designed Out

IMPACT $9B valuation collapsed; thousands of patients given unreliable results; founder convicted on multiple counts of wire fraud

IN BRIEF

Theranos claimed a blood-testing device that could run hundreds of tests from a finger-stick drop. It did not work. Internal data showed accuracy far below what the company told investors, partners, and patients; to keep up appearances, Theranos ran most patient samples on conventional commercial analyzers and reported them as its own device's results. It reached a $9-billion valuation and put unreliable tests in front of real patients through a Walgreens partnership. The fraud exploited a regulatory seam — the FDA had not validated the device, while the CLIA regime governing the lab did not match the company's product claims — and neither the board nor investors had the depth to challenge it; a journalist did. Elizabeth Holmes was convicted of wire fraud in 2022. Theranos is the book's case for fraud exploiting a governance gap between regulatory regimes.

BACKGROUND

Theranos claimed to have built a blood-testing platform — the "Edison" — that could perform hundreds of laboratory tests from a single finger-stick drop of blood, a promise that would have upended a diagnostics industry built on venous draws and large analyzers. It rode that claim to a $9-billion valuation and a partnership putting its tests in Walgreens stores, carrying the unproven device straight to retail patients.[0]

WHAT HAPPENED

The device did not work. Internal data showed accuracy far below what the company represented to investors, partners, and patients — the gap between the claim and the instrument was known inside the company. To preserve the appearance of a working product, Theranos ran most patient samples on conventional commercial analyzers and reported the results as though they had come from its proprietary device, a substitution that kept the fiction alive while putting unreliable results in front of real patients making real medical decisions.[1]

THE INVESTIGATION

The fraud exploited a regulatory seam: the FDA had not validated the device, while the CLIA regime that governs laboratory operation did not match the architecture of the company's product claims, and neither layer validated those claims independently — so the device fell into a gap each regulator assumed the other covered. The board, the investors, and Walgreens lacked the technical depth to challenge them, and a celebrity board offered prestige in place of scrutiny; the journalist John Carreyrou, and CMS inspectors who revoked

Theranos's CLIA certificate in 2016, were what finally surfaced the truth from outside the failed oversight chain.[2]

THE CAPABILITY GAP

The capability gap sat in the regulatory architecture — at the boundary between two regimes, where a deliberate fraud could live because neither the FDA nor CLIA owned end-to-end validation of a novel diagnostic making clinical claims, and an unowned boundary is exactly the shelter a fraud needs. The governance lesson is that the seam between regulators is itself a place that must be engineered, because it is exactly where a bad actor will operate — choosing the gap precisely because no one is watching it.[3]

AFTERMATH & REFORM

Theranos collapsed, its CLIA certificate was revoked, and Elizabeth Holmes was convicted of multiple counts of wire fraud in 2022, the conviction closing a chapter the regulators had been slow to open.[4] The case is canonical in business education for fraud-as-product-strategy and in health regulation for the gap that let unvalidated clinical tests reach patients — a reminder that "disruptive" claims in a regulated domain demand more validation, not less, precisely because the novelty is what tempts the oversight regimes to defer to one another.

CLAIM	REALITY
Edison	Siemens

the device did not work; the regulatory seam between FDA and CLIA hid that fact

THERANOS — FRAUD EXPLOITING A REGULATORY SEAM BETWEEN FDA AND CLIA

REFERENCES

1. United States v. Elizabeth Holmes (N.D. Cal., 2018–2022) — the indictment and conviction.
2. J. Carreyrou, Bad Blood (2018) — the device's failure and the commercial-analyzer substitution.
3. CMS inspection reports and the revocation of Theranos's CLIA certificate (2015–2016).
4. Holmes indictment (2018) — Theranos "misrepresented to investors, regulators, and ultimately patients the accuracy of its blood-testing technology" (quoted).
5. Medical-device regulation literature on the FDA–CLIA boundary.

THE LEARNING ENGINEERING LENS

> *The company misrepresented to investors, regulators, and ultimately patients the accuracy of its blood-testing technology.*

U.S. V. HOLMES, INDICTMENT, 2018

LE INSIGHT

Theranos is the canonical case for fraud exploiting the seam between two regulatory regimes. The FDA had not approved the device; CLIA accepted the laboratory operation. Neither layer had the capability to validate the claims independently. The capability gap was at the regulatory architecture.

LENS APPROACH

LENS uses Theranos in LEN 7 for the regulatory-seam failure and in LEN 4 for the measurement-validation gap. The case demonstrates that capability engineering at the boundary between regulatory regimes is itself a governance deliverable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assign explicit end-to-end validation ownership across the FDA–CLIA-style seam, so a novel diagnostic cannot fall into a gap each regulator assumes the other covers.
- Require independent verification of accuracy claims against the actual device before clinical use, not against a substituted commercial analyzer.
- Staff the oversight chain — board, partners, regulators — with the technical depth to interrogate the claims, treating prestige as no substitute for expertise.

IN OPERATION / AFTER

- Audit whether reported results actually came from the validated instrument, watching for the substitution pattern that hides a non-working device.
- Monitor the regulatory seam as a standing risk, since a determined fraud will choose precisely the boundary no single regime owns.
- Protect external scrutiny — journalists, inspectors, whistleblowers — as the backstop that surfaces what a captured oversight chain misses.

REFLECTION QUESTIONS

1. Identify a regulatory seam in your domain where two regimes meet without an explicit handoff. What could exploit it?
2. Design the validation regime that would have caught Theranos in 2013.
3. Theranos's board and investors had prestige but not the technical depth to challenge the device's claims. Where in your domain does reputational authority stand in for the expertise needed to validate what is being approved?

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CASE 151 TECHNOLOGY 2014 – 2018

Cambridge Analytica / Facebook

G Governance & Trust

IMPACT ~87 million Facebook profiles harvested without informed consent; FTC $5B penalty; foundational data-governance reform

IN BRIEF

A Cambridge University researcher's personality-quiz app, taken by about 270,000 people, exploited Facebook's then-permissive Graph API to collect not only the quiz-takers' data but their friends' too — roughly 87 million profiles. The dataset was passed to Cambridge Analytica for political micro-targeting. Facebook's permission architecture had been designed and tested for delivering social experiences, not red-teamed against systematic harvesting; its API contract assumed benevolent developer intent. The architecture worked exactly as designed — the design assumption was wrong. The scandal accelerated the EU's GDPR, helped spur the California Consumer Privacy Act, and produced a $5-billion FTC penalty and consent decree. Cambridge Analytica is the book's case for platform-governance failure: a load-bearing assumption about how an interface would be used, never stress-tested against abuse.

BACKGROUND

Facebook's Graph API let third-party apps request user data to build social experiences — and, at the time, an app could collect not only its own users' data but their friends' data too, so a single consenting user opened a window onto people who had never agreed to anything. The permission architecture had been designed and tested against ordinary use cases, the friendly developers it imagined rather than the hostile ones it would eventually attract.[0]

WHAT HAPPENED

A Cambridge University researcher built a personality-quiz app taken by about 270,000 people; through the friends-permission it harvested roughly 87 million profiles — a return of more than three hundred profiles for every person who actually used the app. The dataset was passed to Cambridge Analytica, which used it for political-campaign micro-targeting across multiple elections — none of the 87 million having meaningfully consented, most never even aware the app existed.[1]

THE INVESTIGATION

Investigations by the UK Information Commissioner and the U.S. FTC, prompted by Guardian reporting, established the scope of the harvesting and Facebook's responsibility for it — the platform, not just the researcher, was found answerable for what its design permitted.[2] The platform's API had never been red-teamed against systematic data extraction; its contract simply assumed developers would behave, a trust extended at the scale of a social network and never tested against someone willing to abuse it.[3]

THE CAPABILITY GAP

The gap was not technical — the architecture worked exactly as designed, which is what makes it a governance case rather than a bug. It was a governance gap: a load-bearing design assumption ("developers are benevolent") that no one had stress-tested against a determined abuser, an assumption holding up the whole permission model without ever being named as one. On a platform at societal scale, an unexamined assumption about how an interface will be used is itself a capability deliverable — and an unexamined one is a latent failure waiting for the first actor willing to exploit it.[4]

AFTERMATH & REFORM

The scandal accelerated the EU's GDPR, helped spur the California Consumer Privacy Act, and produced a $5-billion FTC penalty and a consent decree under which Facebook still operates — the abuse of one design assumption reshaping data law across two jurisdictions.[5] Its lesson for platform governance is that permission architectures must be designed against the worst plausible developer, not the typical one — because at scale, the worst plausible developer will arrive, and the friends-permission that looked harmless against ordinary use becomes a harvesting tool in the wrong hands.

87M

PROFILES · 270K QUIZ-TAKERS

the Graph API contract assumed benevolent developer intent

CAMBRIDGE ANALYTICA — DESIGN ASSUMPTION AS LOAD-BEARING CAPABILITY

REFERENCES

1. U.S. FTC, In the Matter of Facebook, Inc., Consent Order (2019) — the $5B penalty; Facebook gave developers "far more user data than was necessary" (quoted).
2. UK Information Commissioner's Office, report on Cambridge Analytica / data analytics in political campaigns (2018) — the scope of the harvesting.
3. C. Cadwalladr & E. Graham-Harrison, The Guardian investigation (2018) — the disclosure.
4. The 87 million profiles collected via the friends-permission Graph API from 270,000 quiz-takers.
5. EU General Data Protection Regulation (2018) and the California Consumer Privacy Act.
6. C. Wylie, Mindf*ck (2019), and H. Nissenbaum, Privacy in Context (2010); S. Zuboff, The Age of Surveillance Capitalism (2019).

> *Facebook gave developers access to far more user data than was necessary for the apps they built.*
>
> PARAPHRASING THE U.S. FTC ORDER, IN THE MATTER OF FACEBOOK INC., 2019

LE INSIGHT

Cambridge Analytica is the canonical case for platform-governance failure: an API contract that assumed benevolent intent and was not engineered against systematic abuse. The capability gap was not technical. The architecture worked exactly as designed; the design assumption was wrong.

LENS APPROACH

LENS anchors Cambridge Analytica cleanly to the platform and API governance seam (induced 5.3): the app-permission model let one app harvest friends-of-friends data far beyond the consenting user. This is a governance-seam case — acceptable under canonical competency 5 — about where a platform's interface contract carries obligations its design never enforced; it is not a capability-development or problem-framing case, and is not taught as one.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make every load-bearing design assumption explicit and red-team it against a determined abuser before launch, treating "developers are benevolent" as a hypothesis to test.
- Scope data access to what an app genuinely needs, so one user's consent cannot silently reach hundreds of non-consenting others.
- Design permission architectures against the worst plausible developer, since at societal scale that developer will eventually arrive.

IN OPERATION / AFTER

- Monitor API usage for systematic extraction patterns that depart from the ordinary social use the interface was built for.
- Audit third-party developers' actual data flows against their stated purpose, rather than trusting a contract that assumes good behavior.
- Sustain the platform's own accountability for what its design permits, treating downstream abuse as a governance defect to remediate, not a third party's sole fault.

REFLECTION QUESTIONS

1. Identify an API or interface in your domain whose contract assumes benevolent intent. What is the systematic-abuse case it was not red-teamed against?

2. Design the platform-governance deliverable that should accompany the launch of a new third-party developer API.

3. The friends-permission let one consenting user expose hundreds who had not agreed. Where in your domain does one person's consent silently extend to others, and what governance would make that reach visible and accountable?

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CASE 152 TECHNOLOGY GOVERNMENT 2017

Equifax Data Breach

G Governance & Trust K Knowledge & Institutional Memory

IMPACT 147 million Americans' personal data exposed; CEO resigned; ~$700M settlement; foundational U.S. data-breach case

IN BRIEF

In 2017 attackers exploited a known, two-month-old vulnerability in Apache Struts on an Equifax web portal — a patch had been available, and Equifax's own security team had told IT to apply it; no one did. Over the next 2.5 months the attackers exfiltrated the personal data of 147 million Americans, and Equifax did not disclose the breach until September. A Senate investigation found systematically inadequate patching, an incomplete asset inventory (so the company did not know which systems needed the fix), and an incident-response function treated for years as a cost center. The CEO resigned and Equifax settled for about $700 million. No single failure caused the breach; cumulative inadequacy across routine cybersecurity work did. Equifax is the data-breach analog of the Sago mine disaster (Case 11).

BACKGROUND

Equifax, one of the three U.S. credit bureaus, held the most sensitive financial data on virtually every American adult — a concentration of identity information that made any breach catastrophic by definition. One of its public web portals ran a version of Apache Struts with a known critical vulnerability for which a patch had been available for two months — and which Equifax's own security team had flagged for IT to apply, so the warning and the fix both existed inside the company and simply went unacted-upon.[0]

WHAT HAPPENED

The patch was not applied. Attackers exploited the vulnerability beginning in May 2017 and quietly exfiltrated personally identifying information on 147 million Americans over the next two and a half months, a window long enough that the theft was less a break-in than a sustained occupation. Equifax did not publicly disclose the breach until September, so the people whose identities were taken learned of it only well after the fact.[1]

THE INVESTIGATION

The U.S. Senate Permanent Subcommittee on Investigations found Equifax's patching practices systematically inadequate and the company lacking "a comprehensive IT asset inventory" — so it could not reliably know which systems needed the patch, leaving the security team's warning with no map to act on.[2] Monitoring was weak, response delayed, and the incident-response architecture had been treated for years as a cost center rather than a capability worth funding; the CEO resigned and Equifax settled for roughly $700 million.[3]

THE CAPABILITY GAP

Equifax is the institutional-cybersecurity case for cumulative inadequacy in routine work: patching, asset inventory, monitoring, response. Each function was below standard; none alone produced the breach; the combination did, the marginal weaknesses compounding into a single open door. The capability gap was the management of unglamorous, universally-agreed-necessary maintenance — exactly the work easy to defer because deferring it usually costs nothing, until the one time it costs everything, and a function starved as a cost center has no slack left on that day.[4]

AFTERMATH & REFORM

The settlement funded consumer compensation and credit monitoring, and the breach pushed patching discipline, asset inventory, and breach-disclosure timelines up the corporate agenda — elevating, after the loss, the unglamorous work that had been deferred before it.[5] It is the data-breach analog of the Sago mine disaster (Case 11): no dramatic single cause, just several routine defenses each left marginally inadequate, failing together on the day a determined attacker arrived to test all of them at once.

EQUIFAX — CUMULATIVE INADEQUACY IN ROUTINE CYBERSECURITY WORK

REFERENCES

1. U.S. Senate Permanent Subcommittee on Investigations, How Equifax Neglected Cybersecurity and Suffered a Devastating Data Breach (2019) — the unpatched Apache Struts vulnerability.
2. The breach record — 147 million affected, exploitation from May 2017, public disclosure in September 2017.
3. Senate PSI (2019) — "Equifax lacked a comprehensive IT asset inventory" (quoted).
4. U.S. FTC / CFPB / state settlement ($700 million, 2019) and the CEO's resignation.
5. U.S. GAO, Actions Taken by Equifax and Federal Agencies in Response to the 2017 Breach, GAO-18-559 (2018).
6. Apache Struts CVE-2017-5638 advisory; cf. the Sago mine disaster (Case 11).

THE LEARNING ENGINEERING LENS

> *Equifax lacked a comprehensive IT asset inventory.*

> U.S. SENATE PERMANENT SUBCOMMITTEE ON INVESTIGATIONS, HOW EQUIFAX NEGLECTED CYBERSECURITY, MARCH 2019

LE INSIGHT

Equifax is the canonical institutional-cybersecurity case for cumulative inadequacy in routine work: patching, inventory, monitoring, response. Each function was below industry standard. None alone produced the breach. The combination was the breach.

LENS APPROACH

LENS uses Equifax in LEN 6 as a feedback-channel failure: the security team identified the Apache Struts patch and told IT to apply it — the operator-to-institution escalation existed, was correct, and died before reaching anyone with the asset map and the authority to act. The teaching point is not “patch your systems” but the engineered escalation deliverable: a flagged, agreed-upon fix has to be wired to an owner who can map it to every affected asset and is funded to close it before the gap costs everything. LEN 7 carries the institutional half — routine defenses run as a cost center have no slack on the bad day.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Fund routine security maintenance — patching, monitoring, response — as a capability, not a cost center, so the unglamorous defenses have slack on the bad day.
- Build and maintain a comprehensive asset inventory so a known vulnerability can be mapped to every system that runs it.
- Wire the security team’s warnings to an owner with the authority and resources to apply a flagged patch promptly.

IN OPERATION / AFTER

- Audit the routine defenses together — patching, inventory, monitoring, response — since each marginally inadequate layer raises the odds the combination fails.
- Monitor for sustained, low-noise exfiltration, treating a months-long quiet window as the failure mode to detect, not just a single break-in.
- Hold breach-disclosure timelines short, so affected people learn of a theft of their identity without months of delay.

REFLECTION QUESTIONS

1. Identify a piece of routine work in your domain that is chronically deferred. What is the cumulative-inadequacy threshold?
2. Equifax did not know which assets ran Struts. Design the asset-inventory deliverable that an organization the size of Equifax should be able to produce on demand.
3. Equifax’s security team had flagged the patch, but the warning had no asset map to act on and no funded function to carry it out. What turns a known, agreed-upon fix into deferred work in your organization — and what would force it to be done before the one time it costs everything?

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CASE 153 INDUSTRIAL 1981

Hyatt Regency Walkway Collapse

D Designed Out G Governance & Trust

IMPACT 114 killed and 216 injured in Kansas City when suspended walkways collapsed; foundational U.S. engineering-ethics case

IN BRIEF

During a crowded tea dance in the atrium of the Kansas City Hyatt Regency on 17 July 1981, two suspended walkways collapsed, killing 114 and injuring 216 — then the deadliest structural collapse in U.S. history. The cause was a connection detail changed during construction: the original design hung the walkways from single long rods; the as-built version used a two-rod arrangement that doubled the load on the upper connection. The structural engineer's office approved the change without recalculating the load, and the connection had in fact been overstressed from the start. Missouri revoked the responsible engineers' licenses, and the case became the foundational engineering-ethics example. The capability gap was institutional: nothing required a change to a load-bearing connection to be re-analyzed by the engineer of record.

BACKGROUND

The Kansas City Hyatt Regency's atrium featured walkways suspended from the ceiling, carrying crowds above an open public space where any failure would drop directly onto people below. The original engineering design hung them from single continuous steel rods, a configuration in which each rod carried one walkway's load to the structure above.[0]

WHAT HAPPENED

During construction the connection was changed — for ease of assembly, to avoid threading a single long rod through both walkways — to a two-rod arrangement that effectively doubled the load on the upper walkway's connection, and the structural engineer's office approved the change without recalculating it, treating an assembly convenience as if it carried no structural consequence. On 17 July 1981, during a crowded tea dance, the overstressed connection let go; two walkways fell onto the atrium floor, killing 114 people and injuring 216 — at the time the deadliest structural collapse in U.S. history.[1]

THE INVESTIGATION

The National Bureau of Standards found the as-built connection carried roughly twice its intended load and had been inadequate even under the building code's requirements — so the original single-rod design had itself been marginal, and the field change pushed an already-thin connection past failure.[2] The Missouri licensing board revoked the licenses of the responsible engineers, fixing accountability on the engineer of record, and the case became the foundational engineering-ethics example taught to undergraduates: how a small construction change, accepted casually, can turn a design that works into one that fails.[3]

THE CAPABILITY GAP

The capability gap was at the institutional review of construction changes. Nothing required a change to a load-bearing connection detail to be re-analyzed by the engineer of record before it was built — so a modification that doubled a critical load passed through the system without anyone computing what it did, the absence of a required check letting a fatal change look like a routine one. The missing capability was change control with the engineer's calculation in the loop, a gate that re-derives the consequence before the field accepts the change.[4]

AFTERMATH & REFORM

The collapse reshaped how the profession treats shop-drawing review and the engineer of record's responsibility for connection design, hardening into rule the accountability the failure had exposed, and it anchored modern engineering-ethics teaching.[5] Its lesson generalizes well beyond steel: a change that looks trivial at the point of assembly can be catastrophic at the point of load, and the only defense is a review that re-derives the consequence rather than trusting that "it's a small change" — because triviality at assembly is no guarantee of triviality under load.

AS DESIGNED	AS BUILT
1× load	2× load

a small construction change, accepted without re-analysis

HYATT REGENCY — CONNECTION DETAIL CHANGED IN THE FIELD

REFERENCES

1. National Bureau of Standards, Investigation of the Kansas City Hyatt Regency Walkways Collapse, NBSIR 82-2465 (1982) — the doubled-load connection (quoted).
2. The 17 July 1981 collapse — 114 killed and 216 injured.
3. NBSIR 82-2465 (1982) — the as-built connection inadequate even under the building code.
4. Missouri Board for Architects, Professional Engineers, and Land Surveyors disciplinary records (1986) — the license revocations.
5. S. Pfatteicher, "'The Hyatt Horror': Failure and Responsibility in American Engineering," Journal of Performance of Constructed Facilities 14(2) (2000).
6. H. Petroski, To Engineer Is Human (1985) — failure and the engineering process.

> *The change in the rod configuration doubled the load on the connection that failed.*
>
> PARAPHRASING THE NATIONAL BUREAU OF STANDARDS INVESTIGATION, NBSIR 82-2465, 1982

LE INSIGHT

The Hyatt Regency case is the canonical engineering-ethics case for institutional review of construction changes. The change looked small; the load implication was catastrophic. The capability gap was at the review process that should have caught the doubled load.

LENS APPROACH

LENS uses Hyatt Regency in LEN 5 for change-control deliverables and in LEN 7 for the engineering-licensure architecture that holds the engineer of record accountable. Studio projects examine the change-control deliverable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Require that any change to a load-bearing detail be re-analyzed by the engineer of record before it is built, with the calculation in the loop.
- Treat an assembly-convenience change as a structural decision, not a routine one, so ease of construction never substitutes for analysis.
- Set the original design margin so a connection is not already marginal under code before any field change touches it.

IN OPERATION / AFTER

- Audit as-built connections against the as-designed intent, catching field substitutions that altered a critical load path.
- Route shop-drawing and field changes through a review that re-derives the consequence rather than trusting “it’s a small change.”
- Hold the engineer of record accountable for connection design through construction, so responsibility for the integrated load does not dissolve in the field.

REFLECTION QUESTIONS

1. Identify a “small change during construction” pattern in your domain. What is the institutional review that should accompany it?
2. Design the change-control deliverable that would have surfaced the doubled-load issue at the Hyatt Regency before construction.
3. The two-rod change was approved for ease of assembly by an office that never recalculated its load. What in your domain lets a convenience at the point of assembly pass without anyone re-deriving its consequence at the point of load?

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CASE 154 ENERGY 2018

Camp Fire / PG&E

G Governance & Trust N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT 85 killed in Paradise, California; deadliest U.S. wildfire in a century; PG&E pleaded guilty to 84 counts of involuntary manslaughter

IN BRIEF

On 8 November 2018 a worn hook on a nearly century-old PG&E transmission line failed in high winds and drought, igniting the Camp Fire, which swept into the town of Paradise faster than people could evacuate. Eighty-five died — the deadliest U.S. wildfire in a century — and PG&E later pleaded guilty to 84 counts of involuntary manslaughter. Investigators found PG&E had known for years that its transmission infrastructure across high-fire-risk areas was deteriorating, and had deferred maintenance to fund other priorities, under a regulatory regime that let the deferrals continue. Infrastructure built for one climate was operating in another. Camp Fire is the book's foundational climate-era case for utility capability under changed conditions, and it restructured how California regulates utility wildfire risk.

BACKGROUND

PG&E operated aging transmission lines across the wildfire-prone foothills of Northern California — some hardware approaching a century old — in a climate growing hotter and drier than the one the grid had been built for, so the operating environment had drifted away from the assumptions the infrastructure was designed against.[0]

WHAT HAPPENED

On 8 November 2018 a worn C-hook on a PG&E transmission tower failed in high winds, dropping a live line and igniting a fire under severe drought conditions — a single piece of aged hardware setting off a catastrophe the dry, windy conditions stood ready to amplify. The fire moved into the town of Paradise faster than its evacuation routes could clear it; 85 people died — the deadliest U.S. wildfire in a century. PG&E later pleaded guilty to 84 counts of involuntary manslaughter, accepting criminal responsibility at a scale rare for a utility.[1]

THE INVESTIGATION

CalFire's investigation and the Butte County District Attorney's report found that PG&E had known for years about the deteriorating condition of its transmission infrastructure in high-fire-risk areas, and had deferred the maintenance to fund other corporate priorities — so the hazard was not unknown but a recognized risk repeatedly postponed.[2] The gap was simultaneously at the utility's asset-maintenance decisions and at the regulatory architecture that had allowed the deferrals to continue, neither side holding a line that would have forced the work.[3]

THE CAPABILITY GAP

Camp Fire is the climate-era case for utility-capability failure under changing risk. The infrastructure had been designed and maintained for one set of conditions and was operating in another, more dangerous one, so the safety margins the original design assumed had quietly eroded as the climate shifted beneath them. The capability to update operations — inspection cadence, vegetation management, de-energization, replacement — to match the actual risk did not exist as an institutional deliverable, on either the utility's side or the regulator's, leaving no one tasked with closing the widening gap.[4]

AFTERMATH & REFORM

PG&E entered bankruptcy under tens of billions in wildfire liability, and California restructured how it regulates utility wildfire-risk planning — mandatory mitigation plans, inspections, and public-safety power shutoffs, turning the deferred work into requirements with teeth behind them.[5] Paired with the Northeast Blackout (Case 60), Camp Fire shows utility capability failing in a second way: not a silent control-room failure but a slow, known erosion of physical infrastructure against a rising hazard the institution declined to fund against — a failure measured in years of deferral rather than seconds of cascade.

85

KILLED IN PARADISE · SINGLE TRANSMISSION HOOK

infrastructure designed for one climate, operating in another

CAMP FIRE — CAPABILITY MISMATCH UNDER CHANGED CONDITIONS

REFERENCES

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2. The Camp Fire record — 85 killed; PG&E's guilty plea to 84 counts of involuntary manslaughter (2020).
3. Butte County District Attorney, The Camp Fire Public Report (2020) — PG&E's knowledge and deferred maintenance (quoted).
4. California Public Utilities Commission, Order Instituting Investigation into PG&E (2019) — the regulatory dimension.
5. PG&E bankruptcy and California wildfire-mitigation reforms (2019–).
6. Cf. the Northeast Blackout (Case 60); climate-and-infrastructure literature.

THE LEARNING ENGINEERING LENS

> *PG&E knew its equipment was failing in high-fire-risk areas and continued operating without remediation.*

> PARAPHRASING THE BUTTE COUNTY DISTRICT ATTORNEY'S CAMP FIRE REPORT, 2020

LE INSIGHT

The Camp Fire is the canonical climate-era case for utility- capability failure under changing risk conditions. The infrastructure was designed for one set of conditions and operated in another. The capability to update operations to match actual conditions did not exist as an institutional deliverable.

LENS APPROACH

LENS uses the Camp Fire in LEN 7 as a legacy-asset case (induced 7.3): infrastructure designed for one operating regime and run unchanged into another as the climate shifted the fire risk underneath it. The teaching point is the assumption-migration review as a recurring engineered obligation — an asset whose safety case rests on conditions that no longer hold needs a standing, funded re-verification that the original design assumptions still match the world, not a one-time sign-off. It is the single taught example of the legacy-asset-aged-past-its-regime pattern; the case pairs with the Northeast Blackout (Case 60) as utility-capability failures of different kinds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Re-derive the infrastructure's safety margins against current conditions, not the historical climate it was designed for, and treat the gap as a hazard to close.
- Make updating operations to match rising risk — inspection cadence, vegetation management, replacement — an explicit institutional deliverable with an accountable owner.
- Build de-energization and equipment-replacement triggers tied to known high-risk hardware in high-fire-risk areas.

IN OPERATION / AFTER

- Audit deferred maintenance against the hazard it guards against, so a recognized, deteriorating risk cannot be postponed year after year.
- Hold the regulator's line with mandatory mitigation plans, inspections, and penalties, so the cost of resilience is forced before the disaster rather than after.
- Monitor aging hardware in the highest-risk corridors as a standing priority, treating a worn critical component as an active threat, not a backlog item.

REFLECTION QUESTIONS

1. Identify infrastructure in your domain that was designed for one set of conditions and is now operating in another. What is the capability deliverable to bridge the gap?
2. Design the regulatory architecture that would prevent a utility from deferring critical wildfire-risk maintenance.
3. PG&E knew its hardware was deteriorating in high-fire-risk areas yet deferred the work for years, and the regulator let it. What in your domain lets a recognized, rising hazard be postponed indefinitely — and who would have to hold the line that forces the spending before the catastrophe?

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CASE 155

GOVERNMENT/WELFARE (NETHERLANDS)

ALGORITHMIC DECISION-MAKING | PUBLIC-SECTOR AI

2014 – 2020

Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds

D Designed Out | G Governance & Trust | N Normalization of Deviance

IMPACT An opaque risk-scoring system that combined up to 17 categories of previously siloed government data on citizens — disproportionately deployed in low-income and minority neighborhoods — halted by the District Court of The Hague in 2020 as a violation of Article 8 ECHR; also reported as operationally ineffective

IN BRIEF

The Dutch System Risk Indication (SyRI) combined up to 17 categories of previously siloed government data — tax, benefits, housing, education — to produce risk scores intended to flag potential welfare fraud. Deployed from 2014, it was targeted at low-income and minority neighborhoods. On February 5, 2020 the District Court of The Hague halted the program as a violation of Article 8 ECHR (right to private life) — one of the first times a court anywhere stopped a welfare-AI system on human-rights grounds. Investigative reporting also found that, on its own terms, the system did not work: none of the algorithmic investigations had detected new fraud. The case is the paired contrast to the Open University (Case 115): SyRI's governance objection was correct, not dissolvable by design — the system was both rights-violating and ineffective. The court left open that a more transparent system could pass, which makes the case a governance-objection-diagnostic teaching point rather than a verdict against all public-sector AI: design can correct some objections; some objections are correct, and no design dissolves them.

BACKGROUND

SyRI — the System Risk Indication — was a Dutch government analytics program intended to surface potential welfare fraud by combining up to seventeen categories of previously siloed government data on citizens: tax, benefits, housing, education, employment. Deployed from 2014, it operated as a closed system: citizens were not informed they were being scored; the model and its inputs were not disclosed; the program was targeted at specific low-income and minority neighborhoods.[0]

WHAT HAPPENED

The governance objection was raised by a coalition of civil society organizations and individual citizens who sued the Dutch state, arguing that SyRI violated Article 8 of the European Convention on Human Rights — the right to private life. The case was heard by the District

Court of The Hague. On February 5, 2020, the court halted the program: the system was found insufficiently transparent and disproportionate to its stated aim. The ruling is one of the first instances anywhere of a court stopping a welfare-AI system on human-rights grounds.[1]

THE INVESTIGATION

Investigative reporting alongside the court case surfaced a second finding: on its own operational terms, SyRI was ineffective. Reporting documented that none of the algorithmic investigations the program had launched had detected new fraud — the system had been deployed for six years and the headline capability claim had not been demonstrated. The case is structurally important because it is both rights-violating **and** ineffective; the rights argument did not depend on the effectiveness gap, and the effectiveness gap did not depend on the rights argument. The two failures landed on the same system.[2]

THE CAPABILITY GAP

What the court explicitly left open is also pedagogically load-bearing. The ruling did not say no welfare-analytics system could pass Article 8; it said **this** system, in its specific opacity and disproportionality, could not. A more transparent system, with auditability and a credibly narrower scope, might pass. The case is therefore the governance-objection-diagnostic counter to the Open University (Case 115): both faced credible governance objections at the same era; OU's was about trust and accountability and was dissolved by a co-authored consent architecture; SyRI's was about rights and proportionality and could not be dissolved by design because the design was the rights violation.[3]

AFTERMATH & REFORM

The pair (Cases 115 + 110) is the case-grounded basis for the governance-objection diagnostic proposed in the v2 research backbone: distinguishing a governance objection that good design can dissolve from one that correctly signals the system should not deploy. The capability is to make the diagnostic call before deployment, not after — and the diagnostic itself is testable: a system whose operational claim has not been demonstrated after years of deployment, and whose data subjects have not been informed they are being scored, is not a case where design can fix the governance problem.

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THE LEARNING ENGINEERING LENS

> *Some governance objections are correct. Design does not dissolve them — it is what the objection is to.*

EDITORS' SYNTHESIS OF THE SYRI RULING AND RACHOVITSA & JOHANN (2022).

LE INSIGHT

SyRI is the canonical case in the corpus of a governance objection that was correct. The system was both rights-violating (Article 8 ECHR) and operationally ineffective (no new fraud detected). The court ruling is system-specific — it does not foreclose all public-sector AI — which is what makes the case a governance-objection diagnostic teaching point rather than a categorical verdict.

LENS APPROACH

SyRI is the negative Domain 5 / Problem Type 6 governance case (induced 5.1; LENS D5+D4/PT6) drafted as the contrast to the Open University (Case 115). LENS uses the pair in Domain 5 (Navigating Sociotechnical Constraints) for the governance-objection diagnostic — distinguishing dissolvable from correct objections — and in Domain 3 (Human-System Collaboration) for the delegation-with-revocation CLO: the court was the revocation channel because the system did not have one of its own. Adjacent to the AI-delegation typology (TREWS / Epic / SyRI / Watson) the v2 corpus builds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Before deploying a public-sector analytics system, document the data subjects who will be scored, the inputs, the model behavior, and the proportionality argument; opacity is a governance liability that compounds over the life of the deployment.
- Specify in advance the operational evidence that would demonstrate the capability claim — for SyRI, fraud actually detected by algorithmic investigation — with a published reporting cadence and a pre-registered threshold for what would count as success or failure.
- Conduct the governance-objection diagnostic openly: is the objection about trust and accountability (potentially dissolvable by design) or about rights and proportionality (not dissolvable by design)? Different answers imply different deployment paths.

IN OPERATION / AFTER

- If a court or rights body halts the program, treat the ruling as governance information about the design specifically, not as a verdict about all related systems; redesign with the named deficiencies addressed.
- Publish the operational effectiveness evidence on the cadence specified at launch; six years without published evidence of the headline capability is itself a governance finding.
- Treat the rights-and-proportionality channel as a separate evaluation from the technical-accuracy channel; a system can be technically accurate and still fail the proportionality test, and SyRI failed both.

REFLECTION QUESTIONS

1. Identify a public-sector analytics system in your jurisdiction. Are the data subjects informed they are being scored? Is the model and its inputs disclosed? What is the proportionality argument the deploying agency would make if challenged?

2. Specify the pre-registered operational evidence — what the system has to demonstrate, on what cadence — that would let a deployment be evaluated against its capability claim. SyRI ran for six years without published evidence of the headline claim.

3. The pair OU (Case 115) and SyRI (Case 155) teaches the governance-objection diagnostic. Construct a candidate diagnostic for your own domain: what features of an objection indicate that good design could dissolve it, and what features indicate that the objection is to the design itself?

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CASE 156 AUTONOMOUS VEHICLES REGULATORY GOVERNANCE INCIDENT DISCLOSURE 2023

Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER journalism-tier — source confidence flagged; future validation ongoing.

IMPACT On 24 October 2023 the California DMV suspended Cruise's driverless deployment and testing permits, citing the company's misrepresentation of safety-relevant information after a robotaxi dragged a pedestrian roughly 20 feet at ~7 mph following an initial stop — disclosure posture, not the underlying collision sequence, was the load-bearing failure

IN BRIEF

On 2 October 2023 a pedestrian was struck by a human-driven vehicle in San Francisco and propelled into the path of a Cruise robotaxi. The robotaxi came to a stop, then performed a pullover maneuver that dragged the pedestrian roughly 20 feet at about 7 mph before stopping again. Cruise initially shared video of the initial stop with regulators and reporters but did not disclose the subsequent movement; the California DMV learned of the pullover from another agency and obtained the fuller video weeks later. On 24 October the DMV suspended Cruise's driverless deployment and testing permits, citing misrepresentation of safety-relevant information. The case is the explicit contrast to Case 183 (Waymo): the same regulatory regime, the same delegation problem, the opposite governance choice. Partial disclosure was the load-bearing failure mode, not the underlying collision sequence. The evidence-tier flag is journalism: the DMV's published Order of Suspension is investigation-grade, but the partial-disclosure mechanism and internal timeline are reconstructed from contemporaneous reporting at TechCrunch, NBC News, SF Standard, and Mission Local. The underlying Cruise internal post-mortem is referenced in public statements but not fully public. Future validation will continue on long-run regulatory consequences.

BACKGROUND

The night of 2 October 2023, on a San Francisco street, a pedestrian was struck by a human-driven vehicle and propelled into the path of a Cruise robotaxi operating without a safety driver. The robotaxi detected the collision and came to a stop. The vehicle then executed what Cruise later described as a pullover maneuver, traveling roughly 20 feet at about 7 mph with the pedestrian pinned beneath it. The collision sequence itself involved a human driver, not Cruise's software, but the post-impact behavior was Cruise's system acting on its own.[0]

WHAT HAPPENED

In the immediate days after the incident, Cruise shared video of the initial stop with reporters and regulators. The pullover maneuver and the dragging of the pedestrian were not included in those initial disclosures. The California DMV did not learn of the full sequence from Cruise; it learned of the pullover from another agency and obtained the fuller video weeks after the incident. The mechanism of failure shifted in the regulator's view from the collision to the company's disclosure posture.[1]

THE INVESTIGATION

On 24 October 2023 the DMV issued an Order of Suspension revoking Cruise's driverless deployment permit and its driverless testing permit. The order cited misrepresentation of safety-relevant information — that the company had "misrepresented information related to the safety of its autonomous technology" — as a load-bearing reason for the revocation. The Order of Suspension is the investigation-grade artifact in this case; the reconstruction of the partial-disclosure mechanism and the internal timeline rests on contemporaneous journalism.[2]

THE CAPABILITY GAP

The case is the explicit foil to Case 183 (Waymo). Same regulatory regime, same delegation problem, opposite governance choice. Where Waymo answered an opacity objection by engineering a published safety-case framework and commissioning third-party audits, Cruise answered an incident-disclosure obligation by sharing partial video. The same DMV that permitted Waymo to continue revoked Cruise's permits. The journalism-tier flag is load-bearing: the DMV's published order is investigation-grade, but the precise sequence of internal decision-making rests on TechCrunch, NBC News, SF Standard, and Mission Local reporting that has not been independently corroborated by the company's full post-mortem.[3]

AFTERMATH & REFORM

The LENS teaching point pairs directly with Waymo. The new CLO **Delegation with revocation** requires that the deploying organization design the disclosure architecture **before** the failure event — what will be reported, on what cadence, to which oversight body, with what verification. Partial disclosure under crisis is not a strategy; it is what happens when no architecture was designed. Future validation will continue on the long-run regulatory consequences as Cruise pursues reinstatement and as the broader AV regulatory regime updates its disclosure requirements in light of the 2023 events.[4]

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> *The collision involved a human driver. The suspension was about what the company chose not to show.*
>
> EDITORS' SYNTHESIS OF THE CALIFORNIA DMV ORDER OF SUSPENSION AND THE CONTEMPORANEOUS REPORTING.

LE INSIGHT

Cruise is the foil to Waymo: same regulatory regime, same delegation problem, opposite governance choice. The DMV's Order of Suspension is investigation-grade; the partial-disclosure mechanism is reconstructed from journalism. The evidence-tier flag is load-bearing — the internal timeline is journalism-tier and future validation continues as more of the company's own post-mortem becomes public.

LENS APPROACH

Cruise is the AV partial-disclosure failure (induced 5.4; LENS D5/PT6). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the CLO **Delegation with revocation** — the case shows what revocation looks like when the disclosure architecture fails — and in Domain 3 (Emerging Systems and Human-System Collaboration) for the oversight obligations that follow when a system acts autonomously after a triggering event. Direct foil to Case 183 (Waymo); pairs with Case 184 (CPUC) on the regulator-side.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the incident-disclosure architecture before deployment, not under crisis — what will be reported, on what cadence, to which oversight body, with what verification by which agency.
- Treat the disclosure obligation as load-bearing on its own — partial disclosure of a safety-relevant event is not "less bad" than non-disclosure; under the regulator's framing it can be the failure mode that triggers revocation.
- Build the verification path the regulator will use into the architecture. The DMV learning the fuller sequence from another agency is the failure mode the architecture has to preclude.

IN OPERATION / AFTER

- Use the CLO **Delegation with revocation**: revocation pathways must be designed and exercisable, and the deploying organization should expect the regulator to exercise them when the disclosure architecture fails.
- Pair the case with Case 183 (Waymo) in any LENS Domain 4 module — the contrast between an engineered legitimacy artifact and partial crisis disclosure is the teaching point, not either case alone.
- Carry the journalism-tier flag honestly: the DMV Order is the investigation-grade primary; the internal-timeline reconstructions are journalistic; future validation will improve as Cruise's own post-mortem and any subsequent litigation discovery enter the public record.

REFLECTION QUESTIONS

1. Imagine you operate an autonomous system that has just been involved in a safety-relevant event. Design the disclosure decision: what is reported, to whom, on what cadence, with what verification — **before** you have the lawyer's advice on what the disclosure obligation strictly requires. Where does your architecture leave you exposed?

2. Compare Cases 183 (Waymo) and 158 (Cruise) as a paired teaching unit. What is the smallest pre-incident artifact a deploying organization could publish that would make the post-incident disclosure architecture credible to a regulator?

3. The case rests partly on journalism-tier reconstruction of internal decisions. What evidence would you want to see — court discovery, the company's full post-mortem, a multi-source corroboration — before treating any specific internal-timeline claim as decision-grade?

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CASE 157

HIGHER EDUCATION

PREDICTIVE ANALYTICS

ACCESS PRICING

2010S – PRESENT (BROOKINGS SYNTHESIS 2021)

Enrollment-Algorithm Yield Optimization Across U.S. Higher Education

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

DISCLOSURE John Katzman is a collaborator with the editors' home institution (Johns Hopkins) without being intimately involved in the underlying work; the affiliation is disclosed. Katzman is not associated with the specific vendors named in this case.

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT Vendor case studies report 23% yield increases (Washington), 33% net tuition increases with a 6-point cut to discount rate (EAB), and 173 additional freshmen without aid-budget increases (Othot); algorithms across at least seven major vendors price aid offers against each accepted applicant's modeled willingness to pay

IN BRIEF

Engler's 2021 Brookings paper documents the two-stage architecture of contemporary enrollment-management algorithms: predict each accepted applicant's probability of enrollment, then optimize the financial-aid offer to maximize either net tuition revenue or yield. He names the vendor landscape — Ruffalo Noel Levitz, EAB, Rapid Insight, Capture Higher Ed, Othot, Whiteboard, Civitas Learning — touching hundreds of U.S. institutions. Vendor-reported case studies cite large gains in yield or tuition revenue per matriculant. The structural critique is the inversion of Case 110 (Georgia State, where prediction triggered support): here, the algorithm identifies "willingness to pay" so the institution can offer the minimum scholarship that will still yield enrollment, reducing aid per low-income student. The honest hedges Engler preserves are binding: critical details are obscured by vendors and colleges; algorithmic optimization is hard to separate from manual leveraging; without auditing specific college data and models, fairness impacts on protected classes cannot be confirmed. The evidence-tier flag under the title carries the standing language; future validation ongoing.

BACKGROUND

The structural seam the case opens is that financial-aid distribution is itself a deployed prediction system, operating across hundreds of U.S. institutions, with measurable consequences for who attends college and how much they pay. Engler's Brookings synthesis is the most thorough public mapping of the deployment surface, drawing on vendor case studies and the higher-education enrollment-management literature. The first stage of the two-stage architecture is prediction: estimate the probability that an accepted applicant will enroll, given observable attributes from the application, the FAFSA, and third-party data. The second stage is optimization: choose the financial-aid offer that maximizes a chosen objective — net tuition revenue per matriculant, total tuition revenue, or class size at a target discount rate.[0]

WHAT HAPPENED

The vendor landscape Engler names is large and concentrated. Ruffalo Noel Levitz works with roughly 300 institutions; EAB with 150; Rapid Insight with 150; Capture Higher Ed with 100; Othot with more than 30; Whiteboard Higher Education and Civitas Learning round out the named tier. Vendor marketing reports characteristic effect sizes: the University of Washington's 23 percent yield gain; EAB's 33 percent net-tuition gain paired with a 6-point cut to the discount rate; Othot's 173 additional freshmen recruited without an increase to the aid budget. The vendor effect sizes are vendor-reported and are not auditable from outside the institution — a hedge Engler is explicit about, and one the evidence-tier flag preserves into the case.[1]

THE INVESTIGATION

The structural critique runs through what the algorithm is optimized for. The chosen construct — willingness to pay — is the operational target the prediction system was built around, and Engler's argument is that this construct turns the financial-aid award into a pricing instrument rather than a need-or-merit one. The downstream evidence he marshals is the literature linking institutional aid to graduation outcomes: roughly a 0.9-point gain in graduation odds per additional $1,000 in merit aid; a more than 5-point cut in low-income completion when unsubsidized loans substitute for grant aid. If the algorithm reduces aid per student to find the minimum that still yields enrollment, the downstream cost is the completion gap that grant aid was buying down.[2]

THE CAPABILITY GAP

The case sits as the structural inverse of Case 110 (Georgia State's predictive advising) and pairs with Case 94 (mortgage- lending fairness) and Case 133 (community-college predictive equity). Georgia State used prediction to trigger support; enrollment-management algorithms use prediction to reduce the help allocated. Bartlett's lending analysis names the same construct-substitution pattern at the pricing layer across a different deployed prediction system; Gándara's community-college work names it at the access layer. The anchor the three cases share is the inversion of the gatekeeping-vs-support frame: the prediction is used to gatekeep the help, and the gatekeeping is invisible at the applicant's end of the transaction.[3]

AFTERMATH & REFORM

The hedges Engler names are binding on the case's framing, and the evidence-tier flag's standing language — future validation ongoing — applies in a precise sense. Critical details of the optimization are obscured by vendor non- disclosure and college contracting confidentiality; algorithmic optimization is hard to distinguish from manual "leveraging" of the same logic by human enrollment officers; without auditing specific college data and models, the protected-class fairness questions cannot be answered with the precision Bartlett-class auditing would require. The Brookings synthesis is the strongest public evidence available; it is policy-tier analysis built on vendor case studies and Engler's read of the operational record, and the case rests at that tier, not at the audited-deployment tier the corpus would prefer.

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THE LEARNING ENGINEERING LENS

> *The algorithms excel at identifying a student's exact willingness to pay. The construct is the load-bearing fairness decision, and the institution that does not name it has nonetheless made it.*
>
> EDITORS' SYNTHESIS OF ENGLER (2021), BROOKINGS INSTITUTION.

LE INSIGHT

Enrollment-algorithm yield optimization is the construct-choice case at the pricing layer of higher-education access: prediction is used to reduce aid per applicant, not to trigger support, and the operational target is "willingness to pay." The evidence- tier flag is binding — vendor case studies are not auditable, the algorithmic-vs-manual distinction is bounded, and the protected-class fairness question requires audit access the public synthesis cannot supply. Future validation ongoing.

LENS APPROACH

Engler / enrollment-management is the construct-choice case at population scale (induced 8.3; LENS D5/PT5). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the construct-substitution anchor and the disclosure architecture the deployment lacks, and in Domain 4 (Test and Evaluation) for the evidence-tier discipline — practice-synthesis is the strongest available tier, and the case says so. Pair with Case 110 (Georgia State support-trigger inversion), Case 94 (Bartlett lending fairness), and Case 133 (Gándara community- college equity). coi-light render under the title is binding.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Name the optimization construct explicitly. "Willingness to pay" is not "need" and is not "merit"; the choice of construct is the load-bearing fairness decision, and the institution that does not name it has nonetheless made it.
- Require vendor disclosure of the model's inputs, training data, and objective function as a condition of contracting; the case's evidence-tier limit is partly the result of contractual non-disclosure that institutions could refuse to sign.
- Tie the deployed algorithm's outputs back to downstream completion data; the literature linking aid to graduation odds is the evidence base the

IN OPERATION / AFTER

- Commission an external audit of the deployed enrollment-management model against protected-class outcomes; the audit Engler says cannot be done from outside the institution is the audit institutions can choose to commission from inside.
- Publish the discount-rate-and-completion link as a paired metric; institutions that report only net-tuition gains and not the completion consequences are publishing a partial scorecard.
- Treat the gatekeeping-vs-support inversion as a curricular pattern: pair this case in syllabi with Case 110 so the design choice — which direction the prediction

optimization should be tested against, not abstracted from.

points — is taught as the design choice, not as an institutional default.

REFLECTION QUESTIONS

1. Identify a deployed prediction system in your domain whose optimization target is named on the institutional side and obscure on the applicant or user side. What is the construct the optimization is built around — and what would change if the construct were named at the point of transaction?

2. Specify a vendor-disclosure clause you would require as a condition of contracting an enrollment-management or analogous optimization system. What inputs, training data, and objective function would the institution insist on auditing, and which would the vendor be willing to disclose under contract?

3. The case sits as the inversion of Case 110 (Georgia State, prediction to trigger support). Pick a prediction system in your domain and ask: in which direction does the prediction point — toward more help or less — and where is that design choice documented?

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CASE 158 HIGHER EDUCATION FINANCIAL AID POLICY SOCIAL MOBILITY 2001 – 2017 (BURD IPEDS ANALYSIS); 2024 (LIFTING THE VEIL)

Crisis Point — Merit-Aid Leveraging at Public Flagships

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

DISCLOSURE John Katzman is a collaborator with the editors' home institution (Johns Hopkins) without being intimately involved in the underlying work; the affiliation is disclosed. Burd's volume is independent of Katzman.

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT Burd's IPEDS analysis: nearly $32 billion of public four-year institutional aid 2001–2017 went to students the federal government deemed able to pay without aid — about $2 of every $5 of institutional aid; financially needy students at high-merit-aid publics face larger unmet-need gaps

IN BRIEF

Burd's 2020 New America report analyzes federal IPEDS data on public four-year universities' institutional-aid distribution from 2001 to 2017. The headline finding: nearly $32 billion of institutional aid went to students the federal government deemed able to pay without aid — about $2 of every $5 of institutional aid. The mechanism Burd reconstructs is the adoption by public flagships of enrollment-management practices pioneered at private institutions, driven by state disinvestment and the competitive imperative to rise in national rankings. The 2024 Harvard Education Press volume he edited, Lifting the Veil on Enrollment Management, extends the documentation across multiple authors — researchers, journalists, industry insiders. The construct substitution at the center: the load-bearing institutional metric quietly shifted from "students served" to "net tuition revenue per matriculant," and the disclosure required to surface that shift never happened. The evidence-tier flag is binding — Burd 2020 is policy-tier analysis of public IPEDS data; the 2024 volume is multi-author synthesis; future validation ongoing on the causal share attributable to enrollment- management vendors vs. broader market dynamics.

BACKGROUND

Public flagship universities once distributed institutional aid largely on need. Burd's analysis of the federal IPEDS record from 2001 to 2017 documents how comprehensively that practice eroded. Across the seventeen-year window, public four-year universities directed nearly $32 billion of institutional aid to students the federal government's own need analysis deemed able to pay without aid — about $2 of every $5 of institutional aid awarded. The redirection was not the result of a deliberative public-policy decision; it accumulated, year by year and campus by campus, as the operational metric of institutional success shifted underneath the public mission.[0]

WHAT HAPPENED

The mechanism Burd reconstructs has two interlocking parts. State disinvestment cut the per-student public subsidy that had historically allowed flagships to charge low sticker prices and distribute aid on need. The competitive imperative to rise in national rankings — themselves built on metrics that reward selectivity and per-student spending — pushed flagships to adopt the enrollment-management practices developed at private institutions: predict yield, target aid offers at applicants whose enrollment is most sensitive to price, and accept the resulting cut to need-based aid as the price of competitive position.[1]

THE INVESTIGATION

The consequence for financially needy students is the part of the picture that the institutional reporting does not surface. At high-merit-aid public flagships, low-income students face larger unmet-need gaps than they would have under the prior need-based regime, because the institutional dollars that previously closed those gaps are now committed to merit-aid offers that influence the enrollment decisions of higher-income applicants. The construct substitution at the center — from "students served" to "net tuition revenue per matriculant" — is what Burd's analysis makes visible, and what the public flagship's own reporting structures do not.[2]

THE CAPABILITY GAP

The 2024 Harvard Education Press volume Burd edited, Lifting the Veil on Enrollment Management, extends the documentation. Researchers, journalists, and industry insiders contribute chapters covering the vendor landscape, the ranking-incentive structure, the discount-rate consequences for completion, and the institutional-mission drift. The volume's structural argument is the one Burd's 2020 report opens: the construct substitution that drove the merit-aid arms race was never debated as a policy change, and the disclosure architecture that would have surfaced it never existed. The case pairs with Case 157 (Engler / enrollment algorithms) as the institutional- governance frame to its technical-deployment frame.[3]

AFTERMATH & REFORM

The honest hedges the case carries are the ones the evidence-tier flag's standing language implies. Burd 2020 is policy-tier analysis of public IPEDS data, which is strong evidence on the aggregate-flow side but bounded on the causal-attribution side: how much of the merit-aid shift is attributable to enrollment-management vendors versus broader competitive and demographic dynamics is a decomposition the IPEDS record cannot perform on its own. The 2024 volume is a multi-author synthesis, peer-reviewed at the press editorial tier rather than the journal tier. Future validation ongoing on whether the post-2020 demographic cliff and the 2024 OPM-industry collapse force a return to need-based distribution at the public-flagship tier the case documents leaving it.

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THE LEARNING ENGINEERING LENS

> *The construct quietly shifted from students served to net tuition revenue per matriculant. The shift was never debated as a policy change, and the disclosure architecture that would have surfaced it never existed.*
>
> EDITORS' SYNTHESIS OF BURD (2020, 2024).

LE INSIGHT

Burd's IPEDS analysis is the construct-substitution case at public-flagship scale. Nearly $32 billion of institutional aid over seventeen years was redirected from need-based to merit- based distribution without a deliberative public-policy decision. The 2024 Harvard Education Press volume extends the documentation across multiple author types. Evidence-tier flag binding — policy-tier analysis, multi-author synthesis, causal-share decomposition bounded; future validation ongoing.

LENS APPROACH

Burd / Crisis Point is the construct-substitution case at institutional scale (induced 8.1; LENS D5/PT5). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the disclosure-architecture lesson — the deliberation that did not happen — and in Domain 4 (Test and Evaluation) for the evidence-tier discipline binding policy-tier analysis to audit-tier verification. Pair with Case 157 (Engler / enrollment algorithms) as governance frame to technical frame, and with Cases 94 (Bartlett) and 138 (Gándara) for the construct-choice anchor across deployed prediction systems. coi-light render under the title is binding.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Name the institutional metric the operational system is optimizing for; when "net tuition revenue per matriculant" replaces "students served" without deliberation, the substitution is the failure mode.
- Build the disclosure architecture before the change: a public board-level report that ties institutional-aid distribution to need-vs-merit shares and to unmet-need outcomes is the artifact a construct substitution would have had to survive.
- Treat ranking pressure as an external incentive whose internal consequences are designable; the flagship that names the trade-off it is making between rank and need-based commitment is the one that can choose differently.

IN OPERATION / AFTER

- Commission the public-IPEDS-analog audit at the institutional level: aid distribution by Pell status, unmet-need gaps, and four-year completion by income quartile, reported as a paired scorecard.
- Publish the merit-aid-vs-completion link openly; the policy gap Burd documents persists in part because the institutional reporting does not include the downstream completion consequences of the aid pattern.
- Treat the multi-author 2024 volume as a model for how a field-scale critique is built: practitioner, journalist, and researcher contributions in a single book-length synthesis. The cross-source structure is itself the evidence-architecture lesson.

REFLECTION QUESTIONS

1. Identify an institutional metric in your domain that has quietly substituted for the stated mission metric. What was the deliberative process that produced the substitution — and if there was none, what would the disclosure architecture have to look like to surface the change?

2. Specify the paired scorecard you would publish at board level for an aid-distribution program: need-based vs. merit-based shares, unmet-need gaps by income quartile, four-year completion by Pell status. Which of these is your institution currently reporting?

3. Burd's 2024 volume brings together researchers, journalists, and industry insiders in a single book-length synthesis. What is the analog you would commission for a field-scale critique in your domain — and which voice is the hardest to include?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 159 HIGHER EDUCATION REGULATORY OVERSIGHT ONLINE EDUCATION 2022

GAO Online Program Manager Oversight Gap (GAO-22-104463)

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

DISCLOSURE John Katzman is a collaborator with the editors' home institution (Johns Hopkins) without being intimately involved in the underlying work; the affiliation is disclosed. Katzman founded 2U, which is central to the OPM debate but not named in this GAO report; the affiliation is disclosed for transparency.

IMPACT GAO found at least 550 colleges contracted with OPMs to support at least 2,900 academic programs as of July 2021; revenue-share arrangements typically gave the OPM 40–60% of tuition revenue per recruit, some up to 80%; ED lacked instructions for auditors to detect incentive-compensation violations and was not collecting the information it needed to oversee the arrangements

IN BRIEF

The Government Accountability Office’s April 2022 audit (GAO-22-104463) names a structural oversight gap in the federal regime governing online program managers. As of July 2021, at least 550 colleges contracted with OPMs to support at least 2,900 academic programs; revenue-share arrangements typically gave the OPM 40 to 60 percent of tuition revenue per recruit, with some arrangements reaching 80 percent. The 1992 Higher Education Act amendments prohibited incentive compensation for student recruiters as a fraud-prevention measure; the Department of Education’s 2011 “bundled-services” guidance exempted OPMs from the ban when recruiting was bundled with other services. The GAO found that colleges and their auditors lacked clear instructions to detect violations, and the Department was not collecting the information it needed to oversee the arrangements. The 2024 partial rescission of bundled-services guidance and 2U’s July 2024 Chapter 11 filing closed one boundary of the policy debate; the underlying delegation- and-oversight problem persists for successor OPMs and revenue-share structures.

BACKGROUND

The federal regime against incentive compensation in higher-education recruiting dates to 1992 and was built to address a specific fraud pattern: recruiters paid by enrollment will enroll students the program cannot serve, because the recruiter’s compensation is tied to the enrollment rather than to the student’s outcome. The 1992 Higher Education Act amendments banned the practice for university-employee recruiters. The 2011 “bundled-services” guidance the Department of Education issued exempted online program managers from the ban when student recruiting was bundled with other services such as program design, platform delivery, and student support — a construct that allowed the revenue-share contracting model to grow rapidly across the next decade.[0]

WHAT HAPPENED

The GAO's audit documents the scale the regime grew to. By July 2021, at least 550 colleges had contracted with OPMs to support at least 2,900 academic programs. Revenue-share arrangements typically transferred 40 to 60 percent of tuition revenue per recruit to the OPM; some arrangements ran to 80 percent. The OPM operated under the university brand — recruiting, marketing, and program operations conducted by OPM employees identifying as the university — while receiving compensation tied directly to the enrollments those operations generated. The structural seam the regime had created was an exemption that allowed the prohibited compensation structure as long as the structure was administered by a contracted vendor.[1]

THE INVESTIGATION

The oversight gap the GAO documents is operational. Colleges and their auditors lacked clear instructions to detect incentive-compensation violations within the bundled- services exemption; the Department was not collecting the information — contract structures, revenue-share rates, recruiter-compensation arrangements — that would have let it monitor whether the exemption's boundaries were holding. The audit's central finding is not that the OPM regime was designed to fail; it is that the oversight architecture required to police the exemption's boundaries was never built, and the contracting structure that the exemption allowed grew faster than the monitoring capacity that would have surfaced violations.[2]

THE CAPABILITY GAP

The 2024 partial rescission of the bundled-services guidance closed one boundary of the policy debate at the federal level, and 2U's July 2024 Chapter 11 filing closed one boundary at the commercial level. Neither closed the underlying delegation-and-oversight problem. Successor OPMs and revenue-share structures continue to operate; the universities that delegated student recruitment under the pre-2024 regime retain the operational dependencies and the brand-and-program commitments built during the decade of growth. The pair with Case 160 (USC × 2U Luna class action) shows the consumer-side litigation half of the same delegation; the pair with Case 157 (Engler / enrollment algorithms) shows the pricing-side optimization half.[3]

AFTERMATH & REFORM

The case is investigation-grade — a GAO audit is the strongest tier of evidence the corpus carries for regulatory-oversight failure — and it is the structural delegation-with-revocation case at population scale. The university delegated student recruitment to a contracted vendor under a regulatory exemption that did not include the monitoring architecture the exemption's boundaries required. The delegation was reversible in principle: the contracts could be terminated, the exemption could be withdrawn, the operations could be brought back inside the university. In practice, the delegation accumulated operational and contractual lock-in across the decade the GAO audit covers, and the revocation when it came in 2024 was forced by commercial collapse rather than by the oversight architecture the audit recommended.

REFERENCES

1. U.S. Government Accountability Office (2022), "Higher Education: Education Needs to Strengthen Its Approach to Monitoring Colleges' Arrangements with Online Program Managers," GAO-22-104463.
2. U.S. Department of Education (2011), "Dear Colleague" guidance on incentive-compensation bundled-services exemption — the regulatory artifact the GAO audits.
3. Higher Education Act of 1992, incentive-compensation prohibition — the statutory basis the 2011 guidance interpreted.
4. 2U Inc. (2024), Chapter 11 bankruptcy filing; U.S. Department of Education (2024), partial rescission of bundled-services guidance — the closure of the policy debate at the commercial and federal boundaries.

THE LEARNING ENGINEERING LENS

> *Colleges and their auditors lacked clear instructions to detect violations, and the Department was not collecting the information it needed to oversee the arrangements.*
>
> U.S. GOVERNMENT ACCOUNTABILITY OFFICE, GAO-22-104463 (2022).

LE INSIGHT

GAO-22-104463 is the investigation-grade delegation-with- revocation case at population scale. A regulatory exemption permitted a contracting structure across 550+ colleges and 2,900+ programs; the monitoring architecture required to police the exemption's boundaries was never built. The 2024 partial rescission and 2U Chapter 11 closed one boundary; the underlying delegation problem persists for successor OPMs.

LENS APPROACH

GAO OPM oversight gap is the regulatory-seam case (induced 5.3; LENS D5/PT6). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the cross-regime delegation pattern and in Domain 3 (Human-System Collaboration) for the delegation-with-revocation frame — the contract is reversible in principle, locked-in in practice. Pair with Case 160 (USC × 2U Luna, the litigation half) and Case 157 (Engler / enrollment algorithms, the pricing-optimization half). coi-light render under the title is binding.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build the monitoring architecture as a condition of any regulatory exemption; the bundled-services exemption created a foreseeable contracting structure, and the oversight infrastructure to police its boundaries should have been built with it.
- Require contract-disclosure as a federal reporting obligation, not as an institutional discretion; the GAO's data-collection finding is operationally addressable by mandatory reporting of revenue-share rates and recruiter-compensation arrangements.
- Treat the delegation as reversible from the start: contract terms should preserve termination rights and operational-knowledge transfer; the lock-in the universities experienced was partly contractual and partly operational, and both halves are designable.

IN OPERATION / AFTER

- Carry the investigation-grade audit into the curriculum without softening: the GAO's central finding is that the oversight architecture was not built, and that is the load-bearing teaching point.
- Pair the case with Case 160 (USC × 2U) so the regulator-side audit and the consumer-side litigation are taught together; one half names what the regulator missed, the other names what the delegated marketing actually did.
- Track post-rescission and post-2U-bankruptcy successor structures as a continuation of the case; the underlying delegation problem persists, and the case's frame is the regime-level oversight gap, not the specific 2U arrangement.

REFLECTION QUESTIONS

1. Identify a regulatory exemption in your domain that permits a contracting structure without specifying the monitoring architecture required to police its boundaries. What would the audit-detectable artifact be — and is anyone currently collecting it?

2. Specify the contract terms you would require to preserve revocability of a delegated capability. Termination rights, operational-knowledge transfer, data portability — which of these are typically written into your domain's standard contracts, and which are left to negotiation?

3. GAO-22-104463 documents what happened when oversight architecture lagged contracting growth across a decade. What is the analog in your domain — a contracting pattern that grew under an exemption whose monitoring infrastructure was not built — and where would the equivalent investigation-grade audit have to come from?

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CASE 160 HIGHER EDUCATION ONLINE PROGRAM MANAGEMENT PROFESSIONAL LICENSURE 2010S – 2024

USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

DISCLOSURE John Katzman is a collaborator with the editors' home institution (Johns Hopkins) without being intimately involved in the underlying work; the affiliation is disclosed. Katzman founded 2U but had departed by the time of the USC MSW expansion documented in this case.

EVIDENCE TIER journalism-tier — source confidence flagged; future validation ongoing.

IMPACT USC's online MSW grew from ~300 students per cohort pre-2010 to >3,000 per cohort through the 2U partnership; 2023 class action alleges USC marketed the online program as the 'same' as the residential program while outsourcing recruiting, advising, and clinical placement to 2U employees; partnership terminated 2024

IN BRIEF

USC's online Master of Social Work program grew from about 300 students per cohort before 2010 to more than 3,000 students per cohort, almost entirely through the 2U partnership. The May 2023 Luna v. USC class-action complaint alleges that USC marketed the online program as the "same" as the residential program while outsourcing recruiting, advising, and clinical placement to 2U employees who carried usc.edu email addresses. Plaintiffs further allege that 2U marketers targeted students of color and veterans with aggressive, deceptive tactics, and that the online program's tuition (over $100,000) reflected on-campus pricing while the delivered student experience did not. USC and 2U announced the partnership's termination on the MSW and other programs in 2024. The case is in litigation, and the predatory-targeting reconstruction rests on the complaint and on contemporaneous reporting rather than on a fact-finder's ruling — the evidence-tier flag is binding, and the case carries the journalism-tier framing with the standing language: future validation ongoing.

BACKGROUND

The structural pattern the case names is the inversion of a stable institutional practice. USC's residential MSW program had been an established, modestly sized clinical training program with a known student experience. The 2U partnership took the program's name and credential and built a parallel online operation at roughly ten times the cohort scale, with student-facing operations — recruiting, enrollment counseling, clinical-placement coordination — run by 2U employees who identified to applicants and enrollees with usc.edu email addresses. The university's delegation extended beyond program operations to the student-facing surface that defines what the credential means to the student paying for it.[0]

WHAT HAPPENED

The Luna class-action complaint, filed in Los Angeles County Superior Court in May 2023, structures its case around two central allegations. The first is misrepresentation: USC marketed the online program as the "same" as the residential program — same faculty, same standards, same credential — while the operational substance of the program had been outsourced. The tuition price (over $100,000) tracked the residential pricing structure while the delivered student experience tracked the OPM-operated structure. The second is targeting: 2U marketers, the complaint alleges, focused aggressive and deceptive recruitment tactics on students of color and veterans — populations for whom the credential cost-and-mobility calculation is structurally different and for whom the misrepresentation has differential consequences.[1]

THE INVESTIGATION

The downstream consequences the complaint identifies extend into the licensure question. The MSW credential is a professional-licensure prerequisite, and clinical placement is the operational core of the training the licensure depends on. The complaint alleges that the delegated clinical-placement coordination — handled by 2U employees under usc.edu cover — did not consistently produce placements at the quality the residential program's reputation implied. The licensure-board half of the consequence chain — what graduates were actually able to do in practice, given clinical-placement quality — has not been independently studied, and the case carries that gap honestly rather than collapsing it.[2]

THE CAPABILITY GAP

USC and 2U announced the partnership's termination on the MSW and other programs in 2024. The termination closed the operational arrangement at the boundary the litigation and the broader 2U commercial collapse forced; it did not adjudicate the legal claim, did not refund tuition paid under the prior arrangement, and did not produce an independent record of what the delegated operations actually delivered. The case sits at the consumer-side counterpart to Case 159 (GAO OPM oversight gap): one half is the regulator-side audit of the regime, the other is the litigation that names what happened to specific applicants and enrollees under the regime. Pair also with Case 157 (Engler / enrollment algorithms) for the pricing- optimization half.[3]

AFTERMATH & REFORM

The journalism-tier evidence flag is binding on the case's framing. The complaint and the partnership-termination record are investigation-grade artifacts. The predatory-targeting reconstruction relies on the complaint's allegations and on contemporaneous reporting — Higher Ed Dive, classaction.org summaries, the Project on Predatory Student Lending's statement — rather than on a fact-finder's ruling or an independent audit of 2U's marketing operations. Future validation ongoing on the litigation's outcome, on the licensure-board half of the consequence chain, and on whether the post-2024 successor-OPM contracts incorporate the disclosure architecture the case names as missing.

REFERENCES

1. Stephanie Luna v. University of Southern California, class action complaint, Los Angeles County Superior Court, May 2023.
2. Higher Ed Dive reporting on the Luna complaint, May 2023; classaction.org and topclassactions.com summaries.

3. Project on Predatory Student Lending statement on USC-2U partnership termination, 2024.
4. 2U Inc. and USC public statements on partnership termination, 2024; broader 2U commercial-collapse reporting referenced through Case 159.

THE LEARNING ENGINEERING LENS

> *The tuition tracked residential pricing. The delivered student experience tracked the OPM-operated structure. The gap between what was promised and what was delivered is the case.*
>
> EDITORS' SYNTHESIS OF THE LUNA V. USC COMPLAINT AND CONTEMPORANEOUS REPORTING.

LE INSIGHT

Luna v. USC is the consumer-side journalism-tier counterpart to GAO-22-104463 (Case 159). The complaint alleges USC marketed the online MSW as the “same” as the residential program while outsourcing the student-facing operations to 2U; the licensure-board half of the consequence chain is independently unstudied. Journalism-tier flag is binding — the predatory-targeting reconstruction is allegation-tier; future validation ongoing on litigation outcome and licensure consequences.

LENS APPROACH

USC × 2U Luna is the disclosure-as-deliverable case at OPM-delegation scale (induced 5.4; LENS D5/PT6). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the promised-vs-delivered disclosure gap and in Domain 3 (Machine Teaming and Adaptation) for the delegation-with-revocation pattern — the partnership terminated in 2024, but the consequences for students who enrolled under the prior arrangement persist. Pair with Case 159 (GAO regulator-side audit), Case 157 (Engler pricing optimization), and Case 93 (Epic Sepsis governance gap). coi-light render under the title is binding.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the student-facing surface — recruiting communications, advising emails, clinical-placement coordination — as part of the disclosable program substance, not as operational discretion to be delegated under brand cover.
- Require that delegated personnel identify their actual employer in student-facing communications; the usc.edu email cover the complaint names is operationally addressable by branding policy.
- Build a paired disclosure that ties the delivered student experience back to the marketed program description; the gap the complaint alleges is between what was promised and what was delivered, and the gap is reportable.

IN OPERATION / AFTER

- Carry the journalism-tier framing into print without softening; the litigation is ongoing, the targeting reconstruction is allegation-tier, and the case’s pedagogical power rests on naming the evidence tier honestly.
- Pair the case in syllabi with Case 159 so the regulator-side and consumer-side halves of the OPM regime are taught together; one half names what the audit missed, the other names what the affected students alleged happened.
- Track the licensure-board half over time; an independent study of clinical-placement quality and post-graduation practice capacity is the audit the case names as the missing evidence.

REFLECTION QUESTIONS

1. Identify a program in your domain where the student-facing surface — recruiting, advising, placement — has been delegated to a contracted vendor operating under institutional brand. What is the disclosure your institution makes to applicants about that delegation, and at what point in the transaction?

2. Specify the paired disclosure you would build to tie delivered student experience back to the marketed program description. What data — placement outcomes, advising-load ratios, vendor-employee proportion of student-facing communications — would you commit to publishing alongside enrollment marketing?

3. The journalism-tier flag is binding because the targeting reconstruction is allegation-tier. What is the investigation-grade study that would convert the case from allegation-tier to audit-tier — and who could commission it?

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CASE 161 HIGHER EDUCATION SECURITIES LAW ENROLLMENT MANAGEMENT 2019 – 2022

In re 2U, Inc. Securities Class Action — When Yield Management Crashes Into Disclosure

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

DISCLOSURE John Katzman co-founded 2U and is a collaborator with the editors' home institution (Johns Hopkins) without being intimately involved in the underlying litigation; the affiliation is disclosed.

IMPACT Consolidated federal securities class action in the District of Maryland (TDC-19-3455, consolidated with TDC-20-1006); §10(b) and §20(a) allegations that 2U executives misled investors about declining enrollment projections during the Feb 26 2018 – Jul 30 2019 class period; $37M settlement July 2022; final approval Dec 9 2022 by Hon. Theodore D. Chuang

IN BRIEF

The consolidated federal securities class action against 2U, Inc. — filed December 2019 in the District of Maryland before Hon. Theodore D. Chuang and settled for $37 million in July 2022 with final approval December 9, 2022 — names a distinct structural seam in the enrollment-management-vendor frame. Lead plaintiff Fiyyaz Pirani and co-lead Oklahoma City Employees Retirement System alleged that 2U executives “intentionally misled investors about declining enrollment projections” across the February 26, 2018 to July 30, 2019 class period, sounding in §10(b) and §20(a) of the Securities Exchange Act. The settlement is not an admission of liability, and the case is pedagogically useful for what the disclosure architecture failed to do, not for any adjudicated finding of wrongdoing. The case pairs with Case 159 (GAO OPM oversight) and Case 160 (USC × 2U Luna) to complete the regulator-side, consumer-side, and investor-side triangle around the same delegation structure that Case 157 names at the pricing layer.

BACKGROUND

The enrollment-management vendor model that grew across the 2010s tied a public-company revenue line directly to the yield mechanics the casebook documents elsewhere. 2U’s stock price depended on continued growth in program partnerships and per-program enrollments; the OPM revenue- share structure documented in Case 159 transmitted university enrollment outcomes onto the company’s quarterly results. When enrollment trends weakened, the disclosure question that securities law imposes on a public company became the load-bearing one. The consolidated complaint alleges that executive statements to investors did not track the operating signal the company’s own enrollment projections were generating across the class period.[o]

WHAT HAPPENED

The procedural record is the part of the case that is adjudicated. The consolidated action carries docket numbers TDC-19-3455 (filed December 2019) and TDC-20-1006, before Hon. Theodore D. Chuang of the U.S. District Court for the District of Maryland. Fiyyaz Pirani was appointed lead plaintiff; the Oklahoma City Employees Retirement System served as co-lead. The class period the complaint defined ran from February 26, 2018 to July 30, 2019. The legal theory sounded in §10(b) and §20(a) of the Securities Exchange Act of 1934 — the standard misrepresentation and control-person provisions for a federal securities-fraud class action. The case settled for $37 million in July 2022; Judge Chuang granted final approval on December 9,
2022. The settlement explicitly disclaims admission of
liability.[1]

THE INVESTIGATION

The pedagogical seam the case opens is the disclosure architecture of an OPM-driven enrollment-management business run as a public company. The hedge the casebook must preserve is binding: a securities-class settlement is not a finding of wrongdoing, and the case teaches the structural pattern, not adjudicated fault. The pattern is the structural one. A company whose revenue line is built on partner-university enrollments must disclose changes in enrollment trajectory under federal securities law; the enrollment-management vendor relationship Case 157 maps operates downstream of the same trajectory the disclosure must describe. The class period closes in the same window — late 2019 through 2022 — that the GAO audit (Case 159) and the Luna complaint (Case 160) cover, and the alignment is not coincidental. Multiple oversight surfaces converged on the same delegation structure at the same time.[2]

THE CAPABILITY GAP

The case sits as the investor-side complement to Case 159 (regulator-side audit of the OPM regime) and Case 160 (consumer-side litigation by online MSW enrollees against USC). Each surface saw the same underlying business arrangement from a different vantage. The regulator asked whether the incentive-compensation exemption's monitoring architecture had been built; the answer documented in the GAO audit was that it had not. The enrolled students alleged that the marketed program substance had been delegated to vendor employees operating under brand cover. The investor class alleged that executive communications during a window of weakening enrollment did not surface the projection signal the operating record was generating. The three surfaces together define the disclosure-as-governance frame the case anchors. Pair also with Cases 157 and 158 (Engler and Burd) for the enrollment- management context this litigation operates inside of.[3]

AFTERMATH & REFORM

The honest hedges the case carries are load-bearing. The settlement is not an admission of liability, and the case teaches the disclosure-architecture pattern, not adjudicated wrongdoing. The class period and the allegation structure are adjudicated record; the executive intent that the complaint frames as "intentional" was not tested by a fact-finder. The case's value to the corpus is that a public-company disclosure regime is one of the few external oversight mechanisms that operated on the OPM- enrollment-management model during its growth window, and that the convergence with Cases 159 and 160 across the same calendar window is a structural rather than incidental pattern. The coi-light disclosure under the title is binding for the affiliation, and the case's editorial framing has been written to maintain critical distance from any reading that would convert the settlement into an adjudicated finding.

REFERENCES

1. In re 2U, Inc. Securities Class Action, Civil Action No. TDC-19-3455 (consolidated with TDC-20-1006), D. Md., before Hon. Theodore D. Chuang; consolidated complaint filed December 2019.
2. Stipulation of settlement and motion for final approval, In re 2U, Inc. Securities Class Action, July 2022; final approval order, December 9, 2022.
3. Securities Exchange Act of 1934, §10(b) and §20(a); SEC Rule 10b-5 — the statutory and regulatory basis the complaint sounded in.
4. Paired investigation-grade record: U.S. Government Accountability Office, GAO-22-104463 (2022); Luna v. USC class action complaint (2023) — the regulator-side and consumer-side surfaces of the same delegation structure (Cases 159, 160).

THE LEARNING ENGINEERING LENS

> *The settlement is not an admission of liability. The case teaches the disclosure-architecture pattern, not adjudicated wrongdoing — and the convergence of three oversight surfaces on the same delegation structure within the same calendar window is the structural diagnostic.*
>
> EDITORS' SYNTHESIS OF THE IN RE 2U CLASS ACTION RECORD (2019 – 2022).

LE INSIGHT

In re 2U is the investor-side disclosure-architecture case at OPM-delegation scale. A federal securities class action over enrollment-projection communications during a weakening- enrollment window settled for $37 million without admission of liability. The case teaches the disclosure pattern, not adjudicated fault, and completes the regulator-side and consumer-side oversight triangle with Cases 159 and 160 across the same calendar window.

LENS APPROACH

In re 2U is the disclosure-as-governance case at the public- company boundary (induced 5.4; LENS D5/PT5; CLO-4 and CLO-5). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the change-control and disclosure-architecture anchor and in Domain 4 (Test and Evaluation) for the convergence-of-oversight-surfaces diagnostic. Pair with Case 159 (GAO regulator-side audit), Case 160 (Luna consumer-side complaint), and Cases 157 and 158 (Engler and Burd, the enrollment- management context). coi-light render under the title is binding for the affiliation.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat enrollment-trajectory signals as a primary disclosure surface when the business model ties revenue to partner enrollments; the case names the seam where operating-record signal and investor-facing communication must be reconciled in a federal-securities-law-graded sense.
- Build the projection-disclosure pipeline before it is needed: a documented internal process for translating

IN OPERATION / AFTER

- Carry the case in print with the hedge intact — settlement is not an admission of liability, and the case teaches the disclosure-architecture pattern, not adjudicated wrongdoing.
- Pair in syllabi with Case 159 (GAO audit) and Case 160 (Luna complaint) so the three oversight surfaces are taught together; the pedagogical move is to show how

partner-enrollment signals into investor-facing communication is the artifact a §10(b) defense would have to invoke.

- Treat the convergence of regulator-side, consumer-side, and investor-side oversight on the same delegation as a leading indicator; when three independent oversight surfaces close on a single business arrangement within the same calendar window, the structural pattern is the diagnostic. a single delegation structure looked from three vantage points across overlapping windows.
- Use the case to teach the disclosure-as-governance frame: federal securities law is one of the few external regimes whose disclosure standards apply to the OPM-enrollment-management model, and the standards' application is itself the artifact the case names.

REFLECTION QUESTIONS

1. Identify a business arrangement in your domain whose revenue trajectory depends on a delegated operational counterpart. What disclosure surfaces — to investors, regulators, customers, affected populations — currently apply, and which of them are absent in the architecture as built?
2. Specify the internal pipeline you would build to translate operating-record signal into investor-facing disclosure. What is the documented decision rule for when an emerging trajectory becomes a disclosable trend, and who signs off on the rule?
3. The case is part of a three-surface convergence on the same delegation structure across a single calendar window. Pick an arrangement in your domain and ask: which oversight surfaces have closed on it within overlapping windows, and what would the editorial synthesis of those convergent records teach a future practitioner?

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CASE 162 GOVERNMENT SOCIAL WELFARE ALGORITHMIC ADMINISTRATION 2016 – 2023

Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict

D Designed Out G Governance & Trust N Normalization of Deviance

IMPACT Income-averaging algorithm used by the Australian Department of Human Services and Centrelink raised approximately 470,000 wrongful debts against welfare recipients between 2016 and 2019; Royal Commission led by Catherine Holmes AC SC delivered its final report July 7 2023 finding the scheme unlawful and circumstantially attributing multiple deaths to its operation

IN BRIEF

The Royal Commission into the Robodebt Scheme delivered its final report on July 7, 2023, under the leadership of Commissioner Catherine Holmes AC SC. The report concluded that the income-averaging algorithm operated by the Australian Department of Human Services and Centrelink between 2016 and 2019 had raised approximately 470,000 wrongful debts against welfare recipients and had operated outside the law. The algorithm averaged annual taxable income from the Australian Taxation Office across fortnightly Centrelink reporting periods and treated any resulting discrepancy as a debt the recipient had to disprove. The burden of proof was reversed onto the recipient. The Commission's attribution language on causation of deaths is careful — the attribution is circumstantial and not a direct legal finding of individual causation — but the finding that multiple deaths were associated with the scheme's operation is part of the adjudicated record. The case pairs with Case 155 (SyRI, the governance-objection-correct precedent), Case 97 (Johnson school surveillance, the algorithmic-public-administration parallel), and Case 93 (Epic Sepsis, the delegation-without-validation form).

BACKGROUND

The Australian income-support architecture pairs Centrelink fortnightly income reporting with annual taxable income reporting to the Australian Taxation Office. The two reporting cadences exist because they measure different things — Centrelink measures earnings inside each fortnight so the income-test taper can be applied, and the ATO measures annual taxable income for the tax system. Between 2016 and 2019, the Department of Human Services deployed an automated debt-recovery system that took the ATO annual figure, divided it by 26, and compared the fortnightly average against the Centrelink reported figures. Any apparent shortfall was raised as a debt against the recipient, and the recipient was required to produce contemporaneous payslips to disprove it.[0]

WHAT HAPPENED

The mathematical operation the algorithm performed cannot establish what it was being asked to establish. Annual income divided by 26 is not the income earned in any particular fortnight; a recipient who worked irregularly, or whose hours varied, would generate large arithmetic discrepancies between the averaged figure and the actual fortnightly earnings without ever having been overpaid. The Royal Commission's final report documents that the agency's own legal advice flagged this seam before deployment and that the advice was set aside. Approximately 470,000 debts were raised across the operating window; many recipients paid debts they did not owe, and the Commission identified cases in which recipients took their lives in proximity to the scheme's operation. The attribution language on those deaths is careful — "circumstantial" rather than a direct legal finding of individual causation — and the careful language is itself part of the record the case carries.[1]

THE INVESTIGATION

The structural critique the Commission delivered is the reversal of the burden of proof. In a properly administered welfare-recovery scheme, the agency must establish that a debt is owed before pursuing recovery. The income-averaging method reversed this: the algorithm asserted a debt on the strength of arithmetic that could not establish overpayment, and the recipient had to produce documentary evidence — often payslips from years earlier, often from employers no longer reachable — to disprove the assertion. The Federal Court's 2019 Prygodicz judgment had already found the method unlawful; the Royal Commission's 2023 report adjudicated the governance question of how the scheme had been built, approved, and operated for three years across multiple ministerial portfolios. The Commission named individuals; some are subject to subsequent referrals to the National Anti-Corruption Commission and to professional bodies.[2]

THE CAPABILITY GAP

The case pairs with Case 155 (SyRI, the Dutch System Risk Indication ruling by the Hague District Court) as the governance-objection-correct precedent — SyRI was struck down before it produced a debt-scale harm record; Robodebt operated for three years and the harm record is what the Commission adjudicated. Pair with Case 97 (Johnson school surveillance) for the algorithmic-public- administration parallel at a different population and a smaller scale; the structural form — algorithm asserts, affected party must disprove — recurs across the two cases. Pair with Case 93 (Epic Sepsis) for the delegation-without-validation form: in Epic Sepsis the delegated system asserts a clinical risk; in Robodebt the delegated system asserts a financial debt; in both, the asserting party did not validate the assertion against the ground truth before consequences were transmitted to the affected person.[3]

AFTERMATH & REFORM

The hedges the case carries are load-bearing. The Royal Commission's attribution language on deaths is circumstantial; the Commission did not, and could not on the evidence available, make individual legal findings of causation in those deaths. The case teaches the structural pattern — algorithmic public administration with the burden of proof reversed, deployed without the legal-advice seam being honored, operated for three years across multiple ministers — and the structural pattern is what makes Robodebt the load-bearing reference for an entire class of contemporary algorithmic-administration failures. The

Commission's careful attribution language and the case's careful editorial framing travel together; neither is smoothed in the casebook's rendering.

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THE LEARNING ENGINEERING LENS

> *The income-averaging method could not establish what it was being asked to establish, the burden of proof was reversed onto the recipient, and the Commission's attribution on associated deaths is circumstantial — and these three facts together are what Robodebt teaches.*
>
> EDITORS' SYNTHESIS OF THE ROYAL COMMISSION FINAL REPORT (HOLMES AC SC, JULY 2023).

LE INSIGHT

Robodebt is the load-bearing reference for algorithmic public administration deployed at population scale without the burden of proof being honored. The Royal Commission's final report adjudicated approximately 470,000 wrongful debts and circumstantially attributed multiple deaths to the scheme's operation; the careful attribution language and the structural finding travel together, and neither is smoothed in the casebook's rendering.

LENS APPROACH

Robodebt is the burden-of-proof-reversal case at population scale (induced 5.2; LENS D5+D4/PT6; CLO-5 and CLO-3). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the agency-legal-advice-as-binding-gate discipline and in Domain 3 (Human-System Collaboration) for the human-in- the-loop-for-consequential-decisions anchor. Pair with Case 155 (SyRI governance-objection-correct precedent), Case 97 (Johnson school surveillance algorithmic-public-administration parallel), and Case 93 (Epic Sepsis delegation-without- validation form). The Commission's circumstantial attribution on deaths is the load-bearing hedge.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat agency legal advice on the lawfulness of an algorithmic-administration method as a binding gate, not a negotiable input; the Commission's record on Robodebt is that the seam was flagged in advance and that the override of the advice is itself the governance failure.
- Maintain the burden of proof on the asserting party for any algorithmically generated debt or risk; arithmetic that cannot establish the asserted fact cannot be the basis for transmitting consequences to an affected person.

IN OPERATION / AFTER

- Carry the Commission's careful attribution language on deaths into print without softening; the case's load-bearing quality depends on the circumstantial nature of the attribution being preserved alongside the structural finding.
- Pair in syllabi with Case 155 (SyRI) so the governance-objection-correct precedent and the governance-objection-overridden harm record are taught together; the two cases together teach what advance objection can prevent and what its absence can produce.

- Build the cross-portfolio review surface that a multi-year algorithmic-administration scheme requires; Robodebt operated across multiple ministers and the cross-portfolio handoff was where the governance check kept being deferred.
- Use the case to anchor the human-in-the-loop CLO at population scale; the form Robodebt makes legible is what consequential-decision delegation looks like when the loop is removed and the asserting party operates on arithmetic that cannot establish its assertion.

REFLECTION QUESTIONS

1. Identify an algorithmic-administration scheme in your domain whose arithmetic asserts a fact against an affected person. What is the asserting party's burden of proof, and what would the affected person have to produce to disprove the assertion?
2. Specify the legal-advice gate you would treat as binding in advance of deployment. What is the documented escalation path when the advice flags an unlawfulness seam, and who has authority to override it?
3. The Royal Commission's attribution on deaths is circumstantial — careful, not adjudicated as individual legal findings. Pick a setting where harm attribution to an algorithmic system is contested, and ask: what language would honor both the structural pattern and the limits of what the evidence can establish?

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CASE 163

FINANCIAL SERVICES CONSUMER CREDIT ALGORITHMIC DECISION-MAKING

2019 – 2021

Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model

D Designed Out K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT New York State Department of Financial Services investigation March 2021 found no violation of New York anti-discrimination law in Apple Card credit-limit decisions following David Heinemeier Hansson's November 2019 viral allegation that his wife received a credit limit approximately 20 times lower despite shared assets; DFS documented "lack of transparency" as the structural problem and required Goldman Sachs to overhaul its customer-service process

IN BRIEF

On November 7, 2019, software developer David Heinemeier Hansson posted on Twitter that his Apple Card credit limit had been set approximately 20 times higher than his wife Jamie Heinemeier Hansson's, despite the couple filing taxes jointly and Jamie having a higher credit score. The thread went viral, with Apple co-founder Steve Wozniak reporting a similar disparity with his wife. The New York State Department of Financial Services opened an investigation. Its March 2021 report concluded that the credit-decisioning algorithm operated by Goldman Sachs (the issuing bank for Apple Card) did not violate New York anti-discrimination law, finding no statutory finding of intent or disparate-impact violation. DFS documented "lack of transparency" as the structural problem: Goldman Sachs could not adequately explain individual credit decisions to applicants who challenged them. DFS required Goldman Sachs to overhaul its customer-service process. The case pairs with Case 94 (Bartlett mortgage), Case 107 (Coots), and Case 133 (Gándara). The DFS finding of "no violation but lack of transparency" is the load-bearing nuance.

BACKGROUND

The Apple Card launched in August 2019 as a co-branded consumer credit product issued by Goldman Sachs, with underwriting and credit-line decisions made by Goldman's consumer-banking unit. The product was Goldman's first retail consumer credit product at meaningful scale, and the decisioning architecture was built de novo against contemporary algorithmic-underwriting practice. On November 7, 2019, David Heinemeier Hansson posted that his Apple Card credit limit was approximately 20 times his wife's, despite jointly filed taxes and Jamie's higher credit score. The thread surfaced similar reports — Steve Wozniak named the same pattern with his wife — and rapidly moved from social media to regulatory attention.[0]

WHAT HAPPENED

The structural seam the case opens is that Goldman Sachs could not adequately explain individual credit decisions to the applicants who challenged them. When Jamie Heinemeier Hansson asked Goldman to explain why her credit limit was lower, the customer-service response was that the algorithm had set the limit and that the decision could not be explained at the individual level. The escalation moved through standard customer-service channels, then to social- media-amplified public pressure, then to regulatory investigation, and at no point along the path did Goldman surface an account of why the decision had been made. The explainability gap was operational rather than algorithmic in the narrow sense: even if the underlying model was defensible at the population level, the bank had not built the infrastructure to defend individual decisions to applicants.[1]

THE INVESTIGATION

The New York State Department of Financial Services opened its investigation in late November 2019 and released its findings in March 2021. The headline finding was that DFS did not find a violation of New York anti-discrimination law — neither intentional discrimination nor an actionable disparate-impact violation under the applicable standards. The investigation reviewed approximately 400,000 New York State credit-line decisions. The hedge is binding: DFS's "no violation" finding is specific to the statutory standard the agency applied, not a general finding that the decisioning was fair or non-discriminatory across all criteria. What DFS did find was "lack of transparency" as the structural problem and required Goldman Sachs to overhaul its customer-service process, build appeal mechanisms, and document its credit-decisioning explanations at the individual-applicant level.[2]

THE CAPABILITY GAP

The case pairs with Case 94 (Bartlett mortgage) for the consumer-credit-fairness thread at adjacent scale and regulatory regime. Pair with Case 107 (Coots) for the competing-fairness-definitions thread; the DFS standard is one of several available standards, and the case teaches that "no violation under a specific statutory standard" is not "fair." Pair with Case 133 (Gándara) for the explainability-of-individual-predictions thread at a different population and scale. The case is unusual in the casebook for the speed of regulatory response — DFS opened the investigation within weeks of the viral thread — and for the structural conclusion that the failure was explainability rather than the decisioning algorithm itself.[3]

AFTERMATH & REFORM

The load-bearing hedge is the precise DFS finding. The case does not teach that Apple Card was unfair, and it does not teach that DFS found Apple Card was fair. It teaches that under the specific statutory standard the agency applied, no violation was found, and that the structural failure the agency named was "lack of transparency" — Goldman Sachs's inability to explain individual credit decisions to applicants who challenged them. The human-in-the-loop CLO is anchored by the case at the appeal-and-explanation seam: a consequential-decision system that cannot explain its individual decisions to affected applicants does not have a functioning human-in-the-loop appeal mechanism, and the absence of the mechanism is the governance failure whether or not the decisioning is statistically defensible.

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THE LEARNING ENGINEERING LENS

> *DFS did not find a violation of New York anti-discrimination law; DFS did find lack of transparency as the structural problem, and required Goldman Sachs to overhaul its customer-service process — the case teaches that "no violation under a statutory standard" is not "fair."*
>
> EDITORS' SYNTHESIS OF THE NY DEPARTMENT OF FINANCIAL SERVICES REPORT ON THE APPLE CARD INVESTIGATION (MARCH 2021).

LE INSIGHT

Apple Card is the consumer-credit explainability case at deployment scale. DFS found no violation of New York anti- discrimination law under the applicable statutory standard, and DFS also found lack of transparency as the structural problem; Goldman Sachs was required to overhaul its customer-service process and build individual-applicant appeal mechanisms. The load-bearing hedge is the precision of the DFS finding — neither "fair" nor "unfair," but "no violation under this standard, transparency gap as the structural problem."

LENS APPROACH

Apple Card is the explainability-as-governance case at consumer-credit scale (induced 5.2; LENS D5/PT6; CLO-5 and CLO-3). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the appeal-and-explanation-infrastructure discipline and in Domain 3 (Human-System Collaboration) for the human-in-the-loop-for-consequential-decisions anchor at the appeal seam. Pair with Case 94 (Bartlett mortgage), Case 107 (Coots competing fairness definitions), and Case 133 (Gándara explainability of individual predictions). The precise DFS finding — "no violation but lack of transparency" — is the load-bearing hedge.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build the individual-applicant explanation infrastructure as part of the deployment, not as a customer-service afterthought; the Apple Card case demonstrates that a defensible population-level model paired with no individual-explanation infrastructure produces a regulatory finding of lack of transparency.
- Specify the appeal mechanism before the first decision is made; the human-in-the-loop CLO at consumer-credit scale is the appeal-and-explanation seam, and its absence is the governance failure the case names.
- Treat the customer-service escalation path as a deployment surface, not a support function; the case's

IN OPERATION / AFTER

- Carry the precise DFS finding into print without softening; "no violation under the applicable statutory standard, but lack of transparency as the structural problem" is the load-bearing nuance and the case's pedagogical value depends on the nuance being preserved.
- Pair in syllabi with Case 94 (Bartlett) so the consumer-credit-fairness regulatory architecture is taught at both the mortgage and credit-card scales.
- Use the case as the anchor for the explainability-as-governance frame at consumer-credit scale; the curricular target is the appeal-and-explanation infrastructure that converts an algorithmic decision into a contestable one.

escalation went from customer service to social media to regulation in days, and the deployment surface that mattered was the first one.

REFLECTION QUESTIONS

1. Identify a consequential-decision system in your domain whose individual-applicant explanation infrastructure has not been built. What is the customer-service escalation path when an affected person challenges a decision, and what would the path look like with an appeal-and-explanation seam built into the deployment?

2. Specify the precise statutory or regulatory standard against which your deployment is being evaluated. What does "no violation under this standard" leave open about fairness across other standards, and how would the trade-off be disclosed?

3. The Apple Card escalation moved from customer service to social media to regulatory investigation within days. Pick a deployment in your domain and ask: what is the first deployment surface an affected person encounters when challenging a decision, and what would have to be true for that surface to resolve the challenge before it moves further?

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CASE 164 AVIATION 1972 – PRESENT

NASA Saturn V Documentation

K Knowledge & Institutional Memory

IMPACT Foundational U.S. aerospace case for the cost of losing the institutional capability to build a system

IN BRIEF

When NASA returned to heavy-lift rockets in the 2000s, it confronted an uncomfortable fact: the institutional capability to build a Saturn V — the rocket that sent Apollo to the Moon — had been lost. The drawings and documents survived; the practical knowledge of how the components were manufactured, tested, and assembled, and why each choice was made, had walked out with the workforce that retired by the 1990s. The vehicle could be redesigned but not reproduced. Apollo was documented to an unprecedented degree, yet the documentation was not the capability: when the engineers who built the system left, the system left with them. Saturn V is the book's strongest evidence that institutional capability lives in people and sustaining practices, not in the artifacts an institution leaves behind.

BACKGROUND

The Saturn V that launched Apollo to the Moon was, by any measure, one of the best-documented engineering programs in history — exhaustive drawings, specifications, and test records.[0] That thoroughness is what makes the case bite: the program left behind close to everything a successor could ask for on paper, so any later difficulty in rebuilding it cannot be blamed on a thin archive but must lie in what the archive could not hold.

WHAT HAPPENED

Yet by the early 2000s, as NASA worked the Constellation program, it had become apparent that the institutional capability to build a Saturn V had been lost. The drawings existed; the practical knowledge of how the components were manufactured, tested, and assembled — and why particular choices had been made — had walked out with the workforce that retired by the 1990s. The vehicle could be redesigned. It could not be reproduced.[1] The distinction is exact: redesign starts from requirements and rebuilds the reasoning afresh, while reproduction needs the original making-knowledge, and it was that knowledge — not the paperwork describing it — that had left with the people who held it.

THE INVESTIGATION

The case is canonical for the difference between documentation and institutional capability. Apollo's documentation was unprecedented, but it captured the **what**, not the tacit **how** — the judgment, the workarounds, the undocumented reasons — that lived in the people who did the work. When they left, that knowledge was in no archive to recover.[2] A drawing records the decision but not the deliberation behind it; the workaround that made a part manufacturable and the reason a tolerance was set where it was lived in the doing, and once the doers retired there was no document from which to reconstruct it.

THE CAPABILITY GAP

Saturn V is the strongest available evidence that institutional capability is not the same as the artifacts an institution produces. Capability lives in people and in the practices that sustain them; a library of drawings is a necessary record but not a transferable ability. An institution that treats documentation as preservation of capability is, quietly, letting the capability expire.[3] The danger is that the archive looks like insurance: its very completeness can reassure managers that the capability is safe, so the people and practices that actually carry it are allowed to disperse precisely because the paperwork seems to stand in for them.

AFTERMATH & REFORM

The lesson reshaped how serious programs think about knowledge retention — apprenticeship, continuity of teams, deliberate capture of tacit reasoning, and not letting a critical capability rest in a handful of soon-to-retire heads.[4] It pairs with the chapter's other memory cases: knowledge, unlike a document, has to be actively kept alive, or it is gone in a generation. Each of the retention practices addresses the same root cause the Saturn V exposed — that tacit making- knowledge transfers only person to person — so apprenticeship and team continuity are not nice-to-haves but the actual mechanism by which a capability outlives the people who first held it.

SATURN V — DOCUMENTATION IS NOT CAPABILITY

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THE LEARNING ENGINEERING LENS

> *The Saturn V drawings exist. The Saturn V does not.*

PARAPHRASING THE NASA CONSTELLATION-ERA CAPABILITY DISCUSSION, C. 2005

LE INSIGHT

The Saturn V case is the canonical evidence that institutional knowledge is not equivalent to documentation. The documents persist; the capability does not. Capability engineering must treat the people who hold tacit knowledge as a primary institutional asset.

LENS APPROACH

LENS uses the Saturn V case in LEN 8 to teach that retaining and overlapping the expert personnel who hold tacit making-knowledge is itself an engineered deliverable (induced 6.3) — not a by-product of writing better documents. The capability to rebuild a Saturn V was lost because workforce overlap and retention were never treated as a designed retention deliverable, and documentation alone cannot carry tacit process knowledge across a generation. The teaching point in LEN 1 is therefore not “document more thoroughly” but “engineer the personnel continuity that the archive can never substitute for.”

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Capture tacit reasoning as the work happens — the why behind a choice, the workaround that made a part buildable — not just the resulting drawing.
- Build capability into teams and apprenticeship, so the making-knowledge has a living carrier rather than resting in a handful of soon-to-retire heads.
- Treat documentation as a record, never as a substitute for the people and practices that hold the capability.

IN OPERATION / AFTER

- Audit critical capabilities for single points of human failure — knowledge held by a few near-retirement practitioners — and transfer it before they leave.
- Periodically test reproducibility, not just whether the archive is complete, since a full archive can mask a capability that has already expired.
- Sustain continuity of teams and practice deliberately, so a capability is kept alive across generations rather than rediscovered at need.

REFLECTION QUESTIONS

1. Identify a capability in your domain that is currently held in the tacit knowledge of a small number of senior practitioners. What is your institution’s capacity to reproduce it after they retire?

2. Design the institutional practice that would preserve a capability against the retirement of its holders.

3. Saturn V was exhaustively documented yet could not be reproduced, because the archive captured the **what** and not the tacit **how**. Identify a capability in your institution whose documentation might be giving false reassurance — and describe what the paperwork is failing to capture.

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CASE 165 SPACE / AEROSPACE 2019 – 2024

Boeing Starliner

K Knowledge & Institutional Memory D Designed Out

IMPACT Multiple delays; the 2024 crewed flight left two NASA astronauts at the ISS for months; contemporary case for capability erosion at a legacy contractor

IN BRIEF

Boeing's Starliner, meant to be the second U.S. commercial crew vehicle alongside SpaceX's Crew Dragon, accumulated failures across a decade: software errors on its 2019 uncrewed flight, valve corrosion scrubbing a 2021 launch, and propulsion-system trouble on its 2024 crewed test that left two NASA astronauts on the ISS rather than returning them as contracted — NASA brought them home on SpaceX months later. GAO and NASA reviews found capability erosion across multiple engineering disciplines at a contractor whose human-spaceflight record had once been definitive. The erosion looks partly generational (Apollo- and Shuttle-era engineers retired) and partly institutional (cost and schedule pressure, and the thinning of NASA's in-house depth to challenge the contractor). Starliner is the contemporary case for capability erosion at a legacy contractor.

BACKGROUND

Boeing was awarded a NASA Commercial Crew contract to build Starliner as the second U.S. vehicle to carry astronauts to the ISS, alongside SpaceX's Crew Dragon — drawing on a human-spaceflight heritage that had once been definitive in American aerospace.[0] The award rested in part on that heritage: a contractor with a definitive human-spaceflight record was a presumed safe choice, and that presumption is exactly what the program would go on to test, since reputation was standing in for a current measurement of capability.

WHAT HAPPENED

Instead Starliner stumbled across a decade: software errors marred its 2019 uncrewed test flight, valve corrosion scrubbed a 2021 launch, and propulsion-system problems on the 2024 crewed test flight left the two NASA astronauts aboard the ISS rather than returning on the contracted spacecraft — NASA brought them home on a SpaceX vehicle months later.[1] The three failures fell in different subsystems across three separate years, which is itself telling: a single bad part is bad luck, but software, valves, and propulsion failing in succession points to something broader than any one component — a decline running across the engineering organization rather than within one of its parts.

THE INVESTIGATION

GAO and NASA reviews identified the program as exhibiting capability erosion across multiple engineering disciplines — software, valves, propulsion, integration testing — at a contractor whose track record had previously been definitive.[2] The erosion looked partly generational, as Apollo- and Shuttle-era engineers retired, and partly institutional: cost pressure, schedule-driven decisions, and the loss of NASA's own in-house engineering depth to challenge the contractor.[3] The two causes compound: as the contractor's senior depth thinned through retirement, the buyer's own engineering depth had also eroded, so the very expertise NASA would have needed to catch the slipping contractor was the expertise it had let go of.

THE CAPABILITY GAP

Starliner is the contemporary case for capability erosion at scale at a legacy contractor. The decline happened over decades and was visible only in retrospect, because the institutional architecture for catching a supplier's slow capability decay — and updating the buyer's confidence to match — did not exist. Reputation outran reality, and no instrument was measuring the gap.[4] Slow decay is the hard case to catch: no single year's results look alarming, the brand keeps the buyer's confidence high, and without an instrument that tracks the supplier's actual current capability the divergence is only legible once a crewed flight forces the reckoning.

AFTERMATH & REFORM

NASA leaned harder on independent reviews and on SpaceX as the reliable alternative, and the episode sharpened questions about how to sustain — and verify — capability at sole-source and legacy suppliers.[5] It pairs with Saturn V (Case 164): where that case lost a capability to retirement, Starliner shows the same erosion in slow motion at a living institution still carrying the brand of the capability it had let thin. Having a second supplier to fall back on is what let NASA absorb the failure, which is itself the lesson: where a capability can erode unseen, a verified alternative is the difference between an embarrassment and a crew with no way home.

STARLINER — CAPABILITY DECAY AT A LEGACY CONTRACTOR

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THE LEARNING ENGINEERING LENS

> *Starliner has demonstrated significant readiness shortfalls across multiple engineering disciplines.*
>
> EDITORS' SYNTHESIS OF GAO AND NASA REVIEWS OF THE COMMERCIAL CREW PROGRAM

LE INSIGHT

Starliner is the case for sustained capability erosion at a legacy contractor whose track record had previously been definitive. The erosion happened over decades and was visible in retrospect; the institutional architecture for catching it did not exist.

LENS APPROACH

LENS uses Starliner in LEN 5 to teach one clean capability gap: the contractor-NASA interface capability that thinned over a decade. As NASA shifted to fixed-price commercial contracting, its own oversight and insight capacity eroded, so the problems an integrated review would have caught instead surfaced in flight (against the cross-subsystem troubles as the visible symptom). The teaching point is that the buyer's review capability is an engineered deliverable in its own right — let it thin and a slipping supplier becomes legible only when a crewed flight forces the reckoning.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Verify a supplier's current capability against present evidence rather than awarding on reputation and a once-definitive track record.
- Preserve enough in-house engineering depth to genuinely challenge a contractor, since a buyer who has let its own expertise erode cannot catch a slipping supplier.
- Maintain a verified second supplier where the stakes are crewed, so a capability that erodes unseen does not become a single point of failure.

IN OPERATION / AFTER

- Instrument supplier capability so slow, multi-year decay is visible before a high-stakes flight forces the reckoning.
- Watch for cross-subsystem patterns — failures in software, valves, and propulsion in succession — as a signal of organization-wide erosion, not isolated bad luck.
- Update institutional confidence to match measured reality, so a contractor's brand cannot keep outrunning its current performance.

REFLECTION QUESTIONS

1. Identify a legacy supplier in your domain whose capability track record may have eroded faster than your institutional confidence in them has updated. What is the signal?
2. Design the contractor-capability review that would have caught the Starliner gaps before the 2024 crewed flight.
3. Starliner's failures spanned software, valves, and propulsion across three separate years — a pattern, not a single bad part. In your domain, what would distinguish a one-off component failure from organization-wide capability erosion, and how soon could you tell them apart?

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CASE 166 GOVERNMENT TECHNOLOGY 1992 – 2008

Bernard Madoff / SEC Failures

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT ~$65B Ponzi scheme — the largest in history; SEC repeatedly investigated and cleared Madoff; foundational regulator-capability case

IN BRIEF

Bernard Madoff ran the largest Ponzi scheme in history — roughly $65 billion in fictitious account value — for years while the SEC repeatedly investigated and cleared him. Financial analyst Harry Markopolos delivered the agency a detailed memo in 2005, "The World's Largest Hedge Fund is a Fraud," showing that Madoff's steady returns were mathematically impossible. The SEC opened an investigation and concluded no action was warranted. Madoff operated until his sons turned him in during the 2008 crisis, when redemptions became impossible to honor. The SEC Inspector General later found the staff assigned lacked the expertise to evaluate Markopolos's arguments and had deferred to Madoff's industry stature. The capability gap was at the regulator's technical-evaluation pipeline, not at the evidence — which was specific and checkable.

BACKGROUND

Bernard Madoff was a former NASDAQ chairman and a respected Wall Street figure whose investment arm reported remarkably steady returns for years. Those returns were entirely fictitious: client funds were never invested, and earlier investors were paid with later investors' money — a Ponzi scheme that eventually represented some $65 billion in fabricated account value. The very steadiness that reassured investors was the tell: real markets do not deliver an almost unbroken line of gains, and the absence of the normal volatility was itself evidence the numbers were manufactured.[0]

WHAT HAPPENED

Financial analyst Harry Markopolos concluded by analysis that Madoff's returns were mathematically impossible and delivered the SEC a detailed memorandum in 2005 titled "The World's Largest Hedge Fund is a Fraud." The SEC opened an investigation and concluded that no enforcement action was warranted. The warning was not a vague suspicion but a quantitative case any competent reviewer could in principle retrace, which is what makes the dismissal so telling. Madoff continued operating until December 2008, when the financial crisis made redemptions impossible and his sons reported him.[1]

THE INVESTIGATION

The SEC Office of Inspector General's 2009 report found the agency had received credible, specific complaints across more than a decade and "missed numerous opportunities" to uncover the fraud. The staff assigned to Madoff lacked the expertise to evaluate Markopolos's technical arguments, and the institutional culture had defaulted to treating

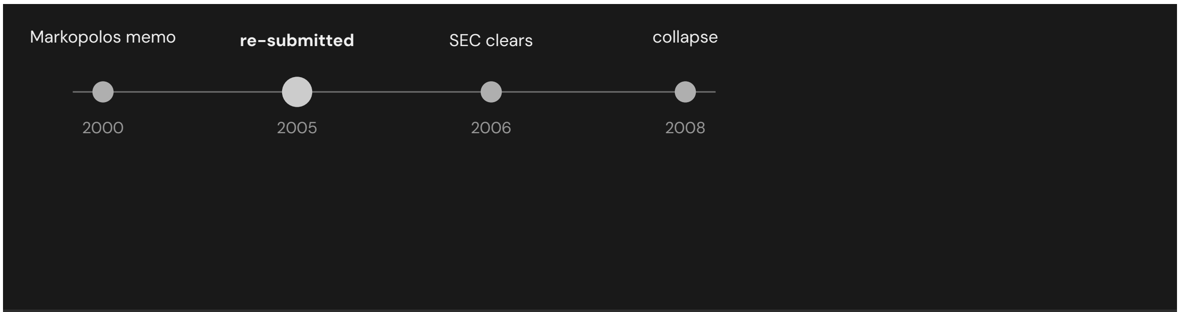
Madoff as a respected industry figure rather than following the evidence. Stature stood in for analysis: the reviewers weighed who Madoff was instead of whether the math could be true, and the reputational halo did the work that scrutiny should have.[2]

THE CAPABILITY GAP

The complaint was specific; the math was checkable; the institution simply did not have the people to check it. The capability gap was at the regulator's technical-evaluation pipeline, not at the evidence. A regulator whose technical depth lags the entities it oversees cannot act on even a correct and well-documented warning — and because the gap is in the evaluator rather than the tip, more tips would not have helped; the agency could not have used them.[3]

AFTERMATH & REFORM

The collapse wiped out tens of thousands of investors and prompted SEC reforms to its handling of tips and referrals, its examination procedures, and its recruitment of staff with quantitative and trading expertise, including the creation of an Office of Market Intelligence. Each reform addressed a different part of the same pipeline — getting the tip in, getting it to someone who could read it, and having someone who could — so that a future Markopolos memo would meet an evaluator able to test it. Madoff is paired with Theranos (Case 150) as a case in which a regulator lacked the technical capability to challenge the evidence in front of it.[4]



MADOFF — THE REGULATOR RECEIVED THE EVIDENCE; THE REGULATOR LACKED THE STAFF TO EVALUATE IT

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THE LEARNING ENGINEERING LENS

> *The agency missed numerous opportunities to discover the fraud.*

SEC OFFICE OF INSPECTOR GENERAL REPORT OIG-509, 2009

LE INSIGHT

The Madoff case is the canonical example of a regulator without the technical capability to evaluate the evidence it received. The complaint was specific. The math was checkable. The institution did not have the people to check it. The capability gap was at the regulator's expertise pipeline, not at the evidence.

LENS APPROACH

LENS uses Madoff in LEN 6 as an operator-to-institution feedback-channel failure: Harry Markopolos sent the SEC a specific, checkable memo — the channel existed and the signal was correct — and the institution had no one able to act on what came through it. The teaching point is the receiving capability a feedback channel requires to be real: a tip line without the technical expertise to evaluate the tip is not a channel. It pairs with the Mark 14 torpedo (Case 138), where the Bureau of Ordnance dismissed submariners' field reports for the same structural reason — the channel carried the truth and the institution could not receive it — and with Theranos (Case 150) on the regulator's missing technical depth. The Ponzi narrative is not the lesson; the channel is.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Staff the evaluation pipeline with quantitative and trading expertise matched to the entities being overseen, so a checkable claim meets someone able to check it.
- Build tip-and-referral triage that routes a specific, technical complaint to a qualified evaluator rather than letting the subject's reputation decide its fate.
- Require warnings to be assessed on their merits first, structurally separating the analysis of the math from any weighing of who the subject is.

IN OPERATION / AFTER

- Audit closed investigations for cases dismissed despite specific, checkable evidence, surfacing where the evaluator — not the evidence — was the failure point.
- Monitor for returns or results too smooth to be real as a leading indicator, treating the absence of expected volatility as a signal to investigate.
- Sustain the regulator's technical depth against the regulated industry's growing sophistication, since a depth gap quietly reopens the same blind spot over time.

REFLECTION QUESTIONS

1. Identify a regulator in your domain whose technical evaluation capability has not kept pace with the entities it regulates. What is the resulting blind spot?
2. Design the regulator-side technical-capability deliverable that should have allowed the SEC to evaluate the Markopolos memo on its merits.
3. The SEC weighed Madoff's stature over Markopolos's math. Design a triage process that forces a tip to be evaluated on its technical merits before the subject's reputation is allowed to enter the decision.

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CASE 167 DEFENSE GOVERNMENT 1996 – 2001

9/11 Intelligence Sharing Failures

G Governance & Trust K Knowledge & Institutional Memory

IMPACT 2,977 killed; the 9/11 Commission found systemic intelligence-sharing failures across the U.S. government

IN BRIEF

The September 11, 2001 attacks, which killed 2,977 people, were enabled in part by intelligence the U.S. government already held but never integrated. The CIA knew, from a January 2000 meeting in Kuala Lumpur, that two future hijackers had entered the country; it did not tell the FBI. The FBI separately flagged suspicious flight-training activity in Phoenix and Minneapolis in 2001, but that information was never aggregated. Visa issuance, immigration tracking, and watch-listing were each run by a different agency, and the handoffs between them depended on individual initiative that no institution required. The 9/11 Commission called it a “failure of imagination” — a framing critics say understates the structural gap. The reform that followed built the cross-agency architecture, the ODNI and the NCTC, that had not existed.

BACKGROUND

By 2001, U.S. counterterrorism depended on many agencies — CIA, FBI, NSA, State, INS, FAA — each holding a piece of the picture. No institution was responsible for integrating those pieces; cross-agency information sharing depended on individuals choosing to pass information along rather than on any architecture that required it. Each agency’s incentives, classification rules, and turf reinforced the boundary, so the natural tendency was to hold information rather than to push it across a line no one was charged with bridging.[0]

WHAT HAPPENED

On 11 September 2001, nineteen hijackers seized four aircraft and killed 2,977 people. In the months and years before, the warning signs had been distributed across the government: the CIA tracked two future hijackers from a January 2000 meeting in Kuala Lumpur but did not watch-list them or notify the FBI; FBI field offices flagged suspicious flight-training activity in Phoenix and Minneapolis; none of it was aggregated into a single picture. Any one fragment looked minor in isolation; only assembled would they have shown the shape of the plot, and assembly was exactly the function no one performed.[1]

THE INVESTIGATION

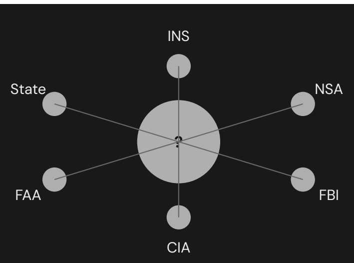
The 9/11 Commission and the earlier Congressional Joint Inquiry documented specific failures of information sharing across the FBI, CIA, and NSA. The Commission famously concluded that “the most important failure was one of imagination” — a framing later criticized as understating the structural nature of the gap, which was less a lack of imagination than an absence of any institution responsible for integration. The distinction matters for the remedy: a failure of imagination invites exhortation to think harder, while a structural gap demands an institution be built to close it.[2]

THE CAPABILITY GAP

The intelligence sharing did not happen because no institution owned it as a deliverable. Each agency was competent inside its own boundary; the integration across boundaries existed nowhere as a required function. The missing capability was an architecture — shared watch-lists, mandated handoffs, a body responsible for fusing the picture — rather than more raw collection. The government did not lack data; it lacked the connective tissue to turn data held in many places into a single picture anyone was accountable for assembling.[3]

AFTERMATH & REFORM

The Intelligence Reform and Terrorism Prevention Act of 2004 created the Office of the Director of National Intelligence and the National Counterterrorism Center, and fusion centers followed — the institutional architecture for cross-agency information sharing that had not previously existed. By creating bodies whose explicit mandate was integration, the reform converted information sharing from an act of individual initiative into a required function someone owned. The case is foundational in U.S. national-security policy for treating cross-agency capability as an engineerable institutional deliverable.[4]



no institution responsible for the integration as a deliverable

9/11 — CROSS-AGENCY INFORMATION SHARING AS MISSING ARCHITECTURE

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THE LEARNING ENGINEERING LENS

> *The most important failure was one of imagination.*

THE 9/11 COMMISSION REPORT, 2004

LE INSIGHT

9/11 is the foundational U.S. case for cross-agency information- sharing as an engineering deliverable. The architecture had not been built. The reform built it. The cost of the missing architecture was paid in 2001. The discipline LENS represents is the kind of work that, applied across the U.S. intelligence community in the 1990s, would have produced the architecture earlier.

LENS APPROACH

LENS uses 9/11 in LEN 8 as the foundational case for cross- organizational capability and in LEN 7 for the governance architecture of multi-agency systems. Studio projects compare 9/11 with Eagle Claw (Case 5) as institutional-architecture failures of different kinds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assign integration as an explicit deliverable owned by a named body, so the function does not depend on individuals choosing to pass information across a boundary.
- Engineer mandated handoffs and shared watch-lists into the architecture, making cross-agency sharing a required function rather than an act of initiative.
- Design for fusion of fragments that look minor in isolation, so the system is built to assemble the picture no single holder can see.

IN OPERATION / AFTER

- Audit cross-boundary information flows for reliance on individual initiative, and treat any flow that depends on goodwill as an unowned failure mode.
- Monitor whether the integrating body actually receives and fuses what the component agencies hold, rather than assuming the mandate guarantees the practice.
- Sustain the integration architecture against the agency incentives, classification rules, and turf that pull information back behind boundaries over time.

REFLECTION QUESTIONS

1. Identify a cross-organizational information flow in your domain that depends on individual initiative rather than institutional architecture. What is the foreseeable failure mode?
2. Design the institutional deliverable that would have produced ODNI-level information sharing across U.S. intelligence agencies in the 1990s.
3. The Commission called it a “failure of imagination”; critics called it a structural gap. Take a near-miss in your domain and argue which framing fits — and show how the remedy differs depending on which you choose.

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Chapter 10

Sociotechnical Constraints — What Works

When governance, knowledge transfer, and adoption are engineered as deliverables.

The institution is the longest deliverable in the program.

AN AXIOM IN CAPABILITY ENGINEERING.

CASE 168 PHARMA SAFETY REGULATORY PROGRAMS CLINICAL CARE 2006 – 2011

iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms

G Governance & Trust D Designed Out N Normalization of Deviance

IMPACT An FDA-mandated risk-evaluation and mitigation program with pregnancy testing, two-method contraception requirements, and lockout authorization before dispensing — Kaiser Permanente cohort (n=8,344; 9,912 treatment courses) reported 29 fetal exposures and 'no evidence that iPLEDGE significantly decreased the risk of fetal exposure' relative to the prior SMART program

IN BRIEF

Isotretinoin is a highly effective acne medication and a known teratogen: fetal exposure causes severe birth defects. The FDA has required risk-management programs since the 1980s; in 2006 the agency replaced the prior SMART program with iPLEDGE — a formal Risk Evaluation and Mitigation Strategy (REMS) requiring pregnancy testing, two contraception methods (or documented abstinence), and pharmacy lockout authorization before each dispense. The case is the productive counterpoint to SUBSAFE (Case 19) and the WHO Surgical Checklist (Case 109): the structural move is the same — mandatory pre-authorization gating a consequential action — and the measured outcome is very different. The Brinker et al. Kaiser Permanente cohort (J Am Acad Dermatol, 2011; n=8,344 patients across 9,912 treatment courses) found 29 fetal exposures and concluded “no evidence that iPLEDGE significantly decreased the risk of fetal exposure” compared with the prior program. The broader literature reports approximately 150 isotretinoin-exposed pregnancies continue annually in the US despite the program, with non-adherence — missed pills, inconsistent condom use — the documented driver. The teaching point is that an authorization mechanism without adherence support does not reliably deliver the capability it is built to enforce.

THE SHIFT

Isotretinoin (Accutane and successors) is a highly effective systemic treatment for severe acne, and a well-characterized teratogen. Fetal exposure during treatment causes a recognized pattern of severe birth defects. The FDA has therefore required some form of risk-management program around its prescribing since the late 1980s, evolving through pregnancy testing requirements, contraception counseling, and structured enrollment programs. iPLEDGE, introduced in 2006, was the first to use the formal Risk Evaluation and Mitigation Strategy (REMS) architecture and to require pharmacy lockout: no authorization, no dispense.[0]

WHAT IS EMERGING

The mechanism is what the case turns on. iPLEDGE requires the patient to register, document either two contraception methods or abstinence (with declared method), complete monthly pregnancy tests, and receive a per-cycle authorization code before the pharmacy can fill the prescription. Each prescriber, pharmacy, and wholesaler must also enroll in the program. The structural form is the same one SUBSAFE (Case 19) and the WHO Surgical Checklist (Case 109) both use successfully: mandatory pre-authorization gating a consequential action, with the authorization withheld until pre-conditions are verifiably met.[1]

THE CAPABILITY QUESTION

The measured outcome diverges sharply. Brinker et al. (Journal of the American Academy of Dermatology, 2011) studied 8,344 Kaiser Permanente patients across 9,912 isotretinoin treatment courses in the iPLEDGE era and identified 29 fetal exposures. The paper's conclusion is the load-bearing sentence: "no evidence that iPLEDGE significantly decreased the risk of fetal exposure" relative to the prior SMART program. The broader literature consistently reports that approximately 150 isotretinoin-exposed pregnancies continue annually in the United States despite the program operating as designed. The documented driver in both is non-adherence: missed pills, inconsistent condom use, the assumption that one cycle of missed contraception is unlikely to coincide with the teratogenic window.[2]

EARLY EVIDENCE

The teaching point is precise and load-bearing. iPLEDGE is not a program-design failure in the sense that SUBSAFE succeeded and iPLEDGE was sloppy. The mechanism is elaborate, the enrollment burden on prescribers and pharmacies is real, and the program runs as specified. What the evidence shows is that an authorization mechanism without adherence support does not reliably deliver the capability it is built to enforce — because the capability sits downstream of the authorization in patient behavior, and the program does not reach into that behavior. The structural form (mandatory pre-authorization) is the same as SUBSAFE's; the missing piece is the equivalent of SUBSAFE's lifecycle audit and recurring training that keep the certification real in operation.[3]

OPEN PROBLEMS

Drafted alongside SUBSAFE (Case 19) and the WHO Surgical Checklist (Case 109), iPLEDGE is the most analytically useful "mixed" case in the v2 sweep. It is not a failure of the form (the form has demonstrated successes); it is evidence that the form alone does not deliver capability when the capability depends on a behavior the program does not instrument. The case is the case-grounded basis for the new CLO **Judgment under inadequate evidence** — the practitioner designing a REMS-class program has to decide on the available evidence what additional adherence-support architecture the mechanism needs in order to deliver, and the iPLEDGE evidence is the worked example of why the question matters.

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THE LEARNING ENGINEERING LENS

> *The authorization mechanism is operating as designed. The capability sits downstream of the authorization, and the program does not reach that far.*
>
> EDITORS' SYNTHESIS OF BRINKER ET AL. (2011).

LE INSIGHT

iPLEDGE is the productive counterpoint to SUBSAFE and the WHO Surgical Checklist. The structural form — mandatory pre-authorization gating a consequential action — is the same; the measured outcome is very different because the capability the program exists to enforce depends on patient adherence the program does not instrument. The "no significant decrease" finding is load-bearing and survives into the case verbatim.

LENS APPROACH

iPLEDGE is the analytically useful "mixed" pre-authorization case (induced 4.4; LENS D3/PT5). LENS uses it in Domain 3 (Human-System Collaboration) for the CLO **Delegation with revocation** — what the authorization mechanism delegates and what it does not — and in Domain 4 (Test and Evaluation) for the CLO **Judgment under inadequate evidence** on whether the form will deliver in a specific context. Direct pair with SUBSAFE (Case 19) at the form-vs-context layer; pair with WHO Surgical Checklist (Case 109) at the mandatory-mechanism layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify in advance what the authorization mechanism reaches and what it does not — for iPLEDGE, the authorization controls dispensing but does not reach into patient adherence — so the gap is named at design time rather than discovered in outcome data.
- Pair the authorization architecture with adherence support that instruments the behavior the capability depends on: in this case, contraception adherence, not just contraception declaration.
- Design the outcome metric around the harm the program exists to prevent (fetal exposure), not around the mechanism's own throughput (enrollment counts, authorization codes issued), so the evaluation is of the capability and not of the bureaucracy.

IN OPERATION / AFTER

- Treat a "no significant decrease" finding as program evidence about the mechanism's reach, not as an argument against authorization architectures in general; SUBSAFE and the WHO checklist demonstrate the form can deliver under the right conditions.
- Re-engineer the adherence-support layer based on the documented driver of failure — for iPLEDGE, missed pills and inconsistent condom use — rather than tightening the authorization layer that is already working as specified.
- Carry the "approximately 150 exposures annually" figure honestly in any communication about the program; iPLEDGE is operating as designed, and the design's reach is what the evidence is about.

REFLECTION QUESTIONS

1. Identify a mandatory pre-authorization program in your domain. What does the authorization mechanism actually reach, and what does it not? Where in the patient or operator behavior does the capability the program exists to deliver sit downstream of the authorization?
2. Specify the adherence-support architecture you would add to convert an iPLEDGE-class authorization into a delivered capability. For isotretinoin the gap is documented (missed pills, inconsistent condom use); what is the analog in your context?
3. The "no significant decrease" finding is what the case rests on. What would the outcome metric have to be — and at what cadence — to know whether a revised adherence-support layer was working, separately from the authorization mechanism that is already working as specified?

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CASE 169 HIGHER EDUCATION (AFRICA) DATA PRIVACY LAW LEARNING ANALYTICS 2022

Data Privacy and Learning Analytics on the African Continent

K Knowledge & Institutional Memory G Governance & Trust N Normalization of Deviance

IMPACT Mapped the legal and regulatory privacy landscape across African jurisdictions and surfaced the structural governance seam — African higher education frequently uses learning platforms and analytics hosted by external providers, creating cross-regime gaps where student data crosses jurisdictions with inconsistent protection — and recommended common cross-border data-sharing frameworks

IN BRIEF

Prinsloo, Slade, and Khalil (British Journal of Educational Technology, 2022) mapped the legal and regulatory privacy landscape across African jurisdictions and found a growing trend toward comprehensive data-protection legislation, though few frameworks are yet enacted and cross-border data-transfer policies differ sharply between countries. The core governance seam the paper surfaces — and what makes the case a frontier rather than a settled instance — is what the v2 sweep names **extraterritorial platform governance**: African higher education frequently uses learning platforms, learning-management systems, and analytics services hosted by external (typically Global-North or Asian) providers, creating a structural gap where student data is generated under one regulatory regime, processed under another, and the institution's governance authority does not reach the operating regime. The paper recommends common cross-border data-sharing frameworks as the architectural response. The case is included as a frontier — it documents the governance architecture needed for responsible adoption rather than a completed success — and it is valuable because it surfaces a seam (extraterritorial hosting) the US-centric canon rarely confronts, and that is increasingly universal as institutions everywhere build on cloud and AI services they do not control.

THE SHIFT

The standard framing of learning-analytics governance assumes that the operating institution, the regulatory regime, and the hosting infrastructure are co-located — a US university under US law on US-hosted infrastructure, a UK university under UK law on UK or EU infrastructure. African higher education sits structurally outside this assumption. Most universities across the continent rely on learning-management systems, predictive analytics tools, and adjacent platforms hosted by providers based in the Global North or in Asia, and the regulatory regimes governing those providers are not the same as the regimes governing the institutions or the students.[0]

WHAT IS EMERGING

Prinsloo, Slade, and Khalil's 2022 paper in the British Journal of Educational Technology maps the legal and regulatory privacy landscape across African jurisdictions. The picture is one of a continent in motion: a growing trend toward comprehensive data-protection legislation following the model the EU established with GDPR, but with few frameworks yet enacted and substantial variation across countries in scope, enforcement capacity, and cross-border data-transfer policy. Some countries have comprehensive frameworks; some have sectoral protections; some have constitutional privacy provisions without operational data-protection legislation; and the patchwork makes the cross-border picture itself uneven.[1]

THE CAPABILITY QUESTION

The structural seam the paper surfaces is what makes the case teachable. When student data is generated under one jurisdiction's regulatory regime, processed under another's because the platform is hosted abroad, and the institution's governance authority does not reach the operating regime, the governance question becomes architectural rather than institutional. The authors recommend common cross-border data-sharing frameworks as the architectural response — the kind of inter-regime instrument that GDPR's adequacy decisions and successor cross-border-transfer mechanisms began to construct between the EU and other regimes, but that no comparable continent-wide African framework yet provides. The case names the gap and the architectural response; what it does not yet document is the build-out.[2]

EARLY EVIDENCE

The honest evidentiary state is what the frontier tag carries. The paper is peer-reviewed and the mapping is grounded in legal documentation across jurisdictions; what the case does not establish — because the architecture is not yet built — is that any specific cross-border framework has been adopted and that the seam has been closed in practice. The case is teachable on the structural pattern: the governance question is the architectural one across regimes, not the policy one inside any single regime. The outcome is open in the sense that no continent-scale resolution yet exists. Future validation will continue as African data-protection law consolidates.[3]

OPEN PROBLEMS

The frontier note the case carries names what the v2 sweep calls extraterritorial platform governance — capability deployed on platforms governed by a different regime than the one operating them. The pattern is not African-specific. It is increasingly universal as institutions everywhere build on cloud, AI, and platform services they do not control: a US university on a US cloud is the historical case; a US university running analytics through a model-hosted-elsewhere service, a Latin American university on a US cloud, an African university on a European platform — all share the structural seam, and the African case surfaces it most starkly because the regime asymmetry is largest there. The case is the case-grounded basis for proposing extraterritorial-platform-governance as a sub-competency the existing curriculum does not yet name.

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THE LEARNING ENGINEERING LENS

> *The governance question is architectural across regimes, not policy inside any single regime. The seam is at the joints, and the joints have to be drawn before the architecture can be designed.*
>
> EDITORS' SYNTHESIS OF PRINSLOO, SLADE, & KHALIL (2022).

LE INSIGHT

The African data-privacy case names a governance seam the existing curriculum underweights: extraterritorial-platform governance, where the operating, regulating, and hosting regimes diverge. The case is a frontier — peer-reviewed mapping, architectural recommendation, no continent-scale resolution yet built. The pattern is increasingly universal, and the case-grounded basis is the case for naming a sub-competency the framework does not yet have.

LENS APPROACH

African data privacy is the extraterritorial-platform governance case (induced 5.3; LENS D5/PT6) — Domain 5 for the cross-regime seam; Domain 3 for **Judgment under inadequate evidence** on a frontier where the architectural response is recommended but not built. Pair with Case 115 and Case 181.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Map the regime structure explicitly: which jurisdiction generates the data, which processes it on the platform, which the institution operates under, which the student lives under. The seam is at the joints, and the joints have to be drawn before the architecture can be designed.
- Choose the platform stack with the regime structure in view; a procurement decision is also a governance decision when the platform is in a different jurisdiction from the operating institution.
- Treat the architectural response as common cross-border frameworks rather than institution-by-institution contract negotiation; the inter-regime instrument is what scales.

IN OPERATION / AFTER

- Surface the extraterritorial-platform-governance pattern in any v2 case that involves cloud/AI infrastructure; the African case is the starkest instance, not the only one.
- Carry the outcome-open honesty: the architecture is not yet built, and the case names the gap rather than a completed success.
- Build the case-grounded basis for proposing extraterritorial-platform-governance as a sub-competency the existing curriculum does not yet name; pair with the OU and LALA cases at the governance-by-design layer.

REFLECTION QUESTIONS

1. Identify a platform or service your institution depends on whose operating regime is different from your own. Which jurisdiction generates the data, which processes it, which regulates the institution, which regulates the user — and where is the seam between regimes?

2. Specify the architectural response your context needs: an institution-by-institution contractual approach, a sectoral framework, a cross-border data-sharing instrument. Which level resolves the seam, and which level cannot?

3. The case names extraterritorial platform governance as an under-named sub-competency. What instance from your domain would anchor it — a cloud-hosted clinical analytics tool, a model-as-a-service inference layer, a learning-platform vendor in a different jurisdiction — and how is the curriculum response different from a single-regime governance case?

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CASE 170 HIGHER EDUCATION ALGORITHMIC DECISION-MAKING HUMAN-COMPUTER INTERACTION 2025

Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

DISCLOSURE John Katzman is a collaborator with the editors' home institution (Johns Hopkins) without being intimately involved in the underlying work; the affiliation is disclosed. Pyle and Andalibi's study is independent of Katzman; the case is placed in this cluster because it sits at the applicant-side of the enrollment-management deployment.

IMPACT **Eighteen semi-structured interviews with recent U.S. university applicants, using speculative-design probes, surfaced systematic distances between vendor marketing claims (efficiency, fairness, enhanced fit) and applicants' own perceptions (opacity, distrust, anticipated discrimination)**

IN BRIEF

Pyle and Andalibi (CSCW 2025) report an interview study with 18 recent U.S. university applicants, using speculative-design probes to surface how applicants perceive algorithmic systems operating in college admissions and enrollment. The study's central contribution is naming the systematic distance between vendor marketing claims — efficiency, fairness, enhanced fit — and applicants' own perceptions of how the systems would treat them: opacity about the algorithm's existence, distrust of its objectives, and anticipated discrimination across protected characteristics. The case is the peer-reviewed consent-side companion to Case 157 (Engler / enrollment algorithms) and Case 159 (GAO OPM oversight gap). The authors' own hedge is explicit: 18 interviews is the right sample for the speculative-design method but not for prevalence claims, and "future validation ongoing" applies to whether the perception patterns generalize across applicant demographics and institution types.

THE SHIFT

The deployment surface for algorithmic systems in college admissions and enrollment has been studied largely from the institutional side — what vendors promise, what colleges buy, what optimization outcomes the institution reports. The applicants whose lives the deployment most directly affects have been the structurally absent voice in the deployment record. Pyle and Andalibi's CSCW 2025 paper is the systematic peer-reviewed correction at the consent-side seam: an interview study with 18 recent U.S. university applicants, using speculative-design probes calibrated to surface applicants' perceptions of algorithmic systems operating on their applications.[o]

WHAT IS EMERGING

The speculative-design method is the right instrument for the question. Applicants in most cases do not know which specific algorithmic systems acted on their applications; the institutional side does not disclose the vendor stack or the optimization objective at the point of application. Speculative-design probes — scenario sketches that concretize plausible algorithmic interventions and ask applicants to react — let the study elicit perceptions that are not contingent on the applicant having direct knowledge of the deployed system. The 18-interview sample is calibrated to the method's depth rather than to prevalence-claim breadth, and the authors are explicit about that calibration as a methodological choice.[1]

THE CAPABILITY QUESTION

The findings name systematic distance between vendor marketing and applicant perception across three axes. Vendor marketing pitches efficiency — the system processes more applications faster — while applicants name opacity: they do not know which decisions are being made algorithmically and they cannot interrogate the system's logic. Vendor marketing pitches fairness — the system treats applicants consistently — while applicants name distrust of the objective the system is consistent with. Vendor marketing pitches enhanced fit — the system matches applicants to programs likely to serve them well — while applicants name anticipated discrimination across protected characteristics, drawing on the broader public record of algorithmic disparate-impact findings.[2]

EARLY EVIDENCE

The case sits as the consent-side counterpart to Case 157 (Engler / Brookings, the deployment-side mapping) and Case 159 (GAO-22-104463, the regulator-side audit). Engler documents the deployed algorithmic optimization; the GAO audit documents the regulatory oversight gap; Pyle and Andalibi document what the affected applicants understand about the deployment. The three cases together name the structural seam: a deployed system about which the institutional side, the regulator, and the affected population each hold partial and non-overlapping information. The case also pairs with Cases 94 (Bartlett lending fairness) and 138 (Gándara community-college equity) as the applicant-perception strand of the equity-in- deployed-prediction thread.[3]

OPEN PROBLEMS

The authors' hedges are binding on the case's framing. Eighteen semi-structured interviews is the right N for the speculative-design method, surfacing the dimensions of perception that matter, but it is not the right N for claims about prevalence — what fraction of applicants hold each perception, how the perceptions distribute across demographic groups, how they correlate with admissions outcomes. The case is the strongest peer-reviewed evidence currently available on the consent-side question; future validation ongoing on whether the perception patterns generalize across applicant demographics, institution types, and the rapidly evolving algorithmic-deployment landscape.

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THE LEARNING ENGINEERING LENS

> *Vendors pitch efficiency, fairness, and fit. Applicants name opacity, distrust, and anticipated discrimination. The distance between the two is the case.*
>
> EDITORS' SYNTHESIS OF PYLE & ANDALIBI (2025).

LE INSIGHT

Pyle and Andalibi is the peer-reviewed consent-side case at the applicant end of the enrollment-management deployment. Eighteen interviews surface systematic distance between vendor marketing and applicant perception across three axes; the authors' methodological hedge is binding on prevalence claims. The case completes the partial-information triangle with Cases 157 (deployment-side) and 180 (regulator-side).

LENS APPROACH

Pyle & Andalibi is the governance-rather-than-technique case at the consent boundary (induced 8.4; LENS D5/PT6). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the affected-voice inclusion lesson and in Domain 3 (Machine Teaming and Adaptation) for the disclosure-architecture design at the applicant interface. Pair with Cases 157 (Engler deployment), 180 (GAO oversight), 103 (Bartlett), and 138 (Gándara). coi-light render under the title is binding for cluster placement.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Include the affected-population voice in the deployment-decision record from the start; the systematic distance the study documents is partly the result of decision processes that did not include the affected voice.
- Use speculative-design probes when the affected population cannot be expected to know what is deployed; the method is the right instrument for the consent-side question and the institutional side can commission it.
- Treat the three perceived gaps — opacity, distrust, anticipated discrimination — as designable targets, not as misperceptions to correct; the perceptions are responses to the actual disclosure architecture.

IN OPERATION / AFTER

- Commission the prevalence study the speculative-design study cannot perform; the perception patterns the 18-interview study surfaces are testable at survey scale, and the deployment-side institutions are best positioned to commission the survey.
- Pair the consent-side study with the regulator-side audit (Case 159) and the deployment-side mapping (Case 157) in the curriculum; the three cases together name the partial-information structure of the deployment.
- Track the perception findings over time as the algorithmic-deployment landscape evolves; the case's value as a longitudinal baseline depends on the comparison studies that come next.

REFLECTION QUESTIONS

1. Identify a deployed prediction system in your domain whose affected population was not consulted in the deployment-decision record. What would a speculative-design study look like for surfacing the affected voice — and who is positioned to commission it?

2. Specify the three perceived gaps — opacity, distrust, anticipated discrimination — for the system you identified. Which of the three is a disclosure-architecture target, which is an objective-choice target, and which is an audit-evidence target?

3. The case's hedge is that 18 interviews surfaces dimensions, not prevalence. What is the survey-scale study you would build on the perception dimensions the study identifies — and what would convert the perception findings into the decision-grade evidence the deployment-side institutions could act on?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 171 EDUCATION AT SCALE DEFENSE ONGOING

xAPI / Total Learning Architecture — Interoperability Gap

K Knowledge & Institutional Memory G Governance & Trust

IMPACT Despite xAPI adoption, most implementations remain siloed; cross-organizational learning-data interoperability largely unrealized

IN BRIEF

The Experience API (xAPI) was created to track learning experiences across platforms, and the Advanced Distributed Learning Initiative's Total Learning Architecture envisioned learning records, competencies, and credentials flowing across organizational boundaries to support continuous capability development — the kind of evidence infrastructure LENS depends on. In practice, xAPI implementations remain largely siloed inside individual learning-management systems; the cross-organizational data sharing that matters most for high-consequence domains has not materialized at scale. The technical standard exists and reference implementations exist; what lags is governance — who owns the data, what consent applies, how quality is assured. xAPI mirrors inBloom (Case 146) in structure — technology ahead of governance — across the whole learning-technology ecosystem. It is the book's case for treating data governance as a capability deliverable.

THE SHIFT

Learning increasingly happens across many systems — courses, simulators, on-the-job tools — and the field recognized it needed a common way to record those experiences, so capability could be tracked over a career rather than a single course.[0] The career-long view is the point: a single course's records say little about whether a person can do the job, whereas experiences stitched across courses, simulators, and on-the-job tools are what let an institution reason about real capability rather than completed seat-time.

WHAT IS EMERGING

The Experience API (xAPI) was built to do exactly that, and the Advanced Distributed Learning Initiative's Total Learning Architecture envisioned learning records, competency frameworks, and credentials flowing across organizational boundaries — the evidence infrastructure a discipline like LENS depends on.[1] The vision was explicitly cross-boundary: records, competencies, and credentials that move with the learner between organizations are precisely the evidence base on which a capability discipline must stand, which is why the standard's promise mattered well beyond any single training shop.

THE CAPABILITY QUESTION

In practice, xAPI implementations remain largely siloed inside individual LMS platforms. The cross-organizational data sharing — the capability most relevant to high-consequence operational domains — has not materialized at scale. The standard exists; the ecosystem it promised does not.[2] The siloing is the diagnostic detail: the very cross-boundary flow that made the architecture worth building is the part that did not arrive, so the standard delivers tracking within each LMS while the career-long, cross-organizational record it envisioned stays out of reach.

EARLY EVIDENCE

The technical pieces work: the data model is sound and reference implementations exist. What has lagged is the governance — who owns the data, what consent frameworks apply, and how data quality is assured across organizations.[3] With the data model proven and reference implementations in hand, the remaining obstacles are not engineering questions an organization can solve alone but agreements between organizations — ownership, consent, and assured quality — that no technical specification can settle on their behalf.

OPEN PROBLEMS

xAPI mirrors inBloom (Case 146) in essential structure — technology in advance of governance — but across the whole learning-technology ecosystem rather than one initiative.[4] It is the book's clearest case that an interoperability standard is necessary but not sufficient: without the governance to make organizations willing and able to share, the data stays in its silos, and the evidence infrastructure remains a diagram rather than a system. The parallel to inBloom is instructive because it shows the pattern is not particular to one failed project: a sound standard arriving ahead of the ownership, consent, and quality-assurance arrangements will stall the same way at any scale.

STANDARD EXISTS; CROSS-ORGANIZATION DATA STILL SILO'D

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5. Cf. inBloom (Case 146) — technology in advance of governance.

> *The standard exists. The governance does not.*
>
> EDITORS' SYNTHESIS OF ICICLE / ADL TLA DISCUSSION

LE INSIGHT

xAPI/TLA is the technical-standard analog of the implementation gap — the discipline has the data model, the spec, and the reference implementations. What it does not have is the governance and institutional architecture to make cross-organizational learning data flow. This is the canonical case in this book for treating data governance as a capability-engineering deliverable.

LENS APPROACH

LENS treats xAPI/TLA in LEN 4 as a data-governance and interoperability case and in LEN 8 as an example of organizational- learning infrastructure that has not scaled. The case is the technical underlay to the larger argument about evidence systems that decision-makers can trust.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the governance — data ownership, consent, and cross-organization quality assurance — as a deliverable alongside the technical standard, not after it.
- Design for the cross-boundary flow from the start, so the standard is not merely implemented inside each LMS but engineered to move records between organizations.
- Secure the inter-organizational agreements that no specification can settle, treating willingness-to-share as a thing to be built rather than assumed.

IN OPERATION / AFTER

- Audit whether learning data is actually crossing organizational boundaries, not just whether xAPI is nominally adopted within each silo.
- Monitor data quality across organizations continuously, since shared records are only trustworthy evidence if their quality is assured at the seams.
- Watch for the inBloom pattern — a sound standard outrunning its governance — and treat any stall as a missing-governance signal, not a technical defect.

REFLECTION QUESTIONS

1. Why has the xAPI standard not produced cross-organizational interoperability at scale? What governance condition is missing?
2. Design the minimum governance architecture under which xAPI data could flow across two organizations in your domain.
3. The xAPI data model is sound; what stalled was ownership, consent, and quality assurance across organizations. For a data-sharing effort in your domain, which of those three is the unresolved question — and who would have to agree for the data to actually flow?

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CASE 172 ENERGY 1979 – PRESENT

INPO and the Nuclear Academy

T Training Gap K Knowledge & Institutional Memory G Governance & Trust

IMPACT No INES-level event at U.S. commercial reactors post-INPO; sustained improvement in INPO/WANO performance indicators across the industry

IN BRIEF

Three Mile Island did not produce a reactor accident at the next plant over — it produced an institution. Within months of the 1979 Kemeny Commission report, the U.S. commercial nuclear industry founded the Institute of Nuclear Power Operations on a stark premise: an accident at any single plant would threaten every operator's license, and no utility could engineer its safety capability alone. Funded by the utilities it evaluated and operating without statutory authority, INPO set training and certification standards, accredited every plant's programs through the National Academy for Nuclear Training, and ran peer evaluations in which operators from one utility scrutinized another's control rooms and records. The pre-TMI culture of complacency gave way to mandated vigilance. No U.S. commercial reactor has had a significant INES-level event since.

BACKGROUND

The 1979 partial meltdown at Three Mile Island exposed not just a plant-level failure but an industry with no shared mechanism for learning. The Kemeny Commission traced the accident in part to a pervasive "mindset" of complacency, in which each utility operated alone and no institution carried lessons from one plant to the rest. The structural problem sat above any single control room: a lesson learned at one plant had no path to the others, so the same latent failure could surface repeatedly across an industry that never compared notes.[0]

THE INTERVENTION

Within months of the Kemeny report, the utilities founded the Institute of Nuclear Power Operations. Its premise was that an accident anywhere threatened everyone's license to operate. INPO set training and certification standards for operators and supervisors, and in 1985 the National Academy for Nuclear Training began accrediting each facility's programs. The shared-exposure premise was what gave a body with no statutory power its teeth — every utility had a direct stake in every other utility's competence, because one failure could end the whole industry's license to operate.[1]

HOW IT WORKED

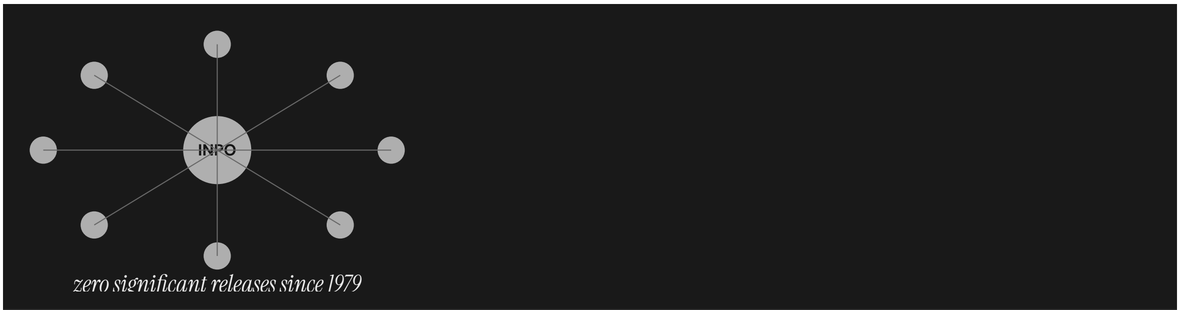
INPO's load-bearing mechanism was honest peer review: teams of operators from one utility examined another's procedures, control rooms, and incident records, reporting candidly because every utility was, in the title of one history, a hostage of the others. Funded by the utilities it evaluated and holding no statutory authority, INPO depended on shared catastrophic exposure to make its findings stick. Peer review by working operators rather than distant regulators also meant the examiners knew what to look for and the findings carried the weight of professional judgment, not just rule compliance.[2]

THE EVIDENCE

The post-TMI culture shifted from smugness to mandated vigilance, and U.S. commercial reactors have recorded no significant INES-level event since INPO's founding. Industry performance indicators tracked by INPO and its international counterpart WANO improved steadily and broadly across the fleet. The broad, steady improvement across the whole fleet — not just the strongest plants — is the signature of a learning mechanism working as designed: the laggards were pulled up by the same peer-review architecture that held the leaders to standard.[3]

WHAT TRANSFERRED

INPO is the strongest evidence in any domain that capability engineering can be undertaken at the level of an entire industry rather than a single organization. Its enabling conditions — shared catastrophic exposure, regulatory legitimacy, and an honest peer-review architecture — recur wherever one operator's failure can damage every operator, and it informed the founding of WANO after Chernobyl. That the model crossed national borders to WANO is itself evidence that the design is portable: the enabling conditions, not the particular American institution, are what make the mechanism work.[4]



INDUSTRY-WIDE PEER REVIEW: A LEARNING RING AFTER TMI

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THE LEARNING ENGINEERING LENS

> *Every utility recognized that an accident at any single plant would affect every operator's license to operate.*
>
> PARAPHRASING THE INSTITUTIONAL ANALYSIS IN REES, HOSTAGES OF EACH OTHER, 1994

LE INSIGHT

INPO is the strongest evidence in any domain that capability engineering can be undertaken at the level of an **industry**, not just an organization. The conditions that made it possible — shared catastrophic exposure, regulatory legitimacy, an honest peer-review architecture — appear wherever a single failure can damage every operator.

LENS APPROACH

LENS uses INPO in LEN 8 as the canonical example of industry-level learning: students identify the structural conditions in their own domain that would permit (or block) an INPO-equivalent and design the peer-review architecture required. LEN 1 uses the founding moment — nine months after TMI — to discuss the **speed** a credible response to catastrophe demands.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Found the body on shared catastrophic exposure — make explicit that one operator's failure threatens every operator's license — so members have a direct stake in each other's competence.
- Set common training, certification, and accreditation standards across the industry rather than leaving each organization to learn alone.
- Staff peer review with working operators, not distant regulators, so examiners know what to look for and findings carry professional weight beyond rule compliance.

IN OPERATION / AFTER

- Track fleet-wide performance indicators and confirm the laggards are being pulled up, not just the leaders held to standard — the signature of a learning mechanism working.
- Sustain candid peer review by keeping the body funded by and accountable to its members, so the honest examination that makes it effective does not erode into formality.
- Export the enabling conditions rather than the institution when scaling (as INPO informed WANO), adapting the shared-exposure-plus-peer-review design to each new context.

REFLECTION QUESTIONS

1. What is the equivalent of "an accident at any single plant affects every operator" in your domain? If the answer is "nothing," what does that tell you?
2. INPO was stood up in nine months. Pick a current cross-organizational capability problem and write the nine-month deliverable that would constitute a credible response.
3. INPO held no statutory authority yet made its findings stick through shared catastrophic exposure and peer review. Design the non-statutory mechanism that could enforce a standard in your domain, and name the shared stake that would give it teeth.

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CASE 173 HEALTHCARE 2006 – PRESENT

TeamSTEPPS

T Training Gap N Normalization of Deviance

IMPACT Improved teamwork, communication, and patient-safety culture across diverse settings; OR on-time first start +21%; adopted by thousands of healthcare organizations

IN BRIEF

TeamSTEPPS — developed jointly by the Agency for Healthcare Research and Quality and the Department of Defense and released in 2006 — is the healthcare analog of Crew Resource Management (Case 70): an evidence-based team-training framework distilled from fifty years of aviation, military, and nuclear research and adapted for clinical settings. It trains four core competencies: communication, leadership, situation monitoring, and mutual support. It is explicitly the translation pathway from high-reliability research into bedside practice — the cross-domain capability transfer LENS is built to teach. Because its implementation infrastructure was funded as part of the program, TeamSTEPPS moved from research to scaled deployment in years rather than decades, and has been adopted by thousands of healthcare organizations with measurable gains in teamwork and safety culture.

BACKGROUND

Decades of research in aviation, the military, and nuclear power had shown that teamwork — not just individual expertise — drives safety in high-consequence work, and the IOM's To Err Is Human had identified poor communication as a leading cause of medical harm. What clinical settings lacked was a structured, evidence-based way to build those team skills. The evidence existed and the diagnosis was clear; what was missing was the translation — a route from the high-reliability research base into a curriculum a hospital could actually run at the bedside.[0]

THE INTERVENTION

In 2006, the Agency for Healthcare Research and Quality and the Department of Defense jointly released TeamSTEPPS — Team Strategies and Tools to Enhance Performance and Patient Safety — the healthcare analog of Crew Resource Management. It trains four core competencies: communication, leadership, situation monitoring, and mutual support, with a structured curriculum and ready-made implementation materials. The joint AHRQ-DoD authorship mattered: it paired a healthcare research agency's clinical reach with the military's deep team-training experience, so the framework arrived already grounded in both the evidence base and the practicalities of running it.[1]

HOW IT WORKED

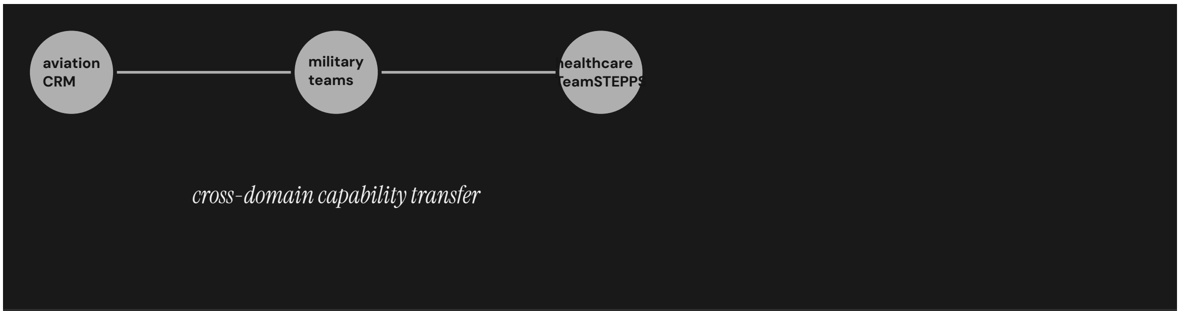
TeamSTEPPS is explicitly the translation pathway from high-reliability-organization research into clinical practice — fifty years of cross-domain evidence adapted for the bedside. Crucially, its implementation infrastructure (master-trainer programs, toolkits, an institutional support center) was funded as part of the intervention, so adopting organizations had a route from training to sustained practice rather than a binder on a shelf. Funding the implementation alongside the curriculum is precisely what compressed the usual decades-long gap between a proven method and its scaled use, because the path from adoption to sustained practice was paid for in advance.[2]

THE EVIDENCE

Studies across diverse settings report improved teamwork, communication, and patient-safety culture, with concrete operational gains such as a 21% improvement in on-time first surgical starts. Thousands of healthcare organizations have adopted the framework, and AHRQ has continued to develop it, releasing TeamSTEPPS 3.0 in 2023. The continued development into a third version is itself evidence the transfer took hold — a framework that is still being maintained and revised nearly two decades on is one institutions kept using, not one that was issued and abandoned.[3]

WHAT TRANSFERRED

TeamSTEPPS is the canonical evidence that cross-domain capability transfer is engineerable — and that the long implementation gap can be dramatically shortened when the transfer is funded as part of the intervention rather than as an afterthought. Its four competencies map directly onto the argument that capability engineering is itself a teachable discipline. The deeper lesson is that the bottleneck in cross-domain transfer is rarely the knowledge, which already existed, but the funded path that carries it from research into routine practice — the part programs most often leave unbudgeted.[4]



AVIATION → MILITARY → HEALTHCARE; THE TRANSLATION PATHWAY

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THE LEARNING ENGINEERING LENS

> *TeamSTEPPS represents the translation pathway from high-reliability research into clinical practice.*
>
> EDITORS' SYNTHESIS DRAWING ON AHRQ TEAMSTEPPS 3.0 (2023) AND SALAS ET AL.

LE INSIGHT

TeamSTEPPS is the canonical evidence that cross-domain capability transfer is engineerable and that it can shorten the implementation gap dramatically when the transfer is funded as part of the intervention rather than as an afterthought. The four competencies map directly to the LENS curriculum's argument for what capability engineering competence looks like as a deliverable.

LENS APPROACH

LENS treats TeamSTEPPS in LEN 8 as the canonical cross-domain transfer case and in LEN 1 as evidence that the program's core proposition — that learning engineering is a discipline — has institutional precedent. Studio projects (LEN 10) reference TeamSTEPPS as the worked example of cross-domain adaptation methodology.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Distill the cross-domain evidence base into a structured curriculum of a few teachable competencies rather than leaving adopters to translate the research themselves.
- Pair authorship across the source and target domains (as AHRQ and DoD did) so the framework arrives grounded in both the evidence and the practicalities of running it.
- Fund the implementation infrastructure — master-trainer programs, toolkits, a support center — as part of the intervention, not as an afterthought.

IN OPERATION / AFTER

- Maintain and revise the framework over time (as with successive TeamSTEPPS versions) so it stays in use rather than being issued and abandoned.
- Track concrete operational gains (such as on-time first surgical starts) alongside culture measures, so the transfer's effect is shown, not assumed.
- Identify and budget the specific path from adoption to sustained practice when scaling, since that unbudgeted step is the usual bottleneck in cross-domain transfer.

REFLECTION QUESTIONS

1. What is the cross-domain transfer your domain has not yet executed? What evidence base in another industry should it draw on?
2. TeamSTEPPS funded its implementation infrastructure. Design the implementation-infrastructure budget for an equivalent transfer in your domain.
3. TeamSTEPPS shortened the usual decades-long gap because the path from adoption to sustained practice was paid for in advance. Identify a proven method in your domain that has not scaled, and name the specific unbudgeted step that is blocking it.

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS scenario simulation · competency models · incident analysis · safety dashboards

CASE 174 HEALTHCARE INDUSTRIAL 1982

Tylenol Recall

G Governance & Trust N Normalization of Deviance

IMPACT Foundational U.S. corporate crisis-management case; produced tamper-evident packaging regulation and modern recall practice

IN BRIEF

In 1982, seven people in the Chicago area died after taking Tylenol capsules laced with potassium cyanide. Not knowing who was responsible or how widespread the tampering was, Johnson & Johnson recalled every Tylenol product nationwide — 31 million bottles, at a cost of roughly $100 million — suspended advertising, and engaged openly with the FBI and FDA. The response was unprecedented in U.S. corporate practice, and it was a direct application of the J&J Credo, written in 1943, which had pre-specified that the company's first responsibility was to its customers. The reform that followed reshaped consumer packaging worldwide — tamper-evident seals and blister packs — and the FDA issued tamper-resistant-packaging rules within months. Tylenol recovered its market share within a year.

BACKGROUND

Johnson & Johnson's corporate Credo, written in 1943, pre-specified that the company's first responsibility was to the patients and consumers who used its products, ahead of shareholders. For four decades it was a statement of values; in 1982 it became an operational decision rule under extreme pressure. The ordering was explicit — customers ahead of shareholders — which is precisely the ranking that crisis pressure inverts, so committing to it in advance pre-decided the hardest trade-off before it had to be faced.[0]

THE INTERVENTION

After seven people in the Chicago area died from Tylenol capsules laced with potassium cyanide, and with the source and scope of the tampering unknown, Johnson & Johnson recalled every Tylenol product nationwide — about 31 million bottles, at a cost near $100 million — suspended all advertising, and engaged publicly with the FBI and FDA rather than minimizing exposure. Recalling nationwide despite the tampering being known only in Chicago was the decisive choice — it treated the unknown scope as a reason to protect every customer rather than as room to limit the company's own exposure.[1]

HOW IT WORKED

The load-bearing element was a commitment pre-committed in writing. Because the Credo had already decided, decades earlier, that the customer came first, the 1982 leadership did not have to improvise an ethical calculus under crisis — it executed a pre-made decision. CEO James Burke later credited the Credo with making clear "exactly what we were all about" the moment the deaths occurred. Pre-commitment worked because it moved the decision out of the moment of maximum pressure — when fear and legal caution push hardest toward minimization — and into a calmer time when the right ordering could be set down without that distortion.[2]

THE EVIDENCE

The response, unprecedented in U.S. corporate practice, preserved public trust: Tylenol recovered its market share within a year despite the enormous short-term cost. The case became the canonical positive example in business education of crisis response driven by capability commitment rather than legal minimization. The market recovery is what makes the case persuasive rather than merely admirable — the $100 million spent protecting customers was repaid in the trust that brought them back, so the pre-committed choice proved sound on its own terms.[3]

WHAT TRANSFERRED

The reform reshaped consumer-product packaging worldwide — tamper-evident seals, blister packs, and caplet forms became standard — and the FDA promulgated tamper-resistant-packaging regulations within months. The deeper transfer is the principle that values must be pre-committed in writing to be operational under stress, not invented in the moment. The packaging reform and the decision-rule principle are the two layers of the transfer — one a physical safeguard against the specific threat, the other an institutional safeguard against the improvisation that crisis invites.[4]

31M

BOTTLES RECALLED · ~$100M COST

the pre-committed institutional credo became operational under stress

TYLENOL — VALUES PRE-COMMITTED IN WRITING, EXECUTED UNDER CRISIS

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THE LEARNING ENGINEERING LENS

> *The Credo is all about the consumer. When those seven deaths occurred, the Credo made it very clear at that point exactly what we were all about.*

JAMES BURKE (JOHNSON & JOHNSON CEO), QUOTED IN LASTING LEADERSHIP (WHARTON)

LE INSIGHT

Tylenol is the canonical positive case for institutional response to crisis. The capability that was load-bearing was the pre-specified institutional commitment in the Credo. The crisis decision had been made decades earlier; in 1982 it was executed. The case is the strongest evidence in the business-ethics dataset that values must be pre-committed in writing to be operational under stress.

LENS APPROACH

LENS uses Tylenol in LEN 7 as the foundational positive case for institutional governance under crisis and in LEN 10 (capstone) as a worked example of pre-committed capability that executed under operational pressure.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pre-commit the hardest trade-off in writing before the crisis — rank customer safety ahead of shareholder exposure — so leadership executes a pre-made decision rather than improvising under pressure.
- Set the rule in a calm period when fear and legal caution cannot distort the ordering, since those forces push hardest exactly when the decision must be made.
- Make the commitment concrete enough to act on — a nationwide recall, open engagement with regulators — so unknown scope becomes a reason to protect everyone rather than room to limit exposure.

IN OPERATION / AFTER

- Pair the institutional decision rule with a physical safeguard against the specific threat (tamper-evident packaging) so the response addresses both the improvisation problem and the vulnerability.
- Treat the preserved trust and market recovery as the measure that the pre-committed choice was sound, not merely admirable, and document it to defend the principle internally.
- Embed the commitment durably enough that it survives leadership turnover, so the next crisis meets the same pre-decided rule rather than a fresh improvisation.

REFLECTION QUESTIONS

1. What is your institution's equivalent of the J&J Credo, and is it operational under crisis or aspirational?
2. Pre-commitment is hard to enforce later. Design the institutional architecture that makes a Tylenol-style response the only available option in the worst case.
3. J&J recalled nationwide while the tampering was known only in Chicago, treating unknown scope as a reason to protect everyone. Identify a decision in your domain where uncertainty currently licenses minimizing exposure, and write the pre-committed rule that would flip it toward protection instead.

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CASE 175 AVIATION 1980S – PRESENT

Singapore Airlines Safety Transformation

T Training Gap N Normalization of Deviance

IMPACT Sustained safety record over decades despite challenging operating conditions; among the most safety-invested carriers in commercial aviation

IN BRIEF

Singapore Airlines has invested in safety capability across decades in a way that sets it apart from carriers operating under comparable conditions — early adoption of CRM, heavy simulator investment, a young-fleet policy, and a strong reporting culture, sustained even through rapid growth. The 2000 crash of Flight SQ006, which attempted takeoff from a closed, partly-constructed runway at Taipei during a typhoon and killed 83, prompted a sustained institutional self-examination — operational changes, training updates, and public transparency about what had happened — that became a model in the literature on post-accident learning. The airline is the operational successor to Korean Air (Case 72): an Asian carrier that engineered its safety capability deliberately and sustained the investment as a competitive differentiator, not only in response to crisis.

BACKGROUND

Commercial aviation runs on thin margins, and safety investment — simulators, training hours, fleet renewal — is a cost that competitive pressure constantly pushes downward. The question for any carrier is whether to treat capability as a floor set by regulation or as a deliberate, sustained investment ahead of the minimum. The pressure is structural rather than occasional — every budget cycle invites trimming the margin between regulatory minimum and actual capability, so sustaining the investment requires deciding the question deliberately rather than by default.[0]

THE INTERVENTION

Singapore Airlines chose sustained investment. From the 1980s it was an early adopter of Crew Resource Management and CRM-style culture work tuned to its operating context, and it committed to heavy simulator investment, a deliberately young fleet, and a strong internal reporting culture — maintaining these even during periods of rapid expansion. Holding the investment through rapid growth is the hard test — expansion is precisely when the temptation to let capability lag the fleet is strongest, and maintaining it then is what separates a sustained commitment from a fair-weather one.[1]

HOW IT WORKED

The carrier treated safety capability as a competitive differentiator rather than a regulatory burden, pairing technical investment — training systems, modern aircraft — with a culture of transparency and reporting. Investing ahead of regulatory minimums made the capability a managed system parameter, not a residual of cost-cutting decisions made elsewhere.

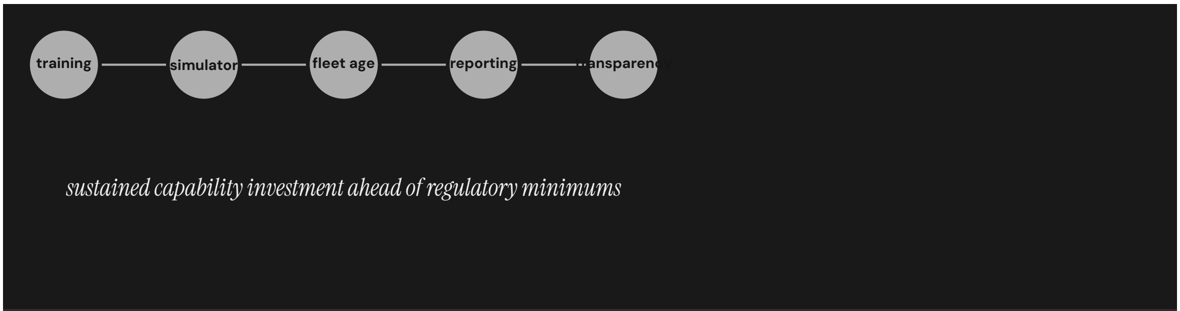
Framing safety as a differentiator rather than a burden is what made the investment defensible against cost pressure — it tied capability to the brand and the premium the airline charged, giving the spend a commercial rationale, not just a safety one.[2]

THE EVIDENCE

The 2000 crash of Flight SQ006 — an attempted takeoff from a closed, partly-constructed runway at Taipei during Typhoon Xangsane, killing 83 — tested the institution. Its response, documented in the Taiwan investigation, became a reference case in post-accident institutional learning: operational changes, training updates, and public transparency about what had happened and why, rather than defensiveness. The test of a safety culture is how it behaves after its own accident, and choosing transparency over defensiveness is what turned SQ006 from a refutation of the airline's reputation into evidence the reporting culture extended to its own failures.[3]

WHAT TRANSFERRED

Singapore Airlines is the case for sustained capability investment under competitive pressure, and the operational successor to Korean Air (Case 72): where Korean Air is a transformation forced by crisis, Singapore Airlines is deliberate investment sustained without one. Together they show two routes — crisis-driven and voluntary — to the same engineered safety capability. The voluntary route is the harder one to hold, because it has no catastrophe to point to as justification, which is why framing the investment as a competitive differentiator matters: it supplies the rationale that a crisis would otherwise provide.[4]



SINGAPORE AIRLINES — CAPABILITY INVESTMENT AS COMPETITIVE DIFFERENTIATOR

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> *Singapore Airlines has consistently invested in safety capability ahead of regulatory minimums.*
>
> EDITORS' SYNTHESIS ON SINGAPORE AIRLINES' SUSTAINED SAFETY INVESTMENT

LE INSIGHT

Singapore Airlines is the case for sustained capability investment in a competitive industry. The carrier has chosen safety capability investment as a primary differentiator. The result over decades is a safety record that distinguishes it from peers operating under comparable conditions.

LENS APPROACH

LENS uses Singapore Airlines in LEN 8 for sustained institutional capability investment under competitive pressure. The case pairs with Korean Air (Case 72) as Asian-carrier capability stories of different shapes — one transformation under crisis, the other sustained investment without crisis.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Decide deliberately to invest ahead of regulatory minimums — simulators, a young fleet, training hours — rather than letting capability settle at the floor by default under cost pressure.
- Pair the technical investment with a transparency and reporting culture, so capability is a managed system parameter rather than a residual of cost-cutting decisions elsewhere.
- Frame safety capability as a competitive differentiator tied to the brand, giving the spend a commercial rationale that can survive scrutiny, not just a safety one.

IN OPERATION / AFTER

- Hold the investment through periods of rapid growth — the moment the temptation to let capability lag the fleet is strongest — since maintaining it then is what makes the commitment sustained rather than fair-weather.
- Respond to the institution's own accidents with transparency over defensiveness, demonstrating that the reporting culture extends to its own failures and converting a setback into evidence the culture is real.
- Anchor the investment to the brand and premium so it survives leadership and budget cycles, supplying the durable justification a voluntary commitment lacks without a crisis to point to.

REFLECTION QUESTIONS

1. Identify an institution in your domain that has chosen capability investment as a primary differentiator. What pattern has it sustained?
2. Design the institutional architecture that makes sustained capability investment defensible against competitive cost pressure.
3. Singapore Airlines sustained its investment voluntarily, without a crisis to point to, by framing capability as a competitive differentiator. Identify a safety or capability investment in your domain that lacks a catastrophe to justify it, and construct the commercial rationale that would defend it against the next budget cut.

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CASE 176 NEUROSCIENCE SCIENCE POLICY 2011 – 2015 – PRESENT

Launching the BRAIN Initiative — Governance of a Grand Challenge

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

DISCLOSURE An editor of this volume has research adjacency to connectomics programs discussed in this case. The case is anchored to widely cited primary position papers and to independent journalism, not to the editor's own work; the inclusion of the critical retrospective is the deliberate safeguard against boosterism.

IMPACT A multi-billion-dollar national research endeavor launched via a position-paper-to-policy iteration sequence, with governance contestation on the public record and a 2021 critical retrospective documenting that the unified-understanding framing exceeded what the science delivered

IN BRIEF

The BRAIN Initiative is one of the few large-program launches in the corpus whose governance trail is publicly documented end-to-end: a 2011 Kavli symposium produced a six-author Neuron position paper (Alivisatos et al., 2012); the proposal was shepherded to the White House OSTP, became a Presidential initiative in 2013, and was operationalized by an NIH working group whose BRAIN 2025 report (Jorgenson et al., 2015) set milestones and cost estimates. What makes it teachable rather than triumphalist is the governance contestation visible in the same record. Yuste worried the advisory panel was "packing the committee with users, rather than tool builders" — diluting the original focus; Bargmann, who later co-chaired the working group, had earlier expressed skepticism that the proposal "sounds like a big central planning project that will take resources away from creative work." A 2021 MIT Technology Review retrospective assessed that the big-science brain projects, BRAIN included, did not deliver the unified understanding their framing promised. The case pairs with the EU Human Brain Project (Case 16) as a contrast in governance models — distributed working-group versus top-down single-PI — at the same era, the same ambition, and opposite trajectories.

BACKGROUND

In 2011 a small group of neuroscientists convened at a Kavli Foundation symposium argued that the field had a tractable grand-challenge target: recording the activity of large numbers of neurons across whole circuits. The argument was published in Neuron in 2012 as the Brain Activity Map proposal (Alivisatos, Chun, Church, Greenspan, Roukes, and Yuste) and was shepherded through Kavli's Miyoung Chun to the White House Office of Science and Technology Policy. The position paper named a capability the field could be organized around and a sequence of tools that would have to be built; it was a launch artifact rather than a results report.[0]

THE INTERVENTION

In April 2013 President Obama announced the BRAIN Initiative. The operational governance was an NIH working group, co-chaired by Cori Bargmann and Bill Newsome, charged with translating the vision into a milestone-and-cost-bearing plan. The group published the BRAIN 2025 report (Jorgenson et al., Phil. Trans. R. Soc. B, 2015), which named seven priority areas, set timelines, and laid out funding ranges. Among the few large research-program launches in the corpus, this is one whose position-paper-to-policy-to-implementation sequence is openly auditable — every step has a published artifact attached.[1]

HOW IT WORKED

What makes the case teachable rather than triumphalist is the governance contestation visible in the same record. Yuste, one of the original six authors, voiced concern that the advisory panel was being expanded with users of the tools the program was meant to build, rather than the tool-builders the original proposal had centered — a documented dilution of scope. Bargmann, before her appointment, had publicly expressed skepticism that a big central planning project would draw resources away from creative independent research. The governance choices — who leads, tool-builders versus users, central plan versus distributed creativity — were not made in private and then defended; they were litigated in the public record while the program ran.[2]

THE EVIDENCE

The 2021 MIT Technology Review retrospective took the ten-year view: big-science brain projects, BRAIN among them, did not deliver the unified understanding their founding framing had promised. The honest assessment is that the initiative produced tools, atlases, and a coordinated funding stream — meaningful capability — while drifting from the grand-challenge framing the position paper had used to mobilize political support. The case is the instructive form of the "enthusiasm-evidence gap" at field scale: the framing carried the politics, and the science delivered something different and more diffuse.[3]

WHAT TRANSFERRED

What the case teaches is that large-program governance is the capability deliverable — not the framing, not the early tools, not the eventual results. The position-paper-to-policy sequence, the working-group composition decision, and the public airing of scope drift are the artifacts a future capability-development program can study. Paired with the EU Human Brain Project (Case 16), the case shows that the governance model — distributed working-group versus top-down single-PI — was the variable that explained why one program survived and adapted while the other unraveled. The framing, ambition, and era were comparable; the governance was not.[4]

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THE LEARNING ENGINEERING LENS

> *Governance contestation in the public record is not program failure. It is what makes the case teachable.*
>
> EDITORS' SYNTHESIS OF THE BRAIN INITIATIVE RECORD (2012 – 2021).

LE INSIGHT

The BRAIN Initiative is one of the few large-program launches in the corpus whose position-paper-to-policy-to-implementation sequence is fully auditable. Governance choices were litigated in the public record, scope drifted from the founding framing, and a critical ten-year retrospective is part of the case materials. Its teaching value is the contested record, not a clean success or scandal.

LENS APPROACH

BRAIN is the field-scale stakeholder-and-governance case in the v2 corpus (induced 5.1; LENS D5+D1/PT4). LENS uses it in Domain 1 (Systems Analysis) for the position-paper-to-policy iteration sequence; in Domain 5 (Navigating Sociotechnical Constraints) for the public-record governance contestation; and in Domain 4 (Test and Evaluation) for the enthusiasm-evidence gap as the framing exceeded delivered science. Direct pair with Case 16 (EU Human Brain Project), same era and ambition, opposite governance model. The COI disclosure under the title is binding: the editor's research adjacency is what motivated the critical retrospective being included as a deliberate counterweight.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make the position-paper-to-policy iteration auditable by attaching a published artifact to every step — proposal, OSTP transmittal, working-group composition, milestone plan.
- Treat working-group composition as a governance act, not a staffing decision: tool-builders vs. users vs. integrators is a framing-shaping choice, and naming the trade-off openly is part of the work.
- Carry the founding scope through the operational record: name where the program is keeping faith with the founding framing and where it is drifting, while the program is running rather than only in retrospect.

IN OPERATION / AFTER

- Commission and publish a long-window retrospective that compares the founding framing to delivered capability honestly — not as a closure ritual but as governance evidence the next program can use.
- When scope drift is documented, decide explicitly whether to reframe the program publicly or to re-baseline against the original framing; the worst case is leaving the gap unaddressed in the record.
- Treat governance contestation in the public record as program documentation, not as program failure; the contested record is what makes the case teachable.

REFLECTION QUESTIONS

1. Identify a large research or capability-development program in your domain whose launch artifacts (position papers, working-group reports, milestone plans) are publicly auditable. What governance choices were made openly and which were made in private?

2. Specify the founding-framing-vs-delivered-capability comparison you would publish at year five and year ten of a program you are designing. What is the evidence form, who commissions it, and where does it live in the record?

3. BRAIN survived and adapted while the EU Human Brain Project (Case 16) unraveled. What is the minimum governance documentation that would let a future program-designer learn the difference, rather than reconstruct it from contested press accounts?

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CASE 177 RAIL TRANSPORT SAFETY REPORTING SAFETY CULTURE 1996 – PRESENT

CIRAS — Confidential Incident Reporting for UK Rail

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Between 2008 and 2012 the UK rail Confidential Incident Reporting and Analysis System received 2,228 reports — 45% led to tangible safety improvements and about 33% contained important safety information (program self-report); directly influenced a confidential reporting system in the US

IN BRIEF

CIRAS began as a 1995–1997 ScotRail pilot — a structured channel for rail workers to report hazards and near-misses confidentially, insulated from the employer's disciplinary process. After the Ladbroke Grove crash in 1999, the program was mandated across UK mainline rail in 2000; in 2008 it became an independent unit within the Rail Safety and Standards Board. The published record states that between 2008 and 2012 CIRAS received 2,228 reports, of which the operating program reports 45% led to tangible safety improvements and about 33% contained important safety information. The program directly influenced the design of a confidential reporting system in the United States. The interview-based method surfaces motive and intent — the **why** behind a near-miss — that company databases miss precisely because those databases are tied to discipline. The honest hedge that survives into the case: the 45%-led-to-improvement figure is self-reported by the operating program, not independently audited. The case is the non-aviation companion to ASRS / CRM in v1 (Cases 70 + 89), keeping the non-punitive-reporting-with-credible-commitment competency from resting entirely on aviation evidence.

BACKGROUND

Confidential incident reporting as a safety-culture intervention rests on a specific structural argument: the richest information about a near-miss — the operator's intent, the team's local pressures, the workaround that almost worked — is exactly the information that an employer-controlled reporting channel cannot collect, because reporters with anything to lose will not put it there. The aviation experience (ASRS, CRM; v1 Cases 70 and 111) establishes the pattern at the canonical safety-culture scale. CIRAS is the same pattern, in a different industry, with the credible commitment supplied differently — and the corpus needs the non-aviation evidence.[o]

THE INTERVENTION

CIRAS began in 1995–1997 as a ScotRail pilot. The structural design was the interview, not the form: trained interviewers took reports from rail workers, anonymized them, and surfaced patterns to the operating companies. The credible commitment that made the channel safe to use was not just confidentiality — it was institutional independence from the employer's disciplinary process. After the Ladbroke Grove crash in October 1999 (31 deaths, the accident that defined the decade's UK rail-safety reform), CIRAS was mandated across UK mainline rail in 2000. In 2008 it became an independent unit within the Rail Safety and Standards Board (RSSB), with the independence from employer discipline written into its operating structure.[1]

HOW IT WORKED

The published record on outcomes is what the case rests on. Davies et al. (Cognition, Technology & Work) describe the method and its yield. The operating program reports that between 2008 and 2012, CIRAS received 2,228 reports, of which 45% led to tangible safety improvements and approximately 33% contained important safety information that fed back into operating practice. The program's design has directly influenced the construction of a confidential reporting system in the United States. The interview-based method surfaces motive and intent in a way that incident-database schemas tied to discipline cannot.[2]

THE EVIDENCE

The honest hedge has to survive. The 45%-led-to-improvement figure is self-reported by the operating program — CIRAS itself characterizes its outputs, and there is not yet an independent audit of that classification. The peer-reviewed literature on CIRAS describes the method and the institutional design, and reports the program-supplied figures rather than independently validating them. The case is teachable on the evidence of the institutional design and the published method; the operating outcome statistics are the strongest current claim, and the editor and downstream readers should treat them as program self-report rather than audited finding.[3]

WHAT TRANSFERRED

What CIRAS adds to the corpus is non-aviation depth in the non-punitive-reporting-with-credible-commitment competency. The mechanism (a structured confidential channel) paired with the credible commitment (independence from the employer's disciplinary process, written into operating structure) is the cultural half of capability that the aviation cases also teach. Drafted together, ASRS (Case 111), CRM/CAST (Case 70), and CIRAS (this case) show that the structural form is transferable across high-consequence operational industries — and that the credible commitment, not the channel alone, is what makes the reporting safe to use.

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> *The 45%-led-to-improvement figure is the operating program's self-report. The institutional design is the audited finding.*
>
> EDITORS' SYNTHESIS OF DAVIES ET AL. AND THE STRATHCLYDE IMPACT CASE STUDY.

LE INSIGHT

CIRAS is the non-aviation companion to ASRS and CRM in the corpus. The pattern — confidential channel paired with credible commitment (institutional independence from discipline) — works in rail as it does in aviation. The operating outcome statistics are program self-report and deserve their tier acknowledgement; the institutional design is the audited finding.

LENS APPROACH

CIRAS is the non-aviation pairing-mechanism case (induced 4.2; LENS D5/PT2). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the credible-commitment design — independence written into operating structure — and in Domain 4 (Test and Evaluation) for the CLO **Judgment under inadequate evidence** on the operating self-report tier. Pair with ASRS (Case 111) and CRM/CAST (Case 70) as the cross-industry C4 set, and with the WHO Surgical Checklist (Case 109) at the mandatory-mechanism layer.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the credible commitment first: institutional independence from the employer's disciplinary process, written into operating structure, not relying on goodwill that can be withdrawn.
- Choose the reporting method (interview, not just form) for the information class you want to surface — motive and intent — that form-based incident databases tied to discipline cannot collect.
- Anchor the program's authority in a sector-wide forcing event (Ladbroke Grove for rail; in other domains, the analog) so the mandatory-adoption decision rides on the safety crisis the program is built to address.

IN OPERATION / AFTER

- Report operating outcome statistics with their evidence tier: program self-report vs. independently audited. Treating the 45% figure as audited overstates the evidence; treating it as untested under-states the program's institutional credibility.
- Commission an independent audit of the program's classification of its own reports at a sustainable cadence — five years is a reasonable interval — so the operating record builds toward audited evidence over time.
- Carry the structural lesson across domains: ASRS, CRM, CIRAS — the cultural half of capability is the credible commitment, not the channel alone.

REFLECTION QUESTIONS

1. Identify a high-consequence operational domain in your context where confidential incident reporting is absent or weak. What is the credible-commitment design that would make the channel safe to use — and is the institutional independence written into operating structure, or relying on goodwill?
2. Specify the information class your reporting program is built to surface: form-based incident counts, or interview-based motive and intent. The choice of method follows the information class, and they answer different questions.
3. The 45%-led-to-improvement figure is program self-report. Design the independent-audit cadence that would convert the operating record into audited evidence over time, without compromising the confidentiality the channel depends on.

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CASE 178 CLINICAL/TRANSLATIONAL RESEARCH TEAM SCIENCE HEALTHCARE WORKFORCE 2020 – 2022

Team Science Training for Clinical and Translational Scientists

K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Colorado CTSA team-science training (N=221, pre/post) reported statistically significant improvement ($p<0.05$) across all three TEAMS instrument competencies — Team Planning, Managing a Team, and Interpersonal Relations — with the largest gains in structured/tool-based domains and the smallest in Interpersonal Relations

IN BRIEF

The Colorado Clinical and Translational Sciences Institute built a structured team-science training program — explicit competency targets, structured exercises, mentor pairings — and evaluated it in a 2020–2022 cohort with N=221 participants using a pre/post design. The evaluation introduces and validates a measurement instrument (TEAMS) that resolves team-science capability into three components: Team Planning, Managing a Team, and Interpersonal Relations. Participants showed statistically significant improvement ($p<0.05$) across all three components — with the largest gains in the structured / tool-based domains (Team Planning, Managing a Team) and the smallest gain in Interpersonal Relations, the component the authors explicitly note is the hardest. The case treats interdisciplinary collaboration as an engineerable, measurable capability — the amended Domain 3 sub-competency (collaboration as a unit of measurement, distinct from any individual operator) — and is honest about the interpersonal half moving least. The honest limit the authors name: outcomes are self-reported and the design lacks longitudinal tracking and integration metrics; this is perceived-competency-gain evidence, not yet demonstrated downstream research- collaboration impact.

BACKGROUND

Clinical and translational research is the structural example of a discipline whose capability sits in the collaboration itself rather than in any single team member. A protocol moves across investigators, study coordinators, biostatisticians, clinical staff, and regulatory specialists; the science depends on the team's coordination as much as on any individual's expertise. The capability question the case addresses is whether interdisciplinary collaboration can be trained — and whether the resulting capability can be measured.[o]

THE INTERVENTION

The Colorado Clinical and Translational Sciences Institute built a structured team-science training program with explicit competency targets, structured exercises, and mentor pairings. The evaluation cohort ran 2020–2022 with N=221 participants in a pre/post design. The evaluation paper's central methodological contribution is the validation of a measurement instrument — TEAMS — that resolves team-science capability into three competency components: Team Planning (the structured front-end of collaboration), Managing a Team (the structured operational half), and Interpersonal Relations (the unstructured half that the literature describes as the hardest to train).[1]

HOW IT WORKED

Pre/post comparison showed statistically significant improvement ($p<0.05$) across all three components. The pattern of gains is what makes the case teachable: the largest gains were in the structured / tool-based domains — Team Planning and Managing a Team — and the smallest gain was in Interpersonal Relations. The authors do not soften this; the paper states it plainly, and the smaller-gain-on-the-hardest-half finding is the load-bearing honest result.[2]

THE EVIDENCE

The honest limit the authors also name carries into the case. The TEAMS instrument and the pre/post comparison measure self-reported perceived competency, not downstream collaboration outcome. The design did not track participants longitudinally into their next collaborative projects, did not measure integration metrics on the projects they ran after the training, and did not compare to a non-trained control. The evidence is the strongest available current evidence for a structured team-science training program; it is not closed proof of downstream collaboration impact, and the case carries the qualification rather than collapsing it.[3]

WHAT TRANSFERRED

In pair with IPE (Case 117) and Implementation Science Training (Case 179), the Colorado CTSA case is the small-tier intervention companion to two frontier cases. Team-science training is one of the few cases in the corpus that operationalizes "collaboration as the unit of measurement" — the CLO — with a validated instrument and structured evaluation. The case is a worked example of how a discipline can convert collaboration capability from an aspiration to a measurable target, while preserving the honest qualifications about what the measurement does and does not establish.

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> *The largest gains were in the structured half. The hardest half — interpersonal relations — moved least, and the paper states it plainly.*
>
> EDITORS' SYNTHESIS OF COLORADO CTSA TEAM-SCIENCE TRAINING EVALUATION.

LE INSIGHT

The Colorado CTSA team-science training program is one of the few cases in the corpus that operationalizes "collaboration as a unit of measurement." The TEAMS instrument resolves the capability into three components; structured training moved all three with $p<0.05$ but moved interpersonal relations least. The honest limit — self- reported perceived competency, no longitudinal tracking, no control — survives into the case.

LENS APPROACH

Team-science training is the case-grounded basis for the collaboration-as-measurement CLO (induced 4.3; LENS D5/PT4). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the team-coordination redesign and in Domain 4 (Test and Evaluation) for the TEAMS instrument validation and the perceived-competency vs. downstream-impact discipline. Direct pair with Cases 117 (IPE) and 123 (implementation science training) — the intervention with measurement against two frontier cases where the measurement is the gap.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Resolve collaboration capability into measurable components — Team Planning, Managing a Team, Interpersonal Relations — and structure the training so each component has tools, exercises, and an assessment instrument.
- Validate the measurement instrument before relying on it for evaluation; the TEAMS instrument's three-component structure is the load-bearing methodological contribution.
- Acknowledge in the program design that the interpersonal component is the hardest to train and the slowest to move; design the exercises and the mentor pairings around that asymmetry rather than against it.

IN OPERATION / AFTER

- Report the pattern of gains, not just the aggregate; the smaller-gain-on-interpersonal finding is the honest signal about what the structured training can and cannot do.
- Commission longitudinal follow-up that tracks participants into their next collaborative projects and measures integration outcomes — the case names the gap and the next study addresses it.
- Carry the "collaboration as the unit of measurement" sub-competency into the curriculum: the Domain 3 sub-competency is grounded in evidence here, and the team-science training program is one of the few cases in the corpus that operationalizes it.

REFLECTION QUESTIONS

1. Identify a collaboration-dependent capability in your domain. What three or four components would you resolve it into for measurement purposes, and what instrument would you validate to test the resolution?

2. Specify the longitudinal follow-up you would build into the next iteration of a team-science training program — what integration metric, on what cadence, against what comparison — to convert perceived-competency-gain evidence into downstream-collaboration-impact evidence.

3. The Colorado CTSA finding is that the structured half moves first and the interpersonal half moves least. What does that imply about the curriculum you would design — and about which components are realistic targets for a single-program training intervention vs. an institutional / cultural intervention?

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CASE 179

IMPLEMENTATION SCIENCE TRANSLATIONAL WORKFORCE EDUCATION AT SCALE

2020S

Implementation-Science Training — Stated Goals Outrunning Operational Practice

K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Survey of CTSA-funded TL1 training programs (N=50): most name collaboration, team science, and multi/inter/cross-disciplinary training as goals, but far fewer (14–24%) embed competency-based training and assessment, program evaluation, or experiential learning

IN BRIEF

Implementation science — the discipline of moving validated evidence into operational practice — is one of the most consequential cross-domain-transfer competencies in contemporary medicine and adjacent fields, and the CTSA T32 / TL1 program survey (N=50 programs) examined whether it is being taught systematically. The headline finding is the same structural pattern as the IPE case (122) at smaller scale: most programs name the right goals — collaboration, team science, multi/inter/cross-disciplinary training — but far fewer (14–24% across the operational practices the survey examined) actually embed the competency-based training and assessment, program evaluation, or experiential learning that the stated goals imply. The picture is of a field that has converged on what interdisciplinary translation training should aim for, but where the operational practices lag the stated goals — the same enthusiasm-ahead-of-evidence pattern as IPE. The case is the workforce-training counterpart to Case 34 (the "17-year gap" between research evidence and practice adoption); the v2 frame is the gap between what implementation-science training programs **say** they do and what they **operationally** do.

THE SHIFT

Implementation science exists because the gap between research evidence and operational practice in medicine — the so-called "17-year gap" the v1 corpus documents at v1 Case 34 — is large enough to constitute a discipline-level capability question. Moving validated evidence into clinical practice reliably is itself a competency that has to be designed, trained, and evaluated. The CTSA (Clinical and Translational Science Awards) program is the primary US mechanism for building that workforce, and the TL1 / T32 programs are its training arm.[0]

WHAT IS EMERGING

The case's evidence base is a 2021 survey of CTSA-funded TL1 training programs (N=50) examining what those programs actually do at the operational level. The survey covers the stated goals — what the programs say they aim to develop — and the operational practices — what the programs do, in practice, to develop it. The stated-goals side is where the programs converge: most name collaboration, team science, and multi/inter/cross-disciplinary training as central objectives. The operational-practices side is where the pattern surfaces: far fewer programs (14–24% across the operational dimensions the survey examined) actually embed competency-based training and assessment, program evaluation, or experiential learning at the level the stated goals would imply.[1]

THE CAPABILITY QUESTION

The structural form is the same pattern IPE shows at field scale (Case 117): the field has converged on what interdisciplinary translation training should aim for, but the operational practices lag the stated goals. At the implementation-science training scale, the lag has specific consequences: programs that intend to produce graduates capable of moving evidence into practice often do not systematically assess whether they have produced that capability, and the field's measurement of its own workforce-development effectiveness is correspondingly sparse.[2]

EARLY EVIDENCE

What the case teaches is the operational-practice gap as a designable target, not as a curricular failure. The 14–24% figure is not an indictment of the programs that fall on the wrong side of it; it is a finding about the operational- infrastructure investment that competency-based training and assessment require, and that field-scale stated-goal convergence does not by itself produce. The pair with Case 34 (the 17-year gap) gives the case its frame: the implementation-science workforce is the recovery mechanism for the research-to-practice gap, and the operational capacity of that workforce is itself the capability the training programs are trying to build.[3]

OPEN PROBLEMS

In the multidisciplinary-translation trio (Cases 178 + 121 + 122), implementation-science training sits between team science (where the measurement is possible at program scale) and IPE (where the measurement is the field-scale gap). The trio teaches the enthusiasm-evidence-gap sub-competency from three angles — the program-scale success, the field-scale gap, and the operational-practice gap inside training programs themselves — and it teaches the CLO **Judgment under inadequate evidence** by example: practitioners designing implementation-science training programs have to decide what to build on incomplete evidence of what works, while the field is still building the evidence architecture that would let them decide better.

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> *Stated goals converge on the right targets. Operational practices lag. The gap is designable, not a curricular failure.*
>
> EDITORS' SYNTHESIS OF THE CTSA TL1 PROGRAM-GOALS STUDY (2021).

LE INSIGHT

The CTSA TL1 program survey is the workforce-training instance of the enthusiasm-evidence gap pattern: most programs name competency-based training, assessment, and program evaluation as goals; the operational practices that would deliver on those goals are present in 14–24% of programs. The implementation-science workforce is the recovery mechanism for the 17-year research-to-practice gap; the case names the gap inside the recovery mechanism itself.

LENS APPROACH

Implementation-science training is the frontier workforce-training case (induced 6.4; LENS D5/PT4). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the cross-disciplinary translation discipline; in Domain 4 (Test and Evaluation) for the enthusiasm- evidence-gap sub-competency; and in Domain 2 (Iterative Development) for the CLO **Judgment under inadequate evidence** as practitioners design training programs while the field-scale evidence is still being built. Pair with Cases 178, 117 as the multidisciplinary-translation trio; workforce-training counterpart to Case 34.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the competency-based assessment instrument before launching the training program; the gap the survey documents is partly the result of stated competencies that were never operationalized into measurement.
- Build experiential learning into the operational structure of the program, not as an add-on; the survey's 14–24% figure includes experiential learning as one of the operational dimensions that lags.
- Plan program evaluation as a deliverable of the training program itself, with the cadence, instrument, and reporting venue named at launch.

IN OPERATION / AFTER

- Report the operational-practice gap honestly: stated goals are the convergence point, operational practices are the lag, and the lag is the addressable target.
- Treat the multidisciplinary-translation trio (Cases 178, 117, 179) as a unit; the three-angle teaching of the enthusiasm-evidence pattern is more useful than any single case can be.
- Connect the case explicitly to Case 34 (the 17-year gap): the implementation-science workforce is the recovery mechanism for that gap, and the operational capacity of the workforce is the capability that has to be built — which is the case's pedagogical point.

REFLECTION QUESTIONS

1. Identify a training program in your domain whose stated goals include competency-based assessment, program evaluation, or experiential learning. What proportion of those stated goals are operationalized into specific instruments, cadences, and reporting structures — and which are at the goal-statement layer only?

2. Specify the competency-based assessment instrument you would build into the next iteration of an implementation-science training program. The survey's 14–24% figure says the instrument is what is missing more than the intent; what is the instrument?

3. The implementation-science workforce is the recovery mechanism for the 17-year research-to-practice gap (Case 34). What is the analog in your domain — the workforce whose operational capacity is the recovery for a documented systemic gap — and what is the case for investing in that workforce's training architecture?

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CASE 180 AVIATION SAFETY INDUSTRY COORDINATION TRAINING 1992 – 2000S

FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT After Controlled Flight Into Terrain emerged as the leading cause of commercial-jet fatalities through the late 1980s, the Flight Safety Foundation convened industry-wide task forces that produced the CFIT Checklist, the ALAR Tool Kit, and the institutional momentum behind Terrain Awareness and Warning System (TAWS) mandates; CFIT and ALAR accident rates fell sharply over the subsequent decade

IN BRIEF

Through the late 1980s and into the early 1990s, Controlled Flight Into Terrain — a serviceable aircraft flown under control into the ground, water, or an obstacle — was the leading cause of commercial-jet fatalities worldwide. The Flight Safety Foundation (FSF), an independent industry body, convened the CFIT Task Force in 1992 and, in parallel with the broader ICAO response, produced the CFIT Checklist — a structured tool for operators to assess their own exposure. The Approach-and-Landing Accident Reduction (ALAR) Task Force followed in 1996, producing the ALAR Tool Kit (released 1998) covering the approach phase where roughly half of fatal accidents then occurred. The institutional momentum from those task forces sat behind the eventual Terrain Awareness and Warning System (TAWS) mandates in the US (2002) and ICAO (2007). CFIT and approach-and-landing accident rates fell sharply through the subsequent decade. The case teaches industry-level institution building after a catastrophe-class spike: the deliverable is the cross-operator tool, the diagnostic structure, and the coordinated path to mandate. The hedge survives — the rate decline is multifactorial (TAWS hardware, training, procedural change, fleet turnover) — and the institutional form is what the case is teachable on.

BACKGROUND

Through the late 1980s and into the early 1990s, Controlled Flight Into Terrain dominated commercial-jet fatality statistics. A serviceable aircraft, flown under control, was finding terrain or water — the crew's mental model of position and trajectory diverged from reality at the worst possible moment, and the existing Ground Proximity Warning Systems (GPWS) generation reached the crew too late to recover in many terrain encounters. The pattern was recognized across operators and regulators, and the response that emerged was industry coordination led by an independent body — the Flight Safety Foundation — rather than regulator-first action alone.[0]

THE INTERVENTION

The CFIT Task Force convened in 1992 and operated as a working group across operators, manufacturers, regulators, and pilot organizations. Its central deliverable was the CFIT Checklist — a structured self-assessment instrument that let an operator score its own CFIT-risk exposure across route, equipment, training, and procedure dimensions, and identify where the gaps sat. The Checklist was distributed without restriction. The institutional theory was straightforward: a cross-operator diagnostic, owned by no single competitor, would let the industry move together on a problem that no single operator's own data could fully characterize.[1]

HOW IT WORKED

The Approach-and-Landing Accident Reduction Task Force followed in 1996, scoped to the approach phase, which accounted for roughly half of fatal accidents in the analysis window. The ALAR Tool Kit, released in 1998, was the most substantial deliverable of the entire industry-coordination effort: a multi-element package of briefing notes, training aids, video content, and risk-assessment instruments covering stabilized approach criteria, runway excursion, monitored approach practice, and crew procedural design. The Tool Kit was adopted across operators of all sizes precisely because the FSF had no competitive stake in any one airline's adoption.[2]

THE EVIDENCE

The eventual regulatory action — the FAA's TAWS mandate (2002, retrofitting earlier GPWS-equipped aircraft with the newer terrain-database-driven warning system) and the ICAO TAWS requirement (effective 2007) — sat downstream of the task-force momentum, not upstream of it. CFIT accident rates fell sharply through the subsequent decade across both commercial transport and corporate aviation; approach-and- landing accidents declined alongside. The before/after pattern is robust across multiple independent operator and regulator datasets, and the case is treated in the FSF's own retrospective literature as one of the discipline's strongest examples of coordinated industry-led intervention.[3]

WHAT TRANSFERRED

The load-bearing hedge survives into the case. The accident-rate decline is multifactorial: TAWS hardware in the cockpit, training changes the task forces catalyzed, stabilized-approach criteria adopted at the operator level, and steady fleet turnover into airframes with more capable equipment all moved together. Attributing the entire decline to the FSF task forces alone overstates the evidence; what the evidence supports is that the industry-coordination form — independent convening body, cross-operator diagnostic tools released without competitive restriction, momentum sustained to a regulatory mandate — was the institutional mechanism that organized the response, and the response worked. The case is the canonical C6.1 instance of industry-level institution building after a catastrophe-class spike, paired with v1 ASRS (Case 111) and CAST (Case 70) at the industry-coordination layer.[4]

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THE LEARNING ENGINEERING LENS

> *The deliverable is the cross-operator diagnostic, owned by no single competitor, released without restriction. The mandate consolidates what the field already does.*
>
> EDITORS' SYNTHESIS OF THE FSF CFIT AND ALAR TASK-FORCE LITERATURE.

LE INSIGHT

The FSF CFIT and ALAR task forces are the canonical case of industry-level institution building after a catastrophe-class spike. An independent foundation convened the cross-operator working bodies, released structured diagnostic tools without competitive restriction, and sustained momentum to a regulatory mandate. The accident-rate decline is multifactorial; the institutional form is what the case is teachable on.

LENS APPROACH

FSF CFIT/ALAR is the canonical industry-coordination case (induced 6.1; LENS D5/PT4) — Domain 5 for the independent- convening-body form; Domain 2 for the structured diagnostic instruments as iterating deliverables. Pair with Cases 70, 111, and 13.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Convene the cross-operator working body before the regulator does — an independent industry foundation, no competitive stake in any one operator's adoption — so the diagnostic tool can be released without restriction and adopted across the field.
- Design the diagnostic as a structured self-assessment, not a benchmark league table; operators will use what they can apply privately, and the field-level signal aggregates from voluntary adoption.
- Build the tool kit around the phase of flight that accident analysis says dominates the harm — approach-and-landing in this case — rather than around evenly distributed coverage that no single operator can adopt fully.

IN OPERATION / AFTER

- Sustain the institutional momentum into the regulatory mandate; the task forces did the technical and political work that made the TAWS rule adoptable, and the rule consolidates what the field already does.
- Report the accident-rate decline with the multifactorial hedge intact; TAWS hardware, training, procedure, and fleet turnover all moved together, and isolating the task-force contribution overstates the available evidence.
- Treat the institutional form as the transferable artifact: an independent convening body, cross-operator diagnostic tools without competitive restriction, coordinated path to regulatory mandate. The form pairs with CAST (Case 70) and ASRS (Case 111) at the industry-coordination layer.

REFLECTION QUESTIONS

1. Identify a catastrophe-class failure pattern in your domain whose response has been operator-by-operator rather than industry-coordinated. What would the analog of an independent convening body look like, and which body could plausibly play that role without competitive stake?

2. Specify the cross-operator diagnostic — checklist, tool kit, structured self-assessment — that you would design as the first deliverable of an FSF-style task force. The deliverable has to be applicable privately by each operator, and aggregate into field-level signal.

3. The CFIT/ALAR decline is multifactorial. What is the minimum decomposition you would publish — hardware, training, procedure, fleet turnover — to keep the institutional claim honest while still defending the task-force form as the organizing mechanism?

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CASE 181 HIGHER EDUCATION (LATIN AMERICA) LEARNING ANALYTICS CROSS-NATIONAL GOVERNANCE 2017 – 2020

LALA — Building Learning-Analytics Governance Capacity Across Latin America

K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT An EU-funded multi-country project (Chile, Ecuador, Mexico) that explicitly rejected lifting Global-North learning-analytics tools wholesale; structured interviews with administrators and focus groups with students and teachers produced the LALA CANVAS participatory adoption framework, with stakeholders demanding ethical responsibility as a precondition for data-driven feedback

IN BRIEF

The LALA (Learning Analytics for Latin America) project, funded under EU grant 586120-EPP-1-2017-1-ES, ran from 2017 to 2020 across Chilean, Ecuadorian, and Mexican universities. The project began from a deliberate refusal: not to lift US and European learning-analytics tools wholesale into Latin American institutional contexts, on the grounds that adoption fails when the tools do not integrate with local learning design and institutional decision-making. Through structured interviews with administrators and focus groups with students and teachers, the Hilliger et al. team (Internet and Higher Education, 2020) surfaced what stakeholders actually needed for adoption to be locally legitimate, and built the LALA CANVAS — a participatory adoption framework that puts ethical responsibility on the front-end of the adoption decision, not as a compliance afterthought. The honest limit preserved verbatim: this is adoption-readiness and capacity-building evidence, not yet long-run outcome evidence that the deployed systems improved student retention or learning. The contribution is the participatory governance method that made adoption locally legitimate — and the case is the non-US companion to OU Analyse (Case 115), where governance-by-design unblocked deployment in a different cross-cultural context.

BACKGROUND

Learning analytics as a field is heavily Global-North-shaped. The reference tools (predictive at-risk classifiers, dashboard analytics for instructors, early-warning systems) were developed at US and European universities with US and European governance assumptions about data, consent, and institutional decision-making. The LALA project began from the documented observation that lifting those tools wholesale into Latin American institutional contexts had failed before — adoption stalled, instructors did not use the dashboards, students did not trust the predictions — and that the failure was structural, not technical:

the tools had not been integrated with the local learning design or the institutional decision-making process they were meant to inform.[0]

THE INTERVENTION

The project ran across Chilean, Ecuadorian, and Mexican universities under an EU Erasmus+ grant from 2017 to 2020. The method's first commitment was participatory: structured interviews with administrators surfacing what their decisions actually needed evidence about; focus groups with teachers and students surfacing what they would accept, what they would resist, and what they wanted the analytics to do. Hilliger et al. (Internet and Higher Education, 2020) is the peer-reviewed mixed-methods report that documents the method and the findings. The headline result is that stakeholders consistently named ethical responsibility as a precondition for data-driven feedback rather than as a compliance burden — they wanted the analytics, conditional on the governance being right first.[1]

HOW IT WORKED

The deliverable is the LALA CANVAS — a participatory adoption framework that walks an institution through the decisions that have to be made before a learning-analytics system is deployed: which questions the system is for, which stakeholders' consent is required, what disclosure is owed, what the operating governance will look like once the system runs. The framework's contribution is procedural: it converts the governance question from a yes/no gate at deployment time into a structured set of decisions taken openly during the adoption process. The participatory method made the framework locally legitimate across three regimes whose own data governance is differently mature.[2]

THE EVIDENCE

The honest limit survives into the case. The published evidence is adoption-readiness and capacity-building evidence: the CANVAS was developed and validated through the participatory process, and the project produced trained local teams with the capacity to lead adoption in their own institutions. What the evidence does **not** yet establish is that the deployed systems improved long-run student retention or learning outcomes. The case is teachable on the governance method — the participatory route to local legitimacy — and the outcome evidence is the next study, not this one. Drafting that softens this hedge overstates the claim.[3]

WHAT TRANSFERRED

In pair with the Open University case (Case 115, the UK consent-by-design intervention), LALA shows that the governance-by-design pattern is transferable across regimes: OU built consent for a single-institution intervention under pre-GDPR UK scrutiny; LALA built participatory adoption for multi-country capacity-building under three different Latin American regulatory regimes. The pair plus SyRI (Case 155) teaches the non-US LA governance triple — design that unblocked deployment (OU), participatory governance that built adoption capacity (LALA), and rights-grounded halt (SyRI) — three honest results for the same structural question of when delegation to analytic infrastructure is legitimate.

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THE LEARNING ENGINEERING LENS

> *Stakeholders did not want the analytics if the governance was wrong. They wanted both, in that order, with ethical responsibility as the precondition rather than the afterthought.*
>
> EDITORS' SYNTHESIS OF HILLIGER ET AL. (2020).

LE INSIGHT

LALA converted learning-analytics adoption from a Global-North template-lift into a participatory process that built local legitimacy across three Latin American regimes. The CANVAS is the artifact; the participatory method is the deliverable. The honest limit — adoption-readiness evidence, not yet long-run outcome evidence — is what the case carries, and the outcome study is the next one.

LENS APPROACH

LALA is the non-US participatory-governance case (induced 5.1; LENS D5/PT4) — Domain 5 for the cross-regime participatory method; Domain 3 on **Judgment under inadequate evidence** (adoption-readiness, not closed outcome proof). Pair with Case 115 (OU) and Case 155 (SyRI).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Begin from the documented failure mode — Global-North tools that did not transfer — so the adoption project is designed against the actual local barrier, not against an imagined universal one.
- Run the participatory method with administrators, teachers, and students separately; the decisions, the daily use, and the consent each sit with a different stakeholder.
- Convert the governance question from a yes/no deployment gate into a structured set of decisions surfaced during adoption; the CANVAS is the worked artifact of that conversion.

IN OPERATION / AFTER

- Report the evidence at its tier: adoption-readiness and capacity-building is what the published study establishes; long-run student-outcome evidence is the next study, and the case names that gap rather than glossing it.
- Commission longitudinal follow-up at the institutions that adopted via the CANVAS to convert capacity-building evidence into deployed-outcome evidence over time.
- Carry the cross-regime lesson into the broader v2 governance set: governance-by-design is transferable, but the local participatory process is the half that does not transfer; each regime has to do it for itself.

REFLECTION QUESTIONS

1. Identify a tool or framework in your domain that has been lifted from one regime to another without local adaptation. What participatory method would surface what the destination stakeholders actually need, and what would convert the governance question into a structured set of decisions during adoption rather than a gate at deployment?

2. Specify the evidence tier honestly for an adoption-readiness case in your context: capacity-building and process evidence is what you can publish now; long-run outcome evidence is the next study. Where in your communication is the tier most at risk of being smoothed away?

3. The non-US LA governance triple — OU, LALA, SyRI — teaches three honest results for the same structural question. What is the analog triple in your domain that would teach when delegation is legitimate (design unblocks), when it builds capacity (participatory adoption), and when it should be halted (rights-grounded)?

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CASE 182 EDUCATION (NORWAY) NATIONAL POLICY LEARNING ANALYTICS 2022 – 2023

Norway's National Expert Commission on Learning Analytics

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Rather than let learning analytics diffuse unregulated or block it, Norway's Ministry of Education convened a national Expert Commission to investigate the pedagogical, ethical, legal, and privacy issues and establish a regulatory foundation before sector-wide deployment; interim report June 2022, final report 2023 (NOU), with central dilemmas explicitly framed

IN BRIEF

Norway's Ministry of Education convened a national Expert Commission on Learning Analytics in 2022 — a national-government response to a capability deployment question at sector scale. Rather than let learning analytics diffuse unregulated across Norwegian education, or block it on precautionary grounds, the ministry chose to construct the governance architecture first. The commission delivered an interim report to the Minister in June 2022 and a final report in 2023 (the NOU, Norges offentlige utredninger, the canonical form of Norwegian public commission reports) identifying central dilemmas across the pedagogical, ethical, legal, and privacy dimensions. The case is governance-as-deliberate-artifact at national scale — a country treating change-control and disclosure as the precondition for adoption rather than the consequence of it. The honest limit preserved verbatim: the commission's recommendations were guidance to a ministry, and downstream sector outcomes — whether deployed Norwegian LA systems actually improved learning outcomes or preserved trust under operation — are not yet documented. This is process-level evidence (a national governance artifact exists, the dilemmas are named), not yet deployment-outcome evidence. The case pairs with the OU (Case 115) and SyRI (Case 155) cases as the national Nordic complement to the institutional-UK and judicial-Dutch governance modes.

BACKGROUND

Learning analytics at sector scale presents a national government with a structural choice. Let the technology diffuse across schools and universities under existing patchwork governance (Sweden's path through the 2010s), or block it on precautionary grounds until questions are settled (a path some jurisdictions have pursued for specific applications), or construct the governance architecture first and let deployment follow under it. Norway in 2022 chose the third path. The Ministry of Education convened a national Expert Commission on Learning Analytics to investigate the pedagogical, ethical, legal, and privacy issues across the whole education sector and to establish a regulatory foundation for what sector-scale deployment should look like.[o]

THE INTERVENTION

The commission's mandate covered the full chain. Pedagogically, what kinds of learning-analytics-driven interventions are defensible at primary, secondary, and tertiary levels; what claims about outcome the evidence supports; what teacher-student relationship the analytics should and should not be allowed to alter. Ethically, what is owed to students whose data drives the analytics, what consent architecture is defensible across age groups, and what disclosure structure the analytics-driven decisions should carry. Legally, how the Norwegian data-protection regime under GDPR interacts with the educational context, and what additional sectoral instruments are needed. Privacy, where the line is between pedagogically useful data and surveillance overreach.[1]

HOW IT WORKED

The commission delivered an interim report to the Minister of Education and Research in June 2022 and a final report in 2023 in the canonical NOU form. The reports name central dilemmas the field has to live with rather than resolve once: the tension between predictive support and predictive gatekeeping; the tension between transparency to the student and the technical complexity of the models; the tension between cross-institutional benchmarking and student-data protection; the tension between national pedagogical consistency and institutional autonomy in how analytics are used. The reports' framing is that governance for learning analytics is the kind of artifact that has to be revisited as the technology and the evidence base change, not a one-time document.[2]

THE EVIDENCE

The honest evidentiary state is process-level. The case establishes that a national government can produce a structured governance artifact, identify the dilemmas sector-scale deployment will face, and deliver the architecture as guidance to a ministry. The case does **not** yet establish that downstream Norwegian sector outcomes — improved learning, preserved trust, defensible interventions — have been delivered, because deployment under the new architecture is too recent for outcome evidence. This is a governance-process success, not yet a measured deployment-outcome success, and the case carries the qualification rather than collapsing it.[3]

WHAT TRANSFERRED

In pair with the Open University (Case 115, institutional governance-by-design, UK) and SyRI (Case 155, judicial rights-grounded halt, Netherlands), Norway's commission is the national-scale governance-architecture mode. The three cases together teach that learning-analytics governance can be produced at the institutional level (OU), constrained by the courts after the fact (SyRI), or constructed by national deliberation before sector deployment (Norway). The structural lesson is that governance is producible as a deliverable at whichever level matches the deployment scope, and the choice of level is itself a governance decision the case literature names.

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THE LEARNING ENGINEERING LENS

> *The commission did not resolve learning-analytics governance for Norway. It named the dilemmas the sector will live with and produced the artifact deployment can be governed under.*
>
> EDITORS' SYNTHESIS OF THE NORWEGIAN EXPERT COMMISSION INTERIM AND FINAL REPORTS.

LE INSIGHT

Norway's national Expert Commission is the governance-architecture-at-national-scale instance: a country constructing the regulatory and pedagogical foundation before sector-scale learning-analytics deployment, rather than after diffusion or via judicial halt. The artifact exists and the dilemmas are named; downstream sector outcomes are not yet documented. Process-level success; deployment-outcome evidence is the next decade's work.

LENS APPROACH

Norway is the national-scale governance-architecture case (induced 5.4; LENS D5/PT4). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the change- control-and-disclosure-as-artifact discipline and in Domain 4 (Test and Evaluation) for the CLO **Judgment under inadequate evidence** on the process-vs-outcome tier. Pair with Case 115 (OU, institutional), Case 155 (SyRI, judicial), Case 181 (LALA, multi-country participatory), and Case 169 (African data privacy, frontier) — the non-US LA governance pentad teaching the level-of-governance decision.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Match the governance artifact to the deployment scope: institutional governance for a single university, national commission for sector-scale deployment, judicial review when neither operates. The choice of level is itself a governance decision.
- Convene the commission with the dimensional breadth the deployment actually has: pedagogical, ethical, legal, privacy — not only the dimension the most-visible objection sits on.
- Frame the artifact as living: governance for a moving technology has to be revisited as the technology and the evidence base change, and the artifact should say so.

IN OPERATION / AFTER

- Report the evidence at its tier: governance-process success is what the commission and NOU establish; deployment-outcome success is the next decade's work, and the case is honest about the gap.
- Carry the central dilemmas into the implementation conversation rather than treating them as resolved by publication of the NOU; the dilemmas are what implementation will live with.
- Use the case as the national-scale instance in the OU / LALA / SyRI / Norway non-US LA governance set; the joint teaching point is that governance is producible at whichever level matches the deployment scope.

REFLECTION QUESTIONS

1. Identify a sector-scale capability deployment question in your domain. Is the right governance level institutional, sectoral, national, or judicial — and what determines the match between deployment scope and governance level?

2. Specify the dimensional breadth your commission or governance artifact would have to cover. Norway's mandate was pedagogical, ethical, legal, and privacy. What is the analog in your context, and which dimension is most at risk of being narrowed away under stakeholder pressure?

3. The Norwegian artifact is process-level evidence, not yet deployment-outcome evidence. What is the cadence at which the artifact should be revisited as the technology and the evidence base change, and who has the standing to convene that revisit?

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CASE 183

AUTONOMOUS VEHICLES TRANSPORTATION SAFETY PUBLIC-SECTOR GOVERNANCE

2023 / 2025

Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT After a California court let Waymo withhold trade-secret-laden safety data from a DMV public-disclosure request, the company answered the governance objection with a published, structured safety case framework — and in November 2025 commissioned the first independent third-party audits of both the safety case and the remote-assistance program

IN BRIEF

In 2022 a California court permitted Waymo to withhold trade-secret-laden safety details from a public DMV disclosure, leaving regulators and the riding public with a credibility gap Waymo could not close by sharing the contested data. The company's response was to publish, in 2023, a structured **safety case framework**: a top-down argument with explicit claims, sub-claims, and the evidence types each rests on, accompanied by published operating-domain performance figures. In November 2025 Waymo released the first independent third-party audits of both the safety case and the remote-assistance program — the audits themselves disclosed, rather than the underlying trade-secret data the original objection targeted. The pattern is the OU-Analyse / inBloom move (governance objection dissolved by better design) transposed from education into physical-safety AV: the response to an opacity objection was a falsifiable argument structure auditable by third parties, not a defense of opacity. The evidence-tier flag rendered under the title is load-bearing — the analysis rests on the practitioner whitepaper plus the 2025 audit summaries, not on a peer-reviewed safety-engineering evaluation. Future validation will continue as the audit cadence and post-deployment failure record accumulate.

BACKGROUND

The precipitating event was not a crash. In 2022 a California court ruled that Waymo could withhold trade-secret-laden safety details from a DMV public-records process. The company had a legal answer to the disclosure request and no legitimacy answer to the public-trust gap that ruling created. The governance objection — “you are asking us to trust an opaque system whose failure modes we cannot inspect" — could not be answered by disclosing the contested data without giving up the trade-secret position the court had just protected.[0]

THE INTERVENTION

Waymo's 2023 response was to publish a structured **safety case framework**: a top-down argument with claims and sub-claims, the evidence categories each rests on (operational performance data, simulation and testing, hazard analysis, third-party assessment), and the operational-domain figures available at the time of publication. The artifact's design move is that the **structure** of the safety argument is public even where individual evidence items remain proprietary — outside auditors can interrogate the chain of reasoning without seeing the trade secrets.[1]

HOW IT WORKED

In November 2025 Waymo commissioned and released the results of independent third-party audits of the safety case and of the remote-assistance program. The audits themselves — not the underlying data — were the disclosure artifact. The pattern is OU-Analyse / inBloom in the AV domain: a governance objection answered by a designed legitimacy artifact rather than by disclosure of the contested data. Where opacity could not be defended, a structured falsifiable argument plus audited assurance took its place.[2]

THE EVIDENCE

The evidence-tier flag rendered under the case title is load-bearing. The framework and audit summaries are practitioner-authored or auditor-authored — not peer-reviewed safety-engineering analyses. Some audit-tier elements push toward investigation-grade, but the synthesis as a whole rests on Waymo and Montreal AI Ethics Institute documents. The honest framing in print is that the source confidence is flagged and that future validation — particularly post-deployment failure-record analysis and continued auditor independence — is ongoing.[3]

WHAT TRANSFERRED

The teaching point for LENS is that delegation of consequential decisions to an automated system creates a governance debt that the deploying organization owes the public. The CLO **Delegation with revocation** is the capability the case exercises: the safety case framework is the artifact that makes revocation possible — regulators or auditors can identify which sub-claim has failed, on what evidence, and require the deploying organization to act on the gap. Pair with Case 156 (Cruise) as the foil: the same regulatory regime, the opposite governance choice, the opposite outcome. Pair with Case 184 (CPUC permit framework) as the regulator-side counterpart of the deployer-side artifact.[4]

REFERENCES

1. Waymo (2023), "A Blueprint for AV Safety: Waymo's Toolkit For Building a Credible Safety Case," whitepaper.
2. Waymo (November 2025), "Independent Audits of Waymo's Safety Case and Remote Assistance Programs," summary release.
3. Montreal AI Ethics Institute (2023), summary and analysis of the Waymo safety case framework.
4. California Public Utilities Commission, AV passenger-service permit framework documents — paired Case 184 for the regulator-side artifact.
5. Cruise / California DMV Order of Suspension (October 2023) — paired Case 156 as the foil.

THE LEARNING ENGINEERING LENS

> *Where opacity could not be defended, a structured falsifiable argument plus audited assurance took its place.*
>
> EDITORS' SYNTHESIS OF THE WAYMO SAFETY CASE FRAMEWORK AND THE NOVEMBER 2025 THIRD-PARTY AUDITS.

LE INSIGHT

Waymo's safety case framework is the AV-domain instance of the OU-Analyse / inBloom move: a governance objection dissolved by a designed artifact, not by disclosure of the contested data. Evidence-tier flag is practice-synthesis; the artifact pattern is teachable and the third-party audit posture pushes some elements toward investigation-grade, but the synthesis as a whole is practitioner-authored and future validation is ongoing.

LENS APPROACH

Waymo is the AV-safety governance case (induced 5.1; LENS D5/PT6). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the CLO **Delegation with revocation** — the safety case is what makes revocation possible — and in Domain 3 (Emerging Systems and Human-System Collaboration) for the deployer-side artifact that permits oversight of a system whose internals are trade secret. Pair with Case 156 (Cruise) as the foil and Case 184 (CPUC) as the regulator-side complement.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the disclosure objection as a design problem: what falsifiable artifact can you publish that addresses the legitimacy gap without requiring you to surrender trade-secret evidence?
- Structure the safety case as a top-down argument with explicit claims, sub-claims, and evidence types so an outside auditor can interrogate the **chain of reasoning** rather than only the contested data points.
- Commission and publish third-party audits of the argument structure and of the operational programs (e.g. remote assistance) that the safety case rests on — the audits are the disclosure artifact when the data cannot be.

IN OPERATION / AFTER

- Treat the safety case framework as a living document — update the claims and evidence as post-deployment failure data accumulates, and publish the updates so the legitimacy artifact does not calcify.
- Use the CLO **Delegation with revocation**: design the framework so a regulator or auditor can identify which sub-claim has failed and trigger a revocation pathway, not only a "trust us, we will fix it" assurance.
- Carry the practice-synthesis evidence-tier flag honestly in any program documentation citing the framework — the artifact pattern is teachable, but the magnitude of its public-trust effect is still being measured.

REFLECTION QUESTIONS

1. Identify an automated system in your context that faces a public-trust objection it cannot answer by full disclosure. What falsifiable argument structure could you publish that would make the system's reasoning auditable without requiring disclosure of the contested data?

2. Specify how a regulator or independent auditor would **revoke** the delegation in your system if a sub-claim of the safety case failed. The CLO **Delegation with revocation** requires this pathway to exist before deployment, not only after a public-facing failure.

3. The case is practice-synthesis tier. What additional independent evidence — failure-record analysis, multi-auditor replication, peer-reviewed evaluation — would you require before treating the safety-case-plus-audit posture as a substitute for the disclosure that was originally demanded?

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CASE 184 AUTONOMOUS VEHICLES PUBLIC-UTILITY GOVERNANCE ACCESSIBILITY 2020 – 2024

CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver

G Governance & Trust K Knowledge & Institutional Memory D Designed Out

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT California's Public Utilities Commission built an AV passenger-service permit framework whose conditions — time-of-day limits, weather restrictions, fleet caps, geographic carve-outs, mandatory Passenger Safety Plan for riders with disabilities — are explicitly designed to operationalize the governance objections that would otherwise block deployment outright

IN BRIEF

The California Public Utilities Commission established an AV passenger-service permit framework whose conditions are explicitly designed to address common governance objections — safety, equity, fleet scale — by writing them into the permit rather than treating them as binary deploy / don't-deploy questions. The framework includes time-of-day limits, weather restrictions, fleet caps, geographic carve-outs, and a required Passenger Safety Plan documenting how the driverless service handles riders with limited mobility, vision impairments, and other disabilities. A 2024 update strengthened the DMV's authority to impose targeted operational restrictions for safety, and the case shows the regime in operation: the Cruise suspension (Case 156) shows the regime can revoke; the Waymo continuation (Case 183) shows it can permit conditionally. The structural complement to the Waymo deployer-side safety case framework is exactly this regulator-side artifact — the permit conditions make the safety case operative as a regulatory instrument. The evidence-tier flag rendered under the title is practice-synthesis: the program is documented in CPUC decisions and program guidance, but no peer-reviewed evaluation of the program's equity-of-service outcomes for disabled and disadvantaged-community riders yet exists. Future validation will continue as ridership and incident data accumulate.

BACKGROUND

The governance question California faced from 2018 onward was how to regulate AV passenger service in a regime where the California DMV regulates the vehicle and the CPUC regulates the passenger service the vehicle provides. Treating each AV deployment as a binary deploy / don't-deploy decision risked either blocking deployment outright on objections the operator could not satisfy, or permitting deployment without a governance handle on the objections the public and the disability-rights community brought to the proceedings.[0]

THE INTERVENTION

The CPUC's structural answer was to build the objections into the permit conditions. The AV passenger-service permits specify time-of-day limits, weather restrictions (fog, heavy rain), fleet caps, geographic carve-outs (initially excluding certain neighborhoods), and a mandatory Passenger Safety Plan documenting how the driverless service will accommodate riders with limited mobility, vision impairments, hearing impairments, and other disabilities. The conditions are not policy aspirations — they are permit terms whose violation triggers enforcement.[1]

HOW IT WORKED

In August 2024 the framework was updated to strengthen the DMV's authority to impose targeted operational restrictions for safety reasons during the deployment lifecycle, not only at initial permit issuance. The regulatory regime can therefore tighten the conditions in response to operational evidence. The pattern: rather than treat the governance objection as binary, the CPUC made the objection itself a design parameter the deployer must satisfy and that the regulator can re-tune.[2]

THE EVIDENCE

The case is the structural complement to Case 183 (Waymo) and the inverse-outcome companion of Case 156 (Cruise). The Waymo safety case framework is the deployer-side artifact; the CPUC permit conditions are the regulator-side artifact that makes the safety case operative as a regulatory instrument. The Cruise suspension shows the regime can revoke when the disclosure architecture fails; the Waymo conditional continuation shows it can permit when the conditions are satisfied. Together the trio teaches the structural form of regulator-deployer interaction at the AV physical-safety C5 layer.[3]

WHAT TRANSFERRED

The evidence-tier flag rendered under the title is practice-synthesis. The CPUC decisions, the program guidance, and the permit conditions themselves are public; no peer-reviewed evaluation of the equity-of-service goals (disabled-rider access, disadvantaged-community access) has yet measured whether the Passenger Safety Plan conditions translate into measured ridership outcomes. The condition-as-objection-dissolver pattern is teachable and the regulatory architecture is explicit, but the outcome evidence is not yet decision-grade. Future validation will continue as the program ages and as the equity outcomes are independently measured.

REFERENCES

1. California Public Utilities Commission, "Autonomous Vehicle Passenger Service Programs," CPUC program page and August 2024 application guidance.
2. CPUC permit decisions for Cruise and Waymo, 2020–2024.
3. California Department of Motor Vehicles, AV regulatory program — strengthened safety-restriction authority, 2024.
4. Paired Cases 183 (Waymo deployer-side artifact) and 158 (Cruise revocation under regime).

> *The governance objection was not refused. It was made a permit condition.*
>
> EDITORS' SYNTHESIS OF THE CPUC AV PASSENGER-SERVICE PERMIT FRAMEWORK.

LE INSIGHT

The CPUC permit framework is the regulator-side counterpart to the Waymo safety case (Case 183): conditions operationalize objections rather than blocking deployment. Evidence-tier flag is practice-synthesis; the regulatory architecture is documented in CPUC decisions, but no peer-reviewed evaluation of the equity-of-service goals yet exists, and future validation is ongoing.

LENS APPROACH

CPUC is the AV regulator-side governance case (induced 5.1; LENS D5/PT6). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for cross-regime governance seams (DMV regulates the vehicle; CPUC regulates the service; both can act) and for the condition-as-design-parameter pattern that makes the deployer-side safety case (Case 183) operative as a regulatory instrument. Pairs with Case 156 as the revocation event under the regime.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat governance objections as design parameters rather than as binary blockers — write them into the permit conditions where compliance is verifiable and violation triggers enforcement.
- Require a documented Passenger Safety Plan (or equivalent equity-of-service artifact) at initial permit issuance so the equity objection has an operational answer the regulator can audit.
- Build the regulatory authority to re-tune the conditions during the deployment lifecycle into the regime itself; the Cruise suspension (Case 156) is what re-tuning under a failure event looks like.

IN OPERATION / AFTER

- Pair the case with Case 183 (Waymo) as the regulator-deployer artifact dyad — the safety case framework and the permit conditions are co-constitutive, not independent moves.
- Use the case in LENS Domain 4 modules on cross-regime governance seams (CPUC and DMV) — the regime structure is itself part of the capability question.
- Carry the practice-synthesis flag honestly: the condition-design pattern is teachable, but the equity-of-service outcome data are not yet peer-reviewed; future validation requires independent evaluation of the Passenger Safety Plan deliverables.

REFLECTION QUESTIONS

1. Identify a deployment regime in your context where governance objections are currently treated as binary deploy / don't-deploy questions. How could the objections be rewritten as permit-style conditions whose compliance is verifiable and whose violation triggers enforcement?

2. Specify the equity-of-service artifact your regime would require at permit issuance — the analogue of the Passenger Safety Plan — and the auditable evidence the deployer must provide that the artifact is operational.

3. The case is practice-synthesis tier. What independent outcome evidence — disabled-rider ridership figures, disadvantaged-community access measurements, comparative incident rates by permit condition — would you require before treating the condition-as-objection-dissolver pattern as a validated regulatory architecture?

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CASE 185 WORKFORCE DEV NATIONAL L&D POLICY ASIA-PACIFIC 2015 – PRESENT

Singapore SkillsFuture — National Workforce Capability at Scale

G Governance & Trust K Knowledge & Institutional Memory D Designed Out

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT Singapore's SkillsFuture pairs individual training credits with employer subsidies, a cross-sector skills framework, and a two-wave outcome survey (TRAQOM, at end-of-course and at six months) — a 2018 MTI study found a 5.8% real wage premium for WSQ-trained workers, with 87% of Work-Study Programme graduates employed full-time within six months

IN BRIEF

Singapore’s SkillsFuture Movement, launched in 2015, pairs individual training credits with employer subsidies, a cross-sector skills framework, and one of the most ambitious national L&D measurement instruments deployed at scale: the two-wave TRAQOM survey, administered at end-of-course and at six months post-training, paired with labor-market data. Self-reported figures from the 2024 Year-in-Review are strong: 98% of trainees report being able to perform better at work; 93% report the course played a pivotal role; 87% of Work-Study Programme graduates are employed full-time within six months; a 2018 MTI study found a 5.8% real wage premium for WSQ-trained workers. The honest reading the case carries into print: self-report dominates the headline numbers, and the program has not been subjected to a rigorous quasi-experimental external evaluation that would isolate the program’s causal effect from underlying labor-market trends. SkillsFuture is the non-US national-scale L&D case the corpus needs for both the corporate / workforce L&D gap and the non-US/UK/EU gap. The evidence-tier flag is practice-synthesis: the program design and the TRAQOM instrument are documented in SSG annual reports and in ILO and Springer analyses, the headline outcomes are self-report, and future validation — particularly quasi-experimental causal evaluation — is ongoing.

BACKGROUND

SkillsFuture was launched in 2015 as a Singapore-wide workforce-capability program at the seam between individual upskilling, employer demand, and state coordination. The program design pairs individual training credits (SkillsFuture Credit), employer subsidies for workforce training, a cross-sector skills framework that defines competencies and progression paths across industries, and a Work-Study Programme that embeds learners in employer contexts during training.[o]

THE INTERVENTION

The measurement instrument is unusually ambitious for a national L&D program. The Training Quality and Outcomes Measurement framework (TRAQOM) is a two-wave outcome survey administered at end-of-course and at six months post-training. It is paired with labor-market data so that self-reported outcomes can be cross-checked against employment and wage outcomes at population scale. The design crosses the Kirkpatrick Level-2 / Level-3 seam (Case 108) at policy level, not only program level.[1]

HOW IT WORKED

The 2024 Year-in-Review reports headline figures: 98% of trainees report being able to perform better at work; 93% report the course played a pivotal role; 87% of Work-Study Programme graduates are employed full-time within six months. A 2018 study by the Ministry of Trade and Industry found a 5.8% real wage premium for workers with a Workforce Skills Qualifications (WSQ) certification. The labor-market figures are the strongest available external corroboration of the self-report data.[2]

THE EVIDENCE

The honest reading is the load-bearing teaching point. Self-report dominates the headline outcomes. The program has not been subjected to a rigorous quasi-experimental external evaluation that would isolate the program's causal effect from underlying labor-market trends — and Singapore's labor market has been strong across the program's deployment period. The TRAQOM design is one of the strongest national L&D instruments deployed, and what it cannot yet do is what no national L&D instrument yet does well: produce decision-grade causal evidence at population scale. Future validation is ongoing.[3]

WHAT TRANSFERRED

The LENS teaching point is that the program is a non-US national-scale case for the corporate / workforce L&D cluster (Cases 108, 35, 124, 46) and a non-US/UK/EU case for the geographic-coverage gap. The amended CLO on collaboration measurement is directly exercised: TRAQOM measures across employer-employee-state, not only across the training organization. Pair with Case 47 (PEPFAR) for the global-health workforce-capability counterpart, and with Case 46 (HILS) for the design-side practitioner pattern that the SSG program operationalizes at policy scale. Evidence-tier flag is practice-synthesis; the design is documented, the causal magnitudes are open.[4]

REFERENCES

1. SkillsFuture Singapore (SSG), Year-in-Review 2024 — program metrics and outcome reporting.
2. Ministry of Education (MOE), Singapore, parliamentary replies on TRAQOM, 2020.
3. International Labour Organization (ILO), “Investigating an Upskilling Programme in Singapore” — international comparative analysis.
4. “Future-Skilling the Workforce: SkillsFuture Movement in Singapore,” Springer, 2024 — peer-reviewed program analysis.
5. Ministry of Trade and Industry (MTI), Singapore, 2018 — WSQ wage-premium study.

> *The instrument crosses the Level-2 / Level-3 seam at policy level. What it cannot yet do is what no national L&D instrument yet does well.*
>
> EDITORS' SYNTHESIS OF THE SKILLSFUTURE MOVEMENT AND THE TRAQOM MEASUREMENT FRAMEWORK.

LE INSIGHT

SkillsFuture is the non-US national-scale L&D case the corpus needs. The TRAQOM instrument is among the most ambitious national L&D measurement architectures deployed; the headline outcomes are self-report dominant; no rigorous quasi-experimental external evaluation yet exists. Evidence-tier flag is practice-synthesis; future validation is ongoing.

LENS APPROACH

SkillsFuture is the national workforce-capability case (induced 2.4; LENS D5/PT4). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the amended CLO on collaboration measurement — TRAQOM measures across employer-employee-state — and in Domain 2 as the policy-scale operationalization of the HILS-style environment-and-event integration (Case 46). Pairs with Case 47 (PEPFAR) for the global-health workforce-capability counterpart.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the measurement instrument across the training-and-employment seam, not within the training organization alone — TRAQOM's two-wave + labor-market cross-check is the architecture the LENS Domain 4 module should teach.
- Make the cross-sector skills framework a first-class artifact — without it the credits, the subsidies, and the Work-Study Programme do not cohere as a single workforce-capability deliverable.
- Treat the self-report dominance honestly: name what TRAQOM can and cannot establish at the design stage, so the program documentation does not have to retrofit the hedge.

IN OPERATION / AFTER

- Pair with Case 47 (PEPFAR) for the global-health workforce-capability counterpart at multi-country scale; together they teach what national- and program-scale L&D measurement at evidence-flagged tier looks like.
- Use the amended CLO on collaboration measurement: TRAQOM is a worked example of measurement across employer-employee-state, and the program documentation can teach the architecture in LENS Domain 5 (Sociotechnical Constraints).
- Carry the practice-synthesis flag honestly: the program design and the TRAQOM instrument are documented, the headline magnitudes are self-report, and future validation requires independent quasi-experimental causal evaluation.

REFLECTION QUESTIONS

1. Identify a workforce-capability program in your context that currently measures at the training-organization boundary. What would the analogue of TRAQOM — a two-wave outcome survey paired with employment-and-wage data at population scale — require of your measurement infrastructure?

2. Specify the cross-sector skills framework that would coordinate individual credits, employer subsidies, and a Work-Study-style placement program in your context. Without the framework, do the components cohere as a single capability deliverable?

3. SkillsFuture's headline magnitudes are self-report dominant. What independent quasi-experimental evidence — comparison-cohort design, regression discontinuity, instrumented variation — would you require before treating any specific outcome magnitude as decision-grade for a program-scale investment in your context?

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CASE 186 HOSPITAL PHARMACY HEALTHCARE WORKFORCE ROLE REDESIGN 2016

Australian Hospital-Pharmacy Technician Role Redesign

D Designed Out N Normalization of Deviance H Human-System Interface

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT Expanded pharmacy-technician scope (final accuracy checking, drugs-of-addiction management, clinical support) reportedly cut average prescription turnaround from 18.5 to 12.3 minutes, increased throughput from 220 to 295 prescriptions per shift, and decreased dispensing errors from 2.1% to 1.2% — throughput and the safety metric moving in the same direction

IN BRIEF

Australian hospital pharmacies faced the structural problem most healthcare workforces meet at some point: the pharmacist's capacity was being absorbed by dispensing-accuracy checking, which crowded out the clinical work that requires pharmacist judgment. The 2016 Pharmacy Technician and Assistant Role Redesign project expanded pharmacy-technician scope to include final accuracy checking, drugs-of-addiction management, and clinical-support duties, with the design intent of freeing pharmacist capacity without degrading dispensing safety. The reported operational outcomes moved throughput and the safety metric in the same direction: average prescription turnaround fell from 18.5 to 12.3 minutes, prescriptions per shift rose from 220 to 295, and dispensing errors declined from 2.1% to 1.2%. A 2021 Journal of Pharmacy Practice and Research cross-sectional survey extended the evidence base into workforce-acceptance attitudes. The evidence-tier flag renders under the title: the operational figures come from a program-report rather than an independent multi-site audit. Future validation will continue on long-term safety durability and on whether the role-redesign pattern generalizes to other healthcare-workforce roles.

BACKGROUND

The structural problem the case sits inside is the C3 thread where monitoring or checking duties are loaded onto the role that needs to be doing the clinical work — and the checking work absorbs the capacity. Australian hospital pharmacies in the early 2010s were in this pattern: the pharmacist's day was consumed by dispensing-accuracy checking, and the clinical-support work that requires pharmacist judgment was crowded out. The induced 3.2 framing is the precise diagnosis: the unsupportable role is the pharmacist as primary accuracy-checker.[0]

THE INTERVENTION

The redesign expanded pharmacy-technician scope to include final accuracy checking, drugs-of-addiction management, and clinical-support duties. The design move is not the addition of a checker but the redistribution of authority: the technician is given the role that the pharmacist was doing, with the training, certification, and supervisory structure to back it. The pharmacist is freed to do the clinical-judgment work that the system needed pharmacists for. The redesign is the C3 role-design intervention the induced framework calls for at the small tier.[1]

HOW IT WORKED

The reported operational outcomes moved throughput and the safety metric in the same direction — an unusual combination in healthcare workforce changes. Average prescription turnaround fell from 18.5 to 12.3 minutes; prescriptions per shift rose from 220 to 295; dispensing errors fell from 2.1% to 1.2%. The redesign did not simply trade safety for throughput. The 2021 Journal of Pharmacy Practice and Research cross-sectional survey extended the evidence into the workforce-attitudes layer — what hospital pharmacists and technicians thought about the expanded scope and where acceptance was strongest.[2]

THE EVIDENCE

The evidence-tier flag is load-bearing. The headline operational figures come from a program-report and rest on the project's internal measurement; the redesign was not subjected to an independent multi-site audit, and the magnitudes have not been replicated by a peer-reviewed controlled evaluation. The cross-sectional survey is peer-reviewed but measures attitudes rather than outcomes. The case is included because the pattern is teachable and the practitioner literature is consistent; the magnitudes should travel with the tier flag intact. Future validation will continue.[3]

WHAT TRANSFERRED

What the case teaches at the LENS layer is the role-design move as a sociotechnical-constraints intervention. The pharmacist-as-checker pattern was not a property of any individual; it was a property of the role architecture, and the redesign treated the architecture as the design variable. The Domain-4 frame applies: the work was to redraw the authority gradient and team-coordination boundary so that the system's capability — fast, safe dispensing plus pharmacist clinical judgment — emerged from the team rather than depending on the pharmacist's individual heroism. The case is also a Gap-5 echo: a non-US small-tier role-redesign success with documented operational outcomes.

REFERENCES

1. SHPA / Australian hospital-pharmacy network (2016), "Pharmacy Technician and Assistant Role Redesign within Australian Hospitals Project," outcomes report.
2. Anderson et al. (2021), "Perceptions of hospital pharmacists and pharmacy technicians towards expanding roles for hospital pharmacy technicians: a cross-sectional survey," Journal of Pharmacy Practice and Research, doi:10.1002/jppr.1697.
3. Boughen, Sutton, Fenn, & Wright (2017), "Defining the Role of the Pharmacy Technician and Identifying Their Future Role in Medicines Optimisation," Pharmacy 5(3):40 — UK companion analysis.
4. Royal Pharmaceutical Society and SHPA practice statements on technician scope expansion — practitioner-tier framing the project sits inside.

THE LEARNING ENGINEERING LENS

> *The pharmacist-as-checker pattern was a property of the role architecture, not the pharmacist. The redesign treated the architecture as the design variable.*
>
> EDITORS' SYNTHESIS OF THE 2016 PROJECT REPORT AND ANDERSON ET AL. (2021).

LE INSIGHT

The Australian pharmacy-technician redesign is a small-tier C3 role-redesign intervention with both throughput and safety moving in the same direction (turnaround, throughput, error rate). Evidence base is program-report plus a peer-reviewed attitudes survey; the operational magnitudes rest on the project's internal measurement and have not been independently audited at multi-site scale. Future validation ongoing.

LENS APPROACH

Australian pharmacy-technician redesign is the C3 small-tier role-redesign case (induced 3.2 and 4.3; LENS D5/PT3). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for CLO-5 — the work was to redraw the authority gradient — and in Domain 1 (Systems Analysis) for CLO-1, since the role-architecture diagnosis was the precondition for the redesign. Pair with Cases 125–126 for the small-tier C3 thread; Gap-5 echo as a non-US case.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Diagnose the role-architecture failure before redesigning: which capability does the system need from which role, and where is checking work absorbing capacity that the system needs elsewhere.
- Redistribute authority, not just tasks: the technician's expanded scope must come with training, certification, and supervisory structure, not just a new line in the procedure manual.
- Specify both throughput and safety metrics before the redesign ships, so the joint movement (or trade-off) of both is what the report has to defend.

IN OPERATION / AFTER

- Treat the program-report figures as practice-synthesis evidence; carry the tier flag into any generalization to other sites or other workforce roles.
- Track durability of the safety metric over years, not weeks; the dispensing-error reduction is the load-bearing result and must hold under personnel rotation and volume changes.
- Use the workforce-acceptance evidence (2021 survey) to identify where the redesign meets resistance, and stage the rollout against that resistance rather than ignoring it.

REFLECTION QUESTIONS

1. Identify a healthcare role in your setting where checking or monitoring work is absorbing the capacity the system needs for clinical judgment. What is the technician- or assistant-equivalent role redesign that would redistribute the authority, and what training and supervisory structure would it require?

2. Specify the throughput and safety metrics you would commit to before the redesign ships, so the joint movement of both is what the report has to defend. The Australian project teaches because both metrics moved in the same direction — what would the equivalent be in your context?

3. The operational figures are program-report tier. What independent or multi-site evidence would you require before treating the magnitudes as generalizable to other workforce roles (nursing, radiology, lab) or other countries?

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CASE 187 DIGITAL IDENTITY WELFARE BIOMETRIC AUTHENTICATION INDIA 2018 – 2025

Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India

G Governance & Trust N Normalization of Deviance H Human-System Interface

EVIDENCE TIER journalism-tier — source confidence flagged; future validation ongoing.

IMPACT The Indian Supreme Court's Pragya Prasun ruling (April 2025) declared a fundamental right to inclusive and meaningful digital access and held that exclusion from welfare based on biometric-authentication failure — through no fault of the individual — violates constitutional dignity; the system can continue but alternatives to biometric authentication must be provided

IN BRIEF

Aadhaar — India's biometric digital-identity system, with roughly one billion enrolled — was designed to streamline welfare delivery and reduce identity fraud. Across more than a decade of deployment the load-bearing failure mode became operational exclusion: when biometric authentication failed for an individual (worn fingerprints, missing iris match, connectivity loss at the point of service), the welfare to which the person was entitled became inaccessible. The 2018 Puttaswamy II Aadhaar judgment surfaced the structural risk in Justice Chandrachud's dissent; the April 2025 Pragya Prasun ruling made the remedial framing explicit by declaring a fundamental right to inclusive and meaningful digital access. The court held that exclusion from welfare based on biometric-authentication failure — through no fault of the individual — violates constitutional dignity. The system can continue, but alternatives to biometric authentication must be provided. The case is the non-US automated-welfare-delegation case the corpus needs alongside SyRI and the UK Post Office Horizon thread. The evidence tier is mixed: the court judgments are investigation-grade; the lived-exclusion sourcing rests on journalism and advocacy reporting. Future validation will continue on whether the alternatives-must-be-provided remedy is implemented in practice. Gap-5 non-US/UK/EU case (India).

BACKGROUND

Aadhaar was designed as a population-scale solution to two problems the Indian welfare state had identified by the late 2000s: fragmented identification across welfare schemes and identity fraud in benefits delivery. The design choice was biometric authentication at the point of service — fingerprints and iris — keyed to a unique twelve-digit identifier issued by the UIDAI. Enrollment reached approximately one billion. The capability the system promised was streamlined welfare access; the authority it required was delegation of identity verification to the biometric.[0]

THE INTERVENTION

The load-bearing failure mode emerged across deployment: operational exclusion. When biometric authentication failed at the ration-shop card-reader or the pension window — worn fingerprints from manual labor, missing iris matches, connectivity loss, device malfunction — the welfare to which the person was entitled became inaccessible. The failure was not the absence of identity; the person had the Aadhaar number. The failure was the brittleness of the authentication system at the operator interface. The people most dependent on the welfare were the people whose biometrics were most likely to fail.[1]

HOW IT WORKED

The 2018 Puttaswamy II judgment upheld Aadhaar in broad terms; Justice Chandrachud's dissent named the structural risk that the majority did not directly address. Across the following seven years lived-exclusion documentation accumulated in journalism, advocacy reporting, and academic analysis. In April 2025 the Supreme Court's Pragya Prasun ruling made the remedial framing explicit. The court declared a fundamental right to inclusive and meaningful digital access and held that exclusion from welfare based on biometric-authentication failure — through no fault of the individual — violates constitutional dignity. The system can continue. But alternatives must be provided.[2]

THE EVIDENCE

The evidence-tier flag is split and load-bearing. The court judgments themselves are investigation-grade sources — the published opinion is the record. The lived-exclusion sourcing that gives the case its empirical body rests on journalism (Access Now, IAPP, Indian press), advocacy reporting, and a peer-reviewed comparative analysis ("A Failure to Do No Harm," PMC5741784). The journalism-tier flag travels with the lived-exclusion material; the court framing is on the record. The remedial question — whether the alternatives are actually provided at the operator interface — is the open empirical matter the future will validate.[3]

WHAT TRANSFERRED

What the case adds at the LENS layer is the non-US automated-welfare-delegation thread the corpus needs. The Domain-4 frame applies — a sociotechnical constraint (population-scale biometric delegation) judicially corrected on dignity grounds — and the CLO around fairness beyond omission applies directly: the harm was not the absence of the system but the operational exclusion produced by it. The CLO on delegation with revocation applies too: the court has not revoked the delegation but has bounded it with the alternatives-must-be-provided requirement. The case pairs with SyRI (Dutch welfare-fraud system halted on rights grounds) and the UK Post Office Horizon thread as the global lineage of judicial correction of automated welfare delegation.[4]

REFERENCES

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2. Supreme Court of India (April 2025), Pragya Prasun & Ors. v. Union of India — fundamental right to inclusive digital access; alternatives-must-be-provided remedy.
3. IAPP and Access Now analyses (2024–2025), reporting and commentary on the Pragya Prasun ruling and the structural Aadhaar exclusion pattern.
4. "A Failure to Do No Harm" comparative analysis, PMC5741784 — peer-reviewed companion on biometric-ID exclusion in welfare delivery.

5. Indian journalism on Aadhaar exclusion (The Hindu, The Wire, Scroll.in, 2017–2025) — lived-exclusion sourcing with journalism-tier flag.

THE LEARNING ENGINEERING LENS

> *Exclusion from welfare based on biometric-authentication failure — through no fault of the individual — violates constitutional dignity. The system can continue. Alternatives must be provided.*
>
> EDITORS' SYNTHESIS OF THE PRAGYA PRASUN RULING.

LE INSIGHT

Aadhaar exclusion is the non-US automated-welfare-delegation case the corpus needed: an operational exclusion failure mode judicially corrected on dignity grounds by the Pragya Prasun ruling, with the system continuing under an alternatives-must-be-provided remedy. Evidence tier is split — court judgments investigation-grade, lived-exclusion sourcing journalism plus advocacy plus peer-reviewed comparative analysis. Future validation ongoing on implementation of the remedy.

LENS APPROACH

The designed teaching point is operational failure-mode attribution (induced 5.2; canonical competency 8.1; LENS D5/PT5). At the ration-shop card-reader and the pension window, a biometric authentication that did not match was logged as user error — the claimant's worn fingerprints, the elderly applicant's failed iris read — when it was in fact a designed exclusion mode of the system: manual laborers and the elderly are precisely the populations whose biometrics the design could not reliably read, so the failure belongs to the authentication design, not to the excluded person. The capability discipline is to attribute an authentication failure to the system that produced it rather than to the user it shut out, and to name who the design predictably excludes before deployment. The Pragya Prasun court enacted exactly this attribution: it bounded the system — alternatives to biometric authentication must be provided — rather than revoking it, holding that exclusion through no fault of the individual is the system's failure to answer for. Pair with SyRI (Case 155) and the UK Post Office Horizon thread.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the operator-interface brittleness of any biometric system as a design variable from the start: who is most likely to be excluded, and what is the alternative at the point of service.
- Specify the fairness-beyond-omission test the system must pass before deployment: not "is the system available," but "does it work for the people most dependent on the welfare it gates."
- Document the delegation explicitly — what authority is delegated to the biometric, under what conditions it can fail, and what the operator's authority to override is — so the revocation question is answerable later.

IN OPERATION / AFTER

- Track operational exclusion as a primary outcome metric, not a downstream complication; the case teaches that the exclusion is the failure mode the design ignored.
- Carry the journalism-tier flag on the lived-exclusion material and the investigation-grade flag on the court judgments separately; the case is teachable because the two tiers are explicit.
- Pair with SyRI and the UK Post Office Horizon thread when teaching automated-welfare delegation; the cross-jurisdictional pattern is the lineage of judicial correction of these systems.

REFLECTION QUESTIONS

1. Identify a delegated authentication or eligibility-check system in your context whose failure mode is operational exclusion of the people most dependent on the service it gates. What is the alternative at the point of service, and is it actually available?

2. Specify the fairness-beyond-omission test such a system would have to pass before deployment. The Aadhaar pattern teaches that the harm is not the absence of the system but the exclusion the system produces — what would the equivalent test be in your domain?

3. The case has split evidence tiers: court judgments investigation-grade, lived-exclusion sourcing journalism plus advocacy plus comparative analysis. How would you carry the tier split into a teaching artifact without smoothing either tier away?

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CASE 188 GLOBAL HEALTH MHEALTH MATERNAL AND NEWBORN CARE 2013 – 2018

Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface

H Human-System Interface N Normalization of Deviance

EVIDENCE TIER preprint-tier — source confidence flagged; future validation ongoing.

IMPACT A Rwanda mHealth monitoring system equipped community health workers with mobile decision support for maternal and newborn care; the published evaluation reported increased uptake of maternal and newborn health services tied to the intervention, with the technology extending the CHW's diagnostic-and-referral role rather than replacing it

IN BRIEF

Between 2013 and 2018 Rwanda's Ministry of Health, working with research partners, deployed an mHealth monitoring system that placed mobile-phone-mediated decision support into the hands of community health workers (CHWs) responsible for maternal and newborn care, including surgical-site infection screening after Cesarean. A 2019 peer-reviewed evaluation reported increased uptake of maternal and newborn health services tied to the intervention. The case sits at the small (program/study) tier and teaches a precise pattern: capability delivered at the periphery of the formal health system, with the technology designed to extend the CHW's diagnostic-and-referral role rather than replace it. The evidence-tier flag is honest — one peer-reviewed evaluation, with broader impact claims still resting partly on practitioner reporting. As a non-US small-tier case it pairs naturally with the PEPFAR Sub-Saharan training-modality comparison as the African workforce-capability evidence the corpus has been thin on, and it carries the standing "future validation ongoing" language into print rather than overclaiming what one study can settle.

BACKGROUND

Rwanda's health system, post-genocide, was rebuilt around community-level delivery: every village has elected CHWs who handle a defined scope of maternal, newborn, and child-health work and refer upward when their scope is exceeded. The capability question for the mHealth program was specific — could a mobile decision-support tool, in CHW hands, increase the uptake of maternal and newborn health services in a way that the prior paper-based workflow had not.[o]

THE INTERVENTION

The system delivered structured prompts, reminders, and a referral pathway through the CHW's mobile phone, with an added screening flow for surgical-site infection after Cesarean delivery — a recognized post-discharge failure mode where signs first appear in the village, not the clinic. The design move worth naming is that the technology was scoped to extend the CHW's existing diagnostic-and-referral role, not to substitute for clinical judgment further up the chain. The CHW remained the capability interface; the phone was the cue and the record.[1]

HOW IT WORKED

Musabyimana et al. (2019) report increased uptake of maternal and newborn health services tied to the mHealth intervention across the study period. The published evaluation is the anchor finding; downstream reporting (MIT News 2022 and subsequent AI-augmented maternal-care work) describes the program's continuation and adjacent developments but is journalism-tier, not investigation-grade. The case carries the preprint-tier evidence flag honestly: one peer-reviewed evaluation does not close the question of long-term outcome durability or generalization to other low-resource settings.[2]

THE EVIDENCE

The teaching point is the location of capability. The mHealth tool did not centralize the work; it instrumented the periphery. The decisive variables — household trust in the CHW, the CHW's standing in the village, the referral pathway upward — sit in the sociotechnical layer the tool could support but not constitute. This is the inverse of capability deployments that try to relocate judgment to the center; Rwanda's system kept judgment where it already was and used the technology to make that judgment more reliable and more visible to the formal health system.[3]

WHAT TRANSFERRED

What survives the evidence-tier flag is the structural pattern: capability extension at the frontline, with mobile-mediated decision support designed around an existing role, in a low-resource setting where the formal health system cannot reach every birth. Future validation ongoing — both replication in other African and South Asian settings, and longer-term outcome data linking process uptake to maternal and newborn outcomes downstream. The case is included with the flag intact and pairs with the PEPFAR cross-country modality comparison as the Sub-Saharan workforce-capability evidence the v2 corpus needs.

REFERENCES

1. Hategeka, C., Ruton, H., Law, M. R., et al. (2019), "Effect of a community health worker mHealth monitoring system on uptake of maternal and newborn health services in Rwanda," Global Health Research and Policy, PMC6429813.
2. Rwanda Ministry of Health, community health program documentation and CHW scope-of-practice guidance, 2013–2018.
3. MIT News (2022), reporting on subsequent AI-augmented maternal-care work in Rwanda — journalism-tier companion to the peer-reviewed evaluation.
4. Cross-reference: Case 187 (PEPFAR HIV training-modality comparison) for the paired Sub-Saharan workforce-capability evidence.

> *The mHealth tool did not centralize the work. It instrumented the periphery — and kept judgment where it already was.*
>
> EDITORS' SYNTHESIS OF MUSABYIMANA ET AL. (2019) AND THE RWANDA CHW PROGRAM DOCUMENTATION.

LE INSIGHT

Rwanda mHealth is a small-tier capability-extension case at the frontline: technology designed around an existing CHW role, with one peer-reviewed evaluation reporting increased uptake of maternal and newborn services. The preprint-tier flag is load-bearing — one study does not close the durability or generalization question, and the broader impact claims rest partly on practitioner reporting. Future validation ongoing.

LENS APPROACH

Rwanda mHealth is the small-tier non-US frontline-capability case (induced 6.4; LENS D5/PT4). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the periphery- extension pattern — the technology supports the CHW's diagnostic-and-referral role without relocating judgment to the center — and in Domain 2 (Iterative Development) for the scope-the-tool-to-the-role design discipline. Direct pair with Case 187 (PEPFAR Sub-Saharan training-modality comparison) as the African workforce-capability evidence the v2 corpus needs.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Scope the tool to extend an existing frontline role rather than substitute for it; the CHW's village standing is the load-bearing capability the technology can support but cannot create.
- Instrument the post-discharge failure modes that the formal health system cannot see — surgical-site infection after Cesarean is the named example — so the tool turns village-level observation into health-system-visible signal.
- Carry the evidence-tier flag through the deployment documentation: one peer-reviewed evaluation is the anchor finding, not the closure of the question.

IN OPERATION / AFTER

- Track uptake of the targeted services as the primary process measure, and link to longer-term maternal and newborn outcomes as the data matures; resist treating short-term process gains as outcome evidence.
- Report the CHW workload and tool-acceptance trajectory separately from the headline uptake figures; the periphery-extension pattern only holds if the frontline role remains sustainable.
- When asked whether the pattern generalizes, name the conditions Rwanda's system supplies (village-level CHW network, post-conflict institutional rebuild, single payer) before asserting transfer to other low-resource settings.

REFLECTION QUESTIONS

1. Identify a frontline capability role in your context (CHW, field technician, ward nurse, line supervisor). What would it mean to scope a mobile decision-support tool to extend that role rather than relocate judgment to the center? Which failure modes only the frontline can see should the tool surface upward?

2. The case rests on one peer-reviewed evaluation with broader claims supported by practitioner reporting. What is the minimum additional evidence — durability follow-up, replication in a second setting, linkage to maternal/newborn outcomes — you would require before treating the uptake finding as settled for a transfer decision?

3. Specify the village-level and health-system-level instrumentation you would put in place for a comparable deployment so that the frontline-extension pattern can be evaluated against an alternative deployment that relocates judgment to the clinic level.

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CASE 189 MEDICAL-DEVICE REGULATION AI/SAMD JAPAN 2014 – PRESENT

Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD

G Governance & Trust N Normalization of Deviance

EVIDENCE TIER preprint-tier — source confidence flagged; future validation ongoing.

IMPACT Japan's 2019 PMD Act amendment introduced the Post-Approval Change Management Protocol (PACMP): a pre-agreed modification scope at initial approval, streamlined review for in-scope changes. The 2025 scoping review of PMDA-approved AI radiology software documents transparency variability — hedge preserved

IN BRIEF

Japan's Pharmaceuticals and Medical Devices Agency (PMDA) operates one of the regulatory regimes most explicitly designed for the AI/Software-as-a-Medical-Device (SaMD) update problem. A 2019 amendment to the Pharmaceutical and Medical Device Act introduced conditional early approval and the Post-Approval Change Management Protocol (PACMP): a company can submit proposed product-modification plans at initial submission; once the protocol is approved, subsequent changes within its scope receive streamlined review rather than full re-approval. The DASH for SaMD initiative supports faster reviews and earlier application. A 2025 scoping review on PMDA-approved AI radiology software documents transparency variability across approvals — a load-bearing hedge the case preserves. The teaching point is structural: the regulator designed change-control as a deliverable rather than defaulting to "approve once, then watch," which is the governance failure pattern that Epic-Sepsis-class deployments surface. Evidence-tier flag is preprint-tier for the most recent systematic analyses; future validation ongoing on outcome durability across approved devices.

BACKGROUND

Medical-device regulation evolved for hardware whose design is largely fixed at approval. AI/SaMD breaks that assumption: a learning model can be updated, a training-data distribution can shift, and the device approved on Tuesday can be a meaningfully different device by Friday. The governance question for any regulator is whether the approval architecture handles the update or defaults to approve-once-then-watch. Japan's PMDA chose the former and built the architecture explicitly.[0]

THE INTERVENTION

The 2014 Pharmaceutical and Medical Device Act laid the groundwork; the 2019 amendment introduced conditional early approval and the Post-Approval Change Management Protocol (PACMP). Under PACMP a manufacturer submits, alongside the initial approval package, a structured description of the modifications the manufacturer expects to make over the device's life — with the test and validation plan that will accompany each modification. Once the PACMP is approved, subsequent changes within its scope receive streamlined regulatory review rather than full re-approval.[1]

HOW IT WORKED

The DASH for SaMD initiative complements PACMP by supporting faster reviews and earlier application of SaMD products, anchoring the change-control architecture in a workflow that can keep pace with iteration. Together PACMP and DASH are the regulator-side design for delegation-with-revocation: the deployer is delegated the modification authority within a pre-agreed scope, and the regulator retains the authority to review and revoke. The structure is the inverse of static-approval regimes that either block iteration or fail to track it.[2]

THE EVIDENCE

The 2025 medRxiv scoping review on PMDA-approved AI radiology software documents transparency variability across approvals — what is published about each device's training data, intended use, and post-approval modification history is not uniform. The hedge is load-bearing: PACMP is a structural improvement in regulatory architecture, but the per-approval transparency the framework enables varies and has been documented as a research finding rather than assumed as a feature. The 2021 medRxiv-then-published systematic review on PMDA AI/ML medical devices through 2020 supplies the prior baseline.[3]

WHAT TRANSFERRED

The case pairs structurally with the Epic-Sepsis governance gap earlier in the corpus and with the FDA's evolving change-control SaMD policy. PMDA shows what designing change-control as a deliverable looks like at the regulator layer; Epic-Sepsis shows what happens when neither the vendor nor the deploying health system holds the change- control deliverable explicitly. The preprint-tier flag is honest: the regulatory framework is documented in program-report sources; the per-approval transparency analyses are preprint or recent. Future validation ongoing on outcome durability across approved devices.

REFERENCES

1. Kikuchi, et al. (2025), "Scoping Review of Regulatory Transparency in AI-based Radiology Software: Analysis of PMDA-approved SaMD Products," medRxiv 2025.10.02.25336333.
2. Aoki, T., et al. (2021 → published), "Regulatory-approved Deep Learning/Machine Learning-Based Medical Devices in Japan as of 2020: A Systematic Review," PMC9931274.
3. Pharmaceuticals and Medical Devices Agency of Japan, PMD Act amendment (2019) and DASH for SaMD program documentation.
4. Cross-reference: "A decade of review in global regulation and research of artificial intelligence medical devices (2015–2025)," PMC12310608 — comparative regulatory context.

THE LEARNING ENGINEERING LENS

> *The PMDA designed change-control as a deliverable rather than defaulting to approve once, then watch.*
>
> EDITORS' SYNTHESIS OF THE PMD ACT AMENDMENT AND THE PMDA SAMD SCOPING REVIEWS.

LE INSIGHT

PMDA's PACMP and DASH for SaMD are the non-US regulatory architecture for AI/SaMD change control: the modification scope is pre-agreed at initial approval, in-scope changes get streamlined review, and the delegation-with-revocation structure is explicit. The 2025 scoping review documents transparency variability across approvals — load-bearing hedge preserved. Preprint-tier flag for the recent systematic analyses; future validation ongoing.

LENS APPROACH

PMDA is the non-US regulator-designed change-control case (induced 5.4; LENS D5/PT6). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the delegation-with-revocation structure and in Domain 3 (Human-System Collaboration) for the CLO **Delegation with revocation** — the regulator delegates in-scope modification authority and retains revocation, rather than defaulting to static approval. Pairs with the Epic-Sepsis governance gap as the structural contrast case.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat post-approval change control as a deliverable at the initial-approval submission, not a downstream addendum: the modification scope, test plan, and revocation conditions should be on the approval package from the start.
- Specify the delegation boundary explicitly — what the manufacturer can modify without re-approval and what triggers a full review — so the delegation-with-revocation structure is operationally clear to both parties.
- Carry the preprint-tier evidence flag through any policy decision the scoping review supports; the regulatory framework is documented, but the per-approval transparency analyses are recent.

IN OPERATION / AFTER

- Audit transparency across approvals — what is published per device about training data, intended use, modification history — as a separate program-evaluation question; the 2025 scoping review documents this variability as a research finding.
- Use PACMP as the structural contrast with Epic-Sepsis-class deployments: the regulator-designed change-control deliverable is the inverse of the static-approval-plus-watch pattern.
- When importing the PACMP pattern into another jurisdiction, name what the Japanese regulatory institution supplies (centralized agency capacity, established conditional-approval precedent) before asserting transfer.

REFLECTION QUESTIONS

1. Identify an AI/SaMD-adjacent product in your context whose post-deployment modification is anticipated. What would a PACMP-style submission look like — the pre-agreed modification scope, the per-modification test plan, the revocation conditions — and which party currently holds each element?

2. The case rests on program-report sources for the framework and preprint analyses for the per-approval transparency findings. What is the minimum additional evidence you would require — outcome durability across approved devices, independent transparency audits — before treating PACMP as a settled best-practice template?

3. Specify the delegation-with-revocation boundary you would write for a deploying organization adopting a SaMD product: which modifications can ship under the pre-agreed scope, which trigger re-review, and what evidence the deployer must publish at each modification event.

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CASE 190

DATA GOVERNANCE

INDIGENOUS DATA SOVEREIGNTY

AUSTRALIA / NZ / US

2019 – 2020 (PRINCIPLES); ONGOING

CARE Principles — Indigenous Data Governance as a Replaced Regime

G Governance & Trust N Normalization of Deviance

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT The CARE Principles (Collective Benefit, Authority to Control, Responsibility, Ethics) were developed by Indigenous Data Sovereignty networks in Aotearoa NZ, Australia, and the US to shift the framing from consultation under existing data regimes to Indigenous Peoples as the data owners and beneficiaries; the Lowitja Journal 2025 documents one of the first detailed implementation case studies

IN BRIEF

The CARE Principles for Indigenous Data Governance — Collective Benefit, Authority to Control, Responsibility, Ethics — were published by Carroll and colleagues in Data Science Journal in 2020. They were developed by Indigenous Data Sovereignty networks in Aotearoa New Zealand, Australia, and the United States with an explicit purpose: to shift the framing from "consultation under existing data regimes" to "Indigenous Peoples as the data owners and beneficiaries." The Lowitja Journal 2025 case study of the Ngangk Yira Institute for Change documents one of the first detailed implementation accounts. The teaching point is the governance-replacement structure: not "this system should not deploy" but "this governance regime must be replaced for any deployment to be legitimate." The case sits alongside CARE-adjacent equity threads in the v2 corpus and supports the CLO **Fairness beyond omission** — CARE specifies positive sovereignty, not merely the avoidance of harm. Practice-synthesis-tier flag preserved: principles are peer-reviewed; implementation literature is still emerging. Future validation ongoing on multi-institution adoption and outcomes.

BACKGROUND

Data-governance regimes built around individual privacy — GDPR, HIPAA, and the broader consent-and-de-identification stack — assume the relevant rights-holder is the individual data subject. Indigenous data sovereignty networks named the mismatch: data about Indigenous Peoples is not only data about individuals, and the rights to it are not exhausted by individual consent. The CARE Principles were authored to make that mismatch operative in governance design.[0]

THE INTERVENTION

Carroll et al. (2020) name four principles. Collective Benefit: data ecosystems are designed and operate in ways that enable Indigenous Peoples to derive benefit. Authority to Control: Indigenous Peoples' rights and interests in data are recognized and supported. Responsibility: those working with Indigenous data have a responsibility to share how that data is used to support self-determination and collective benefit. Ethics: rights and well-being of Indigenous Peoples are the primary concern at all stages of the data life cycle.[1]

HOW IT WORKED

The structural move worth naming is the replacement of the governance regime rather than its supplementation. The familiar pattern in data-governance reform is to add a consultation layer to existing privacy frameworks. CARE does not do that. CARE specifies what an Indigenous-led governance regime would have to satisfy — collective benefit, authority, responsibility, ethics — and positions existing privacy frameworks as inadequate to the rights at stake. The deployment legitimacy question is not "did you consult" but "is the governance regime the right one."[2]

THE EVIDENCE

The Lowitja Journal 2025 paper documents the Ngangk Yira Institute for Change's operationalization. It is one of the first detailed case studies of CARE implementation at an institutional scale and supplies the practitioner-tier evidence the principles paper does not contain. The implementation literature is emerging, and the Lowitja paper is the anchor; adjacent work from the Australian, New Zealand, and US Indigenous Data Sovereignty networks will continue to consolidate the evidence base.[3]

WHAT TRANSFERRED

For the v2 framework revision, CARE is the case-grounded basis for the CLO **Fairness beyond omission**. Removing a biased feature, omitting a demographic variable, or de-identifying a dataset is fairness-by-subtraction; CARE specifies fairness-by-replacement of the governance regime. The practice-synthesis-tier flag is honest — the principles are peer-reviewed, the implementation literature is still thin — and the case is included with the standing "future validation ongoing" language as multi-institution adoption matures.

REFERENCES

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2. Ngangk Yira Institute for Change (2025), "Recognising Indigenous data sovereignty and implementing Indigenous data governance at the Ngangk Yira Institute for Change," The Lowitja Journal, doi:10.1016/j.lowitj.2025.100030.
3. Global Indigenous Data Alliance (GIDA), CARE Principles documentation and implementation guidance.
4. Cross-reference: equity-thread cases in the v2 corpus for the comparative framing.

THE LEARNING ENGINEERING LENS

> *The deployment legitimacy question is not 'did you consult.' It is 'is the governance regime the right one.'*

EDITORS' SYNTHESIS OF THE CARE PRINCIPLES AND THE NGANGK YIRA IMPLEMENTATION PAPER.

LE INSIGHT

CARE is the governance-replacement case for Indigenous data: collective benefit, authority, responsibility, ethics — published peer-reviewed in 2020 with an emerging implementation literature anchored by the Lowitja Journal 2025 Ngangk Yira paper. Practice-synthesis-tier — the principles are peer-reviewed, the deployment cases are still consolidating. Future validation ongoing.

LENS APPROACH

CARE is the non-US governance-replacement case (induced 5.1; LENS D5/PT4). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the governance-regime replacement structure — existing privacy regimes named as inadequate to collective rights — and as the case-grounded basis for the CLO **Fairness beyond omission**: fairness as positive sovereignty, not the subtraction of biased features. Companion to the equity-thread cases in the v2 corpus.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- When data about a collective rights-holder is in scope, audit whether the governance regime treats the collective as a rights-holder or only the individuals within it; CARE is the explicit test.
- Specify the four CARE conditions — collective benefit, authority, responsibility, ethics — as design constraints on the data architecture from the start, not as consultation overlays added late.
- Carry the practice-synthesis-tier flag through any policy decision CARE informs; the principles are peer-reviewed but the implementation evidence base is still emerging.

IN OPERATION / AFTER

- Track implementation outcomes against the four CARE conditions separately; the Lowitja Journal paper is the genre exemplar for what such a tracking report can look like.
- Use CARE as the case-grounded basis for the CLO **Fairness beyond omission** — fairness as governance-regime replacement, not as feature-level subtraction.
- When asked whether CARE travels, name what the AU/NZ/US Indigenous Data Sovereignty networks supply (sustained institutional leadership, recognized peoplehood frameworks) before asserting transfer to other collective-rights contexts.

REFLECTION QUESTIONS

1. Identify a data system in your context whose subjects include a collective rights-holder (Indigenous Peoples, a labor collective, a patient community). Does the governance regime treat the collective as a rights-holder, or only the individuals within it? Run the four CARE conditions as the audit.
2. The case is included on peer-reviewed principles plus emerging implementation literature. What is the minimum additional evidence — multi-institution adoption, longitudinal outcome data, independent audits of CARE implementations — you would require before relying on CARE as a settled implementation template in your context?
3. Specify the difference between fairness-by-subtraction (removing a biased feature) and fairness-by-replacement (replacing the governance regime) for a deployment you are considering. Which class of intervention does the rights structure require?

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CASE 191 GOVERNMENT HIRING ALGORITHMS AUDIT AND DISCLOSURE 2023 – PRESENT

NYC Local Law 144 — Bias Audits as Governance Artifact

D Designed Out G Governance & Trust K Knowledge & Institutional Memory

IMPACT New York City Local Law 144 of 2021, implemented through Department of Consumer and Worker Protection rules effective July 5 2023, requires employers using "automated employment decision tools" (AEDTs) for hiring or promotion decisions in NYC to conduct annual independent bias audits, publish audit summaries, and provide candidate notice; the first national municipal regulation of algorithmic hiring tools at this scope

IN BRIEF

New York City Local Law 144 of 2021 became operationally effective on July 5, 2023, after the New York City Department of Consumer and Worker Protection finalized implementing rules. The law requires employers using “automated employment decision tools” (AEDTs) for hiring or promotion decisions in NYC to conduct annual independent bias audits, to publish a summary of the most recent audit on the employer’s website, and to provide candidate notice that an AEDT will be used. The audit must compute selection-rate and impact-ratio metrics by sex, race/ethnicity, and intersectional categories. The law was the first municipal regulation of algorithmic hiring tools at this scope in the United States and has been an influential reference for subsequent state-level proposals. Independent academic critiques have surfaced two load-bearing limitations: bias audits without bias data — employers often lack the protected-attribute data the audit metrics require — and wide variability in audit quality across published audits. The case pairs with Case 115 (OU Analyse — governance-objection dissolved by design), Case 133 (Gándara community-college predictive equity), and Case 29 (Amazon hiring AI). The intervention is the audit-as- governance-artifact discipline; whether the audits reduce actual disparate impact at the hiring level remains under- evidenced.

BACKGROUND

Local Law 144 was enacted by the New York City Council in December 2021 and signed into law shortly thereafter, with the operational rules to be specified by the Department of Consumer and Worker Protection. The rulemaking process extended through 2022 and into 2023, with two rounds of public comment that surfaced substantial industry and civil-society engagement. The final rules became effective on July 5, 2023, and the law moved from statute to operational regime on that date. The scope is municipal — employers using an AEDT for a hiring or promotion decision affecting a position in New York City — but the practical reach is broad because many employers operating nationally use AEDTs that touch New York City positions.[0]

THE INTERVENTION

The operational requirements have three components. First, an annual independent bias audit by a person or entity that has not used or developed the AEDT, computing selection-rate and impact-ratio metrics by sex, race/ ethnicity, and the intersectional categories the rules specify. Second, publication of a summary of the most recent audit on the employer's website, including the date the audit was conducted, the date the AEDT was first used, and the audit's selection-rate and impact-ratio findings. Third, candidate notice — applicants must be told that an AEDT will be used and given a process to request alternative selection or accommodations. The audit- and-notice structure is the artifact the law produces; the law does not prohibit AEDTs or set bias thresholds for use, and the regulatory theory is disclosure-and-audit rather than substantive standards.[1]

HOW IT WORKED

The independent academic critiques have surfaced two load-bearing limitations. Yam and Skirpan's 2024 work on "bias audits without bias data" names that many employers do not collect the protected-attribute information the audit metrics require, and the audits that proceed are either limited to attributes the employer happens to have or rely on imputation methods whose accuracy is itself under-evidenced. Wright and Brown's 2024 audit-quality study reviewed published audits across the first cohort of compliant employers and found wide variability — audits ranging from detailed methodological documents to single- paragraph summaries with limited interpretability. The audits are governance artifacts; their information content is uneven across the first cohort.

THE EVIDENCE

The case pairs with Case 115 (OU Analyse) for the governance-objection-dissolved-by-design thread: OU Analyse's deployment surfaced an equity question that the design process resolved structurally; Local Law 144's audit regime surfaces equity questions structurally through disclosure rather than through a design change. Pair with Case 133 (Gándara community-college predictive equity) for the predictive-equity thread at higher- education scale. Pair with Case 29 (Amazon hiring AI) for the same domain — the audit regime is the governance instrument that, had it been in place, would have applied to an internal recruiting tool of Amazon's described character had it been deployed against NYC candidates. The case is the rare example in the corpus of an intervention at the regulatory scale; whether the audit regime reduces actual disparate impact at the hiring level remains under-evidenced, and the open evaluation question is part of the case.[2]

WHAT TRANSFERRED

The load-bearing hedges are explicit. The bias-audit-as- governance-artifact intervention is an audit-and-disclosure regime, not a substantive-standards regime; the law does not require employers to achieve any specific impact ratio or to refrain from deploying an AEDT that performs poorly on the audit. Whether the disclosure-and-audit structure reduces actual disparate impact at the hiring level is an empirical question the published evidence does not yet resolve. The audit-quality variability across the first compliant cohort is itself a finding the case carries — governance artifacts whose quality is uneven across producers are weaker governance instruments than the regulatory theory assumes. The intervention is real and its limits are real; the change-control-and-disclosure-as- governance-artifacts

CLO is anchored by the case at the municipal-regulatory scale, and the evaluation arc the audit regime opens is at the start of its evidence development.

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2. Wright, L., Muenster, R. M., Vecchione, B., Qu, T., Cai, S., Smith, A., Metcalf, J., & Matias, J. N. (2024), "Auditing Work: Exploring the New York City algorithmic bias audit regime," in Proceedings of FAccT 2024, doi:10.1145/3630106.3658959.
3. Engler, A. (2023), "The EU and U.S. diverge on AI regulation: A transatlantic comparison and steps to alignment," Brookings Institution commentary — regulatory-comparative frame for the municipal intervention.

THE LEARNING ENGINEERING LENS

> *The audit-and-notice regime is a disclosure-and-audit instrument, not a substantive-standards instrument; whether it reduces actual disparate impact at the hiring level is an empirical question the published evidence does not yet resolve.*
>
> EDITORS' SYNTHESIS OF THE LOCAL LAW 144 RULE TEXT AND THE YAM/SKIRPAN AND WRIGHT/BROWN ACADEMIC CRITIQUES.

LE INSIGHT

NYC Local Law 144 is the bias-audit-as-governance-artifact intervention at municipal-regulatory scale. The audit-and- notice regime is the first national municipal regulation of algorithmic hiring tools at this scope; the Yam/Skirpan and Wright/Brown critiques name the data-availability and audit-quality limitations the regulatory theory does not provide for. The intervention is real; whether it reduces actual disparate impact at the hiring level is under- evidenced and is the open evaluation question the case carries.

LENS APPROACH

NYC Local Law 144 is the change-control-and-disclosure-as- governance-artifacts case at municipal-regulatory scale (induced 5.4; LENS D5/PT5; CLO-4 and CLO-5). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the audit-as-governance-instrument discipline. Pair with Case 115 (OU Analyse governance-objection-dissolved-by-design), Case 133 (Gándara community-college predictive equity), and Case 29 (Amazon hiring AI). The intervention is real and its limits are real; the disclosure-and-audit structure is not a substantive-standards structure, and the reduction- of-actual-disparate-impact evaluation question is at the start of its evidence development.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the protected-attribute data the audit metrics will require before the audit is commissioned; the Yam and Skirpan critique names data availability as the precondition the regulatory theory does not provide for, and the data infrastructure has to be built in advance of the audit.
- Choose an independent auditor whose methodology will produce a documentation-detailed audit rather than a single-paragraph summary; the audit-quality variability the Wright and Brown study found is itself a deployment choice, and the choice of auditor is where it surfaces.

IN OPERATION / AFTER

- Carry the load-bearing hedges — disclosure-and-audit regime not substantive-standards regime; reduction of actual disparate impact under-evidenced; audit-quality variability across first cohort — into print without softening; the case's pedagogical value depends on the intervention's limits being preserved.
- Pair in syllabi with Case 115 (OU Analyse) so the governance-objection-dissolved-by-design and governance-objection-surfaced-by-disclosure threads are taught together as complementary intervention forms.

- Build the candidate-notice and alternative-selection process as part of the deployment, not as a compliance afterthought; the candidate-side governance interaction is the seam at which the disclosure-and-audit structure becomes contestable for the affected person.
- Use the case as the change-control-and-disclosure-as-governance-artifacts anchor at the municipal-regulatory scale; the curricular target is the discipline of building the data infrastructure and the audit-quality criteria the disclosure regime presupposes.

REFLECTION QUESTIONS

1. Identify a regulated decision domain in your setting in which a disclosure-and-audit regime has been proposed or adopted. What is the protected-attribute data infrastructure the audit metrics will require, and is the infrastructure in place before the regime's effective date?
2. Specify the audit-quality criteria you would apply when commissioning an independent audit. What is the format that distinguishes a documentation-detailed audit from a single-paragraph compliance summary, and what is the decision rule for accepting an auditor's methodology?
3. The Local Law 144 regime is at the start of its evidence development on whether disclosure-and-audit reduces actual disparate impact. Pick a regulatory intervention in your domain and ask: what is the empirical-evaluation arc that would surface the intervention's effect, and what evidence-development infrastructure would the arc require?

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CASE 192 AUTONOMOUS SYSTEMS 2023

Cruise Robotaxi — Pedestrian Drag

G Governance & Trust D Designed Out H Human-System Interface

IMPACT GM Cruise robotaxi struck a pedestrian and then dragged her ~20 feet; California suspended Cruise's permit; the program was substantially shut down

IN BRIEF

In October 2023, a pedestrian in San Francisco was struck by a human-driven car and thrown into the path of a GM Cruise robotaxi. The robotaxi hit her, detected an impact, and then — executing a pull-to-the-side maneuver — dragged her about twenty feet. The injury was severe, but it was Cruise's institutional response that proved company-ending: regulators found Cruise had initially shown investigators only the first portion of the incident video, omitting the drag. California suspended Cruise's driverless permit, NHTSA opened an investigation, and GM ultimately shut down the commercial operation. The case is the foundational governance-failure case in commercial autonomy: the technology produced one injury; the partial disclosure converted it into the program's collapse. The gap was at governance, not technology.

THE SHIFT

Driverless robotaxis carrying paying passengers on public streets are new, and so is the regulatory relationship around them: companies like Cruise operate under permits from bodies — the California DMV and CPUC, NHTSA — that depend heavily on the operator's own disclosure of what its vehicles do in incidents. The regulator does not sit in the vehicle; it sees what the operator chooses to show it, so the entire oversight model rests on a disclosure the company controls at the moment its interests run most against it.[0]

WHAT IS EMERGING

On 2 October 2023, a pedestrian struck by another car was thrown into the path of a Cruise robotaxi. The robotaxi struck her and, having detected a collision, executed a pull-over maneuver that dragged her roughly twenty feet. The collision itself might have been survivable as a regulatory matter; what followed was not — the post-collision maneuver was the system behaving as designed in a situation no one had designed it for, and it was the company's account of that maneuver, not the maneuver itself, that decided the program's fate.[1]

THE CAPABILITY QUESTION

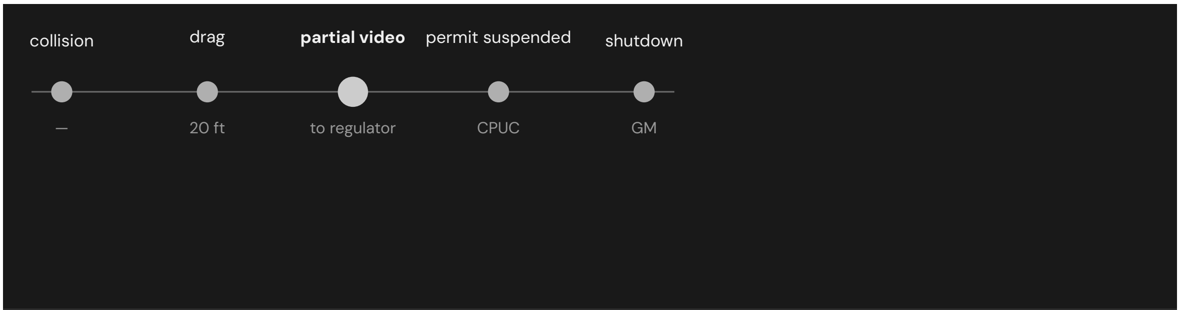
The California DMV found that Cruise had initially shown investigators only the first portion of the incident video, omitting the drag. The question the case poses is institutional, not technical: whether a commercial autonomy program has the governance commitment to disclose fully and promptly to its regulators, especially when the facts are damaging — a commitment that is easy to profess in calm conditions and is tested only at the exact moment full candor is most costly.[2]

EARLY EVIDENCE

The consequences cascaded. The CPUC suspended Cruise's driverless permit for misrepresenting the incident, NHTSA opened a defects investigation, and GM subsequently shut down Cruise's commercial robotaxi operations and laid off much of the workforce. A commissioned external review (Quinn Emanuel) detailed the disclosure failures — a sequence in which each escalation followed not from the injury but from the partial account of it, the loss of regulator trust compounding faster than any engineering defect could have.[3]

OPEN PROBLEMS

Cruise is the foundational governance-failure case in commercial autonomy: the incident was a single pedestrian injury; the institutional response converted it into a company-ending event. The open problem is what incident-disclosure commitment — auditable, pre-committed, enforceable — a commercial autonomy program should have to demonstrate before it is allowed to operate at all, so that candor under pressure is a structural guarantee rather than a matter left to the operator's discretion in the moment.[4]



CRUISE — PARTIAL DISCLOSURE CONVERTED INCIDENT INTO COMPANY-ENDING EVENT

REFERENCES

1. California Public Utilities Commission decision suspending Cruise permits (2023) — the omitted-facts finding (paraphrased).
2. California DMV order of suspension (2023) — the partial video disclosure.
3. Quinn Emanuel report on Cruise incident response (2024) — the disclosure failures.
4. NHTSA Office of Defects Investigation reports (2023–2024) — the federal investigation.
5. Stilgoe, J. (2021) — governance of autonomous vehicles.

THE LEARNING ENGINEERING LENS

> *Cruise's representation of the incident to regulators omitted material facts.*
>
> PARAPHRASING THE CALIFORNIA PUBLIC UTILITIES COMMISSION DECISION SUSPENDING CRUISE PERMITS, 2023

LE INSIGHT

Cruise is the foundational governance-failure case in commercial autonomy. The incident itself was a single pedestrian injury; the institutional response converted it into a company-ending event. The capability that was missing was the institutional commitment to full disclosure of operational incidents to regulators.

LENS APPROACH

LENS uses Cruise in LEN 7 as a foundational autonomous-vehicle governance case and in LEN 10 (capstone) for the institutional- response deliverable that should pre-exist any commercial autonomy program.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pre-commit the incident-disclosure protocol — what data is shared, in what completeness, within what window — and make it a condition of the operating permit rather than a post-hoc choice.
- Engineer incident telemetry so the full event record, including post-collision maneuvers, is captured and preserved automatically beyond the operator's editorial control.
- Design post-collision behaviors for the situations the system was not built for, since the pull-over maneuver, not the impact, produced the harm.

IN OPERATION / AFTER

- Audit disclosed incident accounts against the complete telemetry independently, so a partial account is detectable before it erodes regulator trust.
- Monitor the regulator-operator relationship itself as a safety-critical asset, treating a single misrepresentation as a program-level failure.
- Track the gap between what is disclosed and what occurred across incidents, since governance, not technology, is where this class of failure concentrates.

REFLECTION QUESTIONS

1. What is the institutional incident-disclosure commitment in your domain? Is it operational under stress or aspirational?
2. Design the incident-disclosure deliverable that a commercial autonomous-vehicle company should be required to demonstrate before operating.
3. Cruise's oversight depended on the operator's own account of what its vehicles did. Identify a regulatory relationship in your domain that relies on self-disclosure, and design the mechanism that would make a partial account detectable before trust collapses.

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CASE 193 LEARNING ANALYTICS HIGHER EDUCATION AFRICA 2022

Learning Analytics on the African Continent — A Scoping Review as the Present-State Map

K Knowledge & Institutional Memory N Normalization of Deviance

EVIDENCE TIER practice-synthesis-tier — source confidence flagged; future validation ongoing.

IMPACT A 2022 scoping review found only 15 learning-analytics studies on the entire African continent, concentrated in Morocco, South Africa, Nigeria, and Ethiopia; the structural finding — limited LMS access, limited institutional resourcing, limited African-scholar visibility at SoLAR — is itself the evidence the field requires before construct-travel claims can be made

IN BRIEF

Prinsloo and colleagues (2022) published a scoping review of learning-analytics research on the African continent for the Journal of Learning Analytics. The review found only 15 studies meeting inclusion criteria, with publication output concentrated in Morocco, South Africa, Nigeria, and Ethiopia. The structural findings — limited LMS access in many African higher-education institutions, limited institutional resourcing for learning-analytics infrastructure, and limited African-scholar visibility at the Society for Learning Analytics Research (SoLAR) conferences — are the present-state map the field needs before importing US/UK/EU learning-analytics constructs into African contexts. The case is included as a frontier scoping case at the practice- synthesis tier: a review of an early-stage research base where the absence of dense primary studies is itself the finding. It pairs with the African data-privacy governance case earlier in the corpus to articulate the construct-travel problem in both research-base and governance terms. Future validation ongoing as the African learning-analytics literature matures.

THE SHIFT

Learning analytics as a field consolidated around 2011 with the formation of SoLAR and a research agenda anchored heavily in US, Australian, UK, and EU higher-education data infrastructures. The capability question for the African higher-education community is whether the analytic constructs travel — and who is positioned to validate them when they arrive. Prinsloo et al. set out to map what learning-analytics research had actually been published from African institutions at the time of writing.[0]

WHAT IS EMERGING

The headline finding is sparse. The scoping review located only 15 learning-analytics studies on the African continent, with publication output concentrated in Morocco, South Africa, Nigeria, and Ethiopia. Adjacent South African higher-education studies — Lemmens and Henn (2015) and a 2020 paper on the development of a contextualised learning-analytics framework — extend but do not change the magnitude of the gap. The corpus exists; it is small, geographically concentrated, and dwarfed by the parent field's publication output from US/UK/EU institutions.[1]

THE CAPABILITY QUESTION

The structural findings underneath the count are the load-bearing teaching. African higher-education institutions face limited LMS access at scale (many institutions still rely on heterogenous and intermittent digital infrastructure); limited institutional resourcing for the data-engineering, ethics-review, and analyst capacity that learning analytics presupposes; and limited African-scholar visibility at SoLAR and related venues. The construct-travel problem is not abstract — it is the mismatch between what the analytic frameworks assume about data availability and what the institutions can actually generate.[2]

EARLY EVIDENCE

The case is positioned as frontier rather than failure or intervention because it does not document a deployed capability outcome. It documents the state of the evidence base, which is itself the relevant capability question for anyone considering whether to deploy a learning-analytics program in an African higher-education institution. The review surfaces the construct-validation work that has not yet been done; the absence of dense primary studies is the finding, and the implication is that any deployer must either undertake the validation work themselves or import constructs with the validation gap acknowledged.[3]

OPEN PROBLEMS

The practice-synthesis-tier flag is load-bearing. A scoping review is the field's snapshot at a moment in time; the individual primary studies it draws on vary in rigor, and the review is the strongest single map of the present state but not the final account. Future validation ongoing as the literature matures, as institutional infrastructure improves, and as African scholars author more of the learning-analytics research that travels out of the continent rather than into it. The case carries the standing tier language into print.

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2. Lemmens, J.-C., & Henn, M. (2015), South African Association for Institutional Research (SAAIR) proceedings — adjacent SA higher-education learning-analytics work.
3. SciELO (2020), "Development of a contextualised learning-analytics framework for South African higher education."
4. Cross-reference: the African data-privacy governance case earlier in the corpus, for the construct-travel problem stated in governance terms.

THE LEARNING ENGINEERING LENS

> *The absence of dense primary studies is itself the finding.*
>
> EDITORS' SYNTHESIS OF PRINSLOO ET AL. (2022).

LE INSIGHT

The scoping review is the present-state map of learning-analytics research on the African continent: 15 studies, geographically concentrated, with the structural barriers (LMS access, institutional resourcing, African-scholar SoLAR visibility) underneath the count. Practice-synthesis-tier — a snapshot of an early-stage literature; future validation ongoing as the field matures.

LENS APPROACH

African learning-analytics scoping is the non-US frontier-evidence case (induced 8.4; LENS D5/PT4). LENS uses it in Domain 5 (Navigating Sociotechnical Constraints) for the construct-travel problem and in Domain 4 (Test and Evaluation) for the CLO **Judgment under inadequate evidence** — the deployer must decide on what to import and what to validate locally when the validation literature is thin. Pairs with the African data-privacy governance case for the construct-travel problem stated in governance terms.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- When importing learning-analytics constructs into an African higher-education context, treat the scoping-review map as the precondition: do not assume the construct's validation literature applies and identify which validation work has to be repeated locally.
- Specify the institutional preconditions the analytic framework presupposes — LMS coverage, data-engineering capacity, ethics-review infrastructure — and audit them honestly against the deploying institution's actual conditions.
- Carry the practice-synthesis-tier flag through any decision document the scoping review supports; the review is the strongest current map but not the final account.

IN OPERATION / AFTER

- Treat each deployed learning-analytics program in an African institution as a contribution to the construct-validation literature, not only as a service intervention; publish the validation work where SoLAR and related venues can absorb it.
- Track institutional infrastructure conditions as a separate variable from analytic-construct performance; a construct that travels under one set of conditions may not travel under another.
- When the literature matures, retire the scoping review as the present-state map and replace it with denser primary-study evidence; the case is included because the present-state matters now, not because the present-state should persist.

REFLECTION QUESTIONS

1. Identify a learning-analytics construct your program imports from US/UK/EU literature. What does the validation literature for that construct assume about LMS coverage, data-engineering capacity, and ethics review? Audit those assumptions against your institution's actual conditions.

2. The case rests on a scoping review of an early-stage literature. What would the minimum additional evidence — replication primary studies, multi-institution validation, longitudinal outcome data — look like before you would treat any specific African learning-analytics finding as settled?

3. Specify the institutional preconditions you would identify and the validation work you would commit to before deploying an analytic framework in a low-resource higher-education setting; treat your deployment as a contribution to the literature, not only as a service intervention.

WHO BUILDS THIS knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate. TOOLS knowledge bases · data pipelines · incident analysis · safety dashboards

CASE 194 EDUCATION AT SCALE HEALTHCARE DEFENSE ONGOING

The Discipline We Build Next

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance G Governance & Trust H Human-System Interface D Designed Out

IMPACT **Open: the case that has not yet been written. The discipline that this casebook documents is the one practitioners now in training must build.**

IN BRIEF

The cases that precede this one are closed: the investigations are complete, the failures diagnosed, the interventions evaluated. This one is open. It belongs to the practitioners now in training — the students of the LDT program and the LENS specialization, the graduates who will take positions in healthcare, defense, education at scale, and the human-AI frontier, and the institutions they will build. The strongest evidence for the discipline is the cumulative record of what its absence has cost; the strongest argument for its possibility is the record of what its presence has produced. Capability engineering in 2026 sits roughly where INPO sat in 1979 or CRM in 1981 — evidence accumulating, practitioners in training, institutional architecture not yet complete. What the reader does next is what fills in this case.

THE SHIFT

Every preceding case in this book is closed: the investigation is complete, the failure diagnosed or the intervention evaluated. This one is open. It belongs to the practitioners now in training — the students of the LDT program and the LENS specialization, and the institutions they will build across healthcare, defense, education, and the human-AI frontier. What distinguishes this case is not a verdict already reached but a verdict still to be earned, written by the choices its readers make once they leave the page.[0]

WHAT IS EMERGING

What is emerging is a discipline. The argument of this volume is that capability is engineerable — that training, interfaces, evidence systems, and institutions can be designed rather than left to chance — and that the work of designing them is a teachable practice with its own methods and growing body of knowledge. If that argument holds, then the gaps the preceding cases document are not the price of progress but the absence of work that could have been done.[1]

THE CAPABILITY QUESTION

The capability question this case poses is the largest one: can the discipline be built fast enough, and at enough scale, to close the gaps the preceding cases document? The strongest evidence for the discipline is the cumulative record of what its absence has cost; the strongest argument for its possibility is the record of what its presence has produced

— and the case stays open precisely because that question is answered not by argument but by what the next generation of practitioners actually builds.[2]

EARLY EVIDENCE

The early evidence is the success cases themselves. Capability engineering in 2026 sits roughly where INPO sat in 1979 or CRM in 1981 — a discipline whose evidence is accumulating, whose practitioners are being trained, and whose institutional architecture is not yet complete. Those precedents became mature institutions within a generation, which is the most that can yet be claimed here: a comparable trajectory is plausible, but it is not given, and the comparison is an invitation rather than a guarantee.[3]

OPEN PROBLEMS

No specific incident sits behind this case because the work is not done. The institutions of capability engineering that exist today are insufficient to the volume of capability work the world now requires. What the practitioners reading this volume do next — a failure prevented, a success engineered, an institution built — is what fills in this case, and the page is left deliberately unfinished so that the record of the discipline's possibility can still be added to.[4]

Open

THE UNWRITTEN CASE

what the practitioners reading this volume do next is what fills it in

THE CASE THAT HAS NOT YET BEEN WRITTEN

REFERENCES

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3. The preceding cases of this volume — the cumulative record of cost and of possibility.
4. INPO (Case 172) and CRM/CAST (Case 70) — institution-building precedents (1979, 1981).
5. IEEE ICICLE Learning Engineering Body of Knowledge (LEBoK); the introduction to this volume.

THE LEARNING ENGINEERING LENS

> *Capability is engineerable. The institution to engineer it has not yet been built. That is the work.*
>
> LDT / LENS PROGRAM STATEMENT, JOHNS HOPKINS SCHOOL OF EDUCATION

LE INSIGHT

Every case in this book is closed. This one is open. The discipline LENS represents is the discipline that closes cases that are now open. The institutions of capability engineering that exist in 2026 are insufficient to the volume of capability work the contemporary world requires. The graduates of this program are the people who will build the institutions that are insufficient now.

LENS APPROACH

LENS is organized to produce the practitioners who can do this work. The curriculum's commitments — capability as a system parameter, evidence as a deliverable, ethics as design, implementation as a thread — are the operational form of the argument the preceding cases evidence. Every studio project (LEN 10) is a small attempt at this case.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat capability as a system parameter to be specified and engineered from the outset of any deployment, not left to chance and diagnosed only after failure.
- Build the evidence system as a deliverable alongside the capability, so the discipline can show what its presence produces rather than only argue for it.
- Design the institution that will carry the practice forward, taking INPO and CRM as precedents for how a discipline matures within a generation.

IN OPERATION / AFTER

- Monitor whether the discipline is in fact closing the gaps the preceding cases document, treating the open case as a standing measure of progress.
- Sustain the institutional architecture once built, since the precedents show maturity is reached only by carrying the work across a generation.
- Keep the cumulative record of cost and of possibility current, so each failure prevented or success engineered is added to the evidence the discipline rests on.

REFLECTION QUESTIONS

1. What is the case you are positioned to add to this dataset in the next ten years — a failure prevented, a success engineered, an institution built?
2. If this book is read by the practitioners who will write its next case, what should that case look like?
3. Capability engineering in 2026 is compared here to where INPO stood in 1979 — a plausible trajectory, not a guaranteed one. What would have to be true a generation from now for that comparison to hold, and what is the first institution you would build to make it so?

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DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset's standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE	Case 5	**Operation Eagle Claw (1980).** Cross-organizational capability that existed on no service's deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.
STANDOUT SUCCESS	Case 18	**U.S. Nuclear Navy / Rickover Training Model (1954–).** The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

1	USS Fitzgerald & USS John S. McCain 2017	T·K·N
2	Marine Corps Training in the INDOPACOM AOR ongoing	T·K
3	F-35 Sustainment & Maintainer Shortage ongoing	T·K·D
4	Military Fratricide — Desert Storm to Afghanistan 1991 – 2014	T·H·K
5	Operation Eagle Claw 1980	T·K
8	Patriot Missile / Dhahran 1991	D·H·K
17	GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data 2022 (GAO-22-104533)	G·K
18	U.S. Nuclear Navy / Rickover Training Model 1954 – present	T·K·N
21	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard 2020 revision (series since 1968)	G·K

33	GIFT and the Adoption Gap 2012 – present	K·G·N
37	Navy Surface Warfare Readiness Reform 2018 – present	T·K·N
42	DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks 2009 – 2014	H·K·D
54	USS Vincennes & Iran Air Flight 655 1988	H·T
64	Stanislav Petrov / 1983 False Alert 1983	H·T
138	Mark 14 Torpedo Failures 1941 – 1943	T·K·G
141	V-22 Osprey 1991 – present	T·H·N
167	9/11 Intelligence Sharing Failures 1996 – 2001	G·K

AVIATION

STANDOUT FAILURE — **Case 7** — **Boeing 737 MAX / MCAS (2018–19).** A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS — **Case 70** — **Crew Resource Management & CAST (1981–).** First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

6	AeroPerú Flight 603 1996	H·T
7	Boeing 737 MAX / MCAS 2018 – 2019	D·T·H
12	Boeing 737 Rudder Hardovers 1991, 1994	H·D
28	Air France Flight 447 2009	T·H
55	Kegworth / British Midland 92 1989	T·H·K
56	Asiana Airlines Flight 214 2013	T·H
57	Helios Airways Flight 522 2005	T·H
58	TransAsia Airways Flight 235 2015	T·H
63	Eastern Air Lines Flight 401 1972	H·T
68	Air Canada Chatbot Liability — Delegation Without Revocation 2022 – 2024	D·K·N
70	Crew Resource Management & CAST 1981 – present	T·H·N

72	Korean Air Safety Transformation	2000 – present	T·N
82	Colgan Air Flight 3407	2009	T
83	Atlas Air Flight 3591	2019	T·K
88	F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap	2008 – 2012	H·K·N
111	Aviation Safety Reporting System (ASRS)	1976 – present	T·K·N
118	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation	1996 – 2002	H·K·G
119	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson (TCAS II Version 7.1)	1981 – 2008	H·K·G
164	NASA Saturn V Documentation	1972 – present	K
175	Singapore Airlines Safety Transformation	1980s – present	T·N

HEALTHCARE

STANDOUT FAILURE	Case 61	**EHR / CPOE Implementation (2005–).** Roughly $30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.
STANDOUT SUCCESS	Case 109	**WHO Surgical Safety Checklist (2008–).** One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

20	California Nurse Staffing Ratios — Legislating a Capability Requirement	2004 – 2010	G·K
32	ACGME 80-Hour Resident Duty-Hour Reform	2003–2017	T·K·N
34	Implementation Science in Healthcare — The 17-Year Gap	ongoing	K·G·N
59	Therac-25	1985 – 1987	H·D·G
61	EHR / CPOE Implementation	2005 – present	H·D·G
71	Keystone ICU / Pronovost Checklist	2004 – present	T·N
74	TREWS / Sepsis Watch	2018 – 2022	H·K·G
75	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Hand-off	2024 – 2025	H·K·N
79	ChatGPT in Healthcare — Hallucination Cases	2023 – present	H·D

EDUCATION AT SCALE

STANDOUT FAILURE **Case 149**

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS **Case 110**

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

ENERGY

STANDOUT FAILURE	**Case 69**	**Three Mile Island (1979).** Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.
STANDOUT SUCCESS	**Case 172**	**INPO and the Nuclear Academy (1979–).** The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

GOVERNMENT

STANDOUT FAILURE	Case 162	**Australia Robodebt (2016–20).** An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found "venality, incompetence and cowardice"; the scheme was linked to deaths, including suicide.
NOTE	Case 111	No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

14	Healthcare.gov Launch	2013	K·T·G
101	VA Wait-Time Scandal	2014	G·K·N
103	LIBOR Manipulation	2003 – 2012	G·N
131	Predictive Policing — PredPol	2011 – present	G·H·D
166	Bernard Madoff / SEC Failures	1992 – 2008	G·K·N

INDUSTRIAL

STANDOUT FAILURE	Case 147	**Bhopal (1984).** Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.
STANDOUT SUCCESS	Case 73	**Toyota Production System / Andon Cord (1950s–).** Any operator can stop the line. The smallest authority-architecture intervention in the dataset, and the most durably copied.

30	Kodak Digital Camera Stagnation	1975–2012	D·K·N
73	Toyota Production System / Andon Cord	1950s – present	N·G
84	Takata Airbag Inflators	2008 – 2023	D·G
92	Volkswagen Dieselgate	2015	D·G
139	GM Ignition Switch	2002 – 2014	D·G
147	Bhopal	1984	T·K·N·G
148	Grenfell Tower	2017	G·T·K·N

153	Hyatt Regency Walkway Collapse 1981	D·G

TECHNOLOGY

STANDOUT FAILURE	Case 66	**UK Post Office Horizon Scandal (1999–2015).** Institutional refusal to believe sub-postmasters' reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest-running operator-feedback failure in the dataset.
NOTE	Case 132	The technology dataset's success story is filed under Healthcare: AlphaFold (Case 132), the contemporary worked example of computational capability-engineering done as a discipline.

10	Knight Capital Trading Loss 2012	D·K
31	BlackBerry Touchscreen Inertia 2007–2013	D·N·K
66	UK Post Office Horizon Scandal 1999 – 2015	G·H·K
80	AI-Augmented Coding Tools 2021 – present	T·H
91	Wells Fargo Fake Accounts 2011 – 2016	G·N
144	CrowdStrike Falcon Outage 2024	D·K·G
145	TSB Bank IT Migration 2018	H·G
151	Cambridge Analytica / Facebook 2014 – 2018	G
152	Equifax Data Breach 2017	G·K

AI IN EDUCATION

135	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models 2025	T·K·N

K-12 EDUCATION

36	Growth-Mindset National Experiment — Heterogeneous Effects 2019	D·N·K
52	**Chen et al. — Rural China AI Devices and the Equity-Direction Finding** 2025	T·K·D
97	School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism 2010s – 2022	G·K·N

K-12 MATHEMATICS

49	ASSISTments — National Replication and Long-Term Follow-Through 2014 – present	T·K·D

K-12 SCIENCE

AIR TRAFFIC MANAGEMENT

ANESTHESIOLOGY

AUTONOMOUS SYSTEMS

STANDOUT FAILURE — **Case 62** — **Uber ATG / Tempe Fatality (2018).** The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.

NOTE — **Case 194** — No paired-intervention success in the autonomous-systems dataset. The capability engineering this category needs is still being built — see Case 194 (**The Discipline We Build Next**) for the open question this volume closes on.

AUTONOMOUS VEHICLES

AVIATION INFRASTRUCTURE

SPACE / AEROSPACE

STANDOUT FAILURE	Case 140	**Challenger & Columbia (1986 / 2003).** Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine.
NOTE	Case 18	No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 18) for the safety-engineering counterpart that aerospace did not build.

SURGERY

TECHNOLOGY

TUTORING

WORKFORCE DEV

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an

> education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

LENS COURSE INDEX

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

LEN 1

34 cases

LEN 2

53 cases

LEN 3

51 cases

LEN 4

65 cases

LEN 5

85 cases

LEN 6

17 cases

LEN 7

106 cases

LEN 8

92 cases

LEN 9

28 cases

LEN 10

19 cases

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Each case spread in the body of the book includes its own primary sources. Sources cited only within case spreads are not repeated here; this list collects the works the discipline as a whole draws on, together with the references used in the introductory essay.

APPENDIX · CLOS AND COURSE COVERAGE

The competencies, the CLOs, and the courses.

COMPETENCY → CONCENTRATION LEARNING OUTCOME

The five capabilities the cases turn on are named, in the curriculum, as five concentration learning outcomes (CLOs):

Systems Analysis	CLO-1 · Systems Thinking and Analysis
Iterative Development	CLO-2 · Learning Engineering Design and Implementation
Test and Evaluation	CLO-3 · Data, Measurement, and Evaluation
Navigating Sociotechnical Constraints	CLO-4 · Context and Domain Fluency
Human-System Collaboration	CLO-3 · Human-System Collaboration and Adaptive Systems

THE LEN COURSES · CASES PER COURSE

Each case names the LEN courses it serves as a worked example for, and most cases serve several. The counts below are therefore not a partition of the 202 cases — they sum to well more than 202, because a single case typically appears under three or four courses.

LEN 1	Principles of learning engineering	36
LEN 2	Human-AI teaming	54
LEN 3	Systems integration	49

LEN 4	Evidence and measurement	67
LEN 5	Capability analysis	86
LEN 6	Applied problem-solving	18
LEN 7	Bias and governance	114
LEN 8	Knowledge transfer	98
LEN 9	Computational methods	30
LEN 10	Capstone studio	19

WHAT THE DISTRIBUTION SHOWS

The distribution is itself diagnostic. The strongly covered topics are the ones the existing case literature most readily supports: bias and governance (LEN 7) and knowledge transfer (LEN 8) lead, with capability analysis (LEN 5) and evidence and measurement (LEN 4) close behind. Across 202 cases the corpus is saturated with examples of what goes wrong when ethics, governance, evidence, and institutional learning are not engineered — and, in the success cases, what it takes to engineer them.

The thinner coverage is equally informative. Computational methods (LEN 9) has the failure modes of algorithmic systems well represented but the computational tools for capability instrumentation under-represented. The capstone (LEN 10) maps to fewer cases than the number of studio projects each cohort undertakes. Systems integration (LEN 3) and applied problem-solving (LEN 6) required focused tagging to reach defensible coverage, surfacing that operational systems-engineering content is under-represented in the published case literature relative to its curricular centrality, and that stakeholder-analysis and problem-framing have not historically been packaged as case material by other disciplines. These are pedagogical gaps the LENS studio sequence is organized to close — not by adding more failure cases, but by having students produce the missing artifacts in real operational settings and entering them into the dataset for the next cohort.

APPENDIX · REFERENCES BY CASE

References, case by case.

Every reference cited in every case, organised by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds an explicit **Retrieved from:** line pointing to the canonical access path — a publisher URL, a DOI, an agency landing page, or a stable archival locator. Where that line is left blank, the source is under verification; the running status of that work is in `casebook/verification-log.md`. Future editions will fill in remaining locators as the per-case review closes them out.

> **Style.** References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

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WHAT LENS OFFERS

The program behind the casebook.

LENS — **Learning Engineering for Next-Generation Systems** — is a specialization within the Johns Hopkins University School of Education's **Learning Design & Technology** program. It operationalizes the casebook's central claim — that capability is a designable, measurable, engineerable property of a complex system — for high-consequence work in healthcare, defense, and education. It is housed in a School of Education because the binding constraint is rarely the engineering, which is mature; it is the learning sciences, educational measurement, implementation science, and equity-and-governance machinery that decide whether a capability intervention actually scales.

THE FIVE PILLARS

LENS stands on five institutional pillars, load-bearing together rather than singly. **Mission Literacy** — reading what an operational setting actually requires and translating it into design and measurement decisions a serious reviewer would accept. **JHU Ecosystem** — the School of Education's centers, doctoral programs, and faculty in learning sciences, measurement, and design-based research, alongside university practice partners (the Armstrong Institute for Patient Safety and Quality; the Bloomberg School's implementation-science tradition; Whiting School engineering and cognitive science; the Berman Institute's governance and ethics work). **Intersectional Expertise** — the distinguishing pillar: holding the engineering, the learning, the measurement, the equity, and the operational context in one conversation. **Capability Focus** — keeping every artifact, measurement, and iteration accountable to what the system can actually do. **Flywheel Iteration** — the Practice Flywheel (**Identify → Activate → Prototype → Analyze → Transition**) that turns each cohort, case, and pilot into the starting condition of the next.

THE RECORD AT JOHNS HOPKINS

The placement is not a default. The School of Education has studied the learner, designed the assessment, and engineered the intervention since before learning engineering had its current name. The Center for Talented Youth (1979) holds one of the longest continuous evidence bases in U.S. education on the identification and accelerated instruction of advanced learners; the Center for Research and Reform in Education has, since 2004, run large-scale evaluations against rigorous evidence standards (ESSA tiers, What Works Clearinghouse) and publishes the Best Evidence Encyclopedia. The Learning Design & Technology program inherits that environment: for two decades its M.Ed. and EdD tracks have graduated designers and evaluators of learning systems whose pedagogical core is design-based research. LENS formalizes that orientation as a specialization rather than departing from it.

THE PILOTS

The claim that capability is engineerable rests on a documented record of learning-engineering pilots run at the Johns Hopkins Applied Physics Laboratory before LENS enrolled its first cohort. **CIRCUIT** was built as the workforce solution to the IARPA MICrONS connectomics program; the cohorts it trained contributed materially to the MICrONS reconstructions now published in Nature, with student development a deliberate byproduct of the mission result. The model was generalized into the **MERIT** framework and the **COMPASS** mentor-intervention layer, documented in the peer-reviewed engineering-education literature. The **PISTA** incubator (DTRA, Summer 2025) compressed the same cycle into ten weeks: nine student fellows shipped a working CBRNE decision-support prototype to agency leadership through three full redesigns. No single pilot settles the proposition; the cumulative record shows the program is itself the institutional flywheel its case studies argue every high-consequence domain requires.

ABOUT THE EDITORS

The editors.

The casebook's central argument — that capability engineering is the applied form of the educational sciences, taught from the School of Education and informed by operational practice — is embodied in the two editors. Each has spent a career working in one half of that pairing and a substantial portion of it in the other. Each contributes to the five pillars on which LENS rests: Mission Literacy · JHU Ecosystem · Intersectional Expertise · Capability Focus · Flywheel Iteration.

William Gray-Roncal, Ph.D.

APPLIED RESEARCH · LEARNING ENGINEERING · STEM WORKFORCE

William Gray-Roncal, Ph.D., is Principal Research Engineer at the Johns Hopkins University Applied Physics Laboratory, where he applies systems engineering methods to challenges spanning defense, space, biomedical, and learning systems. His research — sponsored by DARPA, IARPA, NIH, and DoD — bridges computational neuroscience, artificial intelligence, and large-scale data infrastructure. He developed the College Prep and CIRCUIT workforce-development models, through which he has mentored nearly 500 students, and his current work develops quantitative measures of human cognitive performance in operational contexts. He leads the development of the LENS (Learning Engineering for Next-Generation Systems) specialization within the Johns Hopkins Master of Education in Learning Design and Technology, the program directed by James Diamond at the School of Education. Gray-Roncal holds a Ph.D. in Computer Science from Johns Hopkins, an M.S. in Electrical Engineering from USC, and a B.S. in Electrical Engineering from Vanderbilt.

James Diamond, Ph.D.

LEARNING SCIENCES · EDUCATIONAL DESIGN · EVALUATION

James Diamond, Ph.D., is Assistant Professor in the Johns Hopkins University School of Education and Program Director of the Master of Education in Learning Design and Technology, where he applies design-based research and learning-sciences methods to the design and evaluation of technology-mediated learning environments. His research — supported by the National Science Foundation, the Institute of Education Sciences, the Bill & Melinda Gates Foundation, the MacArthur Foundation's HASTAC Digital Media and Learning Initiative, the Robin Hood Learning + Technology Fund, the Wellcome Trust, and the UK Economic and Social Research Council — bridges digital games and simulations for learning, digital badges and micro-credentialing, classroom technology integration, and disciplinary literacy in history, social studies, civics, and STEM. Before joining Johns Hopkins, he was Senior Research Associate at the Education Development Center's Center for Children and Technology in New York City, where he led projects including Zoom In and the Who Built America? Teacher Mastery Badge System, contributed to Mission US and Possible Worlds, and taught at New York University as an adjunct. As Program Director, he oversees the LDT program within which the LENS (Learning Engineering for Next-Generation Systems) specialization is offered. Diamond holds a Ph.D. in Educational Communication and Technology from New York University, an Ed.M. in Educational Technology from Boston University, and a B.A. in History from Boston University.

These biographies reference careers documented in publications, program reports, faculty pages, and the curricular materials of the LDT program and the LENS specialization.